
MARROW ED8

Physiology

Comprehensive Question Bank

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Homeostasis and cellular physiology

Question 1:

Homeostasis is called as dynamic equilibrium because_____

- a) All variables are given equal importance
- b) Can achieve a constancy in extreme environmental challenge
- c) The set-point for all homeostatic variables is always fixed
- d) Body maintains an internal balance within fluctuating limits

Question 2:

Which is a component of homeostatic control system?

- a) Stimulus
- b) Receptor
- c) Effector
- d) All of the above

Question 3:

Feedback regulation is seen in _____

- a) The increased heart rate that occurs in an athlete just before a sprint begins
- b) Smell of food triggers secretion of salivary and gastric juices
- c) An increase in insulin before glucose absorption has raised blood glucose levels
- d) The increased heart rate in response to a decrease in blood pressure

Question 4:

Positive feedback is seen in all except_____

- a) LH surge
- b) Action potential
- c) Gastric secretion

d) Thrombus formation

Question 5:

On infusion of a large volume of blood, the MAP rises from 100 mmHg to 175 mmHg. Due to the baroreceptor feedback mechanism, the MAP falls down to 125 mmHg. Calculate the gain of the feedback system.

- a) -2
- b) -3
- c) +2
- d) +3

Question 6:

Plasma membrane in eukaryotes is made up of all except _____

- a) Cholesterol
- b) Lecithin
- c) Sphingomyelin
- d) Triglycerides

Question 7:

Membrane fluidity is increased by _____.

- a) Stearic acid
- b) Palmitic acid
- c) Myristic acid
- d) Linoleic acid

Question 8:

Which of the following is an example of peripheral protein?

- a) Sodium potassium ATPase pump
- b) Insulin receptor

- c) G-Protein coupled receptors
- d) Cell surface receptors

Question 9:

Which of the following membranes has the highest protein content per gram tissue?

- a) Myelin sheath
- b) Outer mitochondrial membrane
- c) Sarcoplasmic reticulum
- d) Inner mitochondrial membrane

Question 10:

The sequence of vesicular transport in a cell is

- a) ER - Trans - Cis - Mitochondria
- b) ER - Cis - Trans - Mitochondria
- c) ER - Trans - Cis - Cell membrane
- d) ER - Cis - Trans - Cell membrane

Question 11:

All of the following are enzyme markers of mitochondria except_____

- a) ATP synthase
- b) 5'-nucleotidase
- c) Glutamate dehydrogenase
- d) Creatine kinase

Question 12:

Which of the following will test negative for all mitochondrial markers?

- a) Spermatozoa
- b) Ova
- c) Mature red blood cells

- d) White blood cells

Question 13:

Which is the characteristic enzyme present in peroxisome?

- a) Arylsulfatases
- b) Ribonuclease
- c) Glycosidases
- d) Catalase

Question 14:

Gap junctions are found in all except _____

- a) Neurons
- b) Cardiac muscle
- c) Smooth muscle
- d) Skeletal muscle

Question 15:

Tight junctions are not seen in _____.

- a) Cardiac muscle
- b) Renal tubules
- c) Sertoli cells
- d) Choroid plexus

Question 16:

Which of the following is not a cell adhesion molecule (CAM)?

- a) Integrin
- b) Selectin
- c) Cadherin
- d) Spectrin

Question 17:

Identify the incorrectly matched pair of intercellular connections and their structural units

- a) Desmosome - Desmoglein
- b) Gap junctions - Cadherins
- c) Hemidesmosome - Integrins
- d) Zona occludens - Claudins

Question 18:

Integrin connects with:

- a) Cadherin
- b) Vitronectin
- c) Fibronectin
- d) Selectin

Question 19:

The force-generating proteins are _____.

- a) Myosin and myoglobin
- b) Dynein and kinesin
- c) Calmodulin and kinesin
- d) Troponin and dynamins

Question 20:

An absent dynein arm in cilia can lead to all of the following except:

- a) Chronic sinusitis
- b) Bronchiectasis
- c) Ciliocytophthoria
- d) Situs inversus

Question 21:

In dividing cells, mitotic spindles are formed by ____.

- a) Ubiquitin
- b) Tubulin
- c) Vimentin
- d) Keratin

Answer Key

Question No.	Correct Option
1	d
2	d
3	d
4	c
5	a
6	d
7	d
8	d
9	d
10	d
11	b
12	c
13	d
14	d
15	a
16	d
17	b
18	c
19	b
20	c
21	b

Detailed Explanations

Solution to Question 1:

Homeostasis is called dynamic equilibrium because the body maintains an internal balance within fluctuating limits. It describes the maintenance of nearly constant conditions in the internal environment despite changes in the outside world.

A number of variables such as body temperature, blood sugar levels, blood pH, etc. are regulated within a range of values (homeostatic ranges) and not at fixed values.

These variables are not given equal importance. There is a hierarchy of importance so that the constancy of certain variables may be altered markedly to maintain others within their normal range. For example, respiratory or metabolic compensation to maintain blood pH.

The disruption of homeostasis either due to extreme environmental challenge or a breakdown of internal homeostatic mechanisms is termed as a disease.

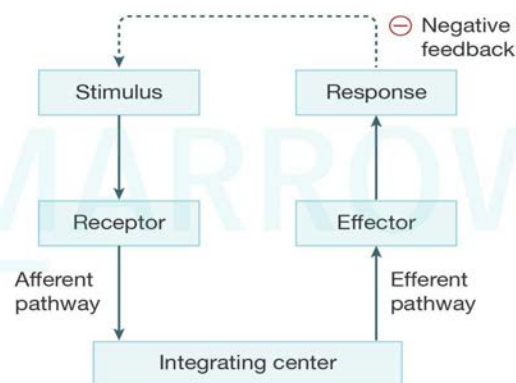
Solution to Question 2:

Stimulus, receptor and effector are all components of a homeostatic control system.

Components of homeostasis include:

- Stimulus - It is a detectable change in the internal or external environment, such as change in temperature.
- Receptor - It detects environmental changes.
- Integrating centre
- Effector - It is an organ or cell that acts in response to a stimulus.

Components of homeostasis



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Solution to Question 3:

Feedback regulation is seen in the increased heart rate in response to a decrease in blood pressure. This is because, in feedback regulation, the homeostatic response (increased heart rate) occurs after the change (decrease in blood pressure) has occurred.

On the other hand, in feedforward regulation, the homeostatic responses take place before the change has occurred. This is done in anticipation of the change to minimize fluctuations.

Examples of feedforward regulation include:

- The increased heart rate that occurs in an athlete just before a sprint begins.
- The smell of food triggers the secretion of salivary and gastric juices.
- An increase in insulin before glucose absorption has raised blood glucose levels.

Solution to Question 4:

Positive feedback is seen in all except gastric secretion. It is under negative feedback control.

Negative feedback consists of certain changes that affect a deficient or excessive variable to pull it back to a certain mean value to maintain homeostasis. Most processes in the body are under negative feedback. For example, an increase in gastric acids leading to a decrease in gastric secretion.

Positive feedback consists of changes that affect a variable to push it further away from a certain mean value. For example, an increase in LH leading to a further increase in LH. Instances, where positive feedback plays an important role are:

- Clotting of blood
- Calcium release from the sarcoplasmic reticulum
- LH Surge
- Action potential
- Shock
- Parturition

Solution to Question 5:

The gain of the feedback system in the above case is -2.

The gain of the system is then calculated by using the following formula:

$$\text{Gain} = C/E$$

Here, C is correction achieved from the deviated value and E is the remaining error post-correction.

Calculation

- The original value in the given case is 100 mmHg.
- The deviated value in the given case is 175 mmHg.
- The corrected value in the given case is 125 mmHg.
- C is given by correction achieved from deviated value, Corrected value - Deviated value = 125 - 175 = -50 mmHg.
- E is given by the remaining error post-correction, Corrected value - Original value = 125 - 100 = 25 mmHg.
- Gain = $-50/25 = -2$.

The negative symbol indicates that the homeostatic mechanism acts in a direction opposite to the initial deviation.

Solution to Question 6:

Plasma membrane in eukaryotes does not contain triglycerides.

The plasma membrane is composed almost entirely of proteins and lipids. The major lipids include:

- Phospholipids: Lecithin, cephalin, phosphatidylserine
- Sphingolipids: Sphingomyelin
- Cholesterol

Solution to Question 7:

Membrane fluidity is increased by linoleic acid, an unsaturated fatty acid. All other options are saturated fatty acids.

Longer and saturated fatty acids decrease the fluidity because their side chains are straight and interact more strongly with each other. Conversely, unsaturated fatty acids increase the fluidity because their side chains are kinked and less compact.

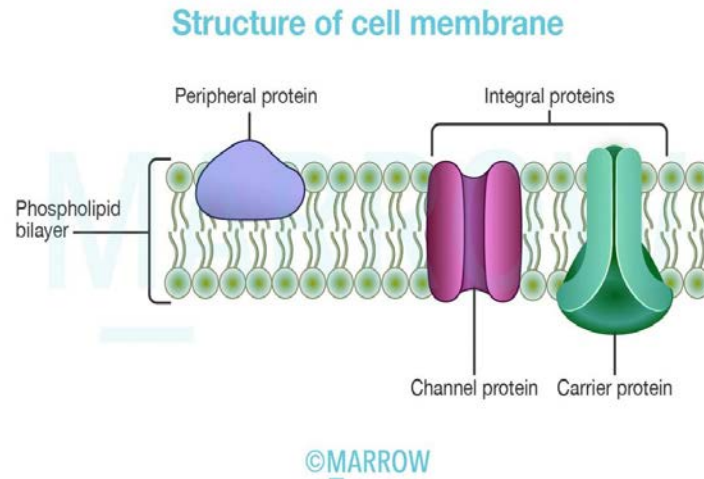
Cholesterol, also known as fluidity buffer helps to maintain the fluidity of the cell membrane.

Solution to Question 8:

Cell surface receptors are peripheral proteins.

Proteins in the Plasma Membrane

Integral Proteins	Peripheral Proteins
Transmembrane proteins i.e. , they span the thickness of the membrane.	Present either on the inner or outer surface of the membrane
Examples -Insulin receptor, pumps, transport proteins (carriers), G-protein coupled receptors	Examples -Cell surface receptors and enzymes



Solution to Question 9:

The inner mitochondrial membrane has the highest protein content per gram tissue.

Myelin has the lowest protein content per gram tissue.

Protein to Lipid Ratio in Different Membranes

Membranes	Protein: Lipid Ratio
Myelin	0.23
Human RBC	1.1
Mitochondrial outer membrane	1.1
Sarcoplasmic reticulum	2.0
Mitochondrial inner membrane	3.2

Solution to Question 10:

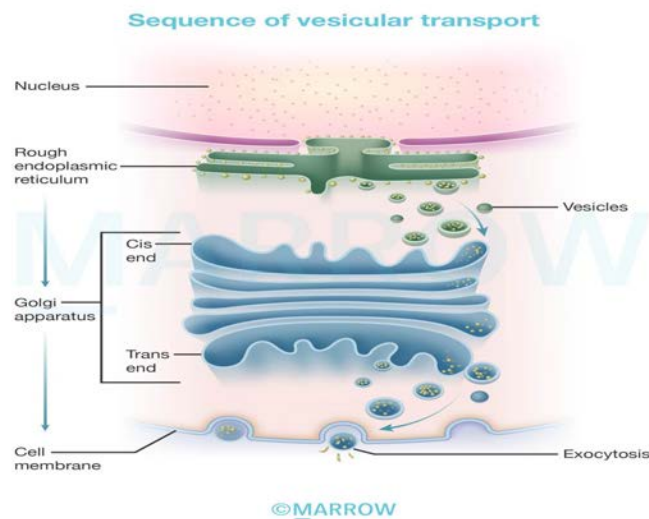
The sequence of vesicular transport in a cell is ER - Cis - Trans - Cell membrane.

Rough endoplasmic reticulum (RER) - It is the site of protein synthesis. Proteins are taken to the Golgi apparatus through transport vesicles.

Cis end of Golgi apparatus - Also called as receiving end, it is closest to RER

Trans end of Golgi apparatus - Also called as releasing end, it is furthest away from RER.

Cell membrane - Synthesised proteins within secretory vesicles are released into the extracellular space through the process of exocytosis.



Solution to Question 11:

5' Nucleotidase is an enzyme marker for plasma membrane. All the other options are mitochondrial enzyme markers.

Solution to Question 12:

Mature red blood cells will test negative for all mitochondrial markers as they do not have mitochondria.

Lack of mitochondria is the reason why red blood cells exclusively use anaerobic glycolysis to generate ATP.

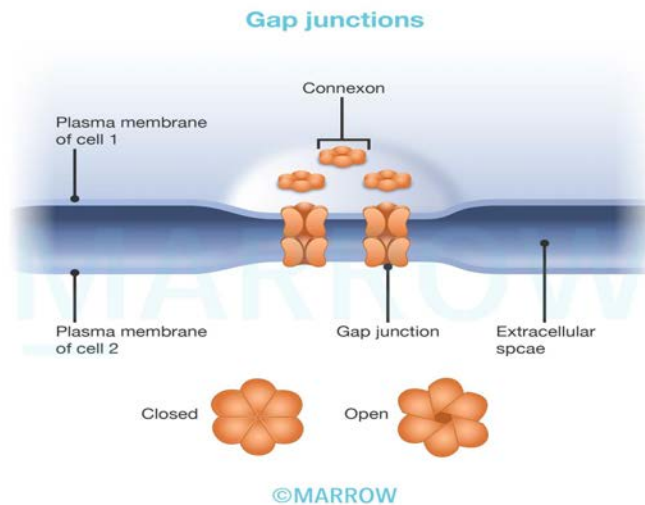
Solution to Question 13:

Catalase is the characteristic enzyme of peroxisomes. All other are lysosomal enzymes.

Solution to Question 14:

Gap junctions are absent in adult skeletal muscles.

Gap junctions are intercellular connections with a channel in between. They are present in excitable tissues such as cardiac muscle, smooth muscle and neurons but not in skeletal muscles.



Solution to Question 15:

Tight junctions are not seen in cardiac muscle.

Tight junctions form a water-tight seal between cells. They are characteristically seen in the following:

- Intestinal mucosa
- Renal tubules
- Choroid plexus
- Sertoli cells

Functions of tight junctions are:

- Prevents the diffusion of macromolecules between cells.
- Maintains cell polarity.
- Seals the paracellular pathway.

Solution to Question 16:

Spectrin is a cytoskeletal protein but not a cell adhesion molecule.

Cells are attached to each other or to the basal lamina by cell adhesion molecules (CAMs). Some important classes of CAMs are:

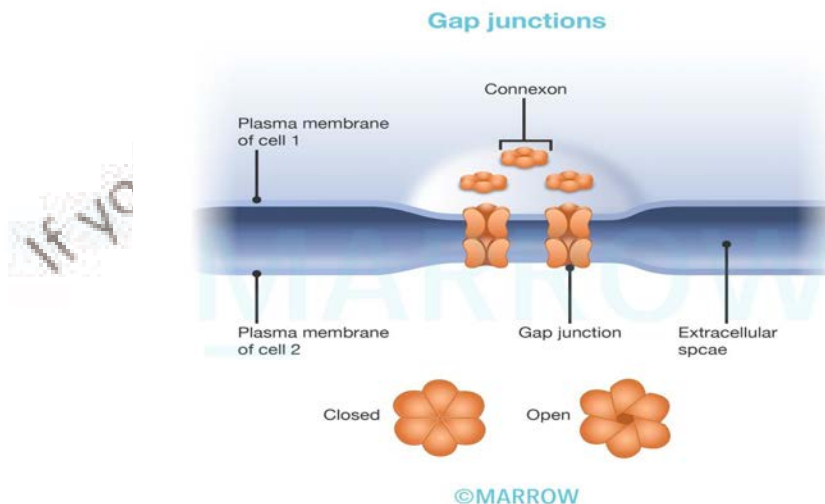
- Integrins: heterodimers that bind to various receptors
- IgG superfamily of immunoglobulins
- Cadherins: Ca^{2+} -dependent molecules that mediate cell-to-cell adhesion by homophilic reactions
- Selectins which have lectin-like domains that bind carbohydrates

Clinical significance

- Cancer cells have abnormal CAMs and hence cause decreased cell adhesion and allow individual cancer cells to detach from the parent cancer.
- The decrease in the amount of E-cadherin maybe the reason for early metastasis of numerous tumors.

Solution to Question 17:

Gap junctions are made of connexons. Connexons are in turn made of units called connexins.



The different intercellular connections and their structural units are given below:

Intercellular connections	Structural units
Tight junctions or zona occludens	Occludins, claudins and junctional adhesion molecules
Gap junctions	Connexons
Zona adherens	Cadherins

Intercellular connections	Structural units
Desmosome	Cadherins like desmoglein
Hemidesmosome	Integrins

Solution to Question 18:

Integrin connects with fibronectin.

Integrins are transmembrane glycoproteins (integral proteins) that allow cells to attach to extracellular matrix constituents such as laminin, fibronectin and collagen. It helps in linking the intracellular cytoskeleton functionally and structurally with the outside world.

Solution to Question 19:

The force-generating proteins are dynein and kinesin.

There are three superfamilies of proteins namely kinesin, dynein, myosin referred to as molecular motors.

The molecular motors are the force generating proteins as they move proteins, organelles, and other cell parts (collectively referred to as "cargo") to all parts of the cell.

Solution to Question 20:

An absent dynein arm in cilia leads to primary ciliary dyskinesia. Kartagener syndrome is a primary ciliary dyskinesia characterized by the classical triad of situs inversus, chronic sinusitis, and bronchiectasis.

Ciliocytophthoria is the degradation of ciliated cells due to viral infections. It is not seen in primary ciliary dyskinesia.

Solution to Question 21:

Tubulins constitute the microtubules that form the mitotic spindle in a dividing cell.

Cytoskeletal elements of the cell.

Ubiquitin helps in protein degradation.

Cytoskeletal element	Protein subunits
Microfilament(smallest diameter)	Actin

Cytoskeletal element	Protein subunits
Intermediate filament	Keratin, vimentin, desmin, lamin etc.
Microtubule(largest diameter)	Kinesin, dynein and tubulin

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Cellular messengers & Receptors

Question 1:

Which of the following is an example of a polypeptide hormone?

- a) Parathormone
- b) Adrenaline
- c) Cortisol
- d) Thyroxine

Question 2:

Which hormone crosses the plasma membrane for its action?

- a) Epinephrine
- b) Thyroxine
- c) Glucagon
- d) Insulin

Question 3:

Which of the following hormone is not stored before release?

- a) Insulin
- b) Cortisol
- c) Thyroxine
- d) Renin

Question 4:

Downregulation of receptors occurs due to all except_____

- a) Inactivation of receptor
- b) Internalisation
- c) Desensitisation

- d) Decreased level of stimulating hormone

Question 5:

Receptors on cell membrane that activate ion channel after binding with agonists are_____

- a) Nicotinic cholinergic
- b) Muscarinic cholinergic
- c) Opioid μ receptors
- d) GABAB receptors

Question 6:

Adrenaline, noradrenaline and dopamine act through_____

- a) 1 pass receptor
- b) 2 pass receptor
- c) 4 pass receptor
- d) 7 pass receptor

Question 7:

Which of the following statements is true about the G-protein-coupled receptor (GPCR)?

- a) G proteins bind to hormones on the surface of the nucleus.
- b) All the three subunits α , β , and γ should bind to each other for G proteins to act.
- c) G proteins act as inhibitory or excitatory because of differences in α subunit.
- d) G protein is bound to GTP in resting state.

Question 8:

Action of α subunit of G protein is_____

- a) Breakdown of GTP to GDP
- b) Conversion of GDP to GTP
- c) Internalization of receptors
- d) Binding of agonist

Question 9:

PDGF acts through which of the following signaling pathways?

- a) cAMP
- b) IP₃-DAG
- c) cGMP
- d) Tyrosine kinase

Question 10:

Which hormone acts on JAK-STAT kinase receptor?

- a) TSH
- b) Thyroxine
- c) GH
- d) FSH

Question 11:

All of the following act through cell surface receptors except_____

- a) Cortisol
- b) Insulin
- c) FSH
- d) TSH

Question 12:

Steroid hormone receptors have an attachment site for all except_____

- a) Steroid hormone
- b) Transcription repressors
- c) Hormone-responsive element
- d) Transcription activators

Question 13:

True statement about second messengers is _____

- a) Are substances that interact with first messengers outside cells.
- b) Mediate the intracellular responses to many different hormones and neurotransmitters
- c) Are hormones secreted by cells in response to stimulation by another hormone
- d) Are substances that bind to first messengers in the cell membrane.

Question 14:

Which one of the following acts as a second messenger?

- a) Mg^{2+}
- b) Cl^-
- c) Ca^{2+}
- d) PO_4^{3-}

Question 15:

A primigravida with 38 completed weeks of gestation has gone into labor. Oxytocin was started to augment labor. The second messenger system through which oxytocin acts is _____

- a) cAMP
- b) Phospholipase c
- c) cGMP
- d) Tyrosine kinase

Question 16:

cAMP is the exclusive second messenger for all the following except _____

- a) Corticotropin
- b) Dopamine
- c) Glucagon
- d) Vasopressin

Question 17:

cGMP is the second messenger of

- a) Somatostatin
- b) Angiotensin II
- c) ADH
- d) NO (nitric oxide)

Question 18:

Which of the following is not a second messenger?

- a) Guanylyl cyclase
- b) cAMP
- c) IP₃
- d) DAG

Question 19:

Activation of protein kinase C causes _____

- a) Increased adenylyl cyclase
- b) Phosphorylation of proteins
- c) Increased cAMP
- d) Increased ion channel conduction

Question 20:

A 7-year-old girl presents with skin lesions as shown in the image. The parents also complain of early breast development, growth acceleration, and vaginal bleeding. On genetic studies, it was found that there is a gain of function mutation of G Protein. What is the most probable diagnosis?



- a) Color blindness
- b) Hirschsprung disease
- c) McCune-Albright syndrome
- d) Familial hypothyroidism

Question 21:

Nuclear signalling pathways are all except

- a) Caveolins
- b) NF-kB
- c) Importins
- d) Rat proteins

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	d
5	a

6	d
7	c
8	a
9	d
10	c
11	a
12	b
13	b
14	c
15	b
16	d
17	d
18	a
19	b
20	c
21	d

Detailed Explanations

Solution to Question 1:

Parathyroid hormone (PTH) is an example of a polypeptide hormone.

Cortisol is a steroid hormone. Adrenaline and thyroxine are derivatives of the amino acid tyrosine.

Solution to Question 2:

Thyroxine crosses the plasma membrane for its action.

Thyroid hormones enter the nucleus and bind to thyroid hormone receptors. The hormone-receptor complex in the nucleus binds to DNA via zinc fingers. This increases the expression of a variety of different genes that code for proteins regulating cell function. Although thyroxine is an AA derivative, there are transporters to move it into the cell, hence they are intracellular receptors acting on the nucleus.

Epinephrine, glucagon, and insulin bind to cell surface receptors.

Solution to Question 3:

Cortisol is not stored before release.

Usually, the polypeptide and protein hormones such as insulin, thyroxine or renin are stored in secretory vesicles before release. They are synthesized as much larger polypeptide chains (preprohormones) that are biologically inactive and then processed intracellularly by specific proteases to yield the final hormone molecule which is stored in the secretory vesicle until needed.

On the other hand, steroid hormones such as cortisol are synthesised as needed and are not stored. They are lipid soluble and easily diffuse across cell membrane to enter the interstitial fluid and then blood.

Solution to Question 4:

Downregulation of receptors occurs due to internalisation, inactivation of receptors or desensitisation.

A decrease in the level of stimulating hormones generally leads to an increase in the level of active receptors. This is called upregulation of receptors.

On the other hand, an increase in the level of stimulating hormones generally leads to a decrease in the level of active receptors. This is called downregulation of receptors.

Downregulation of receptors can occur by one of the following mechanisms:

- Desensitisation - Receptors undergo chemical modifications that make them less responsive.
- Internalisation - It is the temporary sequestration of the cell membrane receptors to the inside of the cell, away from the action of hormones. Following internalisation, the receptors are either destroyed or recycled.
- Inactivation of receptors.
- Decreased production of receptors.

Solution to Question 5:

Nicotinic cholinergic receptors activate an ion channel after binding with its agonist. These receptors are called ligand-gated channels or ionotropic receptors.

Receptors for neurotransmitters are classified into two broad families based on structure and function. They are:

- Ligand-gated channels (Ionotropic receptors)
- When bound to these receptors, a membrane channel is opened leading to a brief increase in ionic conductance.
- Important for fast synaptic transmission.
- Examples - GABAA, GABAC, Nicotinic cholinergic, Glycine and Serotonin (5HT₃ subtype).
- G-protein coupled receptors (Metabotropic receptors)

- When bound to these receptors, secondary messengers are released that modulate voltage-gated channels on neuronal membranes.
- Examples - GABAB, Muscarinic cholinergic, Catecholamines, Serotonin (all subtypes except 5HT₃) and opioid receptors.

Solution to Question 6:

Adrenaline, noradrenaline and dopamine act through a 7 pass receptor called G-protein coupled receptor (GPCR).

The catecholamines act through GPCRs which are cell membrane receptors that span across the membrane 7 times hence called seven helix or serpentine receptors.

Solution to Question 7:

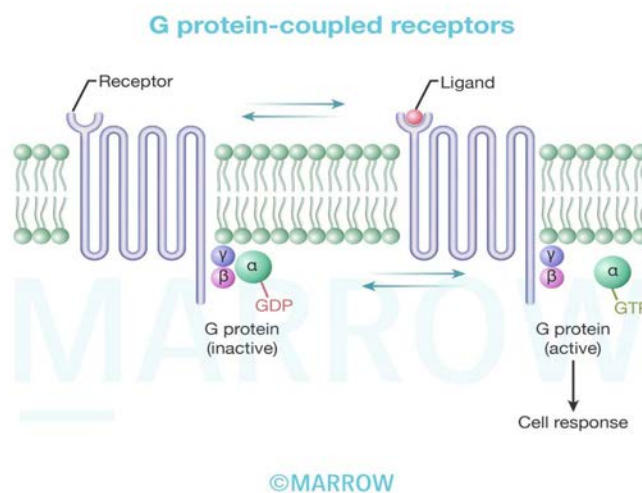
G proteins act as inhibitory or excitatory because of differences in α subunit.

G protein-coupled receptors (GPCRs) are cell membrane receptors that span across the membrane 7 times hence called seven helix or serpentine receptors.

G proteins have 3 subunits (α , β and γ). In its resting state, α subunit is bound to GDP.

Upon ligand binding to GPCR, it undergoes a conformational change that activates G protein within the cell. The α subunit dissociates and in turn alters the activity of second messengers such as cAMP, cGMP or Ca²⁺.

Once the ligand is unbound, the intrinsic GTPase activity of the α subunit converts GTP to GDP. This leads to reassociation of α with the β - γ subunit and termination of effector activation.



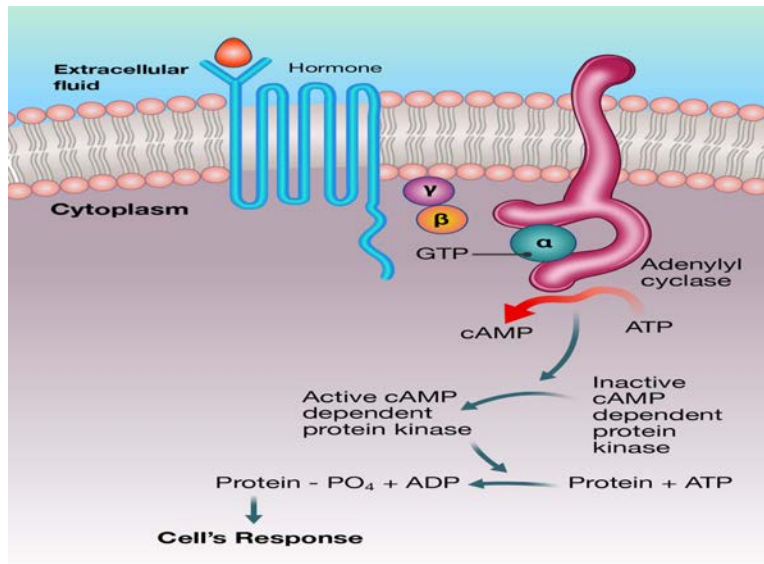
Solution to Question 8:

The action of α subunit of G protein is the breakdown of GTP to GDP as it has an intrinsic GTPase activity.

Guanine exchange factors (GEFs) convert GDP to GTP by nucleotide exchange.

G proteins can be either stimulatory (G_s) or inhibitory (G_i) with respect to the production of the 2nd messenger cAMP by the enzyme adenylyl cyclase. Stimulatory or inhibitory activity resides in the α subunit.

- Cholera toxin acts on α subunit of G_s
- Pertussis toxin acts on α subunit of G_i



Solution to Question 9:

PDGF (platelet-derived growth factor) acts through the activation of tyrosine kinases.

All growth factors that foster the multiplication or development of various types of cells act through the activation of tyrosine kinases.

Solution to Question 10:

Growth hormone acts on JAK-STAT receptors.

Janus tyrosine kinase receptors (JAK-STAT) serves as receptors for:

- Growth hormone
- Prolactin
- Erythropoietin

- Leptin

Thyroxine acts on nuclear receptors. TSH and FSH act on G protein-coupled receptors.

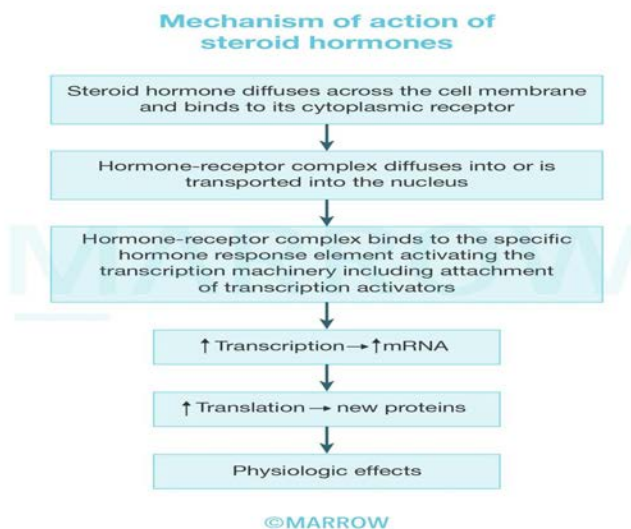
Solution to Question 11:

Cortisol is a steroid hormone and acts on intracellular (cytoplasmic) receptor.

Solution to Question 12:

Steroid hormone receptors have an attachment site for steroid hormone, hormone-responsive element, and transcription activators, but not transcription repressors.

Transcription repressors are proteins that bind to specific sites on DNA and prevent transcription of nearby genes.



Solution to Question 13:

Second messengers mediate the intracellular responses to many different hormones and neurotransmitters.

Cell surface receptors are substances that interact with the first messengers which are hormones and neurotransmitters.

The second messengers are the intracellular signalling molecules. They bring about many short-term changes in cell function by altering enzyme function, triggering exocytosis, and other physiological effects. They can also lead to the alteration of transcription of various genes (long term effect).

Examples of second messengers are cAMP, cGMP, Ca^{2+} and IP₃-DAG (Inositol triphosphate and diacylglycerol).

Solution to Question 14:

Among the following calcium (Ca^{2+}) acts as a second messenger.

Increased cytoplasmic Ca^{2+} binds to and activates calcium-binding proteins (calmodulin, troponin) or activate protein kinases (or phospholipase C) to bring about physiological effects.

Cytoplasmic Ca^{2+} is increased by:

- Direct entry of Ca^{2+} from the ECF through Ca^{2+} channels that could be ligand-gated, voltage-gated or stretch-activated (mechanically gated).
- Mobilisation of Ca^{2+} that are stored in high concentrations in the endoplasmic reticulum by other second messengers such as IP₃-DAG (Inositol triphosphate and diacylglycerol).

Solution to Question 15:

The second messenger system through which oxytocin acts is the phospholipase C system.

It activates the enzyme phospholipase C which breaks down some phospholipids in the cell membrane into IP₃ (phosphoinositol triphosphate) and DAG (diacylglycerol). DAG then activates protein kinase C and IP₃ mobilises Ca^{2+} from mitochondria and endoplasmic reticulum to bring about smooth muscle contraction and other cellular responses.

Solution to Question 16:

Vasopressin does not use cAMP exclusively.

Vasopressin (ADH) acts on at least 3 kinds of receptors: V_{1A}, V_{1B} and V₂

- All the 3 are G-protein coupled receptors
- V_{1A} and V_{1B} receptors (G_q) act through IP₃ as a second messenger, to increase the intracellular calcium concentration
- V₂ receptors (G_s) use cAMP as a second messenger

cAMP is the exclusive second messenger for dopamine, corticotropin and glucagon.

Dopamine has 5 receptors (D₁, D₂, D₃, D₄, D₅)

- all use cAMP as second messengers
- D₁, D₅ act by increasing cAMP
- D₂, D₃, D₄ act by decreasing cAMP

Solution to Question 17:

Vasodilatory actions of NO are mediated by cGMP.

NO is a hormone derived from the amino acid arginine. It activates guanylyl cyclases in vascular smooth muscle cells resulting in the conversion of cGTP to cGMP (2nd messenger). cGMP activates cGMP dependent protein kinase, which causes blood vessels to relax.

cGMP is the second messenger for nitric oxide and atrial natriuretic factor.

Solution to Question 18:

Guanylyl cyclase is the enzyme that catalyzes the formation of cGMP. It is cGMP that acts as the second messenger.

cAMP, IP₃ and DAG are all second messengers.

Solution to Question 19:

Activation of protein kinase C causes phosphorylation of proteins

Cellular Phosphorylation is under the control of two groups of proteins:

- Kinases
- enzymes that catalyse the phosphorylation of tyrosine or serine residues in protein
- Example- Protein Kinase A, Protein Kinase C, Ca²⁺/Calmodulin Kinase
- Phosphatases
- that catalyses the dephosphorylation of tyrosine or serine residues in the protein

Kinases play important roles in regulating cellular physiology, including cell growth and cell death. Philadelphia chromosome, t(9:22) or the bcr-abl fusion gene codes for an abnormal kinase that leads to chronic myeloid leukaemia (CML). The drug imatinib was developed to block the tyrosine kinase activity of bcr-abl protein and is an effective treatment option for CML.

Solution to Question 20:

Among the given options, the most probable diagnosis is McCune-Albright syndrome as it is the only gain-of-function mutation of G Protein.

G proteins or G protein-coupled receptor complexes can undergo a loss- or gain-of-function mutation. Some examples are given below:

In McCune-Albright syndrome, a gain-of-function mutation in Gs^o is associated with the rough-bordered areas of skin pigmentation and hypercortisolism. This mutation occurs during fetal development, creating a mosaic of normal and abnormal cells.

Receptor	Type of mutation	Disease
Cone opsin	Loss	Colour blindness
Rhodopsin	Loss	Night blindness or retinitis pigmentosa
Endothelin receptor	Loss	Hirschsprung disease
TSH receptor	Loss	Familial hypothyroidism
TSH receptor	Gain	Familial hyperthyroidism
LH receptor	Gain	Familial male precocious puberty
Gs protein	Gain	Pituitary adenomas with acromegaly
Gs protein	Gain (mosaic)	McCune-Albright syndrome

Solution to Question 21:

Rat proteins are not nuclear signalling pathways. Recombinant rat IFN- is a bioactive protein intended for use in cell culture applications.

Importin:

- Transporting protein molecules into nucleus by binding to specific recognition sequences, called Nuclear Localization Sequences (NLS).
- 2 subunits, importin α and importin β .

Caveolins:

- Family of integral membrane proteins that are principal components of caveolae membranes
- Involved in receptor-independent endocytosis.
- Acts as scaffolding proteins within caveolar membranes by compartmentalizing and concentrating signaling molecules.

NF- κ B

- Part of a prototypical proinflammatory signaling pathway
- based on the role of NF- κ B in the expression of proinflammatory genes including cytokines, chemokines, and adhesion molecules

Transport Across Cell Membrane

Question 1:

Transport of lipid-soluble substances across the cell membrane occurs via which of the following?

- a) Porins
- b) Ionophore
- c) Lipopolysaccharides
- d) Diffusion

Question 2:

Which of the following will diffuse more rapidly through phospholipid membranes?

- a) Oxygen
- b) Sodium
- c) Chloride
- d) Water

Question 3:

Which of the following is explained by the Fick's law of diffusion?

- a) Active diffusion along concentration gradient
- b) Passive diffusion along concentration gradient
- c) Active diffusion against concentration gradient
- d) Passive diffusion against concentration gradient

Question 4:

For a membrane permeable substance, the intracellular concentration (IC) is 10 units and extracellular concentration (EC) is 5 units. If the intracellular concentration, surface area of the membrane and thickness of the membrane are doubled, and the EC remains the same, What will be the net effect on the diffusion?

- a) Diffusion becomes 2x
- b) Diffusion becomes 3x
- c) Diffusion becomes 1/2x
- d) Diffusion becomes 1/3x

Question 5:

Which of the following does not follow transport saturation kinetics?

- a) Facilitated diffusion
- b) Na⁺-Ca²⁺ exchanger
- c) Simple diffusion
- d) Coupled transport

Question 6:

Which statement about the facilitated diffusion is true?

- a) Form of active transport
- b) Requires carrier protein
- c) Doesn't follow saturation kinetics
- d) Requires creatinine phosphate

Question 7:

The movement of which of the following substance across the membrane is most affected by osmotic pressure?

- a) Sodium
- b) Potassium
- c) Water
- d) Glucose

Question 8:

Osmolarity is defined as _____

- a) Number of osmole per litre
- b) Number of osmole per kg
- c) Weight of solute per litre
- d) Weight of solvent per litre

Question 9:

Passive transfer of only solvent occurs in _____

- a) Diffusion
- b) Active transport
- c) Osmosis
- d) Pinocytosis

Question 10:

A 47-year-old diabetic man takes a low-dose aspirin daily due to cardiovascular risk. The absorption of aspirin in the stomach is an example of which type of transport?

- a) Nonionic diffusion
- b) Facilitated diffusion
- c) Simple diffusion
- d) Secondary active transport

Question 11:

Energy is required in which of the following mode of transport.

- a) Osmosis
- b) Diffusion
- c) Facilitated diffusion
- d) Active transport

Question 12:

Movement of calcium from the cytosol to the extracellular space takes place by

- a) Primary active transport
- b) Facilitated diffusion
- c) Simple diffusion
- d) None of the above

Question 13:

Na⁺-K⁺ ATP pump uses _____

- a) ATPase
- b) GTPase
- c) Acetyl CoA
- d) NADPH

Question 14:

Primary active transport is mediated by which one of the following enzymes?

- a) G-Protein
- b) Na⁺-K⁺ ATPase
- c) Carrier protein
- d) Channel protein

Question 15:

For Na⁺-K⁺ ATPase coupling ratio is _____

- a) 1:1
- b) 1:2
- c) 2:3
- d) 3:2

Question 16:

A 40-year-old diabetic man has been prescribed empagliflozin. Which of the following processes will most likely be affected by this drug?

- a) Secondary active transport
- b) Primary active transport
- c) Facilitated diffusion
- d) Simple diffusion

Question 17:

Glucose is absorbed in intestine by?

- a) Primary active transport
- b) Facilitated diffusion
- c) Simple diffusion
- d) Secondary active transport

Question 18:

A 40-year-old lady is diagnosed with Lambert- Eaton syndrome. Radioimmunoassay reveals the presence of serum antibodies against voltage-gated calcium channels. Which of the following process will be affected as a result of this pathology?

- a) Facilitated Diffusion
- b) Endocytosis
- c) Exocytosis
- d) Simple Diffusion

Question 19:

Which pathway is called regulated pathway?

- a) Constitutive exocytosis
- b) Constitutive endocytosis
- c) Receptor mediated endocytosis
- d) Non Constitutive exocytosis

Question 20:

A 30-year-old gentleman is diagnosed with elevated blood cholesterol levels. A defect in a process that involves the protein 'clathrin' is being suspected. Which of the following process will be most likely affected by this defect?

- a) Receptor mediated endocytosis
- b) Exocytosis
- c) Cell to cell adhesion
- d) Plasma membrane

Question 21:

Which of the following is true about secretory vesicle transport?

- a) Along concentration gradient
- b) No relation with concentration gradient
- c) Against the concentration gradient
- d) Passive method of membrane transport

Question 22:

Transcytosis occurs in which of the following cell types?

- a) Epithelial cells of intestine
- b) Neuroglia
- c) Neuron
- d) Axolemma

Question 23:

Which of the following ion is used in exocytosis?

- a) K^+
- b) Na^+
- c) Ca^{2+}
- d) Mg

Question 24:

A 35-year-old diabetic man is started on an insulin secretagogue. Which of the following ions is required for endocytosis or reverse pinocytosis during the insulin secretion process?

- a) Na⁺
- b) K⁺
- c) Ca²⁺
- d) Mg²⁺

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	b
5	c
6	b
7	c
8	a
9	c
10	a
11	d
12	a
13	a
14	b
15	d
16	a
17	d
18	c
19	d
20	a
21	b
22	a
23	c
24	c

Detailed Explanations

Solution to Question 1:

Lipid-soluble substances are transported across the cell membrane via diffusion.

- The rate of diffusion depends directly on lipid solubility.
- Characteristics of simple diffusion
- The only form of transport that is not carrier-mediated
- Occurs down an electrochemical gradient ("downhill")
- Does not require metabolic energy and therefore is passive
- Permeability
- Describes the ease with which a solute diffuses through a membrane
- Depends on the characteristics of the solute and the membrane
- Small hydrophobic solutes (e.g., O_2) have the highest permeabilities in lipid membranes.
- Hydrophilic solutes (e.g., Na^+) must cross cell membranes through water-filled channels, or pores. If the solute is an ion, then its flux will depend on both the concentration difference and the potential difference across the membrane.

Solution to Question 2:

Among the following substances, oxygen diffuses more rapidly through phospholipid membranes as it is a lipid-soluble substance.

Phospholipid membranes are:

- Permeable to lipid-soluble substances (e.g., O_2 , CO_2 , NH_3 , steroid hormones)
- Impermeable to water-soluble substances (e.g., Na^+ , Cl^- , glucose)
- Semipermeable to water

Solution to Question 3:

Fick's law of diffusion explains passive diffusion along the concentration gradient.

Fick's law of diffusion: The magnitude of the diffusing tendency is directly proportional to the cross-sectional area across which diffusion is taking place and the concentration gradient, or chemical gradient, which is the difference in concentration of the diffusing substance divided by the thickness of the membrane.

Solution to Question 4:

The net effect on the diffusion is, it becomes 3x.

The magnitude of the diffusion:

where,

- J = net rate of diffusion
- D = diffusion coefficient
- A = area of the membrane
- Δx = thickness of the membrane or the distance the molecule must travel across for diffusion
- Δc = concentration gradient

In the condition mentioned above, the surface area, the thickness of the membrane and intracellular concentration becomes doubled. This means,

- A is doubled = $2A$
- Δx is doubled = $2\Delta x$
- IC is doubled. $10 \times 2 = 20$ units, EC is 5 units.

So, $\Delta c = 20 - 5 = 15$. Δc was 5 with the original concentrations. Hence Δc has been tripled.

Now,

$$J = -D \times 2A \times 3\Delta c / 2\Delta x = 3 \times [DA \Delta c / \Delta x]$$

Therefore the diffusion of the substance increases by three-fold.

2nd method :

$$J = -DA (\Delta c / \Delta x)$$

After applying changes, $\Delta c = \text{intracellular conc.} - \text{extracellular conc.} = 10 - 5 = 5$

$$\text{Therefore, } J_1 = -DA (5 / \Delta x)$$

After appropriate doublings, New intracellular conc. = $10 \times 2 = 20$ New extracellular conc. = 5 (since no change) $\Delta c = 20 - 5 = 15$

$$\begin{aligned} \text{Therefore, } J_2 &= -D \cdot 2A (15 / 2\Delta x) \\ &= -DA (15 / \Delta x) \end{aligned}$$

$$\text{Now, } J_2 / J_1 = [-DA (15 / \Delta x)] / [-DA (5 / \Delta x)]$$

$$J_2 / J_1 = 15 / 5$$

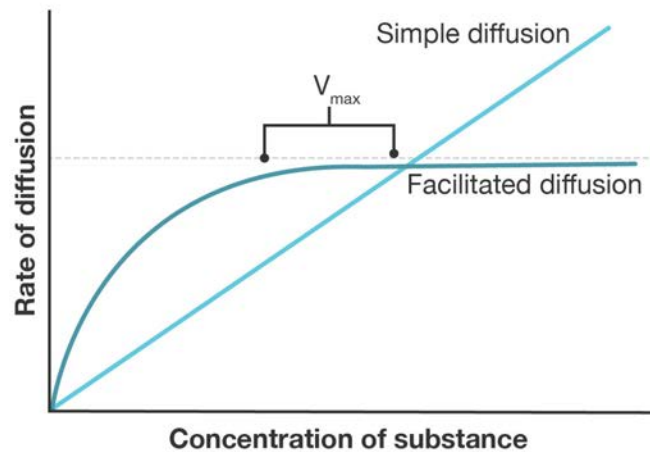
$$= 3$$

Therefore the diffusion of the substance increases by three-fold.

Solution to Question 5:

Simple diffusion is not carrier-mediated transport hence do not follow saturation kinetics.

- Facilitated diffusion, primary active transport, and secondary active transport all involve integral membrane proteins and are called carrier-mediated transport.
- All carrier-mediated transport system follows saturation kinetics.
- The figure shows that as the concentration of the diffusing substance increases, the rate of simple diffusion continues to increase proportionately, but in the case of facilitated diffusion, the rate of diffusion cannot rise greater than the transport maximum (T_m). T_m is analogous to the maximum velocity (V_{max}) level in enzyme kinetics.



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Solution to Question 6:

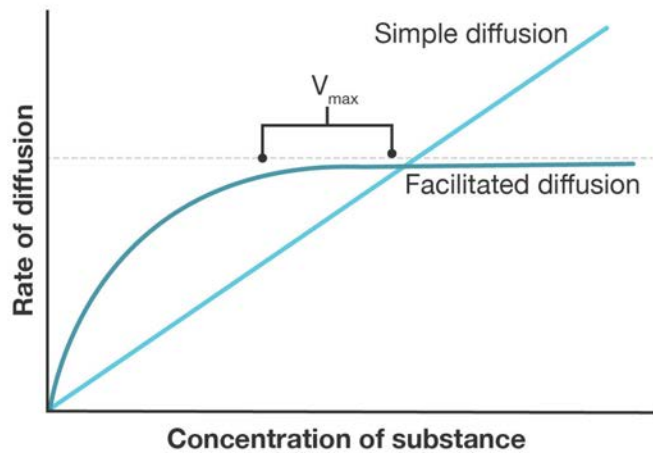
When carrier proteins help in the movement of substances in the direction of their chemical or electrical gradients (downhill movement), no energy input is required and the process is called facilitated diffusion.

Examples

- GLUT
- Urea Transporters :
- UTA 1, 2, 3, 4 in Kidney
- UTB in RBC
- Amino Acid Transporters

Other options are wrong because:

- Facilitated diffusion is a type of passive transport.
- It follows saturation kinetics.
- No energy input is required in facilitated diffusion.



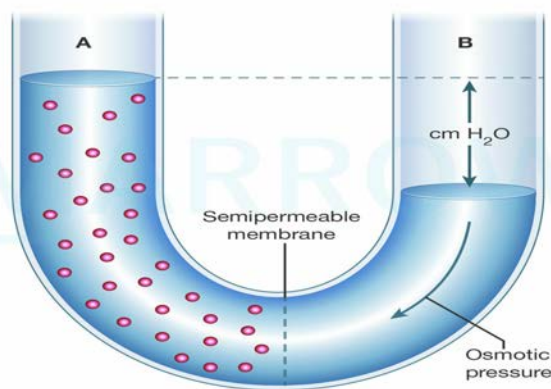
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Solution to Question 7:

The movement of water across the membrane is most affected by osmotic pressure.

Osmotic pressure is the pressure that would have to be applied to prevent the movement of water molecules to the more concentrated solution i.e. to a region in which there is higher concentration of a solute to which the membrane is impermeable.

Osmotic pressure



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- Osmotic pressure is calculated by Vant Hoff's law.

Solution to Question 8:

Difference between osmolarity & osmolality:

- Osmolarity is the number of osmoles per litre of solution,
- Osmolality is the number of osmoles per kilogram of solvent.
- Therefore, osmolarity is affected by the volume of the various solutes in the solution and the temperature, while the osmolality is not.

Solution to Question 9:

Passive transfer of the only solvent occurs in Osmosis.

The diffusion of solvent molecules into a region in which there is a higher concentration of a solute to which the membrane is impermeable is called osmosis.

Note: Membrane is impermeable to the solute.

Solution to Question 10:

The absorption of salicylates like aspirin in the stomach is an example of nonionic diffusion.

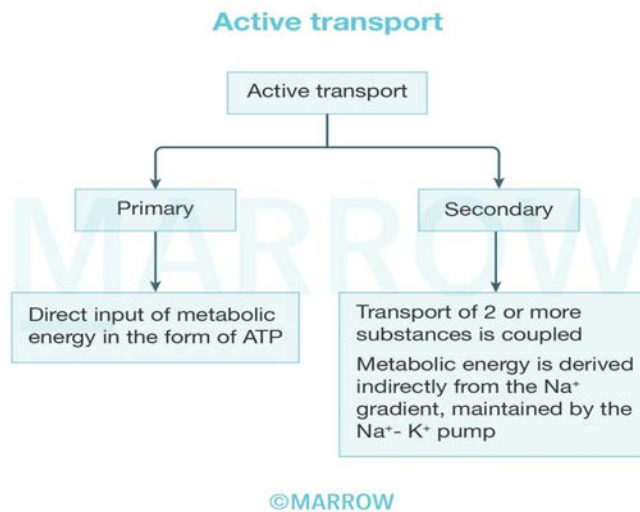
- Some weak acids and bases are quite soluble in cell membranes in the undissociated form, but they cannot cross membranes in the charged (i.e. dissociated) form. Consequently, molecules of the undissociated substance diffuse from one side of the membrane to the other and then dissociate. This phenomenon is called nonionic diffusion.

Examples

- Ammonia excretion in collecting ducts
- Excretion of weak acids and bases in the kidney
- Absorption of salicylates in the stomach
- Absorption of bile acids in the distal ileum

Solution to Question 11:

When a carrier protein transports substances against their electrical and chemical gradients (uphill movement) it requires energy and this type of transport is called active transport.



Solution to Question 12:

The movement of calcium from the cytosol to the extracellular space takes place by primary active transport.

Two primary active transport calcium pumps maintain low intracellular calcium levels. One pumps calcium ions to the extracellular space and another to the sarcoplasmic reticulum.

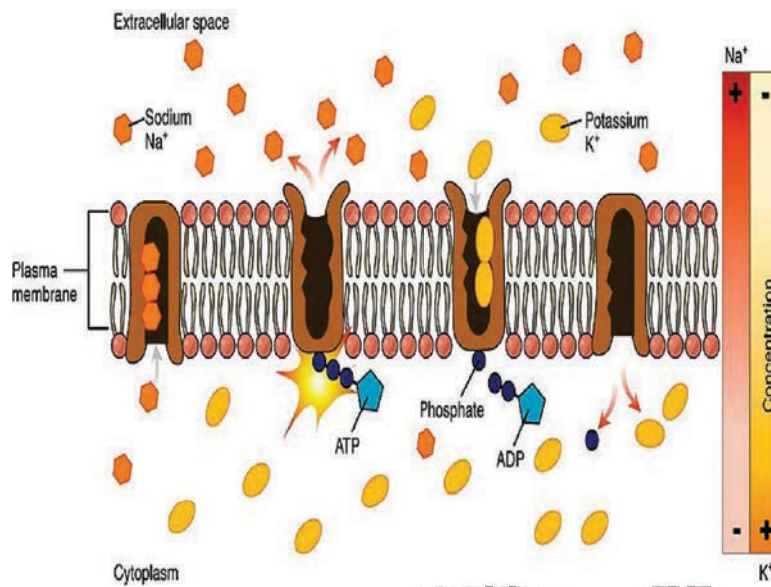
Calcium transport also takes place via secondary active transport. A sodium-calcium exchanger/antiporter moves sodium ions to the interior and calcium ions to the exterior.

Solution to Question 13:

Na⁺- K⁺ ATPase or Na⁺-K⁺ ATP pump catalyzes the hydrolysis of ATP to ADP and uses the energy to extrude three Na⁺ from the cell and take two K⁺ into the cell for each molecule of ATP hydrolyzed.

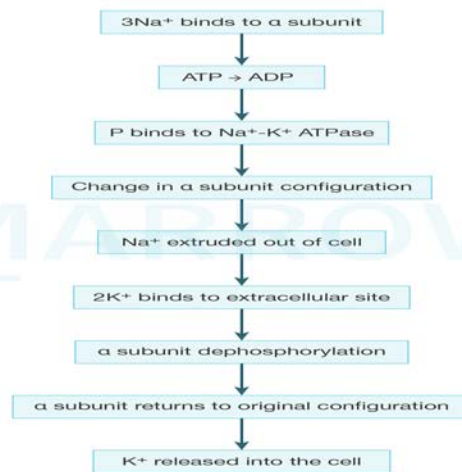
Sodium-Potassium Pump/ Na⁺- K⁺ Pump:

- Carrier protein located on the plasma membrane of all cells
- Pumps 3 Na⁺ out of the cell & 2 K⁺ into the cell (coupling ratio 3:2)
- 2 separate globular proteins:
- Larger- α subunit & smaller- β subunit, α subunit concerned with Na⁺- K⁺ transport
- Binding sites on α subunit:
- 3 intracellular sites- for Na⁺, ATP, and one phosphorylation site
- 2 extracellular sites- for K⁺ and ouabain



Mechanism Of Action

Sodium-potassium pump



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Functions:

- Controlling Cell Volume.
- Electrogenic Nature- maintains Resting membrane potential
- The energy from the Na⁺-K⁺ pump also indirectly serves as the energy source for secondary active transport.

Regulation:

- Pump activity is altered by second messenger molecules eg. cAMP, DAG
- Stimulated by thyroid hormones, insulin and aldosterone
- Inhibition by dopamine, digitalis glycosides, ouabain

Solution to Question 14:

In active transport the energy is provided almost exclusively by the hydrolysis of ATP. Hence many carrier molecules are ATPases (eg. Na⁺-K⁺-ATPase), enzymes that catalyze the hydrolysis of ATP.

PRIMARY ACTIVE TRANSPORT	SECONDARY ACTIVE TRANSPORT
Requires direct input of metabolic energy in the form of adenosine triphosphate (ATP)	The transport of two or more solutes is coupled and metabolic energy is provided indirectly from the Na ⁺ gradient that is maintained across cell membranes by Na ⁺ -K ⁺ -ATPase pump
a) Na ⁺ -K ⁺ -ATPase or sodium potassium pump b) Ca ²⁺ -ATPase c) H ⁺ -ATPase or proton pump d) H ⁺ -K ⁺ -ATPase	a) Na ⁺ -glucose cotransport b) Na ⁺ -Ca ²⁺ counter transport

Solution to Question 15:

Na⁺-K⁺ ATPase or Na⁺-K⁺ ATP pump is an electrogenic pump in that it moves three positive charges out of the cell for each two that it moves in, and it is therefore said to have a coupling ratio of 3:2.

Note:

- Na⁺-K⁺ ATP pump is found in almost all parts of the body, and it catalyzes the hydrolysis of ATP to ADP and uses the energy for its activity
- It is one of the major energy using process in the body. On an average, it accounts for 24% of the energy utilized by the cells, and in neurons it accounts for 70%. Thus, it accounts for large share in basal metabolism

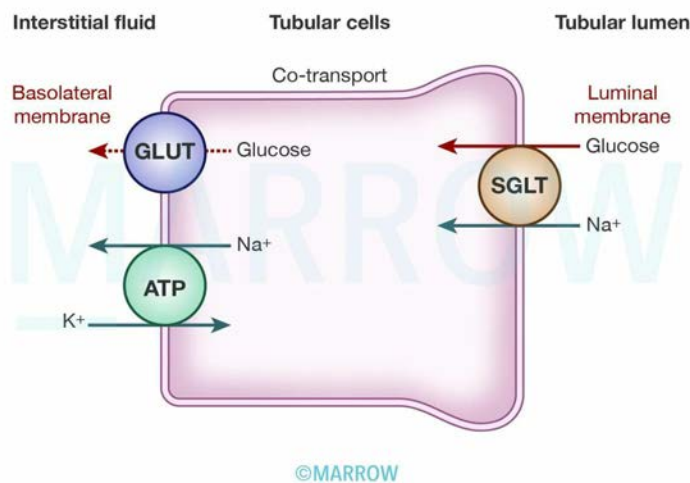
Solution to Question 16:

Empagliflozin is an SGLT-2 inhibitor. SGLT-2 or sodium-glucose transport protein 2 is an example of a cotransporter that plays a key role in glucose absorption from renal tubule by secondary active transport. This process is blocked by SGLT-2 inhibitors, leading to renal loss of glucose and improved glycemic control.

- A special property of the transport protein involved in secondary active transport is that a conformational change to allow Na⁺ movement to the interior will not occur until a glucose

molecule also attaches to the transport protein. When they both become attached, the conformational change takes place automatically, and the Na^+ and glucose are transported to the inside of the cell at the same time.

- The figure shows glucose is transported uphill and Na^+ is transported downhill. Energy derived from the downhill movement of Na^+ is maintained by the Na^+ - K^+ ATPase pump on the basolateral side.



Solution to Question 17:

Glucose is absorbed in the intestine by secondary active transport (sodium co-transport).

The energy for glucose transport is provided indirectly, by the active transport of Na^+ out of the cell. This maintains the concentration gradient across the luminal border of the cell so that more Na^+ and consequently more glucose enter.

Though both secondary active transport and facilitated diffusion play a role in glucose absorption, it is the initial active transport of sodium through the basolateral membranes of the intestinal epithelial cells that provide the eventual force for moving glucose through the membranes as well. So, secondary active transport is the single best answer for this question.

Solution to Question 18:

Exocytosis of neurotransmitters at the nerve terminals will be affected as a result of antibodies against voltage-gated calcium channels.

After an action potential reaches the terminal endings of a nerve, the influx of Ca^{2+} via voltage-gated calcium channels leads to the fusion of intracellular vesicles with cell membranes. This results in the release of neurotransmitters into the synaptic space. This process will be impaired in the case of Lambert-eaton syndrome.

Exocytosis is the extrusion of membrane-bound vesicles. In exocytosis, intracellular vesicles move to the cell membrane, fuse with it and then release their contents to the extracellular fluid. Solute and water can be released from the cell via vesicles by the process of exocytosis.

Exocytosis can be:

Constitutive exocytosis:

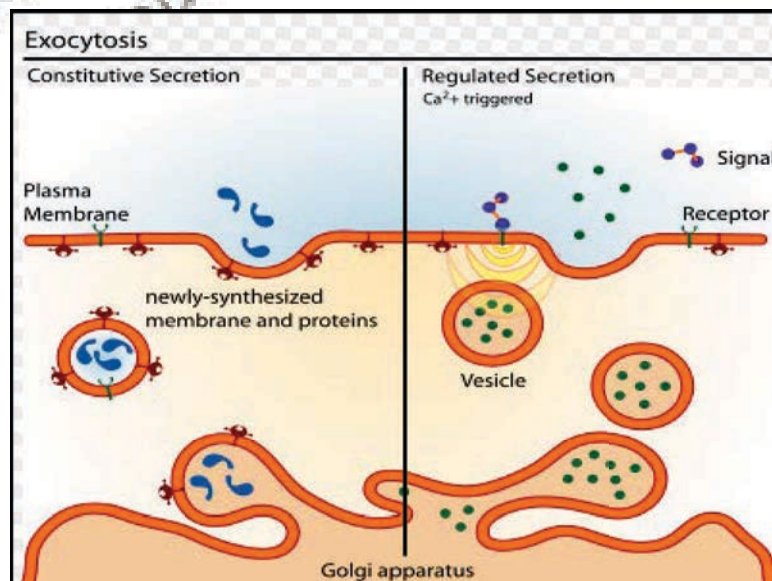
- Involves prompt transport of proteins to the cell membrane in vesicles, with little or no processing or storage
- Examples
- Goblet cells in the intestine continuously to release mucus
- Fibroblasts in connective tissue release collagen

Regulated/ Non constitutive exocytosis:

- Here the proteins from golgi apparatus initially enter the secretory granules, where the processing of prohormones to mature hormones occurs before exocytosis
- Examples
- Endocrine cells, hormones are stored in secretory vesicles and released in response to a signal from outside the cell.
- To insert proteins into the cell membrane like aquaporin insertion

Note:

- The term 'regulated exocytosis' is misleading because even constitutive exocytosis is regulated.
- Exocytosis involves two families of proteins: Rabs, which help vesicles dock onto the membrane, and SNAREs, which facilitate membrane fusion.
- This is Ca^{2+} -dependent process
- Exocytosis, like endocytosis, requires energy in the form of ATP.



Solution to Question 19:

The non constitutive pathway of exocytosis is called the regulated pathway.

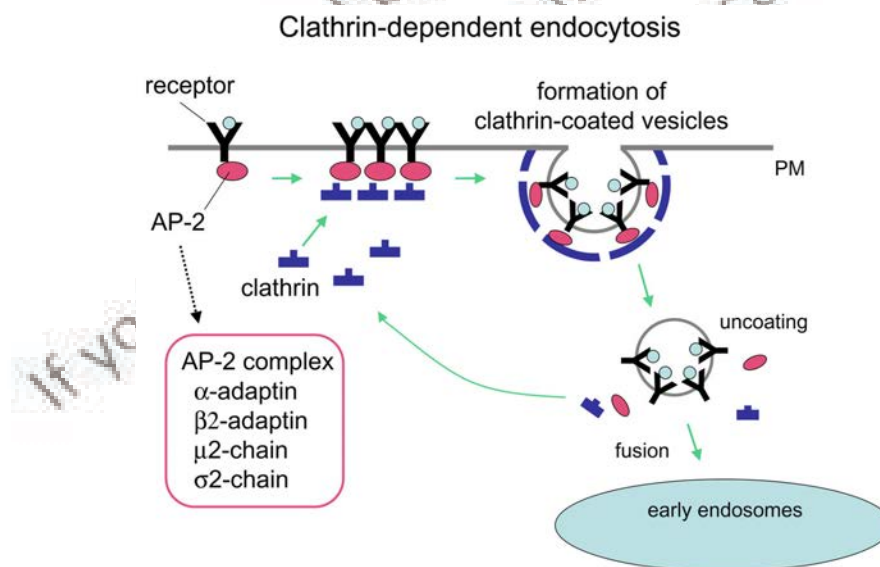
Exocytosis can be either constitutive or non constitutive(regulated):

- Regulated or nonconstitutive pathway: Proteins from the Golgi apparatus initially enter secretory granules, where processing of prohormones to the mature hormones occurs before exocytosis.
- Constitutive pathway: Involves the prompt transport of proteins to the cell membrane in vesicles, with little or no processing or storage.

Solution to Question 20:

Receptor-mediated endocytosis takes place in coated pits or indentations where the cytoplasmic side of the membrane has high concentrations of protein. The most common protein found in coated pits is clathrin.

A GTP binding protein dynamin is involved in pinching off the vesicle formed in clathrin-mediated endocytosis



Importance:

- Internalisation of many receptors and ligands bound to them eg. LDL, Nerve Growth Factor.
- Receptor-mediated endocytosis transports a variety of substances including protein hormones, growth factors, antibodies, and plasma proteins that serve as carriers for iron and cholesterol.
- Synaptic function
- Abnormalities in receptor-mediated removal of cholesterol from the blood are associated with elevated plasma cholesterol levels and cardiovascular disease.

Solution to Question 21:

Vesicular transport has no relation to the concentration gradient.

Vesicular transport requires energy expenditure by the cell, so this is an active method of membrane transport. Energy is needed to accomplish vesicle formation and vesicle movement within the cell.

METHOD OF TRANSPORT	SUBSTANCE INVOLVED	ENERGY REQUIREMENT
Endocytosis	Uptake of fluids or macromolecules by the cell	
Pinocytosis	Substances ingested are in the solution	Small molecules are taken up by invaginations of the plasma membrane and pinch off to form a vesicle.
2. Phagocytosis	Multimolecular particles (e.g. bacteria, cellular debris)	Active. The cell extends pseudopods around the particle to form a vesicle.
3. Receptor-mediated	Specific large polar molecule (ex: Transferrin, LDL)	Active. Molecules bind to receptors that are localized in clathrin-coated pits in the plasma membrane, endocytosed in vesicles that fuse with lysosomes
Transcytosis	Intact proteins across epithelial barriers (e.g., ingested antibodies in maternal milk across intestinal epithelia)	Movement of endocytosed vesicles between the apical and basolateral compartments of cells.
Exocytosis	Secretory products (e.g. hormones, enzymes), as well as large molecules that pass through the cell intact.	Active. An increase in cytosolic Ca^{2+} induces the fusion of a secretory vesicle with the plasma membrane; the vesicle opens up and releases its contents.

Solution to Question 22:

Transcytosis is seen in epithelial cells of intestine. Transcytosis is endocytosis across one membrane of the cell followed by exocytosis across the opposite membrane.

Transcytosis across epithelia or endothelia: Some molecules, such as proteins, are too large to diffuse through the membrane or cross the membrane transporters of epithelia. Instead they are moved across epithelia by transcytosis.

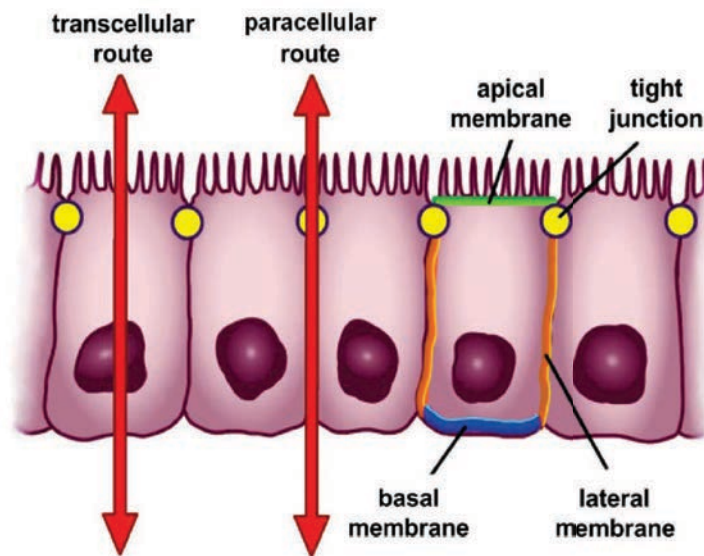
Steps:

- The molecule is brought into the epithelial cell via receptor-mediated endocytosis.

- The resulting vesicle attaches to microtubules in the cell's cytoskeleton and is transported across the cell by vesicular transport.
- At the opposite side of the epithelium, the contents of the vesicle are expelled into the interstitial fluid by exocytosis.

Significance:

- Transcytosis makes it possible for large proteins to move across an epithelium and remain intact.
- It is the means by which infants absorb maternal antibodies in breast milk. The antibodies are absorbed on the apical surface of the infant's intestinal epithelium and then released into the extracellular fluid.



Solution to Question 23:

As exocytosis, begins → ↑ in intracellular Ca^{2+} → Ca^{2+} interacts with a calcium sensing protein → secretory vesicle docking and fusion → fused area of membrane opens, the vesicle contents diffuse into the extracellular fluid while the vesicle membrane stays behind and becomes part of the cell membrane.

Solution to Question 24:

Endocytosis requires Ca^{2+} directly.

Endocytosis is the movement of secretory granules to the plasma membrane of the cell, fusion of the granule surface membrane with the cell membrane, rupture of membranes, and liberation of granular contents into the pericapillary space.

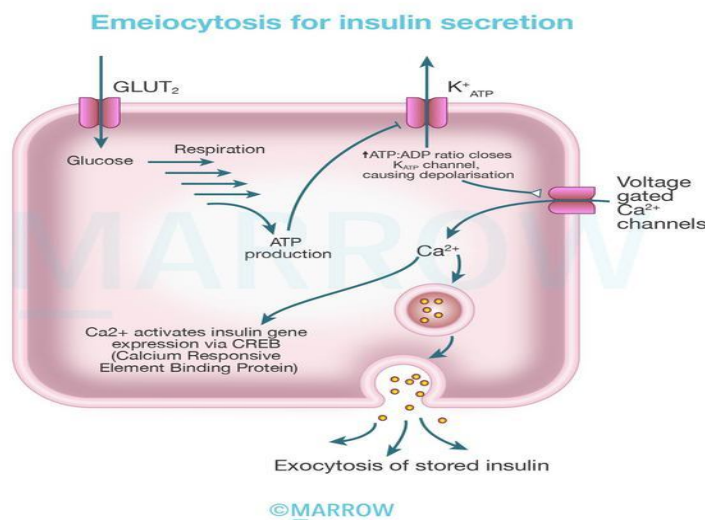
Insulin secretion (as shown in the image below):

- Glucose enters B cells by GLUT-2 transporters.
- It is phosphorylated and metabolized to pyruvate (Pyr) in the cytoplasm.
- The Pyr enters the mitochondria and is metabolized via the citric acid cycle. The ATP formed by oxidative phosphorylation inhibits ATP-sensitive K^+ channels, reducing K^+ efflux.
- This depolarizes the B cell, and Ca^{2+} influx is increased. The Ca^{2+} stimulates the release of insulin by exocytosis. Glutamate (Glu) is also formed, and this primes secretory granules, preparing them for exocytosis.

It is promoted by Ca^{2+} , $\uparrow K^+$

It is inhibited by Mg^{2+} , K^+ depletion.

The process of exocytosis for insulin secretion is depicted in the image below.



Membrane Potentials

Question 1:

Which of the following equations is used to calculate the equilibrium potential for an ion?

- a) Gibbs-Donnan equation
- b) Nernst equation
- c) Goldmann equation
- d) Bernoulli equation

Question 2:

Which of the following is used to calculate the equilibrium potential for diffusion of multiple ions?

- a) Gibbs-Donnan equation
- b) Nernst equation
- c) Goldmann equation
- d) Henderson equation

Question 3:

Gibbs-Donnan equilibrium explains the _____

- a) Effect of one non-diffusible ion on the distribution of other non-diffusible ions
- b) Effect of one permeant ion on the distribution of other permeant ions
- c) Effect of a permeant ion on the distribution of nondiffusible ions
- d) Effect of a non-diffusible ion on the distribution of permeant ions

Question 4:

Extracellular concentration of a positive ion (with valency = 1) is 100mmol/L and its intracellular concentration is 10mmol/L, the potential across the membrane using Nernst equation is (mV)_____

- a) -60

- b) -10
- c) +60
- d) 10

Question 5:

Extracellular concentration of a negative ion (with valency = 1) is 10mmol/L and its intracellular concentration is 100mmol/L. The equilibrium potential for this ion using Nernst equation is (mV) _____

- a) 10
- b) - 10
- c) 60
- d) - 60

Question 6:

What is the Nernst potential for K⁺ in millivolt (mV)?

- a) +90
- b) -90
- c) +70
- d) -70

Question 7:

Which of the following ions have the highest positive equilibrium potential?

- a) Na⁺
- b) Cl⁻
- c) K⁺
- d) Mg

Question 8:

The resting membrane potential of a nerve fibre depends on _____

- a) Diffusion of Na^+ ions
- b) Na^+ - K^+ ATPase pump
- c) Diffusion of K^+ ions
- d) All of the above

Question 9:

Resting membrane potential of the nerve fibre is equal to equilibrium potential of _____

- a) Na^+
- b) K^+
- c) Cl^-
- d) Mg^{2+}

Question 10:

Driving force of a charged ion is _____

- a) Measured membrane potential - calculated equilibrium potential
- b) Measured membrane potential + calculated equilibrium potential
- c) Calculated equilibrium potential - Measured membrane potential
- d) None of the above

Question 11:

If the membrane potential of this cell is -80mV the driving force is greatest for which ion?

Ion	Equilibrium potential (mV)
Ca ²⁺	+120
Na ⁺	+60
K ⁺	-90
Cl ⁻	-70

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- a) Ca²⁺
- b) Cl⁻
- c) K⁺
- d) Na⁺

Question 12:

Which of the following techniques is used to study the current flow across single ion channel?

- a) Patch clamp technique
- b) Voltage clamp technique
- c) Iontophoresis
- d) Galvanometry

Question 13:

What is the effect on the cellular membrane when the extracellular concentration of K⁺ is decreased?

- a) Increased positivity of the membrane
- b) Increased negativity of RMP
- c) Decreased fluctuations in RMP
- d) Decreased negativity of membrane

Question 14:

X, Y, Z are three permeable ions. $X = -50$ and $Y = -30$. At RMP, if there is no net electrogenic transfer, what is the value of Z?

- a) +20
- b) -20
- c) +80
- d) -80

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	c
5	c
6	b
7	a
8	d
9	c
10	a
11	a
12	a
13	b
14	c

Detailed Explanations**Solution to Question 1:**

Equilibrium potential for an ion is calculated using the Nernst equation.

When there is a concentration gradient for an ion across the membrane, the ion tends to diffuse down the concentration gradient. The electric potential that exactly opposes this net diffusion is called the equilibrium potential for that ion. This is calculated using the Nernst equation.

It is expressed as the potential inside the cell (intracellular potential relative to the extracellular potential).

To calculate the value of the equilibrium potential:

Nernst equation

$$\text{EMF (millivolts)} = \pm \frac{61}{z} \times \log \frac{\text{Concentration inside}}{\text{Concentration outside}}$$

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Note: In this formula, the choice of concentration of the ion placed in the numerator and denominator does not matter because the formula is only used to derive the numerical value and not the sign (+/-). Mathematically, $\log (100/10) = 1$ and $\log (10/100) = -1$. Hence, changing the numerator and denominator only changes the sign and the numerical value remains the same.

This equation is used only to derive the numerical value of the potential. It is NOT used to derive the sign/charge (+/-).

The sign/charge is derived from logic:

- In a positive (+) ion, if the concentration of the + ion is greater outside, then the + ion moves from outside to inside. The intracellular potential becomes more + relatively. Hence, the potential will be positive.
- In a positive (+) ion, if the concentration of the + ion is greater inside, then the + ion moves from inside to outside. The intracellular potential becomes more - relatively (as + ion moves outside). Hence the potential will be negative.
- In a negative (-) ion, if the concentration of the - ion is greater outside, then the - ion moves from outside to inside. The intracellular potential becomes more - relatively. Hence the potential will be negative.
- In a negative (-) ion, if the concentration of the - ion is greater inside, then the - ion moves from inside to outside. The intracellular potential becomes more + relatively (as - ion moves outside). Hence the potential will be positive.

Solution to Question 2:

The Goldman equation is used to calculate the diffusion potential when the membrane is permeable to several different ions.

Goldman-Hodgkin-Katz equation

Goldman-Hodgkin-Katz equation

$$\text{EMF (millivolts)} = -61 \times \log \frac{C_{\text{Na}^+}_i P_{\text{Na}^+} + C_{\text{K}^+}_i P_{\text{K}^+} + C_{\text{Cl}^-}_o P_{\text{Cl}^-}}{C_{\text{Na}^+}_o P_{\text{Na}^+} + C_{\text{K}^+}_o P_{\text{K}^+} + C_{\text{Cl}^-}_i P_{\text{Cl}^-}}$$

Here,

- P= permeability of the membrane to each ion, and
- C= concentrations of the respective ions on the inside (i) and outside (o) of the membrane.

The Na⁺, K⁺ and Cl⁻ are the most important ions involved in the development of membrane potentials in nerve fibres and muscle fibres.

Solution to Question 3:

Gibbs-Donnan equilibrium explains the effect of a non-diffusible ion on the distribution of permeant ions.

Consider the situation depicted below,

Gibbs-Donnan equilibrium

X	Y
m	
K ⁺	K ⁺
Cl ⁻	Cl ⁻
Prot ⁻	

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The membrane is permeable to K⁺ and Cl⁻. However, it is impermeable to the proteins. The impermeable protein molecules hinder the movement of K⁺ (negative charge of the proteins holding back positive charge of the K⁺) and favor the movement of Cl⁻ (like charges repel). According to the Gibbs-Donnan equation, at equilibrium, the concentration ratios of diffusible ions are equal across the membrane.

$$\frac{[K^+]_X}{[K^+]_Y} = \frac{[Cl^-]_Y}{[Cl^-]_X}$$

Cross-multiplying,

$$[K^+]_X [Cl^-]_X = [K^+]_Y [Cl^-]_Y$$

Solution to Question 4:

The potential of a positive ion across the membrane using Nernst equation is +60mV.

$$\text{EMF (millivolts)} = \pm \frac{61}{z} \times \log \frac{\text{Concentration inside}}{\text{Concentration outside}}$$

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Where, z is valency in the formula.

Deriving the numerical value:

$$\begin{aligned}\text{EMF} &= 61/1 \times \log 10/100 \\ &= 61\end{aligned}$$

The choice of concentration of the ion placed in the numerator and denominator does not matter because the formula is only used to derive the numerical value and not the sign (+/ -).

Mathematically, $\log (100/10) = 1$ and $\log (10/100) = - 1$. Hence, changing the numerator and denominator only changes the sign and the numerical value remains the same.

Deciding the sign/charge. A positive ion with greater concentration extracellularly will try to diffuse inside the cell. A positive intracellular potential is required to prevent this. Hence the sign/charge (potential) is positive (+).

Hence, the potential across the membrane using the Nernst equation is +60mV (the closest answer chosen).

Solution to Question 5:

The equilibrium potential for this negative ion using the Nernst equation is +60mV.

$$\text{EMF (millivolts)} = \pm \frac{61}{z} \times \log \frac{\text{Concentration inside}}{\text{Concentration outside}}$$

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Where, z is valency in the formula.

Deriving the numerical value:

$$\begin{aligned}\text{EMF} &= 61/1 \times \log 100/10 \\ &= 61\end{aligned}$$

Deciding the sign/charge: A negative ion with greater concentration intracellularly will try to diffuse outside the cell. A positive intracellular potential is required to prevent this. Hence, the potential is positive.

Hence, the Nernst potential is +60mV (the closest answer chosen).

Solution to Question 6:

The Nernst potential for K⁺ is -90mV.

Note:

- For K⁺ ions, the concentration gradient is directed outward and the electrical gradient is directed inward.
- In general, all excitable cell membranes at rest are maximally permeable to K⁺ and thus their RMP is close to equilibrium potential of K⁺ ion.

Solution to Question 7:

The ion with the highest positive equilibrium potential is Na⁺ that is +60mV.

Solution to Question 8:

The resting membrane potential (RMP) of a nerve fibre depends on the diffusion of both K^+ and Na^+ and the Na^+-K^+ ATPase pump.

The most predominant factor, however, is the diffusion of K^+ ions. This is because, at rest, the permeability of the membrane of the nerve fibre to K^+ ions is almost 100 times greater than that to Na^+ .

Solution to Question 9:

Resting membrane potential of a nerve fiber is equal to the equilibrium potential of Cl^- and is $-70mV$.

Solution to Question 10:

The driving force on a charged ion is measured membrane potential (E_m) - calculated equilibrium potential (E_x).

$$\text{Net driving force (mV)} = E_m - E_x$$

where,

E_m = Actual membrane potential (mV)

E_x = Equilibrium potential for X (mV)

When the driving force is negative (i.e., E_m is more negative than the ion's equilibrium potential), then

- A cation will enter the cell
- An anion will leave the cell

Solution to Question 11:

The driving force is greatest for Ca^{2+} .

Net driving force is the difference between the membrane potential and the equilibrium potential for that particular ion. Hence $E_m - E_x$. Where:

E_m = Actual membrane potential (mV) = $-80mV$

E_x = Equilibrium potential for X (mV)

For the respective ions:

$E_{Ca^{2+}} = +120$; Driving force = $-80 - (120) = -200mV$

$E_{Na^+} = +60$; Driving force = $-80 - (60) = -140mV$

$E_{K^+} = -90$, Driving force = $-80 - (-90) = 10\text{mV}$

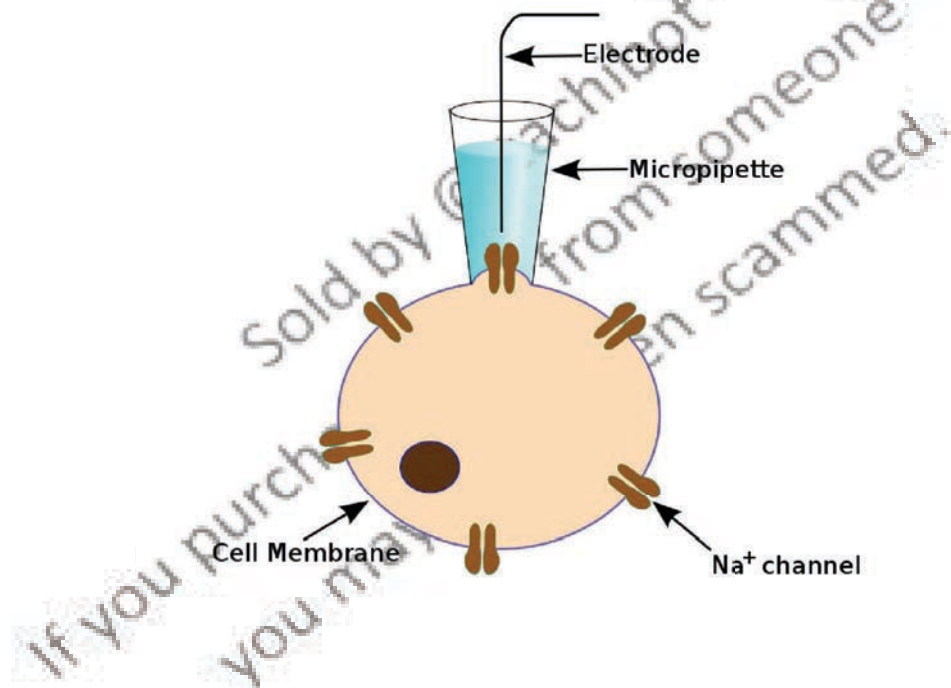
$E_{Cl^-} = -70$, Driving force = $-80 - (-70) = -10\text{mV}$

Thus, Ca^{2+} will have the largest driving force and it will move inwards or into the cell.

Solution to Question 12:

The patch-clamp technique is used to record ion current flow through a single protein channel.

In the patch-clamp experiment, a small pipette is carefully maneuvered to seal off a portion of a cell membrane. The pipette has an electrode bathed in an appropriate solution that allows for the recording of electrical changes through any pore in the membrane.



Solution to Question 13:

When the extracellular concentration of K^+ is decreased (hypokalemia), there is increased negativity of RMP.

RMP predominantly depends on the diffusion of K^+ from inside to outside the cell. This diffusion depends on the concentration gradient with the inside of the cell having a larger concentration of K^+ ions. When the extracellular concentration of K^+ decreases, the concentration gradient for K^+ rises. Therefore, more K^+ ions diffuse out of the cell. This leaves the inside of the cell more negative relative to the extracellular space. This increases the negativity of the RMP, leaving the membrane hyperpolarised.

Solution to Question 14:

No net electrogenic transfer at RMP means $X + Y + Z = 0$

Since $X = -50$ and $Y = -30$, then Z must be $+80$ since $(-80 + 80 = 0)$

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Body Fluids

Question 1:

In an average young adult male, what percentage of the bodyweight is water?

- a) 40
- b) 50
- c) 60
- d) 70

Question 2:

If the total body water is 60% of the body weight, what are the proportions of ECF and ICF?

- a) 20,40
- b) 30, 30
- c) 40, 20
- d) 10, 50

Question 3:

A 60 kg adult male has a haematocrit reading of 40% and a plasma volume of 3L. What is his total blood volume in L?

- a) 4
- b) 5
- c) 6
- d) 7

Question 4:

What is the volume of blood in ml, in a child per kg body weight?

- a) 70-80
- b) 100-150

- c) 150-200
- d) 200-150

Question 5:

What is the water content of lean body mass is (in mL/100g of tissue)?

- a) 40
- b) 50
- c) 60
- d) 70

Question 6:

How do we measure the volume of distribution of body fluid compartments?

- a) Dye dilution method
- b) Kety method
- c) Fick's principle
- d) None

Question 7:

500 mg of mannitol was injected out of which 10% is excreted. After equilibrium, plasma concentration is 3.2 mg/100 ml. Calculate ECF Volume (L).

- a) 10
- b) 14
- c) 42
- d) 52

Question 8:

10 g of inulin is injected IV and after equilibrium the concentration of inulin in plasma is 50mg/100ml. If 10% of the inulin is excreted, what is the ECF volume?

- a) 1.8 L

- b) 18 L
- c) 2.4 L
- d) 24 L

Question 9:

Of the following compartment and indicator combinations which is true?

- a) Plasma volume-Tritium oxide
- b) RBC volume-Evans' blue
- c) ECF volume-Sodium thiosulphate
- d) Total body water-Cannot be measured directly.

Question 10:

A dye X which measures the plasma volume is replaced by dye Y which can diffuse out of the capillaries. Which of the following is true regarding the newly measured plasma volume?

- a) It is falsely high
- b) It is falsely low
- c) It remains the same
- d) Insufficient information given

Question 11:

Interstitial fluid volume is best measured using a combination of which two dyes?

- a) Inulin and Evans blue
- b) Evans blue and D₂O
- c) Tritium oxide and Inulin
- d) D₂O and Mannitol

Question 12:

Which of the following is higher in the ECF compared to ICF?

- a) K⁺

- b) Proteins
- c) Phosphate
- d) pH

Question 13:

Which of the following is the most abundant intracellular anion?

- a) Phosphate
- b) Protein
- c) Chloride
- d) Bicarbonate

Question 14:

How many grams of NaCl make 1 mole of NaCl?

- a) 58.5
- b) 56.5
- c) 57.5
- d) 50.5

Question 15:

How much is one equivalent of calcium (molecular weight = 40)?

- a) 20g
- b) 30g
- c) 40g
- d) 50g

Question 16:

The unit expressing the concentration of osmotically active particles is_____

- a) Mole
- b) Equivalent

- c) Osmole
- d) Valency

Question 17:

A female patient was infused with a solution which leads to rapid lysis of the RBCs. Which solution was most likely infused?

- a) 150 mMol/L NaCl
- b) 300 mMol/L mannitol
- c) 350 mMol/L mannitol
- d) 300 mMol/L urea

Question 18:

Normal osmolarity of plasma is mainly due to _____.

- a) Na⁺
- b) Glucose
- c) Cl⁻
- d) Plasma Proteins

Question 19:

Plasma proteins contribute only slightly to the total number of osmotically active particles because _____

- a) High molar concentration, high molecular weight
- b) Low molar concentration, low molecular weight
- c) Low molar concentration, high molecular weight
- d) High molar concentration, low molecular weight

Question 20:

Daily water loss in sweat during normal activities is (ml) _____

- a) 50-100

- b) 300-400
- c) 500-700
- d) 1000-1200

Question 21:

Which type of disturbance of body fluids is observed in sweating?

- a) Isosmotic volume contraction
- b) Hyperosmotic volume contraction
- c) Hyposmotic volume contraction
- d) Isosmotic volume expansion

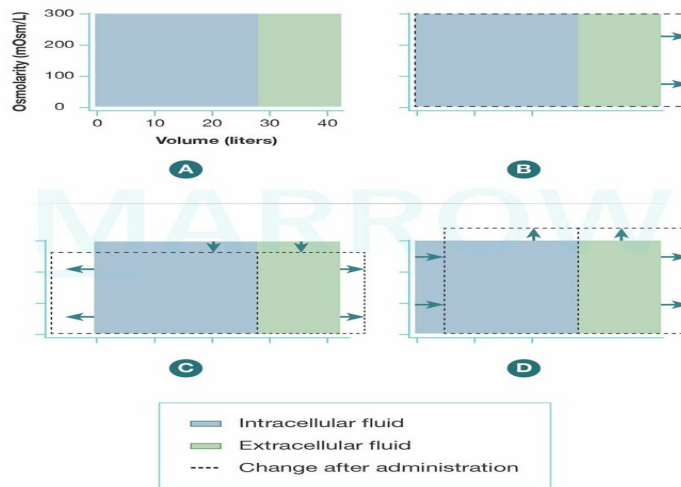
Question 22:

A 73-year-old diabetic man is brought to the emergency with an altered sensorium. He is dehydrated and his blood glucose is 550mg/dl. A hyperosmolar, hyperglycemic state is suspected. Which of the given formulae can correctly evaluate his serum osmolality?

- a) $2 \times \text{sodium} + (\text{BUN}/18) + (\text{glucose}/2.8)$
- b) $2 \times \text{sodium} + (\text{glucose}/18) + (\text{BUN}/2.8)$
- c) $3 \times \text{sodium} + (\text{BUN}/9) + (\text{glucose}/2.8)$
- d) $3 \times \text{sodium} + (\text{glucose}/9) + (\text{BUN}/2.8)$

Question 23:

Addition of 500 mL of hypertonic saline to the extracellular fluid leads to _____

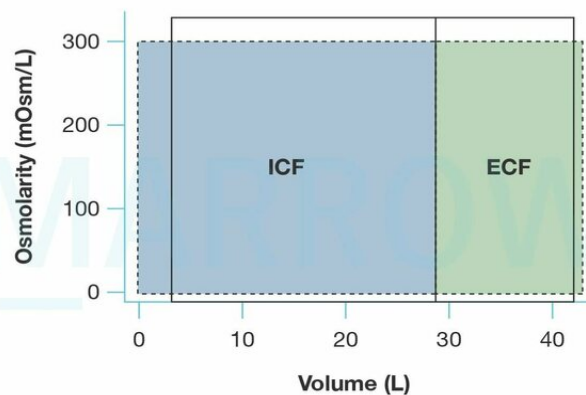


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- a) A
- b) B
- c) C
- d) D

Question 24:

The following Darrow Yannet diagram is suggestive of which condition? (dotted line denotes the change caused by the condition)



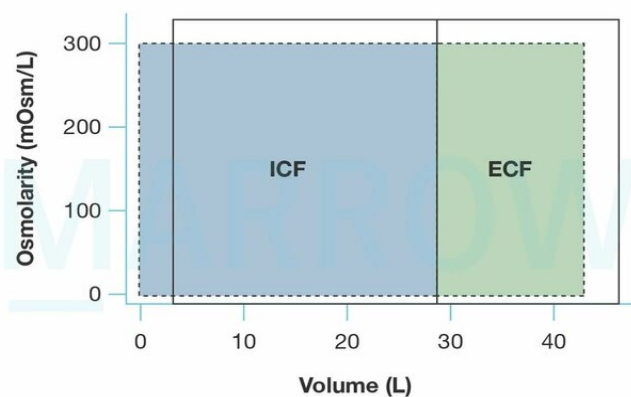
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- a) SIADH

- b) Diarrhoea
- c) Diabetes Insipidus
- d) Adrenal Insufficiency

Question 25:

The following Darrow Yannet diagram is suggestive of which condition? Note : The solid line is the original fluid compartment and the colour shaded compartment within the dotted line denotes the change caused by the condition.



- a) Excessive sweating
- b) Adrenal Insufficiency
- c) Diarrhoea
- d) SIADH

Question 26:

Various safety factors prevent excessive fluid accumulation in the interstitial spaces. Total safety factor against edema is (mmHg)_____

- a) 7
- b) 17
- c) 27
- d) 37

Question 27:

You are an intern and want to determine the quantity of IV fluids needed for a patient. How much amount of transient loss/insensible loss per day would you account for in your calculations?

- a) 150-200 ml
- b) 500-600 ml
- c) 200-350 ml
- d) 350-400 ml

Question 28:

I-125 labelled albumin is used for measurement of:

- a) Plasma volume
- b) Interstitial fluid volume
- c) ICF
- d) ECF

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	a
5	d
6	a
7	b
8	b
9	c
10	a
11	a

12	d
13	a
14	a
15	a
16	c
17	d
18	a
19	c
20	a
21	b
22	b
23	d
24	a
25	b
26	b
27	b
28	a

Detailed Explanations

Solution to Question 1:

In the human body, water constitutes the highest proportion of body weight. The total body water accounts for ~60% of body weight.

Human Body Composition

Component	Percentage of Body Weight
Water	60%
Proteins	18%
Minerals	7%
Fats	15%

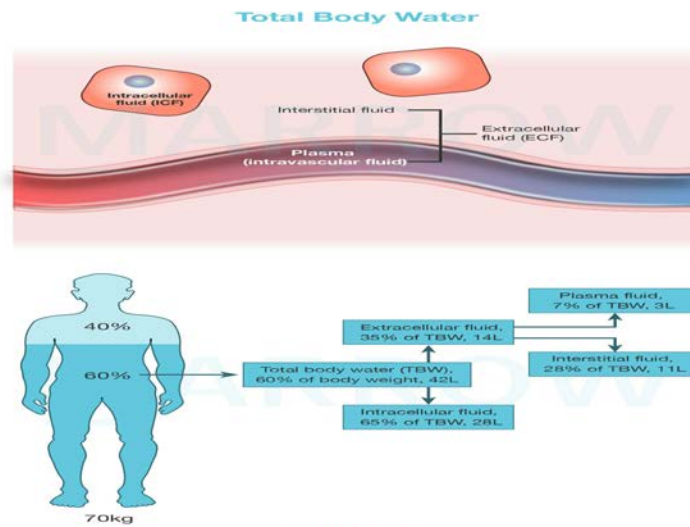
Age (Years)	Male (%)	Female (%)
10-18	59	57

Age (Years)	Male (%)	Female (%)
18-40	61	51
40-60	55	47
Over 60	52	46

Solution to Question 2:

When expressed as the percentage of body weight, 40% of body weight is in ICF (2/3rd of 60%), and 20% of body weight is in ECF (1/3rd of 60%).

The 60-40-20 rule is: 60% of body weight is water, 40% is ICF, and 20% is ECF.



Solution to Question 3:

His total blood volume is 5 litres.

Calculation:

Total blood volume

$$\text{Total blood volume} = \frac{\text{Plasma volume}}{1 - \text{Hematocrit}}$$

$$\text{Total blood volume} = \frac{3\text{L}}{1 - 0.4} = 5\text{L}$$

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Hematocrit is a percentage, as it is 40% which equals $40/100 = 0.4$.

(Normally, Total blood volume is 8% of body weight).

Solution to Question 4:

The total blood volume of a child is around 75–80 ml/kg.

In the neonatal period, it is 85 ml/kg. It rises to a peak of 105 ml/kg by the end of the first month and then drops progressively over ensuing months.

Solution to Question 5:

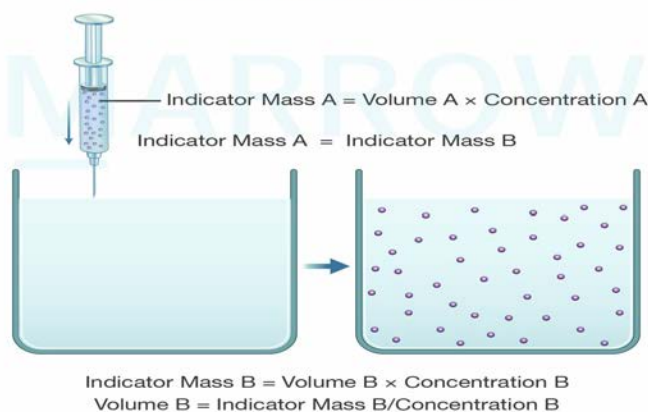
The water content of lean body tissue is constant at 71–72 mL/100 g of tissue.

Lean body mass = Body mass - (Adipose tissue fat mass + Bone Mass + Non-Adipose Tissue Fat Mass)

Solution to Question 6:

The volume of a fluid compartment in the body can be measured by "indicator (dye)-dilution" method. This method is based on the conservation of mass principle, which means that the total mass of a substance after dispersion in the fluid compartment will be the same as the total mass injected into the compartment.

Indicator-dilution method for measuring fluid volumes



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Figure shows→

If none of the substance leaks out of the compartment, the total mass of substance in the compartment (Volume B × Concentration B) will equal the total mass of the substance injected (Volume A × Concentration A). By simple rearrangement of the equation, one can calculate the unknown volume of chamber B as

$$\text{Volume B} = \frac{\text{Volume A} \times \text{Concentration A}}{\text{Concentration B}}$$

Solution to Question 7:

The volume of distribution can be calculated using the formula:

Volume of distribution

$$V = \frac{\text{Amount injected} - \text{Amount excreted}}{\text{Concentration}}$$

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- V = volume of compartment
- Amount injected = 500 mg
- Amount of indicator excreted = 10% of 500 mg = 50mg
- Plasma concentration of the indicator = 3.2 mg/100mL

Calculation:

$$V = \frac{500\text{mg} - 50\text{mg}}{3.2\text{mg}/100\text{ml}} = 14.1\text{L}$$

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2nd method :

Mannitol is a substance which leaves blood vessel but can't enter the cell. Therefore, after attaining equilibrium, conc. of mannitol is same in plasma and interstitial fluid

Therefore, by applying dye-dilution technique, mannitol can be used to measure the volume of ECF (x)

Since, 10% is excreted and 10% of 500mg is 50 mg, the amount of mannitol diluted in ECF is 450 mg (500 - 50) i.e, 450 mg is present in x

From the given values, 3.2 mg is present in 100 ml, which can be re-written as

$$3.2/100 = 450/x$$

Therefore $x = 450 * 100 / 3.2$ mL

$$= 14,062.5 \text{ mL}$$

$$= 14 \text{ L}$$

Clinical :

IV Mannitol is used to treat cerebral oedema (since, it doesn't cross cell membrane, it extracts fluid from brain cells), whereas it is contra-indicated in pulmonary edema (here the fluid collection is in the interstitial space)

Solution to Question 8:

Inulin is the indicator used to measure the ECF volume. (Since, inulin can cross blood vessel but not cell wall)

Calculation:

- Since 10% is excreted, volume of inulin excreted is 1 gm. The amount of inulin diluted in ECF is 9 g (10 - 1) i.e, 9000mg is present in x

From the given values, 50 mg is present in 100ml, which can be re-written as

$$50/100 = 9000/x$$

$$x = 9000 * 100 / 50$$

$$= 9000 * 2$$

$$= 18,000\text{mL}$$

$$= 18\text{L}$$

Second method :

- Since 10% is excreted, volume of inulin excreted is 1 gm.
- Amount injected = 10g
- Amount excreted = 1g

Now, the concentration of inulin after equilibrium is 50 mg/100ml which is equal to 0.05 g/100ml. (Convert the values into same units).

$$V = \frac{\text{Amount injected} - \text{Amount excreted}}{\text{concentration}}$$

Therefore, ECF volume = $(10-1) \times 100 / 0.05 = 18000 \text{ ml} = 18 \text{ L}$

Solution to Question 9:

The ECF volume can be estimated using any of several substances that disperse in the plasma and interstitial fluid but do not readily permeate the cell membrane.

Example- radioactive sodium, thiosulfate ion, and inulin.

Hint : Sodium (Na^+) can cross blood vessel but not cell wall passively (therefore used to measure ECF volume)

Solution to Question 10:

As the dye diffuses out of the capillaries, the measured concentration of the dye after equilibrium will be less. So, as per the formula, the volume measured will be falsely high.

$V = (\text{Amount Injected} - \text{Amount Excreted}) / \text{Plasma concentration of dye}$

Detailed explanation:

- Dye x measures plasma volume:- dye x doesn't have the property of crossing blood vessel. If the dye x have been used properly, for a normal person then the value would come around 3 L
- Dye y crosses capillaries, which means it can cross any blood vessel. This property is used to measure the value of ECF. If the dye y is given, for a normal person then the value would come around 14L
- Thus if dye y is given in place of x, then the value would be 14 instead of 3 which is a false and the obtained value is high when compared to normal value
- Therefore the answer is falsely high.

Solution to Question 11:

Interstitial fluid volume is best measured using a combination of inulin and Evans blue.

Interstitial fluid is measured indirectly. ECF minus plasma volume gives the interstitial fluid value.

- Best measure of ECF - Inulin
- Best measure of Plasma Volume - Evans Blue (T1824)

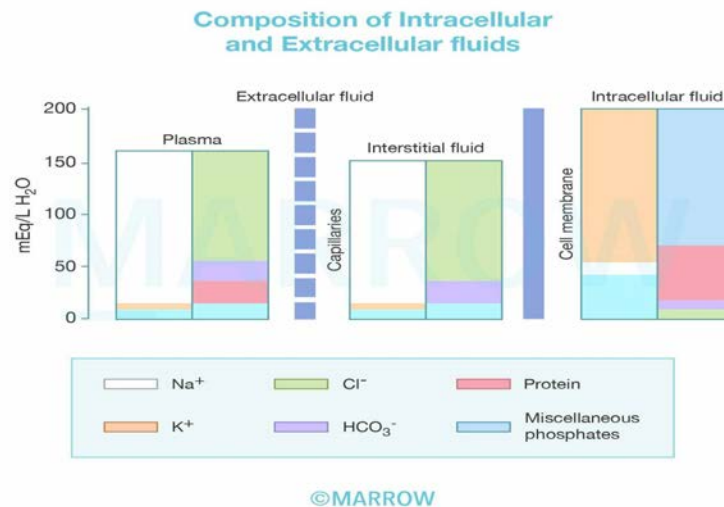
Solution to Question 12:

The pH of ECF is 7.4 while the pH of ICF is 7.0. There is continuous generation of H^+ ions in the ICF due to the various metabolic processes. So, the pH of ICF is a little lower compared to ECF.

Solution to Question 13:

Phosphate is the most abundant intracellular anion.

- ICF
- Major anion- Phosphate
- Major cation- K^+
- ECF
- Major anion- Cl^-
- Cation- Na^+
- Protein content in ICF is ~8 times greater than plasma (ECF)



Solution to Question 14:

1 mol of NaCl is 58.5 g

- A mole is the gram-molecular weight of a substance (molecular weight of a substance in grams)
- In this case, the molecular weight of Na-23 and Cl-35.5.
- 1 mol of NaCl = 23 g + 35.5 g = 58.5 g

Note

- Each mole consists of 6×10^{23} molecules.
- The molecular weight of a substance is the ratio of the mass of 1/12th the mass of an atom of carbon-12. Because molecular weight is a ratio, it is dimensionless.
- The Dalton (Da) is a unit of mass equal to the mass of 1/12th the mass of an atom of carbon-12. Thus molecular mass is expressed in units.

Solution to Question 15:

1 equivalent of calcium is 20 g.

One equivalent is 1 mol of an ionized substance divided by its valency.

For calcium in solution, 1 mol is going to be 40g and its valency is 2. Therefore, 1 equivalent of calcium is $40/2 = 20$ g.

Solution to Question 16:

The concentration of osmotically active particles is usually expressed in osmoles. Normal plasma osmolality is 280 to 290 mosmole/kg H₂O

- Example

- NaCl dissociate into 2 particles- Na⁺ and Cl⁻ in an ideal solution. Hence we can say that for every 1 mole of NaCl, there are 2 osmoles.

- If a solute is a non-ionizing compound such as glucose, the osmotic pressure is a function of the number of glucose molecules present.

- NOTE: The body fluids are not ideal solutions, and although the dissociation of strong electrolytes is complete, the number of particles free to exert an osmotic effect is reduced owing to the interaction between the ions. Thus, 1mol of NaCl per litre body fluids contribute somewhat less than 2 osmoles. The more concentrated the solution, the greater the deviation from the ideal solution.

Solution to Question 17:

Urea is a highly permeable solute, 300mmol/L urea solution behaves as a hypotonic solution.

Urea diffuses to the inside of RBC's via UTB channels very rapidly, while the rest of the solutes in the ECF cannot enter the cell. So the ECF now becomes hypotonic. So, fluid enters the RBC and causes their lysis.

Other Options :

- Option A : The solution is iso-osmotic. (150mmol of NaCl contains 150mmol of Na⁺ and 150mmol of Cl⁻ ions making the total osmolality 300 mOsm/L. This is equivalent to 0.9% NaCl which is isotonic.)

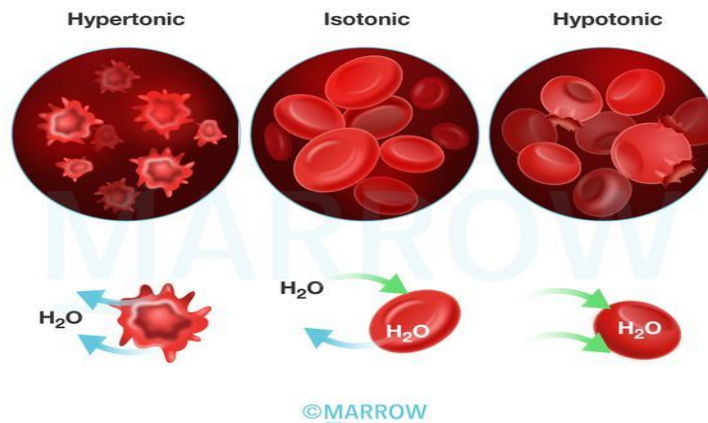
- Option B and C: The solutions are hyper-osmotic. This will lead to shrinking of the RBCs.

Note :

- If a cell is placed into a hypotonic solution that has a lower concentration of impermeant solutes (<282 mOsm/L), water will diffuse into the cell, causing it to swell.

- If a cell is placed in a hypertonic solution having a higher concentration of impermeant solutes, water will flow out of the cell leading to cell shrinkage.

Effects of hypertonic, isotonic and hypotonic solutions on cell volume



Solution to Question 18:

Normal plasma osmolarity (~290 mOsm/L) is mainly due to Na^+ and its accompanying anions, principally Cl^- and HCO_3^- .

- Osmolarity (mOsm/L) = $2 [\text{Na}^+] (\text{mEq/L}) + 0.055 [\text{Glucose}] (\text{mg/dL}) + 0.36 [\text{BUN}] (\text{mg/dL})$
- Because sodium and its associated anions (Cl^- , HCO_3^-) account for about 94 per cent of the solute in the extracellular compartment, plasma osmolarity (Posm) can be roughly approximated as
- $\text{Posm} = 2.1 \times \text{Plasma sodium concentration}$
- With a plasma sodium concentration of 142 mEq/L, the plasma osmolarity would be estimated from the formula above to be about 298 mOsm/L.
- In conditions associated with renal disease, the contribution of two other solutes, glucose and urea, should be included.
- Because urea easily permeates most cell membranes, it exerts little effective osmotic pressure under steady-state conditions. (Ineffective osmoles)
- Contribution to plasma osmolarity:
 - Na^+ and associated anions: 270 mOsm/L
 - Glucose : 5 mOsm/L
 - Urea: 5 mOsm/L
 - Plasma proteins: 2 mOsm/L
 - Other ions: 8 mOsm/L

Solution to Question 19:

Proteins actually contribute only slightly to the total number of osmotically active particles (~1 mOsm) because of their low molar concentration and high molecular weight.

Protein concentration measured as g/L may be high. However, as their molecular weight is high, the protein concentration in terms of moles per liter (molar concentration) will be low. Thus, they contribute only slightly to the total number of osmotically active particles.

Detailed explanation: Terms like high, low are always used for comparison. Here we compare plasma proteins and sodium (sodium is chosen because it contributes the maximum part of osmolarity). Look at the values given below (just for understanding)

	Plasma proteins (Albumin)	Sodium
Molar concentration	0.75mosm/L(Low)	270mosm/L
Molecular weight(1 mole =)	67,000 g(High)	23g

Solution to Question 20:

The amount of water lost by sweating is highly variable, depending on physical activity and environmental temperature. The volume of sweat normally is about 100 ml/day, but in very hot weather or during heavy exercise fluid loss in sweat occasionally increase to 1 to 2 L/hour.

Solution to Question 21:

Sweating → loss of water → Hyperosmotic volume contraction

- The osmolarity of ECF increases because sweat is hyposmotic (relatively more water than salt is lost).
- ECF volume decreases because of the loss of volume in the sweat.
- Water shifts out of ICF; as a result of the shift, ICF osmolarity increases until it is equal to ECF osmolarity, and ICF volume decreases.
- Plasma protein concentration increases because of the decrease in ECF volume. Although hematocrit might also be expected to increase, it remains unchanged because of the water shifts out of the RBCs, decreasing their volume and offsetting the concentrating effect of the decreased ECF volume.
- So, Loss of hypotonic fluid causes hypertonic volume contraction.

Solution to Question 22:

Calculated serum osmolality = $2 \text{ sodium} + (\text{glucose}/18) + (\text{BUN}/2.8)$.

In the given clinical scenario, the patient with untreated type 2 diabetes mellitus has most likely developed a hyperosmolar, hyperglycemic state. His serum osmolality will be significantly raised.

The principal determinants of osmolality of serum are the concentrations of sodium, glucose and urea (blood urea nitrogen or BUN).

Osmolality is defined as a measure of the osmoles (Osm) of solute per kilogram of solvent (osmol/kg or Osm/kg). The osmolality of the intracellular and extracellular fluids is maintained between 290 and 310 mOsm in each compartment.

Solution to Question 23:

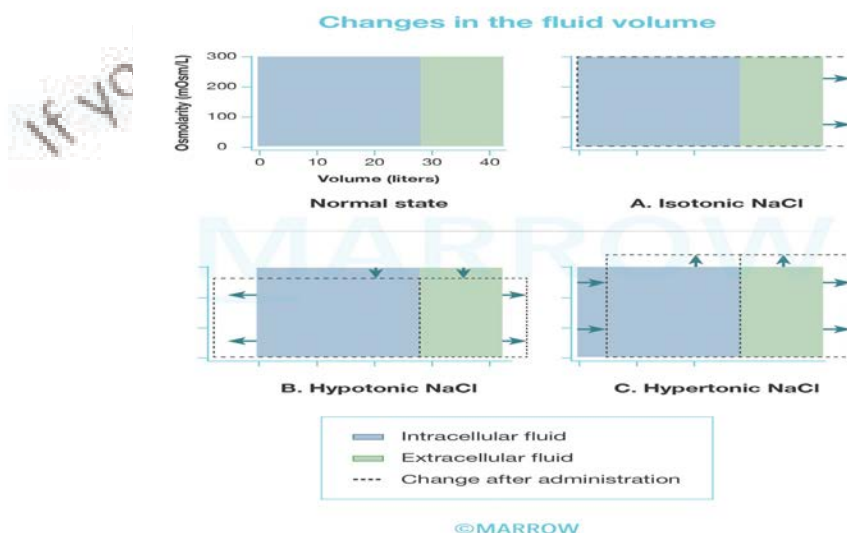
The images shown in the figure are called Darrow Yanner Diagrams. They depict the change in plasma osmolality and plasma volume in various conditions.

If a hypertonic solution is added to the extracellular fluid, the extracellular osmolality increases and causes osmosis of water out of the cells into the extracellular compartment

- ECF osmolality increases. ECF volume increases.
- As, ECF osmolality rises, fluid from ICF moves to ECF till osmotic equilibrium is established. So,
- ICF osmolality rises. ICF volume decreases.

Therefore, the net effect is an increase in extracellular volume (greater than the volume of fluid added), a decrease in intracellular volume, and a rise in osmolality in both compartments.

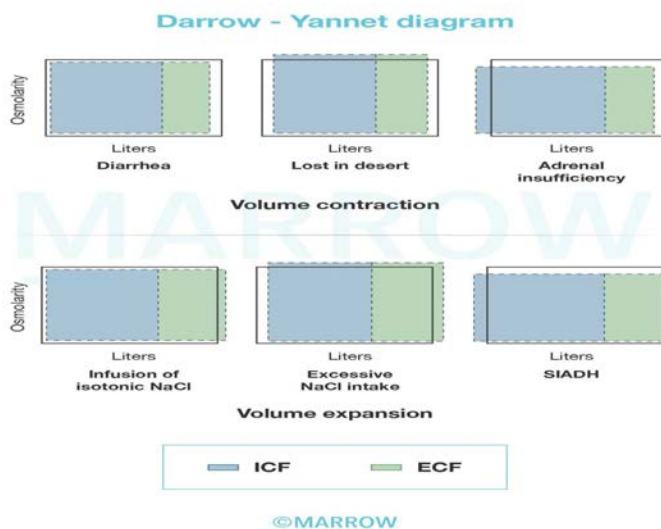
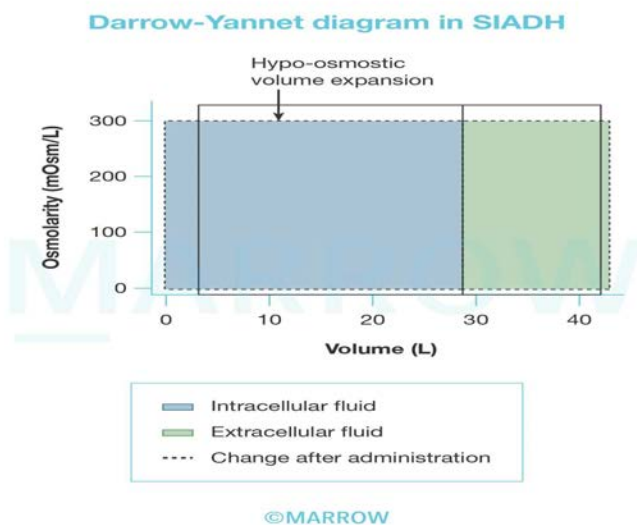
Note : The dotted line in the diagram is the change after either addition or removal of fluid



Solution to Question 24:

In the given darrow yannet diagram, there is hypoosmotic volume expansion which is suggestive of SIADH.

- In SIADH there is excessive secretion of ADH. So, excess water is retained in the body.
- So, ECF volume rises. ECF osmolality decreases.
- ECF fluid enters ICF untill osmotic equilibrium is re-established.
- So, ICF volume rises. ICF osmolality decreases.



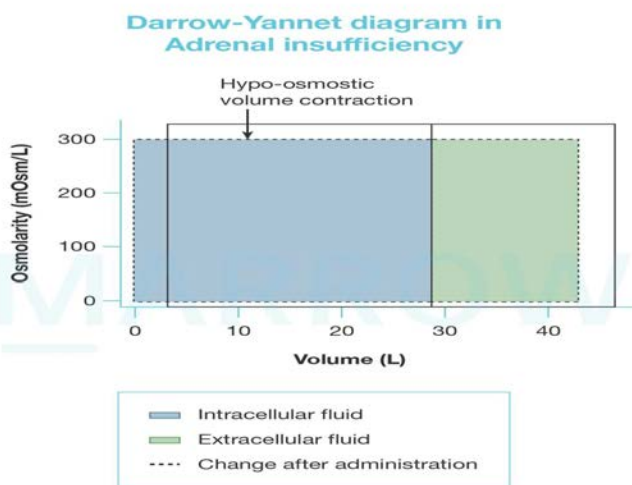
The Darrow Yannet diagrams of various conditions are shown above

Solution to Question 25:

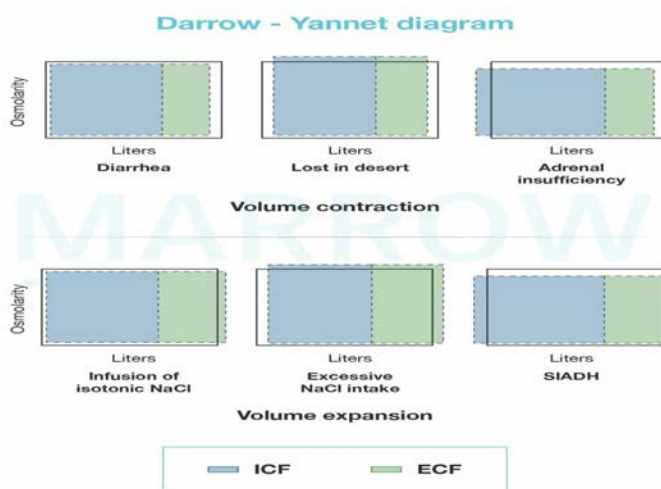
In adrenal insufficiency there is hypo-osmotic volume contraction. Aldosterone deficiency leads to increased sodium loss (loss of hyperosmotic fluid).

- The osmolarity of ECF decreases.
- ECF volume decreases because of the loss of volume.
- Water shifts into ICF; as a result of the shift, ICF osmolarity decreases until it is equal to ECF osmolarity, and ICF volume increases.
- So, loss of hypertonic fluid causes hypotonic volume contraction.

Please note that in the diagram, the horizontal axis represents volume and vertical axis represents osmolarity.



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The Darrow Yannet diagrams of various conditions are shown above.

Solution to Question 26:

The total safety factor against edema is about 17 mmHg.

Summary of safety factors that prevent edema is:

- The safety factor caused by low tissue compliance in the negative pressure range is about 3 mm Hg
- The safety factor caused by increased lymph flow is about 7 mm Hg
- The safety factor caused by washdown of proteins from the interstitial spaces is about 7 mm Hg

Therefore, the total safety factor against edema is about 17 mm Hg.

Solution to Question 27:

500-600 ml of transient loss/insensible loss per day would have to be accounted for in the calculation.

Although water is lost through sweating especially in hot climates or during physical exertion, we tend to lose water at rest as well. Physiologically, water evaporates from the skin and lungs at a rate of about 600 to 700 ml/day. This is known as insensible water loss.

Solution to Question 28:

I-125 labelled albumin is the most commonly used substance for the measurement of plasma volume.

Physiology of Nerve

Question 1:

Which among the following is embryologically and physiologically unrelated to other neural cells?

- a) Astrocytes
- b) Schwann Cells
- c) Oligodendrocyte
- d) Microglia

Question 2:

In which of the following areas is a pseudo-unipolar type of neuron present?

- a) Dorsal root ganglia
- b) Sympathetic ganglia
- c) Parasympathetic ganglia
- d) Cranial nerve nuclei

Question 3:

Where does the initiation of nerve impulse occur in a spinal motor neuron?

- a) Initial node of Ranvier
- b) Axon hillock+ initial segment
- c) Cell body
- d) Dendritic tree

Question 4:

Which part of the neuron contains a maximum number of Na^+ channels per square micrometer?

- a) Cell body
- b) Axon terminal

- c) Surface of myelin
- d) Node of Ranvier

Question 5:

In a cutaneous sensory neuron, at which of the following places is the propagated action potential generated?

- a) First node of Ranvier
- b) Initial segment
- c) Dendrite
- d) Nerve ending

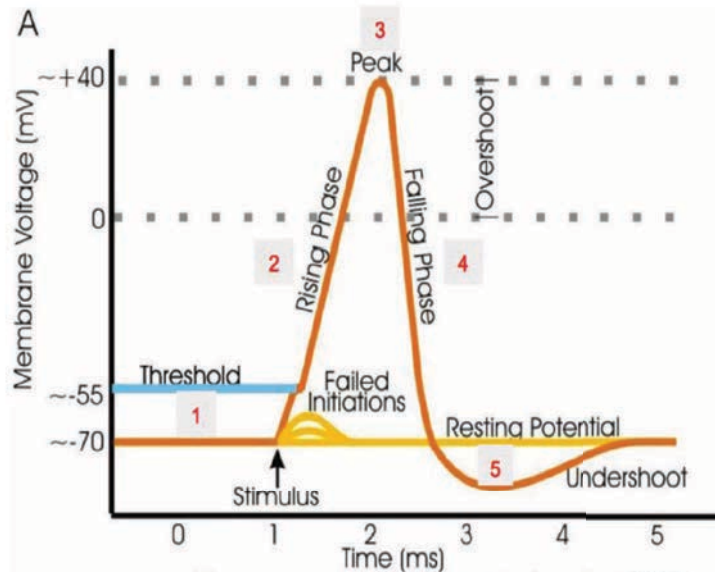
Question 6:

What is the relationship between diameter of the nerve and conduction velocity?

- a) Linear
- b) Parabolic
- c) Hyperbolic
- d) No relation

Question 7:

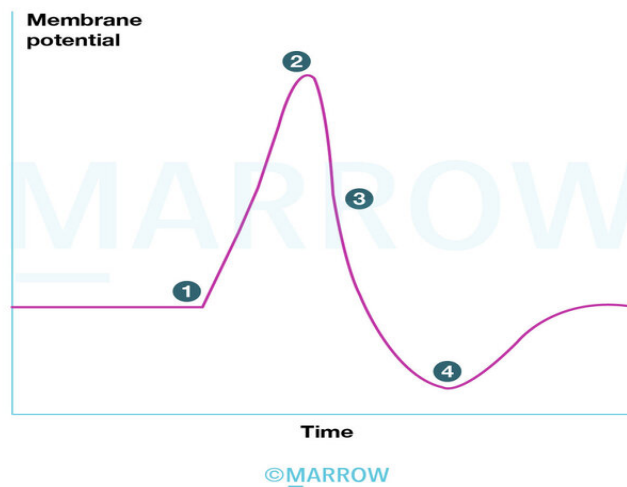
What process is responsible for the change in membrane potential that occurs between point 1 and point 3?



- a) Movement of Na^+ into the cell
- b) Movement of Na^+ out of the cell
- c) Movement of K^+ into the cell
- d) Movement of K^+ out of the cell

Question 8:

The membrane conductance for potassium is maximum at which of the following points in a nerve action potential?



- a) 1

- b) 2
- c) 4
- d) 3

Question 9:

Which of the following is true regarding the upstroke of a nerve action potential?

- a) There is net outward current and the cell interior becomes more negative
- b) There is net outward current and the cell interior becomes less negative
- c) There is net inward current and the cell interior becomes more negative
- d) There is net inward current and the cell interior becomes less negative

Question 10:

Which of the following is correctly matched?

- a) 1
- b) 2
- c) 3
- d) 4

Question 11:

What is known as the time period in which a second stronger impulse can elicit action potential after the application of one stimulus?

- a) Absolute refractory period
- b) Relative refractory period
- c) Latent refractory period
- d) Local response

Question 12:

From where does the relative refractory period start?

- a) Half of repolarization

- b) One third of repolarization
- c) Quarter of repolarization
- d) Two thirds of repolarization

Question 13:

Rheobase is an indicator of which of the following?

- a) Specificity of impulse transmission
- b) Magnitude of current
- c) Rate of discharge
- d) Velocity of nerve conduction

Question 14:

A slowly rising stimulus fails to fire a nerve because of a process known as _____.

- a) Adaptation
- b) Accommodation
- c) Refractoriness
- d) Electrotonus

Question 15:

Which of the following is true?

- a) Type A[°] fibre carry information related to pressure
- b) Type A β fibre has the largest diameter
- c) Type A γ fibre is unmyelinated
- d) Type C fibre has the slowest conduction velocity

Question 16:

You are performing a CNS examination on a patient with a stroke. With his eyes closed, you move his big toe up or down while holding his toe joint. You ask him to state the position of the digit. Which of the following fibers are responsible for the transmission of the sensation tested here?

- a) A°
- b) $A\gamma$
- c) $A\beta$
- d) $A\delta$

Question 17:

In the Erlanger and Gasser classification of peripheral nerve fibers, which type of signal is transmitted by the type B nerve fibers?

- a) Proprioception
- b) Preganglionic autonomic
- c) Motor to muscle spindle
- d) Crude touch

Question 18:

A man falls into a deep sleep with one arm under his head. He finds he is not able to move this arm when he awakens. However, it tingles and pain sensation in the arm is still intact. What is the reason for the loss of motor function without loss of pain sensation is that in the nerves to his arm?

- a) B fibers are more susceptible to pressure than A fibers
- b) A fibers are more sensitive to pressure than C fibers
- c) C fibers are more sensitive to pressure than A fibers
- d) Motor nerves are more affected by sleep than sensory nerves

Question 19:

Spike duration is maximum in which of the following nerve fiber?

- a) A°
- b) B
- c) C
- d) $A\beta$

Question 20:

Golgi tendon organ, being sensory is classified under which subtype of sensory nerve fibres?

- a) Ia
- b) Ib
- c) II
- d) III

Question 21:

Which sensory fibers are least susceptible to hypoxia?

- a) Type I
- b) Type II
- c) Type III
- d) Type IV

Question 22:

All of the following are true regarding a biphasic action potential except _____.

- a) All or none phenomenon
- b) 2 or more positive waves
- c) Refractory period is seen
- d) Recorded on the surface

Question 23:

End plate potential follows all of the following except?

- a) All or none law
- b) Depolarization
- c) Hyperpolarization
- d) Summation

Question 24:

Which of the following can be used to measure a synaptic potential?

- a) Patch clamp
- b) Microelectrode
- c) Voltage clamp
- d) ECG

Question 25:

A man who is fond of exotic cuisine develops perioral numbness after eating the Pufferfish. The given poisoning is due to which of the following toxins?

- a) Saxitoxin
- b) Tetrodotoxin
- c) Curare
- d) Ciguatoxin

Question 26:

Conduction in the chemical synapses in CNS is

- a) Both antidromic and orthodromic
- b) Always antidromic
- c) Always orthodromic
- d) Antidromic under lab conditions

Question 27:

Fast axonal transport is by all except _____

- a) Dynein
- b) Kinesin
- c) Neurofilaments
- d) Microtubules

Question 28:

Which of the following is incorrect regarding axonal transport?

- a) Axonal transport occurs along microtubules
- b) Slow orthograde axonal transport occurs at 100 mm/day
- c) Kinesins are required for orthograde axonal transport
- d) Viruses exploit retrograde transport for invasion

Question 29:

When a nerve is cut, which of the following takes place first?

- a) Myelin degeneration
- b) Regenerative Sprouting
- c) Denervation supersensitivity
- d) Schwann cell proliferation

Question 30:

Which of the following is true regarding the resting membrane potential (RMP) in a nerve fibre?

- a) It is equal to the RMP of inner hair cells of cochlea
- b) It can be measured by surface electrodes
- c) It becomes more positive as extracellular K^+ increases
- d) It depends on the equilibrium potential of Na^+

Question 31:

Which of the following about sensory nerve fibers is not true?

- a) Ia - A alpha
- b) II - A beta
- c) III - A gamma
- d) IV - C

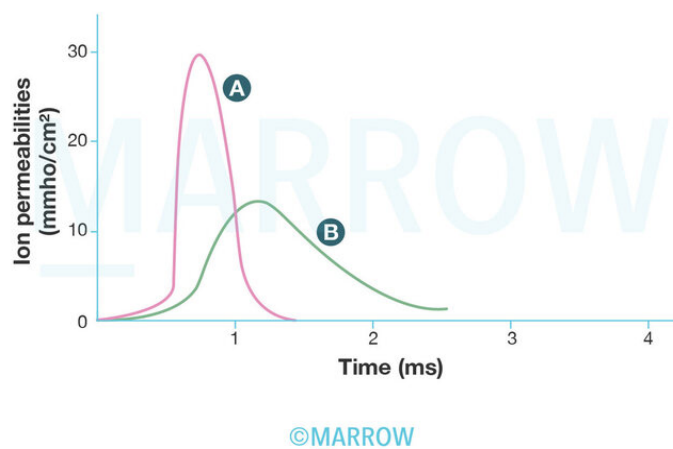
Question 32:

Absolute refractory period is due to:

- a) Opening of calcium channels
- b) Closure of potassium channels
- c) Closure of active gates of sodium channel
- d) Closure of inactive gates of sodium channel

Question 33:

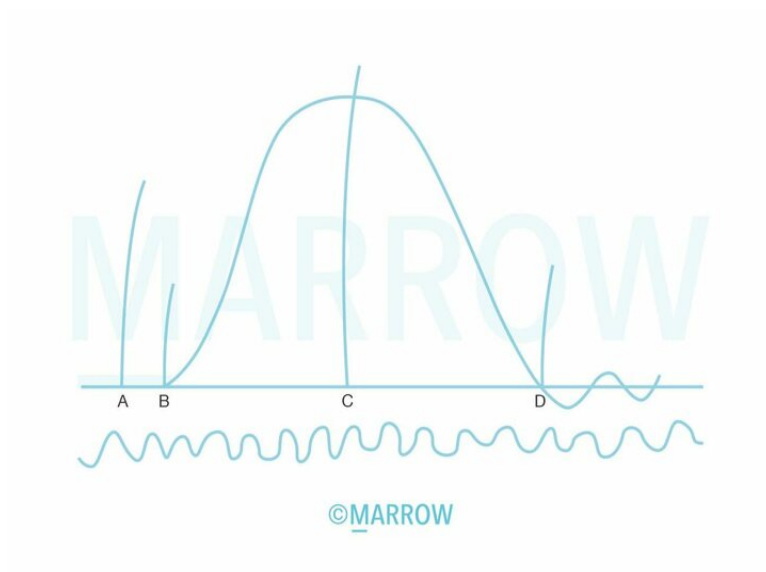
Identify the ions A and B based on conductance graph of a neuron shown below:



- a) Na, K
- b) K, Na
- c) Na, Ca
- d) Ca, Na

Question 34:

The image below shows simple muscle twitch along with the time trace, recorded in a frog gastrocnemius muscle. Calculate the minimum tetanizing frequency. (The time tracer was made with a 100Hz tuning fork).



- a) 25 per second
- b) 15 per second
- c) 17 per second
- d) 20 per second

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	d
5	a
6	a
7	a
8	d
9	d
10	a
11	b
12	b
13	b
14	b

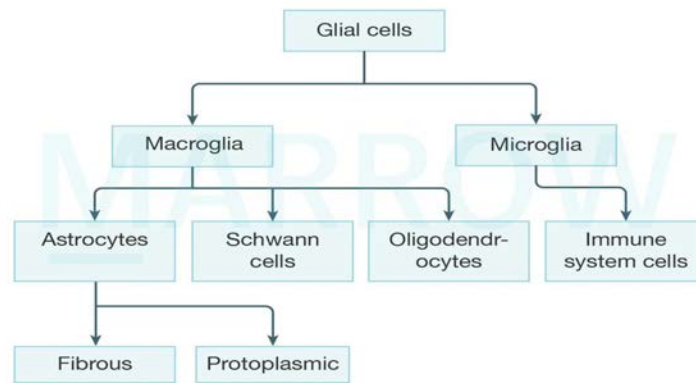
15	d
16	a
17	b
18	b
19	c
20	b
21	d
22	b
23	a
24	b
25	b
26	c
27	c
28	b
29	a
30	c
31	c
32	d
33	a
34	d

Detailed Explanations

Solution to Question 1:

Microglia are physiologically and embryologically unrelated to other neural cell types. They arise from macrophages outside the nervous system.

Types of glial cells



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Solution to Question 2:

The dorsal root ganglion is a classic example of a pseudo-unipolar neuron.

Solution to Question 3:

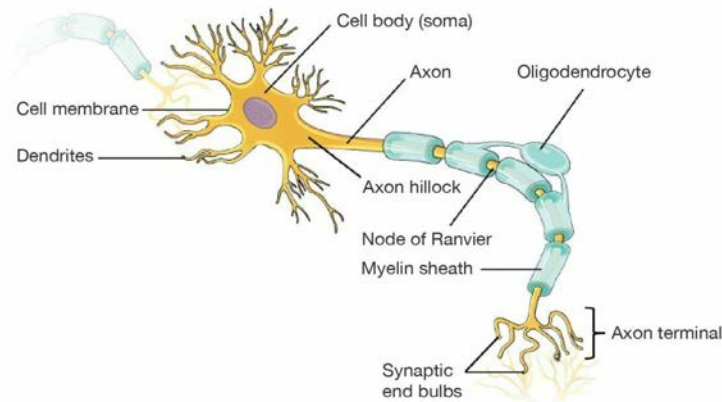
In a spinal motor neuron, initiation of nerve impulse occurs in the axon hillock and initial segment of the axon because it has a very high density of voltage-gated Na^+ and K^+ channels. Hence, this region is also known as the trigger zone for the action potential.

Parts of a Neuron:

- Cell body/perikaryon: contains many Nissl bodies.
- Dendritic zone where multiple local potential changes generated by synaptic connections are integrated.
- Axon hillock: part of the cell body from which the axon arises.
- Initial segment: the first part of the axon where propagated action potentials are generated.
- An axonal process that transmits propagated impulses to the nerve endings. The unmyelinated regions of a myelinated axon are termed as nodes of Ranvier.
- Synaptic knobs/nerve endings where action potentials cause the release of synaptic transmitters.

Note: Action potentials are generated at the initial segment in spinal motor neurons and the initial node of Ranvier in cutaneous sensory neurons.

Structure of a neuron



Solution to Question 4:

The maximum number of Na^+ channels per square micrometer in myelinated neurons is present at the nodes of Ranvier. It contains 2000-12,000 channels per square micrometer.

Voltage-gated Na^+ channels are highly concentrated in the nodes of Ranvier and the initial segment in myelinated neurons.

The spatial distribution of ion channels along the axon plays a key role in the initiation and regulation of action potential.

Solution to Question 5:

In cutaneous sensory neurons, propagated action potentials are generated in the initial node of Ranvier.

From a functional point of view, neurons generally have four important zones :

- A receptor, or dendritic zone, where multiple local potential changes generated by synaptic connections are integrated
- A site where propagated action potentials are generated (the initial segment in spinal motor neurons, the initial node of Ranvier in cutaneous sensory neurons)
- An axonal process that transmits propagated impulses to the nerve endings
- The nerve endings, where action potentials cause the release of synaptic transmitters

The cell body is often located at the dendritic zone end of the axon, but it can be within the axon (eg, auditory neurons)

Solution to Question 6:

The conduction velocity of a myelinated axon increases linearly with diameter.

Conduction velocity is increased by:

-

Increasing the fiber size: Increasing the diameter of a nerve fiber results in decreased internal resistance; thus, conduction velocity down the nerve is faster.

-

Myelination: Myelin acts as an insulator around nerve axons and increases conduction velocity. Myelinated nerves exhibit saltatory conduction. It is a rapid process that allows myelinated axons to conduct up to 50 times faster than the fastest unmyelinated fibers.

-

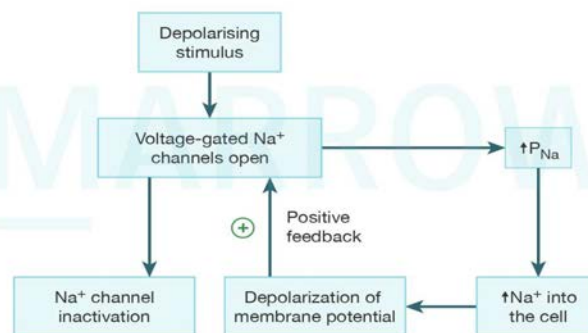
Activity of the Na⁺-K⁺ pump does not directly affect the formation or conduction of action potentials.

Solution to Question 7:

The graph depicts action potential. The movement of Na⁺ into the cell is responsible for the change in membrane potential that occurs between point 1 and point 3.

Once the membrane is depolarized to the threshold, local potential, the upstroke of the nerve action potential is caused by the opening of the Na⁺ channels. The entry of Na⁺ causes the opening of more voltage-gated Na⁺ channels and further depolarization, setting up a positive feedback loop. When the Na⁺ channels open, Na⁺ moves into the cell down its electrochemical gradient, driving the membrane potential toward the Na⁺ equilibrium potential.

Feedback control in voltage-gated Na⁺ channels



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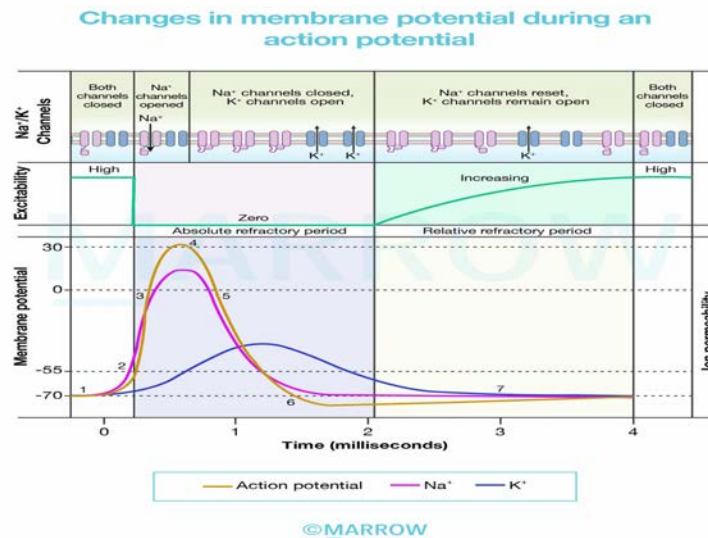
The process responsible for repolarization is the opening of K⁺. The K⁺ permeability becomes very high and drives the membrane potential toward the K⁺ equilibrium potential by the flow of

K⁺ out of the cell.

Solution to Question 8:

The membrane conductance for potassium is maximum during repolarization (marked as point 3 in the above image).

Most of the increase in K⁺ conductance comes after the increase in Na⁺ conductance around the time of repolarization.



During the early portion of the action potential, the sodium conductance is 1000 times greater than potassium conductance. Therefore, far more sodium ions flow to the interior of the fiber than do potassium ions to the exterior. This causes the membrane potential to become positive at the action potential onset.

When the membrane potential rises from -90 millivolts toward zero, this voltage change causes a conformational change in the K⁺ channels which allows increased potassium efflux. Then, the sodium channels begin to close and the potassium channels begin to open, and potassium conductance begins to increase (and sodium conductance decreases).

The increased flow of K⁺ out of the cell causes membrane repolarization. The slow return of the K⁺ channels to the closed state explains the after-hyperpolarization followed by a return to the resting membrane potential.

Solution to Question 9:

Regarding the upstroke of a nerve action potential, there is a net inward current and the cell interior becomes less negative. The direction of the current is either

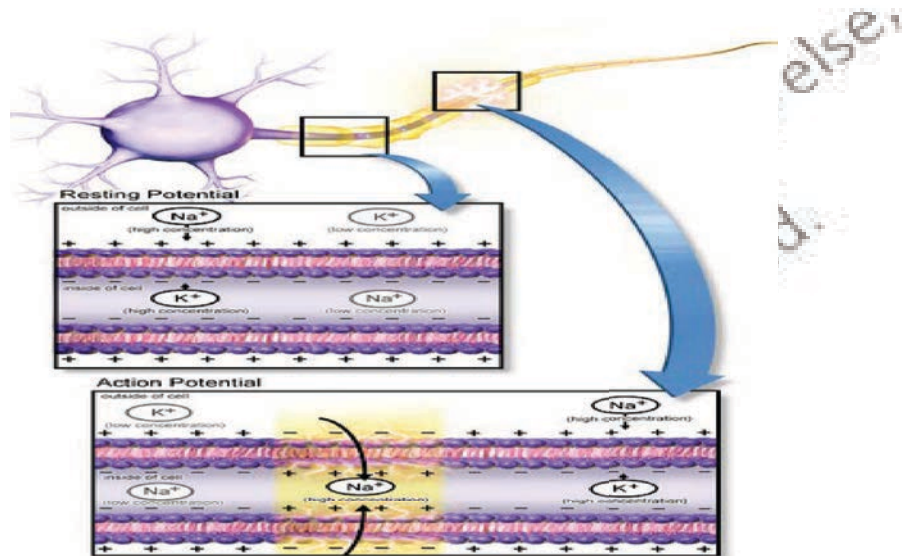
- same as the direction of positive charge (eg, Na⁺)

- opposite to the direction of negative charge (eg, Cl^-)
- opposite to the direction of electrons

Since Na^+ moves in, there is a net inward current.

Since the RMP increases, the cell interior becomes less negative.

In response to a depolarizing stimulus, some of the voltage-gated Na^+ channels open and Na^+ enters the cell and the membrane is brought to its threshold potential. The entry of Na^+ causes the opening of more voltage-gated Na^+ channels and further depolarization, setting up a positive feedback loop. The rapid upstroke in the membrane potential ensues. So, there is a net influx of Na^+ , and the interior of the cell becomes positive during the upstroke of the action potential.



Solution to Question 10:

The slow return of the K^+ channels to the closed state also explains the after-hyperpolarization.

In response to a depolarizing stimulus, some of the voltage-gated Na^+ channels open and Na^+ enters the cell. The entry of Na^+ causes the opening of more voltage-gated Na^+ channels and further depolarization, setting up a positive feedback loop. The rapid upstroke in the membrane potential ensues.

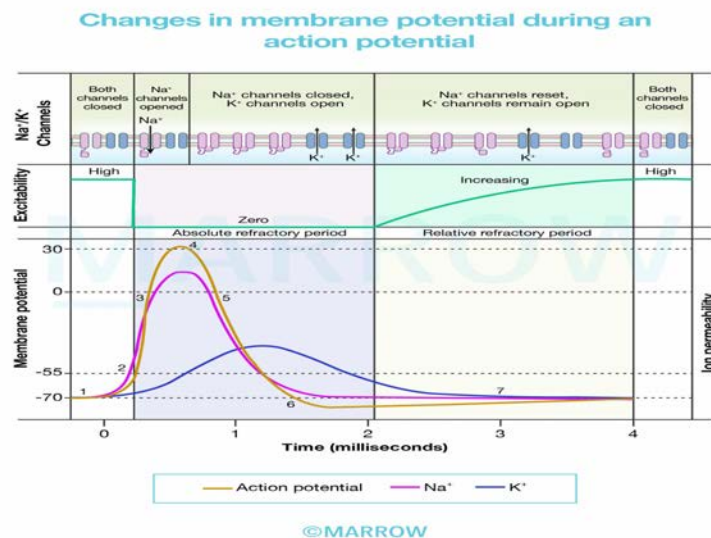
The membrane potential moves toward the equilibrium potential for Na^+ (+60 mV) but does not reach it during the action potential, primarily because the increase in Na^+ conductance is short-lived. The Na^+ channels rapidly enter a closed state called the inactivated state and remain in this state for a few milliseconds.

In addition, the direction of the electrical gradient for Na^+ is reversed during the overshoot because the membrane potential is reversed, and this limits Na^+ influx; also the voltage-gated K^+ channels open. These factors contribute to repolarization. The opening of voltage-gated K^+ channels is slower and more prolonged than the opening of the Na^+ channels. The net movement of positive charge out of the cell due to K^+ efflux at this time helps complete the process of repolarization.

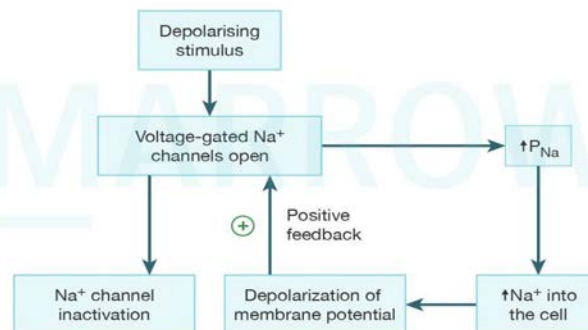
The slow return of the K^+ channels to the closed state also explains the after-hyperpolarization, followed by a return to the resting membrane potential.

Thus, voltage-gated K^+ channels bring the action potential to an end and cause closure of their gates through a negative feedback process.

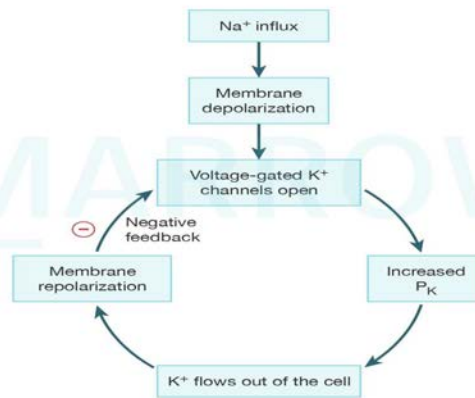
Note: Na^+ channels: Positive feedback loop; K^+ channels: Negative feedback loop.



Feedback control in voltage-gated Na^+ channels



Feedback control in voltage-gated K channels



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Solution to Question 11:

The refractory period is the time period in which a second stronger impulse can elicit action potential after the application of one stimulus. It can be absolute or relative.

During the rising and much of the falling phases of the spike potential, the neuron is refractory to stimulation. This refractory period is divided into an

- Absolute refractory period, corresponding to the period from the time the firing level is reached until repolarization is about one-third complete
- Relative refractory period, lasting from this point to the start of after-depolarization

During the absolute refractory period, no stimulus, no matter how strong, will not excite the nerve. However, during the relative refractory period, stimuli that are stronger than normal can cause excitation.

The absolute refractory period for large myelinated nerve fibers is about 1/2500 second.

Solution to Question 12:

The relative refractory period starts from one-third of repolarization till the start of after-depolarization.

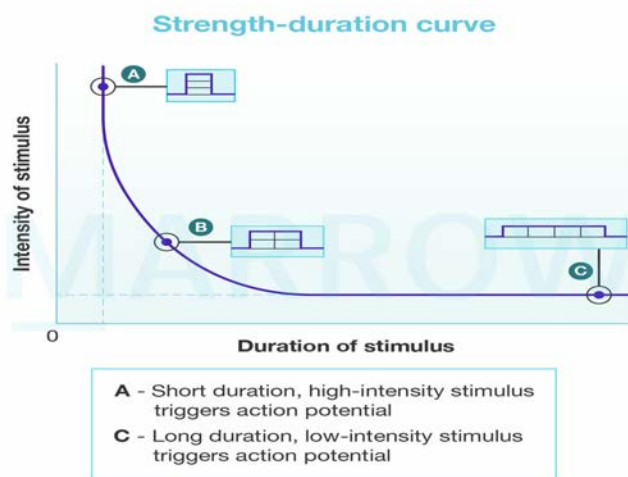
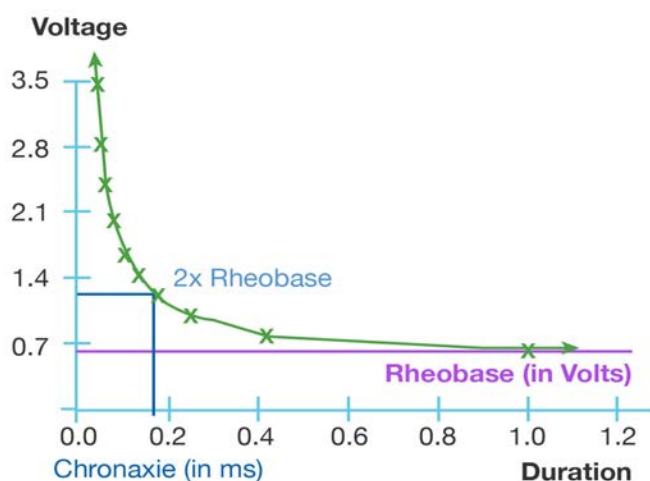
The refractory period is a period during which another normal action potential cannot be elicited in an excitable cell.

Solution to Question 13:

The rheobase is an indicator of the magnitude of the current. It is the minimum intensity (strength/magnitude) of stimulation required to generate an action potential when the stimulation is of infinite duration.

Utilization time is the minimum time (duration) for which the rheobase has to be applied for the excitation of tissue.

Chronaxie is the minimum time (duration) for which a stimulus double the strength of rheobase has to be applied for excitation of tissue.



Strength-duration curve gives the idea of the excitability of a tissue.

Strength-duration relationship arises because the same minimum electrical charge necessary to excite an action potential can come from a current that is either brief but large or prolonged but small. It is the product of strength and duration that determines excitability, and thus these two parameters are inversely related in their effectiveness.

Note: Chronaxie is a better index than rheobase and utilization time for comparing the excitability of a tissue.

Solution to Question 14:

Slowly rising subthreshold currents fail to fire the nerve due to a process called accommodation, which is a type of adaptation.

Accommodation is due to inactivation of voltage-gated Na⁺ channels. Due to the slow rise of stimulus, only a few voltage-gated Na⁺ channels are activated at a time. These few channels are inactivated by the time more channels are activated. So, at no moment, are a sufficient number of channels activated to give an action potential.

When a membrane is depolarized rapidly, a large number of Na⁺ channels open at once, and an action potential is produced.

In contrast, decreased response to a long-lasting and sustained suprathreshold stimulus, in general, is called adaptation.

Solution to Question 15:

Type C fibers are the small unmyelinated nerve fibers that conduct impulses at low velocities.

Greater the diameter of a given nerve fiber, the greater its speed of conduction.

The nerve fibers are divided into types A (myelinated), B (myelinated), and C (Unmyelinated). The type A fibers are further subdivided into α , β , γ , and δ fibers.

Solution to Question 16:

The test performed here is used to check the proprioception. The large axons like A α nerve fibers (12-20 μ m) carry information regarding proprioceptive sensation and somatic motor function.

Solution to Question 17:

The type B fibers transmit the preganglionic autonomic fibers of both the sympathetic and parasympathetic nervous systems.

Erlanger and Gasser classified the fibers into types A, B, and C, and the type A fibers are further subdivided into α , β , γ , and δ fibers.

- Type A fibers are the typical large- and medium-sized myelinated fibers of spinal nerves.
- Type B fibers are smaller fibers with minimal myelination. They transmit impulses of the ANS including the bladder. (B for Bladder)

- Type C fibers are the small unmyelinated nerve fibers that conduct impulses at low velocities. They constitute more than one-half of the sensory fibers in most peripheral nerves, as well as all the postganglionic autonomic fibers.

Solution to Question 18:

The reason for the loss of motor function without loss of pain sensation is that in the nerves to his arm, A-fibers are more sensitive to pressure than C fibers.

Sleeping for a long period with the arms under the head puts pressure on a nerve. It leads to the loss of conduction in large-diameter motor, touch, and pressure fibers, while pain sensation remains relatively intact. A-fibers (includes the motor fibers) are affected.

The susceptibility of the nerve fibers to pressure is A > B > C.

This is responsible for Saturday night or Sunday morning paralysis.

Solution to Question 19:

Spike duration is maximum in C fibers.

The spike duration in decreasing order: C > B > A

Absolute refractory period follows the same order: C > B > A

Fiber s	Spike Duration (ms)	Absolute Refractory Period (ms)
A	0.4-0.5	0.4-1
B	1.2	1.2
C	2	2

Solution to Question 20:

Golgi tendon organ, which contains the receptor for inverse stretch reflex, belongs to Class Ib sensory nerve fibres.

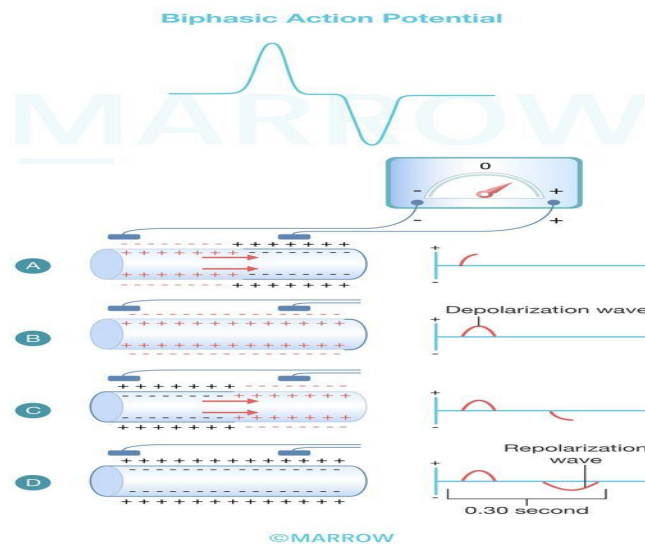
Solution to Question 21:

Susceptibility of nerve fibers to hypoxia is B > A > C, and in the general classification, type IV sensory nerves are Type C fibers.

Solution to Question 22:

Biphasic Action Potential

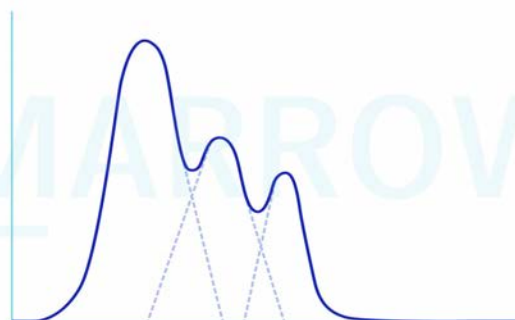
- This type of record is obtained when both recording electrodes are placed on the surface of an axon that has been stimulated.
- The record shows an upward deflection followed by an isoelectric interval and then a downward deflection which denotes to the time taken for the electric potential to travel from one electrode to another.



Compound Action Potential

- It is a property of mixed nerves.
- The record shows multiple peaks in action potential because a mixed nerve is made up of families of fibres with various speeds of conduction. Therefore, when all the fibres are stimulated, the activity in the fast-conducting fibres arrives at the recording electrode sooner than the activity in slower fibers.

Compound Action Potential



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Solution to Question 23:

End plate potential is a graded potential with the following characteristics:

- Magnitude is related to the magnitude of the initiating event
- Amplitude decreases over distance
- Can be summated

Graded Potential	Action Potential
Amplitude proportionate to strength of stimulus. It is graded.	Independent of stimulus intensity – all or none response
Can be depolarising or hyperpolarizing	Depolarizing
Is conducted decrementally - amplitude decreases with distance	Self-propagated unchanged in amplitude
Generated spontaneously or in response to stimuli- physical, chemical	Generated only in response to Membrane Depolarization
Can be summated	Cannot be summated
Has no threshold and refractory period	Has a threshold about 15 mv depolarized relative to the resting potential. Has a refractory period
Not always followed by some action	Always followed by some action

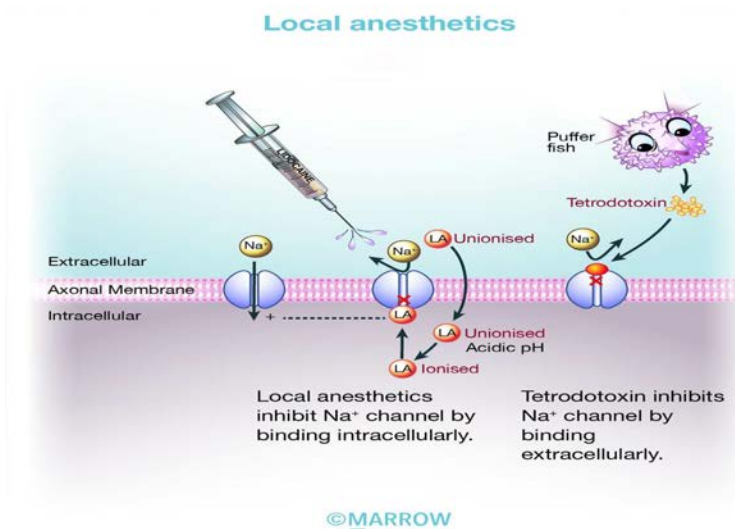
Solution to Question 24:

Instruments required for recording membrane potential/synaptic potential

- Stimulating electrodes: Cathode and anode
- Recording electrodes: Glass microelectrode (tip diameter $< 0.5\mu$)
- Display: Cathode ray oscilloscope (CRO)
- Squid giant axon: It is a large diameter neuron (up to 0.5 mm) from where the recording of potential is easier.

Solution to Question 25:

Pufferfish poisoning is caused by the toxin tetrodotoxin. The sodium channels can be blocked by this toxin when it is applied to the outside of the cell membrane where the sodium activation gates are located.



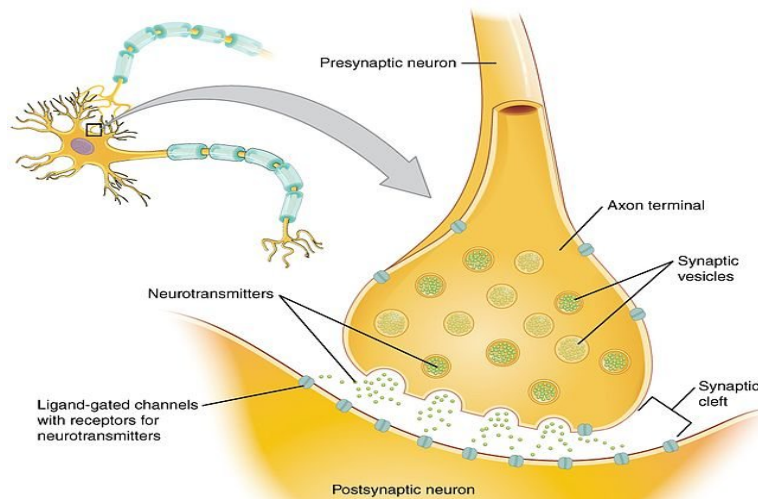
Solution to Question 26:

Conduction in the chemical synapses in CNS is always orthodromic.

When an action potential is set up in an axon, two impulses travelling in the opposite direction is set up by depolarization on either side of the initial current.

In the natural situation, impulses pass in one direction only- from synaptic junctions to axon terminals. Such conduction is called orthodromic. Conduction in the opposite direction is called antidromic. Under natural conditions, impulse conduction is normally only orthodromic. Antidromic conduction is usually only seen when a nerve fibre is stimulated under artificial conditions.

In a chemical synapse, conduction is always orthodromic. As the transmission in a chemical synapse is due to the release of neurotransmitters from the presynaptic membrane which activate receptors on the postsynaptic membrane, antidromic conduction (i.e. from the postsynaptic membrane to presynaptic membrane) is not possible. Any impulse travelling in such direction would die as it encounters the first synapse.



Solution to Question 27:

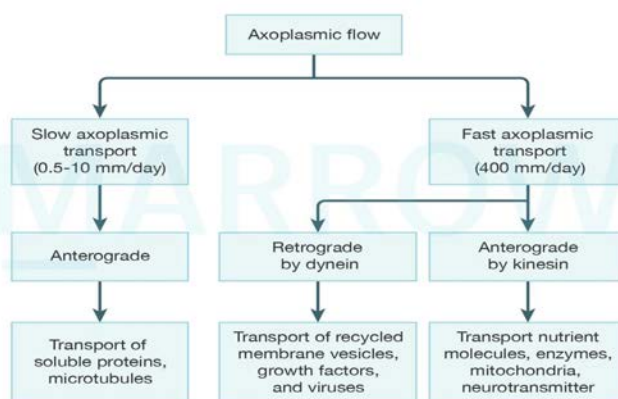
Fast axonal transport is by all except neurofilaments.

Neurofilaments play no role in axonal transport. Fast axonal transport takes place along microtubules which are tubular polymers of protein tubulin.

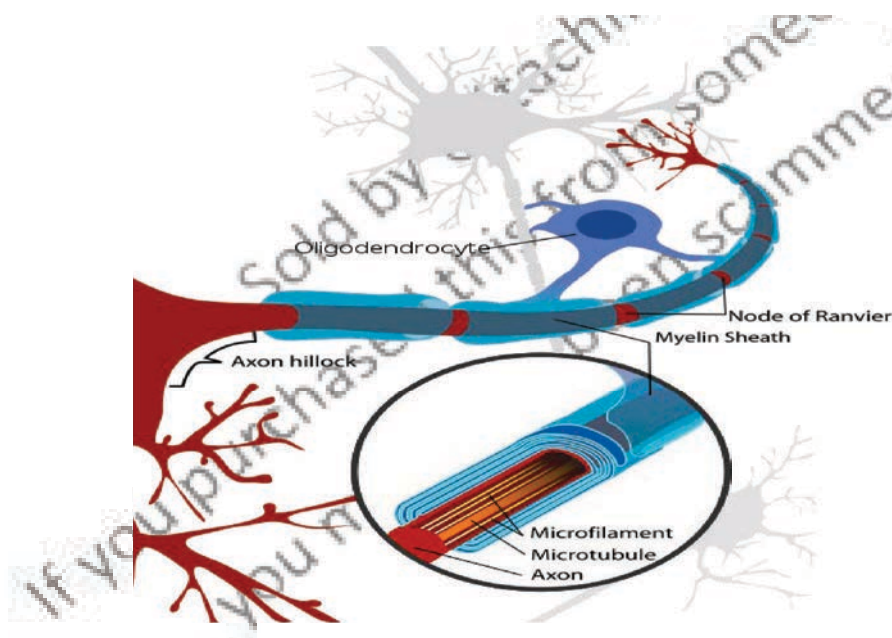
AXONAL TRANSPORT

- Protein synthesis occurs in the cell body of neuron and proteins and polypeptides are transported to axonal endings by the axoplasmic flow.
- Axonal transport can be orthograde or retrograde.
- Orthograde transport
 - Can be either fast or slow axonal transport and it occurs along microtubules that run along the length of the axon away from the cell body.
 - Fast axonal transport occurs at about 400 mm/day, and slow axonal transport occurs at 0.5 to 10 mm/day.
 - Molecular motor involved in orthograde transport is Kinesin.
- Retrograde transport,
 - In the opposite direction (from the nerve ending to the cell body), occurs along microtubules at about 200 mm/day. Synaptic vesicles, nerve growth factor (NGF) and some viruses are transported back to the cell body.
 - Molecular motor involved in retrograde transport is Dyenin.

Axonal transport



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Solution to Question 28:

Slow orthograde transport occurs at a speed of 0.5 to 10 mm/day.

AXONAL TRANSPORT

- Protein synthesis occurs in the cell body of neuron and proteins and polypeptides are transported to axonal endings by the axoplasmic flow.
- Axonal transport can be Orthograde or Retrograde.
- Orthograde transport
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- Molecular motor involved in orthograde transport is kinesin.
- Retrograde transport,
 - In the opposite direction (from the nerve ending to the cell body), occurs along microtubules at about 200 mm/day. Synaptic vesicles, nerve growth factor (NGF) and some viruses (Eg: herpes simplex, rabies, and polio) are transported back to the cell body.
- Molecular motor involved in retrograde transport is dyenin.

Solution to Question 29:

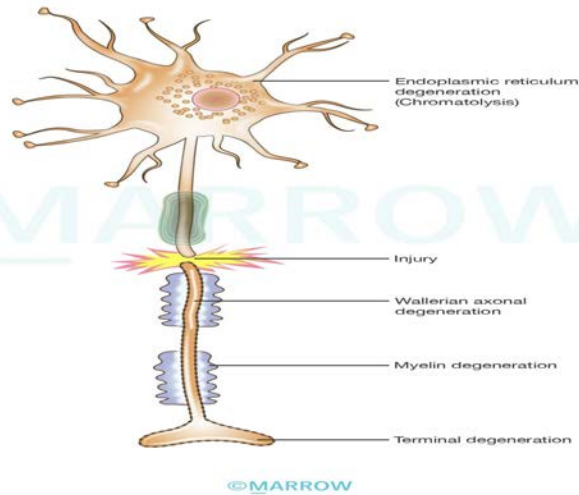
Among the given options, myelin degeneration would be the best answer.

However, the earliest change in Wallerian degeneration after nerve injury is the degeneration of the axon distal to the site of injury (axonal degeneration) where there is a disruption of the retrograde and anterograde flow of signals within the axon.

Wallerian degeneration

- If the axon is cut, the part distal to the cut degenerates; this process is called as Wallerian degeneration.
- Orthograde degeneration (Wallerian degeneration) occurs from the point of injury to the nerve terminal, interrupting neural transmission.
- After the injury, the membrane breaks down and the myelin sheath degenerates. The cell body of the injured neuron swells, the nucleus moves to an eccentric position, and the rough endoplasmic reticulum becomes fragmented (chromatolytic reaction).
- The nerve then starts to regrow, with multiple small branches projecting along the path the axon previously followed (regenerative sprouting).
- When the motor nerve to skeletal muscle is cut and allowed to degenerate, the muscle gradually becomes extremely sensitive to acetylcholine. This is called denervation hypersensitivity or supersensitivity.

Changes occurring in a neuron when its axon is crushed or injured



Solution to Question 30:

The resting membrane potential is close to the equilibrium potential for K^+ . So, changes in the external concentration of this ion can have major effects on the resting membrane potential.

- If the extracellular level of K^+ is increased (hyperkalemia), the resting potential moves closer to the threshold for eliciting an action potential, thus the neuron becomes more excitable.
- If the extracellular level of K^+ is decreased (hypokalemia), the membrane potential is reduced and the neuron is hyperpolarized.

Option a: The RMP of a nerve fiber is -70 mV, while the RMP of inner hair cells is between -45 and -60 mV relative to the fluid that bathes the basal end of the cell.

Option b: The membrane potential can be measured only when one electrode is inserted into the interior of the cell.

Option d: It depends on the equilibrium potential of K^+ .

Solution to Question 31:

Group III fibers carrying temperature, crude touch, and pricking pain sensations are δ -type A fibers, not A-gamma type.

Alternative Classification Used in Sensory Nerves

- Group Ia. Fibers from the annulospiral endings of muscle spindles are α -type A fibers in the general classification.
- Group Ib. Fibers from the Golgi tendon organs are also α -type A fibers.
- Group II. Fibers from most discrete cutaneous tactile receptors and from the flower-spray endings of the muscle spindles are β - and γ -type A fibers in the general classification.

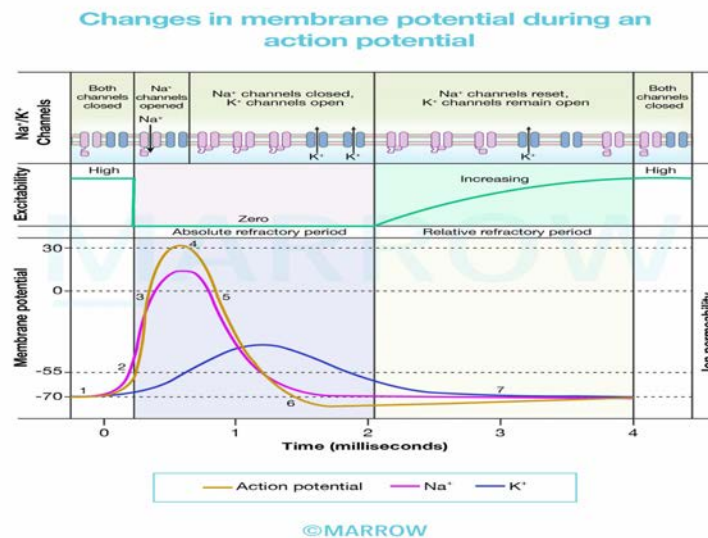
- Group III. Fibers carrying temperature, crude touch, and pricking pain sensations are δ -type A fibers in the general classification.
- Group IV. Unmyelinated fibers carrying pain, itch, temperature, and crude touch sensations are type C fibers in the general classification

Solution to Question 32:

Absolute refractory period is due to closure of inactive gates of sodium channel.

Voltage-gated sodium channels have 3 states that you can find them in. They can either be closed, open, or inactive.

When the stimulus causes activation of the open gates, sodium rushes into the cell causing the membrane potential to become more positive. After 0.5 to 1 millisecond, the inactive gate automatically closes and no stimulus, no matter how strong, can excite the nerve. This is called the absolute refractory period.



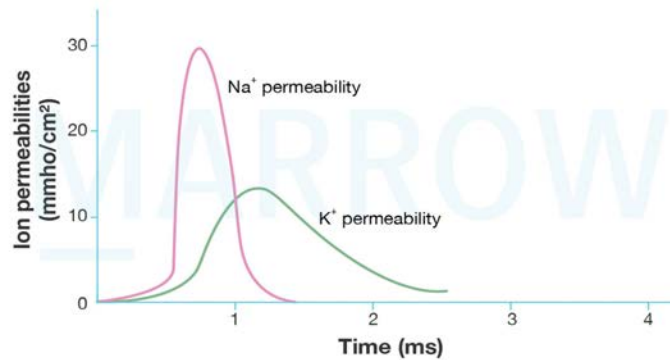
Solution to Question 33:

Channel A: Na^+ , Channel B: K^+

Na^+ channels are fast activating and are responsible for depolarization phase of action potential.

K^+ channels are slow activating and open for prolonged duration than the sodium channels. They are responsible for repolarization phase of action potential. Also, the postassium channels return slowly to their closed state, which is responsible for after-hyperpolarization.

Conductance of neuron

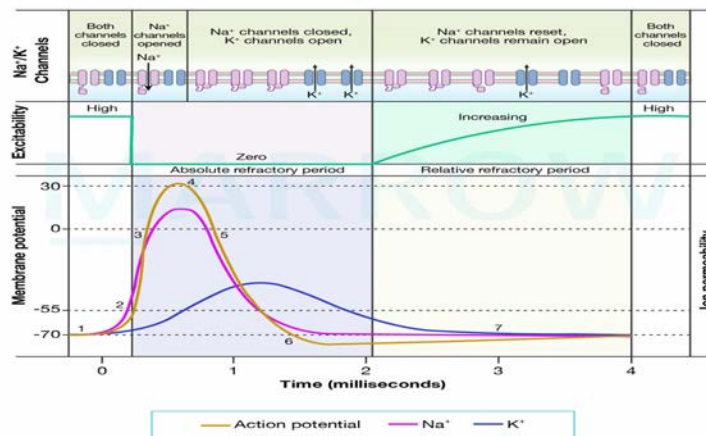


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The sequence of events during the action potential occur in 7 steps as described below.

1. Resting state - Both channels are closed
2. Na⁺ channels open in response to depolarizing stimulus. Sodium enters into the cell. The membrane reaches threshold potential.
3. The sodium influx causes opening of more sodium channels through positive feedback mechanism. Rapid depolarization occurs.
4. Sodium conductance decreases.
5. Na⁺ channels close and K⁺ channels open. Repolarization occurs.
6. Sodium channels are reset, and are ready to respond to another depolarizing stimulus (relative refractory period). During this phase, the potassium channels remain open and slowly return to closed state causing after-hyperpolarization.
7. Both channels are closed. The membrane returns to RMP.

Changes in membrane potential during an action potential



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Solution to Question 34:

The tetanizing frequency of the muscle, based on the given diagram, is 20 per second.

Tetanzing frequency = $1/\text{contraction period}$. It is the frequency at which individual muscle fibres twitch that fuse and cause sustained muscle contraction, where the contraction period is the time taken from latent period to maximum force of contraction.

In this case, the time tracer was taken using a 100Hz tuning fork. Which means the tracer recorded 100 waves per second.

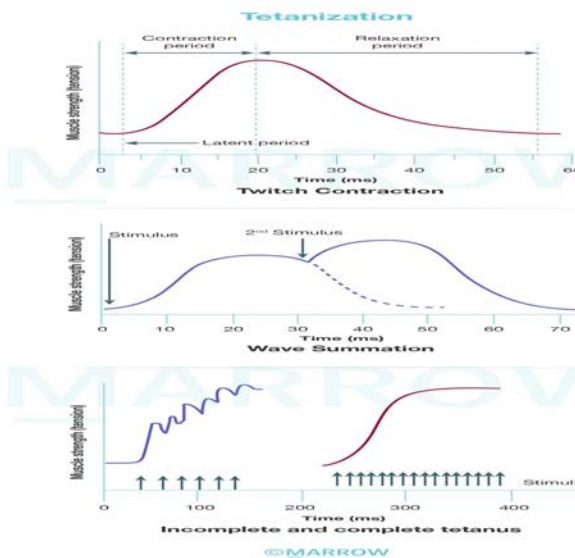
Therefore, the duration of each wave in time tracer = 0.01 seconds.

In the image given in the question, between the point B (end of latent period) to point C (maximum force of contraction), there are about 5 waves.

Hence, the duration of contraction = $5 \times 0.01 = 0.05$ seconds.

Substituting the value in the formula, we get:

Tetanzing frequency = $1/0.05 \text{ sec} = 100/5 = 20/\text{sec}$.



Tetanzation:

The features of muscle contraction can be demonstrated by eliciting single muscle twitches. The adding together of individual twitch contractions to increase the intensity of overall muscle contraction is Summation. Summation occurs in two ways:

- by increasing the number of motor units contracting simultaneously, which is multiple **■**ber summations.
- by increasing the frequency of contraction, which is called frequency summation and can lead to tetanzation.

When the frequency of a contraction reaches a critical level, the successive contractions eventually fuse together, and the whole muscle contraction appears to be completely smooth and continuous,

this is called tetanization. Beyond this point, a further increase in the frequency, will not result in increased force of contraction.

Tetanization occurs in two ways:

- Complete tetanus occurs when no relaxation occurs between stimuli.
- Incomplete tetanus when periods of incomplete relaxation take place between the summated stimuli.

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Muscle Physiology I

Question 1:

All of the following are the connective tissue layers of the skeletal muscle except _____

- a) Endomysium
- b) Sarcolemma
- c) Perimysium
- d) Epimysium

Question 2:

Which of the following is a regulatory muscle protein?

- a) Actin
- b) Tropomodulin
- c) Tropomyosin
- d) Dystrophin

Question 3:

Which of the following does not belong to dystrophin-glycoprotein complex?

- a) Perlecan
- b) Dystrophin
- c) Dystroglycan
- d) Sarcoglycan

Question 4:

Thin filaments consists of all of the following except _____

- a) Actin
- b) Myosin
- c) Troponin

d) Tropomyosin

Question 5:

Which of the following proteins is present in skeletal muscles, but absent in a smooth muscle?

- a) Troponin**
- b) Tropomyosin**
- c) Actin**
- d) Myosin**

Question 6:

Myosin filament has a fixed length of _____.

- a) 16 nanometers**
- b) 1.6 micrometers**
- c) 1.6 nanometers**
- d) 16 micrometers**

Question 7:

The thick filament is made up of _____. 1. Heavy chain of myosin 2. Light chain of myosin 3. Dystrophin 4. Titin

- a) 1, 3 and 4**
- b) 2, 3 and 4**
- c) 1 and 2**
- d) 3 and 4**

Question 8:

Pseudo H zone in a sarcomere extends between the _____

- a) Two Z lines**
- b) Two I bands**

- c) Two sides of M line
- d) A and I band

Question 9:

A cross-sectional view of a skeletal muscle fiber through the H zone would reveal the presence of _____

- a) Actin and Titin
- b) Actin and Myosin
- c) Myosin only
- d) Actin only

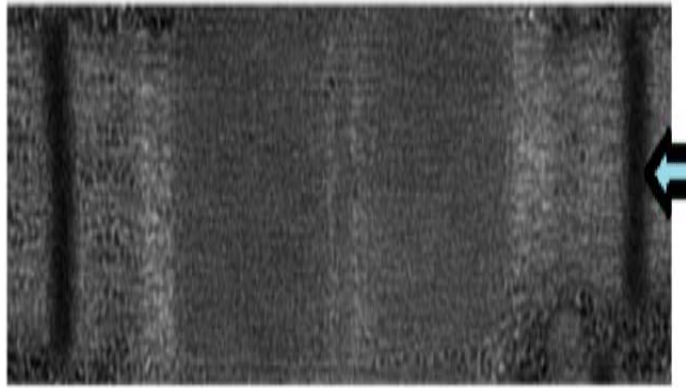
Question 10:

The titin protein present in the muscle fiber binds _____

- a) The tail ends of actin filaments to Z disc
- b) The head ends of myosin filaments to the Z disc
- c) The tail ends of myosin filaments to M disc
- d) The myofibrils to each other, and also to the cell membrane

Question 11:

Which of the following is the constituent of the marked area in the electron microscopic picture of muscle?



- a) Alpha actinin
- b) Actin
- c) Titin
- d) Tropomyosin

Question 12:

Muscle myosin and actin filaments are kept in place by _____

- a) Tropomyosin
- b) Troponin
- c) Actinin
- d) Titin

Question 13:

Choose the true statement regarding the sarcomere in skeletal muscle.

- a) Desmin binds Z line to sarcoplasmic reticulum
- b) Actinin binds M line to actin
- c) Titin binds M line to z line
- d) The length of myosin changes with contraction

Question 14:

Which of the following occurs during skeletal muscle contraction?

- a) A band shorten
- b) Both H and I band shorten
- c) Both A and I band shorten
- d) Both A and H band shorten

Question 15:

The cross-bridges of the sarcomere in skeletal muscle are made up of _____.

- a) Titin
- b) Myosin
- c) Troponin
- d) Tropomyosin

Question 16:

ATPase activity is present in _____.

- a) Myosin Head
- b) Myosin Tail
- c) Actin filaments
- d) Desmin filaments

Question 17:

Myosin head detaches from actin by which of the following mechanism?

- a) Binding of ATP
- b) Entry of calcium into sarcoplasmic reticulum
- c) Change in troponin C configuration
- d) Release of ADP and Pi

Question 18:

In human skeletal muscles, calcium ions are stored in?

- a) Terminal cistern
- b) Sarcolemma
- c) Transverse tubules
- d) All are correct

Question 19:

All of the following are true about excitation-contraction coupling in skeletal muscle except that _____

- a) Acetylcholine is released at the nerve terminal.
- b) Ca^{2+} is released from sarcoplasmic reticulum during contraction.
- c) Ca^{2+} is pumped back into sarcoplasmic reticulum during relaxation.
- d) Ca^{2+} binds to tropomyosin to initiate muscle contraction.

Question 20:

A skeletal muscle contraction ends when _____.

- a) Ions move out of the cytoplasm
- b) Ach is metabolised at the Neuromuscular junction (NMR)
- c) Decreased calcium outside the sarcoplasmic reticulum
- d) Myosin filaments are degraded

Question 21:

A bodybuilder is regularly working out in a gym. Which of the following is true regarding total heat production during muscle contraction?

- a) Activation heat is produced when a muscle is contracting
- b) Initial heat produced is due to the basal metabolic processes.
- c) Relaxation heat is present in both isotonic and isometric contractions
- d) Recovery heat of muscle is greater than the initial heat

Question 22:

In a 100-m dash, most of the energy consumed by skeletal muscles to replenish ATP is derived from _____

- a) Phosphocreatine
- b) Aerobic glycolysis
- c) Oxidation of Fatty acids
- d) None of the above

Question 23:

The length of the muscle fiber at the onset of contraction which is required to attain the maximum active tension is known as

- a) Equilibrium Length
- b) Resting Length
- c) Initial length
- d) Overlapping Length

Question 24:

All of the following are true regarding the resting length of a muscle fiber except _____.

- a) There are maximum cross-bridges between actin and myosin
- b) The passive tension is maximum
- c) The resting length is around $2\mu\text{m}$ in humans
- d) The velocity of contraction is maximum for any given load

Question 25:

Active tension in a muscle depends upon which of the following?

- a) Number of muscle fibres
- b) Number of motor units recruited
- c) Aerobic capacity of muscle
- d) Action potential amplitude

Question 26:

Which of the following have the maximum density of white muscle fibers?

- a) Calf muscles
- b) Extraocular muscles
- c) Back
- d) Hip

Question 27:

All are true about red muscle fibers except -

- a) More mitochondria
- b) High glycolytic metabolism
- c) More myoglobin
- d) More oxidative capacity

Question 28:

A motor unit is made up of _____.

- a) All the muscle fibers within a given muscle
- b) A motor neuron and the muscle fibers it innervates
- c) All the neurons going into an muscle of the body
- d) A fascicle and a nerve

Question 29:

Which of the following is correct about the recruitment of fast fatiguable motor unit ?

- a) Recruited first
- b) Recruited last
- c) During relaxation
- d) For the less demanding tasks

Question 30:

Twitch of a single muscle fiber is called _____

- a) Fibrillation
- b) Fasciculation
- c) Myoclonus
- d) Jerk

Question 31:

A scientist found that the latent period of muscle twitch is 10 ms and the contraction period is 40 ms. What is the tetanizing frequency (Hz)?

- a) 25
- b) 50
- c) 100
- d) 75

Question 32:

Tetanic contraction is due to the accumulation of _____ ions.

- a) Na^+
- b) Ca^{2+}
- c) K^+
- d) Cl^-

Question 33:

The function of the Golgi tendon organ is to _____

- a) Sense changes in muscle direction
- b) Sense changes in muscle tension
- c) Sense changes in position of the muscle
- d) Sense changes in length of the muscle

Question 34:

Which of the following statements are correct regarding mechanisms of depolarisation, repolarisation, and hyperpolarisation in a skeletal muscle action potential?

- a) 2 and 4 are correct
- b) 1, 3, and 5 are correct
- c) 1, 4, and 5 are correct
- d) 2, 4, and 5 are correct

Question 35:

Which of the following statements is true about skeletal muscle contraction?

- a) 2, 3, 4
- b) 3 only
- c) 1, 2, 4
- d) 1, 3, 4

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	b
5	a
6	b
7	c
8	c
9	c
10	b
11	a
12	d
13	c
14	b
15	b
16	a
17	a
18	a

19	d
20	c
21	a
22	a
23	b
24	b
25	b
26	b
27	b
28	b
29	b
30	a
31	a
32	b
33	b
34	b
35	b

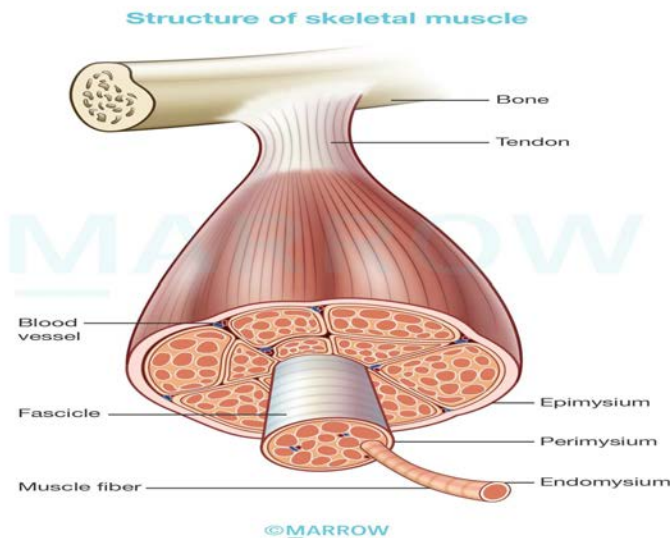
Detailed Explanations

Solution to Question 1:

Sarcolemma is the cell membrane of the muscle cell, while the remaining options are connective tissue layers surrounding the muscle.

A muscle is made up of individual muscle fibers. Each muscle fiber is a single cell that is multinucleated, long, cylindrical, and surrounded by a cell membrane, the sarcolemma.

Arrangement of the connective tissue is shown below.

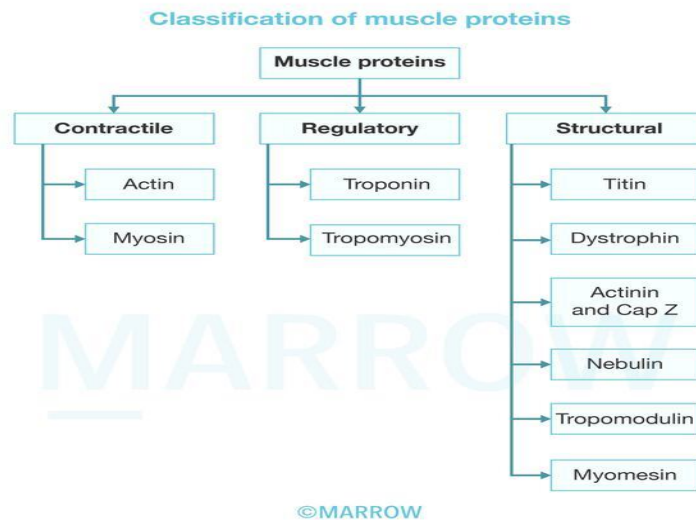


Solution to Question 2:

Tropomyosin is a regulatory muscle protein.

It extends over the entire actin filament and covers the myosin-binding sites on the actin molecules thus preventing the attachment of actin to myosin.

One molecule covers 7 active sites on actin



Troponin is another regulatory protein. Troponin molecules are small globular units located at intervals along the tropomyosin molecules. Each of the three troponin subunits has a unique function:

- Troponin T binds the troponin components to tropomyosin.
- troponin I inhibits the interaction of myosin with actin.

- troponin C contains the binding sites for the Ca^{2+} that helps initiate contraction.

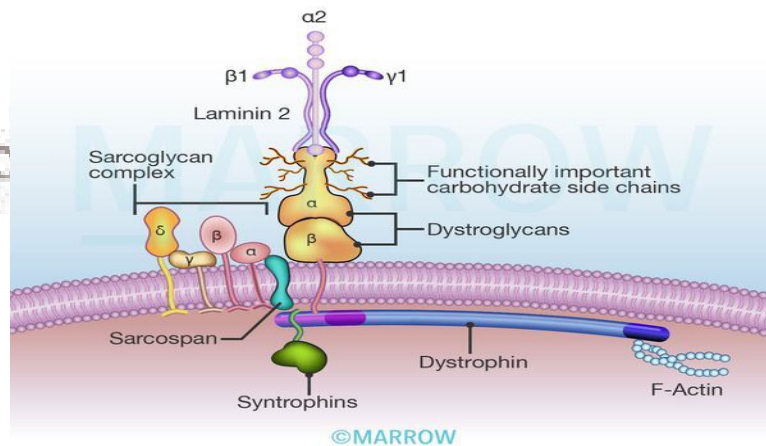
Solution to Question 3:

Perlecan is a large extracellular matrix proteoglycan that plays a crucial role in tissue development and organogenesis.

Dystrophin-glycoprotein complex

- The dystrophin-glycoprotein complex provides a structural link between the cytoskeleton of the muscle cell and the extracellular matrix, which appears to stabilize the sarcolemma, adds strength to the muscle by providing a scaffolding for the fibrils and prevents contraction-induced injury (rupture).
- Organisation of DG complex
- Dystrophin connects F-actin to the transmembrane protein β -dystroglycan via smaller proteins called Syntrophins.
- This β -dystroglycan, in turn connects to the merosin subunit of laminin 211 in the extracellular matrix via α -dystroglycan.
- The dystroglycans are also associated with a complex of four transmembrane glycoproteins, α -, β -, γ -, and δ -sarcoglycan and sarcospan.

Dystrophin-glycoprotein complex



APPLIED ASPECT

- Duchenne Muscular Dystrophy
- X-linked recessive disorder
- Deletion of DMD gene which codes for Dystrophin
- Becker muscular Dystrophy
- Dystrophin is present but altered or reduced in amount

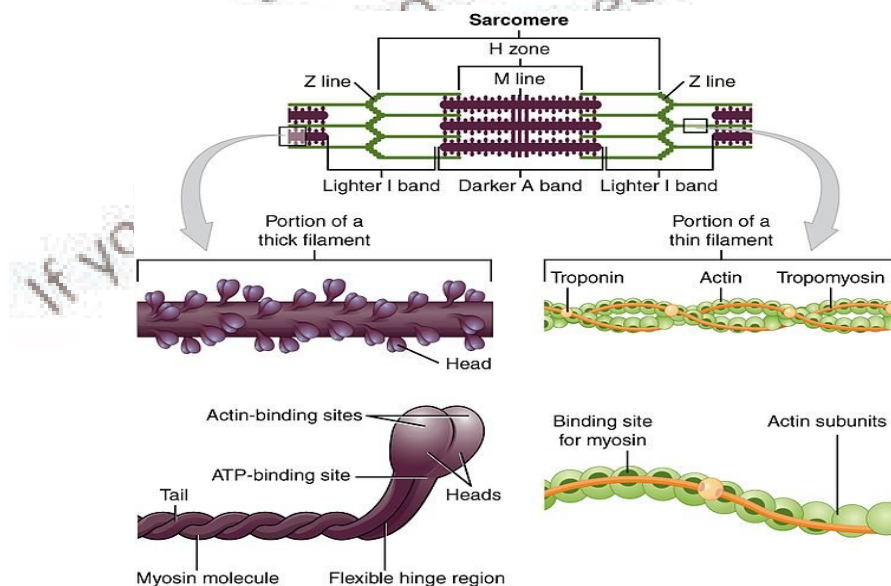
- Limb Girdle Dystrophy
- Sarcoglycan is mutated

Solution to Question 4:

Myosin forms the thick filament, while actin, troponin, and tropomyosin form the thin filament.

Thin Filaments in Sarcomere

- The thin filaments are polymers made up of two chains of actin that form a long double helix.
- Tropomyosin molecules are long filaments located in the groove between the two chains in the actin.
- Each thin filament contains 300–400 actin molecules and 40–60 tropomyosin molecules.
- Troponin molecules are small globular units located at intervals along the tropomyosin molecules.
- Each of the three troponin subunits has a unique function:
- Troponin T binds the troponin components to tropomyosin.
- Troponin I inhibits the interaction of myosin with actin.
- Troponin C contains the binding sites for the Ca^{2+} that helps to initiate contraction.



Solution to Question 5:

Actin, myosin, and tropomyosin are present in both skeletal muscle and smooth muscle. But troponin is absent and calmodulin is present in smooth muscle.

Solution to Question 6:

The total length of each myosin filament is uniform, almost exactly 1.6 micrometers.

Solution to Question 7:

The thick filament or the myosin is made up of the heavy chain of myosin and light chain of myosin.

The myosin molecule is composed of six polypeptide chains- Two heavy chains, each with a molecular weight of about 200,000, and four light chains with molecular weights of about 20,000 each.

The two heavy chains wrap spirally around each other to form a double helix, which is called the tail of the myosin molecule.

One end of each of these chains is folded into a globular polypeptide structure called a myosin head. These heads contain an actin-binding site and a catalytic site that hydrolyzes ATP. Thus, there are two free heads at one end of the double-helix myosin molecule.

The four light chains are also part of the myosin head, two to each head. These light chains help control the function of the head during muscle contraction.

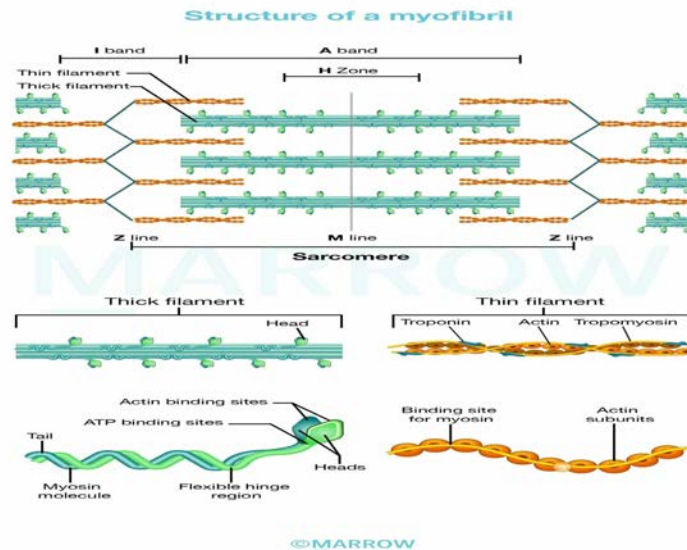
Solution to Question 8:

Pseudo H-zone is the light area between the two sides of M line due to reversal of polarity of the myosin molecules in each of the thick filaments.

Sarcomere

- The sarcomere is the contractile unit of the skeletal muscle fibre.
- Z line: provides attachment to actin filaments. The area between two adjacent Z lines is called the sarcomere.
- It is composed of thick and thin filaments:
 - Thick filament : Myosin
 - Thin filaments : Actin, tropomyosin and troponin
- I band:
 - Isotropic to light.
 - Only thin filaments.
 - Half lies in one sarcomere and half in the next.
- A band:
 - Anisotropic to light.
 - Myosin plus overlapped actin.

- H bands:
- lighter area in the centre of A band, when the muscle is relaxed, the thin filaments do not overlap the thick filaments.
- Only Myosin
- Pseudo H: light portion on two sides of M line because of absence of cross bridges (no myosin head).
- M line: centre of A band. Here there are slender cross-connections that hold the thick filaments in proper array.

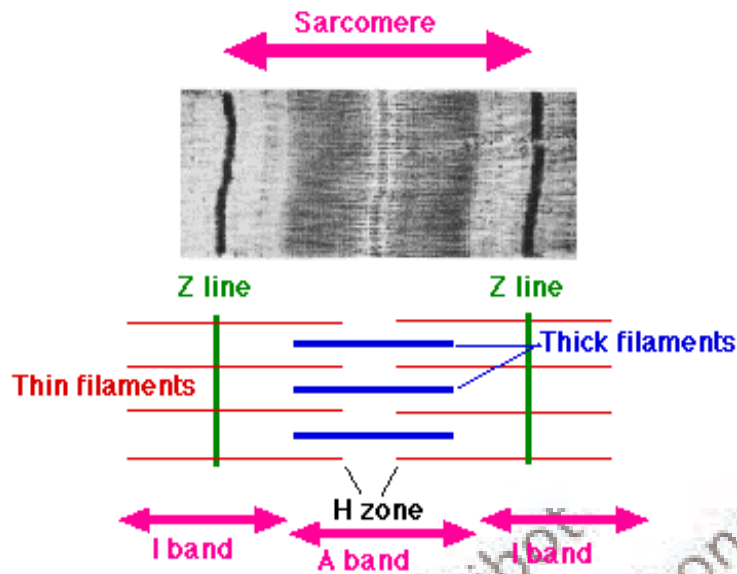


Solution to Question 9:

H zone is the central region of the A band which is lighter than the outer edges of the A band because the H zone is occupied by thick filaments only (No cross bridges)

In the figure, note the section through various levels and the myofilaments observed.

- Section through I band → only actin (thin filament)
- Section through A band (Exlcuding H zone) → Actin and Myosin
- Section through H zone → only myosin (thick filament)
- Section through M line → thick filament linked with accessory proteins
- Section at AI junction → both actin and myosin



Solution to Question 10:

The titin protein present in the muscle fiber binds the head ends of myosin filaments to the Z disc. Several proteins other than actin and myosin are also present in the muscle. Some of them include:

1. Titin:

- It is the largest known protein.
- It runs between the M line and Z disc and binds the head ends of myosin filaments to the Z disc.
- It has a tethered portion in the A band, where it is attached to the myosin filament and an elastic portion in the I band.
- The elastic property of the titin causes elastic recoil of the sarcomere and prevents it from being overstretched.

2. Nebulin:

- It extends from the Z disc and runs along the full length of each actin filament.
- It regulates the length of the actin filaments.

3. α -actinin:

- It is a component of the Z discs
- It binds the plus ends of actin filaments to this disc.

4. Myomesin

- It is present in the region of the M line.
- It maintains the regular lattice arrangement of the myosin filaments in this region.

5. Desmin:

- It is an intermediate filament characteristic of the muscle
- It encircles the myofibrils near the Z disc and binds myofibrils to each other and also to the cell membrane.

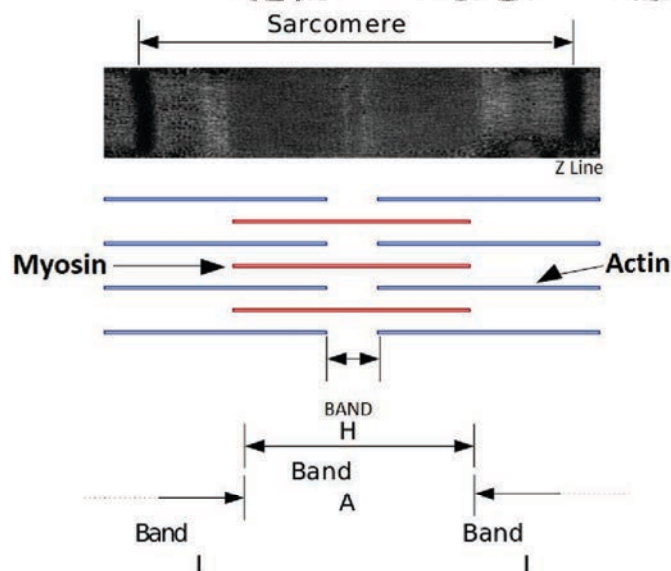
6. Dystrophin:

- It is present in the periphery of the muscle fiber, towards the cytoplasmic face of the sarcolemma.
- It binds to the actin filaments and the glycoprotein complex, consisting of dystroglycan and sarcoglycan that spans the sarcolemma.

Solution to Question 11:

Alpha actinin is the constituent of the marked area.

The image depicts sarcomere and the arrow points towards Z-line. Alpha actinin binds actin to Z-lines.



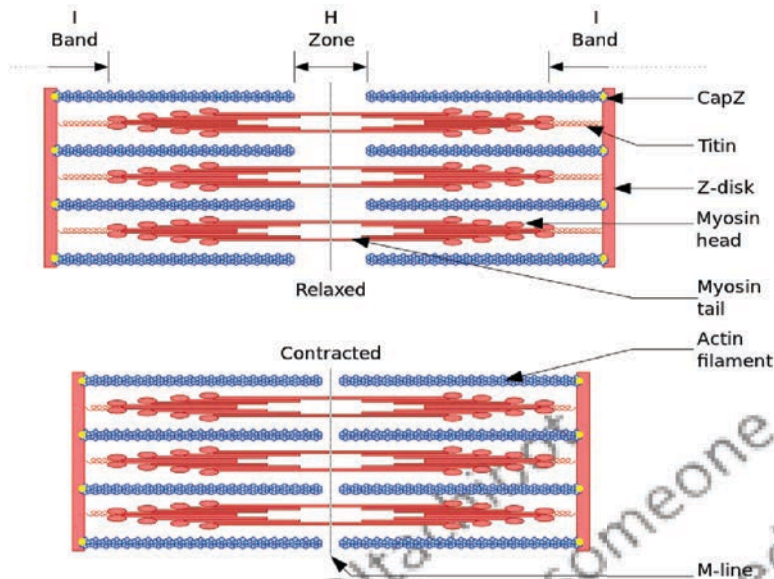
The orderly arrangement of actin, myosin, and related proteins that produces this pattern. The area between two Z-lines is called sarcomere.

The thick filaments, which are about twice the diameter of the thin filaments, are made up of myosin, which are lined to form A bands; the thin filaments are made up of actin, tropomyosin, and troponin. The array of thin filaments extends out of the A band and into the less dense staining I bands.

The lighter H bands in the center of the A bands are the regions where, when the muscle is relaxed, the thin filaments do not overlap the thick filaments. The Z lines allow for anchoring of the thin filaments.

Actinin, titin and desmin are additional structural proteins. Titin connects Z-lines to the M line and provides the scaffolding to the sarcomere. Desmin adds structure to Z-lines by connecting Z-line to plasma membrane.

Note: Some sources say that even titin is a component of Z-disc so if alpha actinin is not given in the options titin is the second best answer.

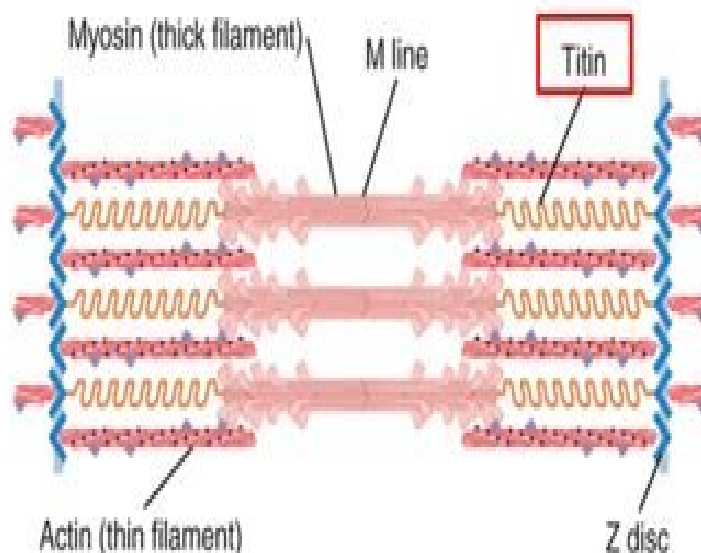


Solution to Question 12:

The side-by-side relationship between the myosin and actin filaments is maintained by a large number of filamentous molecules of a protein called titin.

Titin is a filamentous and springy protein. These springy titin molecules act as a framework that holds the myosin and actin filaments in place so that the contractile machinery of the sarcomere will work.

One end of the titin molecule is elastic and is attached to the Z disk, acting as a spring and changing length as the sarcomere contracts and relaxes. The other part of the titin molecule tethers it to the myosin thick filament.



Solution to Question 13:

Titin binds M line to Z line

Other options

Desmin binds Z line to sarcolemma (not sarcoplasmic reticulum)

Actinin binds Z line to actin (not M line; in fact, one of the two attachments of actin, desmin and titin, is always Z line)

Titin binds M line to z line, forms scaffolding, is responsible for muscle elasticity, is the largest known protein (Largest protein - titin; Largest gene - Dystrophin; Most abundant protein - Collagen)

The length of myosin or actin does not change with contraction; actually, the length of zones I and H get changed (i.e, gets decreased)

Solution to Question 14:

Sliding of thin filament over thick filament (cross bridge formation) reduces the length of I and H band.

- During contraction, the Z lines come closer to each other reducing the length of the sarcomere.
- The A band remains constant and M line becomes prominent.

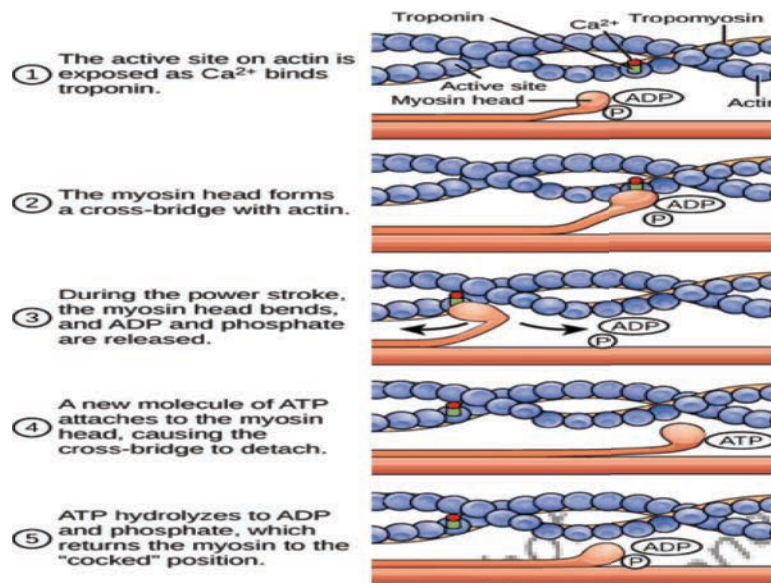
Solution to Question 15:

When the action potential reaches the muscle fiber, there is influx of Ca^{2+} ions. This Ca^{2+} binds to troponin C and tropomyosin is moved out of the way. Myosin heads can now bind to actin and form so-called cross-bridges.

When a head attaches to an active site, this attachment simultaneously causes profound changes in the intramolecular forces between the head and arm of its cross-bridge. This causes the head to tilt toward the arm and to drag the actin filament along with it. This tilt of the head is called the power stroke.

Immediately after tilting, the head then automatically breaks away from the active site.

Formation of cross-bridges is associated with hydrolysis of ATP and generation of force.

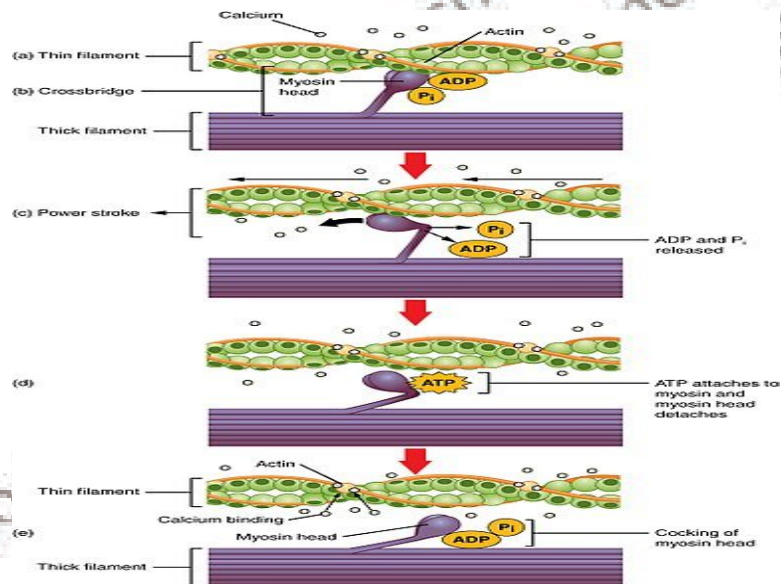
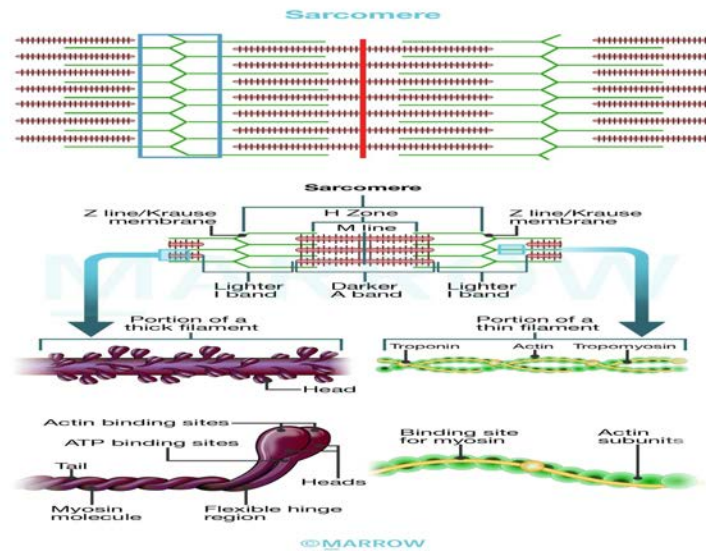


Solution to Question 16:

A feature of the myosin head that is essential for muscle contraction is that it functions as an adenosine triphosphatase (ATPase) enzyme.

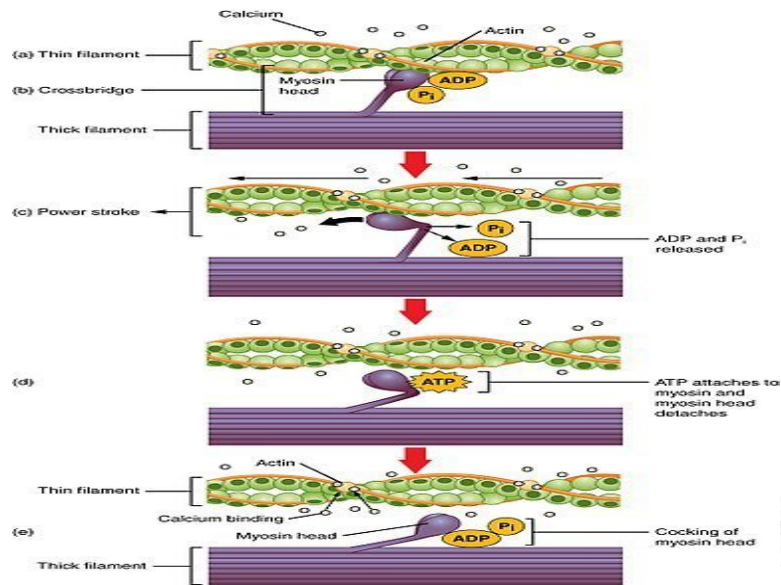
Chemical Events during Muscle Contraction

- Before contraction begins, the heads of the cross-bridges bind with ATP. The ATPase activity of the myosin head immediately cleaves the ATP but leaves the cleavage products, ADP plus phosphate ion, bound to the head. (Point A)
- When the troponin-tropomyosin complex binds with calcium ions, active sites on the actin filament are uncovered and the myosin heads then bind with these sites. (Point B)
- The bond between the head of the cross-bridge and the active site of the actin filament causes a conformational change in the head and the power stroke occurs. (Point C)
- Once the head of the cross-bridge tilts, the ADP and phosphate ion that were previously attached to the head are released and a new ATP molecule binds in that place. (Point D)
- The binding of the new ATP molecule causes detachment of the head of myosin from the actin. (Point D)
- This cycle is then repeated.



Solution to Question 17:

Binding of ATP on the free site of myosin leads to the detachment of myosin head from the thin filament.



Solution to Question 18:

In a skeletal muscle, calcium ions are majorly stored in the terminal cisterns of the Sarcoplasmic Reticulum.

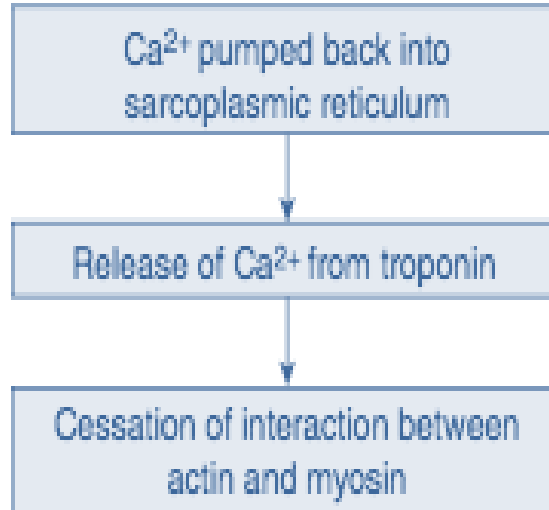
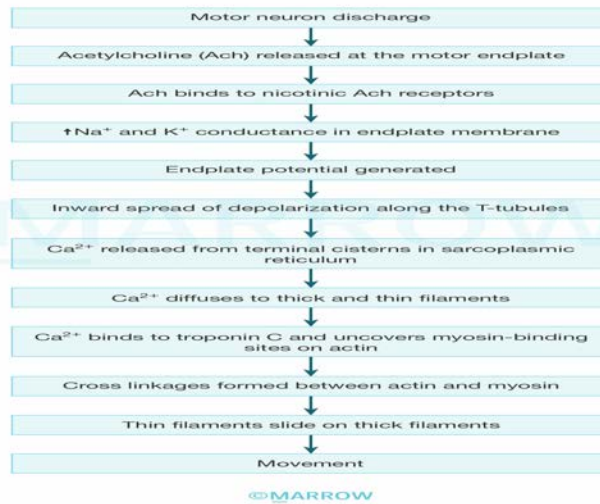
Terminal cisterns are the lateral sacs of the sarcoplasmic reticulum. The spread of action potential triggers the release of Ca²⁺ from the terminal cisterns.

The sarcolemma is the cell membrane of the skeletal muscle. Transverse tubules are the internal extensions of the sarcolemma.

Solution to Question 19:

Ca²⁺ binds to troponin C (not tropomyosin) to initiate muscle contraction.

Excitation-contraction coupling in skeletal muscle



Solution to Question 20:

Once the Ca^{2+} concentration outside the sarcoplasmic reticulum has been lowered sufficiently, chemical interaction between myosin and actin ceases and the muscle relaxes.

- The SERCA pump uses energy from ATP hydrolysis to remove Ca^{2+} from the cytosol back into the terminal cisterns, where it is stored until released by the next action potential.
- Thus during relaxation
- Ca^{2+} concentration in cytoplasmic decreases.
- Ca^{2+} concentration in sarcoplasmic reticulum increases.

Option a: Calcium ions do not move out of the cytoplasm. They are taken up by the sarcoplasmic reticulum.

Solution to Question 21:

Activation heat is the heat that muscle produces whenever it contracts.

Heat Production in Muscle

- Resting heat: The heat given off at rest due to the basal metabolic processes.
- Initial heat: The heat produced in excess of resting heat during contraction. It is made up two subtypes:
 - Activation heat: The heat that muscle produces whenever it is contracting, and
 - Shortening heat: The heat which is proportional in amount to the distance the muscle shortens. Shortening heat is apparently due to some change in the structure of the muscle during shortening.
- Recovery heat: this is the heat liberated by the metabolic processes that restore the muscle to its precontraction state. The recovery heat of muscle is approximately equal to the initial heat.
- Relaxation heat: Heat released in a muscle that has contracted isotonically to restore it to its previous length. This extra heat is in addition to recovery heat.

Solution to Question 22:

In a 100-m dash that takes 10 s, 85% of the energy consumed is derived anaerobically by phosphocreatine, which is hydrolyzed to creatine and phosphate groups with the release of considerable energy.

The ATP concentration in a muscle is low enough (4 millimolar) to sustain contractions for 1-2 seconds after which ATP is dephosphorylated into ADP. The ADP is re-phosphorylated by phosphocreatine for the next 5 to 8 seconds.

ADP is re-phosphorylated by 3 sources.

- Phosphocreatine after 2 seconds upto 10 seconds.
- Glycolysis (anaerobic) after 10 seconds upto a minute
- Oxidative metabolism - after a minute upto 2-4 hours.

Solution to Question 23:

The length of the muscle fiber at the onset of contraction which is required to attain maximum active tension is known as resting length.

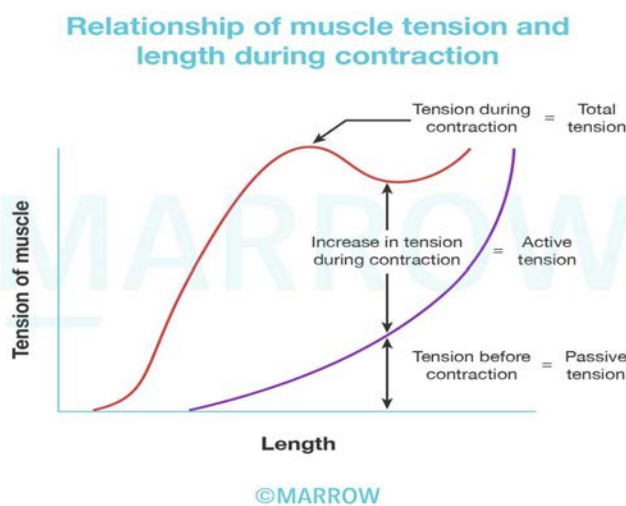
Resting length is also known as the optimum length. It is about $2\mu\text{m}$. At this length, there is a maximal overlap between the actin and myosin fibers and therefore, the maximum number of cross-bridge binding sites are formed.

Passive tension is the tension in a muscle when it is not stimulated. Total tension is the tension developed when a muscle contracts isometrically in response to a maximal stimulus.

Active tension = Total tension - Passive tension

When the muscle fiber contracts, the active tension developed is proportional to the number of cross-bridges between the actin and myosin fibers. Hence, at the resting length, maximum active tension is produced and the muscle fiber contracts with a maximum force of contraction.

In the given image, the purple line represents the passive tension in the muscle (before contraction) and its relation with the muscle length, while the red line represents the total tension present (during contraction) and its relation with the muscle length.



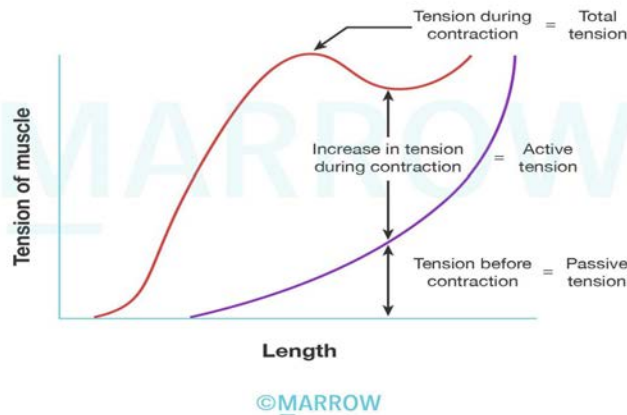
Solution to Question 24:

The passive tension is not maximum at resting length of a muscle fiber.

Passive tension is the tension in a muscle when it is not stimulated. Total tension is the tension developed when a muscle contracts isometrically in response to a maximal stimulus.

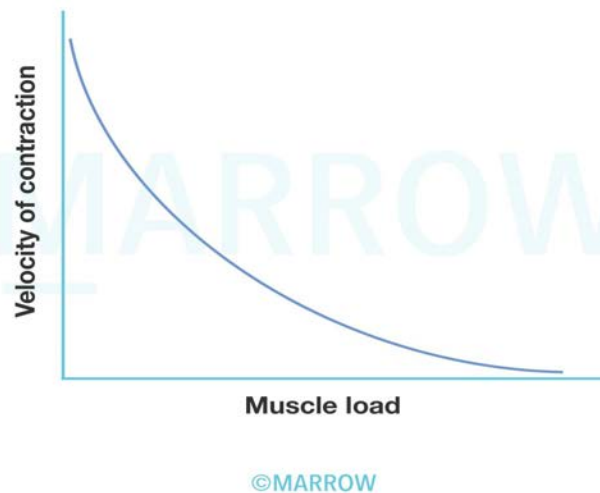
Active tension = Total tension - Passive tension

Relationship of muscle tension and length during contraction



The active tension is maximum at the resting length of a muscle fiber. This is because there is an optimal overlap between the actin and myosin fibers at the resting length and therefore the maximum number of cross-bridges are formed. (Option A) This provides for the maximum force of contraction at this length. The active tension decreases as the muscle is stretched beyond its normal length. The resting length is around $2\mu\text{m}$ in humans. (Option C)

The velocity of muscle contraction varies inversely with the load on the muscle i.e. as the load on the muscle increases, the velocity of contraction decreases. At a given load, the velocity is maximal at the resting length and declines if the muscle is shorter or longer than this length. (Option D)



Solution to Question 25:

Active tension in a muscle depends upon the number of motor units recruited.

The greater the number of motor units recruited, the greater will be the active tension.

Factors affecting muscle tension

Tension developed by each fiber

- Action potential frequency
- Fiber length (length-tension relationship)
- Fiber diameter
- Fatigue

Number of active fibers

- Number of fibers per motor unit
- Number of active motor units (recruitment)

Solution to Question 26:

Extra-ocular muscles have the maximum density of white muscle fibers (type IIB), which contract quickly with peak tension attained within 7.5 msec of stimulation.

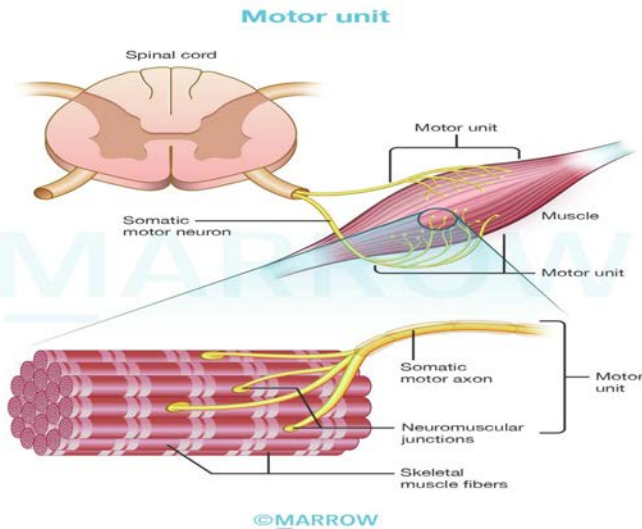
Solution to Question 27:

The red muscle fibers are Type I muscle fibers with low/moderate glycolytic metabolism. Rest of the options are true regarding red muscle fibers.

Solution to Question 28:

Each single motor neuron and the muscle fibers it innervates constitute a motor unit.

- The number of muscle fibers in a motor unit varies.
- In muscles concerned with fine, graded, precise movement, (eg. extraocular muscles) each motor unit innervates very few (three to six) muscle fibers.
- On the other hand, in human leg and back muscles, number of muscle fibers per motor unit can be upto 600.
- Each spinal motor neuron innervates only one kind of muscle fiber, so that all the muscle fibers in a motor unit are of the same type.
- On the basis of the type of muscle fiber they innervate, and thus on the basis of the duration of their twitch contraction, motor units are divided into S (slow), FR (fast, resistant to fatigue), and FF (fast, fatigable) units.



Solution to Question 29:

The fast fatigable motor unit is recruited last.

Each single motor neuron and the muscle fibers it innervates constitute a motor unit. On the basis of the type of muscle fiber they innervate, and thus on the basis of the duration of their twitch contraction, motor units are divided into

- S (slow)
- FR (fast, resistant to fatigue)
- FF (fast, fatigable) units.

S fibers have a low innervation ratio (ie, small units) and FF fibers have a high innervation ratio (i.e., large units).

The recruitment of motor units during muscle contraction follows a general scheme, the size principle.

- In general, a specific muscle action is developed first by the recruitment of S muscle units that contract relatively slowly to produce controlled contraction.
- Next, FR muscle units are recruited, resulting in more powerful response over a shorter period of time.
- Lastly, FF muscle units are recruited for the most demanding tasks.

Solution to Question 30:

Twitch of a single muscle fiber is called fibrillation.

Fibrillation	Fasciculation
Twitch of a single muscle fiber	Jerky, visible contractions of groups of muscle fibers
Occurs due to the degeneration of a single ° motor neuron or its axon.	Occur as a result of a pathologic discharge of spinal motor neurons (motor unit).

Solution to Question 31:

Tetanizing frequency = $1/\text{contraction period} = 1/40 \text{ ms} = 25 \text{ Hz}$.

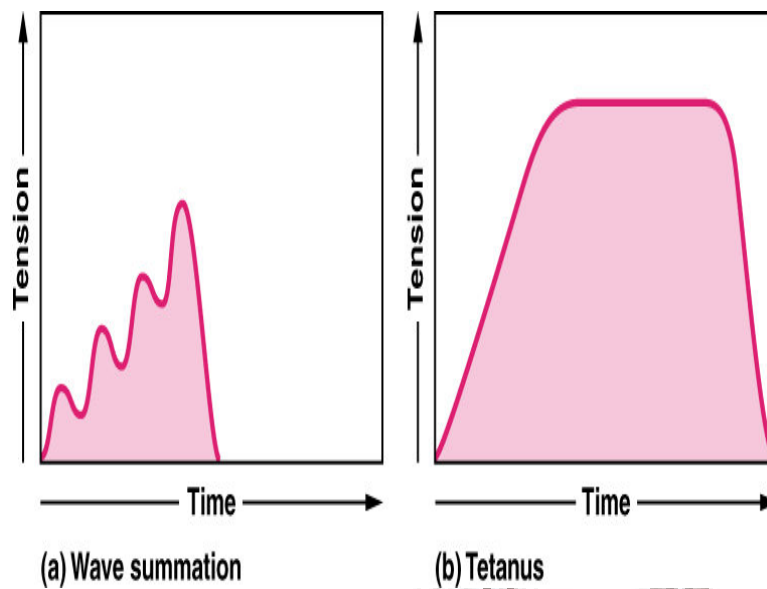
Tetanizing frequency

- As the contractile mechanism does not have a refractory period, on rapid stimulation of a muscle, each new contraction occurs before the preceding one is over. As a result, the second contraction is added partially to the first, and thus the total strength of contraction rises progressively with increasing frequency. Such a response is called a tetanus (tetanic contraction). It is a complete tetanus when no relaxation occurs between stimuli.
- Tetany occurs because enough calcium ions are maintained in the muscle sarcoplasm throughout the period of stimulation.

Solution to Question 32:

Tetanic contraction occurs due to progressive accumulation of Ca^{2+} ions with rapidly repeated stimulation of skeletal muscle.

- With rapidly repeated stimulation, activation of the contractile mechanism occurs repeatedly before any relaxation has occurred, and the individual contractions fuse into one continuous contraction. Such a response is called a tetanus (tetanic contraction).
- A single action potential releases sufficient Ca^{2+} to cause a twitch contraction.
- However, the duration of this contraction is very short because Ca^{2+} is very rapidly pumped back into the SR.
- If the muscle is stimulated a second time before the muscle is fully relaxed, there is progressive accumulation of Ca^{2+} and the force of contraction increases. Thus, twitch forces are amplified as stimulus frequency increases.
- At a high level of stimulation, intracellular Ca^{2+} increases and is maintained throughout the period of stimulation, and the amount of force developed greatly exceeds that seen during a twitch. The response is termed tetany.



Solution to Question 33:

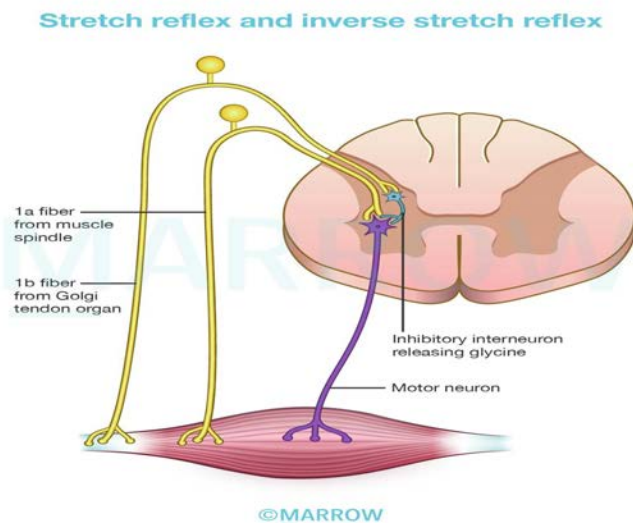
Golgi tendon organ sense changes in muscle tension.

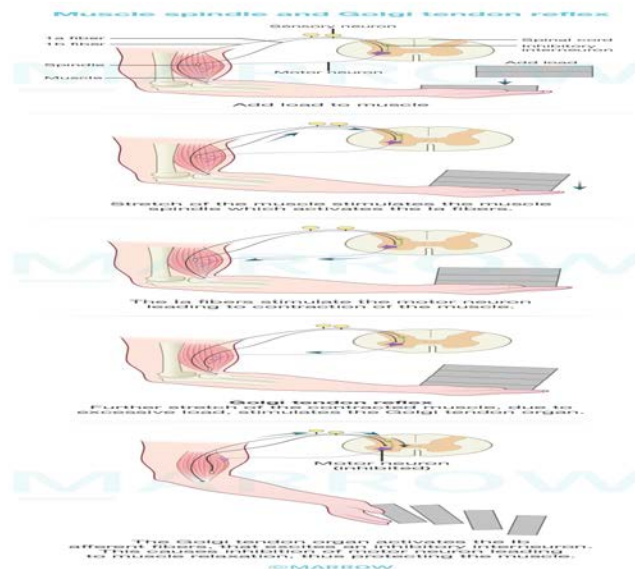
Golgi tendon reflex (Inverse myotatic/ Inverse stretch reflex) is disynaptic, is the opposite of the stretch reflex.

When the tension in a contracted muscle is significantly elevated, the inhibitory effect from the Golgi tendon organ leads to instantaneous relaxation of the entire muscle. This effect is called the lengthening reaction.

It is a protective mechanism to prevent tearing of the muscle or avulsion of the tendon from its attachments to the bone.

Note: Muscle spindles sense changes in muscle length (stretch).



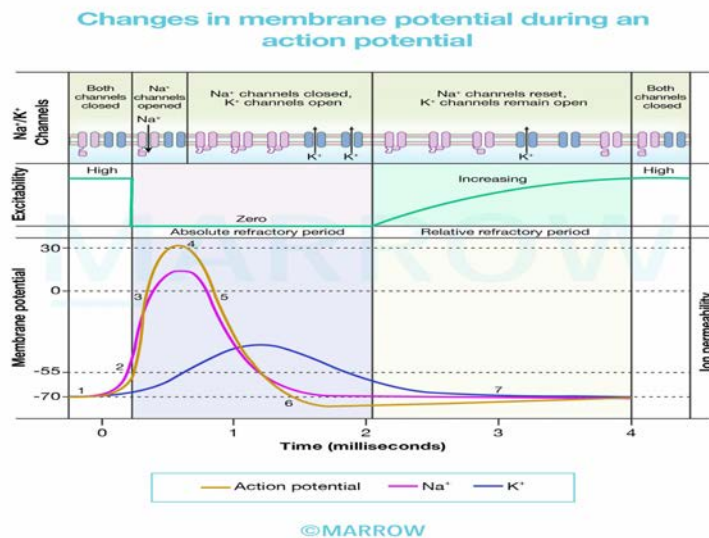


Solution to Question 34:

Statements 1, 3, and 5 are correct.

Action potential in skeletal muscle:

- Motor-neuron discharge causes end-plate potential, which when reaches to a threshold value, causes action potential in the muscle fibre.
- The depolarization of the muscle fibre is caused by the activation of the voltage-gated sodium channels, which causes the influx of sodium.
- During repolarization, the sodium channels are closed, and voltage-gated potassium channels open, which causes efflux of potassium.
- The voltage-gated potassium channels close slowly. As a result, the potassium efflux continues, leading to hyperpolarization of the membrane.



The steps in an action potential in muscle fibre and the neuron, are similar except for the quantitative differences, such as:

- RMP is -80 to -90 mV in skeletal muscle fibres, while it is about -70mV in neurons.
- The duration of the action potential in skeletal muscle is 1 to 5 msec, which is about 5 times longer than in the largest myelinated nerve fibre.
- The velocity of conduction is about 3-5m/s, which is about 1/13th of the velocity of the nerve fibre.

Solution to Question 35:

During the contraction of the skeletal muscle, the Ca²⁺ is released from the sarcoplasmic reticulum. Only statement 3 is correct.

Excitation contraction coupling refers to the process by which the membrane depolarization of the muscle fibre (excitation) leads to the contraction of the muscle.

The steps of excitation-contraction coupling in skeletal muscle are as follows:

- Motor-neuron discharge causes the release of acetylcholine at the motor endplate, which binds to the nicotinic acetylcholine receptors.
- This causes increased Na⁺ and K⁺ conductance in the endplate membrane, leading to the generation of end-plate potential, which is a local potential.
- When the local endplate potential reaches a threshold for the muscle fibre, and an action potential is generated.
- The generated action potential passes inward through the T tubules, leading to the spread of depolarisation.
- The depolarization of the muscle fibre causes the release of Ca²⁺ from terminal cisterns of sarcoplasmic reticulum and diffusion to thick and thin filaments.

- The calcium released, binds to troponin C, uncovering myosin-binding sites on actin.
- The formation of cross-linkages between actin and myosin and sliding of thin on thick filaments causes contraction of the skeletal muscle.

Steps in relaxation:

- The cytoplasmic calcium is pumped back into the sarcoplasmic reticulum, by the action of SERCA (sarcoplasmic endoplasmic reticulum calcium ATPase).
- Once the calcium concentration is lowered, the interaction between actin and myosin ceases, and the muscle relaxes.

Other options:

Statement 1: Incorrect - Extracellular calcium levels play an important role in muscle contraction. When there is a decrease in extracellular calcium as in hypocalcemia, the voltage-gated sodium channels are activated, which causes sodium influx. The increased intracellular sodium concentration causes sustained depolarization, which may lead to tetany.

Statement 2: Incorrect - Ca^{2+} is not released from mitochondria. It is released from the sarcoplasmic reticulum.

Statement 4: Voltage-gated calcium channels on T-tubules (part of sarcolemma) are activated by a depolarizing stimulus, which causes the release of calcium from the sarcoplasmic reticulum. The increased calcium concentration will eventually lead to muscle contraction.

Muscle Physiology II

Question 1:

Human cardiac muscle contains _____.

- a) Alpha heavy chain of Myosin
- b) Beta heavy chain of Myosin
- c) Both a and b
- d) None

Question 2:

Intercalated discs are seen in _____.

- a) Skeletal muscle
- b) Smooth muscle
- c) Cardiac muscle
- d) All of the above

Question 3:

Activation of the sympathetic system increases the myocardial contractility of a patient. Which of the following is responsible for this positive inotropic effect?

- a) Phosphorylation of Phospholamban
- b) Dephosphorylation of Phospholamban
- c) Decreased levels of cAMP
- d) Increased permeability of cell membrane

Question 4:

Which among the following is the function of phospholamban?

- a) Regulation of Na⁺-K⁺ pump
- b) Transport calcium out of mitochondria

- c) Regulates calcium entry into sarcoplasmic reticulum
- d) Binds actin with myosin

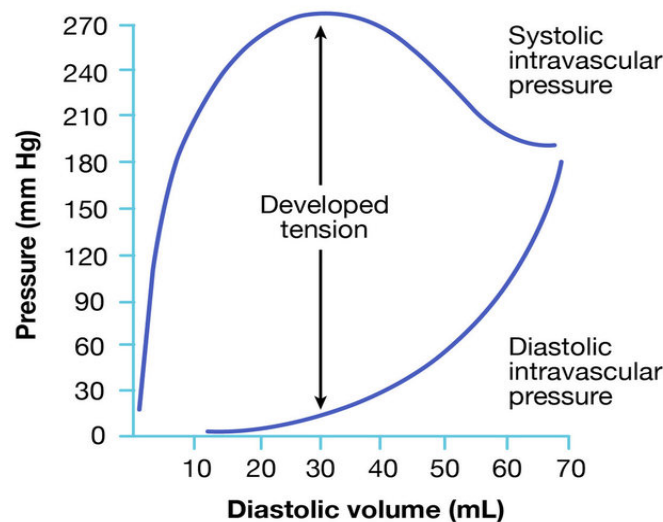
Question 5:

Which of the following is not a feature of cardiac muscle?

- a) All or none phenomenon
- b) Length-tension relationship
- c) Tetanization
- d) Plateau potential

Question 6:

The following graph displays which physiological law of the heart?



- a) Starling Law
- b) Laplace Law
- c) Bainbridge Law
- d) Bell-Magendie law

Question 7:

Muscles in hollow viscera behave as _____

- a) Single unit smooth muscle
- b) Multiunit smooth muscle
- c) Both, depending on action
- d) None

Question 8:

Which of the following is true regarding excitation-contraction coupling in smooth muscle?

- a) Presence of troponin is essential
- b) Sustained contraction occurs with high calcium concentration
- c) Phosphorylation of actin is required for contraction
- d) Cellular calcium concentration is essential to cause contraction

Question 9:

Calmodulin is mostly both structurally and functionally related to _____

- a) Actin
- b) Myosin light chain kinase
- c) Troponin C
- d) Tropomyosin

Question 10:

A sustained contraction with little expenditure of energy is made possible by a latching mechanism in _____.

- a) Skeletal muscle
- b) Smooth muscle
- c) Cardiac muscle
- d) All of the above

Question 11:

The most distinguishing feature between skeletal muscle and smooth muscle is _____

- a) Troponin
- b) Tropomyosin
- c) Myosin
- d) Actin

Question 12:

Which of the following acts to increase the release of calcium ions from the endoplasmic reticulum of a smooth muscle?

- a) Calsequestrin
- b) Diacylglycerol
- c) Inositol Triphosphate (IP₃)
- d) 1, 25 di hydroxy cholecalciferol

Question 13:

Which of the following statements regarding smooth muscle is false?

- a) Unitary smooth muscle has an unstable membrane potential
- b) Dense Bodies are present instead of Z lines
- c) It can exert a variable tension at any given length
- d) Sympathetic stimulation produces excitatory membrane potentials

Question 14:

Given below is the demonstration of tetanization in the gastrocnemius muscle of a frog. Identify the tetanising frequency.



a) 25-20 Hz
b) 30-35 Hz
c) 15-20 Hz
d) 10-15 Hz

Question No.

1

C

2

C

3

a

4

C

5

C

6

a

7

a

8

d

9

C

10

b

11

a

12

C

13

d

14

b

Detailed Explanations

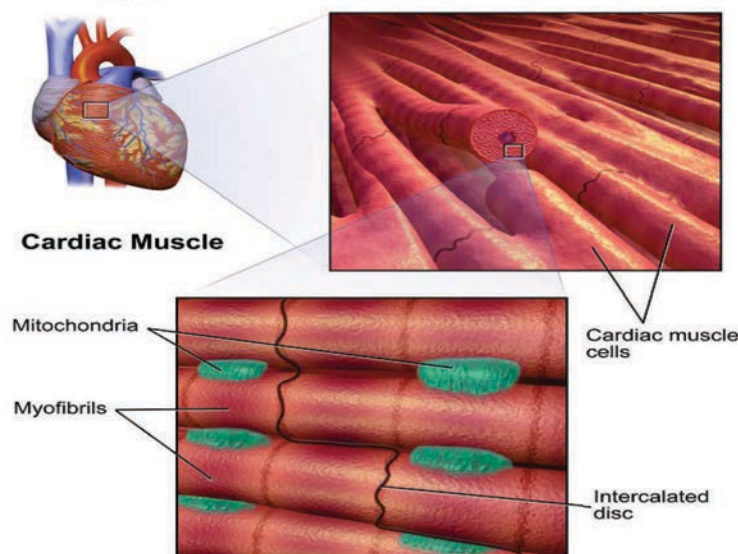
Solution to Question 1:

Human heart contains both alpha and beta isoforms of the myosin heavy chain. β isoform has a lower myosin ATPase activity than the α isoform.

- α isoform predominates in the adult atria.
- β isoform predominates in the adult ventricles.
- The spatial distribution contributes to the well coordinated contraction of the heart.

Solution to Question 2:

- Cardiac muscle fibres are made up of many individual cells connected in series and parallel with one another.
- The areas where the end of one muscle fiber abuts on another, which always occur at Z lines, are called intercalated discs.
- It provides a strong union between fibers, maintaining cell-to-cell cohesion, so that the pull of one contractile cell can be transmitted along its axis to the next.
- It contains Desmosomes to prevent pulling apart of the cells when strain
- At each intercalated disc, the plasma membrane fuses with one another forming gap junction which provide low resistance bridges for the spread of excitation from one fiber to another permitting the cardiac muscle to function as a syncytium.



Solution to Question 3:

Phosphorylation of phospholamban is responsible for the positive inotropic effect on the myocardium observed after the stimulation of the sympathetic nervous system.

Phospholamban is a protein that regulates Ca^{2+} -ATPase in the sarcoplasmic reticulum. When phosphorylated, phospholamban stimulates the Ca^{2+} -ATPase, resulting in greater uptake and storage of Ca^{2+} by the sarcoplasmic reticulum.

Increased Ca^{2+} uptake by the sarcoplasmic reticulum has two effects:

It causes faster relaxation (i.e., briefer contraction).

It increases the amount of stored Ca^{2+} for release on subsequent beats.

This positive inotropic effect has three important features: increased peak tension, increased rate of tension development, and faster rate of relaxation.

Faster relaxation means that the contraction (twitch) is shorter, allowing more time for refilling.

This effect is mediated via activation of β_1 receptors, which are coupled via a G_s protein to adenylyl cyclase.

Activation of adenylyl cyclase leads to the production of cAMP, activation of protein kinases, and phosphorylation of proteins that produce the physiologic effect of increased contractility.

Solution to Question 4:

- Phospholamban is a protein that regulates SR Ca^{2+} -ATPase (SERCA) in the sarcoplasmic reticulum. In its dephosphorylated state, phospholamban inhibits the Ca^{2+} -ATPase, resulting in diminished uptake and storage of Ca^{2+} by the sarcoplasmic reticulum, while in its phosphorylated state, there is disinhibition of the Ca^{2+} -ATPase leading to increased uptake and storage of Ca^{2+} by the sarcoplasmic reticulum.
- Increased Ca^{2+} uptake by the sarcoplasmic reticulum has two effects:
 - It causes faster relaxation (i.e., briefer contraction).
 - It increases the amount of stored Ca^{2+} for release on subsequent beats.
- Phosphorylation of phospholamban is responsible for the positive inotropic effect on the myocardium observed after the stimulation of sympathetic nervous system.
- This effect is mediated via activation of β_1 receptors, which are coupled via a G_s protein to adenylyl cyclase.
- This positive inotropic effect has three important features: increased peak tension, increased rate of tension development, and faster rate of relaxation.
- Faster relaxation means that the contraction (twitch) is shorter, allowing more time for refilling.

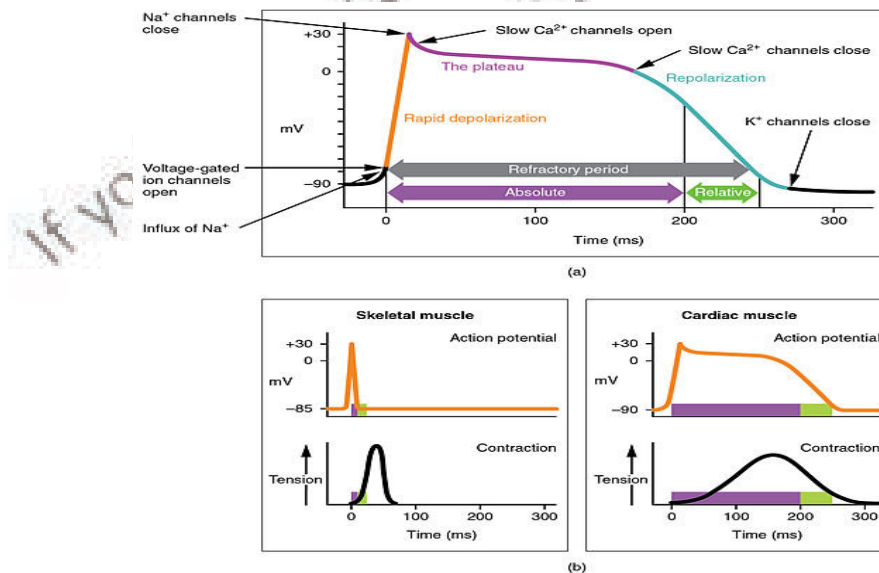
Solution to Question 5:

Cardiac muscle cannot be tetanized due to long refractory period.

In cardiac myocytes, depolarization proceeds rapidly and an overshoot of the zero potential is present. This is followed by a plateau before the membrane potential reaches the RMP (-90mV). This plateau phase lasts for 200 msec or more. Repolarization is therefore not complete until contraction is 50% completed.

Phases of cardiac action potential

- 0 → Depolarisation and overshoot
- Due to opening of voltage-gated Na^+ channels
- 1 → Initial rapid repolarization
- Due to closure of Na^+ channels and opening up of K^+ channels
- 2 → Plateau phase
- Due to slow but prolonged opening of Ca^{2+} channels
- 3 → late rapid repolarization
- 4 → Baseline (RMP)
- Phases 3 and 4 are due closure of Ca^{2+} channels and a slow, delayed increase of K^+ efflux through K^+ channels.
- During phases 0 to 2 and about half of phase 3 of cardiac action potential, cardiac muscle cannot be excited again, that is, it is in its absolute refractory period. It remains relatively refractory until phase 4. Thus, cardiac muscle cannot be tetanized.

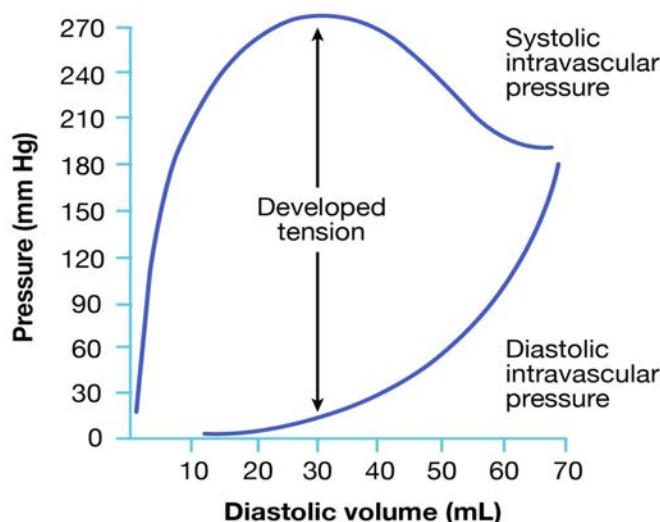


Solution to Question 6:

The diagram displays the Length-Tension relationship for a cardiac muscle explaining the Starling law of the heart.

In the body, the initial length of the fibers is determined by the degree of diastolic filling of the heart, and the pressure developed in the ventricle is proportional to the volume of the ventricle at the end of the filling phase (Starling law of the heart).

- The developed tension increases as the diastolic volume increases until it reaches a maximum, then tends to decrease.
- The decrease is due to beginning disruption of the myocardial fibers and not due to a decrease in the number of cross-bridges between actin and myosin as seen in skeletal muscle.



Solution to Question 7:

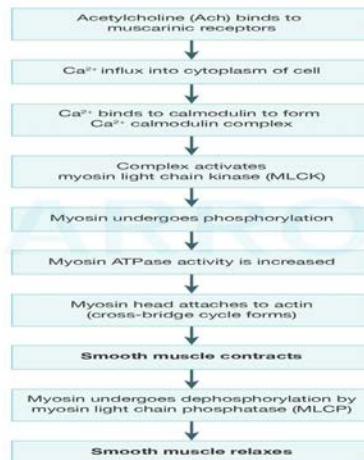
Hollow viscera are primarily composed of single-unit smooth muscles.

Solution to Question 8:

Increased cytosolic Ca^{2+} is essential for contraction of smooth muscles.

Events in Excitation Contraction Coupling in Smooth Muscles.

Smooth muscle contraction and relaxation mechanism



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A unitary smooth muscle is unique in that it can contract when stretched even in the absence of any extrinsic innervation. Stretch is followed by a decrease in membrane potential and an increase in the tone of the muscle.

Other options:

Option a: Troponin is absent in smooth muscle.

Option b: Sustained contraction of the smooth muscle occurs even when the calcium concentration falls. This is mainly because of the latch-bridge mechanism where the myosin cross-bridges remain attached to the actin. This mechanism allows for maintaining the contraction with less expenditure of energy and is particularly important in the vascular smooth muscle.

Option c: Unlike skeletal muscle where troponin binding is essential for contraction, the smooth muscle contraction requires phosphorylation of myosin (not actin).

Solution to Question 9:

Calmodulin is structurally and functionally related to troponin. In place of troponin, smooth muscle cells contain a large amount of the regulatory protein calmodulin.

Calmodulin is similar to troponin; it is different in the manner in which it initiates contraction. Calmodulin initiates contraction by activating the myosin light chain kinase. Troponin, on the other hand, initiates contraction by causing a conformational change in the tropomyosin molecule which then uncovers the active sites on the actin filament.

Solution to Question 10:

- Smooth muscles can maintain prolonged full tonic contraction with little use of energy (as little as 1/300 the energy required for comparable sustained skeletal muscle contraction) and with little

continued excitatory signal due to this mechanism.

- This is due to the fact that even after dephosphorylation of myosin, the cross-bridges continue to cling on for some time.
- The excitation coupling mechanism occurring in smooth muscle is similar to the skeletal muscles but for a few features:
- The process is slow and a low energy mechanism.
- There is no fixed length tension relationship as it shows the property of plasticity (smooth muscles behaving like a viscous mass rather than a rigidly structured tissue).
- This mechanism is specially useful in vascular smooth muscles.

Solution to Question 11:

Actin, myosin, and tropomyosin are present in both skeletal muscle and smooth muscle. But, troponin is absent and calmodulin is present in smooth muscle.

Solution to Question 12:

In response to a stimulus, the efflux of calcium from intracellular stores occurs through an Inositol triphosphate receptor (IP₃R)- Ca⁺² channel.

The source of Ca⁺² in smooth muscle contraction can be different depending on the activating stimulus.

- The influx from the extracellular site through voltage or ligand-gated Ca⁺² channels in the plasma membrane
- Efflux from intracellular stores (eg, sarcoplasmic reticulum) through the RyR and the inositol triosephosphate receptor (IP₃R).

Solution to Question 13:

Sympathetic stimulation (norepinephrine) of a smooth muscle produces inhibitory potentials and relaxation.

Parasympathetic stimulation (or acetylcholine) causes depolarization and contraction in the smooth muscle.

Smooth muscle can be divided into:

- Unitary smooth muscle - has low-resistance gap junctions between individual muscle cells and functions in a syncytial manner
- Multiunit smooth muscle - made up of individual units with few or nil gap junctions.

Blood vessels have both unitary and multiunit smooth muscle in their walls.

Option A: Unitary smooth muscle has an unstable membrane potential with continuous irregular contractions independent of its nerve supply. This state of partial contraction is called tonus or tone.

Option B: Dense bodies are present instead of Z lines in smooth muscle. These are bound to actin filaments by α -actinin.

Option C: Smooth muscle can exert variable tension at any given length. This property is called plasticity. When a unitary smooth muscle is stretched, it first exerts increased tension. However, if it is held at this length after stretching, the tension gradually decreases. This is referred to as plasticity.

Solution to Question 14:

Tetanzing frequency demonstrated by the muscle is 33Hz. In the above graph, we know,

- The speed of the paper is 1000mm/sec.
- The y-axis represents millivolts, the x-axis represents time.
- The scale on the x-axis is 10mm.
- The area under the graph represents the muscle contraction.

Contraction period (latent to maximum force of contraction) includes 3 scales = 30 mm.

As paper speed is 1000mm/sec,

Contraction period = 30mm = $(30/1000) = 30 \text{ msec}$.

Tetanzing frequency = $1/\text{contraction period} = 1000/30 = 33\text{Hz}$.

Tetanzation

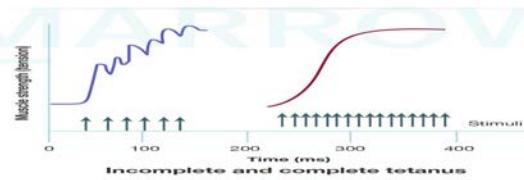
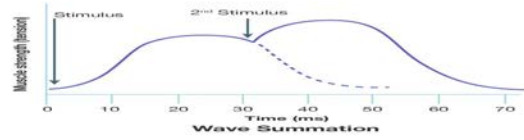
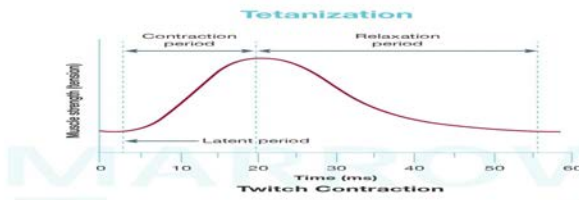
The features of muscle contraction can be demonstrated by eliciting single muscle twitches. The adding together of individual twitch contractions to increase the intensity of overall muscle contraction is Summation. Summation occurs in two ways:

- by increasing the number of motor units contracting simultaneously, which is multiple $\blacksquare$ ber summations.
- by increasing the frequency of contraction, which is called frequency summation and can lead to tetanzation.

When the frequency of a contraction reaches a critical level, the successive contractions eventually fuse together, and the whole muscle contraction appears to be completely smooth and continuous, this is called tetanzation.

Tetanzation occurs in two ways:

- Complete tetanus occurs when no relaxation occurs between stimuli.
- Incomplete tetanus when periods of incomplete relaxation take place between the summated stimuli.



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else,

Synapse and Junctional Transmission

Question 1:

A synapse is a junction between _____.

- a) 1 and 2
- b) 1, 2, 3
- c) All 1, 2, 3, 4
- d) 1, 2, 4

Question 2:

All of the following are true regarding chemical synapses except _____

- a) These are the most common type of synapses
- b) Neurotransmitter is produced by the presynaptic neuron
- c) Protoplasmic continuity between pre- and post- synaptic membrane ensures faster conduction
- d) Conduction is always orthodromic

Question 3:

In a typical chemical synaptic cleft, the extracellular distance between the presynaptic plasma membrane and the postsynaptic plasma membrane is approximately _____

- a) 2-4 nm
- b) 20-40 nm
- c) 200-400 nm
- d) 2-4 μm

Question 4:

Which of the following is true regarding neurotransmitters?

- a) All neurotransmitter are derived from amino acids
- b) Nitric oxide is a neurotransmitter in the CNS

- c) They do not act on ionotropic receptors
- d) They are released on repolarization of the nerve

Question 5:

Which of the following is not a monoamine neurotransmitter?

- a) GABA
- b) Norepinephrine
- c) DOPA
- d) Serotonin

Question 6:

Which of the following is true regarding metabotropic neurotransmitters?

- a) They act through second messenger systems
- b) They bind to cationic channels and increase influx of Na
- c) They bind to anionic channels and increase influx of Cl
- d) They inhibit the fusion of synaptic vesicle with the synaptic cleft

Question 7:

Which of the following is false regarding the neurotransmitter glutamate in the brain?

- a) It is involved in excitatory neurotransmission
- b) It is a minor neurotransmitter in the brain
- c) It acts via both ionotropic and metabotropic receptors
- d) Glycine binding to the NMDA receptor is essential for the receptor to respond to glutamate

Question 8:

Which of the following is a well-known inhibitory brain neurotransmitter?

- a) Glutamate
- b) Acetylcholine
- c) GABA

d) Glycine

Question 9:

Correctly matched with its receptor is _____

- a) Nicotinic – G-Protein Coupled Receptor
- b) GABA-A – ionotropic
- c) GABA-B - Ionotropic
- d) Glycine – G-Protein Coupled Receptor

Question 10:

Which is not true about neuromodulators?

- a) Modify the postsynaptic neuron's response to a specific neurotransmitter
- b) May change the presynaptic neuron's release of a transmitter
- c) May be released by the presynaptic neuron along with the neurotransmitter
- d) Actions are faster than neurotransmitters

Question 11:

A medical student was doing research in a laboratory that studies the neuromuscular junction. Which of the following events at the junction are listed in correct sequential order?

- a) Motor neuron action potential, acetylcholine release, Ca^{2+} entry at the endplate
- b) Ca^{2+} entry into the motor nerve terminal, acetylcholine release, Na^{+} entry at the endplate
- c) Ca^{2+} entry into the motor nerve terminal, Na^{+} entry at the endplate, motor neuron action potential
- d) Na^{+} entry at the endplate, acetylcholine release, muscle fiber action potential

Question 12:

Excitatory postsynaptic potential (EPSP) is due to the influx of _____.

- a) Na^{+}
- b) K^{+}
- c) Mg^{2+}

d) Cl^-

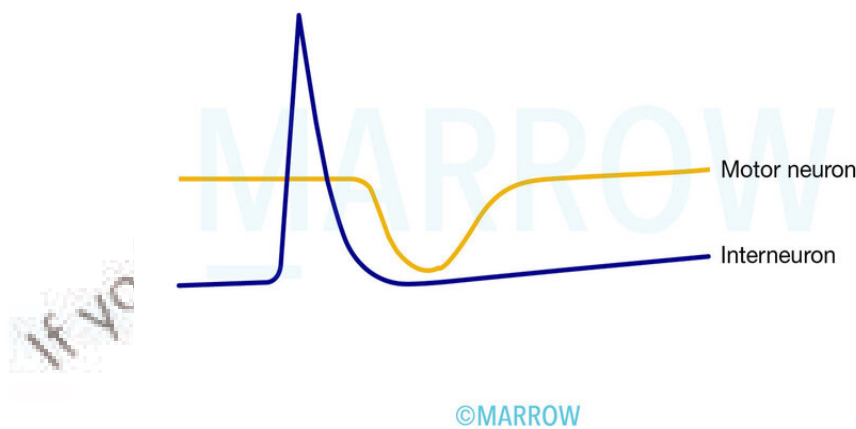
Question 13:

Inhibitory Postsynaptic Potential (IPSP) is due to influx of _____

- a) Na^+
- b) K^+
- c) Ca^{2+}
- d) Cl^-

Question 14:

The following effect in the motor neuron is due to:



- a) Opening of chloride ion channels through the pre-synaptic neuronal membrane
- b) Increase in conductance of potassium ions to the inside of the neuron
- c) Activation of receptor enzymes that increase the number of inhibitory synaptic receptors
- d) Increase in conductance of sodium ions inside the neuron

Question 15:

Slow inhibitory postsynaptic potential (IPSP) is generated by all except _____

- a) Cardiac muscle

- b) Skeletal muscle
- c) Smooth muscle
- d) Autonomic ganglia

Question 16:

Which of the following physiologic processes is not correctly matched?

- a) Presynaptic inhibition : Axo-axonal synapse
- b) Postsynaptic inhibition: GABA produces IPSP
- c) Presynaptic inhibition: GABA-B receptors decrease K^+ conductance.
- d) Presynaptic inhibition: GABA-A receptors increase Cl^- conductance.

Question 17:

Renshaw cell inhibition is _____

- a) Feedback facilitation
- b) Feedback inhibition
- c) Feed forward inhibition
- d) Indirect inhibition

Question 18:

Feedforward inhibition is seen in _____

- a) Medulla
- b) Pituitary gland
- c) Cerebellum
- d) Renshaw cell

Question 19:

Presynaptic facilitation is due to prolonged opening of _____

- a) Na^+ channels
- b) K^+ channels

- c) Ca^{2+} channels
- d) Cl^- channels

Question 20:

A person with myasthenia gravis notes increased muscle strength when he is treated with an acetylcholinesterase (AChE) inhibitor. The basis for his improvement is due to the increase in

- a) The amount of acetylcholine (ACh) released from motor nerves
- b) The level of ACh at the muscle end plates
- c) The number of ACh receptors on the motor end plates
- d) Synthesis of norepinephrine in motor nerves

Question 21:

A 55-year-old woman had an autonomic neuropathy that disrupted the sympathetic nerve supply to the pupillary dilator muscle of her right eye. While having her eyes examined, the ophthalmologist placed a sympathomimetic drug in both her eyes. The right eye became much more dilated than the left eye. This suggests that:

- a) The sympathetic nerve to the right eye had regenerated
- b) The parasympathetic nerve supply to the right eye remained intact and compensated for the loss of the sympathetic nerve
- c) Denervation supersensitivity had developed
- d) The left eye also had nerve damage and so was not responding as expected

Question 22:

Which of the following statements is not true about neurotransmitters?

- a) They must be present at the pre-synaptic membrane
- b) Neurons must be able to synthesize the substance
- c) They are released on repolarization of the nerve
- d) Specific receptors must be present on the post-synaptic membrane

Question 23:

Neurotransmitter can be degraded by all except_____

- a) Can diffuse out of synaptic cleft
- b) Enzymes inactivate it
- c) Reuptake into glial cell
- d) Autoreceptor

Question 24:

Which of the following endogenous opioids is correctly matched with the receptor at which it has maximum potency?

- a) Enkephalins - δ receptors
- b) Dynorphins - μ receptors
- c) Endorphins - κ receptors
- d) Both B and C

Question 25:

Which of the following physiologic processes is not correctly paired with a structure?

- a) Electrical transmission : gap junction
- b) Negative feedback inhibition : Renshaw cell
- c) Synaptic vesicle docking and fusion : presynaptic nerve terminal
- d) Endplate potential : muscarinic cholinergic receptor

Question 26:

Which of the following electrophysiologic events is correctly paired with the change in ionic currents causing the event?

- a) Fast inhibitory postsynaptic potentials (IPSPs) and closing of Cl^- channels
- b) Fast excitatory postsynaptic potentials (EPSPs) and an decrease in Ca^{2+} conductance
- c) Endplate potential and an increase in Na^+ conductance
- d) Presynaptic inhibition and closure of voltage-gated K^+ channels

Question 27:

How is an electrical synapse different from a chemical synapse?

- a) No cytoplasmic continuity between presynaptic and postsynaptic membrane
- b) Direction of impulse is unidirectional
- c) Connexons are involved in electrical synapse
- d) Electrical synapse has more latent period than chemical synapse

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	b
5	a
6	a
7	b
8	c
9	b
10	d
11	b
12	a
13	d
14	c
15	b
16	c
17	b
18	c
19	c
20	b
21	c
22	c
23	d
24	a
25	d

26	c
27	c

Detailed Explanations

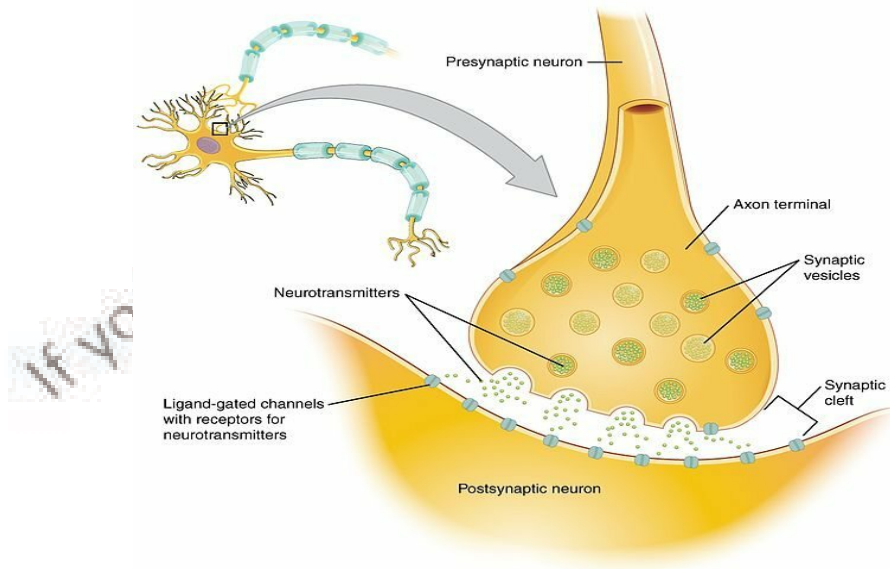
Solution to Question 1:

Impulses are transmitted from one nerve cell to another cell at junctions called synapse. These are the junctions where the axon or some other portion of one cell (the presynaptic cell) terminates on the dendrites, soma, or axon of another neuron or, in some cases, a muscle or gland cell (the postsynaptic cell).

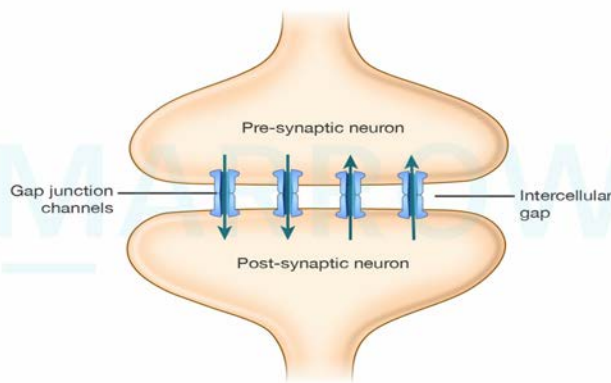
There are two major types of synapses:

- Chemical
- Electrical

When a neuron terminates on a muscle, it is called a neuromuscular junction. This is an example of a chemical synapse.



Electrical synapse



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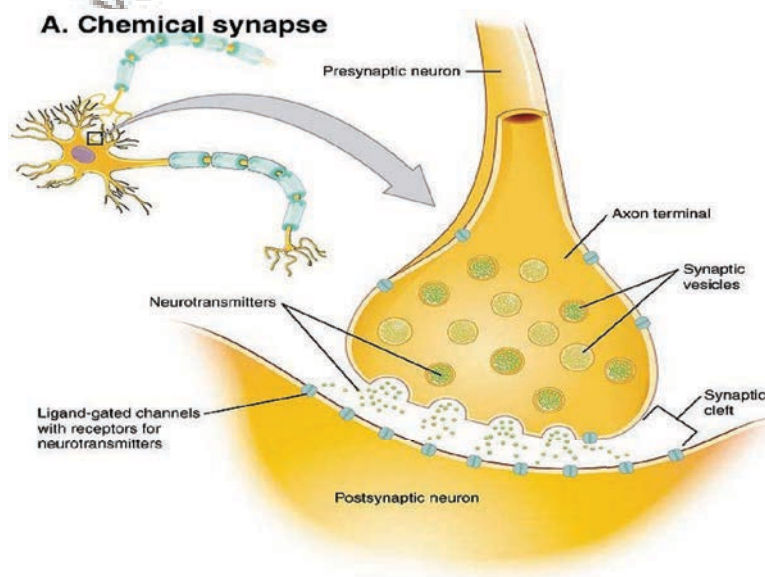
Solution to Question 2:

Chemical synapses are interneuronal junctions without protoplasmic continuity.

Most of the synapses used for signal transmission in the central nervous system of the human being are chemical synapses.

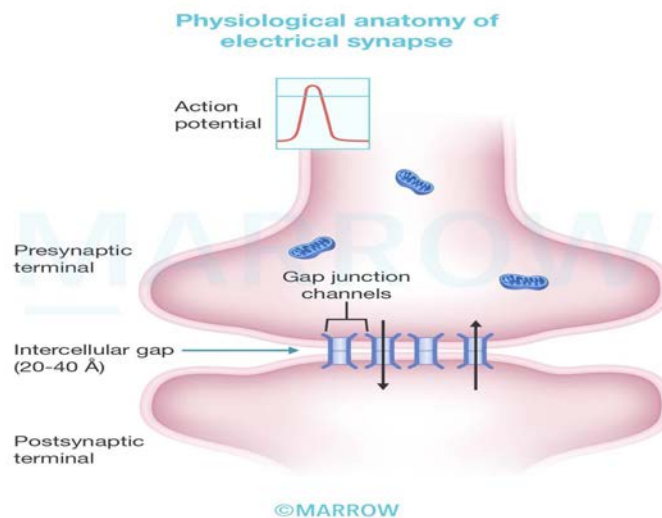
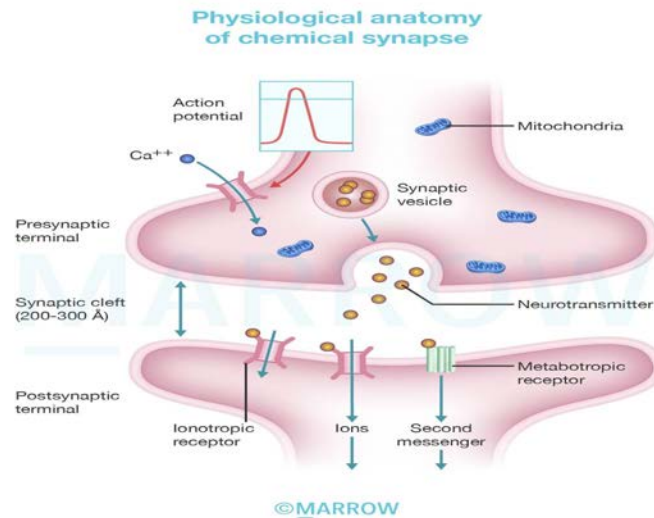
In these synapses, the first neuron (presynaptic neuron) secretes neurotransmitters at its nerve ending synapse. This transmitter, in turn, acts on receptor proteins in the membrane of the next neuron (post-synaptic neuron) to excite the neuron, inhibit it, or modify its sensitivity in some other way.

As neurotransmitters are released only by the pre-synaptic neuron, the synaptic conduction in the CNS is only orthodromic (unidirectional).



Solution to Question 3:

In a typical chemical synaptic cleft, the extracellular distance between the presynaptic plasma membrane and the postsynaptic plasma membrane is approximately 20-40nm.



Solution to Question 4:

Nitric oxide is a neurotransmitter in the CNS.

Neurotransmitters are the substances that mediate chemical signaling between neurons. They can be choline esters, biogenic amines, amino acids, and neuropeptides.

Two main classes of chemical substances that serve as neurotransmitters:

- small-molecule neurotransmitter (acetylcholine)
- large molecule neurotransmitter (enkephalin)

They can act on ionotropic receptors (ligand-gated ion channels) and/or metabotropic receptors on the postsynaptic membrane to produce a change in membrane potential to bring about their action.

Criteria for the substance to be neurotransmitter:

- the substance must be demonstrated to be present in the presynaptic terminal.
- the cell must be able to synthesize the substance.
- should be released on depolarization of the terminal.
- there should be specific receptors for it on the postsynaptic membrane.
- if the substance is applied exogenously to the postsynaptic membrane at physiologic concentration, the response of the postsynaptic cell must mimic the in vivo response.

Solution to Question 5:

GABA is not a monoamine neurotransmitter. It is an amino acid.

Biogenic amines are small, charged molecules that are synthesized from amino acids and contain an amino group ($R-NH_2$). Amines known to act as neurotransmitters are dopamine, norepinephrine (noradrenaline), epinephrine (adrenaline), serotonin (5-hydroxytryptamine [5-HT]).

Amino acids and their derived amines acting as neurotransmitters:

- Tyrosine - catecholamines : dopamine, norepinephrine, and epinephrine .
- Tryptophan - serotonin
- Histidine - histamine

Solution to Question 6:

Neurotransmitter receptors that act through second messenger systems are called metabotropic receptors.

Neurotransmitters bind to receptors on the post-synaptic membrane to activate ion channels in one of the two ways:

- by gating ion channels directly (ionotropic receptors).
- by activating a second messenger (metabotropic receptor) for a prolonged postsynaptic neuronal excitation or inhibition. Most commonly activated is a G-protein coupled membrane receptor.

The following four changes can occur usually after binding to a G-protein coupled receptor:

- Opening specific ion channels through the post-synaptic cell membrane.

- Activation of cyclic adenosine monophosphate (cAMP) or cyclic guanosine monophosphate (cGMP) in the neuronal cell.
- Activation of one or more intracellular enzymes.
- Activation of gene transcription. Activation of gene transcription is one of the most important effects of activation of the second messenger systems because gene transcription can cause formation of new proteins within the neuron, thereby changing its metabolic machinery or its structure.

Solution to Question 7:

Glutamate is a major excitatory neurotransmitter responsible for ~75% excitatory neurotransmission in the CNS.

Glutamate is the best-known example of excitatory amino acid neurotransmitters. It brings about its action via both ionotropic and metabotropic receptors.

Glutamate Receptors :

Type	Mechanism of Action
Ionotropic Glutamate Receptors	
AMPA, Kainate receptors	Primarily permits influx of Na^+ and efflux of K^+ → EPSP (excitatory postsynaptic potential)
NMDA receptors	Permits ↑ influx of Ca^{2+} along with Na^+ → EPSP (excitatory postsynaptic potential) (Glycine binding to the NMDA receptor is essential for the receptor to respond to glutamate)
Metabotropic Glutamate Receptor (mGluR)	Leads to either ↑ in IP_3/DAG or ↓ cAMP

Solution to Question 8:

GABA (gamma-amino-butyric acid) is probably the best known inhibitory neurotransmitter in the brain, especially in the forebrain regions.

GABA mediates both presynaptic and postsynaptic inhibition

GABA Receptors :

Other options :

- Glutamate is a good example of an excitatory brain neurotransmitter.

- Acetylcholine and glycine have been known to play either excitatory or inhibitory neurotransmitter roles depending on the particular brain circuits under study.

Note:

Glycine sometimes potentiates the excitatory action of glutamate at NMDA receptors. So, it might act as an excitatory neurotransmitter as well.

Type	Mechanism of Action
GABAA and GABAC	Ionotropic receptors Activation $\rightarrow$ $\uparrow$ Cl ⁻ influx into the neurons $\rightarrow$ IPSP (inhibitory postsynaptic potential) Benzodiazepines (diazepam), barbiturates and phenobarbital enhance/potentiate the effect of GABA _A receptors (neuromodulation)
GABAB	Metabotropic receptors G-protein coupled receptors G _i $\rightarrow$ inhibits adenylyl cyclase to open K ⁺ channels G _o $\rightarrow$ inhibits or delays Ca ²⁺ influx Net effect = IPSP

Solution to Question 9:

GABA_A is an ionotropic receptor.

GABA is the major inhibitory mediator both pre and postsynaptically in the brain.

GABA receptors :

Type	Mechanism of Action
GABAA and GABAC	Ionotropic receptors Activation $\rightarrow$ $\uparrow$ Cl ⁻ influx into the neurons $\rightarrow$ IPSP (inhibitory postsynaptic potential) Benzodiazepines (eg. Diazepam), barbiturates and phenobarbital enhance/potentiate the effect of GABA _A receptors (neuromodulation)
GABAB	Metabotropic receptors G-protein coupled receptors G _i $\rightarrow$ inhibits adenylyl cyclase to open K ⁺ channels G _o $\rightarrow$ inhibits or delays Ca ²⁺ influx Net effect = IPSP

Solution to Question 10:

Neuromodulators typically have actions that are slower than neurotransmitters.

Neuromodulators are chemicals released by neurons that have little or no direct effects on their own but can modify the effects of neurotransmitters.

They can modify the postsynaptic neuron's response to a neurotransmitter and they can change how much of a neurotransmitter the presynaptic neuron synthesizes or releases.

Neuromodulators also can be released by the presynaptic neuron along with the neurotransmitter.

Solution to Question 11:

Events Occurring at the Chemical Synapse:

The action potential arriving at the end of the motor neuron increases the permeability of its endings to Ca^{2+}



Ca^{2+} enters the endings and triggers a marked increase in exocytosis of the acetylcholine-containing synaptic vesicles



Binding of acetylcholine to nicotinic receptors increases the Na^{+} and K^{+} conductance



Resultant influx of Na^{+} produces a local depolarizing potential, the endplate potential

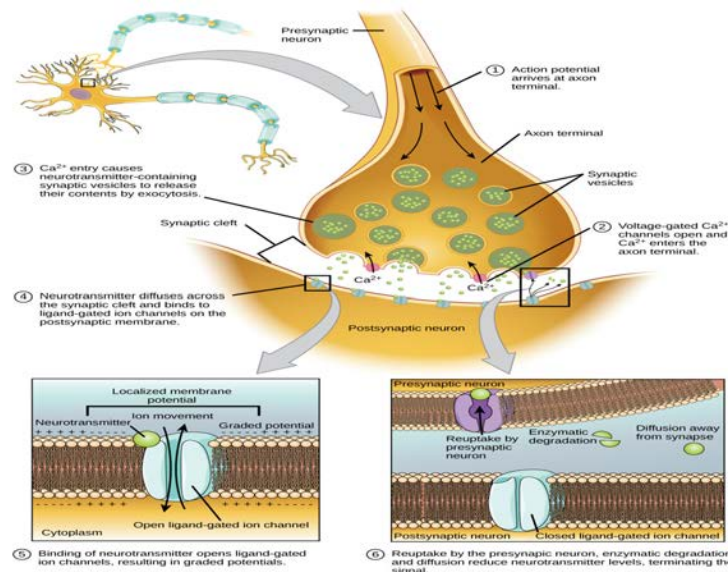


The current created by this local potential depolarizes the adjacent muscle membrane to its firing level



Action potentials are generated on either side of the endplate and are conducted away from the endplate in both directions along the muscle fiber. This muscle action potential, in turn, initiates muscle contraction.

Acetylcholine is then removed from the synaptic cleft by acetylcholinesterase, which is present in high concentration at the neuromuscular junction.



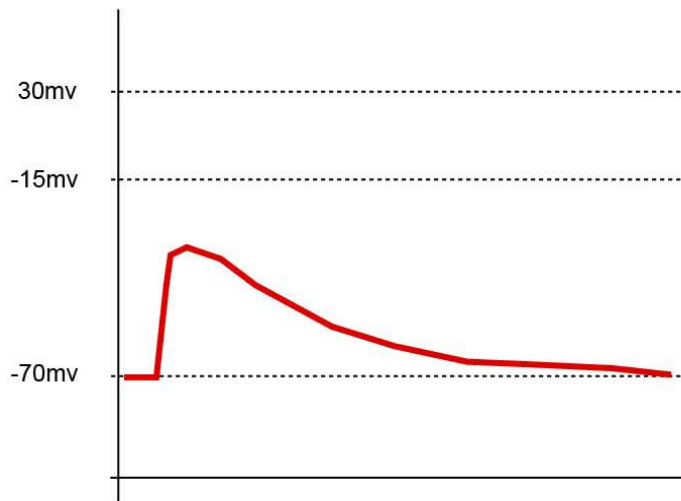
Solution to Question 12:

Excitatory postsynaptic potential (EPSP) is due to the influx of sodium ions (Na^+).

At the post-synaptic membrane, neurotransmitters binding to receptors generate responses that can be excitatory or inhibitory. If it generates an excitatory response (makes the membrane more electrically positive/ depolarized), it is an excitatory postsynaptic potential (EPSP) because it makes the cell more likely to reach an action potential.

EPSP may be due to

- The opening of sodium channels or calcium channels to allow large numbers of positive electrical charges to flow to the interior of the postsynaptic cell.
- Depressed conduction through chloride or potassium channels or both.
- Various changes in the internal metabolism of the postsynaptic neuron to increase the number of excitatory membrane receptors or decrease the number of inhibitory membrane receptors.



As seen in the figure, following the discharge of the sensory neuron, there is positivity in the motor neuron signifying an EPSP.

Solution to Question 13:

Inhibitory Postsynaptic Potential (IPSP) is due to the influx of chloride ions (Cl^-).

If, binding of a neurotransmitter causes the postsynaptic membrane to become more negative (hyperpolarized), it is called an inhibitory postsynaptic potential (IPSP) because hyperpolarization moves the membrane potential farther from the threshold and makes the cell less likely to reach an action potential.

IPSP may be due to :

- Increase in Cl^- influx. This action causes diffusion of negatively charged chloride ions from outside the postsynaptic neuron to the inside, thereby carrying negative charges inward and increasing the negativity inside. This leads to hyperpolarization.
- Increase in K^+ efflux. This action causes positive ions to diffuse to the exterior, which causes increased negativity inside the neuron, which leads to hyperpolarization.
- Activation of receptor enzymes that increase the number of inhibitory synaptic receptors or decrease the number of excitatory receptors.

Solution to Question 14:

Activation of receptor enzymes that increase the number of inhibitory synaptic receptors can cause the effect shown, namely, the generation of IPSP.

In the figure given, there is an increased negativity in the motor neuron following the action potential in the interneuron. So, the synaptic potential is IPSP. This can be caused at the

postsynaptic membrane due to Cl^- influx, K^+ efflux or activation of receptor enzymes that increase the number of inhibitory synaptic receptors or decrease the number of excitatory receptors.

Other options:

- Opening of chloride ion channels through the post-synaptic neuronal membrane.
- Increase in conductance of potassium ions to the outside of the postsynaptic neuron.
- Increase in conductance of sodium ions inside the neuron causes the generation of an EPSP.

Solution to Question 15:

Slow EPSPs and IPSPs are not generated by skeletal muscles.

Slow excitatory postsynaptic potentials (EPSPs) and inhibitory postsynaptic potential (IPSPs) are seen in autonomic ganglia, cardiac, and smooth muscle, and cortical neurons.

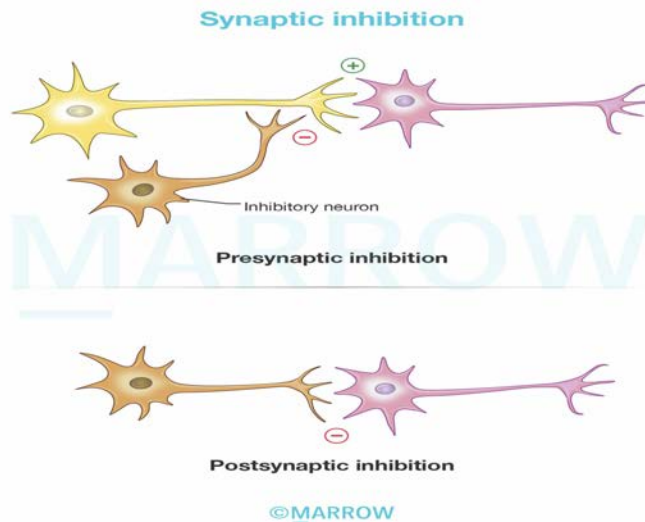
EPSPs and IPSPs occur after a latency of 0.5ms while slow EPSPs and IPSPs occur after a latency of 100–500 ms and last for several seconds. The slow EPSPs are due to a decrease in K^+ efflux, and the slow IPSPs are due to an increase in K^+ efflux.

Solution to Question 16:

GABA-B receptors are present in the spinal cord and appear to mediate presynaptic inhibition via a G-protein that produces an increase in K^+ conductance.

PRE-SYNAPTIC INHIBITION

- Presynaptic inhibition is a process mediated by neurons whose terminals are on excitatory endings, forming axoaxonal synapses.
- The first transmitter shown to produce presynaptic inhibition was GABA. On the pre-synaptic membrane
- Acting via GABA-A receptors, GABA increases Cl^- conductance.
- GABA-B receptors appear to mediate presynaptic inhibition via a G-protein that produces an increase in K^+ conductance.
- This leads to hyperpolarization of the pre-synaptic membrane and inhibition of release of the neurotransmitter from the pre-synaptic neuron.



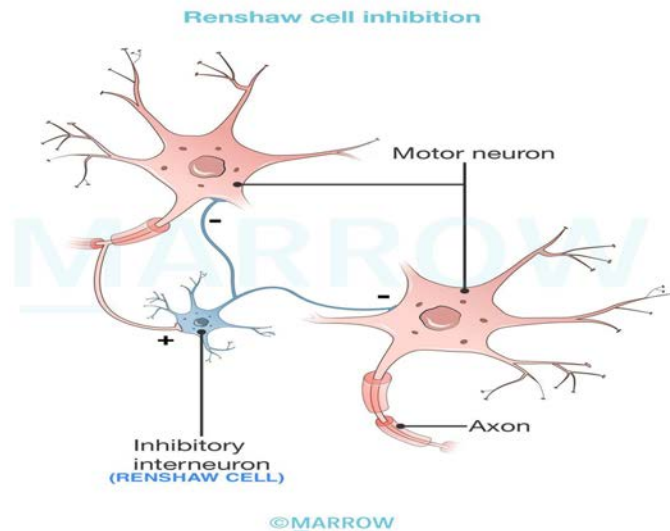
POST-SYNAPTIC INHIBITION

- Postsynaptic inhibition occurs when an inhibitory transmitter such as glycine or GABA is released from a presynaptic nerve terminal onto the postsynaptic neuron to induce an IPSP in the postsynaptic neuron.
- In the figure shown below, an inhibitory transmitter such as GABA is released from the nerve terminals of an inhibitory interneuron (dark) that synapses on a postsynaptic neuron.

Solution to Question 17:

Renshaw cell inhibition is an example of feedback inhibition.

The axon of a spinal motor neuron has a recurrent collateral that synapses on an inhibitory interneuron that terminates on the cell body of the same and other motor neurons. The inhibitory interneuron is called a Renshaw cell and its neurotransmitter is glycine.

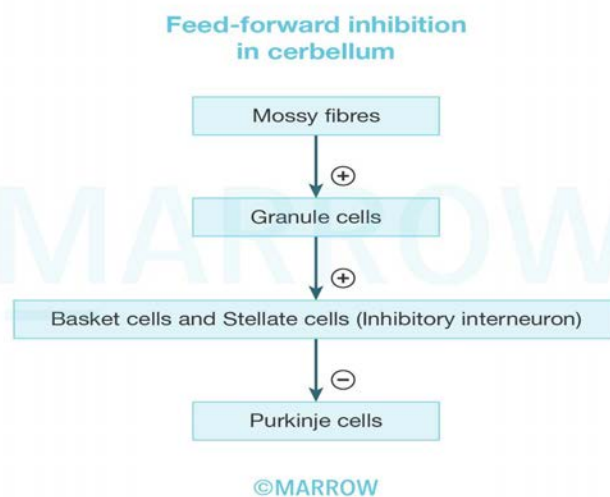


Solution to Question 18:

Feedforward inhibition is seen in the cerebellum. In the cerebellum, stimulation of basket cells / stellate cells produces IPSPs in the Purkinje cells (i.e, inhibits Purkinje cells)

In feedforward inhibition, a presynaptic cell excites an inhibitory interneuron which then inhibits the next follower cell.

In cerebellum (Refer flow chart):



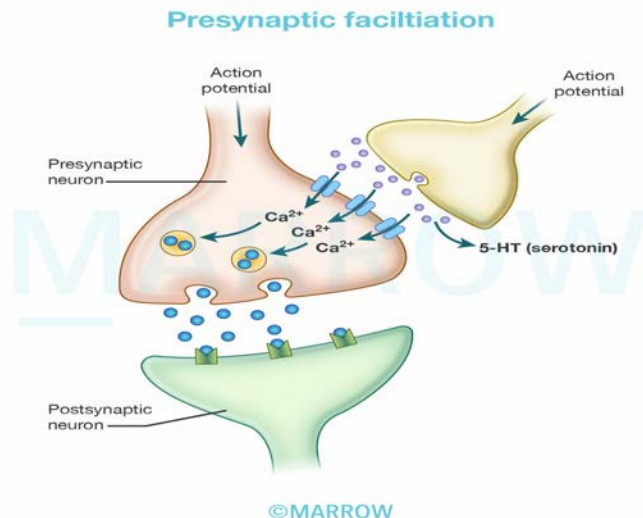
This arrangement is called feed-forward inhibition which presumably limits the duration of excitation produced by any given afferent volley.

Note: Renshaw cell carries out the negative feedback inhibition of a spinal motor neuron.

Solution to Question 19:

Presynaptic facilitation is due to prolonged opening of Ca^{2+} channels .

In presynaptic facilitation, the serotonin released at the axoaxonal ending increases the intraneuronal cAMP levels → phosphorylation and consequent closure of K^{+} channels → slowing of repolarization and prolonging the depolarization by prolonged opening of Ca^{2+} channels



Solution to Question 20:

Treatment with AChE inhibitors (e.g., neostigmine) prevents the degradation of ACh and prolongs the action of ACh at the muscle end plate.

Myasthenia gravis:

- It is caused by the presence of antibodies to the ACh receptor.
- Is characterized by skeletal muscle weakness and fatigability resulting from a reduced number of ACh receptors on the muscle end plate.
- The size of the EPP (end plate potential) is reduced.
- Treatment with AChE inhibitors (e.g., neostigmine) prevents the degradation of ACh and prolongs the action of ACh at the muscle end plate, partially compensating for the reduced number of receptors.

Solution to Question 21:

Denervation supersensitivity develops in the muscle after the nerve supplying it is cut. Since the nerve to her right dilator pupillae is damaged, the right eye becomes excessively sensitive to the sympathomimetic drug. So, the right eye becomes much more dilated than the left.

Denervation Supersensitivity:

- When the motor nerve to skeletal muscle is degenerated, the muscle gradually becomes extremely sensitive to acetylcholine. This is called denervation hypersensitivity or supersensitivity.
- Normally nicotinic receptors are located only in the vicinity of the motor endplate where the axon of the motor nerve terminates. When the motor nerve is severed, there is a marked proliferation of nicotinic receptors over a wide region of the neuromuscular junction.
- Denervation supersensitivity also occurs at autonomic junctions. Smooth muscle, unlike skeletal muscle, does not atrophy when denervated, but it becomes hyperresponsive to the chemical mediator that normally activates it.
- This hyperresponsiveness can be demonstrated by using pharmacologic tools rather than actual nerve section. Prolonged use of a drug such as reserpine can be used to deplete transmitter stores and prevent the target organ from being exposed to norepinephrine for an extended period. Once the drug usage is stopped, smooth muscle and cardiac muscle will be supersensitive to subsequent release of the neurotransmitter.

Option a) If the sympathetic nerve would have regenerated, both right and left eye would have been equally sensitive to the drug and so, would have dilated equally.

Option b) Parasympathetic nerves supply the sphincter pupillae muscle. So, if stimulated they would constrict the eye not dilate it. Also, the drug added was a sympathomimetic agent. So, parasympathetic nerves are not stimulated in the first place

Option d) If the left eye was damaged, it would also have denervation supersensitivity. Again, both eyes would dilate equally, but more than that of normal.

Solution to Question 22:

Neurotransmitters should be released on depolarization of the terminal.

- Neurotransmitters are the substances that mediate chemical signaling between neurons.
- The transmission of information at chemical synapses involves the release of neurotransmitter from a presynaptic cell, diffusion across the synaptic cleft, and binding of the neurotransmitter to specific receptors on the postsynaptic membrane to produce a change in membrane potential.

Criteria for the substance to be neurotransmitter

- the substance must be demonstrated to be present in the presynaptic terminal
- the cell must be able to synthesize the substance.
- should be released on depolarization of the terminal.
- there should be specific receptors for it on the postsynaptic membrane.
- if the substance is applied exogenously to the postsynaptic membrane at physiologic concentration, the response of the postsynaptic cell must mimic the in vivo response

Solution to Question 23:

The final steps in the process lead to termination of the actions of the neurotransmitter and include

- Reuptake into the nerve terminal or glial cell
- Enzymatic degradation.
- Diffusion away from the synaptic cleft

Solution to Question 24:

Enkephalins have maximum potency at δ receptors.

Dynorphins have maximum potency at κ receptors.

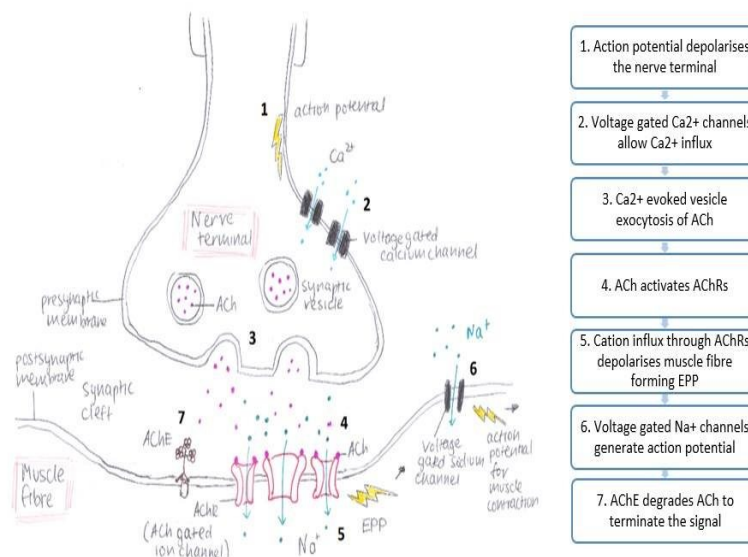
Endorphins have maximum potency at μ receptors.

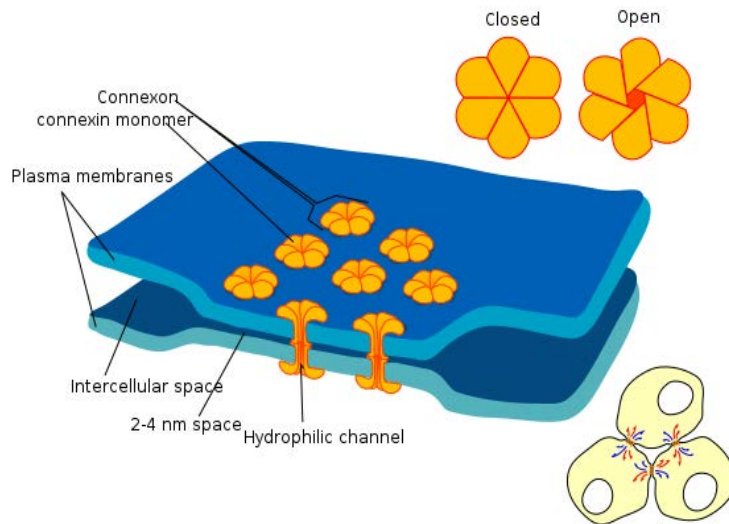
Note: There is a difference of opinion among Goodman & Gillman's textbook and Ganong's review of medical physiology (26th edition pg 151). According to Goodman & Gillman's textbook above answer is correct.

Solution to Question 25:

Endplate potential is generated via nicotinic cholinergic receptor.

The acetylcholine released from the motor neuron diffuses to nicotinic cholinergic (NM) receptors that are concentrated at the tops of the junctional folds of the membrane of the motor endplate.





Solution to Question 26:

Binding of acetylcholine to neuromuscular junction receptors increases the Na^+ and K^+ conductance, and the resultant influx of Na^+ produces a depolarizing potential, the endplate potential.

IPSP may be due to

- Opening of chloride ion channels through the postsynaptic neuronal membrane
- Increase in conductance of potassium ions out of the neuron.
- Activation of receptor enzymes that increase the number of inhibitory synaptic receptors or decrease the number of excitatory receptors.

EPSP may be due to

- The opening of sodium channels to allow large numbers of positive electrical charges to flow to the interior of the postsynaptic cell.
- Depressed conduction through chloride or potassium channels, or both.
- Various changes in the internal metabolism of the postsynaptic neuron to increase the number of excitatory membrane receptors or decrease the number of inhibitory membrane receptors.

Solution to Question 27:

In electrical synapses, the cytoplasm of adjacent cells is directly connected by gap junctions (connexons) that allow free movement of ions from the interior of one cell to the interior of the next cell.

The transmission in electrical synapses is bidirectional and there is no synaptic delay. Unlike chemical synapses, electrical synapse has cytoplasmic continuity.

Note:

Most of the synapses used for signal transmission in the central nervous system are chemical synapses.

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Neurotransmitters

Question 1:

Which of the following neurotransmitter is not a biogenic amine?

- a) Norepinephrine
- b) Dopamine
- c) Serotonin
- d) Glycine

Question 2:

The main excitatory and inhibitory neurotransmitters in the CNS are _____

- a) Glutamate and glycine respectively
- b) Aspartate and glycine respectively
- c) Norepinephrine and GABA respectively
- d) Dopamine and GABA respectively

Question 3:

'A' delta fibers and 'C' fibers carry pain sensation. The neurotransmitter released by 'A' delta fibers is _____

- a) Substance P
- b) Glutamate
- c) Histamine
- d) Norepinephrine

Question 4:

The most important neurotransmitter of reward pathway is _____

- a) Dopamine
- b) Serotonin

- c) Noradrenaline
- d) Acetylcholine

Question 5:

Dopamine beta hydroxylase converts dopamine to _____

- a) Norepinephrine
- b) DOPA
- c) Epinephrine
- d) Tyrosine

Question 6:

Which of these is not a dopaminergic pathway?

- a) Mesocortical pathway
- b) Nigrostriatal pathway
- c) Tuberoinfundibular pathway
- d) Corticostriatal pathway

Question 7:

Find the incorrect match.

- a) Raphe nucleus –Serotonin
- b) Locus coeruleus - Norepinephrine
- c) Basal forebrain complex - GABA
- d) Histamine - hypothalamus

Question 8:

A patient with involuntary jerking and writhing movements is diagnosed with Huntington's disease. Loss of which of the following is responsible for hyperkinetic symptoms in this condition?

- a) Nigrostriatal dopaminergic system

- b) Intraatrial cholinergic pathway
- c) Glutamate and cholinergic system
- d) Intraatrial GABA-ergic and cholinergic system

Question 9:

An infant develops features of botulism following honey intake. This toxin blocks neuromuscular transmission by which of the following mechanisms?

- a) Prevents release of Ach at presynaptic membrane
- b) Opening of calcium channels at the presynaptic membrane
- c) Prevents the binding of Ach at the postsynaptic membrane
- d) Prevents the action of Ach at the postsynaptic membrane

Question 10:

A farmer presents with difficulty swallowing and jaw tightening. He gives a history of getting his finger jammed in the rustic gate a few days ago. He gives no history of previous tetanus immunization. After evaluation, he was diagnosed with tetanus. The release of which of the following neurotransmitters is affected by the toxin responsible for this condition?

- a) Dopamine
- b) Acetylcholine
- c) GABA
- d) Adrenaline

Question 11:

The neurotransmitter released by Renshaw cell interneuron is _____.

- a) Glutamate
- b) Norepinephrine
- c) Acetyl choline
- d) Glycine

Question 12:

Which of the following neurotransmitters is increased during REM sleep?

- a) Acetylcholine
- b) Dopamine
- c) Norepinephrine
- d) Serotonin

Question 13:

Which of the following is the wrong match?

- a) Sympathetic preganglionic fibres - release acetylcholine
- b) Parasympathetic preganglionic fibres - release acetylcholine
- c) Sympathetic post ganglionic fibres - release noradrenaline
- d) Sweat glands - innervated by sympathetic adrenergic fibres

Question 14:

Which part of the basal ganglia is the neurotransmitter glutamate secreted from?

- a) Globus pallidus interna
- b) Globus pallidus externa
- c) Subthalamic nucleus
- d) Putamen

Question 15:

All of the following are excitatory neurotransmitters except:

- a) Acetylcholine
- b) Glutamate
- c) Serotonin
- d) Norepinephrine

Question 16:

Which of the following is not true about serotonin receptor?

- a) 5HT_{2A} receptors mediate platelet aggregation
- b) 5-HT_{2C} receptor agonists increase appetite
- c) 5HT₃ receptors are related to vomiting
- d) 5HT₄ receptors facilitate secretion and peristalsis

Question 17:

Serotonin is present in highest concentration in _____

- a) Basophil
- b) Mast cell
- c) Platelets
- d) Posterior hypothalamus

Question 18:

Which statement regarding the NMDA receptor is incorrect?

- a) Glutamate is its endogenous agonist
- b) Glycine is essential for this receptor's response to its endogenous agonist
- c) It is a chloride ion channel
- d) It is present in hippocampal neurons

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	a
5	a
6	d
7	c
8	d
9	a

10	c
11	d
12	a
13	d
14	c
15	c
16	b
17	c
18	c

Detailed Explanations

Solution to Question 1:

Glycine is not a biogenic amine. It is an amino acid.

Biogenic amines are made from amino acid precursors. They have a nitrogen containing amine (NH₂) group. Unlike the amino acids, the biogenic amines lack a carboxyl (COOH) group.

Norepinephrine, epinephrine, dopamine, serotonin and histamine are biogenic amines.

Solution to Question 2:

Glutamate is the main excitatory neurotransmitter in the CNS. It is responsible for 75% of the excitatory transmission in the brain.

Glycine and GABA are the main inhibitory neurotransmitters in the CNS. Glycine is the main inhibitory neurotransmitter in the spinal cord. GABA is the major inhibitory neurotransmitter in the brain.

Solution to Question 3:

The neurotransmitter released by A delta fibers, which carry sharp pricking pain, is glutamate.

The substance released by C fibers, which carry diffuse, chronic, burning pain is substance P.

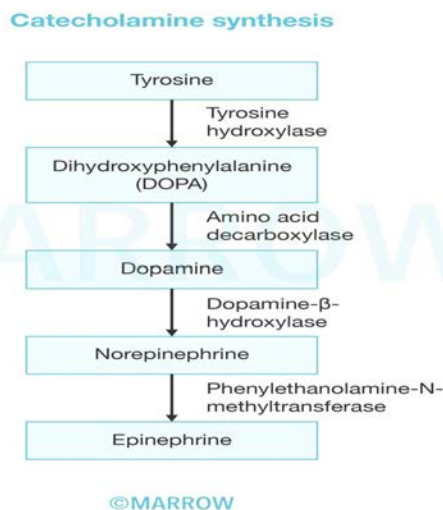
Solution to Question 4:

The most important neurotransmitter of reward pathway is dopamine.

Reward pathway (mesolimbic pathway) connects the ventral tegmental area in the midbrain to the nucleus accumbens and limbic subcortical areas. This pathway is involved in reward behaviour, addiction and in psychiatric disorders such as schizophrenia.

Solution to Question 5:

Dopamine beta hydroxylase converts dopamine to norepinephrine. Vitamin C is required as a cofactor for the enzyme.



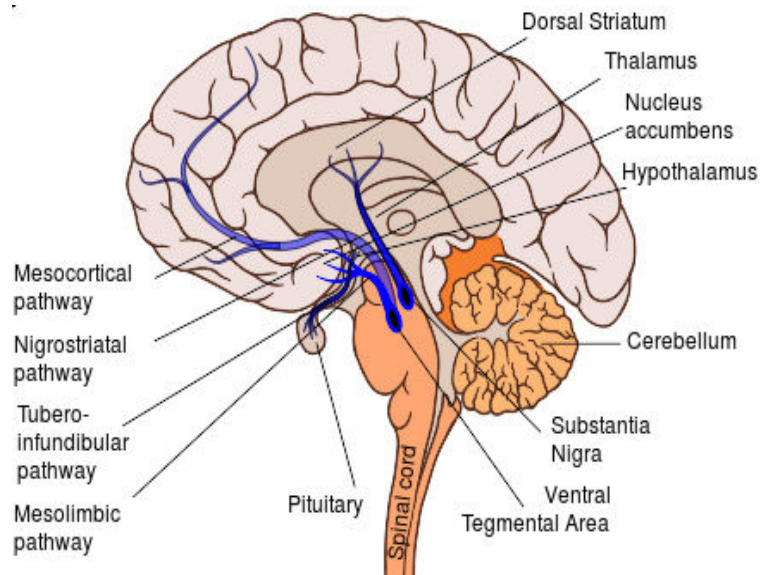
Solution to Question 6:

Corticostratial pathway is not a dopaminergic pathway.

The corticostratial pathway is the primary input pathway for the basal ganglia which uses glutamate as the neurotransmitter.

Important dopaminergic pathways are:

- Mesolimbic pathway - Increased activity responsible for positive symptoms of schizophrenia.
- Mesocortical pathway - Decreased activity responsible for negative symptoms of schizophrenia.
- Nigrostriatal pathway - Decreased activity causes extrapyramidal symptoms.
- Tuberoinfundibular pathway - Decreased activity causes increase in prolactin.



Solution to Question 7:

Basal forebrain complex (nucleus basalis of Meynert) is rich in acetylcholine.

A loss of cholinergic neurons in the nucleus basalis of Meynert is seen in Alzheimer's disease.

Solution to Question 8:

The hyperkinetic feature of Huntington's disease is due to the loss of intrastriatal GABAergic and cholinergic neurons. The loss of the GABAergic pathway (which is an inhibitory pathway) to the globus pallidus external segment and loss of cholinergic neurons in many parts of the brain decreases inhibition, causing the hyperkinetic feature of the disease to develop.

Huntington's disease is an autosomal dominant trinucleotide repeat disorder with multiple CAG repeats. The abnormal gene responsible for the disease is located on chromosome 4.

Solution to Question 9:

Botulinum toxin prevents the release of ACh at the presynaptic membrane.

Botulinum toxins A, B, and E are toxic to humans. They cleave components of SNARE - a complex of proteins necessary for the exocytosis of acetylcholine. Toxins A and E cleave SNAP-25 whereas toxin B cleaves vesicle-associated membrane protein (VAMP) also known as synaptobrevin.

Cleavage of SNARE proteins prevents the release of acetylcholine as the vesicles containing ACh are unable to bind to the presynaptic membrane.

Solution to Question 10:

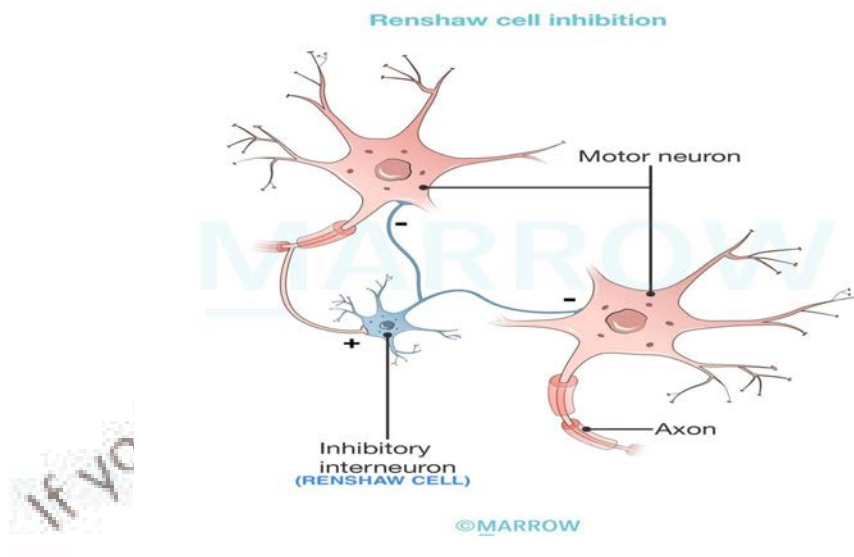
Tetanus toxin affects the release of GABA.

Tetanus toxin escapes normal lysosomal degradation and crosses the synapse to the GABA-ergic presynaptic inhibitory interneuron terminals. The toxin blocks the release of inhibitory neurotransmitters (glycine and gamma-aminobutyric acid) across the synaptic cleft. This blockage of the inhibitory neurotransmitters causes spastic paralysis.

Solution to Question 11:

The neurotransmitter released by Renshaw cell interneuron is glycine.

Renshaw cell is an inhibitory interneuron which is activated by collaterals from spinal motor neurons. Renshaw cell terminates on the cell body of the same and other spinal motor neurons. It reduces or stops the discharge of the motor neuron by releasing glycine.



Solution to Question 12:

Acetylcholine increases during REM sleep.

In REM sleep, there is increase in acetylcholine and decrease in norepinephrine and serotonin. The reverse of this pattern leads to wakefulness.

Neurotransmitters in REM sleep	Levels of neurotransmitter
Acetylcholine	Increased
Serotonin	Decreased
Norepinephrine	Decreased

Solution to Question 13:

Sweat glands are innervated by sympathetic cholinergic fibres (not adrenergic fibres).

Pre-ganglionic fibres of the sympathetic nervous system, pre-ganglionic fibers of the parasympathetic nervous system and post-ganglionic parasympathetic fibres release acetylcholine.

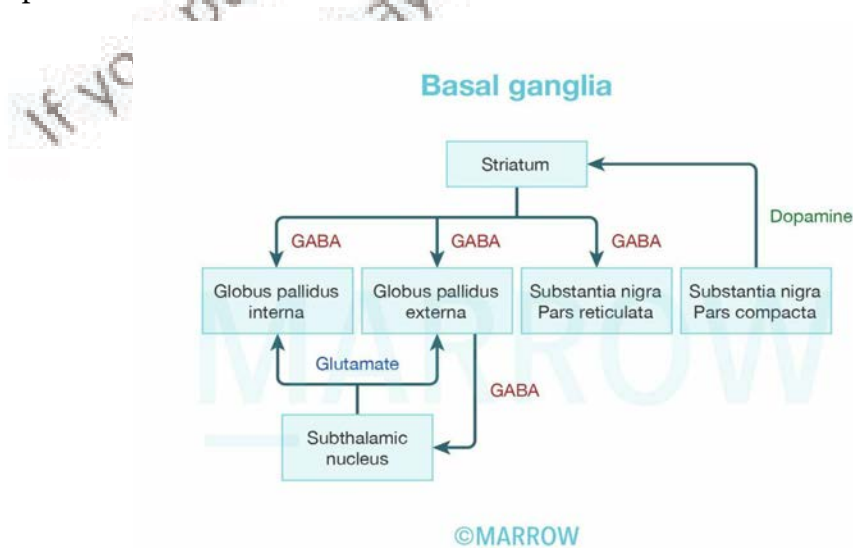
Post-ganglionic sympathetic fibres release noradrenaline in most of the sites, except in certain places such as sweat glands, piloerector muscles and blood vessels in some skeletal muscles where acetylcholine is released.

Solution to Question 14:

Projections from subthalamic nucleus to globus pallidus are glutaminergic.

Neurotransmitters secreted by various regions of basal ganglia:

- Substantia nigra pars compacta = dopamine
- Substantia nigra pars reticulata = GABA
- Subthalamic nuclei = glutamate
- Corpus striatum to globus pallidus = GABA
- Intrastriate neurons = acetylcholine
- Globus pallidus = GABA



Diagrammatic representation of the principal connections of the basal ganglia.

Acetylcholine is the transmitter produced by interneurons in the striatum.

(SNr : substantia nigra pars reticulata; SNc: substantia nigra pars compacta; STN: Subthalamic nucleus ; GPe: globus pallidus externa ; GPi: Globus pallidus interna).

Solution to Question 15:

Serotonin is an inhibitory neurotransmitter.

Acetylcholine, glutamate and norepinephrine are excitatory neurotransmitters.

Solution to Question 16:

5-HT_{2C} receptor agonists increase satiety (state of being fed beyond capacity).

Important serotonin receptors:

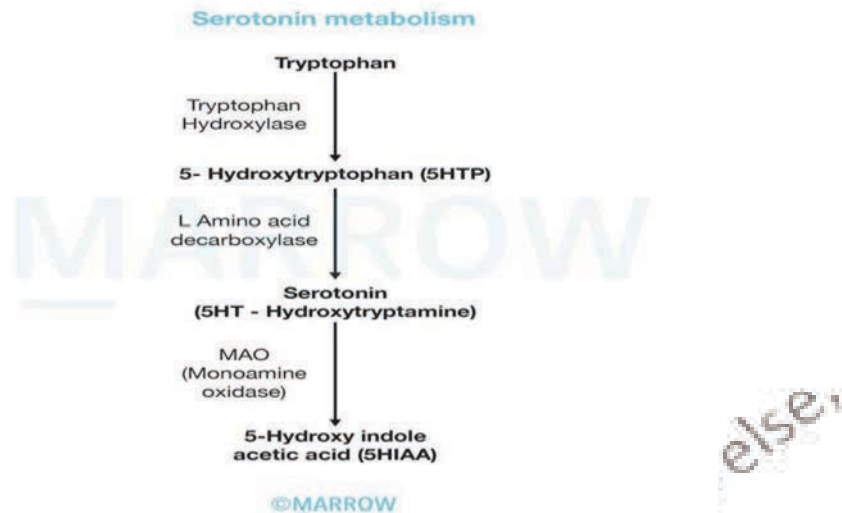
- 5-HT_{2A} receptors mediate platelet aggregation and smooth muscle contraction.
- 5-HT₃ receptors are present in the GIT and the area postrema and are related to vomiting.
- 5-HT₄ receptors are present in the GIT and brain. They facilitate secretion and peristalsis in GIT.
- 5-HT₆ and 5-HT₇ are distributed throughout the limbic system.

Note: All are G protein coupled receptors except 5-HT₃ which is a ligand gated ion channel receptor.

Solution to Question 17:

Serotonin (5-hydroxytryptamine; 5-HT) is present in highest concentration in platelets and in the gastrointestinal tract, where it is found in the enterochromaffin cells and the myenteric plexus. It is also found within the brain stem in the midline Raphe nuclei.

Note: The precursor of serotonin is tryptophan.



Solution to Question 18:

NMDA receptor permits passage of Ca^{2+} and other cations (Na^{+}), not of chloride.

Glutamate is the agonist for NMDA receptor. Glycine binding to the NMDA receptor is essential for the receptor to respond to glutamate. NMDA receptors are present on hippocampal neurons.

Note: GABA and glycine receptors act by increasing Cl^{-} conductance.

Sensory Receptors

Question 1:

Spatial distribution from which a stimulus produces a response is called _____

- a) Receptive field
- b) Motor unit
- c) Dermatome
- d) Sensory neuron

Question 2:

Fast vibration sense is mediated by _____

- a) Merkel's disc
- b) Meissner's corpuscle
- c) Pacinian corpuscles
- d) Ruffini's end organ

Question 3:

The phasic receptor is _____

- a) Merkel's disc
- b) Ruffini's end organs
- c) Meissner's corpuscles
- d) Muscle spindle

Question 4:

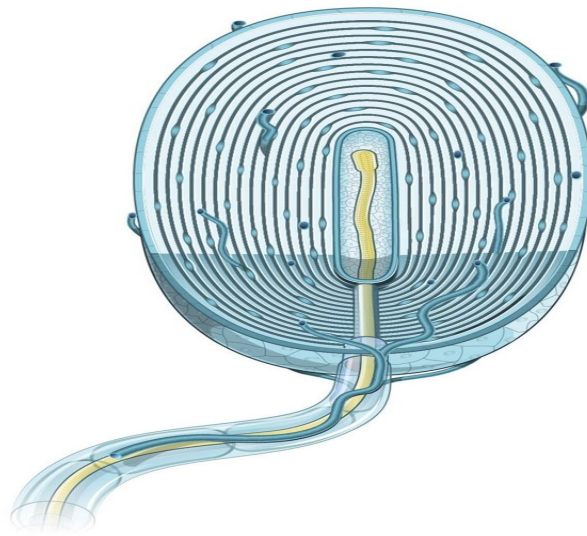
You are doing a CNS examination of a patient with a spinal injury. At which of the following sites would the distance of two-point discrimination be greatest?

- a) Lip
- b) Palm of the hand

- c) Back of scapula
- d) Dorsum of hand

Question 5:

Which of the following is true about the nature of the sensory receptor given below:



- a) Slow adapting and not encapsulated
- b) Fast adapting and encapsulated
- c) Fast adapting and not encapsulated
- d) Slow adapting and encapsulated

Question 6:

Which of the following classification of receptors is the same?

- a) Mechanoreceptors and exteroceptors
- b) Interoceptors and nociceptors
- c) Photoreceptors and proprioceptors
- d) Telereceptors and interoceptors

Question 7:

Which of the following is true regarding sensory receptor properties?

- a) Magnitude of sensation felt is directly proportional to the intensity of the stimulus
- b) A touch receptor will generate a constant rate of discharge of its axon for as long as the receptor is stimulated
- c) Meissner's and Pacinian corpuscles adapt slowly, the others adapt rapidly
- d) The frequency of a train of action potentials in an afferent fibre reflects the intensity of the stimulus given to its receptor

Question 8:

A δ fibers do not transmit _____.

- a) Slow pain
- b) Epicritic pain
- c) Fast pain
- d) First pain

Question 9:

The most potent algogenic substance is _____.

- a) Serotonin
- b) Bradykinin
- c) Histamine
- d) Prostaglandin

Question 10:

Vanilloid receptors are activated by all except _____.

- a) Heat
- b) Thermal
- c) Chemicals
- d) Vibration

Question 11:

Heat-pain fibers are stimulated at temperatures around _____.

- a) 30° C
- b) 49° C
- c) 45° C
- d) 35° C

Question 12:

Weber-Fechner law relates magnitude of sensation to ____.

- a) Strength of muscle contraction
- b) Log of Intensity of stimulus
- c) Amplitude of action potential
- d) Duration of stimulus

Question 13:

A patient's right leg was amputated because it was unsalvageable after the motor vehicle accident. After 2 weeks of her surgery, she started feeling episodes of severe pain in the amputated area. This pain is most likely based on which of the following mechanisms?

- a) Muller's specific nerve energy
- b) Principle of labeled-line
- c) Cortical plasticity
- d) Weber-Fechner law

Question 14:

Perception of normal sensory stimuli as painful is _____

- a) Allodynia
- b) Hyperalgesia
- c) Analgesia
- d) Causalgia

Question 15:

Which of the following is true about anandamide?

- a) It is not produced endogenously
- b) It is a nociceptive agent
- c) There are anandamide-containing neurons in the periaqueductal gray
- d) It binds to opioid receptors

Question 16:

Which of the following is true about visceral pain?

- a) It is poorly localised
- b) It shows rapid adaptation
- c) It is mediated by B fibers
- d) It resembles fast pain

Question 17:

Pain insensitive structure is _____.

- a) Alveoli
- b) Liver capsule
- c) Parietal pleura
- d) Arterial wall

Question 18:

A patient with chronic pain is undergoing transcutaneous electrical nerve stimulation for pain relief. What is the basis of this procedure?

- a) Referred pain
- b) Gate theory of pain
- c) Allodynia
- d) Pain relief by analgesics

Question 19:

Primary cutaneous hyperalgesia_____

- a) Develops in the normal skin region around the area of flare
- b) Is an abnormal condition in the skin in which painful stimuli become less severe
- c) Is due to changes in threshold of pain receptors
- d) Is associated with throbbing type of pain

Question 20:

The transient receptor potential (TRP) channel that is most active at normal skin temperature is _____.

- a) TRPV4
- b) TRPV3
- c) TRPM8
- d) TRPA1

Question 21:

A soldier posted in Siachen is brought to you in the ER. In order to rule out hypothermia, how do you best measure his temperature?

- a) Oral
- b) Rectal
- c) Infrared tympanic thermometer
- d) Axillary

Answer Key

Question No.	Correct Option
1	a
2	c
3	c
4	c
5	b
6	a
7	d

8	a
9	b
10	d
11	c
12	b
13	c
14	a
15	c
16	a
17	a
18	b
19	c
20	b
21	b

Detailed Explanations

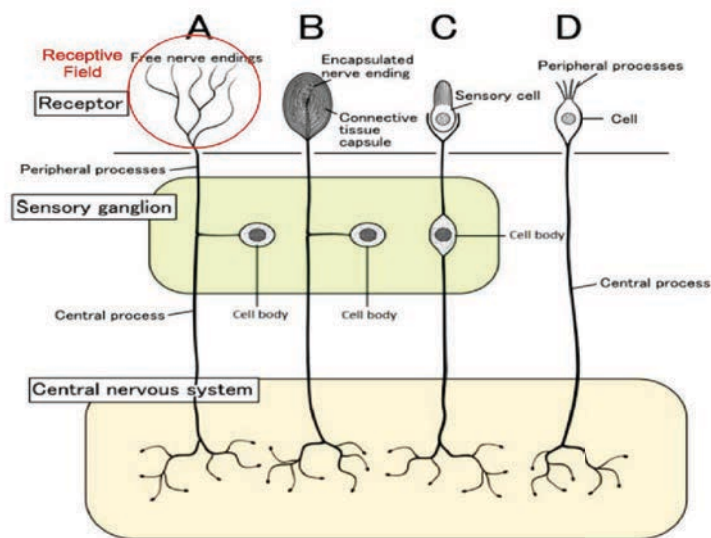
Solution to Question 1:

Spatial distribution from which a stimulus produces a response is called receptive field.

Sensory Unit: A single afferent neuron with all its receptor endings.

Motor unit: Each single motor neuron and the muscle fibers it innervates.

Dermatome: Each spinal nerve innervates a “segmental field” of the skin.



Solution to Question 2:

Fast vibration sense is mediated by Pacinian corpuscles.

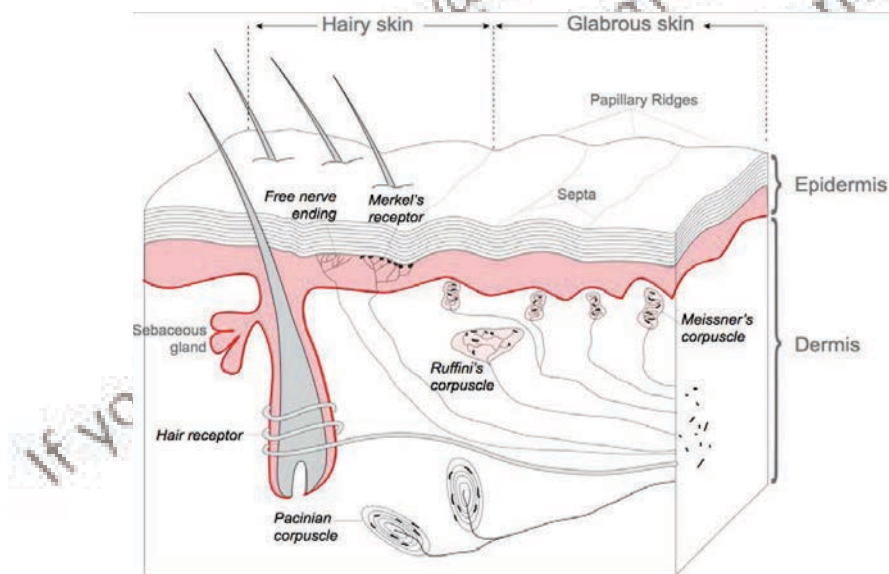
Pacinian corpuscle is the most sensitive mechanoreceptor in the somatosensory system. It is responsive to high-frequency (30–500 Hz) vibratory stimuli and pressure. They are rapidly adapting receptors.

Option A: Merkel's disc sense touch and sustained pressure. They are slow adapting receptors. Present in epidermis.

Option B: Meissner's corpuscle sense tap and flutter. They are present in dermal papillae. They are rapidly adapting receptors.

Option D: Ruffini endings sense skin stretch and fluttering vibration.

The following image shows Merkel's disc and Meissner's corpuscle being more superficial (reponsive to touch) than Ruffini endings and Pacinian corpuscle (which are deeper in the dermis and responsive to pressure).



Solution to Question 3:

Meissner's corpuscles are phasic (rapidly adapting) receptors.

Merkel's disc, Ruffini's end organs, and muscle spindle are tonic (slowly adapting) receptors.

Solution to Question 4:

The distance of two-point discrimination is greatest at the back of the scapula.

The relative tactile acuity of a given region can be determined by the two-point discrimination test. Here, calipers are used to determine the minimum distance between the two points which can be perceived as two points. Greater the density of receptors → Smaller the receptive field → More precise the localization.

For e.g., Stimulus points on the back, for instance, must be separated by at least 65 mm before they can be distinguished as separate, whereas on the fingertips two stimuli are recognized if they are separated by as little as 2 mm. The lips have the greatest acuity (two-point discrimination threshold ~ 1 mm).

Clinical significance:

Blind individuals benefit from the tactile acuity of fingertips to facilitate the ability to read Braille. The dots forming Braille symbols are separated by 2.5 mm.

Two-point discrimination is used to test the integrity of the dorsal column (medial lemniscus) system.

Solution to Question 5:

The sensory receptor shown in the above image is Pacinian corpuscle. The Pacinian corpuscle is a rapidly adapting receptor and is an encapsulated receptor.

When a stimulus of the same strength is continuously given to a sensory receptor, the frequency of the action potentials generated by the receptor gradually decreases. This phenomenon is known as receptor adaptation. Depending on whether the receptor stops responding to a stimulus rapidly or slowly, they are classified as rapidly adapting (phasic) or slow adapting receptors (tonic).

Solution to Question 6:

Mechanoreceptors and exteroceptors have the same classification of receptors.

Exteroceptors are concerned with the external environment or receptors which receive external stimuli. Mechanoreceptors are cutaneous receptors for touch and pressure. Thus mechanoreceptors can be considered to be exteroceptors.

Nociceptors are pain receptors which detect physical and chemical damage occurring in the tissue.

Interoceptors are concerned with the internal environment.

Chemoreceptors are receptors stimulated by a change in the chemical composition of the environment in which they are located. These include receptors for taste and smell as well as visceral receptors such as those sensitive to changes in the plasma level of O₂, pH, and osmolality.

Teleceptors (distance receivers) are concerned with events at a distance, example - photoreceptors.

Proprioceptors provide information about the position of the body in space at any given instant.

Solution to Question 7:

The frequency of a train of action potentials in an afferent fibre reflects the intensity of the stimulus given to its receptor.

Intensity of stimulus applied to a sensory receptor can be determined by the following:

- Variation in frequency of action potentials generated
- Variations in the number of receptors activated

Option A : Weber-Fechner Law states that magnitude of sensation felt is directly proportional to the log of the intensity of the stimulus.

Option B: A touch receptor will not generate a constant rate of discharge, even if the receptor is continuously stimulated. This is called desensitization or adaptation. If the opposite were true (if option B was correct), when we have our clothes on, we would have constant firing of neurons unnecessarily, in response to touch stimulus.

Option C:

Slowadapting(Tonic receptors)	Rapidlyadapting(Phasic receptors)
Merkel's disc, Ruffini's end organs, Joint capsule Muscle spindle, Golgi tendon organs	Pacinian corpuscle, Meissner's corpuscles, Hair end organs

Solution to Question 8:

A δ fibers do not transmit slow pain. They transmit fast pain.

Activation of A δ fibers, which release glutamate, is responsible for first pain (also called fast pain or epicritic pain), which is a rapid response and mediates the ability to localize the site and intensity of the noxious stimulus.

Neuropathic pain can also be burning, dull and poorly localized (protopathic pain, transmitted by C nociceptor fibers (which release substance P).

Dual Pain Pathway

Pain can be carried by either A δ fiber (fast pain/ epicritic pain) or by C fiber (slow pain/protopathic pain)

Feature	A δ fiber	C fiber
Stimuli	MechanicalThermal	Polymodal
Myelination and tract	Myelinatedand carried viaNeospinothalamic tract	Unmyelinatedand carried via paleospinothalamic tract
Conduction velocity	12–30 m/s	0.5–2 m/s
Impulse conduction	Fast componentof pain	Slow componentof pain

Feature	A δ fiber	C fiber
Neurotransmitter	Glutamate	Substance P

Solution to Question 9:

Bradykinin is the most potent algogenic substance.

An algogenic substance is a chemical released at the site of injury which mediates pain. These include bradykinin, serotonin, histamine, potassium ions, acids, acetylcholine, and proteolytic enzymes.

Prostaglandins and substance P enhance the sensitivity of pain endings but do not directly excite them.

Note: This is a controversial question

According to Guyton and Hall, Textbook of Medical Physiology, 13th Edition, Page 622-

"One chemical that seems to be more painful than others is bradykinin. Researchers have suggested that bradykinin might be the agent most responsible for causing pain after tissue damage."

Hence, bradykinin is the better answer (not serotonin)

Solution to Question 10:

Vanilloid receptors are not activated by vibration.

Vanilloid receptors or TRPV1 (transient receptor potential vanilloid1) receptors are a part of family of nonselective cation channels called transient receptor potential (TRP) channels. These channels respond to intense heat, acids, and chemicals such as capsaicin.

Solution to Question 11:

Heat-pain fibers begin to be stimulated at around 45 °C.

Thermal Receptors and their Excitation

Cold receptor	Warm receptor
Peak: 24 °C	Peak: 45 °C.
Cold nociceptor stimulated when temperature is around 5 °C.	Heat nociceptor is stimulated at temperatures around 45 °C.
Cold receptor fibers are A δ and C	Warm receptor fibers are unmyelinated C fibers

Solution to Question 12:

Weber-Fechner Law states that the magnitude of sensation felt is proportionate to the log of the intensity of the stimulus.

Other Laws in Physiology:

- Doctrine of specific nerve energies - When the nerve pathways from a particular sense organ are stimulated, the sensation evoked is that for which the receptor is specialized no matter how or where along the pathway the activity is initiated.
- Law of projection - No matter where a particular sensory pathway is stimulated along its course to the cortex, the conscious sensation produced is referred to the location of the receptor. This principle is called the law of projection.

Solution to Question 13:

This patient is having phantom limb pain. This pain is based on cortical plasticity.

Cortical structures are capable of reorganization after the loss of a body part. The cortical representation of the neighboring parts spreads into the cortical area that was formerly occupied by the representation of the amputated part. This is called cortical plasticity leads to the phenomenon of phantom limb pain

Solution to Question 14:

Perception of normal sensory stimuli as painful is allodynia.

Hyperalgesia is an exaggerated response to a noxious stimulus. An example of primary hyperalgesia is the extreme sensitivity of sunburned skin, which results from sensitization of the skin pain endings by local tissue products released due to burns — histamine, prostaglandins, and others.

Causalgia is a type of neuropathic pain.

Solution to Question 15:

There are anandamide-containing neurons in the periaqueductal gray and other areas concerned with pain.

There are 2 types of endogenous cannabinoids: 2-arachidonyl glycerol and Anandamide. They bind to CB1 and CB2 receptors. They have an anti-nociceptive effect.

Descending pathways from the mesencephalic periaqueductal gray inhibit transmission in nociceptive pathways. This descending pathway includes a synapse in the ventromedial medulla (Raphe nucleus) and the release of endogenous opiates. So periaqueductal gray matter releases endogenous opiates.

Solution to Question 16:

Visceral pain is poorly localized, unpleasant and associated with nausea and autonomic symptoms. It often originates from thoracic and abdominal cavities, transmitted through small type C pain fibers.

It often radiates (or is referred) to other somatic structures perhaps due to convergence of somatic and visceral nociceptive afferent fibers on the same second-order neurons in the spinal dorsal horn that project to the thalamus and then to the primary somatosensory cortex.

When pain is referred, it is usually to a structure that developed from the same embryonic segment or dermatome as the structure in which the pain originates. This principle is called the dermatomal rule.

Example: the heart and the arm have the same segmental origin.

Feature	A δ fiber	C fiber
Stimuli	MechanicalThermal	Polymodal
Myelination and tract	Myelinatedand carried viaNeospinothalamic tract	Unmyelinatedand carried via paleospinothalamic tract
Conduction velocity	12–30 m/s	0.5–2 m/s
Impulse conduction	Fast componentof pain	Slow componentof pain
Neurotransmitter	Glutamate	Substance P

Solution to Question 17:

Alveoli are pain insensitive structures. Alveoli of lungs and parenchyma of liver are almost completely insensitive to the pain of any type.

However, the liver capsule is extremely sensitive to both direct trauma and stretch. The bile ducts, arterial walls, bronchi and the parietal pleura are very sensitive to pain.

Solution to Question 18:

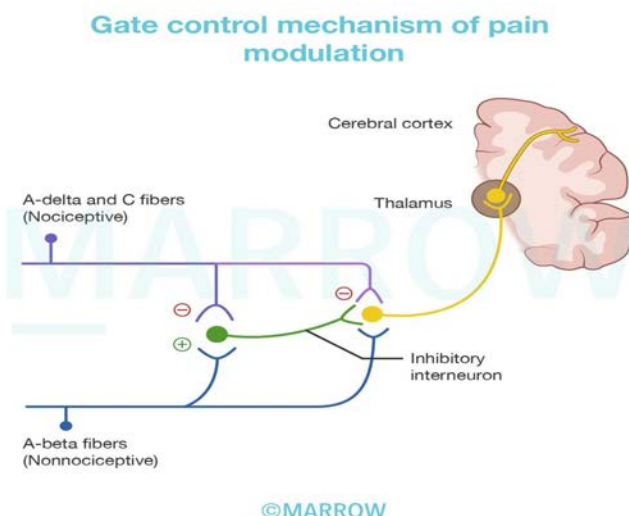
Transcutaneous electrical nerve stimulation (TENS) of pain relief is based on the gate theory of pain. This method uses electrodes to activate A $^\circ$ and A β fibers near the injury to relieve pain.

Gate control theory operates at the substantia gelatinosa of the dorsal horn of the spinal cord.

The relief may result from inhibition of pain pathways (Inhibition of C fibers and A-delta fibers) in the dorsal horn gate, by stimulation of large-diameter touch-pressure afferents(A-beta fibers). This is called the gate-control hypothesis.

Interesting clinical application: Touching or shaking an injured area decreases the pain due to the injury. Stimulation with an electric vibrator at the site of pain also gives some relief.

The same mechanism is probably responsible for the efficacy of counterirritants. Stimulation of the skin over an area of visceral inflammation produces some relief of the pain due to the visceral disease. The old-fashioned mustard plaster works on this principle.



Solution to Question 19:

Hyperalgesia is an exaggerated response to a noxious stimulus.

- Hyperalgesia and allodynia signify increased sensitivity of nociceptive afferent fibers.
- Causes of hyperalgesia
- The excessive sensitivity of the pain receptors, which is called primary hyperalgesia
- Facilitation of sensory transmission, which is called secondary hyperalgesia.

An example of primary hyperalgesia is the extreme sensitivity of sunburned skin, which results from sensitization of the skin pain endings, by local tissue products from the burn — histamine, prostaglandins, and others.

Secondary hyperalgesia frequently results from lesions in the spinal cord or the thalamus.

Solution to Question 20:

At normal skin temperatures, TRPV3 is the most active receptor.

Solution to Question 21:

In the given clinical scenario rectal temperature would be the preferred site to measure the core body temperature.

During low or high temperatures like in Siachen, the axillary and the oral temperatures are highly misleading and should not be used to measure the core body temperatures

Infrared tympanic thermometer measures the temperature of the eardrum which is representative of core body temperature. However, more often the thermometer is not inserted adequately or for safety reasons, they are manufactured in such a way that they do not reach deep into the ear canal, or the ear canal contains cerumen which can disturb the measurement. Hence the infrared thermometer more commonly delivers the temperature of ear canal cerumen and hence not a reliable method of measuring the core body temperature.

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If you purchased this from someone else,
you may have been scammed.

Somatosensory Pathways

Question 1:

The principle that “ In spinal cords, dorsal roots are sensory and ventral roots are motor” is known as _____

- a) Law of projection
- b) Bell – Magendie law
- c) Labelled line principle
- d) Weber – Fechner law

Question 2:

A patient admitted to the ward does not feel a sensation of the heat on touching a hot object. Considering the signal reaches the thalamus, which of the following is not transmitting the signal to an intact postcentral gyrus?

- a) Primary neuron
- b) Receptor
- c) Secondary neuron
- d) Tertiary neuron

Question 3:

Decussation of ascending spinal pathways occurs mainly within the _____

- a) Midbrain and medulla oblongata
- b) Pons and medulla oblongata
- c) Spinal cord and medulla oblongata
- d) Spinal cord only

Question 4:

Conscious proprioception is carried by _____.

- a) Dorsal column fibers

- b) Anterior spinothalamic tract
- c) Lateral spinothalamic tract
- d) Vestibular tract

Question 5:

Which of the following is not a property of fibers in the dorsal column pathway?

- a) Large axonal diameter
- b) Dorsal column carries fine touch
- c) Rapid conduction velocity
- d) Unmyelinated

Question 6:

A patient comes to you for his annual health checkup. He lost his vision due to an accident a few years ago. He now experiences the world through touch. Which of the following Brodmann's areas processes this tactile stimulation?

- a) 3, 1, 2
- b) 4, 6
- c) 44, 45
- d) 41, 42

Question 7:

Cortical representation of body in the cerebrum is _____

- a) Horizontal
- b) Vertical
- c) Sagittal
- d) None

Question 8:

Somatosensory area has the largest representation for _____

- a) Lips
- b) Hand
- c) Thumb
- d) Back

Question 9:

A patient sustains an injury to the lateral portion of the postcentral gyrus in the right hemisphere. Loss of sensation is likely to be seen at which of the following sites in this patient?

- a) Left lower limb
- b) Right lower limb
- c) Left facial area
- d) Right facial area

Question 10:

Lesion of somatosensory area I affects _____

- a) Pain
- b) Localization
- c) Temperature
- d) All of the above

Question 11:

A man was rushed to the casualty after getting stabbed in the back with a knife. On MRI, a complete transection of one side of the spinal cord is suspected. Which of the following is likely to be seen in this patient?

- a) Pain and temperature loss on the opposite side
- b) Fine touch lost on the opposite side
- c) Pain and temperature lost on the same side
- d) Only pain lost on the same side

Question 12:

Which of the following findings are seen in a patient with Brown-Sequard syndrome?

- a) Upper Motor Neuron type of paralysis below the lesion on the opposite side
- b) Impaired fine touch below the lesion on the opposite side
- c) Loss of crude touch below the lesion on the same side
- d) Loss of vibration sense below the lesion on the same side

Question 13:

Itch is carried by_____

- a) Dorsal column
- b) Spinocerebellar tract
- c) Anterolateral system
- d) Corticospinal tract

Question 14:

Following are the endogenous opioids except_____

- a) Dopamine
- b) Endorphin
- c) Enkephalin
- d) Dynorphin

Question 15:

You are examining a patient for a follow-up of severe pain in his right leg. He tells you that he feels much better now that he is taking acupuncture therapy too. Which of the following is responsible for pain relief in acupuncture?

- a) Substance P
- b) Endorphin
- c) Serotonin
- d) Histamine

Question 16:

Fast pain is carried by_____

- a) Neo spinothalamic tract
- b) Paleo spinothalamic tract
- c) Dorsal column
- d) None

Question 17:

Soldiers wounded in the heat of battle often feel no pain until the battle is over which is an example of stress-induced analgesia. This is due to_____

- a) Substance P
- b) Serotonin
- c) Cannabinoids
- d) Cortisol

Question 18:

Which of the following does not carry proprioceptive impulses?

- a) Olivo cerebellar tract
- b) Tecto cerebellar tract
- c) Spino cerebellar tract
- d) Cuneo cerebellar tract

Question 19:

Pain and temperature in the thalamus are controlled by which of the following nuclei?

- a) Ventral posterior lateral nucleus
- b) Ventral posterior medial nucleus
- c) Anterior nucleus
- d) Intralaminar nucleus

Question 20:

Ventrolateral cordotomy for relief of pain in the right lower limb is being performed on a patient. Which of the following is cut in this procedure?

- a) Left ventral spinothalamic tract
- b) Left lateral spinothalamic tract
- c) Right ventral spinothalamic tract
- d) Right lateral spinothalamic tract

Question 21:

Pick the correct statement:

- a) Spinal lemniscus carries fibres of dorsal spinothalamic tract
- b) Trigeminal lemniscus carries pain and temperature sensations from the ipsilateral orofacial region
- c) Lateral lemniscus carries fibres from cochlear nuclei
- d) Medial lemniscus carries fibres from vestibular nuclei

Question 22:

Lesion in the medial lemniscus at the level of pons causes which of the following?

- a) Pain and temperature loss in the same side
- b) Pain and temperature loss in the opposite side
- c) Loss of tactile and proprioception on the same side
- d) Loss of tactile and proprioception on the opposite side

Question 23:

Ablation of somatosensory area 1 leads to_____

- a) Total loss of pain sensation
- b) Total loss of touch sensation
- c) Loss of tactile but not 2 point discrimination
- d) Loss of tactile and 2 point discrimination

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	a
5	d
6	a
7	b
8	a
9	c
10	b
11	a
12	d
13	c
14	a
15	b
16	a
17	c
18	b
19	a
20	b
21	c
22	d
23	d

Detailed Explanations

Solution to Question 1:

The principle that in the spinal cord the dorsal roots are sensory and the ventral roots are motor is known as the Bell–Magendie law.

Laws in Sensory Physiology

- Muller's doctrine of specific nerve energies: When the nerve pathways from a particular sense organ are stimulated, the sensation evoked is that for which the receptor is specialized, no matter how or where along the pathway the activity is initiated.

- Law of projection: No matter where a particular sensory pathway is stimulated along its course to the cortex, the conscious sensation produced is referred to the location of the receptor.
- Weber-Fechner Law: Magnitude of sensation felt is directly proportional to the log of the intensity of the stimulus.
- Labelled line theory: All the sensations from the different parts of the body travel along specified paths. For example, in the posterior column, fibers from the lower parts of the body are placed medially.

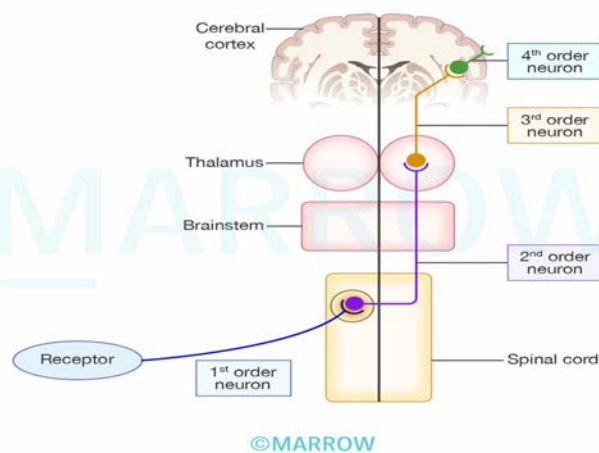
Solution to Question 2:

In this scenario, the tertiary neuron does not transmit the signal to the postcentral gyrus.

In the thalamus, secondary sensory neurons synapse onto tertiary sensory neurons, which in turn project to the somatosensory region of the cerebral cortex.

The question tests your knowledge of this pathway. It is said that the signal reaches the thalamus and the postcentral gyrus is intact. Still, the person cannot perceive the sensation, so the pathway is broken between the thalamus and the gyrus, which has to be at the tertiary neuron.

Sensory pathway in the nervous system

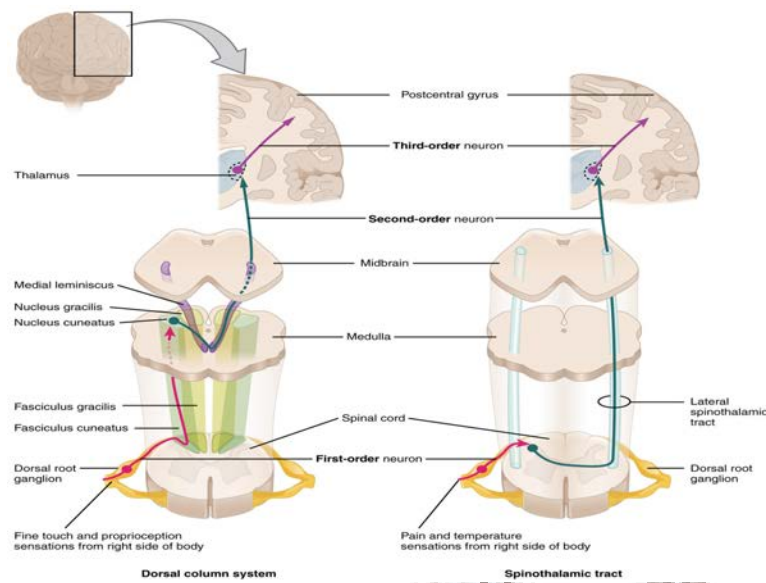


Solution to Question 3:

Decussation of ascending spinal pathways occurs mainly within the spinal cord and medulla oblongata. Most of the ascending pathways and tracts do decussate or cross; however, some remain uncrossed.

Dorsal column pathway crosses in the medulla oblongata.

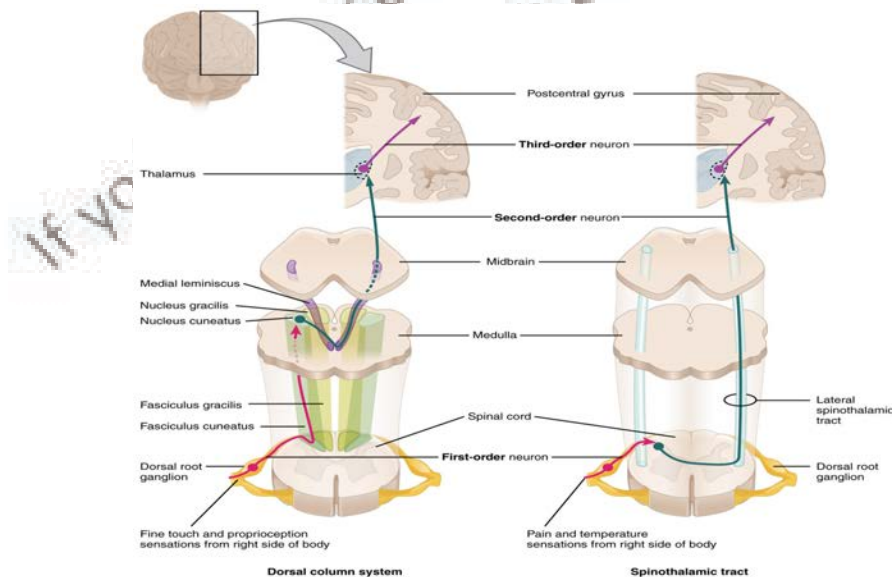
The anterior and lateral spinothalamic tract crosses obliquely in the spinal segments.



Solution to Question 4:

Conscious proprioception is carried by dorsal column. Unconscious proprioception is carried by spinocerebellar tract.

Option B and C: Anterolateral system carries pain, temperature, crude touch.



Solution to Question 5:

The dorsal column–medial lemniscal system is composed of large, myelinated nerve fibers and thus transmit signals to the brain, at high velocities of 30 to 110 m/sec.

Option B: Dorsal column carries fine touch and anterolateral column carries crude touch

Solution to Question 6:

Processing of tactile stimulation occurs in Brodmann's area in 3, 1, 2.

The human cerebral cortex is divided into about 50 distinct areas called Brodmann's areas based on histological structural differences.

Areas 3, 1, 2 which constitute primary somatosensory area I, and areas 5 and 7, which constitute the somatosensory association area process sensory information like tactile stimulation.

Solution to Question 7:

Cortical representation of body in the cerebrum is vertical.

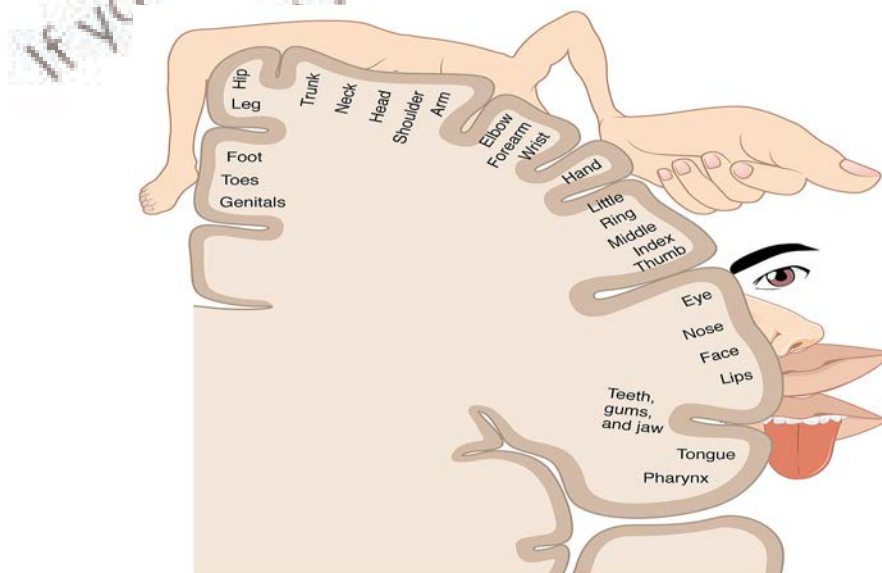
Parts of the body are represented in order along the postcentral gyrus, with the legs on top and head at the bottom of the gyrus.

A cortical homunculus is a distorted representation of the human body, based on a neurological map of the areas and proportions of the human brain dedicated to processing motor functions, or sensory functions, for different parts of the body. The size of the cortical receiving area for impulses from a particular part of the body is proportionate to the use of the part.

Sensory Homunculus is a topographical representation of distribution of sensory input to the somatosensory cortex from different parts of the body.

Motor homunculus is a topographical representation of distribution of motor output from primary motor cortex to different parts of the body.

Sensory homunculus is depicted below:

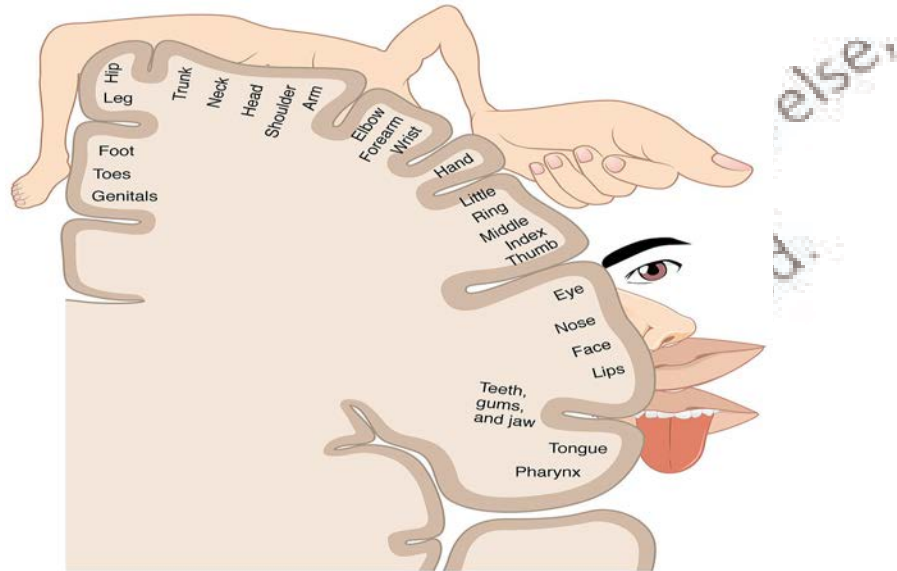


Solution to Question 8:

Somatosensory area has the largest representation for lips.

The relative sizes of the cortical receiving areas of the sensory homunculus are proportionate to the use of the part. The sizes of these areas are directly proportional to the number of specialized sensory receptors in each respective peripheral area of the body.

Some body parts are represented by large areas in the somatic cortex—the lips the greatest of all, followed by the face and thumb, whereas the trunk and lower part of the body are represented by relatively small areas.

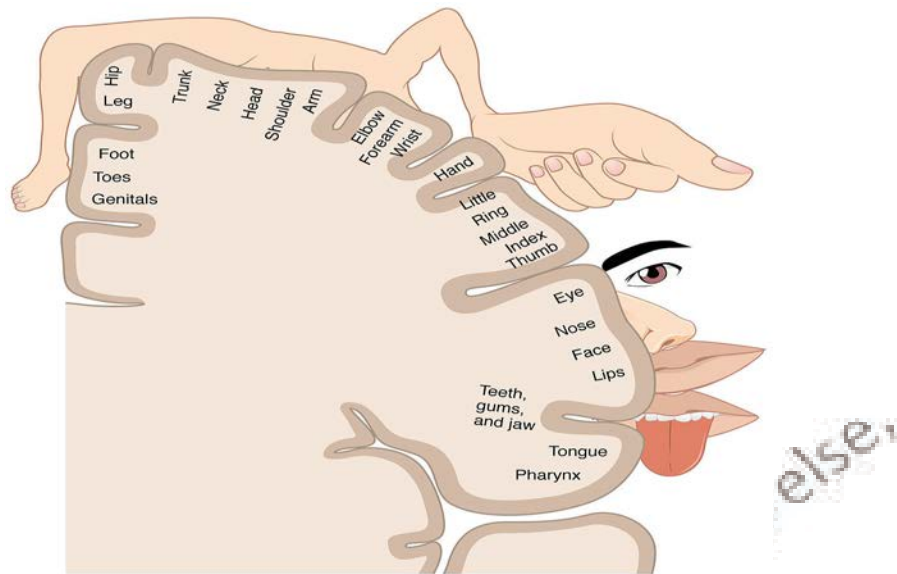


Solution to Question 9:

In this case, injury to the lateral portion of the postcentral gyrus in the right hemisphere would result in loss of sensation in the left facial area.

Each hemisphere receives somatic sensory information from the opposite side of the body in the postcentral gyrus (somatosensory area).

The lateral portion of the postcentral gyrus receives somatic sensory information from the facial and neck region while the medial portion of the postcentral gyrus received somatic sensory information from the trunk and appendages.



Solution to Question 10:

Lesion of the somatosensory area I affects the localization of sensations.

Bilateral lesion of the somatosensory area I leads to the following impairment of sensory judgment:

- Inability to localize different sensations in different parts of the body
- Inability to judge degrees of pressure against the body
- Inability to judge the weight of the objects
- Inability to judge shapes of forms of objects - astereognosis
- Inability to judge the texture of the materials

Pain and temperature are not affected both in quality and intensity in the lesion of the somatosensory area I. However, they are poorly localised. Pain and temperature localization depends greatly on the topographical map of the body in somatosensory area I to localize the source.

Solution to Question 11:

This patient most likely has Brown-Sequard syndrome. It is characterized by loss of pain and temperature on the opposite side.

Hemisection of the spinal cord causes damage to ascending sensory (dorsal column pathway, ventrolateral spinothalamic tract) and descending motor (corticospinal tract) pathways, which is called the Brown-Séquard syndrome.

The lesion to fasciculus gracilis or fasciculus cuneatus leads to ipsilateral loss of discriminative touch, vibration, and proprioception below the level of the lesion.

The loss of the spinothalamic tract leads to contralateral loss of pain and temperature sensation beginning one or two segments below the lesion.

Damage to the corticospinal tract produces weakness and spasticity in certain muscle groups on the same side of the body.

Solution to Question 12:

A patient with Brown Sequard syndrome will have a loss of vibration sense below the lesion on the same side.

Brown-Sequard syndrome is caused due to hemisection of the Spinal cord. It can be caused by fracture-dislocation of the vertebral column, by a bullet or stab wound, or by an expanding tumor.

Clinical features:

- **Pyramidal tract:** Pyramidal weakness (ipsilateral upper motor neuron type of paralysis) below the level of the lesion on the same side as the lesion (pyramidal tract crosses in the medulla).
- **Dorsal columns:** Impaired fine touch, 2 point discrimination, proprioception, vibration below the level of lesion on the same side of lesion (dorsal column fibres cross in the medulla).
- **Spinothalamic tract:** Loss of pain, temperature, crude touch, pressure sensation below the level of the lesion on the contralateral side of the lesion (spinothalamic tract crosses midline after entering the cord).

Solution to Question 13:

Itch-specific fibers have been demonstrated in the anterolateral spinothalamic tract. They are carried by unmyelinated nerve fiber C. Itching can be produced by repeated local mechanical stimulation of skin, variety of chemical agents including histamine and kinins released in the skin in response to tissue damage

Simple scratching relieves itching because it activates large fast conducting afferents ($A\beta$) that gate the transmission in the dorsal horn in a manner analogous to inhibition of pain by stimulation of similar afferents (Gate control theory).

Solution to Question 14:

Endogenous opioids, a group of neuropeptides that includes beta-endorphin, dynorphins, and enkephalins.

Opioids cause both presynaptic and post synaptic inhibition reducing excitatory post-synaptic potential (EPSP). Opioids are released from the descending analgesic pathway and bind with opiate receptors on the afferent pain-fiber terminal. This binding suppresses the release of substance P, blocking further transmission of the pain signal.

Solution to Question 15:

Pain relief in acupuncture is by the release of endorphin.

The acupuncture endorphin hypothesis is the primary mechanism of acupuncture analgesia. According to this hypothesis, Acupuncture is thought to activate afferent neurons leading to spinal cord and midbrain centers that release endogenous opioids implicated in pain relief.

Solution to Question 16:

Fast pain is carried by neospinothalamic tract.

Pain signals take two pathways to the brain:

- Neospinothalamic tract for fast pain/epicritic pain (A δ fibres)
- Paleospinothalamic tract for slow pain/Protopathic pain (C fibres)

Solution to Question 17:

Stress induced analgesia is due to cannabinoids.

The stress-induced analgesia, i.e., reduced pain sensitivity during stressful events might be due to:

- Release of endogenous cannabinoids such as 2-arachidonoylglycerol (2-AG) and anandamide
- Release of Norepinephrine from brainstem catecholaminergic neurons in the amygdala.

Solution to Question 18:

Tectocerebellar tract does not carry proprioceptive impulses. Tectocerebellar tract carry visual and auditory impulses.

Spinocerebellar, olivo cerebellar, cuneo cerebellar tracts carry Proprioceptive impulses.

Vestibulocerebellar tract carry vestibular impulses.

Solution to Question 19:

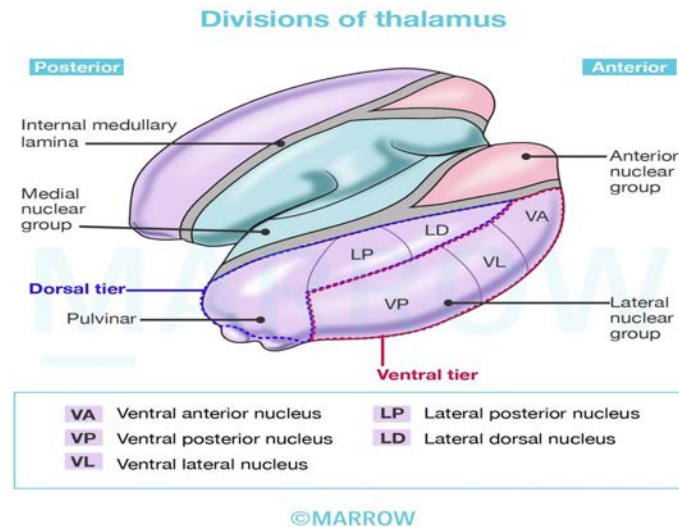
Pain and temperature in thalamus are controlled by the ventral posterior lateral nucleus. Pain and temperature is carried by anterolateral spinothalamic tract.

Fibres from anterolateral spinothalamic tract and dorsal column synapse in the ventral posteriorlateral (VPL) nucleus of thalamus.

Option B: The pathway from taste receptors includes synapses in the nucleus of the tractus solitarius in the brainstem and ventral posterior medial nucleus in the thalamus.

Option C: Anterior nucleus functions as a relay station for the hippocampal impulses. It receives afferents from mammillothalamic tract. It sends efferents to cingulate gyrus.

Option D: Intralaminar nucleus participates in arousal reaction, it receives afferents from reticular formation of brain stem.



Solution to Question 20:

In ventrolateral cordotomy, selected pain-conducting tracts in the spinal cord, i.e. fibers in the lateral spinothalamic pathway, are cut leading to contralateral loss of pain and temperature sensation below the level of the lesion.

In the more recent percutaneous approach, destruction of the lateral spinothalamic at the level of C1-C2 results in a contralateral disruption of painful sensations beyond C4.

Note: The fibers carrying pain sensation ascend in the lateral spinothalamic tract while the fibers carrying crude touch and pressure sensations ascend in the anterior spinothalamic tract.

Solution to Question 21:

Lateral lemniscus carries fibres from cochlear nuclei

Medial lemniscus carries fibres from of dorsal column nuclei

Spinal lemniscus carries fibres of anterolateral spinothalamic tract

Trigeminal lemniscus carries pain and temperature sensations from the contralateral orofacial region

Solution to Question 22:

Lesion in the medial lemniscus at the level of pons causes loss of tactile and proprioception on the opposite side.

Fibers of medial lemniscus ascend ipsilaterally in the dorsal columns of the spinal cord to the medulla, where they synapse in the gracilis and cuneate nuclei.

The second-order neurons from these nuclei cross the midline and ascend in the medial lemniscus to end in the contralateral ventral posterior lateral (VPL) nucleus and related specific sensory relay nuclei of the thalamus.

Solution to Question 23:

Ablation of somatosensory area 1 leads to Loss of tactile and 2 point discrimination.

Widespread bilateral excision of the somatosensory area I causes loss of the following types of sensory judgment:

- The person is unable to localize discretely the different sensations in the different parts of the body. However, he or she can localize these sensations crudely, such as to a particular hand. The brain stem, thalamus, or parts of the cerebral cortex can perform some degree of localization.
- The person is unable to judge critical degrees of pressure/weight/texture.

Special Senses

Question 1:

Light entering the eye first passes through _____ retinal layer.

- a) Inner nuclear layer
- b) Outer nuclear layer
- c) Outer plexiform layer
- d) Retinal ganglion layer

Question 2:

Initiation of visual impulse is associated with _____

- a) Condensation of opsin with Vitamin A
- b) Photoisomerisation and hydrolysis of visual purple
- c) NADP
- d) NAD

Question 3:

Which form of vitamin A combines with opsin to make rhodopsin?

- a) All trans retinaldehyde
- b) 11-cis –retinaldehyde
- c) All trans retinol
- d) 11-cis- retinol

Question 4:

Which of the following statements is true regarding distribution of rods and cones in retina?

- a) Macula lutea is the blind spot
- b) low density of photoreceptors are present in macula
- c) Foveola contains only cones

- d) Peripheral retina has greater cones to rods ratio

Question 5:

The rod receptor potential differs from other surface receptors in that it shows _____.

- a) Depolarization
- b) Decreased negativity
- c) Increased conductance of sodium
- d) Hyperpolarization

Question 6:

Which is the first-order neuron in the optic pathway?

- a) Rods and cones
- b) Ganglion cells
- c) Cells in lateral geniculate body
- d) Bipolar cells

Question 7:

Purely depolarizing cells in optic pathway are _____

- a) Bipolar cells
- b) Cones
- c) Amacrine cells
- d) Rods

Question 8:

Amacrine cells secrete _____

- a) Acetylcholine
- b) Serotonin
- c) Histamine
- d) Glutamate

Question 9:

The only neurons in retina showing action potential are _____

- a) Bipolar cells
- b) Ganglion cells
- c) Amacrine cells
- d) Rods and cones

Question 10:

The fibers from contralateral nasal retina project to the following layers of lateral geniculate nucleus _____

- a) 2,3,5
- b) 1,4,6
- c) 1,2,3
- d) 4,5,6

Question 11:

Parvocellular pathway carries signal for detection of _____

- a) Color contrast
- b) Luminous contrast
- c) Temporal frequency
- d) Saccadic eye movements

Question 12:

The axons from lateral geniculate nucleus terminate in _____ layer of primary visual cortex.

- a) Layer 1
- b) Layer 4
- c) Layer 2 and 3
- d) Layer 5 and 6

Question 13:

An ophthalmologist explains to a patient that he is a protanope. These patients have difficulty in identifying which of the following color?

- a) Red
- b) Green
- c) Blue
- d) Black

Question 14:

Auditory information is relayed through _____ thalamic nucleus.

- a) Dorsomedial nucleus
- b) Lateral geniculate nucleus
- c) Medial geniculate nucleus
- d) Ventral Posterolateral nucleus

Question 15:

In the auditory pathway, which is the first structure that can determine the direction of the perceived sound?

- a) Cochlear nucleus
- b) Inferior colliculus
- c) Lateral lemniscus
- d) Superior olivary nucleus

Question 16:

Which of the following events prompts the auditory system to interpret a sound as loud?

- a) Decreased number of inner hair cells become stimulated
- b) Decreased number of outer hair cells become stimulated
- c) Hair cells excite nerve endings at a diminished rate

- d) Amplitude of vibration of the basilar membrane increases

Question 17:

Umami taste is evoked by_____

- a) Glucose
- b) Quinine
- c) Glutamate
- d) Sodium chloride

Question 18:

Mitral cells are seen in _____

- a) left atrium
- b) Olfactory bulb
- c) Primary visual cortex
- d) Geniculate body

Question 19:

Which of the following sensory systems has the smallest range of intensity discrimination?

- a) Auditory
- b) Visual
- c) Olfactory
- d) Somatosensory

Question 20:

Number of cones in the retina (millions) is_____

- a) 3-6
- b) 5-12
- c) 10-15
- d) 20

Question 21:

What is the site of lesion in Wernicke's hemianopic pupil?

- a) Optic nerve
- b) Optic tract
- c) Optic radiation
- d) Optic chiasma

Answer Key

Question No.	Correct Option
1	d
2	b
3	b
4	c
5	d
6	a
7	c
8	a
9	b
10	b
11	a
12	b
13	a
14	c
15	d
16	d
17	c
18	b
19	c
20	a
21	b

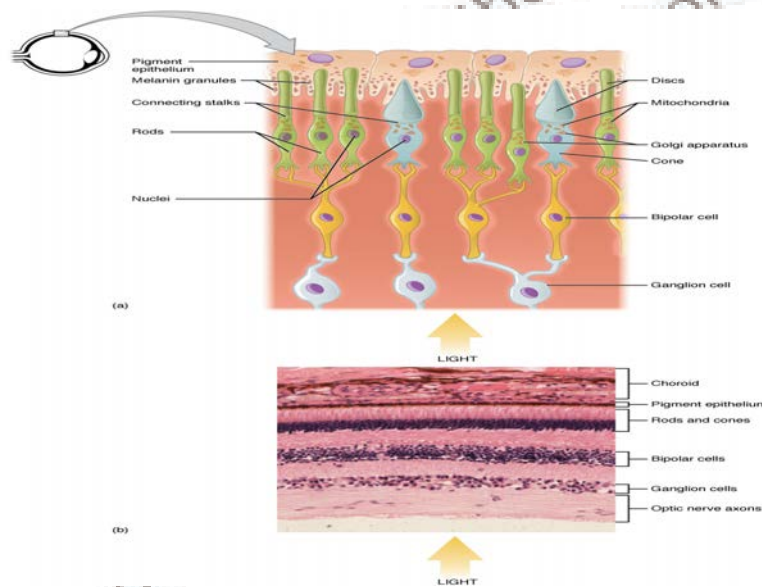
Detailed Explanations

Solution to Question 1:

Light passes first through the retinal ganglion layer, the most anterior/inner layer of the retina. Light then passes through other layers of the retina until it reaches the photoreceptors in the posterior/outer region of the retina.

Retina is organized into 10 layers or zones where distinctive components are clustered together to form continuous strata.

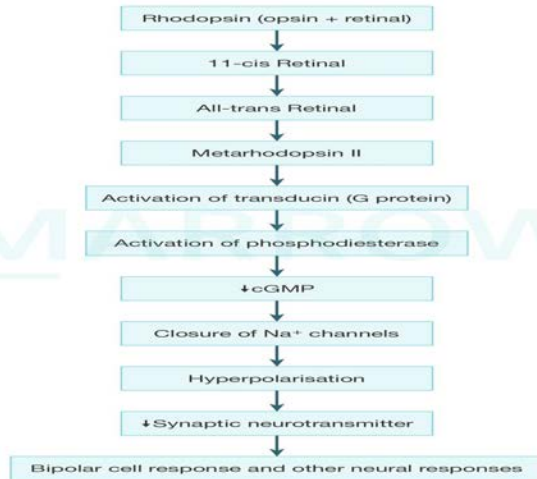
- Light passes in the direction: Vitreous humor(anterior)→ choroid(most posterior) encountering the cell layers in that direction (ganglion cell layer→photoreceptors)
- Visual processing/Phototransduction occurs in the opposite direction (Photoreceptors→ ganglion cell layer)



Solution to Question 2:

Initiation of visual impulse is associated with photoisomerisation and hydrolysis of visual purple. Rhodopsin or visual purple, the photosensitive pigment, is composed of opsin (a protein belonging to the superfamily of G protein–coupled receptors) and retinal (an aldehyde of vitamin A). When light strikes the photoreceptors, retinal is chemically transformed in a process called photoisomerization, which begins the transduction process.

Phototransduction in rods and cones

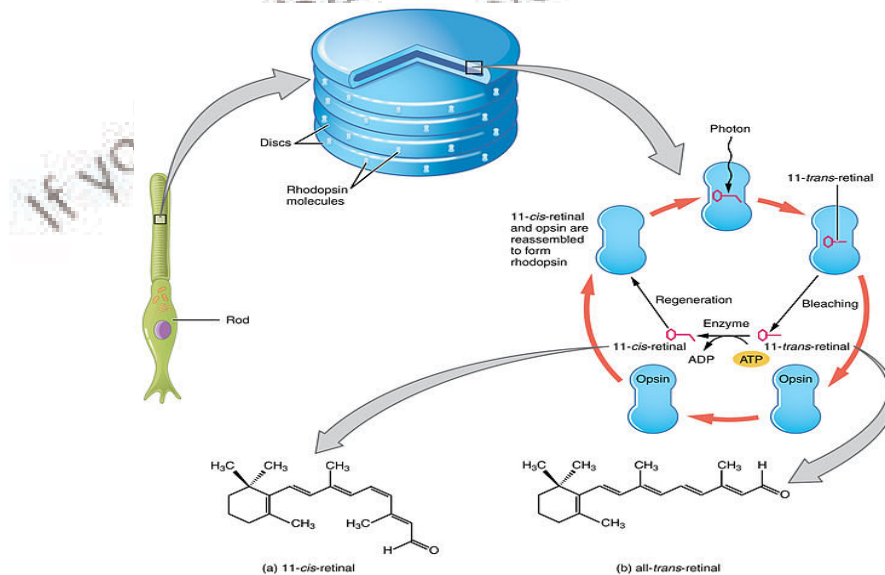


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Solution to Question 3:

11-cis –retinaldehyde combines with opsin to make rhodopsin.

All-trans retinol is one form of Vitamin A which is converted to 11-cis retinol under the influence of Isomerase. Finally, 11-cis retinol is converted into 11-cis retinal which combines with Scotopsin to form rhodopsin.



Solution to Question 4:

Central fovea (foveola) contains only cones (rod-free area) and is a region of highest resolution.

Regional Variations in the Retina

Macula lutea

- Located at posterior pole, temporal to the optic disc
- Area of central vision
- High density of photoreceptors and interneurons

Optic disc

- Lacks photoreceptors
- Is called "Blind spot"
- Normally a person is unaware of blind spot because
- Corresponding part of the visual field can be seen by the contralateral eye
- Neurological processes complete the incomplete image perceptually

Peripheral retina

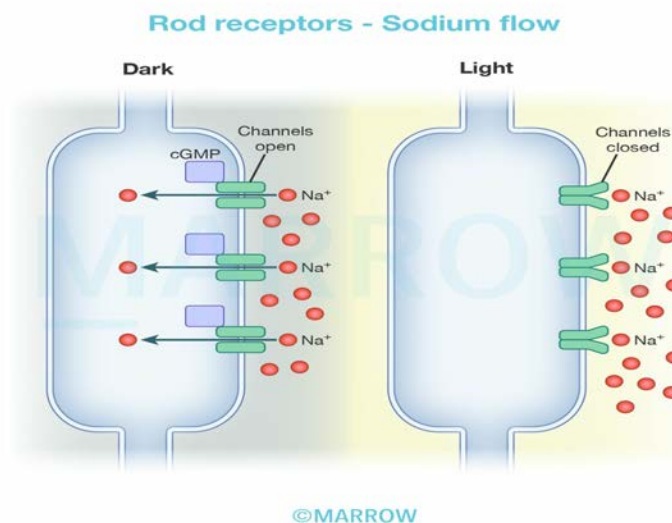
- Higher ratio of rods to cones = Rod to cone $\rightarrow 20:1$
- Greater degree of convergence of photoreceptors to the target interneuron (bipolar cells)
- More sensitivity to light
- Lower visual acuity

Solution to Question 5:

The rod receptor potential differs from other surface receptors in that it shows hyperpolarization.

In most sensory receptors: Arrival of stimulus $\rightarrow$ Ion channels are opened up allowing current to flow $\rightarrow$ depolarisation of the receptor.

In photoreceptors: When the light strikes the outer segment, a cascade of reactions (refer figure) are initiated, that close some of the Na^+ channels in the outer segment and results in hyperpolarization.



Solution to Question 6:

Photoreceptors - rods and cones are the first-order neurons in the visual pathway.

Solution to Question 7:

The responses of the rods, cones, and horizontal cells are hyperpolarizing, and the responses of the bipolar cells are either hyperpolarizing or depolarizing, whereas amacrine cells produce purely depolarizing potential.

Solution to Question 8:

Amacrine cells release acetylcholine.

Amacrine cells are one of the neuronal cell types of the retina. They transmit signals in two directions, either directly from bipolar cells to ganglion cells or horizontally within the inner plexiform layer from axons of the bipolar cells to dendrites of the ganglion cells or to other amacrine cells.

Amacrine cells release GABA, glycine, dopamine, acetylcholine & indolamine.

Other neuronal cell types of the retina are:

- The photoreceptors: rods & cones
- Horizontal cells
- Bipolar cells
- Ganglion cells

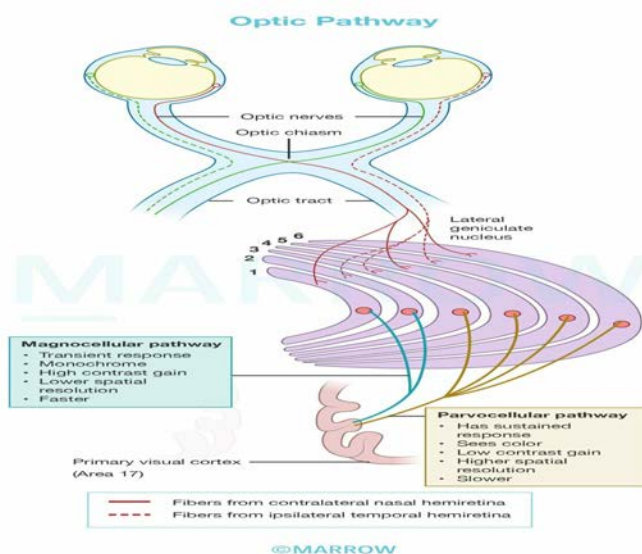
Of these cell types, only neurotransmitters secreted by photoreceptors & amacrine cells have been discovered. Rods and cones release glutamate at their synapses with the bipolar cells.

Solution to Question 9:

All neural elements in the retina exhibit local graded potentials except ganglion cells which shows action potential.

Solution to Question 10:

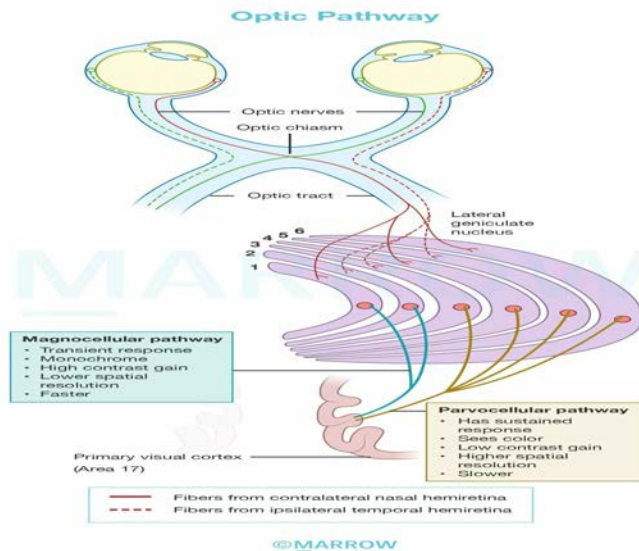
On each side, layers 1, 4, and 6 of the lateral geniculate nucleus receive input from the contralateral eye, whereas layers 2, 3, and 5 receive input from the ipsilateral eye. (Refer figure).



Solution to Question 11:

From the Lateral Geniculate Nucleus (LGN), a magnocellular pathway (from layers 1 and 2) and a parvocellular pathway (from the rest of the layers) project to the visual cortex.

Parvocellular pathway	Magnocellular pathway
Receive input entirely from type X retinal ganglion cells, and transmit color.	Formed by Y ganglion cells project to magnocellular layers of the lateral geniculate nucleus.
Convey accurate point-to-point spatial information.	Colorblind, transmitting only black-and-white information.
Terminate in layer 4C of the primary visual cortex.	Terminate in layer 4C of the primary visual cortex.
Carries signals for color vision, texture, shape, and fine details.	Carries signals for detection of movement, depth, and flicker.



Solution to Question 12:

The axons from the lateral geniculate nucleus (LGN) mostly end in layer 4.

The primary visual receiving area (primary visual cortex) is located principally on the sides of the calcarine fissure and has 6 layers similar to the neocortex. Layer 4 receives most of the input from the LGN, which is divided further into 4A, 4B, 4C^o, and 4C β .

The magnocellular pathways from layers 1 and 2 of the LGN end in layer 4C^o while the parvocellular pathways from layers 3-6 of the LGN end in layer 4C β .

Solution to Question 13:

Protanopes are dichromats with difficulty in identifying red color.

- Anomalous Trichromatic Vision
- Deficiency of red is Protanomaly.
- Deficiency of green is Deuteranomaly.
- Deficiency of blue is Tritanomaly.
- Dichromatic Vision (Individuals with only two cone systems)
- Blindness to red color Protanopia (Protanope).
- Blindness to green color Deuteranopia (Deuteranope).
- Blindness to blue color Tritanopia (Tritanope).
- Monochromatic Vision Have only one cone system.
- Color blindness is X-linked recessive.

- Achromatopsia/ Achromasia- Color blindness is due to lesions of area V8 of the visual cortex.

Solution to Question 14:

The medial geniculate nucleus is the thalamic nucleus that conveys auditory information from the brainstem to the primary auditory cortex.

Lateral geniculate nucleus is the thalamic nucleus that conveys visual information.

Solution to Question 15:

The superior olivary nuclei receive information from both ears and begin the process of detecting the direction from which a sound comes.

The eighth nerves reach their respective cochlear nuclei in the medulla from where the nerves travel to both the ipsilateral and contralateral superior olivary nuclei in the pons and hence giving it the ability to detect the direction of the sound.

The lateral part of superior olivary nucleus does so by comparing the difference in intensities of sound reaching the two ears, whereas the medial part of the superior olivary nucleus detects the time lag between the signals entering the two ears.

Solution to Question 16:

There are at least three ways that the auditory system determines loudness:

- As the sound becomes louder- the amplitude of vibration of the basilar membrane increases so that hair cells excite the nerve endings at more rapid rates.
- As the amplitude increases- more and more hair cells on the fringes of the resonating portion of the basilar membrane become stimulated.
- As the amplitude of vibration reaches high intensity, the outer hair cells become recruited at a significant rate.

Solution to Question 17:

Umami taste is triggered particularly by the monosodium glutamate (MSG) used so extensively in Asian cooking.

Taste Receptors

- Humans have five established basic tastes: sweet, salt, sour, bitter and umami.
- The taste receptors for these five basic tastes are of two major types:
- Ligand-gated channels (ionotropic receptors): for salt and sour taste

- G-Protein Coupled Receptors (metabotropic receptors): for sweet, bitter and umami
- Salt and sour taste are mediated through epithelial sodium channels (ENaC)
- Na⁺ movement through ENaC → salt taste
- H⁺ movement through ENaC → sour taste
- Sweet, bitter, and umami are sensed via GPCRs
- T1R2 and T1R3 families → sweet taste
- T2R family → bitter taste
- Metabotropic glutamate receptor (mGluR4) → Umami

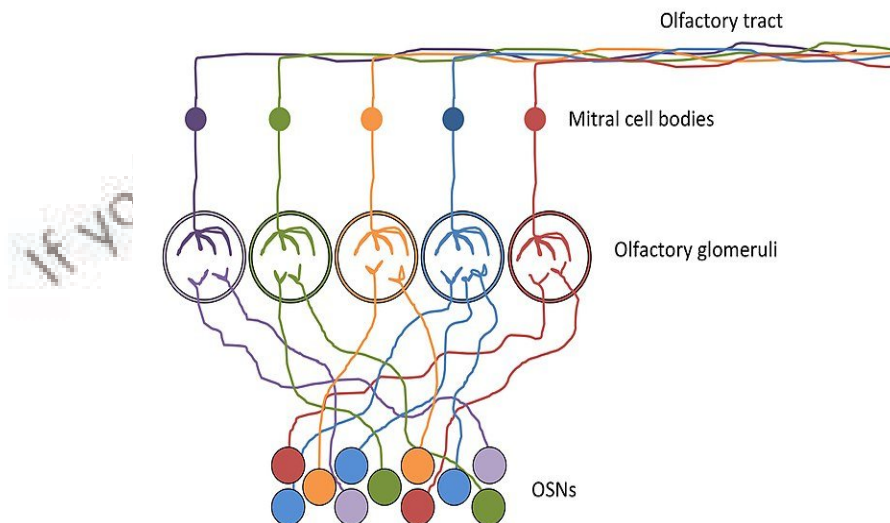
Solution to Question 18:

Mitral cells are seen in Olfactory bulb.

Axons of Olfactory nerves (First cranial nerve) synapse with the mitral cells and tufted cells in the olfactory bulb to form anatomically discrete units called olfactory glomeruli.

The olfactory bulb also contains periglomerular cells that are inhibitory interneurons and granule cells that make reciprocal synapses with mitral and tufted cells.

These mitral and tufted cells (second-order neurons) terminate in Olfactory cortex.



Solution to Question 19:

The olfactory system has the smallest range of intensity discrimination.

Smell is concerned more with detecting the presence or absence of odors, than with the quantitative detection of their intensities. Concentrations that are only 10 to 50 times above threshold values will evoke maximum intensity of smell, which is in contrast to most other

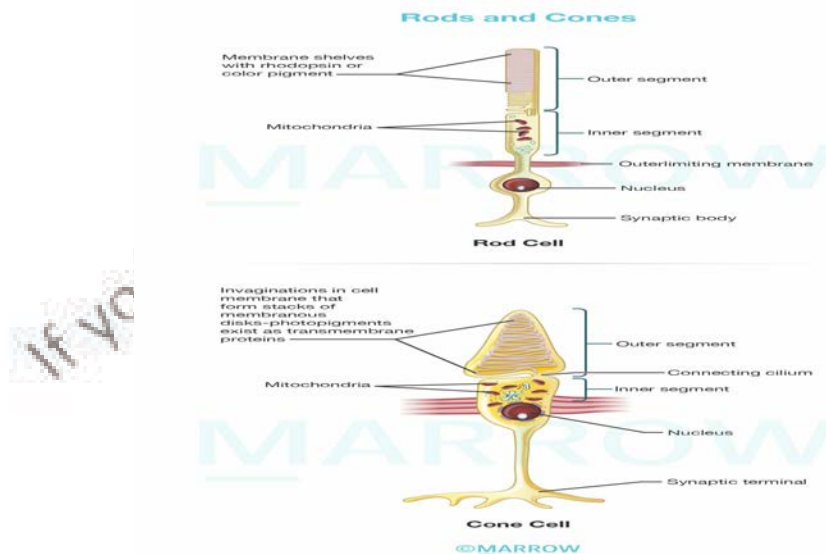
sensory system of the body.

Solution to Question 20:

There are approximately 6 million cones and 120 million rods in each human eye.

Features of Photoreceptors

- Each rod and cone are divided into outer segment, inner segment, nuclear region and a synaptic zone.
- Saccules and discs in the outer segment contain photosensitive pigments.
- If a person spends a considerable length of time in brightly lighted surroundings and then moves to a dimly lighted environment, the retinas slowly become "accustomed to the dark." This decline in visual threshold is known as dark adaptation. It is nearly maximal in about 20 min, much longer than the time taken for light adaptation which is 5 min.
- When one passes suddenly from a dim to a brightly lighted environment, the light seems intensely and even uncomfortably bright until the eyes adapt to the increased illumination and the visual threshold rises. This adaptation occurs over a period of about 5 min and is called light adaptation.



Solution to Question 21:

The site of lesion in Wernicke's hemianopic pupil is optic tract.

Wernicke hemianopic pupil is the light reflex, where, illumination of the blind half of the retina shows no pupillary reaction while stimulating the normal side of the retina shows a pupillary reaction. It is seen in lesions involving the optic tract. It is also seen in cases such as TB, syphilis and tumours of the optic thalamus.

Homonymous hemianopia (HH) involves loss of vision on the same side of the visual field in both eyes. HH is characteristically seen in lesions involving the visual pathway posterior to the optic chiasma.

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Motor Physiology - 1

Question 1:

Each muscle spindle consists of _____ intrafusal fibers.

- a) 12 to 24
- b) 3 to 12
- c) 1 to 3
- d) 24 to 50

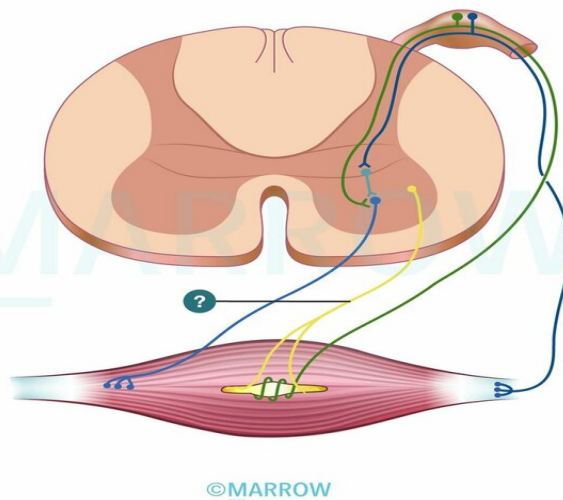
Question 2:

Motor supply to muscle spindle is _____

- a) I°
- b) II
- c) γ
- d) None

Question 3:

Identify the type of nerve fibre in the diagram below.



- a) Gamma Neuron
- b) Alpha Neuron
- c) Beta Neuron
- d) Delta Neuron

Question 4:

Which of the following is false with respect to muscle spindles?

- a) The intrafusal and extrafusal fibers are positioned parallel to each other
- b) The primary endings of intrafusal fibers are most sensitive to a dynamic response
- c) Type II nerve fibers do not innervate static nuclear bag fibers
- d) Two types of γ motor neurons supply the muscle spindle

Question 5:

Which of the following is sensed by the nuclear bag fibers?

- a) Dynamic length of muscle
- b) Muscle tension
- c) Joint position
- d) None

Question 6:

The afferents for deep tendon reflexes innervate _____

- a) Dynamic intrafusal fibers
- b) Golgi tendon organ
- c) Nuclear bag and chain fibers
- d) Golgi tendon and muscle spindle

Question 7:

Stimulus for stretch reflex is _____

- a) Passive stretch

- b) Active stretch
- c) Both
- d) None

Question 8:

Withdrawal reflexes are not_____

- a) Initiated by nociceptive stimuli
- b) Prolonged if the stimulus is strong
- c) An example of a flexor reflex
- d) Accompanied by the same response on both sides of the body

Question 9:

The maintenance of normal posture depends on_____

- a) Integrity of reflex arc
- b) Muscle power
- c) Type of muscle fiber
- d) Joint movement in physiological range

Question 10:

Receptor for inverse stretch reflex_____

- a) Muscle spindle
- b) Extrafusal fiber
- c) Intrafusal fiber
- d) Golgi tendon organ

Question 11:

The number of muscle fibers per Golgi tendon organ is ____.

- a) 3-20
- b) 200-300

- c) 50-60
- d) 80-100

Question 12:

The inverse stretch reflex_____

- a) Occurs when Ia spindle afferents are inhibited
- b) Is a monosynaptic reflex initiated by activation of the Golgi tendon organ
- c) Is a disynaptic reflex with a single interneuron inserted between the afferent and efferent limbs
- d) Is a polysynaptic reflex with many interneurons inserted between the afferent and efferent limbs

Question 13:

All of the following are true about withdrawal reflex except _____

- a) Diverging circuits to spread the reflex to the necessary muscles for withdrawal are present
- b) Circuits to inhibit the antagonist muscles, called reciprocal inhibition circuits are present
- c) Circuits to cause afterdischarge that lasts many fractions of a second after the stimulus is over are present
- d) The pathways for eliciting the flexor withdrawal reflex pass directly to the anterior motor neurons making it a monosynaptic reflex

Question 14:

Muscle tone is maintained by_____

- a) Golgi tendon organ
- b) Muscle spindle
- c) Renshaw cell
- d) None

Question 15:

Which of the following is true during initiation of voluntary movements?

- a) α motor neurons are activated followed by γ motor neuron

- b) A γ motor neurons are activated followed by α motor neuron
- c) Both α and γ motor neurons are activated together
- d) Only α motor neurons are activated

Question 16:

Righting reflex is _____

- a) Stretch reflex
- b) Postural reflex
- c) Spinal reflex
- d) Ocular reflex

Question 17:

In a patient with decerebrate rigidity, on applying pressure on the footpad, extension of that limb occurs. Which of the following statements is false regarding this phenomenon?

- a) It is Magnet Reaction
- b) It is integrated at the Spinal Level
- c) It helps to prevent falling on that side
- d) It is a righting reflex integrated at midbrain level

Question 18:

Pyramids are formed by _____

- a) Arcuate nucleus
- b) Vestibular nuclei
- c) Interstitial cells of Cajal
- d) Corticospinal tract

Question 19:

Which of the following is true about corticospinal tract?

- a) Mostly crossed

- b) Uncrossed
- c) Crossed at midspinal level
- d) 20% crossed

Question 20:

Which of the following is false regarding the origin of corticospinal tract?

- a) Corticospinal tract fibers originate from Betz Cells
- b) 30% from premotor and supplementary motor areas
- c) 40% from somatosensory areas
- d) 90% from the primary motor cortex

Question 21:

Which of the following is not an extrapyramidal tract?

- a) Reticulospinal tract
- b) Rubrospinal tract
- c) Corticospinal tract
- d) Vestibulospinal tract

Question 22:

Which pathway is not involved in maintenance of posture?

- a) Reticulospinal tract
- b) Rubrospinal tract
- c) Tectospinal tract
- d) Vestibulospinal tract

Question 23:

Gamma motor neurons are mainly influenced by _____

- a) Reticulospinal tract
- b) Rubrospinal tract

- c) Tectospinal tract
- d) Vestibulospinal tract

Question 24:

According to Herrington classification, the decerebrate rigidity is characterized by all except _____

- a) Rigidity occurs in all the muscles of the body
- b) Increase in the rate of discharge of the efferent neuron
- c) Increased excitability of the motor neuron pool
- d) Decerebration produces no phenomenon akin to spinal shock

Question 25:

Which of the following is true for decorticate rigidity?

- a) It is produced by the removal of cerebral cortex and basal ganglia
- b) Flexion of lower limbs and extension of upper limbs occurs
- c) Rigidity is pronounced
- d) It is characterized by flexion of upper limbs and extension of lower limbs

Question 26:

Which of the following reflex is absent in a decorticate animal?

- a) Righting reflex
- b) Hopping and Placing Reflex
- c) Labrynthine Reflex
- d) Crossed Extensor Reflex

Question 27:

Destruction of anterior cerebellum in a decerebrate animal leads to _____.

- a) No effect on rigidity
- b) Increased flexor muscle tone

- c) Increased rigidity via alpha motor neuron
- d) Decrease in rigidity

Question 28:

Upper Motor Neurons (UMN) include _____

- a) Pyramidal cells
- b) Peripheral nerves
- c) Glial cells
- d) Anterior horn cells

Question 29:

A 30-year-old man meets with an accident and develops spinal cord transection. Which of the following features will be seen as a result of pyramidal tract injury below the level of motor decussation?

- a) Positive Babinski sign
- b) Loss of deep tendon reflex
- c) Loss of proprioception
- d) Flaccidity

Question 30:

A 36-year-old male patient presented to the ER following a road traffic accident. Which of the following features would not be present if there is a corticospinal tract involvement?

- a) Cog-Wheel Rigidity
- b) Extensor Plantar response
- c) Spasticity
- d) Loss of superficial reflex

Question 31:

While exercising, a 42-year-old woman developed sudden onset of tingling in her right leg and an inability to control movement in that limb. A neurologic exam showed a hyperactive

knee jerk reflex and a positive Babinski sign. Which of the following is not true regarding her condition?

- a) Reflexes can be modified by impulses from various parts of the CNS
- b) Reflexes may involve simultaneous contraction of some muscles and relaxation of others
- c) Reflexes are chronically suppressed after spinal cord transection
- d) Reflexes often occur without conscious perception

Question 32:

A 50-year-old man with a history of medically controlled atrial fibrillation develops sudden onset weakness in the right side of his body. Which of the following Brodman's area is most likely affected?

- a) 4
- b) 1,2,3
- c) 5,7
- d) 16,18

Question 33:

Skilled voluntary movement is initiated by_____

- a) Association cortex
- b) Basal ganglia
- c) Cerebral cortex
- d) Cerebellum

Question 34:

In the homunculus, which of the following has maximum representation?

- a) Low control and skilled movements
- b) More muscle fibres are present
- c) Increased preciseness of movements
- d) None of the above

Question 35:

In the degrees of representation of the different muscle areas as mapped by Penfield and Rasmussen in the primary motor cortex, which of the following has the maximum representation?

- a) Vocalization
- b) Mastication
- c) Salivation
- d) Thumb

Question 36:

While doing a neurosurgical procedure, the paracentral lobules of both hemispheres were damaged. It will lead to which of the following?

- a) Hemiplegia
- b) Monoplegia
- c) Paraplegia
- d) Quadriplegia

Question 37:

Setting posture before planned movement is associated with ____.

- a) Premotor cortex
- b) Motor cortex
- c) Frontal
- d) Supplementary

Question 38:

A quadriplegic man with a high cervical lesion develops sudden onset of flushing, sweating, pounding headache and hypertension. He has a distended bladder due to a blocked urinary catheter. On immediate drainage of the urinary bladder and repositioning him from supine to reclining, his symptoms resolve. Which of the following is false regarding the most likely diagnosis?

- a) SBP may rise over 200 mm Hg

- b)** Major portion of the body's skeletal muscle go into extensor spasm
- c)** Large areas of the body break out into profuse sweating
- d)** Colon and Bladder evacuate

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	c
5	a
6	c
7	a
8	d
9	a
10	d
11	a
12	c
13	d
14	b
15	c
16	b
17	d
18	d
19	a
20	d
21	c
22	b
23	a
24	a
25	d
26	b
27	c
28	a

29	a
30	a
31	c
32	a
33	a
34	c
35	a
36	c
37	a
38	b

Detailed Explanations

Solution to Question 1:

- Each muscle spindle is 3 to 10 millimeters long. It is built around 3 to 12 tiny intrafusal muscle fibers that are attached to the glycocalyx of the surrounding large extrafusal skeletal muscle fibers.
- Each intrafusal muscle fiber is a tiny skeletal muscle fiber. However, the central region of each of these fibers has few or no actin and myosin filaments. Therefore, this central portion does not contract when the ends do. Instead, it functions as a sensory receptor. The end portions that do contract are excited by small gamma motor nerve fibers.

Solution to Question 2:

Motor supply to muscle spindle is γ

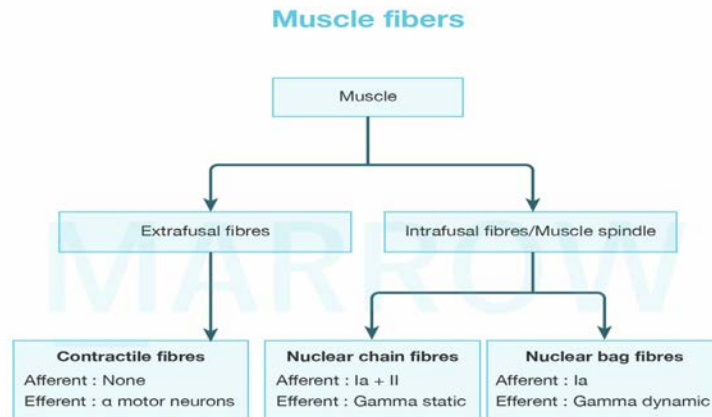
Muscle spindles have a motor nerve supply of their own. These nerves belong to Erlanger and Gasser's A α group. They are just 3–6 μ in diameter (hence called small motor nerve system/■ efferents of Leksell) and go exclusively to muscle spindles.

Note:

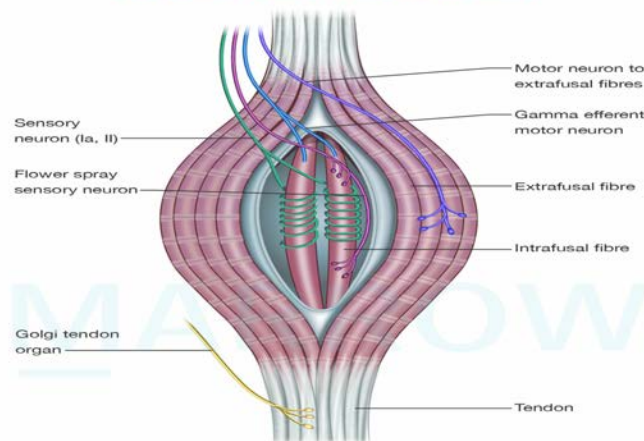
- Motor innervation of muscle spindle is by gamma efferents
- Motor innervation of extrafusal fibers is by alpha motor neurons

Nuclear bag muscle fibers: 1–3 per muscle spindle.

Nuclear chain muscle fibers: 3–9 per muscle spindle.



Structure of a muscle spindle



Solution to Question 3:

In the given figure, the fiber in yellow innervates the ends of the muscle spindle. Therefore, it is a gamma motor neuron.

Many much smaller gamma motor neurons are located in the spinal cord anterior horns (shown as the yellow efferent fiber). These gamma motor neurons average 5 micrometers in diameter, which go to small, special skeletal muscle fibers called intrafusal fibers or the muscle spindle. These fibers constitute the ends of the muscle spindle, which help control basic muscle “tone.”

Option (b):

The alpha motor neurons (shown as the blue efferent fibre) are on an average 14 micrometers in diameter. These fibers branch many times after they enter the muscle and innervate the large skeletal muscle fibers at the motor endplate. Stimulation of a single alpha nerve fiber excites

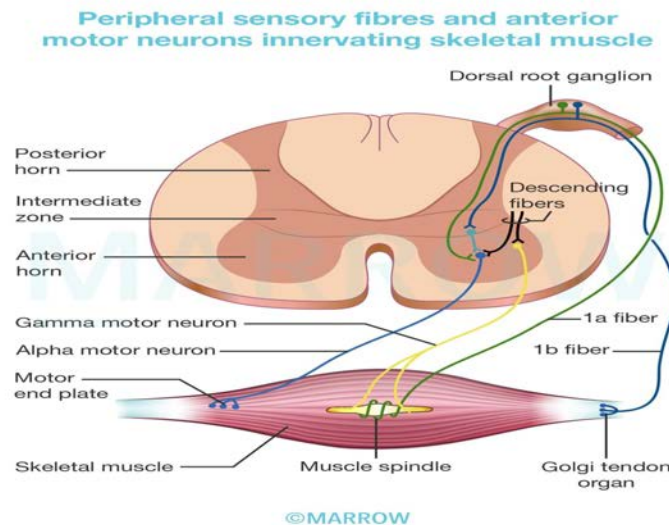
anywhere from three to several hundred skeletal muscle fibers, which are collectively called the motor unit.

Option (c):

A-beta fibers are fibers for touch and pressure.

Option (d):

A-delta fibers are fibers for pain and temperature.



Solution to Question 4:

Type II nerve fibers do innervate the static nuclear bag fibers. They do not innervate dynamic nuclear bag fibers.

A muscle spindle has:

- Intrafusal fibers - these are tiny skeletal muscles which have few or no actin and myosin filaments in the central portion. This portion acts as a sensory receptor.
- Extrafusal fibers

These fibers are positioned parallel to each other (Option A).

There are two types of intrafusal fibers:

- Nuclear bag fibers - subdivided as dynamic and static nuclear bag fibers
- Nuclear chain fibers

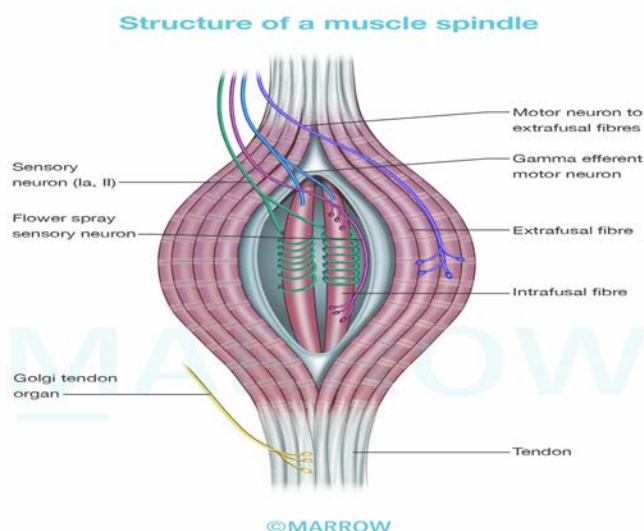
Sensory nerve supply: There are two types of sensory endings:

- Primary endings - (annulospiral ending) from Type Ia nerve fibers. This innervates both the nuclear bag and nuclear chain fibers. They are most sensitive to the velocity of the change in muscle length. This is called a dynamic response. (Option B)

- Secondary endings - from Type II nerve fibers. This innervates static nuclear bag and nuclear chain fibers. They do not innervate dynamic nuclear bag fibers. (Option C)

Both primary and secondary endings show a static response.

Motor nerve supply: The muscle spindles are supplied by γ motor neurons. There are two types of γ motor neurons- dynamic that supply dynamic nuclear bag fibers and static that supply static nuclear bag fibers and nuclear chain fibers. (Option D)



Solution to Question 5:

The nuclear bag fibers sense the dynamic length of the muscle.

Nuclear bag fibers are intrafusal fibers, which are a part of the muscle spindle. Intrafusal fibers serve a purely sensory function in the muscle spindle.

Types of intrafusal fibers in muscle spindles:

Note- Golgi tendon organ senses changes in muscle tension. Muscle spindle also provides sense of joint position.

Feature	Nuclear bag fibers	Nuclear chain fibers
Structure	Nuclei in central dilated area	Nuclei in rows
Types	StaticDynamic	----
Primary sensory afferent	Ia (for both the types)	Ia
Secondary sensory ending	II (only for static bag fiber)	II
Efferent	static & dynamic γ -motor neurons	static γ -motor neurons
Way of detection of the change in muscle length	Static & dynamic changes	Static change

Solution to Question 6:

The afferents for deep tendon reflexes innervate both - nuclear bag and chain fibers.

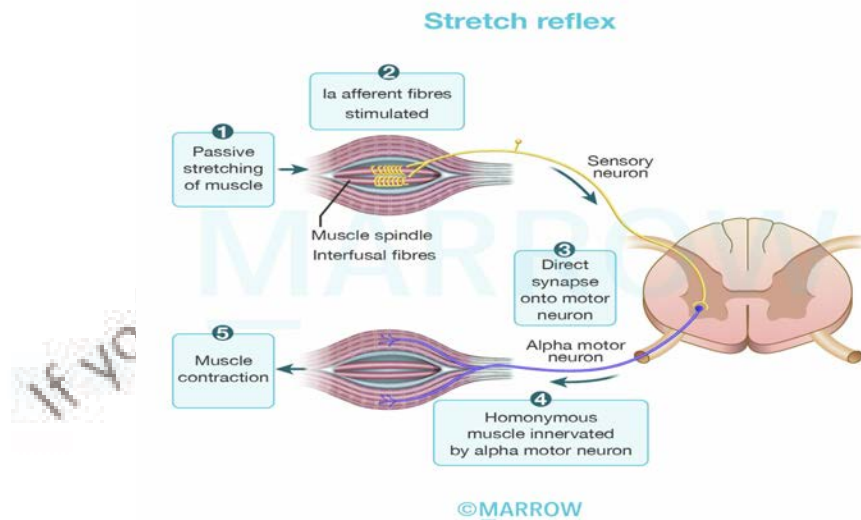
The afferents for the DTRs are the type Ia and type II nerve fibers. There are three types of intrafusal muscle fibers: dynamic nuclear bag, static nuclear bag, and nuclear chain fibers. A single Ia afferent fiber innervates all three types of fibers to form a primary sensory ending. A group II sensory fiber innervates nuclear chain and static bag fibers to form a secondary sensory ending.

Deep tendon reflexes are an example of stretch reflex or myotatic reflex.

Solution to Question 7:

Stretch reflex occurs when the muscle is stretched passively.

Stretch (myotatic) reflex is monosynaptic. (As illustrated in the figure given below, steps 1→5).



Solution to Question 8:

Withdrawal reflexes are not accompanied by the same response on both sides of the body. On the ipsilateral side of the pain stimulus, flexors are stimulated (they contract) and extensors are inhibited (they relax), and the arm is jerked away from the stove. On the contralateral side, flexors are inhibited and extensors are stimulated (crossed extension reflex) to maintain balance.

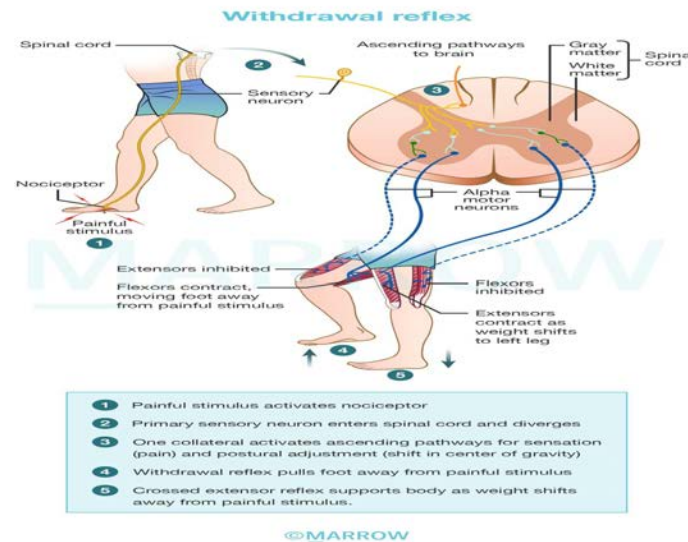
It results in flexion on the ipsilateral side and extension on the contralateral side.

Somatosensory and pain afferent fibers elicit withdrawal of the stimulated body part from the noxious stimulus.

Pain (e.g., touching a hot stove) stimulates the flexor reflex afferents of groups II, III, and IV.

The afferent fibers synapse polysynaptically (via interneurons) onto motoneurons in the spinal cord.

As a result of persistent neural activity in the polysynaptic circuits, an afterdischarge occurs. The afterdischarge prevents the muscle from relaxing for some time.



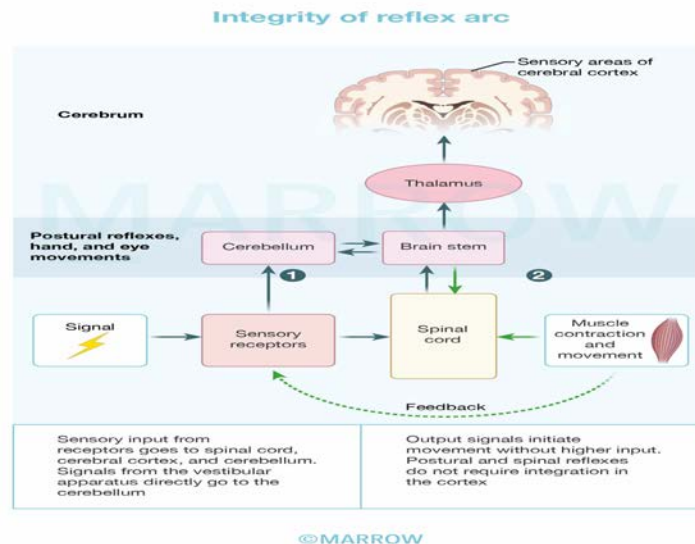
Solution to Question 9:

The maintenance of normal posture depends on the Integrity of reflex arc.

The muscles that maintain upright posture—that is, support the body's weight against gravity—are controlled by the brain and by reflex mechanisms.

Three levels of the nervous system control movement:

- The spinal cord, which integrates spinal reflexes and contains central pattern generators.
- The brain stem and cerebellum, which control postural reflexes and hand and eye movements.
- The cerebral cortex and basal ganglia, which are responsible for voluntary movements.



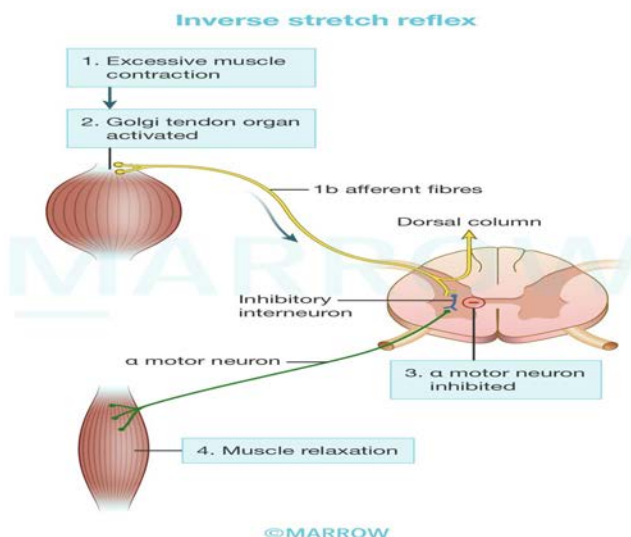
Solution to Question 10:

Golgi tendon organ is the receptor for inverse stretch reflex (refer figure).

The Golgi tendon organs, unlike the spindles, are in series with the muscle fibers; therefore, they are stimulated by both passive stretch and active contraction of the muscle.

The degree of stimulation by contraction of muscles is much greater than that caused by the stretching of muscles.

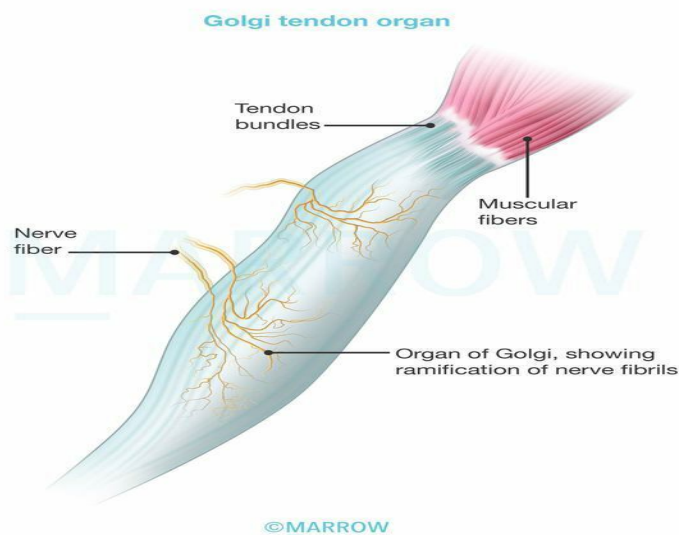
Muscle contraction → Ib fibers from the Golgi tendon organ excite an inhibitory interneuron that releases inhibitory mediator glycine → hyperpolarization (inhibition) of α motor neurons → muscle relaxation.



Solution to Question 11:

There are 3-20 muscle fibers per Golgi tendon organ.

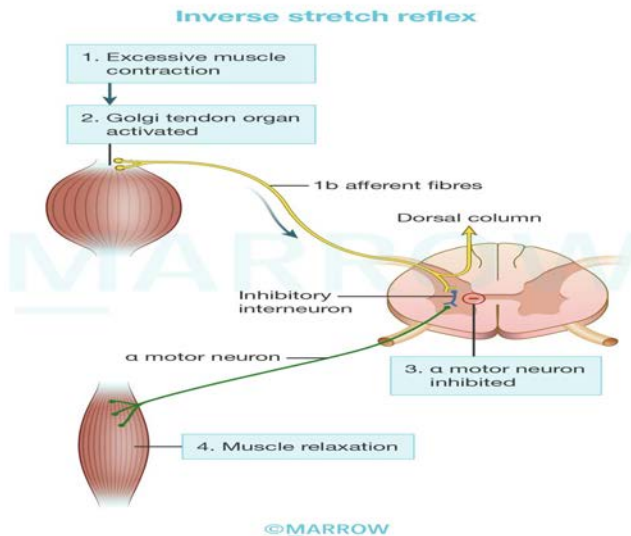
- Golgi tendon organ consists of a netlike collection of knobby nerve endings among the fascicles of a tendon.
- There are 3–25 muscle fibers per Golgi tendon organ.
- The fibers from the Golgi tendon organs make up the Ib group of myelinated, rapidly conducting sensory nerve fibers.
- Golgi tendon organs help in regulating the velocity of the muscle contraction, muscle length, and muscle force.
- They are the receptor for inverse stretch reflex.



Solution to Question 12:

Golgi tendon reflex (Inverse myotatic/Inverse stretch reflex) is disynaptic, is the opposite, or inverse, of the stretch reflex.

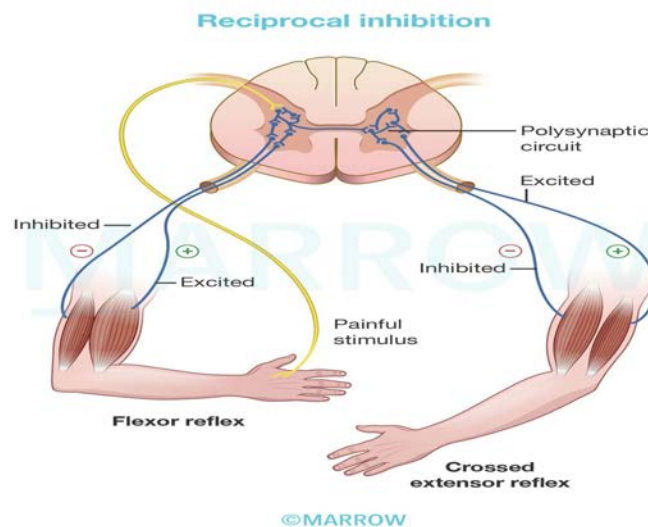
When tension on the muscle—and therefore on the tendon—becomes extreme, the inhibitory effect from the golgi tendon organ can be so great that it leads instantaneous relaxation of the entire muscle. This effect is called the lengthening reaction; it is probably a protective mechanism to prevent tearing of the muscle or avulsion of the tendon from its attachments to the bone.



Solution to Question 13:

The pathways for eliciting the flexor withdrawal reflex do not pass directly to the anterior motor neurons; instead, they first pass into the spinal cord interneuron pool of neurons and only secondarily to the motor neurons. The shortest possible circuit is a three- or four-neuron pathway. The following basic types of circuits are present:

- Diverging circuits to spread the reflex to the necessary muscles for withdrawal;
- Circuits to inhibit the antagonist muscles, called reciprocal inhibition circuits; and
- Circuits to cause after-discharge are present. They prolong the reflex contraction of the muscles that last many fractions of a second after the stimulus is over.



The flexor reflex is organized to withdraw a pained or otherwise irritated part of the body from a stimulus. Further, because of after-discharge, the reflex can hold the irritated part away from the stimulus for 0.1 to 3 seconds after the irritation is over. During this time, other reflexes and actions of the central nervous system can move the entire body away from the painful stimulus.

Solution to Question 14:

The resistance of a muscle to stretch is often referred to as its tone or tonus and muscle spindle helps to control basic muscle tone.

The tonic (static) stretch reflex circuit includes both group Ia and group II afferent fibers from muscle spindles. Normally, there is ongoing activity in the Ia and II afferents that helps maintain a baseline firing of α motor neurons; therefore, the tonic stretch reflex contributes to muscle tone.

Solution to Question 15:

The cell bodies of α and γ motor neurons to a given muscle lie close together. Both types are activated by interneurons in their immediate vicinity and sometimes directly by neurons of the descending pathways. During many voluntary and involuntary movements, they are coactivated, i.e., they are excited at almost the same time. (α - γ coactivation)

The purpose of α - γ coactivation:

- First, it keeps the length of the receptor portion of the muscle spindle from changing during the course of the whole muscle contraction. Hence, the muscle contraction is not opposed by the stretch reflex.
- Second, if the muscle spindle did not contract and relax along with the large muscle fibers, the receptor portion of the spindle would sometimes be flail and sometimes be overstretched.

Coactivation thus ensures that information about muscle length will be continuously available to provide for adjustment during ongoing actions and to plan and program future movements.

Solution to Question 16:

Righting reflexes operate to maintain the normal standing position and keep an animal's head upright. These reflexes are a series of responses integrated for the most part in the nuclei of the midbrain.

Solution to Question 17:

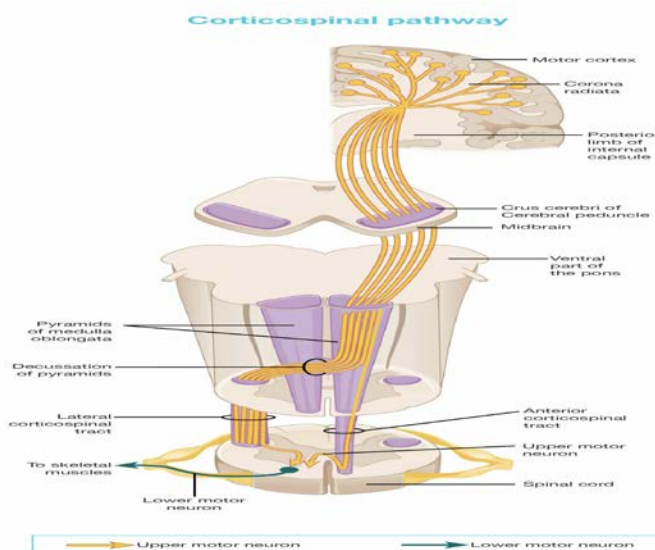
Pressure on the footpad of a decerebrate patient causes the limb to extend against the pressure applied to the foot. This reflex is called the positive supportive reaction and is a righting reflex integrated at the spinal level.

Pressure on one side causes extension in that direction (e.g., on applying pressure on the right footpad, extension of the right lower limb occurs), an effect called the magnet reaction. It is seen in infants and following spinal transection.

From an evolutionary perspective, this reflex helps to prevent falling on the side where the stimulus is given.

Solution to Question 18:

The corticospinal pathways are also called the pyramidal tracts or pyramidal system because of their triangular shape as they pass along the ventral surface of the medulla oblongata.



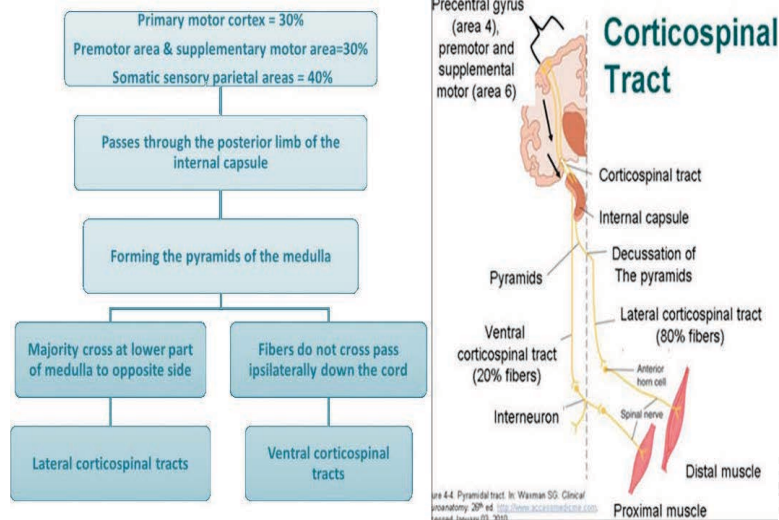
Solution to Question 19:

The corticospinal tract contains crossed fibers (majority).

Fibers of the corticospinal tract originate from the primary motor cortex (about 30%), the premotor and supplementary motor areas (30%), and the somatosensory areas posterior to the central sulcus (40%).

Most fibers (80% fibers) decussate (cross) in the medullary pyramids and descend in the lateral white matter of the spinal cord to form the lateral corticospinal tract. They control the movements of the contralateral limbs.

The remaining 20% make up the ventral corticospinal tract. These fibers remain uncrossed until reaching the spinal cord where axons terminate bilaterally on spinal interneurons antecedent to motor neurons. These fibers control the movements of the axial musculature.



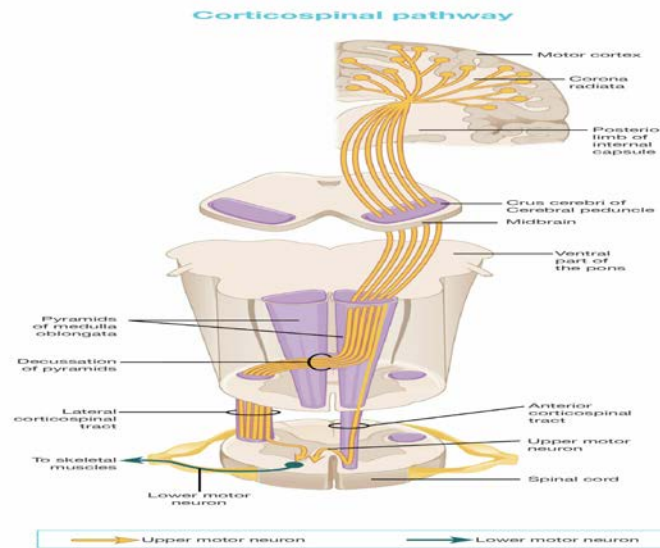
Solution to Question 20:

30% of corticospinal tract originates about from the primary motor cortex, not 90%.

30 % from the premotor and supplementary motor areas. 40 % from the somatosensory areas posterior to the central sulcus.

Betz cells

- Some large myelinated fibres in the corticospinal tract originate from giant pyramidal cells, also known as Betz cells.
- Found in the layer 5 of primary motor cortex.
- These large fibers represent only 3 percent of the total fibers of the corticospinal tract. (The other 97 percent less than 4 micrometers in diameter that conduct background tonic signals to the motor areas of the cord.)



Solution to Question 21:

Corticospinal tract is a pyramidal tract .

Pyramidal tracts (corticospinal and corticobulbar) pass through the medullary pyramids.

All others are extrapyramidal tracts and originate primarily in the the brain stem.

Solution to Question 22:

Rubrospinal tract is not involved in posture maintenance.

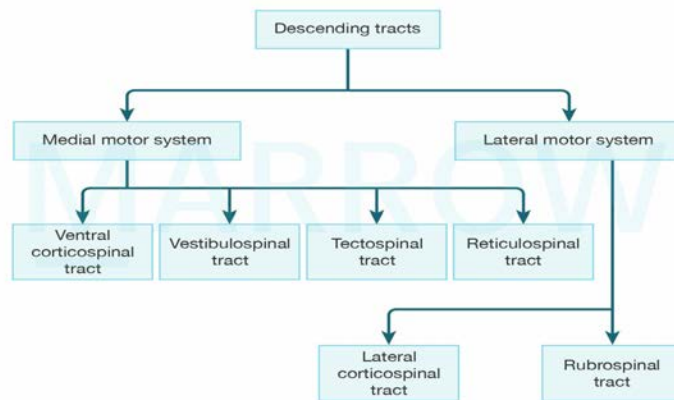
Medial motor system

- Controls the axial muscles (muscles of the trunk)
- Axial muscles are concerned with postural adjustments and gross movements

Lateral motor system

- Controls the distal muscles
- Distal muscles mediate fine, skilled movements

Descending tracts in the spinal cord



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Solution to Question 23:

- The pontine and medullary reticulospinal tracts are involved in maintenance of posture and in modulating muscle tone especially via an input to α -motor neurons.
- Pontine reticulospinal neurons are primarily excitatory and medullary reticulospinal neurons are primarily inhibitory.
- Stimulation of γ -motor neurons does not lead directly to detectable contraction of the muscles because the intrafusal fibers are not strong enough or plentiful enough to cause shortening.
- However, stimulation does cause the contractile ends of the intrafusal fibers to shorten and therefore stretches the nuclear bag portion of the spindles, deforming the endings, and initiating impulses in the Ia fibers. This in turn can lead to reflex contraction of the muscle.
- Thus, muscles can be made to contract via stimulation of the α -motor neurons that innervate the extrafusal fibers or the γ -motor neurons that initiate contraction indirectly via the stretch reflex.

Solution to Question 24:

In decerebrate rigidity, there is rigidity in extensor muscles of all the four extremities (not all muscles of the body).

Decerebration

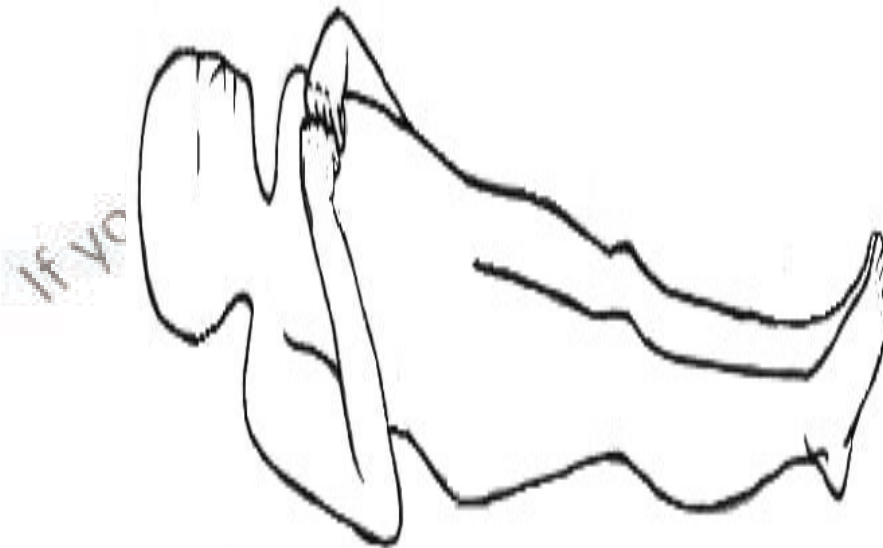
- Complete transection of the brain stem between the superior and inferior colliculi permits the brain stem pathways to function independently of their input from higher brain structures. This is called a mid-collicular decerebration.
- This lesion interrupts all input from the cortex (corticospinal and corticobulbar tracts) and red nucleus (rubrospinal tract), primarily to distal muscles of the extremities.

- The excitatory and inhibitory reticulospinal pathways (primarily to postural extensor muscles) remain intact.
- The dominance of drive from ascending sensory pathways to the excitatory reticulospinal pathway leads to hyperactivity in extensor muscles in all four extremities which are called decerebrate rigidity.
- The excitatory input from the reticulospinal pathway activates γ -motor neurons which indirectly activate α -motor neurons (via Ia spindle afferent activity). This is called the gamma loop.

Option 4. Spinal shock is different from decerebrate rigidity as it is a transient state that can occur immediately after a spinal cord injury. It is characterized by a temporary loss or depression of spinal reflexes below the level of the injury.

Solution to Question 25:

- Removal of the cerebral cortex (decortication) produces decorticate rigidity.
- Characterized by flexion of the upper extremities at the elbow and extensor hyperactivity in the lower extremities.
- The flexion can be explained by active rubrospinal excitation of flexor muscles in the upper extremities; the hyperextension of lower extremities is due to the same changes that occur after midcollicular decerebration.



Solution to Question 26:

The Hopping and Placing reflex is absent in a decorticate animal as it is integrated in the cerebral cortex. Rest of the reflexes are integrated below the cortex.

Solution to Question 27:

Destruction of the anterior cerebellum in a decerebrate animal leads to an increased rigidity via alpha motor neuron.

If the anterior lobe of the cerebellum is removed in a decerebrate animal, extensor muscle hyperactivity is exaggerated (decerebellate rigidity). This cut eliminates cortical inhibition of the cerebellar fastigial nucleus and secondarily increases excitation to vestibular nuclei. Subsequent dorsal root section does not reverse the rigidity, hence proving that the rigidity is due to activation of α -motor neurons independent of the gamma loop.

Solution to Question 28:

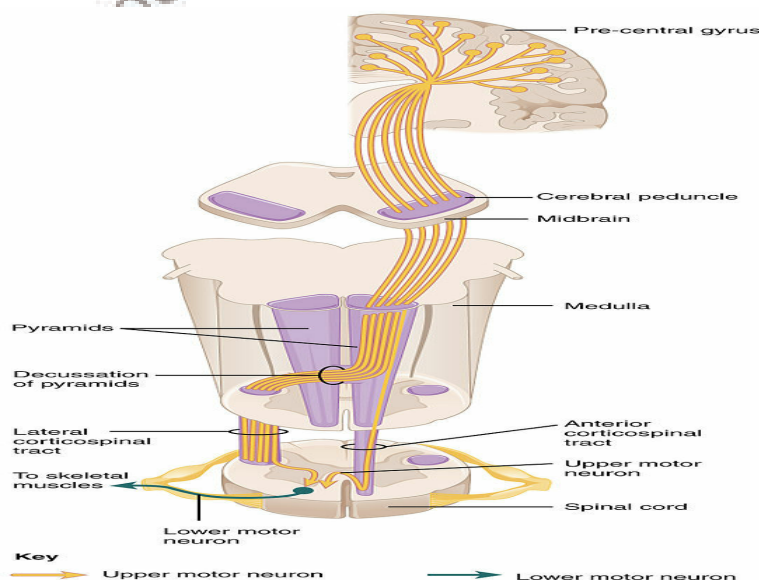
- Upper motor neurons typically refer to corticospinal tract neurons that innervate spinal motor neurons, but they can also include brainstem neurons that control spinal motor neurons.
- Corticospinal tract contain a population of large myelinated fibers with a mean diameter of 16 micrometers. These fibers originate from giant pyramidal cells, called Betz cells, that are found only in the primary motor cortex.

Solution to Question 29:

Cutting the pyramidal tract below the decussation (UMN lesion) causes abnormal plantar extensor reflex (positive Babinski sign).

It initially causes muscles to become weak and flaccid but eventually leads to spasticity, hypertonia (increased resistance to passive movement), and hyperactive stretch reflexes.

The image below shows the pyramidal tract (corticospinal pathway).



Solution to Question 30:

Cog-wheel rigidity is not seen if there is a corticospinal tract involvement. Cogwheel rigidity is seen in extrapyramidal tract involvement when there is a tremor superimposed on hypertonia (e.g., Parkinsonism).

In an upper motor neuron lesion, there is damage to the corticospinal or corticobulbar fibres. If the corticospinal fibres are damaged, it leads to diminished or loss of superficial reflexes, shows an extensor plantar response (Babinski's sign) with loss of fine skilled movement and spasticity.

Solution to Question 31:

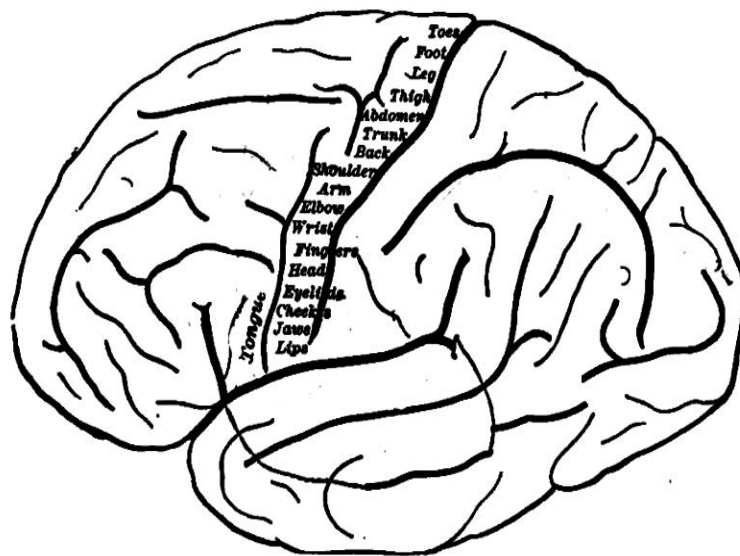
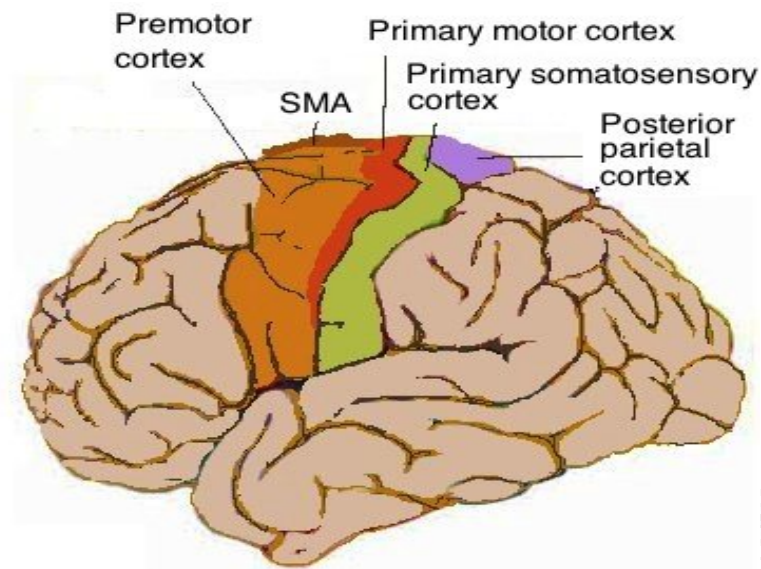
The given clinical vignette is suggestive of spinal cord transection, which is characterized by an exaggeration of reflexes and not suppression.

Upper motor neurons typically refer to corticospinal tract neurons that innervate spinal motor neurons. Damage to these neurons, as in transection of the spinal cord, initially causes muscles to become weak and flaccid but eventually leads to spasticity, hypertonia (increased resistance to passive movement), hyperactive reflexes, and abnormal plantar extensor reflex (positive Babinski sign).

Solution to Question 32:

The given clinical vignette is suggestive of an acute stroke, most probably due to atrial fibrillation induced thromboembolism. Since the patient has developed motor weakness in the right side of his body, he most likely has an infarct affecting the primary motor cortex (Brodmann area 4).

- The primary motor cortex (or just the motor cortex) can be defined as the region of the cortex from which movements are elicited with the least amount of electrical stimulation.
- It is essentially congruent with Brodmann's cytoarchitectonic area 4.
- The primary motor cortex (M1) is in the precentral gyrus of the frontal lobe, extending into the central sulcus.



Solution to Question 33:

Commands for voluntary movements originate in cortical association areas.

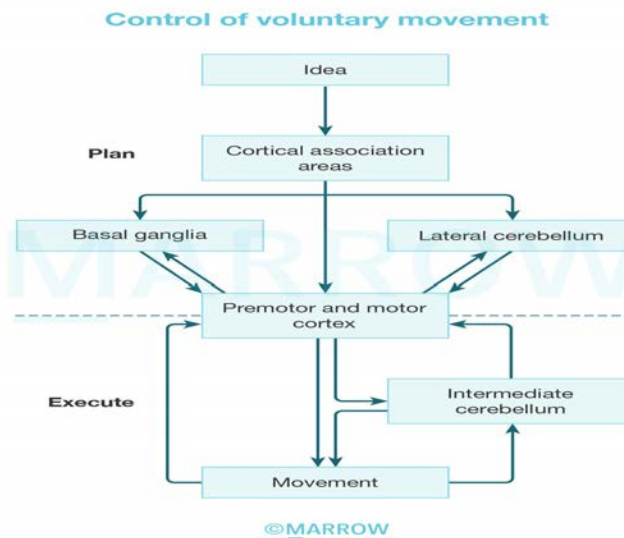
The corticospinal and corticobulbar system (originating in the motor cortex located in precentral gyrus) is the primary pathway for the initiation of skilled voluntary movement.

The movements are planned in the cortex as well as in the basal ganglia and the lateral portions of the cerebellar hemispheres.

The basal ganglia and cerebellum funnel information to the premotor and motor cortex by way of the thalamus.

Motor commands from the motor cortex are relayed in large part via the corticospinal tracts to the spinal cord.

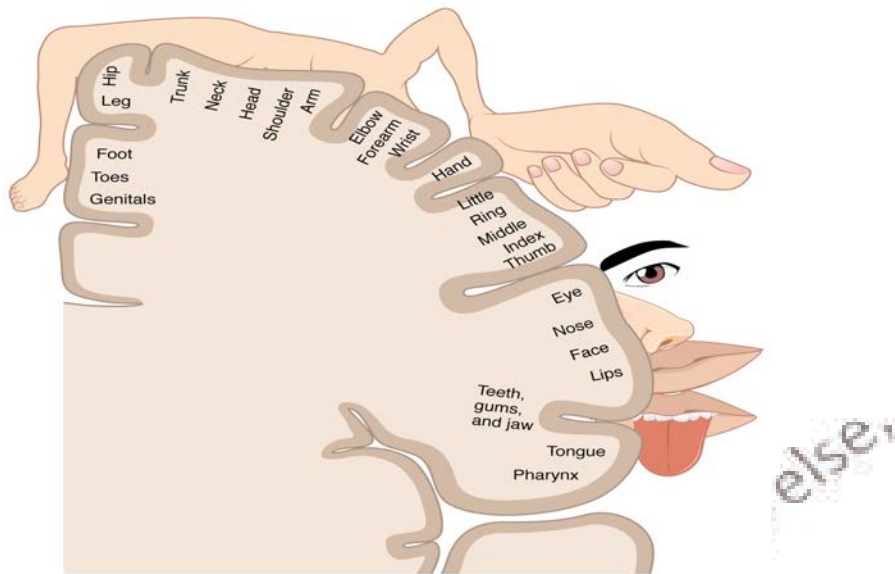
Movement sets up alterations in sensory input from the special senses and from muscles, tendons, joints, and the skin. This feedback information, which adjusts and smooths movement, is relayed directly to the motor cortex and to the spinocerebellum.



Solution to Question 34:

In a homunculus, the more the representation of an area, the more precise the movements are.

- The relative sizes of the cortical receiving areas of the sensory homunculus are proportionate to the use of the part.
- The sizes of these areas are directly proportional to the number of specialized sensory receptors in each respective peripheral area of the body.
- Some body parts are represented by large areas in the somatic cortex—the lips the greatest of all, followed by the face and thumb, whereas the trunk and lower part of the body are represented by relatively small areas.

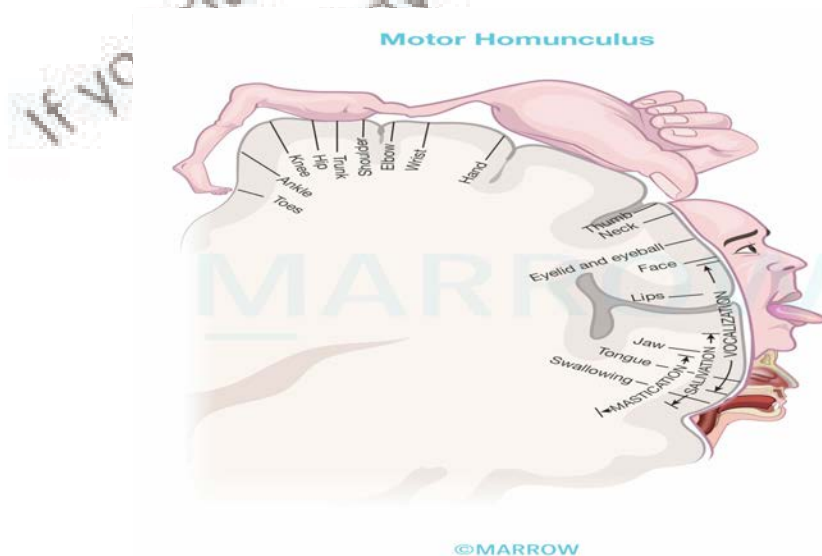


Solution to Question 35:

As seen from the figure given below, Vocalization & Salivation & Mastication & Thumb

The topographical organization is demonstrated graphically in the figure, which shows the degrees of representation of the different muscle areas as mapped by Penfield and Rasmussen. This mapping was done by electrically stimulating the different areas of the motor cortex in human beings who were undergoing neurosurgical operations.

The face is represented bilaterally. The rest of the body has a unilateral representation.



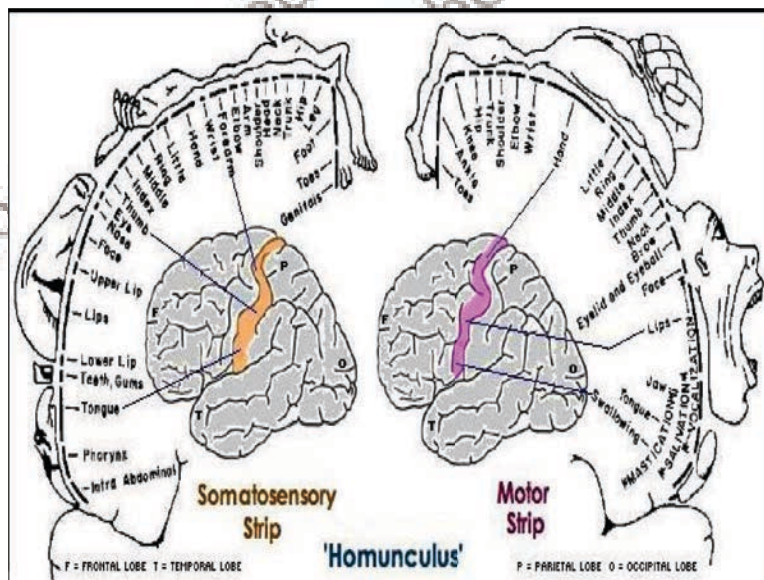
Solution to Question 36:

Damage to both paracentral lobules will cause paraplegia.

The paracentral lobule is the medial continuation of the precentral and postcentral gyri. It controls motor and sensory innervations of the contralateral lower extremity. It is also responsible for the control of defecation and urination.

Solution to Question 37:

- The premotor cortex may be concerned with a setting posture at the start of a planned movement and getting the individual prepared to move. It is most involved in control of proximal limb muscles needed to orient the body for movement.
- It is located anterior to the precentral gyrus, on the lateral and medial cortical surface; it also contains a somatotopic map almost identical to the topographic organization in primary motor cortex with the mouth and face areas located most laterally. As one moves upward, the hand, arm, trunk, and leg areas are encountered. (as seen in the figure below)
- This region receives input from sensory regions of the parietal cortex and projects to primary motor cortex (M1), the spinal cord, and the brainstem reticular formation.
- A special class of neurons called mirror neurons becomes active when a person performs a specific motor task or when he or she observes the same task performed by others. These neurons transform sensory representations of acts that are heard or seen into motor representations of these acts.



Solution to Question 38:

The given clinical vignette is suggestive of spinal mass reflex which is seen in patients with a high spinal lesion. A major portion of the body's skeletal muscles goes into strong flexor spasm, and not extensor spasm, in a spinal mass reflex.

Mass Reflex: In a spinal animal or human being, sometimes the spinal cord suddenly becomes excessively active, causing massive discharge in large portions of the cord. The usual stimulus that causes this excess activity is a strong pain stimulus to the skin or excessive filling of a viscus, such as over-distention of the bladder or the gut. The mass reflex involves large portions or even all of the cord.

The effects are as follows:

- A major portion of the body's skeletal muscles goes into strong flexor spasms;
- The colon and bladder are likely to evacuate;
- The arterial pressure often rises to maximal values, sometimes to a systolic pressure well over 200 mm Hg; and
- Large areas of the body break out into profuse sweating.

The mechanism is similar to the mechanism of epileptic seizures, which involve reverberating circuits that occur in the brain instead of in the cord.

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Motor Physiology - 2

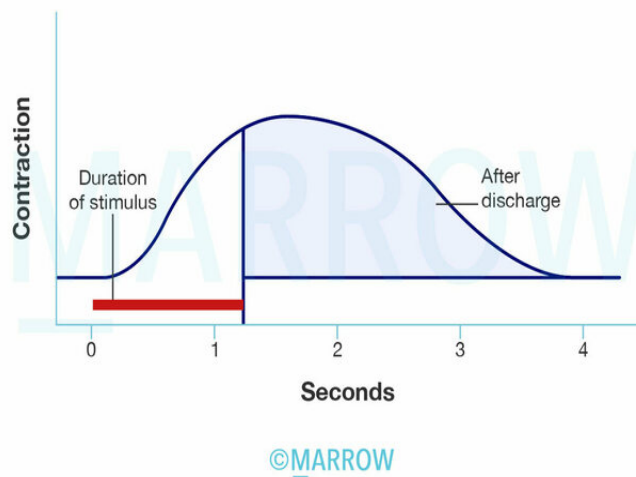
Question 1:

When dynamic γ -motor neurons are activated at the same time as α -motor neurons to muscle,

- a) Prompt inhibition of discharge in spindle Ia afferents takes place.
- b) The muscle will not contract.
- c) The number of impulses in spindle Ia afferents is smaller than when α discharge alone is increased.
- d) The number of impulses in spindle Ia afferents is greater than when α discharge alone is increased.

Question 2:

The myogram below shows a slow onset and a longer after-discharge. Which type of reflex does it represent?



- a) Flexor withdrawal reflex
- b) Stretch reflex
- c) Crossed extensor reflex
- d) Inverse stretch reflex

Question 3:

The bulboreticular facilitatory area of the brain stem excites the gamma efferent system. This bulboreticular area receives impulses from which of the following areas?

- a) Cerebellum
- b) Basal Ganglia
- c) Cerebral Cortex
- d) All of the above

Question 4:

Which of the following describes components of the central pathway responsible for control of posture?

- a) The pontine reticulospinal pathway terminates on neurons in the dorsomedial area of the spinal ventral horn that innervate flexor muscles.
- b) The tectospinal pathway terminates on neurons in the dorsolateral area of the spinal ventral horn that innervate limb muscles.
- c) The medullary reticulospinal pathway terminates on neurons in the ventromedial area of the spinal ventral horn that innervate axial and proximal muscles.
- d) None of the above

Question 5:

Which of the following is not a medial pathway involved in maintenance of posture?

- a) Reticulospinal Tract
- b) Rubrospinal Tract
- c) Vestibulospinal Tract
- d) Tectospinal Tract

Question 6:

Increased neural activity before a skilled voluntary movement is first seen in ____.

- a) precentral motor cortex
- b) midbrain
- c) cerebellum

- d) cortical association areas

Question 7:

Which of the following descending motor tracts is essential for rapid skilled movement?

- a) Anterior corticospinal tract
- b) Rubrospinal tract
- c) Lateral vestibulospinal tract
- d) Medullary Reticulospinal tract

Question 8:

Which of the following is true about experimental sham rage?

- a) Occurs in decorticate animals
- b) Is goal directed
- c) Is caused by removing the hypothalamus
- d) Appears and goes slowly

Question 9:

A 62-year-old known hypertensive woman presented to the emergency with spasticity of both lower limbs. Further neurological examination showed regular, repetitive, rhythmic contractions of a muscle subjected to sudden, maintained stretch. Which of the following is false regarding this phenomenon?

- a) Only sustained clonus with five or more beats is considered abnormal
- b) Clonus may also occur in spinal cord injury that disrupts the descending cortical input to a spinal glycinergic inhibitory interneuron called the Renshaw cell
- c) The stretch reflex–inverse stretch reflex sequence may contribute to this response
- d) This is characteristic of states in which decreased γ -motor neuron discharge is present

Question 10:

Which of the following statements regarding a clasp knife effect is false?

- a) Hypertonia

- b) Golgi Tendon Organ is involved
- c) It is called Lengthening Reaction
- d) Pacinian Corpuscle is involved

Question 11:

Fill in the blanks: Efferent axons from the dentate nucleus terminate in _____ thalamic nucleus and travel through _____ peduncle?

- a) Ventral anterior ---middle cerebellar
- b) Ventral lateral---superior cerebellar
- c) Ventral lateral ---middle cerebellar
- d) Ventral posteromedial ---superior cerebellar

Question 12:

Which of the following does not carry proprioceptive impulses?

- a) Olivo cerebellar tract
- b) Tecto cerebellar tract
- c) Spino cerebellar tract
- d) Cuneo cerebellar tract

Question 13:

A patient has developed an inability to control the distance, speed, and range of motion necessary to perform smoothly coordinated movements. His finger-to-nose test and heel-to-shin tests are abnormal. A lesion in which of the following sites is most likely to be found?

- a) Basal Ganglia
- b) Spinocerebellar tract
- c) Midbrain
- d) Pons

Question 14:

Dysdiadochokinesia refers to which of the following?

- a) Inability to perform the Tandem-Gait test
- b) Inability to stand without swaying with eyes open
- c) Inability to control vestibulo-ocular reflexes
- d) Inability to perform rapid alternating movements with the hands

Question 15:

The patient exhibited an unsteady "drunk like" gait and the physician noted that he swayed while entering the examination room. Which region of the cerebellum was most likely affected by the bleed?

- a) Flocculonodular Lobe
- b) Spinocerebellum
- c) Cerebrocerebellum
- d) Superior Vermis

Question 16:

The efferent fibres of the basal ganglia include which of the following?

- a) 1, 2
- b) 1, 4
- c) 1, 2, 4
- d) 1, 2, 3, 4

Question 17:

Which of the following describes a connection between components of the basal ganglia?

- a) The subthalamic nucleus releases glutamate to excite the globus pallidus, internal segment.
- b) The substantia nigra pars reticulata releases dopamine to excite the globus pallidus, external segment.
- c) The striatum releases acetylcholine to excite the substantia nigra pars reticulata.
- d) The globus pallidus, external segment releases glutamate to excite the striatum.

Question 18:

Which of the following is false regarding the functions of basal ganglia ?

- a) Neurons in the basal ganglia, like those in the lateral portions of the cerebellar hemispheres, discharge before movements begin.
- b) The putamen circuit has its inputs mainly from the parts of the brain adjacent to the primary motor cortex but not much from the primary motor cortex itself.
- c) The caudate nucleus plays a major role in this cognitive control of motor activity.
- d) Substantia nigra pars reticulata is a major source of dopaminergic neurons to the corpus striatum.

Question 19:

In a patient suffering from parkinsons's disease, what is the most likely reason behind the gradual decline in the therapeutic effect of L-dopa?

- a) Antibodies to dopamine receptors develop.
- b) Inhibitory pathways grow into the basal ganglia from the frontal lobe.
- c) There is an increase in circulating α -synuclein.
- d) The dopaminergic neurons in the substantia nigra continue to degenerate.

Question 20:

A 50-year-old gentleman who is a known patient of parkinson's disease, develops refractoriness to L-dopa therapy. An alternative treatment modality is being planned, which employs ablation of certain thalamic regions by neurosurgery. Which thalamic nucleus is the stereotaxic target for the neurosurgeons?

- a) Centromedian
- b) Ventrolateral
- c) Ventroposterolateral
- d) Ventromedial

Answer Key

Question No.	Correct Option
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1	d
2	c
3	d
4	c
5	b
6	d
7	b
8	a
9	d
10	d
11	b
12	b
13	b
14	d
15	a
16	b
17	a
18	d
19	d
20	b

Detailed Explanations

Solution to Question 1:

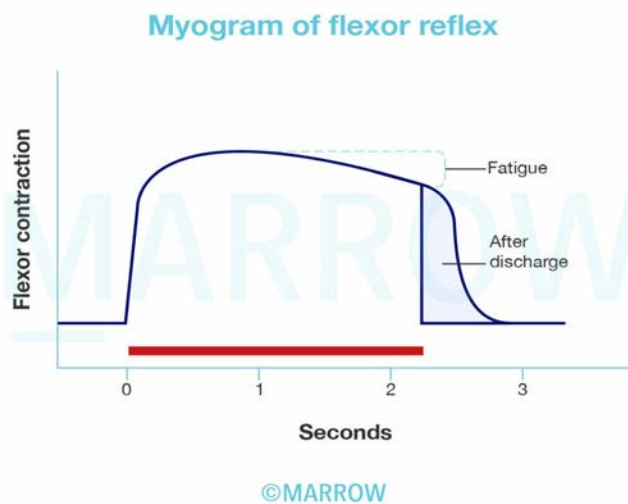
When dynamic γ -motor neurons are activated at the same time as α -motor neurons to muscle, the number of impulses in spindle Ia afferents is greater than when α discharge alone is increased.

- When α – γ coactivation occurs, intrafusal and extrafusal fibers shorten together, and spindle afferent activity can occur throughout the period of muscle contraction.
- Stimulation of γ -motor neurons causes increase discharge in spindle Ia afferent, leading to reflex contraction of the muscle.
- When a muscle contracts it leads to shortening of the muscle spindles and so impulses in the Ia afferents decreases.
- Thus, muscles can be made to contract via stimulation of the α -motor neurons that innervate the extrafusal fibers or the γ -motor neurons that initiate contraction indirectly via the stretch reflex.
- If the whole muscle is stretched during stimulation of the γ -motor neurons, the rate of discharge in the Ia fibers is further increased. Increased γ -motor neuron activity thus increases spindle sensitivity during stretch. This is the basis of the Jendrassik manouver.

Solution to Question 2:

The figure shows the typical myogram recorded from a muscle involved in a crossed extensor reflex. This myogram demonstrates the relatively long latency before the reflex begins and the long after-discharge at the end of the stimulus.

- When a painful stimulus is applied to the body, the flexor withdrawal and the crossed extensor reflexes are stimulated.
- About 0.2 to 0.5 seconds after the flexor reflex in one limb, the opposite limb begins to extend. This reflex is called the crossed extensor reflex.
- After the painful stimulus is removed, the crossed extensor reflex has an even longer period of after-discharge than does the flexor reflex. This prolonged after-discharge results from reverberating circuits among the inter-neuronal cells.



Compare the myograms of a flexor reflex with a crossed extensor reflex.

In the flexor reflex, the contraction occurs immediately after applying the stimulus and the after-discharge is smaller.

Solution to Question 3:

Impulses are transmitted into the bulboreticular area from

- Cerebellum,
- Basal ganglia, and
- Cerebral cortex.

The gamma efferent system is excited by signals from the bulboreticular facilitatory region of the brain stem.

When the bulboreticular facilitatory region transmits excitatory signals through the gamma nerve fibers to the intrafusal muscle fibers of the muscle spindles, it shortens the ends of the spindles and stretches the central receptor regions, thus increasing their signal output. If the spindles on both sides of each joint are activated at the same time, then reflex excitation of the skeletal muscles on both sides of the joint increases, producing tight, tense muscles opposing each other at the joint. The net effect is that the position of the joint becomes strongly stabilized.

Any time a person must perform a muscle function that requires a high degree of delicate and exact positioning, the bulboreticular facilitatory region of the brain stem stabilizes the positions of the proximal joints.

Solution to Question 4:

The medullary reticulospinal pathway terminates on neurons in the ventromedial area of the spinal ventral horn, that innervate axial and proximal muscles.

The medullary reticular nuclei transmit inhibitory signals to the antigravity anterior motor neurons by way of the medullary reticulospinal tract, located in the lateral column of the cord.

Option A: The pontine reticular nuclei transmit excitatory signals through the pontine reticulospinal tract in the anterior column of the cord. The fibers of this pathway terminate on the medial anterior motor neurons that excite the axial muscles of the body, that is the muscles of the vertebral column and the extensor muscles of the limbs.

Option B: The tectospinal tract originates in the superior colliculus of the midbrain. It projects to the contralateral cervical spinal cord to control head and eye movements.

Solution to Question 5:

The rubrospinal tract is a part of the lateral brainstem pathway.

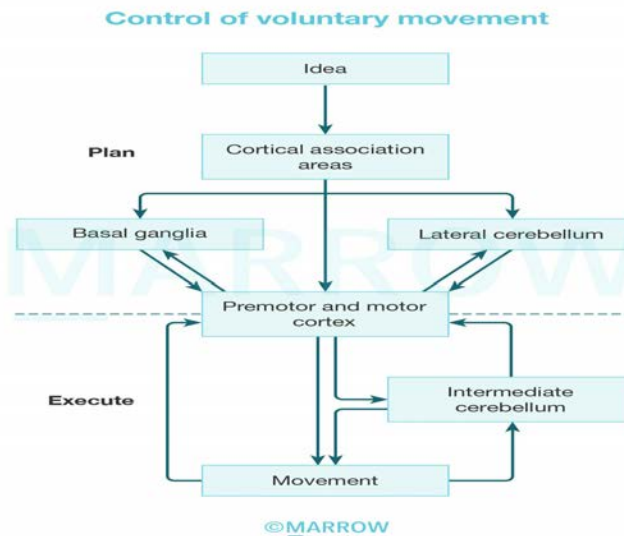
- The medial brainstem pathways : pontine and medullary reticulospinal, vestibulospinal, and tectospinal tracts.
- These pathways descend in the ipsilateral ventral columns of the spinal cord and terminate predominantly on interneurons in the ventromedial part of the ventral horn to control axial and proximal muscles.
- The lateral brainstem pathways : The corticospinal tract and the rubrospinal tract.
- They cross the midline and project to interneurons in the dorsolateral part of the spinal ventral horn to also influence motor neurons that control distal limb muscles.
- The rubrospinal tract excites flexor motor neurons and inhibits extensor motor neurons. This pathway may play a role in the posture typical of decorticate rigidity.

Solution to Question 6:

The command for voluntary movement originates in cortical association areas, then planned in the cortex as well as in the basal ganglia and the lateral portions of the cerebellar hemispheres. Hence, the neural activity first originates in the cortical association areas.

The basal ganglia and cerebellum funnel information to the premotor and motor cortex by way of the thalamus.

Motor commands from the motor cortex are relayed in large part via the corticospinal tracts to the spinal cord.



- Motor Association Area: Idea about voluntary movement
- Basal Ganglia: Planning and Organising the act
- Cerebellum(Neocerebellum) - Planning the act
- Premotor Cortex: Setting the posture before the act
- Primary Motor Cortex: Executing the act
- Spinocerebellum : Smoothening the act

Solution to Question 7:

The lateral corticospinal and rubrospinal tracts are concerned with distal limb muscles and, with skilled voluntary movements.

The ventral corticospinal tract and medial descending brainstem pathways (tectospinal, reticulospinal, and vestibulospinal tracts) are concerned with the axial muscles - postural adjustments and gross movements.

Note: Medial portion of the ventral horn innervate proximal limb muscles, particularly the flexors. Lateral ventral horn innervates distal limb muscles.

Solution to Question 8:

Sham rage occurs in decorticate animals is the true statement.

Sham rage is a controversial term used to describe behavior such as biting, clawing, hissing, arching the back and "violent alternating limb movements" produced in animal experiments by removing the cerebral cortex, which is claimed to occur in the absence of any sort of inner experience of rage.

The reason for this might be that the removal of the neocortex causes a loss of inhibition of the areas involved in the rage reaction, causing those areas to become hyperactive (due to an uninhibited hypothalamus).

Removal of hypothalamus abolishes the sham rage.

Sham rage may be triggered by very mild stimuli such as weak touch. It is also non-goal directed.

Solution to Question 9:

The case scenario indicates clonus. A characteristic of states in which increased γ -motor neuron discharge is present is clonus.

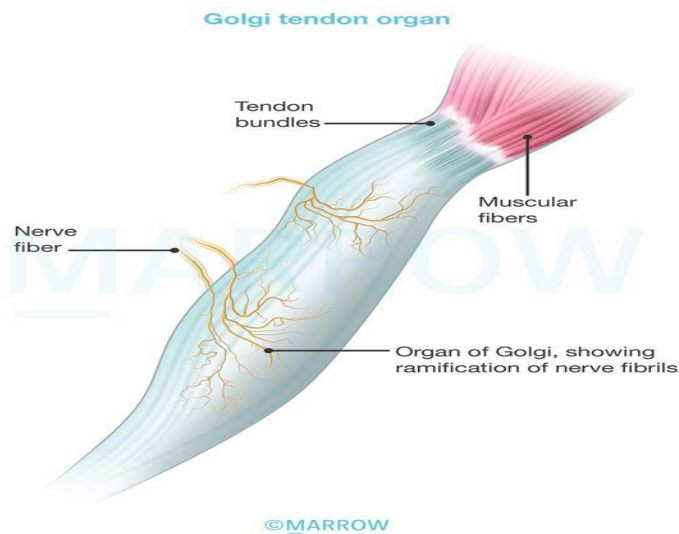
All the other options are correct.

When a muscle is subjected to a sudden and maintained stretch, there occur regular, repetitive, and rhythmic contractions. This is called clonus. Ankle clonus is a typical example.

Solution to Question 10:

Pacinian corpuscle is not involved in the clasp knife effect.

- When the muscles are hypertonic, the sequence of moderate stretch leads to muscle contraction, strong stretch leads to muscle relaxation. Eg. Passive flexion of the elbow meets immediate resistance due to stretch reflex in the triceps muscle. Further stretch activates the inverse stretch reflex. The resistance to flexion suddenly collapses, and the arm flexes. This sequence of resistance followed by a sudden decrease in resistance when a limb is moved passively is known as the clasp-knife effect because of its resemblance to the closing of a pocket knife.
- Lengthening reaction: When the tension on the muscle becomes extreme, the inhibitory effect from the Golgi tendon organ leads to an instantaneous relaxation of the entire muscle. It is a protective mechanism to prevent tearing of the muscle.



Solution to Question 11:

Efferent axons from the dentate nucleus terminate in Ventral lateral nucleus of the thalamus (VL) and travel through superior cerebellar peduncle.

The cerebellar cortex projects to the deep nuclei, which in turn project to the brainstem. The deep nuclei provide the only output for the spinocerebellum and the cerebrocerebellum:

- The medial spinocerebellum projects to the fastigial nuclei and from there to the brainstem.
- The adjacent hemispheric portions of the spinocerebellum project to the emboliform and globose nuclei and from there to the brainstem.
- The cerebrocerebellum projects to the dentate nucleus (Purkinje cells of cerebellum relay into this nucleus) and from there either directly or indirectly to the ventrolateral nucleus of the thalamus.

These fibres pass through the cerebellar peduncles:

- The superior cerebellar peduncle includes fibers from deep cerebellar nuclei that project to the brainstem, red nucleus, and thalamus. It also contains the ventral spinocerebellar tract fibers.
- The middle cerebellar peduncle contains only afferent fibers from the contralateral pontine nuclei.
- Inferior cerebellar peduncle a mixture of afferent fibers from the brainstem and spinal cord and efferent fibers to the vestibular nuclei.

Solution to Question 12:

- The tecto cerebellar tract transmits only the auditory and visual impulses to the cerebellum.

- The olivocerebellar pathway projects to the cerebellar cortex via climbing fibers; the rest of the listed paths project via mossy fibers.

Afferent Tracts	Transmits
Vestibulocerebellar	vestibular impulses from labyrinths, directs & via vestibular nuclei
Dorsal spinocerebellar	Proprioceptive & exteroceptive impulses from muscle spindles, Golgi tendon organ & joint receptors of lower limbs & trunk
Ventral spinocerebellar	Proprioceptive & exteroceptive impulses from muscle spindles, Golgi tendon organ & joint receptors of upper & lower limbs
Cunocerebellar	Proprioceptive & exteroceptive impulses from muscle spindles, Golgi tendon organ & joint receptors of upper limb & upper thorax
Tectocerebellar	Auditory & visual impulses via inferior & superior colliculi, respectively
Pontocerebellar	Impulse from motor & other parts of cerebral cortex via pontine nuclei
Olivocerebellar	Proprioceptive input from whole body via relay in inferior olive

Solution to Question 13:

The given clinical findings are suggestive of dysmetria which is most commonly found due to a lesion involving the cerebellum or the cerebellar pathways. Therefore, a lesion in the spinocerebellar tract is most likely to be found in this patient.

- Dysmetria - In the absence of the cerebellum, the subconscious motor control system cannot predict how far movements will go. Therefore, the movements ordinarily overshoot their intended mark; then the conscious portion of the brain overcompensates in the opposite direction for the succeeding compensatory movement. Dysmetria also results from lesions in the spinocerebellar tracts because feedback information from the moving parts of the body to the cerebellum is essential for cerebellar timing of movement termination.

Solution to Question 14:

Dysdiadochokinesia—Inability to Perform Rapid Alternating Movements.

- In cerebellar lesions, the motor control system fails to predict where the different parts of the body will be at a given time. As a result, the succeeding movement may begin much too early or much too late and no orderly “progression of movement” can occur.
- So, a series of stalled attempted but jumbled movements occurs instead of the normal coordinated movements. This condition is called dysdiadochokinesia.

Solution to Question 15:

Flocculonodular Lobe is involved in the unsteady "drunk like" gait of the patient.

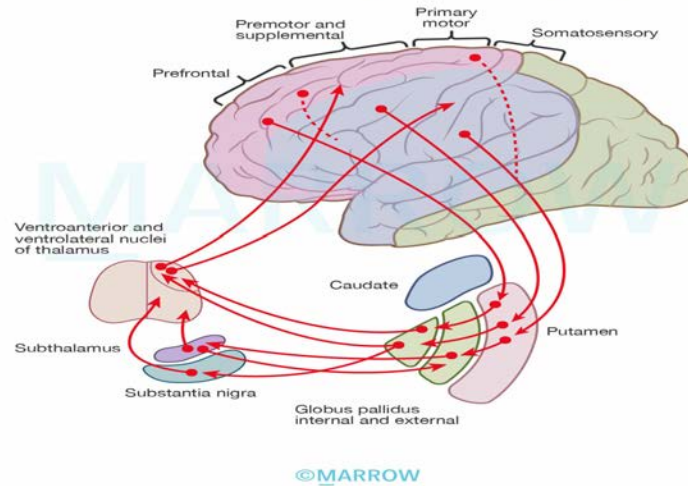
Cerebellum has three parts based on function.

- Vestibulocerebellum (or flocculonodular lobe): The oldest part of the cerebellum which contains nodulus in the vermis and the flanking flocculus. It has vestibular connections and is concerned with equilibrium and gait. Loss of it can cause disturbance of equilibrium and postural movements, especially during rapid movements.
- Spinocerebellum: The region that receives proprioceptive input from the body as well as a copy of the “motor plan” from the motor cortex. By comparing the plan with performance, it smooths and coordinates movements that are ongoing. Superior vermis is a part of spinocerebellum. Loss of it can cause problems with coordination.
- Cerebrocerebellum: They are the newest from a phylogenetic point of view, reaching their greatest development in humans. They interact with the motor cortex in planning and programming movements.

Solution to Question 16:

The two major outputs/ efferents of the basal ganglia are from Globus pallidus interna and substantia nigra pars reticulata. Both are inhibitory (GABAergic) and both project to the ventrolateral and ventroanterior thalamus.

Putamen circuit through the basal ganglia

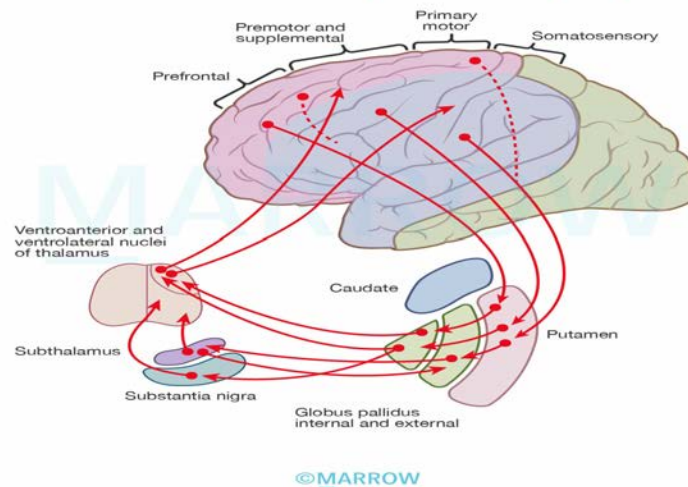


Solution to Question 17:

Intranuclear Connections:

- Dopaminergic nigrostriatal projection from the substantia nigra pars compacta to the striatum. GABAergic projection from the striatum to substantia nigra pars reticulata.
- There is an inhibitory (GABAergic) projection from the striatum to both GPe and GPi.
- The subthalamic nucleus receives an inhibitory input from GPe, and in turn the subthalamic nucleus has an excitatory (glutamate) projection to both GPe and GPi.
- From the ventrolateral thalamus, there is an excitatory (presumably glutamate) projection to the prefrontal and premotor cortex. This completes a full cortical-basal ganglia-thalamic-cortical loop.

Putamen circuit through the basal ganglia



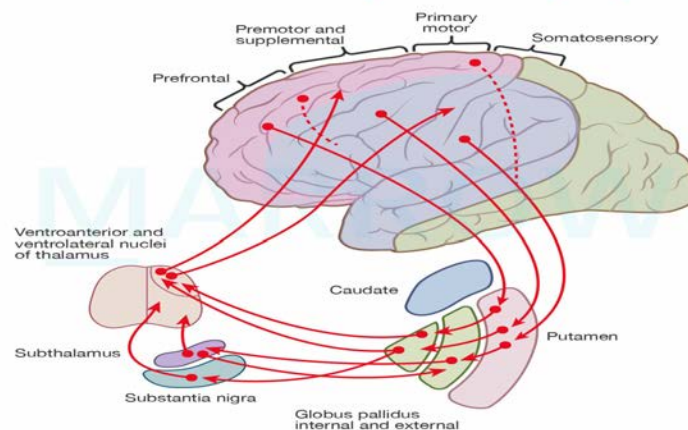
Solution to Question 18:

Substantia nigra pars compacta (not pars reticulata) is the major source of dopaminergic neurons to the striatum.

Neurons in the basal ganglia, like those in the lateral portions of the cerebellar hemispheres, discharge before movements begin since the basal ganglia are involved in the planning and organization of skilled movements.

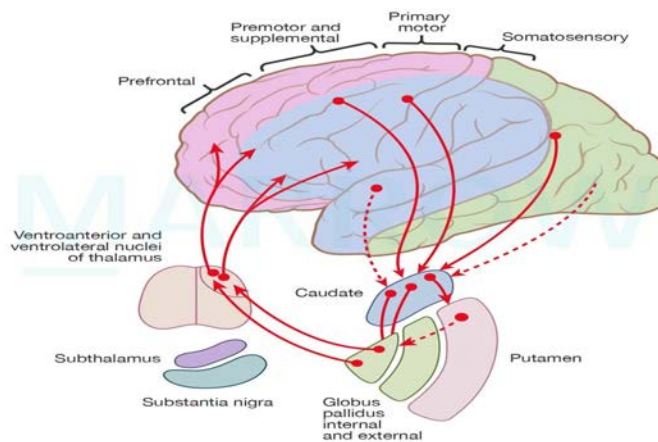
- The putamen circuit has its inputs mainly from the parts of the brain adjacent to the primary motor cortex. Then its outputs do go mainly back to the primary motor cortex and adjacent premotor cortex. It is mainly involved in complex motor activity and skilled movement.
- The caudate circuit plays a major role in the cognitive control of motor activity.

Putamen circuit through the basal ganglia



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Caudate circuit through the basal ganglia



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Solution to Question 19:

The most likely reason behind the gradual decline in the therapeutic effect of L-dopa in a patient suffering from parkinson's disease is the continued degeneration of the dopaminergic neurons in the substantia nigra. Since endogenous production of dopamine continues to decline, the exogenous source of dopamine (i.e. L-dopa) alone fails to maintain the same level of therapeutic effect.

- Parkinson's disease (Paralysis Agitans) was shown to result from the degeneration of dopaminergic neurons in the substantia nigra pars compacta. The fibres to the putamen (part of the striatum) are most severely affected.
- Dopaminergic neurons and dopamine receptors are steadily lost with age in the basal ganglia in healthy individuals, and an acceleration of these losses apparently precipitates parkinsonism. Symptoms appear when 60–80% of the nigrostriatal dopaminergic neurons degenerate.

Solution to Question 20:

The ventrolateral thalamic nucleus is the stereotaxic target for the neurosurgeons to alleviate symptoms in Parkinson's disease refractory to levodopa.

- The two major outputs of the basal ganglia are from GPi and substantia nigra pars reticulata. Both are inhibitory (GABAergic) and both project to the ventrolateral and ventroanterior thalamus. From the thalamus, there is an excitatory (presumably glutamate) projection to the prefrontal and premotor cortex. This completes a full cortical-basal ganglia-thalamic-cortical loop. Therefore stereotactic ablation of ventrolateral thalamic nucleus helps in alleviation of symptoms in parkinson's disease refractory to levodopa.

Basal Ganglia and Cerebellum

Question 1:

Which of the following is a nucleus of basal ganglia?

- a) Dentate
- b) Thalamus
- c) Caudate
- d) Red nucleus

Question 2:

Which part of the basal ganglia receives the afferent input?

- a) Striatum
- b) Globus pallidus
- c) Striatum nigra
- d) Subthalamic nuclei

Question 3:

A patient is found to have a hemorrhage in the basal ganglia region. Which of the following is least likely to be affected in this patient?

- a) Planning of movements
- b) Cognition
- c) Maintenance of equilibrium
- d) Skilled movements

Question 4:

Glutamate is synthesized mainly in which part of basal ganglia?

- a) Striatum
- b) Globus pallidus

- c) Striatum nigra
- d) Subthalamic nuclei

Question 5:

A 56-year-old woman presents with abnormal movements of the limbs. A lesion of the globus pallidus is suspected. Which of the following did she most likely present with?

- a) Chorea
- b) Athetosis
- c) Hemiballismus
- d) Parkinsonism

Question 6:

Cells present in cerebellar cortex are all except

- a) Purkinje
- b) Bipolar
- c) Granule
- d) Golgi

Question 7:

All are cerebellar nuclei except_____.

- a) Globose
- b) Emboliform
- c) Dentate
- d) Red nucleus

Question 8:

Efferent from cerebellar cortex is through which cell:

- a) Purkinje
- b) Bipolar

- c) Granule
- d) Golgi

Question 9:

Flocculonodular lobe has direct connections with _____

- a) Red nucleus
- b) Inferior olivary nucleus
- c) Vestibular nucleus
- d) Dentate nucleus

Question 10:

Which of the following are climbing fibers?

- a) Olivocerebellar fibers
- b) Spinocerebellar
- c) Pontocerebellar
- d) Vestibulocerebellar

Question 11:

Which of the following is correct?

- a) Spino cerebellum constitutes vermis only
- b) Spino cerebellum constitutes vermis and lateral portion of cerebellar hemisphere
- c) Vestibulo cerebellum constitutes nodulus and flocculus
- d) Vestibulo cerebellum constitutes lateral portion of cerebellar hemispheres

Question 12:

Which of the following is a function of the neocerebellum?

- a) Maintenance of equilibrium
- b) Correction of voluntary movements
- c) Planning and programming of voluntary movements

- d) Maintenance of muscle tone

Question 13:

A patient with a cerebellar lesion will show all of the following features except:

- a) Nystagmus
- b) Resting tremor
- c) Ataxia
- d) Past pointing

Question 14:

The vestibulo-ocular reflex is being performed in a patient with vertigo. Which of the following is associated with this reflex?

- a) Spinocerebellum
- b) Flocculonodular lobe
- c) Neocerebellum
- d) Occipital lobe

Question 15:

Purkinje fibers are inhibitory for _____.

- a) Deep cerebellar nuclei
- b) Climbing fibers
- c) Granule cells
- d) Spino cerebellar tracts

Question 16:

Which of the following fibres reach the Purkinje cell directly?

- a) Inferior olivary nucleus
- b) Raphe nucleus
- c) Vestibular nucleus

d) Locus coeruleus

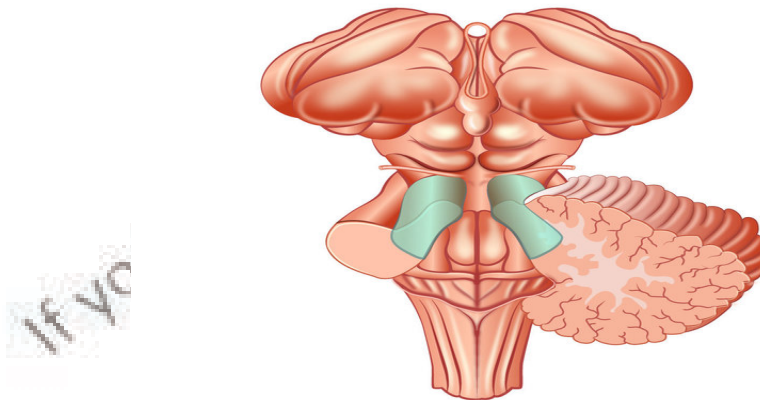
Question 17:

Which of the following is true?

- a) Molecular layer contain stellate and basket cells
- b) Molecular is the innermost layer of cortex
- c) Purkinje layer contain Golgi cells
- d) Axons of purkinje cell secrete Glutamate on deep cerebellar nuclei

Question 18:

Which of the following ascending pathways enters the cerebellum through the peduncle highlighted in green in the image?



- a) Dorsal spinocerebellar tract
- b) Cuneo cerebellar tract
- c) Anterior spinocerebellar tract
- d) Ponto cerebellar fibers

Question 19:

Concerning the circuitry between mossy fibers, climbing fibers and cells of the cerebellar cortex, which of the following pairs provide excitatory input?

- a) Mossy fibers and Purkinje cells
- b) Golgi cells and climbing fibers
- c) Granule cell parallel fibers and mossy fibers
- d) Climbing fibers and basket cells

Question 20:

Inhibition of Purkinje cells by stellate cells is an example of

- a) Presynaptic inhibition
- b) Feedback inhibition
- c) Feedforward inhibition
- d) Stellate cells doesn't inhibit purkinje cells

Answer Key

Question No.	Correct Option
1	c
2	a
3	c
4	d
5	b
6	b
7	d
8	a
9	c
10	a
11	c
12	c
13	b
14	b
15	a

16	a
17	a
18	c
19	c
20	c

Detailed Explanations

Solution to Question 1:

The basal ganglia has five nuclei on each side of the brain.

- Caudate nucleus
- Putamen
- Globus pallidus: Divided into globus pallidus external (GPe) and globus pallidus internal (GPi)
- Substantia nigra: Divided into pars compacta and pars reticulata
- Subthalamic nucleus

The caudate nucleus and putamen together form the striatum.

The putamen and globus pallidus collectively form the lentiform nucleus.

Solution to Question 2:

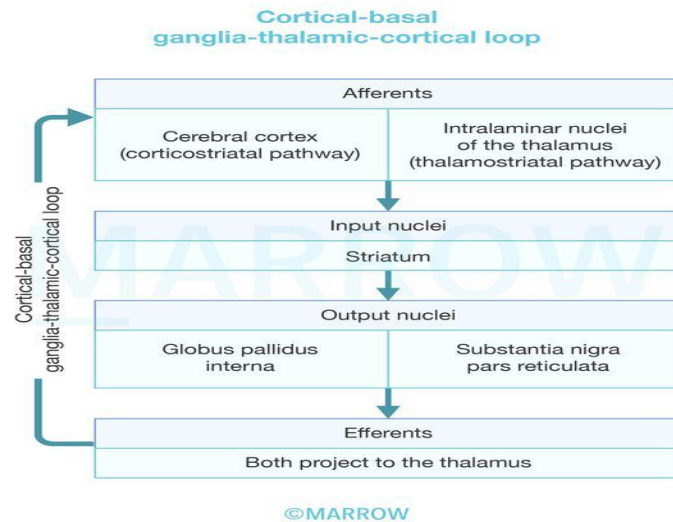
The striatum receives the afferent input.

Note:

There are two main inputs to the basal ganglia: From the cerebral cortex (corticostriate pathway) and from the thalamus (thalamostriate pathway). They are both excitatory (glutamate) and they both terminate in the striatum.

The two major outputs of the basal ganglia are from GPi and substantia nigra. Both are inhibitory (GABAergic) and both project to the thalamus.

From the thalamus, there is an excitatory (glutamatergic) projection to the prefrontal and premotor cortex. This completes the cortical-basal ganglia-thalamic-cortical loop.



Solution to Question 3:

This patient will have dysfunction of the basal ganglia. Maintenance of equilibrium is a function of the cerebellum and vestibular apparatus, hence will not be affected in this patient.

Functions of basal ganglia:

- Planning of movements
- Regulation of muscle tone
- Control of reflex muscular activity & posture
- Programming of movement - speed, direction, and intensity
- Role in the cognitive process (especially caudate nucleus)

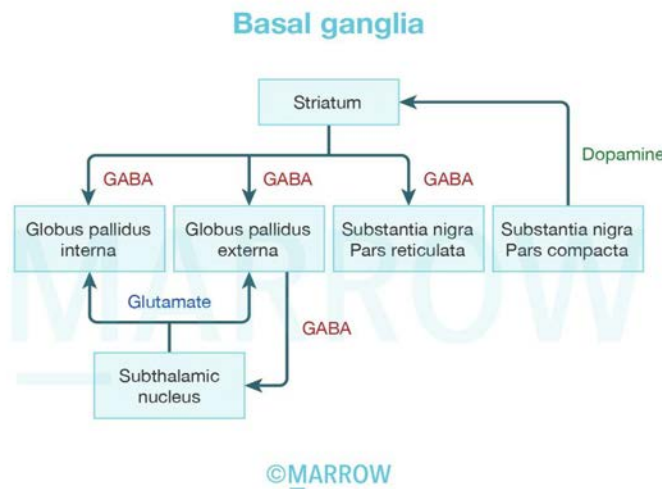
Solution to Question 4:

Glutamate is synthesized mainly in the subthalamic nuclei of basal ganglia.

Connections within the basal ganglia and the respective neurotransmitters:

Connections between	Neurotransmitter
Substantia nigra pars compacta to striatum (Nigrostriatal projection)	Dopamine
Striatum to substantia nigra pars reticulata	GABA
Striatum to GPe and GPi	GABA

Connections between	Neurotransmitter
GPe to subthalamic nuclei	GABA
Subthalamic nuclei to GPe and GPi	Glutamate

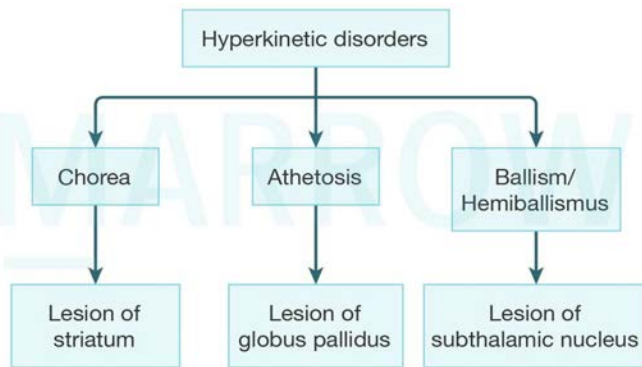


Solution to Question 5:

This woman with a lesion at the globus pallidus most likely presented with athetosis: Slow, writhing movements of a hand/arm/the face.

Condition	Site of lesion	Description
Chorea	Striatum	Flicking movements in the hands, face, and other parts of the body
Athetosis	Globus pallidus	Spontaneous, continuous writhing movements of a hand, an arm, the neck, or the face
Hemiballismus	Subthalamic nucleus	Sudden flailing movements of an entire limb
Parkinsonism	Substantia nigra	Severe rigidity, akinesia, and tremors

Hyperkinetic disorders



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Solution to Question 6:

Bipolar cells are not present in cerebellum.

Cerebellum has 3 layers, 4 deep nuclei and 5 types of neurons.

The cerebellar cortex has three layers:

- External molecular layer
- Purkinje cell layer
- Internal granular layer

There are four deep nuclei: D, E, F, G

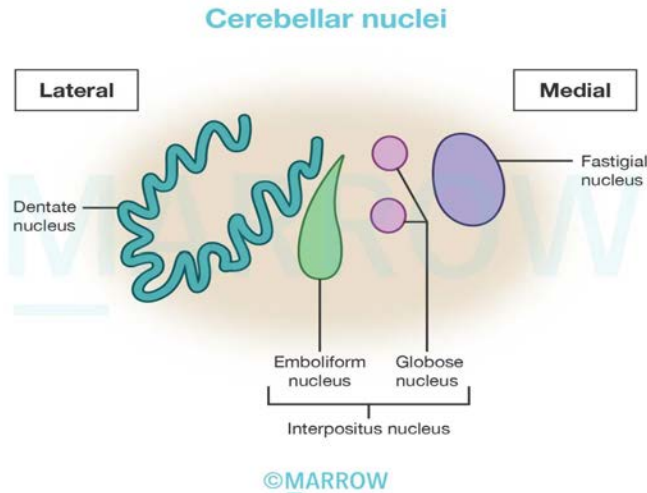
- Dentate
- Emboliform
- Fastigial nuclei
- Globose

Interpositus nucleus- Globose and the emboliform nuclei together.

There are five types of neurons in the cortex:

- Purkinje
- Granule
- Basket
- Stellate
- Golgi cells

Solution to Question 7:



Interpositus nucleus= globose+emboliform nuclei

Note: Red nucleus is situated in the midbrain. It crosses the midline and projects to interneurons in the dorsolateral part of the spinal ventral horn to also influence motor neurons that control distal limb muscles. This rubrospinal tract excites flexor motor neurons and inhibits extensor motor neurons. This pathway is not prominent in humans, but it may play a role in the posture typical of decorticate rigidity.

Solution to Question 8:

Purkinje cells are the only output neurons (efferent neurons) from the cerebellar cortex.

The Purkinje cells are the largest neurons in the cerebellum. Their axons, which are the only output from the cerebellar cortex, project to the deep cerebellar nuclei, especially the dentate nucleus, where they form inhibitory synapses.

Solution to Question 9:

The flocculonodular lobe has connections with the vestibular apparatus.

The flocculonodular lobe forms the vestibulocerebellum, which projects to the vestibular nuclei and is concerned with equilibrium and eye movements.

Solution to Question 10:

Climbing fibers are olivocerebellar fibers. They originate from the inferior olivary nucleus.

Climbing fibers and mossy fibers are the two main excitatory inputs to the cerebellum.

Note: Mossy fibres provide direct input via the pontine nuclei.

Climbing fibres	Mossy fibres
They come from one single source via the inferior olivary nuclei.	They come from many sources and the fibers synapse with granule cells to form glomeruli.
End on single Purkinje cells. Convey proprioceptive inputs from all over the body.	Axons of granule cells bifurcate and form parallel fibers which excite multiple Purkinje cells.

Solution to Question 11:

Vestibulocerebellum (archicerebellum) constitutes nodulus and flocculus and is hence called flocculonodular lobe.

Spinocerebellum (paleocerebellum) constitutes vermis and adjacent medial portions of cerebellar hemispheres.

Cerebrocerebellum (neocerebellum) constitutes the lateral portions of hemispheres.

Solution to Question 12:

The function of the neocerebellum is planning and programming of voluntary movements.

The cerebellum can be divided into:

The Vestibulocerebellum/Archicerebellum: Consists principally of the flocculonodular lobe and adjacent portions of the vermis. It participates in balance, vestibular reflexes, and eye movements.

The Spinocerebellum/Paleocerebellum: Consists of most of the vermis plus the adjacent intermediate zones on both sides of the vermis. It coordinates movements of the limbs. Vermis: Proximal muscles, Intermediate zones: Distal muscles

The Cerebrocerebellum/neocerebellum: Consists of the large lateral zones of the cerebellar hemispheres. It receives all its input from the cerebral cortex (motor, premotor and somatosensory). It transmits its output back to the brain, functioning in a feedback manner to plan sequential voluntary body and limb movements.

Solution to Question 13:

A patient with a suspected cerebellar lesion will not have resting tremors. Resting tremors are characteristic features of basal ganglia lesions. (Parkinson's disease).

Intentional tremors are seen in cerebellar diseases.

Features of cerebellar disease

- Ataxia – Incoordinated, intermittent jerky movements with drunken gait.
- Hypotonia or atonia – Reduced or absence of muscle tone.
- Dysmetria – Inability to judge the force and range of movement resulting in past pointing.
- Dysdiadokokinesia – Failure to perform rapid alternating movements.
- Decomposition of movement – Lack of smooth flow of movement.
- Speech defect – Staccato speech or slurring speech.
- Pendular knee jerk (a feature of hypotonia).
- Intention tremors.
- Nystagmus (jerky movements of the eye).
- Rebound phenomenon.

Solution to Question 14:

The vestibulo-ocular reflex is controlled by the vestibulo-cerebellum, which constitutes the flocculonodular lobe.

Solution to Question 15:

The input from the Purkinje cells to the deep nuclei is inhibitory.

However, the deep nuclei output is always excitatory.

Solution to Question 16:

The fibres of inferior olivary nucleus (climbing fibres) reach the Purkinje cell directly.

Note:

The input from the Purkinje cells to the deep nuclei is inhibitory.

However, the deep nuclei output is always excitatory.

Solution to Question 17:

The cerebellar cortex has three layers:

External molecular layer: Contains Basket and Stellate Cells.

Purkinje cell layer: Only one cell thick, containing the Purkinje cells.

Internal granular layer: Contains Granule and Golgi Cells

Solution to Question 18:

The peduncle highlighted in the image is the superior cerebellar peduncle. The anterior spinocerebellar tract enters the cerebellum through it and terminates in both sides of the cerebellum.

Dorsal spinocerebellar tract enters the cerebellum through the inferior cerebellar peduncle and terminates on the same side as its origin.

Solution to Question 19:

The excitatory fibers in the cerebellum are climbing fibers, mossy fibers, and granule cell parallel fibers.

The fundamental circuits of the cerebellar cortex:

- Climbing fibers exert a strong excitatory effect on single Purkinje cells. Mossy fibers exert an excitatory effect on granule cells.
- Each granule cell sends an axon to the molecular layer, where the axon bifurcates to form a T. The branches of the T are straight and run long distances; thus, they are called parallel fibers. The parallel fibers form excitatory synapses on the dendrites of many Purkinje cells.
- The other neurons (basket cells, stellate cells, and Golgi cells) are inhibitory interneurons. Basket cells and stellate cells receive excitatory input from the parallel fibers and each inhibits many Purkinje cells. Golgi cells receive excitatory input from the parallel fibers and mossy fibers. Their axons project to the dendrites of the granule cells where they form an inhibitory synapse.
- The output of the Purkinje cells is ultimately inhibitory to the deep cerebellar nuclei.

Solution to Question 20:

Inhibition of Purkinje cells by stellate cells is an example of feedforward inhibition.

Mossy fibers exert a weak excitatory effect on many Purkinje cells via the granule cells. The granule cells also excite the basket and stellate cells via their parallel fibers. The basket and stellate cells, in turn, inhibit the Purkinje cells (Feed forward inhibition).

Renshaw cells exhibit feedback inhibition.

Hypothalamus and Limbic System

Question 1:

Which of the following is true about thalamus?

- a) Thalamus is the sensory gateway to cerebral cortex
- b) Thalamus is the largest part of telencephalon
- c) Y-shaped band of fibers divides the thalamus into 4 subdivisions
- d) Thalamus is the motor gateway from the cortex

Question 2:

Which thalamic nuclei transmit pain?

- a) Lateral dorsal
- b) Intralaminar
- c) Ventral posterior
- d) Pulvinar

Question 3:

Which of the following is called "the window of limbic system" ?

- a) Hypothalamus
- b) Hippocampus
- c) Amygdala
- d) Thalamus

Question 4:

Which of the following is a component of the Papez circuit?

- a) Lateral dorsal nucleus
- b) Anterior nucleus of thalamus
- c) Ventral anterior nucleus

d) Pulvinar

Question 5:

Stimulation of which of the following regions promotes a 'sense of reward' in human beings?

- a) Insula
- b) Putamen
- c) Medial forebrain bundle
- d) Aqueduct of Sylvius

Question 6:

In response to an increase in plasma osmolality, ADH is released in blood. Which of the following sites contains the receptor to detect a change in plasma osmolality?

- a) Preoptic
- b) Arcuate
- c) Anterior hypothalamus
- d) Mammillary

Question 7:

The main organ for biological clock is _____.

- a) Thalamus
- b) Hypothalamus
- c) Cerebellum
- d) Basal ganglia

Question 8:

Which of the following clinical features will be most likely seen in a patient with an anterior hypothalamic lesion?

- a) Hyperphagia
- b) Hypothermia

- c) Hyperglycemia
- d) Hyperthermia

Question 9:

Diurnal variation of ACTH is controlled by which of the following regions?

- a) Paraventricular hypothalamus
- b) Anterior hypothalamus
- c) Lateral hypothalamus
- d) Supra chiasmatic nucleus

Question 10:

Which of the following is not a function of the categorical hemisphere?

- a) Mathematic ability
- b) Language
- c) Writing ability
- d) Music ability

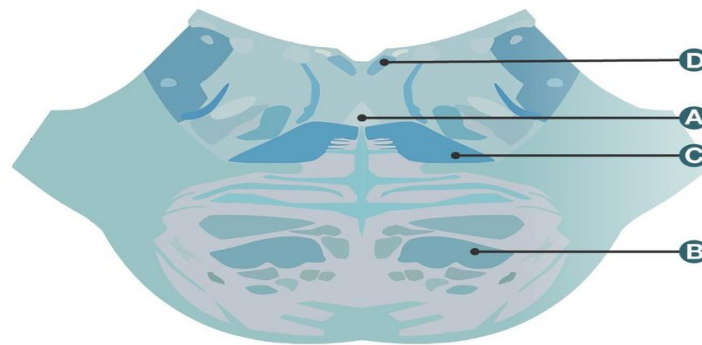
Question 11:

Lesions in which nucleus of the hypothalamus produce a voracious appetite and rage?

- a) Upper region of the lateral hypothalamus
- b) Mammillary region
- c) Ventromedial nucleus of the hypothalamus
- d) Anterior region of the hypothalamus

Question 12:

Identify the site which is responsible for relieving pain when pressure is applied to the painful area.



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- a) A
- b) B
- c) C
- d) D

Question 13:

The factors that can cause a person to drink water due to stimulation of thirst mechanism is/are _____.

- a) Dehydration
- b) Hemorrhage
- c) Prandial Drinking
- d) All of the above

Answer Key

Question No.	Correct Option
1	a
2	c
3	c
4	b

5	c
6	c
7	b
8	d
9	d
10	d
11	c
12	a
13	d

Detailed Explanations

Solution to Question 1:

The thalamus is being called the sensory gateway to the cerebral cortex as all information that reaches the cortex is processed by the thalamus.

The thalamus is a large collection of neuronal groups within the diencephalon; it participates in sensory, motor, and limbic functions.

Other Options:

Option B: Thalamus is the largest part of the diencephalon.

Option C: Y shaped band of fibers divide the thalamus into 3 subdivisions- anterior, medial, and lateral group.

Option D: Thalamus is the sensory gateway to the cortex.

Solution to Question 2:

The ventroposterior nuclear complex of the thalamus represents the main termination site for ascending somatosensory information (which includes pain) in the diencephalon.

Thalamic nuclei have been divided into

A. Specific (Relay) nuclei because they have a specific and selective relationship with a particular portion of neocortex

- Medial geniculate body for auditory information
- Lateral geniculate body is for information from retina
- Ventroposterior group for somatosensory information

B. Nonspecific nuclei project to several cortical and subcortical regions

Solution to Question 3:

The amygdala has been called the “window” through which the limbic system sees the place of the person in the world.

The amygdala receives neuronal signals from all portions of the limbic cortex, as well as from the neocortex of the temporal, parietal, and occipital lobes, especially from the auditory and visual association areas.

Components of the limbic system

- The term limbic lobe, or limbic system, is applied to the part of the brain that consists of a rim of cortical tissue around the hilum of the cerebral hemisphere and a group of associated deep structures—the amygdala, the hippocampus, and the septal nuclei.
- The papez circuit is a closed neural circuitry that starts and ends in the hippocampus, which forms the basal ganglia. It is also known as the medial limbic circuit. The components of papez circuit are-

Mnemonic-He Man Ate Cat

- He-Hippocampus
- Man-Mamillary body
- Ate-Anterior thalamus
- Cat-Cingulate gyrus

Solution to Question 4:

Anterior nucleus of the thalamus is a component of Papez circuit, which is important for short term memory.

Papez Circuit (refer figure).

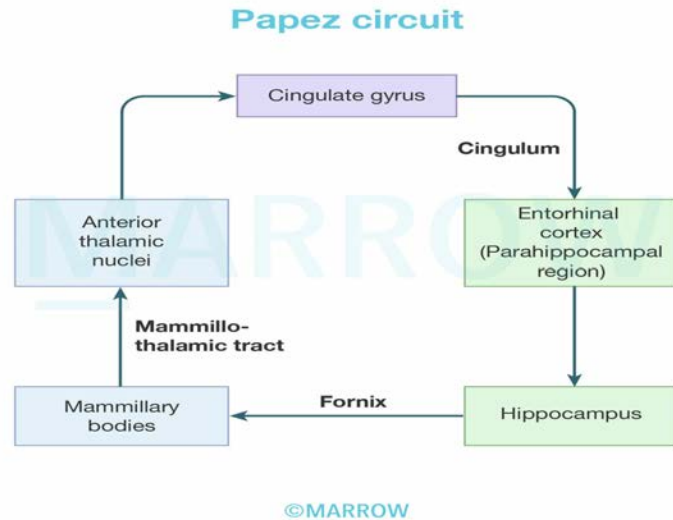
The limbic system was studied by James Papez who described the Papez circuit. It consists of:

Mnemonic for components of Papez circuit-He Man Ate Cat

- He-Hippocampus
- Man-Mamillary body
- Ate-Anterior thalamus
- Cat-Cingulate gyrus

The fornix connects the hippocampus to the mammillary bodies, which are in turn connected to the anterior nuclei of the thalamus by the mammillothalamic tract. The anterior nuclei of the thalamus project to the cingulate cortex, and from there, connections to the hippocampus complete a complex closed circuit.

The boundaries of the limbic system were subsequently expanded to include the amygdala, septum, basal forebrain, nucleus accumbens, and orbitofrontal cortex.



Solution to Question 5:

Major reward centres are located along the course of the medial forebrain bundle, especially in the lateral and ventromedial nuclei of the hypothalamus. Therefore, stimulation of those areas will promote a sense of reward.

It is interesting to note that in the lateral nucleus of the hypothalamus, a weak stimulus produces a sense of pleasure whereas a strong stimulus produces rage. This phenomenon occurs in many areas where a weak stimulus produces a sense of reward while stronger ones produce a sense of punishment.

Solution to Question 6:

Osmoreceptors that detect any change in plasma osmolality and control ADH secretion, are located in anterior hypothalamus.

Solution to Question 7:

In the hypothalamus, the suprachiasmatic nucleus functions as the principal pacemaker, or time clock, for circadian rhythms.

The pacemaker receives input from the eyes and many other parts of the nervous system, and these inputs mediate the effects exerted by the external environment (retinohypothalamic tract).

One output of the pacemaker goes to the pineal gland, a gland within the brain that secretes the hormone melatonin. These neural signals cause the pineal gland to secrete melatonin during darkness but not during daylight.

Solution to Question 8:

Lesions involving the anterior and preoptic hypothalamic regions cause paradoxical vasoconstriction, tachycardia, and hyperthermia.

The anterior hypothalamus activates mechanisms of cooling the body when the temperature increases via the reflex response.

Stimulation of the anterior hypothalamus causes cutaneous vasodilation and sweating, thereby cooling the body. Lesions in this region cause hyperthermia, with rectal temperatures sometimes reaching 43°C (109.4°F).

Solution to Question 9:

Diurnal variation of ACTH depend on suprachiasmatic nucleus.

The secretory rates of ACTH and cortisol are high in the early morning but low in the late evening. This effect results from a 24-hour cyclical alteration in the signals from the suprachiasmatic nucleus of hypothalamus that cause cortisol secretion. When a person changes daily sleeping habits, the cycle changes correspondingly.

Solution to Question 10:

Functions of the categorical hemisphere are logical, analytical, sequential thinking, and verbal tasks, such as math, language forms, and philosophy.

Human language functions depend more on one cerebral hemisphere than on the other. This hemisphere is concerned with categorization and symbolization and has often been called the dominant hemisphere. However, the other hemisphere is specialized in the area of spatiotemporal relations.

The concept of "cerebral dominance" and a dominant and non-dominant hemisphere has been replaced by, one hemisphere for sequential-analytic processes (the categorical hemisphere) and one for visuospatial relations (the representational hemisphere).

The representational cerebral hemisphere excels in nonlanguage skills, especially spatial perception, and artistic and musical talents.

Solution to Question 11:

A lesion in the ventromedial nucleus of the hypothalamus, the satiety center, is responsible for producing a voracious appetite and rage.

Lesions in the upper region of the lateral hypothalamus produce a profound loss of appetite.

Solution to Question 12:

It is a cross section of brain stem at pons.

- A is a periaqueductal gray matter which also consists of raphe nucleus.
- Descending pathways from the mesencephalic periaqueductal gray matter inhibit transmission in nociceptive pathways. This descending pathway includes a synapse in the ventromedial medulla (Raphe nucleus) and the release of endogenous opiates. So periaqueductal gray matter releases endogenous opiates.

Other marked structures are

- B: Corticospinal tract
- C: Medial lemniscus
- D: Medial longitudinal fasciculus

Solution to Question 13:

Dehydration, hemorrhage, and prandial drinking: All three can cause increased thirst resulting in water drinking.

Increased osmolality acts via osmoreceptors in the anterior hypothalamus.

The decreased plasma volume, as in hemorrhage, acts via

1. Renin-angiotensin system
2. Baroreceptors in heart and blood vessels.

The intake of liquids during a meal is known as prandial drinking. The increase is known as learned or habit response but is not understood completely.

The following factors are thought to play a role

1. Increase in plasma osmolality due to food absorption
2. Action on hypothalamus by one or more gastrointestinal hormones

Higher Mental Functions

Question 1:

Which of the following statements about the reticular activating system is false?

- a) Involved in arousal
- b) Polysynaptic pathway
- c) Receives collateral from sensory pathway
- d) Does not produce EEG alerting response

Question 2:

Which is the EEG wave recorded in a subject who is awake but eyes closed?

- a) Alpha waves
- b) Beta waves
- c) Theta waves
- d) Delta waves

Question 3:

Which of the following structures regulates the sleep-wake cycle?

- a) Basal ganglia
- b) Hypothalamus
- c) Medulla
- d) Cerebellum

Question 4:

In which stage of sleep are K complex and sleep spindles seen?

- a) REM
- b) Stage 1 NREM
- c) Stage 2 NREM

d) Stage 3 NREM

Question 5:

Which stage constitutes the largest amount of sleep time of a young adult?

- a) REM
- b) NREM 1
- c) NREM 2
- d) NREM 3

Question 6:

Which of the following is observed during sleep induction?

- a) ↑ Norepinephrine and serotonin
- b) ↑ Acetyl choline
- c) ↑ Histamine
- d) ↓ GABA

Question 7:

In NREM sleep, which part of brain is not involved?

- a) Dorsal Raphe nucleus
- b) Thalamus
- c) Hypothalamus
- d) Basal forebrain

Question 8:

Delta waves are seen in_____

- a) Deep sleep
- b) REM Sleep
- c) Awake state
- d) NREM 1

Question 9:

Which of the following is associated with slow-wave sleep?

- a) Nightmares
- b) Cardiac arrhythmias
- c) Penile tumescence
- d) Delta activity

Question 10:

Which of the following is the source of melatonin?

- a) Pineal gland
- b) Hypothalamus
- c) Adrenal cortex
- d) Melanocytes

Question 11:

A businessman who takes frequent long trips to overseas destinations visits the clinic asking for "something that helps with falling asleep and adapt with the local time". He is given a low dose of melatonin as a part of his treatment plan. Which of the following best describes its action?

- a) Facilitates ACTH secretion
- b) Prevents sleep induction
- c) Regulates circadian rhythm
- d) Release of melanin

Question 12:

In which type of memory are basal ganglia involved?

- a) Conditioning
- b) Procedural
- c) Explicit

d) None

Question 13:

A 50-year-old lady develops severe anterograde amnesia. On further evaluation, a suprasellar craniopharyngioma invading adjacent structures is seen on T1-weighted MRI. Compression of which of the following structures can explain her symptoms?

- a) Prefrontal cortex
- b) Neocortex
- c) Mammillary body
- d) Amygdala

Question 14:

Which of the following neural mechanism is not involved in synaptic plasticity and learning?

- a) Post-tetanic potentiation
- b) Habituation
- c) Desensitization
- d) Long-term depression

Question 15:

The type of potentiation in hippocampus is _____

- a) Long-term potentiation
- b) Post tetanic potentiation
- c) Long-term depression
- d) Habituation

Question 16:

Wernickes area is in _____

- a) Superior temporal gyrus
- b) Precentral gyrus

- c) Post central gyrus
- d) Inferior frontal gyrus

Question 17:

Which of the following is false about Broca's aphasia?

- a) Lesion lies in frontal lobe
- b) Fluency is impaired
- c) Neologisms are absent
- d) Repetition is preserved

Question 18:

A lesion in the Broca's area will affect which of the following?

- a) Word formation
- b) Word comprehension
- c) Word memorization
- d) Reading

Question 19:

A patient is able to recognize a person by name but not face. Which of the following is the most likely site of the lesion?

- a) Frontal lobe
- b) Occipital lobe
- c) Post parietal region
- d) Occipitotemporal

Question 20:

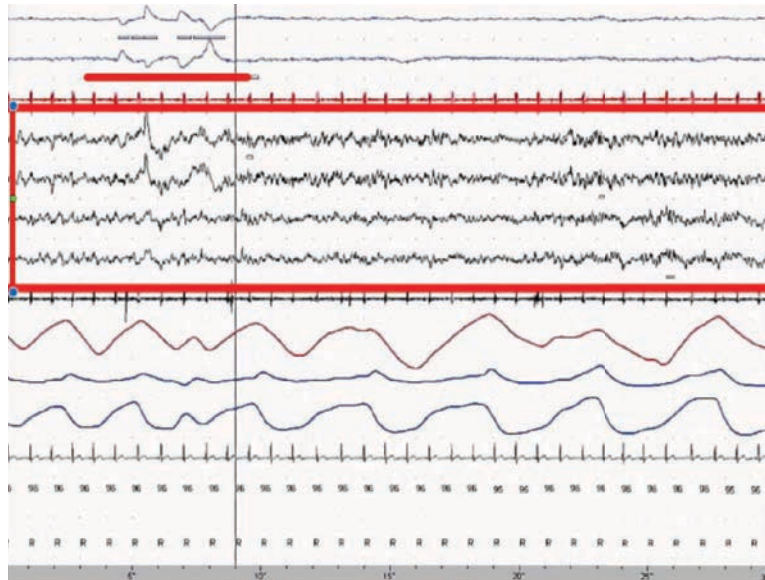
Which of the following is the test for immediate memory?

- a) Digit span forward up to 7 digits with 2 skips allowed
- b) Serial (100-7) subtraction test up to 5 steps

- c) Digit span backwards up to 5 digits with 2 skips allowed
- d) Serial (20-1) subtraction test up to 5 steps

Question 21:

The marked area given in the sleep polysomnography is of _____



- a) REM sleep
- b) NREM stage 1
- c) NREM stage 2
- d) NREM stage 3

Question 22:

In which phase of sleep bruxism occurs?

- a) NREM2
- b) NREM 3
- c) REM
- d) NREM 4

Question 23:

Which of the following sleep disorder does not occur in stage 3 of NREM?

- a) Night terror
- b) Nocturnal enuresis
- c) Bruxism
- d) Sleep walking

Question 24:

A 30-year-old male, who met with an RTA, regained consciousness after 36 hours. It was then discovered that the patient is unable to create new memories. Which of the following is the probable site of the lesion?

- a) Amygdala
- b) Neocortex
- c) Hippocampus
- d) Hypothalamus

Question 25:

Which of the following is correctly paired to show the relationship between a brain area and a type of memory?

- a) Neocortex and associative learning
- b) Medial temporal lobe and declarative memory
- c) Angular gyrus and procedural memory
- d) Striatum and priming

Question 26:

A 35-year-old man is severely injured in a road traffic accident. Brain MRI shows significant damage to the bilateral frontal lobes. Which of the following neurological features will be least likely seen in this patient?

- a) Slowed thinking
- b) Decreased curiosity
- c) Social withdrawal
- d) Retained judgement

Question 27:

A lady presents with vascular injury to the inferior frontal gyrus. Which functional area is likely to be affected?

- a) Motor speech area
- b) Wernicke's area
- c) Auditory area
- d) Visual area

Question 28:

What is Prosopagnosia?

- a) Impairment of consciousness
- b) Being unaware of one's problems
- c) Difficulty in identifying known faces
- d) Failure to identify objects

Question 29:

Emotion is regulated by:

- a) Temporal lobe
- b) Frontal lobe
- c) Limbic system
- d) Parietal lobe

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	c
5	c

6	b
7	a
8	a
9	d
10	a
11	c
12	b
13	c
14	c
15	a
16	a
17	d
18	a
19	d
20	a
21	a
22	a
23	c
24	c
25	b
26	d
27	a
28	c
29	b

Detailed Explanations

Solution to Question 1:

High-frequency stimulation of the midbrain reticular formation (the RAS) produces the EEG alerting response and arouses a sleeping animal.

The Reticular Activating System (RAS) is a complex polysynaptic pathway arising from the brainstem reticular formation and hypothalamus with projections to the intralaminar and reticular nuclei of the thalamus which, in turn, project diffusely and nonspecifically to wide regions of the cortex including the frontal, parietal, temporal, and occipital cortices.

Collaterals funnel into it from the long ascending sensory tracts, the trigeminal, auditory, visual, and olfactory systems.

The reticular formation plays an important role in determining the level of arousal; thus, it is called the ascending reticular activating system (ARAS).

Solution to Question 2:

Alpha-waves are recorded best in the parieto-occipital region in adults at rest, i.e. awake but relaxed with eyes closed.

Solution to Question 3:

The hypothalamus and several non-specific nuclei in ascending reticular activating system (ARAS) located in the midbrain are involved in the regulation of the sleep-wake cycle.

Sleep Centres

Stimulation of the midbrain reticular formation (RAS) produces the EEG alerting response and promotes the waking state.

The nucleus coeruleus (locus coeruleus) is rich in norepinephrine and the raphe nuclei of ARAS is rich in serotonin. These regions when stimulated induce wakefulness.

The cholinergic neurons in the pontine reticular formation, on stimulation, induce sleep.

Stimulation of the histaminergic neurons of the posterior hypothalamus produces arousal.

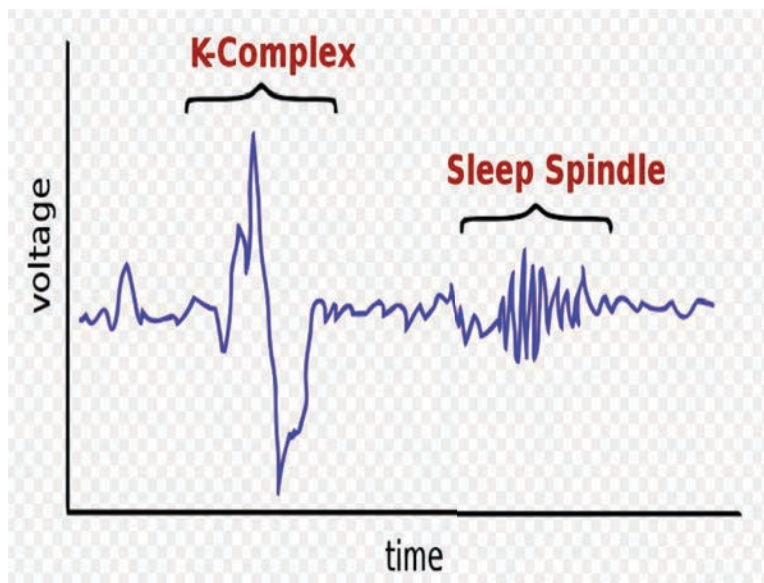
Electrical stimulation of the anterior hypothalamus and the adjacent basal forebrain region rapidly induces sleep. This sleep-inducing effect is thought to be mediated by GABA-ergic inhibitory neurons.

Orexin produced by hypothalamic neurons also appears to have a role in sleep-wake transition by regulating the changes in activity in these brainstem neurons.

wakeful state	Sleep
Norepinephrine	Acetylcholine
Serotonin	GABA
Histamine	

Solution to Question 4:

Stage 2 of NREM sleep is marked by the appearance of sinusoidal waves called sleep spindles (12–14 Hz) and occasional high-voltage biphasic waves called K complexes.



Solution to Question 5:

- In young adults, the largest amount of sleep time (50–60%) is spent in stage 2 NREM sleep; REM phases constitute 20–25% of total sleep time, stages 3 and 4 NREM about 15–20%, and stage 1 non-REM about 5%.
- REM sleep occupies 80% of total sleep time in premature infants and 50 % in full-term neonates. In most adults, it is 25 %. In the elderly, it falls to about 20%.

Solution to Question 6:

The cholinergic neurons in pontine reticular formation on stimulation, induce sleep.

When the activity of norepinephrine and serotonin containing neurons (locus coeruleus and raphe nuclei) is dominant, there is a reduced level of activity in acetylcholine-containing neurons in the pontine reticular formation leading to wakefulness. The reverse of this pattern leads to REM sleep. A more even balance in the activity of these groups of neurons is associated with NREM sleep.

Solution to Question 7:

The nucleus coeruleus (locus coeruleus) is rich in norepinephrine and the Raphe nuclei of ARAS is rich in serotonin. These regions when stimulated induce arousal. Thus, the dorsal raphe nucleus is not involved in NREM sleep.

Genesis and Regulation of NREM sleep

- Electrical stimulation of the anterior hypothalamus and the adjacent basal forebrain region rapidly induces sleep mediated by GABA-ergic inhibitory neurons called the NREM-on cells.
- These cells are thought to produce sleep by inhibiting the histaminergic cells in the posterior hypothalamus as well as cells of the nucleus reticularis pontis oralis in the midbrain that mediate arousal.
- They are maximally active in NREM sleep and inactive during waking and REM sleep.
- NREM sleep is produced by synchronized synaptic potentials in cortical neurons generated by the rhythmic firing of thalamic relay neurons that project to the cortex. (Thalamocortical loop)

Solution to Question 8:

Delta waves are predominant in N3 (formerly called stage 3 and 4) of NREM. This is characteristic of deep sleep, also referred to as slow wave sleep.

(Delta waves for Deep sleep)

Solution to Question 9:

In slow-wave sleep, (Stages 3 and 4 of NREM) delta waves predominate the EEG.

Solution to Question 10:

Melatonin is released by the pineal gland (pinealocytes) that plays an important role in sleep mechanisms.

In infants, the pineal gland is large and the cells are arranged in alveoli. It begins to involute before puberty and small concretions of calcium phosphate and carbonate appear (radiopaque structure) = physiological calcification.

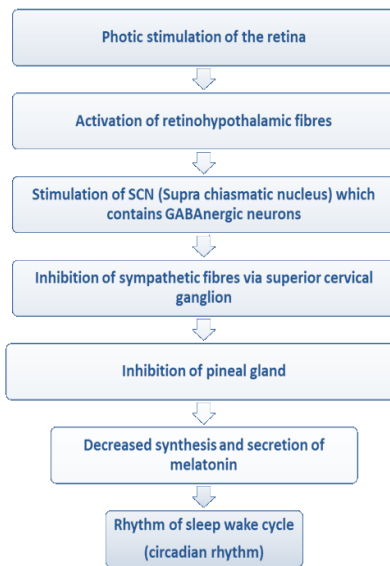
Solution to Question 11:

Melatonin plays a key role in regulating the circadian rhythm. Therefore it is helpful in restoring the usual sleep wake pattern in patients with jetlag.

Diurnal variation in melatonin secretion may function as a timing signal to coordinate events with the light-dark cycle in the environment (circadian rhythm).

Melatonin synthesis and secretion are increased during the dark period of the day and maintained at low levels during the daytime.

Increased melatonin levels induce sleepiness.



Solution to Question 12:

Procedural memory includes skills and habits, which, once acquired, become unconscious and automatic. This type of memory is processed in the striatum.

Implicit memory/Non-declarative memory: It does not require being conscious or awareness. E.g., Remembering how to brush your teeth. They are of 4 types:

- Procedural memory (skills and habits) in striatum, cerebellum, motor cortex
- Priming in neocortex
- Non-associative learning (includes habituation and sensitization, e.g., Byhearting facts)
- Associative learning: classical and operant conditioning

Explicit/declarative memory: Memory of words, rules, language, etc. Associated with consciousness/awareness. E.g., recalling first day in school.

They are of two types:

- Semantic memory (facts) e.g., Knowing capital of India: Parts of brain involved are temporal cortex and prefrontal cortex.
- Episodic memory (events) e.g., Recalling first day in college: hippocampus, medial temporal lobe and neocortex.

Solution to Question 13:

Compression or a lesion affecting the mammillary body will cause anterograde amnesia.

The Hippocampus, mammillary body and mammillothalamic tract are the primary structures involved in the conversion of short-term memory to long-term memory. Any lesion involving this

pathway will affect the ability to convert short-term memory to long-term memory, leading to anterograde amnesia.

Solution to Question 14:

Sensitization is a neural mechanism involved in synaptic plasticity and learning (not desensitization).

Neural basis of synaptic plasticity and learning

- Post-tetanic Potentiation
- Production of enhanced postsynaptic potential, which occurs after a brief tetanizing train of stimuli in the presynaptic neuron
- This enhancement lasts up to 60 seconds.
- Habituation
- A simple form of learning in which neutral stimulus is repeated many times until the subject becomes habituated to the stimulus and ignores it
- A classic example of non-associative learning (e.g., learning a fact by heart)
- Sensitization: The opposite of habituation
- Long-term Potentiation (LTP)
- A rapidly developing persistent enhancement of the post-synaptic potential response to presynaptic stimulation, after rapidly repeated stimulation of presynaptic neuron
- Characteristically observed in the hippocampus
- Long-term Depression
- Is the opposite of LTP
- Characterized by a decrease in synaptic strength
- Produced by slower stimulation of presynaptic neuron
- May be involved in the neural mechanism by which learning occurs in the cerebellum

Solution to Question 15:

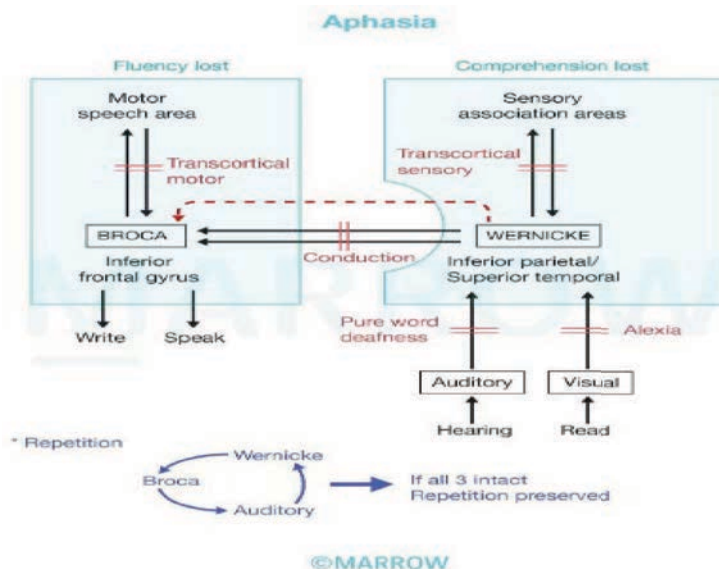
Long-term potentiation occurs in the hippocampus.

LTP is a rapidly developing persistent enhancement of the post-synaptic potential response to presynaptic stimulation, after rapidly repeated stimulation of presynaptic neuron.

NMDA receptors are involved in LTP.

Solution to Question 16:

- Wernicke's area, lies behind the primary auditory cortex in the posterior part of the superior gyrus of the temporal lobe.
- Wernicke's area is concerned with language comprehension.



Solution to Question 17:

- Repetition is impaired in Broca's aphasia.
- Aphasias are abnormalities of language functions that are not due to defects of vision or hearing or motor paralysis. They are caused by lesions in the categorical hemispheres.

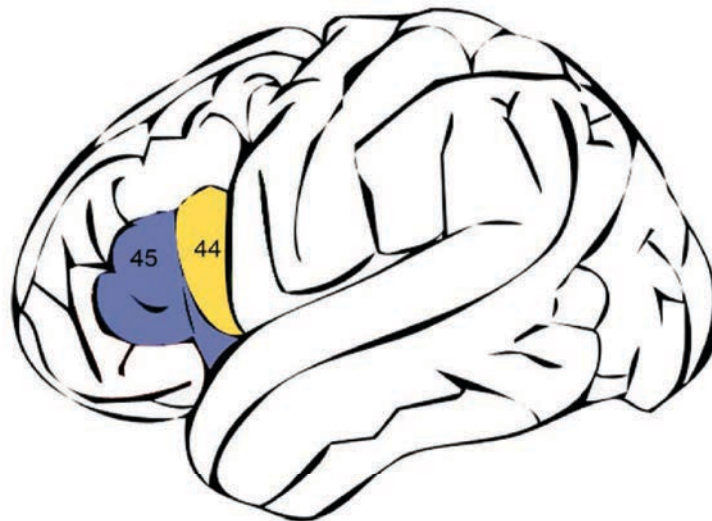
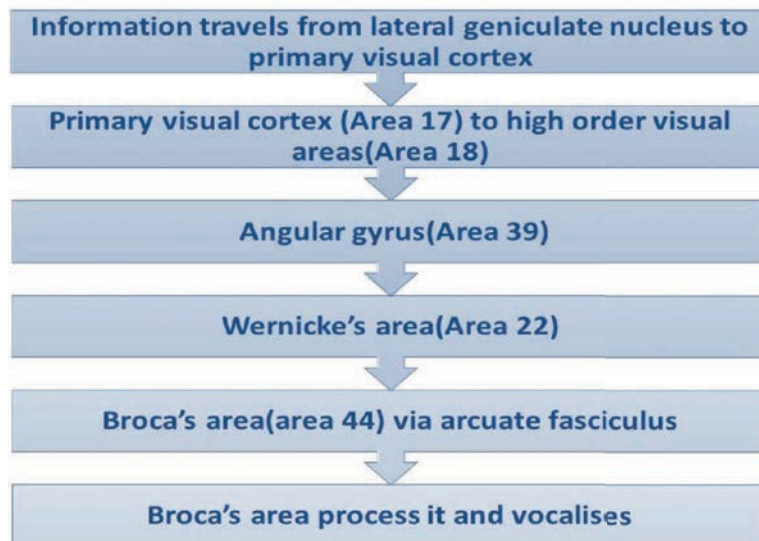
Solution to Question 18:

A lesion in Broca's area will lead to an impairment of word-formation.

Broca's area is in the inferior frontal lobe.

Broca area processes the information received from the Wernicke area into a detailed and coordinated pattern for vocalization and then projects the pattern via a speech articulation area in the insula to the motor cortex. This then initiates the appropriate movements of the lips, tongue, and larynx to produce speech.

The probable sequence of events that occurs when a subject names a visual object is shown in the figure and flowchart.



Solution to Question 19:

This condition is prosopagnosia, which occurs in the lesions of the inferior occipitotemporal lobe.

In humans, storage and recognition of faces are more strongly represented in the right inferior temporal lobe in right-handed individuals (i.e non-dominant inferior temporal lobe).

Damage to this area can cause prosopagnosia, the inability to recognize faces. Patients with this abnormality can recognize people by their voices, body language, etc. However, they cannot identify the familiar people by seeing their faces. The left hemisphere is also involved, but the role of the right hemisphere is primary.

Solution to Question 20:

Forward digit span test is the best test among the given options.

Memory is of 3 types:

- Immediate
- Recent
- Remote

Immediate memory is attention dependent. Tests of attention and memory include

- Digit span test: Most important and best test for immediate memory
- Serial (100-7) subtraction test is the best for concentration
- Days of the week backwards
- Spell WORLD backwards
- (20-3) test

Digit span test

The examiner recites numbers randomly without any sequence and asks the patient to repeat the same. The examiner starts with a 2-number sequence and the digit span increases from 2 numbers to 3 numbers and so on. The patient is asked to repeat until he/she fails to repeat all numbers correctly.

Digit forward

- Normally a person can repeat up to 7 digits (+/- 2)
- E.g., Examiner says 2 - 4 - 8 - 1 - 5 - 7 - 9. The person has to repeat in the same order 2 - 4 - 8 - 1 - 5 - 7 - 9.

Digit backward

- Reproduction of longer digits in backward direction.
- Normally a person can repeat up to 5 digits (+/- 1) in backward direction (making option C wrong)
- Eg : Examiner says 2 - 4 - 8 - 1 - 5 - 7 - 9. The person has to repeat it backwards. 9 - 7 - 5 - 1 - 8 - 4 - 2.

Serial (100-7) subtraction test

- 5 scores / steps are taken in to account
- 1st score – When the patient tells 93 ($100 - 7 = 93$)
- 2nd score - When the patient tells 86 ($93 - 7 = 86$)
- And so on . . 100 , 93 , 86 , 79 , 72 , 65

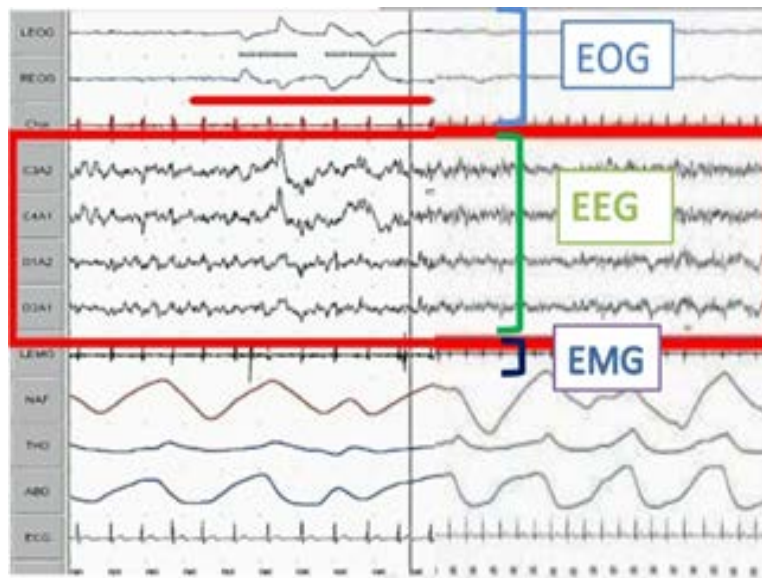
Hence among the given options :

Digit span forward up to 7 digits with 2 skips allowed would be the answer of choice as it is the best test for immediate memory

Solution to Question 21:

The marked area in the given polysomnography represents REM sleep.

The three zones in the given image represent EOG, EEG and EMG, respectively.



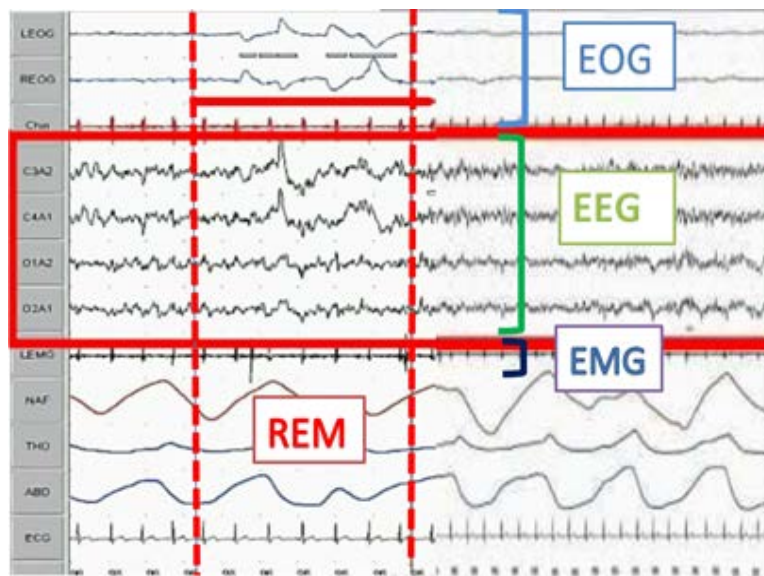
The following table helps identify the state of wakefulness from the graph :

In the marked area,

- EOG shows Positive activity
- EEG shows Mixed waves
- EMG shows Minimal activity

This is characteristic of REM sleep.

	EEG	EOG	EMG
Active state	Fast	+	+
REM sleep	Mixed	+	-
NREM sleep	Slow	-	+



Solution to Question 22:

Sleep-related bruxism is diagnosed when an individual grinds or clenches the teeth during sleep. Bruxism occurs more commonly during NREM 2.

Parasomnias refer to abnormal behavior or experiences that arise from or occur during sleep.

Solution to Question 23:

Bruxism occurs in stage 2 of NREM, whereas other sleep disorders given in the option occur in stage 3 of NREM.

Solution to Question 24:

From the history, it is clear that the patient is suffering from anterograde amnesia, most probably due to a lesion in the hippocampus.

Hippocampus is the primary structure within the temporal lobe involved in conversion of short-term memory to long-term memory.

Hippocampal cognitive map is inappropriately reactivated during a déjà vu experience.

Hippocampus does not store memory but only converts short-term memory to long-term memory. Hence, hippocampal lesions cause only anterograde amnesia.

Solution to Question 25:

Declarative memory involves the hippocampus and the medial temporal lobe for retention.

Priming - neocortex.

Procedural memory - striatum.

Associative learning- amygdala for its emotional responses and the cerebellum for the motor responses.

Nonassociative learning - various reflex pathways.

Solution to Question 26:

Retained judgement will be least likely seen if the patient survives such a devastating injury.

Injury to the frontal lobe may lead to frontal lobe syndrome, which consists of poor judgment, slowed thinking, decreased curiosity, social withdrawal, and irritability.

Frontal lobe injury:

- Impairs the executive functions: motivation, attention, and sequencing of actions
- Aphasia, apraxia, constructional apraxia, neglect, and memory deficits can be seen
- Characterized by changes in personality
- Apathetic indifference and impulsive behaviour.
- Commonly urinary incontinence and faecal incontinence is seen
- Commonly produced by trauma, infarcts, tumours, lobotomy, multiple sclerosis, or Pick's disease.

Frontal lobe pathology may become apparent only under unstructured, stressful, real-life situations and are difficult to detect in intelligence scales, as IQ mostly requires parietal lobe activation.

Solution to Question 27:

Broca's motor speech area is likely to be affected in the vascular injury of the inferior frontal gyrus

A lesion in this region leads to motor/Broca's aphasia. Speech is non-fluent and interrupted by many word-finding pauses. The speech is impoverished in function words that contribute to the syntax but is enriched in meaningful nouns and verbs. However, comprehension of speech is intact and the patient knows what he wants to say.

For example, a patient with Broca's aphasia may say "walk dog" instead of "I want to take the dog for a walk".

Solution to Question 28:

Prosopagnosia is difficulty in identifying known faces.

In humans, storage and recognition of faces is more strongly represented in the right inferior temporal lobe in right-handed individuals, though the left lobe is also active. Damage to this area can cause prosopagnosia.

Patients with this abnormality can recognize forms and reproduce them. They can recognize people by their voices, and many of them show autonomic responses when they see familiar as opposed to unfamiliar faces. However, they cannot identify the familiar faces they see. The left hemisphere is also involved, but the role of the right hemisphere is primary.

The presence of autonomic response to a familiar face in the absence of recognition has been explained by postulating the existence of a separate dorsal pathway for processing information about faces that leads to recognition at only a subconscious level.

Note:

- Autotopagnosia - inability to recognize or to orient any part of one's own body
- Simultagnosia - inability to recognize multiple elements in a visual presentation, one object or some elements of a scene can be appreciated but not the display as a whole
- Somatagnosia - inability to identify any part of the body, either one's own or another's body
- Tactile agnosia (astereognosis) - inability to recognize objects by touch, in the presence of intact cutaneous and proprioceptive hand sensation; caused by a lesion in the contralateral parietal lobe

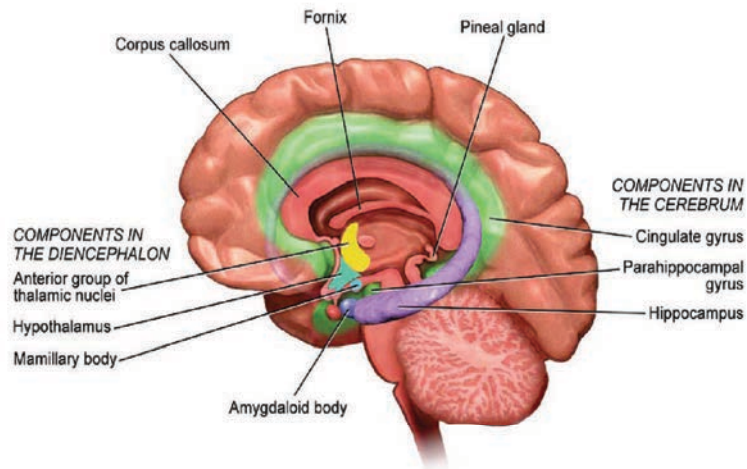
Solution to Question 29:

Emotion is regulated by the frontal lobe.

There are two aspects of emotion:

- Genesis of emotion and rating of emotional importance –from the limbic system – amygdala.
- The limbic system houses the emotional association areas, which direct the hypothalamus to express the motor and endocrine components of the emotional state.
- The amygdala rates the emotional importance of an experience and activates the hippocampus (memory) accordingly. Thus an emotionally intense experience is etched in memory, which is the basis for learned fear responses.
- Damage to the amygdala ablates the ability to distinguish fear and anger in other persons' voices and facial expressions.
- Regulation of emotion - frontal lobe.

The Limbic System



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Functional Anatomy

Question 1:

What is the total number of airway divisions present in between trachea and alveolar sacs?

- a) 21
- b) 22
- c) 23
- d) 24

Question 2:

Which among the following has the highest airway resistance?

- a) Alveolar duct
- b) Bronchi
- c) Respiratory bronchioles
- d) Small bronchioles

Question 3:

A 14-year-old boy with shortness of breath was diagnosed with chronic inflammation of conducting airways. Which of the following is not a feature of the affected structure?

- a) Includes first 16 generations starting from trachea
- b) Smooth muscles and cartilage predominate
- c) Secrete variety of molecules that aid in lung defence
- d) Membrane bound organelles namely lamellar bodies are present

Question 4:

Maximal smooth muscle relative to wall thickness is seen in _____.

- a) Terminal bronchiole
- b) Trachea

- c) Bronchi
- d) Respiratory bronchioles

Question 5:

An elderly smoker who presented with an acute exacerbation of COPD is administered salbutamol. Which one of the following substances has a similar mechanism of action?

- a) Substance P
- b) Vasoactive intestinal polypeptide
- c) Leukotrienes
- d) Cholinergic stimulation

Question 6:

Which of the following cells secrete surfactant in lungs?

- a) Type 1 pneumocyte
- b) Alveolar macrophage
- c) Dust cells
- d) Type 2 pneumocyte

Question 7:

Surfactant is composed mainly of_____

- a) Phosphatidyl glycerol
- b) Cephalin
- c) Lecithin
- d) Protein

Question 8:

All are true of surfactant except_____

- a) Helps prevent pulmonary edema
- b) Deficiency of surfactant causes Infant Respiratory Distress Syndrome

- c) Helps prevents alveolar collapse according to Bernoulli's principle
- d) Decrease in surfactant increases the work of breathing

Question 9:

A young lady at 35 weeks of gestation is at high risk of preterm delivery in the next 7 days. She is started on antenatal corticosteroid therapy to increase surfactant production in the fetus. What is the mechanism of action of this substance?

- a) Breaks the structure of water in alveoli
- b) Lubricate the flow of CO₂ diffusion
- c) Binds oxygen
- d) Makes the capillary surface hydrophilic

Question 10:

Stability of alveoli is maintained by_____.

- a) Lung compliance
- b) Negative intrapleural pressure
- c) Increased alveolar surface area by surfactant
- d) Residual air in alveoli

Question 11:

A woman at risk for preterm delivery is planned for assessment of fetal lung maturity. At what gestational age will the surfactants be detectable in the amniotic fluid?

- a) 28 weeks
- b) 34 weeks
- c) 35 weeks
- d) 36 weeks

Question 12:

A woman on multiple medications came for an antenatal check-up. Which among the following medications that she takes decreases surfactant production?

- a) Thyroid hormones
- b) Beta adrenergic agonists
- c) Inadequate insulin
- d) Calcium supplements

Question 13:

Which is the best metabolic function of the lung?

- a) Inactivation of serotonin
- b) Inactivation of bradykinin
- c) Metabolism of basic drugs
- d) Conversion of angiotensin I to angiotensin II

Answer Key

Question No.	Correct Option
1	c
2	b
3	d
4	a
5	b
6	d
7	c
8	c
9	a
10	c
11	a
12	c
13	d

Detailed Explanations

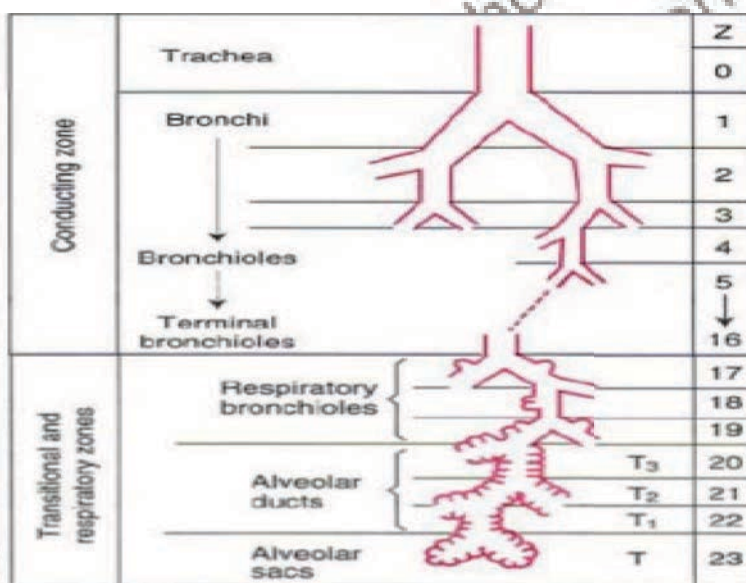
Solution to Question 1:

The total number of airway generations present in between the trachea and alveolar sac is 23.

The tracheobronchial tree is divided into multiple generations based on Weibel's lung model. It begins at the trachea (designated as zero) and ends with alveoli (last generation). There are 2 zones:

- Conducting zone
- Consists of the first 16 generations
- Incapable of gas exchange
- Respiratory zone
- Consists of the last 7 generations (17-23)
- Capable of gas exchange

Terminal bronchiole is the transition point between the conducting and respiratory zone.



Solution to Question 2:

Among the given options, bronchi (medium-sized airway) offer maximum resistance.

The best answer here is bronchi, as all the other options suggest smaller airways, while bronchi are the only medium-sized airway.

Airflow can be derived from Poiseuille-Hagen formula:

$$Q = \pi \Delta P r^4 / 8 \eta L$$

Where

- Q is the flow rate
- π is pie symbol
- ΔP is the pressure difference

- r is radius
- η is viscosity
- L is length

Also,

Flow rate (Q) = pressure difference (ΔP) / resistance (R).

Therefore, substituting this in Poiseuille- Hagen formula,

$$\Delta P / R = \pi \Delta P r^4 / 8 \eta l$$

$$1/R = \pi r^4 / 8 \eta l$$

$$R = 8 \eta l / \pi r^4$$

Therefore it can be derived that airway resistance is inversely proportional to the radius by the power of four.

With Poiseuille- Hagen formula we might feel that the smallest airways are the major site of resistance. However, the contribution by the smallest airways is very little to the overall total resistance. This is because, the airway branches in each generation exist in parallel rather than in series. As the parallel branching increases, the resistance decreases. (Total cross-sectional diameter of small airways put together is high). Thus the overall resistance of the small airways is very small. Also, airflow velocity decreases substantially as the effective cross-sectional area increases (i.e., the flow becomes laminar).

Other factors affecting airway resistance :

- Lung volume, as lung volume increases the resistance decreases and vice versa. Lung volume increases the transairway pressure thus increasing the caliber of the airway.
- Airway mucus, edema, and contraction of bronchial smooth muscle increase the airway resistance. Its clinical application is seen in asthma, bronchitis, and emphysema
- The density and viscosity of inspired gas also contribute to airway resistance. A gas with more density imparts more resistance.

Solution to Question 3:

Chronic inflammation of conducting airways is suggestive of asthma. Lamellar bodies are storage bodies of surfactant present in the alveoli and not the conducting airway.

The conducting airways form the first 16 generations from the trachea to the terminal bronchial. It is lined by pseudostratified ciliated columnar epithelium. It also consists of smooth muscles which are responsible for bronchoconstriction and dilatation. It also consists of other specialized cells such as:

- Basal cells - stem cells
- Clara/Club cells - specialized epithelial cells (non-ciliated cuboidal cells) that:
- Functions as stem cells
- Helps in lung defense.

The conducting airways secrete several substances that in lung defense such as:

- Secretory immunoglobins - IgA
- Collectins - surfactant protein(SP)-A and SP-D
- Defensins
- Reactive oxygen species
- Reactive nitrogen species.

Solution to Question 4:

Smooth muscle is maximum in terminal bronchioles.

Solution to Question 5:

Salbutamol, a short-acting beta (adrenergic) agonist (SABA) is a bronchodilator. Similarly, vasoactive intestinal polypeptide (VIP) is also a bronchodilator. Along with dynorphin, it is part of the non-adrenergic non-cholinergic (NANC) inhibitory system.

Bronchial innervation:

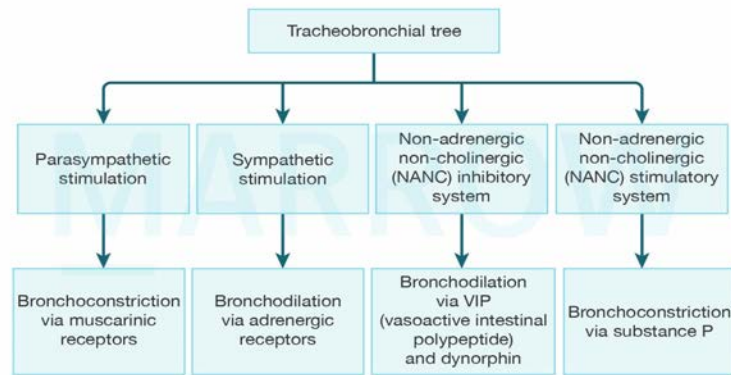
Other potent bronchoconstrictors are histamine and leukotrienes.

Clinical relevance:

- Methacholine sensitivity test or Airway hyperresponsiveness test: Methacholine which is similar to acetylcholine causes exaggerated bronchoconstriction in case of bronchial asthma.
- 5-Lipoxygenase inhibitor (5-LOX inhibitor-Zileuton), selective CysLT-receptor antagonists (Cys leukotriene receptor antagonist-zafirlukast, pranlukast, and montelukast), and β_2 receptor agonists are used in the treatment of asthma.

Site	Type of receptor	Effect
Sympathetic system		
Tracheal and bronchial smooth muscle	β_2	Relaxation
Bronchial glands	$\alpha_1\beta_2$	Reduce secretion Increase secretion
Parasympathetic system (more predominant)		
Tracheal and bronchial smooth muscle	M2, M3	Contraction
Bronchial glands	M2, M3	Stimulation

Innervation of the tracheobronchial tree



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Solution to Question 6:

Type 2 pneumocytes secrete surfactant in the lungs.

Alveolar Lining Epithelial Cells

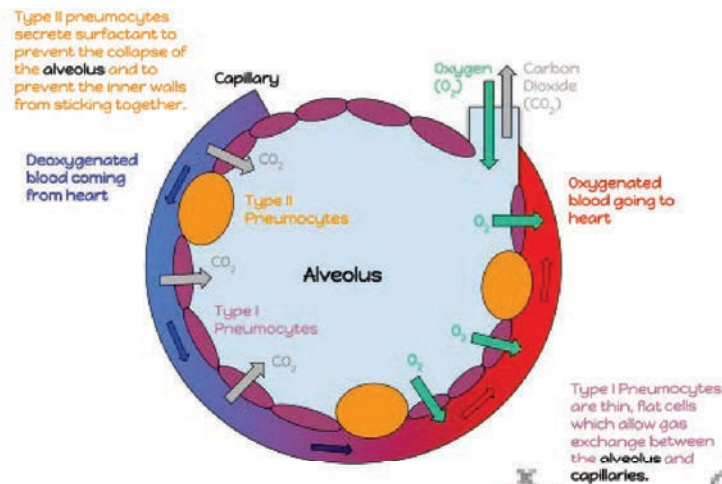
Two types

- Type I cells – The primary lining cells of the alveoli (squamous epithelium); Helps in gas exchange
- Type II cells – Granular pneumocytes produce surfactant

Type II cells make 60% of epithelial cells in alveoli but type I cells form 95% of the total surface area because of its flat surface.

Type II cells are stem cells which help in replenishing damaged Type 1 pneumocytes in conditions like drug-induced pulmonary toxicity by amiodarone, bleomycin etc.

Other cells – Pulmonary alveolar macrophages (PAMs/type 3 alveolar cells), lymphocytes, plasma cells, neuroendocrine cells (APUD cells-Amine precursor uptake and decarboxylation), and mast cells.



Solution to Question 7:

Major surfactant is dipalmitoylphosphatidylcholine i.e Lecithin. Sphingomyelin is a minor surfactant.

The surfactant is a surface-active agent in water, which means that it greatly reduces the surface tension of water.

Pulmonary surfactant is produced by type 2 pneumocytes.

Most important components are

- Phospholipid - dipalmitoylphosphatidylcholine (DPPC) also known as Dipalmitoyl Lecithin. Lecithin is the most abundant lipid in surfactant.
- Phosphatidylglycerol- minor component
- Surfactant apoproteins
- Calcium ions.
- 4 unique apoproteins of surfactant: surfactant protein (SP)-A, SP-B, SP-C, and SP-D.

Clinical relevance:

- Tests to assess fetal lung maturity:
- L/S ratio i.e. Lecithin/Sphingomyelin ratio: Ratio ≥ 2 is considered normal.
- Shake test/Bubble test/Clement test: Formation of stable foam in the presence of surfactant.
- Phosphatidylglycerol levels in Amniotic fluid
- Autosomal dominant familial interstitial lung disease has absent surfactant protein SP-C.
- Pulmonary alveolar proteinosis - Autoantibody against GM-CSF which is involved in clearance of surfactant.

Solution to Question 8:

Surfactant helps to prevent alveolar collapse according to Law of Laplace.(Not Bernoulli's)

Physiologic advantages of surfactant

- By reducing surface tension it increases lung compliance
- Reduces the lung's tendency to recoil
- Helps to keep the alveoli dry
- Helps to prevent pulmonary edema (Option A)

The surface tension generates a pressure that tends to collapse the sphere.

In accordance with Law of Laplace - $P=2T/r$, where

P - Collapsing pressure on alveolus (dynes/cm²) or Pressure required to keep alveolus open (dynes/cm²)

T = Surface tension (dynes/cm)

r = Radius of the alveolus (cm).

Collapsing pressure is inversely related to radius i.e. large alveolus (one with a large radius) will have a low collapsing pressure and therefore will require only minimal pressure to keep it open. On the other hand, a small alveolus (one with a small radius) will have a high collapsing pressure and require more pressure to keep it open.

By reducing surface tension, surfactant reduces the collapsing pressure for a given radius.

Option B: Surfactant deficiency is an important cause of infant respiratory distress syndrome (IRDS) also known as hyaline membrane disease, the serious pulmonary disease that develops in infants born before their surfactant system is functional.

- In IRDS, there is high surface tension in the lungs resulting in atelectasis
- Also, there is retention of fluid in the lungs.
- During fetal life, Cl⁻ is secreted with fluid by the pulmonary epithelial cells. At birth, there is a shift to Na⁺ reabsorption by these cells via ENaC and fluid is absorbed along with Na⁺. Prolonged immaturity of ENaC contributes to pulmonary edema in IRDS.

Option D: Surfactant increases lung compliance, which reduces the work of expanding the lungs during inspiration. Increasing the compliance of the lungs reduces the collapsing force at any given volume so that it is easier for the lungs to expand.

Solution to Question 9:

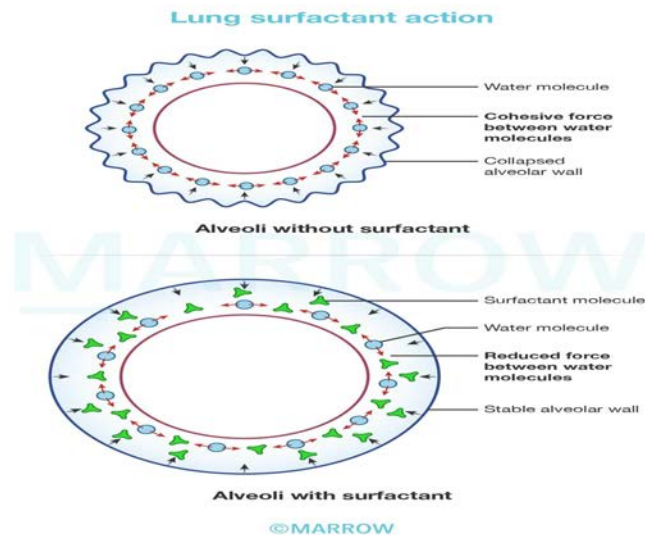
The mechanism of action of surfactant is by breaking the structure of water in alveoli.

When water forms a surface with air, the water molecules on the surface of the alveoli have a strong attraction for one another. This pulls the molecules together, leading to high surface

tension.

The surfactant is amphipathic in nature (i.e., hydrophobic on one end and hydrophilic on the other). They align themselves on the alveolar surface, with their hydrophobic portions facing the alveolar lumen.

The cohesive force between a water molecule and an adjacent pulmonary surfactant molecule is very low. So, they break up the attracting forces between water molecules lining the alveoli (which had been responsible for the high surface tension). Thus, they reduce surface tension.



Solution to Question 10:

Stability of alveoli is maintained by increased alveolar surface area by surfactant.

Surfactant reduces the lung's tendency to recoil so that they do not collapse easily and so the stability of the alveoli is maintained.

Factors that maintain alveolar stability

- Pulmonary surfactant: ↓ surface tension within alveoli
- Mechanical interdependence: prevents alveolar collapse

Interdependence

The tendency of one alveolus to collapse is opposed by the traction exerted by the surrounding alveoli. Thus, the collapse of a single alveolus stretches and distorts the surrounding alveoli, which in turn are connected to other alveoli. This support offered to lung units by those surrounding them is termed Interdependence.

Solution to Question 11:

Surfactant appears in the amniotic fluid between 28 and 32 weeks of gestation.

By week 20-24, primitive alveoli form and surfactant production begins but it does not reach the surface of the lungs until later.

Mature levels of pulmonary surfactant are present usually after 35 wk of gestation.

Clinical aspects :

- Preterm infants develop respiratory distress syndrome or hyaline membrane disease due to surfactant deficiency.
- Antenatal corticosteroids are given to increase the production of surfactants.
- Lecithin/Sphingomyelin ratio is a test for fetal lung maturity. L/S ratio ≥ 2 is suggestive of a mature fetal lung.

Solution to Question 12:

Inadequate Insulin intake in diabetic pregnant causes inadequate glucose control. This causes excess glucose in maternal plasma, crossing Placenta. Excess glucose in fetal plasma causes fetal pancreas to hypersecrete fetal Insulin.

Excess insulin in fetal plasma decreases surfactant production. This is the reason for the increased incidence of hyaline membrane disease in infants of diabetic mothers.

Factors increasing surfactant synthesis:

- Glucocorticoids
- Thyroxine
- Stretching of lungs in infancy
- Beta adrenergic agonist
- Calcium
- Regular exercise.

Factors decreasing surfactant synthesis:

- Smoking
- Long term inhalation of 100% oxygen
- Occlusion of main bronchus/pulmonary artery.

Solution to Question 13:

Conversion of angiotensin I to angiotensin II is the most significant metabolic function of the lungs.

This is mediated by angiotensin-converting enzyme (ACE), located on the surface of endothelial cells in the pulmonary capillaries.

- Synthesized and used in the lungs

- Surfactant
- Synthesized or stored and released into the blood
- Prostaglandins
- Histamine
- Kallikrein
- Partially removed from the blood
- Prostaglandins
- Bradykinin
- Adenine nucleotides
- Serotonin
- Norepinephrine
- Acetylcholine
- Activated in the lungs
- Angiotensin I → Angiotensin II

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Lung Mechanics

Question 1:

A healthy 25-year-old software engineer participates in a 10-km charity run. Which of the following muscle does this person use during expiration?

- a) Diaphragm
- b) Internal intercostals and abdominal recti
- c) Diaphragm and External intercostals
- d) Diaphragm and Internal intercostals

Question 2:

The intrapleural pressure is _____

- a) Always positive
- b) Sometimes positive and sometimes negative
- c) Negative except in forced expiration
- d) Negative at apex but positive at base

Question 3:

The intrapleural pressure is negative both during inspiration and expiration because _____

- a) Intrapulmonary pressure is negative
- b) Thoracic cage and lung recoil in opposite direction
- c) Transpulmonary pressure determines negativity
- d) Surfactant prevents lung collapse

Question 4:

The intrapleural pressure (cm of H₂O) at the beginning of inspiration is around

- a) -3.4
- b) -2.5

- c) 0
- d) +2.5

Question 5:

With reference to atmospheric pressure, what is the alveolar pressure at the end of normal inspiration (in cms of H₂O)?

- a) +1.0
- b) 0
- c) -1.0
- d) -6.0

Question 6:

The transpulmonary pressure is the pressure difference between _____

- a) Intrapleural and atmospheric pressure
- b) Intra alveolar and intrapleural pressure
- c) Intrapulmonary and atmospheric pressure
- d) None of the above

Question 7:

A man inspires 1000ml from a spirometer. The intrapleural pressure was -4cm H₂O before inspiration and -12cm H₂O at the end of inspiration. What is the compliance of lungs?(ml/cm H₂O)?

- a) 50
- b) 100
- c) 125
- d) 150

Question 8:

A 22-year-old woman has pulmonary compliance of 0.2 L/cm H₂O and an alveolar pressure of 4 cmH₂O and an intrapleural pressure of -5 cmH₂O at the beginning of inspiration. What

is the transpulmonary pressure (in cmH₂O) when this woman inhales 600mL of air ?

- a) 5
- b) 1
- c) 12
- d) 10

Question 9:

Compared to a gas-ventilated lung, a liquid-ventilated lung _____

- a) Has reduced airway resistance
- b) Has increased residual volume
- c) Is more compliant
- d) Has more hysteresis

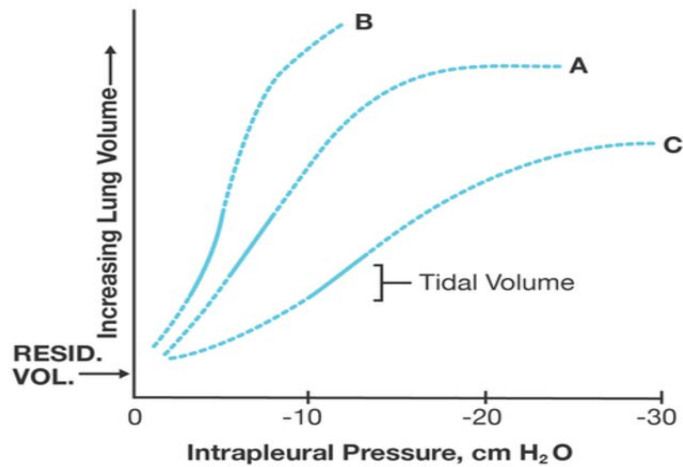
Question 10:

The factor responsible for hysteresis is _____

- a) Elastic recoil
- b) Surfactant
- c) Negative intrapleural pressure
- d) All of the above

Question 11:

The diagram below shows the pressure-volume curves for 3 different lungs. Which of the following is the best representation of compliance between the three lungs?

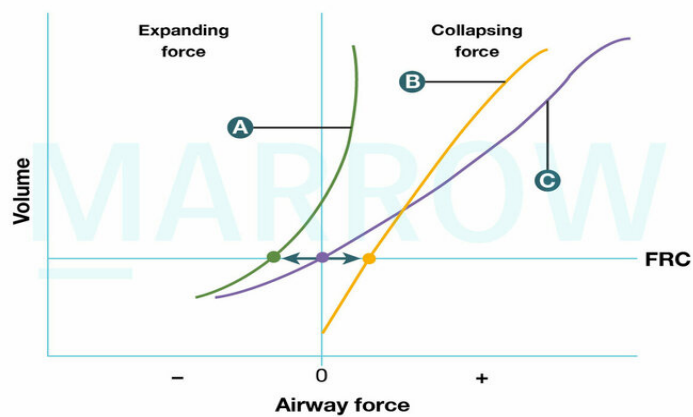


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- a) $A < B < C$
- b) $B > A > C$
- c) $B > C > A$
- d) $C > A > B$

Question 12:

In the pressure–volume graph shown below, compliance of lung is represented by _____.



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- a) A

- b) B
- c) C
- d) None of the above

Question 13:

What is specific compliance of lung?

- a) Compliance/FRC
- b) FRC/Compliance
- c) Compliance/TLC
- d) Compliance/TV

Question 14:

Work done in quiet breathing is (kg-m/min)_____

- a) 0.1
- b) 0.5
- c) 10
- d) 2

Question 15:

Work of breathing is mostly due to_____

- a) Surface tension elasticity
- b) Tissue elasticity
- c) Viscous resistance
- d) Airway resistance

Question 16:

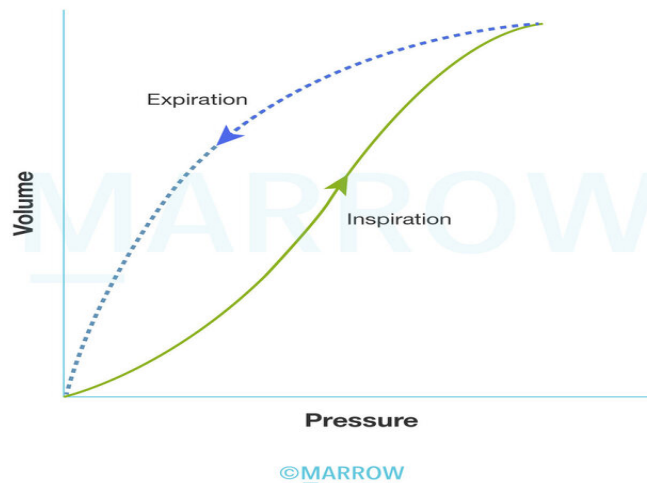
If the lung is allowed to undergo elastic recoil without the chest wall, the resultant volume of the lung would be?

- a) Minimum volume

- b) Zero
- c) Functional residual capacity
- d) Residual volume

Question 17:

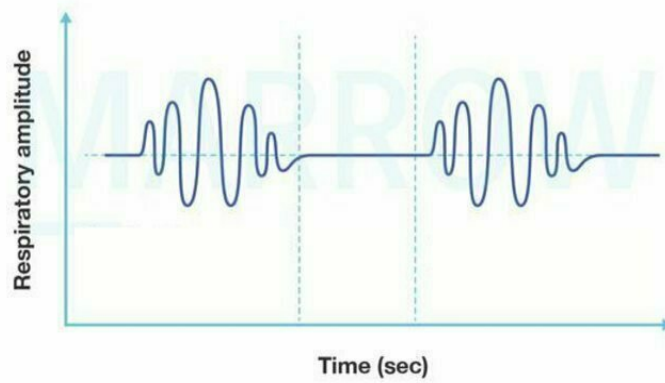
Difference in trajectory between inspiratory loop and the expiratory loop in the curve is due to:



- a) Difference in alveolar pressure during inspiration and expiration
- b) Difference in conc. of surfactant during inspiration and expiration
- c) Difference in airway resistance during inspiration and expiration
- d) Inspiration is active and expiration is passive

Question 18:

The following diagram shows the depth of respiration of a 45-year old man who suffered a head injury in an automobile accident. Identify the pattern of breathing.



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- a) Apnea
- b) Biot breathing
- c) Cheyne-Stokes breathing
- d) Hyperpnea

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	a
5	b
6	b
7	c
8	c
9	c
10	b
11	b
12	b
13	a
14	b

15	a
16	a
17	b
18	c

Detailed Explanations

Solution to Question 1:

In this scenario, the person has performed a strenuous exercise, so he will be using accessory muscles of respiration along with primary muscle of respiration. Internal intercostals and abdominal recti are the muscles involved.

Muscles involved in Respiration	
Primary inspiratory muscles	Diaphragm External intercostal muscles
Accessory inspiratory muscles	Sternocleidomastoid Scalene Lae nasi Small muscles of the head & neck
Accessory expiratory muscles	Internal intercostal muscles Rectus Abdominis Internal & external obliques Transversus abdominis

Clinical correlation:

- The diaphragm is the chief muscle of inspiration.
- It is supplied by the phrenic nerve (C3-C5)
- Hence transection of the spinal cord above the third cervical segment is fatal without artificial respiration, but transection below the fifth cervical segment is not fatal because, it leaves the phrenic nerves that innervate the diaphragm.

Solution to Question 2:

Intrapleural pressure is usually negative. (normally -2.5 mmHg)

During inspiration, it becomes more negative. (-6 mmHg).

During forced expiration, contraction of the expiratory muscles raises intrapleural pressure to a positive value.

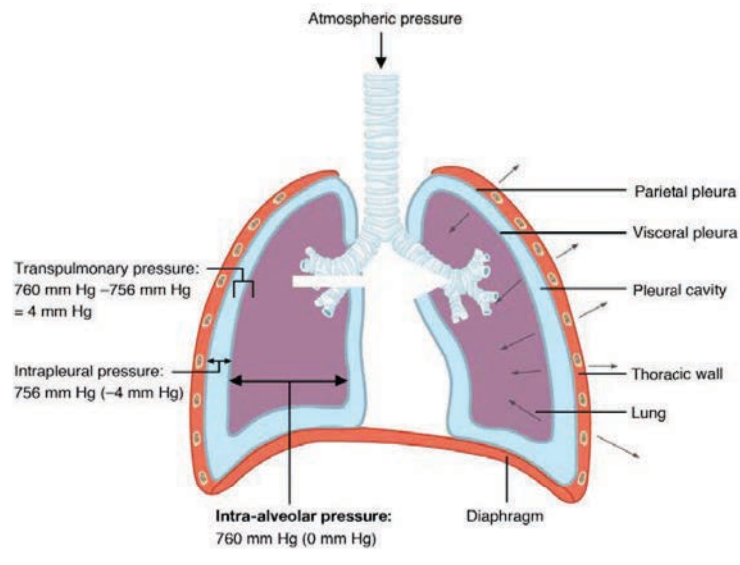
Intrapleural pressure is the pressure of the fluid in the thin space between the visceral and parietal pleura.

Solution to Question 3:

The intrapleural pressure is negative both during inspiration and expiration because the thoracic cage and lung recoil in opposite direction.

The 2 factors that lead to a negative or subatmospheric intrapleural pressure are:

- The lungs, with their elastic properties, tend to collapse, and the chest wall, with its elastic properties, tends to spring out. When these two opposing forces pull on the intrapleural space, a negative pressure, or vacuum, is created.
- The continuous suction of excess fluid into the lymphatic channels.



Applied aspect:

In cases of pneumothorax, intrapleural pressure suddenly becomes equal to atmospheric pressure. Thus instead of its normal negative value, intrapleural pressure becomes zero. Hence lungs collapse and the chest wall spring out.

Solution to Question 4:

The normal intrapleural pressure at the beginning of inspiration is about $-3.4 \text{ cm H}_2\text{O}$ (-2.5 mmHg).

Intrapleural pressure is the pressure of the fluid in the thin space between the lung pleura and the chest wall pleura. During normal inspiration, expansion of the chest cage pulls outward on the lungs with greater force and creates more negative pressure.

At the peak of normal inspiration, it is about $-8.15 \text{ cm H}_2\text{O}$ (-5.4 mm Hg).

Note: $1 \text{ mm Hg} = 1.36 \text{ cm H}_2\text{O}$

Solution to Question 5:

At the end of inspiration, the alveolar pressure is zero with reference to the atmospheric pressure.

Whenever alveolar pressure equals atmospheric pressure, alveolar pressure is said to be zero. This happens when no air is moving in or out of the lungs.

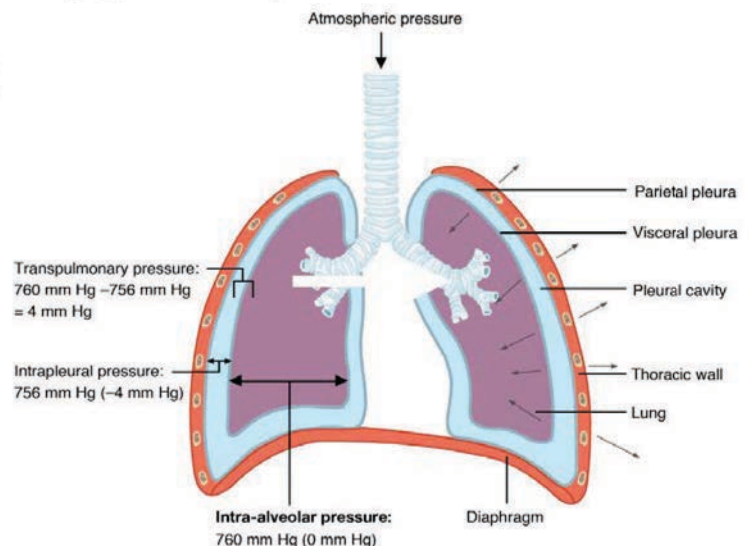
During inspiration, the diaphragm contracts, causing the volume of the thorax to increase. As lung volume increases, the pressure in the lungs must decrease. (Boyle's law states that $P \times V$ is constant at a given temperature.) Halfway through inspiration, the alveolar pressure falls below atmospheric pressure ($-1 \text{ cm H}_2\text{O}$). The pressure gradient between the atmosphere and the alveoli drives airflow into the lung. Air flows into the lungs until, at the end of inspiration, alveolar pressure is once again equal to the atmospheric pressure; the pressure gradient between the atmosphere and the alveoli has dissipated, and airflow into the lungs ceases.

During expiration, alveolar pressure rises to about $+1 \text{ cm H}_2\text{O}$.

Solution to Question 6:

Transpulmonary pressure is intra-alveolar pressure minus intrapleural pressure. (Mnemonic TAP: Transpulmonary pressure = Alveolar pressure - Pleural pressure)

It is the pressure difference between that in the alveoli and that on the outer surfaces of the lungs, and it is a measure of the elastic forces in the lungs that tend to collapse the lungs at each instant of respiration, called the recoil pressure.



Solution to Question 7:

Compliance is change in volume per unit change in pressure.

$$C = \Delta V / \Delta P$$

In this case,

$$\Delta V = 1000 \text{ mL}$$

$$\Delta P = -12 - (-4) = -8 \text{ cm of H}_2\text{O}$$

The minus sign indicates that the pressure has fallen by 8 cm of H₂O at the end of inspiration

Therefore the compliance is,

$$C = 1000 / 8 = 125 \text{ mL/cm of H}_2\text{O}$$

Solution to Question 8:

The transpulmonary pressure would be 12 cmH₂O when she inhales 600 mL of air.

Transpulmonary pressure (P_{tp}) is the difference between alveolar pressure (P_{alv}) and intrapleural pressure (P_{ip}). It is given by the formula: P_{tp} = P_{alv} - P_{ip}. Inspiration causes a negative change in both the P_{alv} and P_{ip}. Lung compliance is the extent to which the lungs will expand for each unit increase in transpulmonary pressure.

In a normal adult, the pulmonary compliance is 0.2 L/cmH₂O or 200 mL/cmH₂O of both lungs together. Hence, expanding the lungs by 200 mL of inspired air will require an increase in P_{tp} by 1 cm H₂O.

Given:

$$\text{Pulmonary compliance} = 0.2 \text{ L/cm H}_2\text{O} = 200 \text{ mL/cm H}_2\text{O}$$

$$\text{Alveolar pressure at the beginning of inspiration} = 4 \text{ cmH}_2\text{O}$$

$$\text{Intrapleural pressure at the beginning of inspiration} = -5 \text{ cmH}_2\text{O}$$

$$\text{Volume of air inhaled} = 600 \text{ mL (0.6L)}$$

Step 1: Calculate the initial transpulmonary pressure (P_{tp}).

$$\text{Transpulmonary pressure} = \text{Alveolar pressure} - \text{Intrapleural pressure}; \text{ Initial transpulmonary pressure} = 4 \text{ cmH}_2\text{O} - (-5 \text{ cmH}_2\text{O}) = 9 \text{ cmH}_2\text{O}$$

Step 2: Use the lung compliance formula to find the change in transpulmonary pressure.

$$\text{Lung compliance} = \text{Change in volume} / \text{Change in transpulmonary pressure.}$$

Pulmonary compliance is 0.2 L/cmH₂O or 200 mL/cmH₂O. Hence, to expand the lungs by 200 mL of inspired air, it will require an increase in P_{tp} by 1 cm H₂O. So, for 600 mL inspired air, the compliance is = 600 mL / 200 mL/cm H₂O = 3 cmH₂O

Step 3: Calculate the transpulmonary pressure (P_{tp}) after inhaling 600 mL of air.

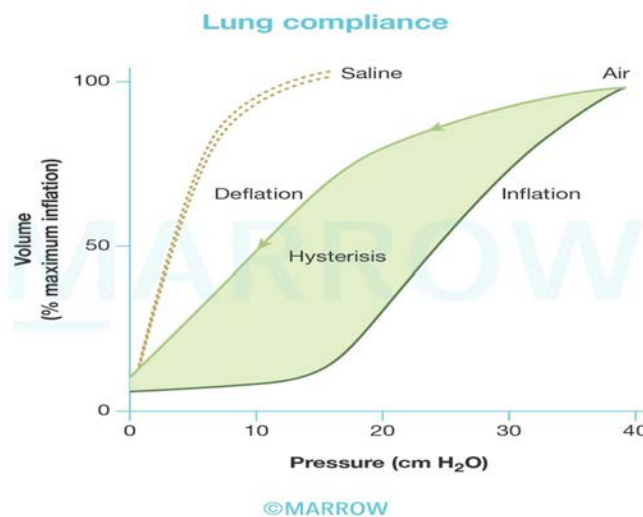
During inspiration, the alveolar pressure (P_{alv}) decreases, and the intrapleural pressure (P_{ip}) becomes more negative.

$$\text{Final transpulmonary pressure} = \text{Initial transpulmonary pressure} + \text{change in transpulmonary pressure}; \text{ Final transpulmonary pressure} = 9 \text{ cmH}_2\text{O} + 3 \text{ cmH}_2\text{O} = 12 \text{ cmH}_2\text{O}.$$

Solution to Question 9:

A liquid-ventilated lung is more compliant compared to a gas-ventilated lung.

Compliance is a measure of distensibility. In a gas-filled lung, surface tension and elastic recoil forces act to collapse the lung and thus reduce compliance.



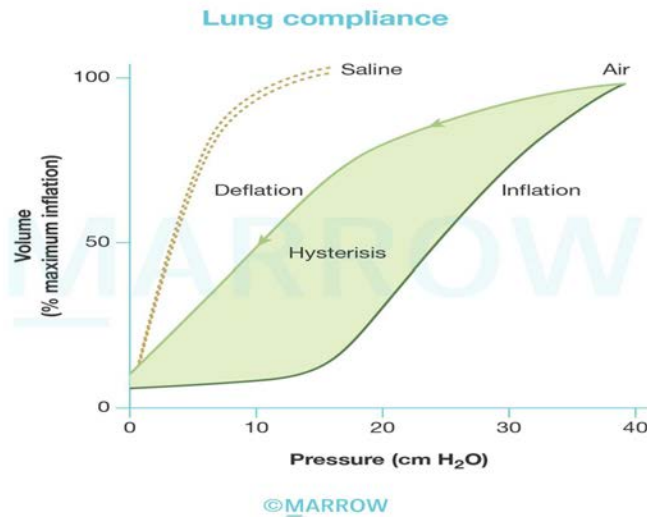
- When the lungs are filled with air, there is an interface between the alveolar fluid and the air in the alveoli; so, there is surface tension.
- But, in the case of the saline solution-filled lungs, there is no air–fluid interface and there is no surface tension, making it more compliant.
- Hence, the pressure–volume curve obtained with saline measures only the tissue elasticity.
- The difference in the curves generated during inflation and deflation, termed hysteresis, is not present in the saline-generated curves.

Solution to Question 10:

Surfactant is the factor responsible for hysteresis.

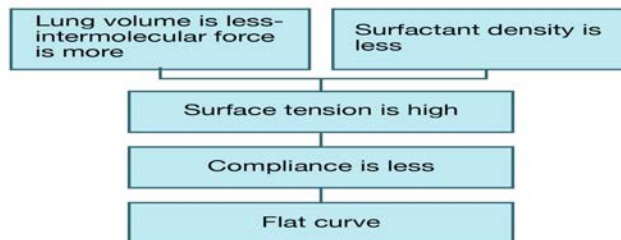
Difference between the curve of inflation (inspiration) and deflation (expiration) in a pressure–volume curve of the lung is termed as hysteresis.

The alveolar environment and, specifically, the secreted factors that help reduce surface tension and keep alveoli from collapsing, contribute to hysteresis.



Inspiration curve:

Beginning of inspiration:

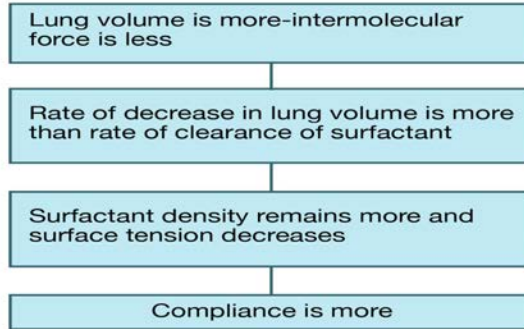


As inspiration continues:

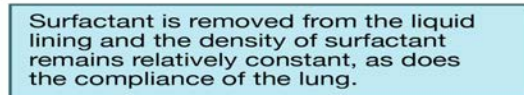


Expiration curve:

Beginning of expiration:



As expiration continues:

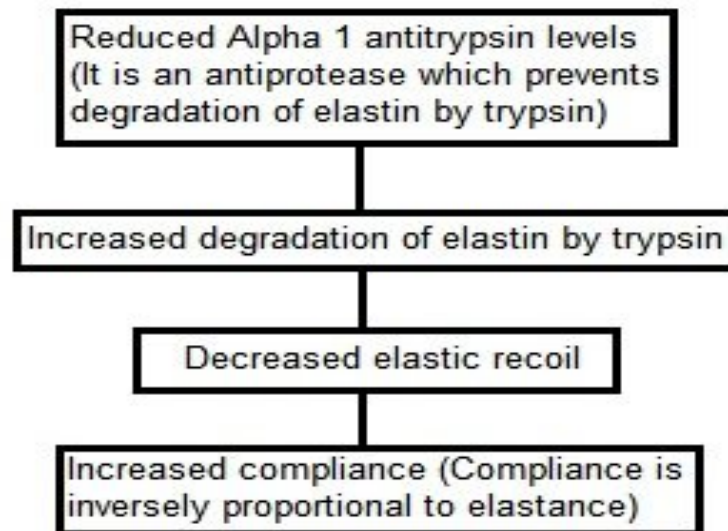


Solution to Question 11:

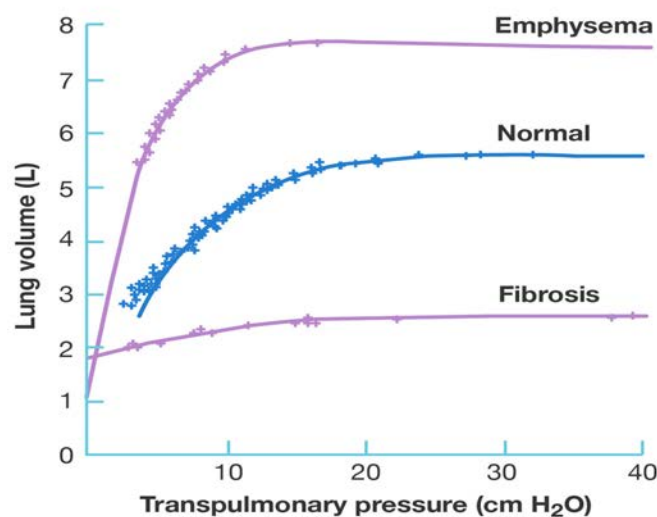
- The pressure-volume relationship is the fundamental relation between the work of breathing and lung mechanics.
- The diagram depicts the change in volume with respect to the change in intrapleural pressure, which is compliance. Here, the change is more (steep curve) with B and least (flat curve) with C.
- The "stiffer" lung displays a more horizontal curve (C), meaning that it takes a greater pressure to achieve the same inflation volume compared with a normal lung (A). These lungs are called less compliant.
- Curve (B) is more compliant as it takes lesser pressure to achieve the same inflation volume compared to normal (A).

Applied aspect:

Compliance is increased in emphysema. The mechanism is shown in the flow chart below.



Compliance is decreased in case of pulmonary edema or fibrosis.



Solution to Question 12:

In the given pressure–volume graph:

- A – Compliance of chest
- B – Compliance of lung
- C – Compliance of lung and chest wall

The equilibrium position is at functional residual capacity (FRC), where the expanding force on the chest wall is exactly equal to the collapsing force on the lungs. FRC is also known as relaxation volume and represents the amount of air remaining in the lungs at the end of normal expiration.

At the point where volume = FRC,

- Intraalveolar pressure is equal to atmospheric pressure and considered as zero.
- Intrapleural pressure is negative (-2.5 mmHg)
- Transpulmonary pressure = Intraalveolar pressure – intrapleural pressure = $0 - (-2.5 \text{ mmHg}) = +2.5 \text{ mmHg}$.

For compliance of lung – Pressure considered is Transpulmonary pressure. So at FRC, Transpulmonary pressure is positive (+2.5 mmHg). Hence, B indicates compliance of lung.

For compliance of chest wall – Pressure considered is intrapleural – atmospheric pressure. At FRC, this value is -2.5 mmHg. Hence, A represents compliance of the chest wall.

When the volume is FRC, the combined lung and chest-wall curve intersects the X-axis at an airway pressure of zero. Hence, C represents the combined chest wall and lung compliance.

Solution to Question 13:

Specific compliance is obtained by dividing compliance by functional residual capacity.

Types of compliance:

- Static compliance: It is measured at rest.
- Dynamic compliance: It is measured during breathing. It considers airflow resistance or impedance into account. It is maximum in medium-sized bronchi.
- Specific compliance:
- Lung compliance is volume-dependent.
- It is greater at low lung volumes than at high lung volumes. For this reason, the term specific compliance is often used to denote compliance with reference to the original lung volume.
- The total compliance of a normal person near the normal end-expiratory lung volume (functional residual capacity [FRC]) is about 0.1 L/cm H₂O.
- The compliance of the lungs is about 0.2 L/cm H₂O.

Applied aspect:

In case of emphysema, compliance is increased but specific compliance decreases (because FRC is severely elevated).

Solution to Question 14:

Work done in quiet breathing is 0.5 kg-m/min.

Estimates of the total work of quiet breathing range from 0.3 up to 0.8 kg-m/min.

The work of breathing can be calculated from the pressure-volume curve.

The value rises markedly during exercise, but the energy cost of breathing in normal individuals represents less than 3% of the total energy expenditure during exercise.

Applied aspect:

In the case of COPD, airway resistance increases (obstruction to outflow). Hence, the patient should take slow, deep breathing.

In the case of restrictive lung disease, viscous resistance increases. Hence, the patient is advised rapid, shallow breathing.

Solution to Question 15:

Most of the work of breathing is due to surface tension elasticity.

Work of Breathing:

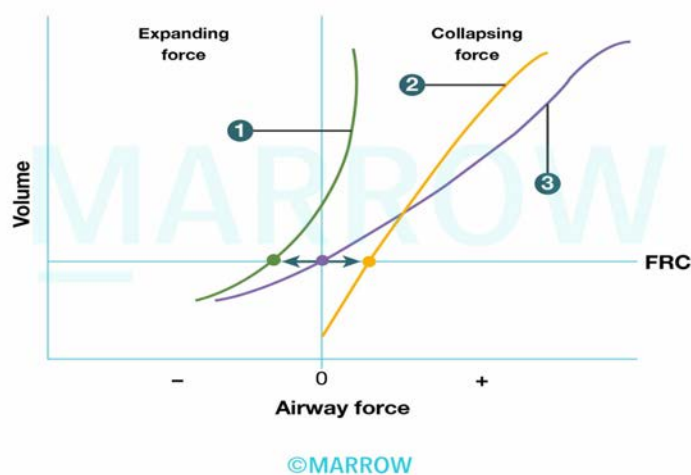
- Elastic/Compliance work (65%): Surface tension (2/3rd) + Tissue elasticity (1/3rd)
- Non elastic work (35%): Tissue/Viscus resistance (7%) + Airway resistance (28%)

Work of quiet breathing = 0.3 - 0.8 kg-m/min

Maximum work of breathing = 10kg-m/min

Solution to Question 16:

If lung is allowed to undergo elastic recoil without the chest wall, the resultant volume of the lung would be minimum volume as the opposing force by the chest wall would be non existent.



The yellow line in the above graph denotes the compliance of lungs alone. According to the graph, when the pressure (x axis) inside the lung is zero, there is some volume of air lesser than FRC still present (y axis) inside the lungs which is less than the functional residual capacity and residual

volume which can be called as minimum volume.

Solution to Question 17:

The difference between the inflation and the deflation paths is called hysteresis.

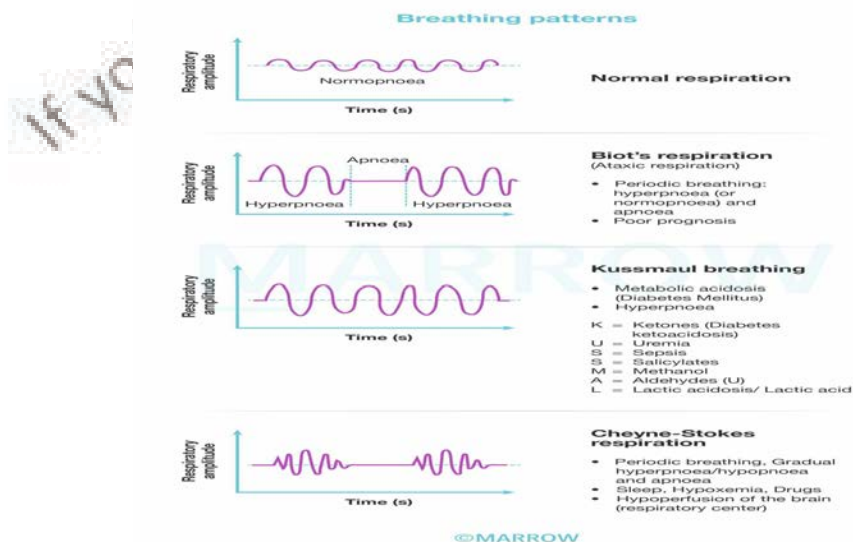
It exists because a greater transpulmonary pressure is required to open a previously closed airway, due to a deficit of surfactant at the air-water interface, than to keep an open airway from closing, as then it has abundant surfactant.

Solution to Question 18:

The pattern of breathing seen in the image is Cheyne–Stokes breathing. In this case, the person breathes deeply for a short interval and then breathes slightly or not for an additional interval. This pattern repeats itself about every minute.

It is the most common type of periodic breathing seen most commonly in patients with heart failure and uremia, also seen in patients with brain disease and during sleep in normal individuals.

This increased response occurs apparently due to the disruption of neural pathways that normally inhibit respiration. In such an individual, increased sensitivity to CO_2 causes relative hyperventilation, lowering the arterial Pco_2 . During the resultant apnoea, the arterial Pco_2 rises again to normal, but the respiratory mechanisms again over-responds to CO_2 . Breathing ceases, and the cycle repeats.



Alveolar Gas Exchange

Question 1:

Oxygen comes from alveoli to blood by _____.

- a) Osmosis
- b) Active transport
- c) Diffusion
- d) Receptor mediated

Question 2:

What is the time taken (in seconds) for blood to traverse the pulmonary capillaries at rest?

- a) 1.0
- b) 0.1
- c) 0.3
- d) 0.75

Question 3:

What is the percentage of systemic cardiac output which bypasses the pulmonary circulation (physiological shunt)?

- a) 5
- b) 2
- c) 10
- d) 15

Question 4:

A 35-year-old man with shortness of breath is undergoing tests to assess lung function. Which of the following is used to measure lung diffusion capacity?

- a) CO
- b) N₂O

- c) CO₂
- d) O₂ (under normal condition)

Question 5:

The partial pressure of H₂O at the body temperature (37° C) is (in mm Hg)_____

- a) 21
- b) 47
- c) 40
- d) 100

Question 6:

What is the partial pressure of O₂ in the alveolus?

- a) 100 mmHg
- b) 45 mmHg
- c) 60 mmHg
- d) 40 mmHg

Question 7:

If the tidal volume is 500 mL, P_eCO₂ is 30 mmHg and P_aCO₂ is 45 mmHg, calculate the dead space?

- a) 150 mL
- b) 160 mL
- c) 166 mL
- d) 145 mL

Question 8:

Which of the following is true regarding dead space volume in a healthy individual?

- a) Anatomical dead space > physiologic dead space
- b) Anatomical dead space < physiological dead space

- c) Anatomical dead space and physiologic dead space are almost equivalent
- d) Anatomical dead space is not related to physiologic dead space

Question 9:

From base to the apex of the lung during standing, all the following decrease except _____

- a) Pulmonary blood flow
- b) Ventilation per unit volume
- c) V/Q ratio
- d) Intrapleural pressure (with respect to atmospheric pressure)

Question 10:

Waterfall effect is seen in _____

- a) Lower and middle portions of the lung
- b) Upper portion of the lung
- c) Middle portion of the lung
- d) Lower portion of the lung

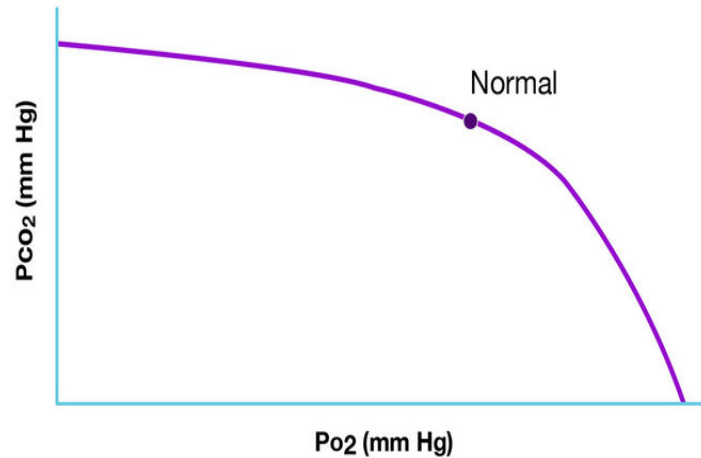
Question 11:

V/Q = infinity holds true in which of the following conditions?

- a) Foreign body obstruction in the bronchus
- b) Equal to dead space
- c) When O₂ and CO₂ ratio is equal
- d) The PO₂ of alveolar air is 159 mmHg and PCO₂ is 40 mmHg

Question 12:

The O₂-CO₂ diagram shown below shows a ventilation-perfusion ratio curve for the normal lung. Which of the following options best describes the effect of decreasing the ventilation-perfusion ratio on the alveolar Po₂ and Pco₂?



- a) $\downarrow$ O_2 tension, $\downarrow$ CO_2 tension
- b) $\uparrow$ O_2 tension, $\downarrow$ CO_2 tension
- c) O_2 tension no change, $\downarrow$ CO_2 tension
- d) $\downarrow$ O_2 tension, $\uparrow$ CO_2 tension

Question 13:

A person with normal lung at sea level (760 mm Hg) is breathing 50% oxygen. With a normal respiratory quotient of 0.8, what is the approximate alveolar P_{O_2} (in mmHg)?

- a) 713
- b) 159
- c) 306
- d) 365

Question 14:

Which of the following decreases in a patient with emphysema?

- a) Alveolar P_{CO_2}
- b) Cardiac output
- c) Diffusion area
- d) Pulmonary artery pressure

Question 15:

Which of the following best describes the mean arterial oxygen tension (pO₂), carbon dioxide tension (pCO₂) and pH in a healthy athlete during moderate exercise?

- a) ↓ Arterial pO₂, ↓ Arterial pCO₂, ↓ Arterial pH
- b) ↓ Arterial pO₂, ↑ Arterial pCO₂, ↓ Arterial pH
- c) ↑ Arterial pO₂, ↓ Arterial pCO₂, ↑ Arterial pH
- d) No change in Arterial pO₂, Arterial pCO₂ or Arterial pH

Answer Key

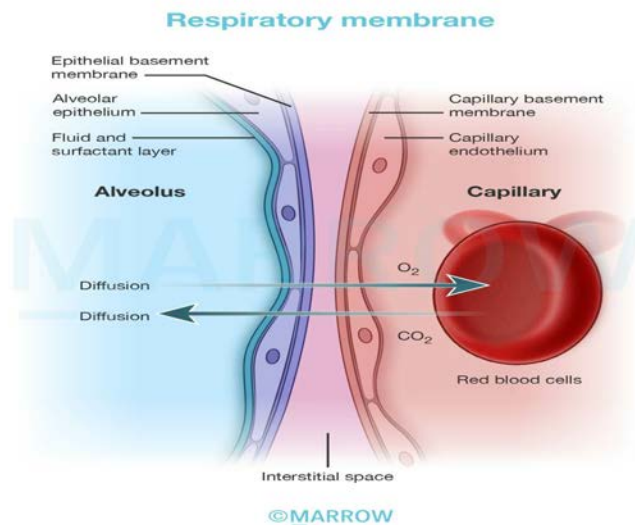
Question No.	Correct Option
1	c
2	d
3	b
4	a
5	b
6	a
7	c
8	c
9	c
10	c
11	b
12	d
13	c
14	c
15	d

Detailed Explanations

Solution to Question 1:

Oxygen diffuses from alveoli to blood along the concentration gradient i.e. passive process (ATP is not used).

It diffuses across respiratory membrane



It is governed by Fick's law of diffusion which is given in equation:

Fick's law of diffusion

$$D \propto \frac{\Delta P \times A \times S}{d \times \sqrt{MW}}$$

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Where

- D is the diffusion rate
- ΔP is the partial pressure difference between the two ends of the diffusion pathway
- A is the cross-sectional area of the pathway
- S is the solubility of the gas
- d is the distance of diffusion
- MW is the molecular weight of the gas.

(Diffusion coefficient of the gas, which is proportional to $S/\sqrt{MW}$).

Solution to Question 2:

Time taken by a red cell is about 0.75 seconds to traverse the pulmonary capillaries at rest and 0.3 seconds or less during exercise.

Solution to Question 3:

The percentage of cardiac output that bypasses the pulmonary circulation or physiological shunt is 2%. The fraction of mixed venous blood that does not become oxygenated as it moves from the right atrium to the left ventricle is called physiologic shunt.

Causes of physiological shunt:

- Extrapulmonary R-&L Anatomic Shunt (normally ~ 1% of cardiac output)
- Intrapulmonary "True" Shunt
- Flow past unventilated or collapsed alveoli (e.g., due to edema, pneumonia, atelectasis)
- Anatomic arterio-venous pathways - Mixing of unoxygenated blood in bronchial veins with oxygenated blood in pulmonary capillaries or vein reduces the final partial pressure of O₂ in the arterial blood (PaO₂).
- Intrapulmonary regions with low ventilation/perfusion ratio: Alveolar ventilation is not zero, but blood is not completely oxygenated (A virtual shunt, not a "true" shunt).

Solution to Question 4:

The diffusing capacity for CO (DLCO) is measured as an index of the diffusing capacity of the lung because its uptake is diffusion-limited.

The diffusing capacity of the lung for a given gas is directly proportional to the surface area of the alveolo-capillary membrane and inversely proportional to its thickness.

Diffusing capacity for CO (DLCO)

$$DL_{CO} = \frac{\text{Amount of CO entering the blood}}{\text{Partial pressure of CO in the alveoli} - \text{Partial pressure of CO in the blood}}$$

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The normal value of DLCO at rest is about 25 mL/min/mm Hg.

It increases up to threefold during exercise because of capillary dilation and an increase in the number of active capillaries.

Solution to Question 5:

The partial pressure of water vapor at a normal body temperature of 37°C is 47 mm Hg.

Humidification of air:

- When dry air enters respiratory system it is humidified i.e. water molecules are continually escaping from the water surface (tissues) into the gas phase.
- The partial pressure that the water molecules exert to escape through the surface is called the vapor pressure of the water.
- The vapor pressure of water depends entirely on the temperature of the water.
- The greater the temperature, the greater the kinetic activity of the molecules and, therefore, the greater the likelihood that the water molecules will escape from the surface of the water into the gas phase.
- For instance, the water vapor pressure at 0°C is 5 mm Hg, and at 100°C it is 760 mm Hg.
- The most important value to remember is the vapor pressure at body temperature 47 mm Hg. (Once the equilibrium is attained after diffusion, vapour pressure is called partial pressure).

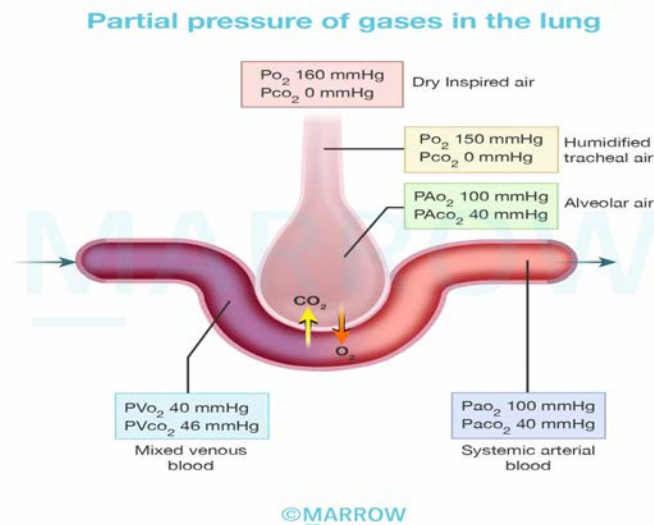
Gas	Inspired Air	Alveolar Air	Expired Air
O ₂	158.0	100.0	116.0
CO ₂	0.3	40.0	32.0
H ₂ O	5.7	47.0	47.0

Gases	Inspired Air	Alveolar Air	Expired Air
N ₂	596.0	573.0	565.0
Total	760.0	760.0	760.0

Solution to Question 6:

The partial pressure of O₂ in the alveolus is 100 mmHg.

The partial pressure of gases as they enter and leave the lungs is depicted below.



Solution to Question 7:

The Bohr-enghaff equation is used to calculate physiological dead space/total dead space.

The original Bohr equation required measurement of the alveolar partial pressure of CO₂ (PACO₂).

Bohr equation

$$\frac{V_d}{V_t} = \frac{P_A \text{ CO}_2 - P_e \text{ CO}_2}{P_A \text{ CO}_2}$$

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Later Enghoff modified the equation and replaced the $P_A \text{ CO}_2$ with $P_a \text{ CO}_2$ (partial pressure of CO_2 in arterial blood). Enghoff's modification of Bohr equation is:

Enghoff's modification of Bohr equation

$$\frac{V_d}{V_t} = \frac{P_a \text{ CO}_2 - P_e \text{ CO}_2}{P_a \text{ CO}_2}$$

©MARROW

Where,

- V_D is the physiological dead space
- V_T is the tidal volume
- $P_a \text{ CO}_2$ is the partial pressure of CO_2 in the arterial blood
- $P_e \text{ CO}_2$ is the average partial pressure of CO_2 in the entire expired air.

Substituting the values:

$$V_d = (45 - 30/45) \times 500 = 166.66 \text{ ml}$$

Solution to Question 8:

In healthy individuals, anatomical dead space and physiologic dead space are almost equivalent, and can be estimated by body weight.

Dead space is the volume of the airways and lungs that do not participate in gas exchange.

Types

- Anatomic dead space- Volume of the air-occupying conducting zone where no exchange of gases takes place.
- Alveolar/functional dead space- the volume of gas not equilibrating with blood (wasted ventilation)
- Total dead space/Physiological dead space = Anatomical + Alveolar dead space.

In healthy individuals:

- Anatomical dead space = 150 mL.
- Alveolar dead space is usually zero (increased in diseased conditions).

Hence,

- Physiological dead space = Anatomical + Alveolar = 150 + 0 = 150 mL.
- Physiological dead space = Anatomical dead space.
- Radford's formula: Anatomical dead space in mL is approximately equal to the body weight in pounds.
- Dead space can be measured by the following:
 - Single Breath N₂ method (Fowler's method)
 - Bohr's equation

Solution to Question 9:

V/Q ratio does not decrease from the base to apex of the lung while standing.

- Normal differences in ventilation and perfusion of the lung in the upright position (refer figure).

Blood flow distribution in the lung

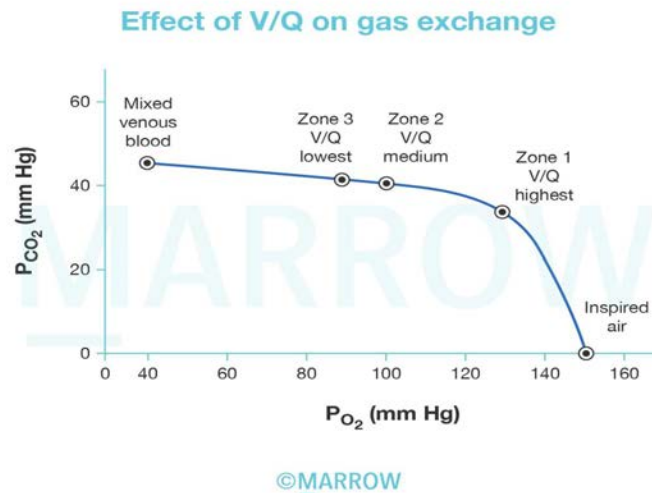
PA: Alveolar pressure

Pa: arterial pressure

PV: venous pressure

Zones of lung	Blood flow	Pressure driving blood flow
Zone 1 (near Apex)	Lowest	PA > Pa > Pv
Zone 2	Medium	Pa > PA > Pv

Zones of lung	Blood flow	Pressure driving blood flow
Zone 3	Highest	$P_a > P_v > P_A$



Ventilation-perfusion ratio is not the same in all part of the lungs because of the effect of gravity.

At the apex (Against gravity):

Both ventilation and perfusion decrease since the flow is against gravity. But, a decrease in perfusion is higher than the decrease in ventilation. Hence, ratio increases (3.2).

At the base of lung:

Both ventilation and perfusion increase since the flow is towards gravity. Increase in perfusion is more than increase in ventilation, i.e., denominator increases. Hence, ratio decreases to 0.6.

The average value of V/Q ratio throughout lungs is 0.8.

Important points:

- If the effect of gravity is eliminated, as in the supine position, the value is the same in all parts of the lung = 0.8.
- V/Q ratio is infinity when perfusion is zero as in case of pulmonary embolism.
- V/Q ratio is zero when ventilation is zero like in case of foreign body obstruction.
- High V/Q ratio at apex leads to high pA_{O_2} which favors the growth of microorganisms like tubercle bacilli.

Intra-pleural pressure decreases as we move from the base to the apex of the lung i.e- becomes more negative as we move from the base to the apex of the lung.

Solution to Question 10:

Water fall effect has been used to describe the blood flow in the middle portion of lung.

In this region blood flows in pulses. At first there is no flow because of obstruction at the collapsed venous end of the capillary bed. Pressure from the arterial side builds up until it exceeds alveolar pressure and flow resumes. This dissipates the capillary pressure and returns to the start of the cycle.

The blood flow in this region is determined by the pulmonary artery–alveolar pressure difference. Blood “falls” into the pulmonary veins, which are compliant and take whatever amount of blood can flow into them. This has been called the waterfall effect.

Blood flow distribution in the lung

PA: Alveolar pressure

Pa: arterial pressure

PV: venous pressure

Zones of lung	Blood flow	Pressure driving blood flow
Zone 1 (near Apex)	Lowest	$PA > Pa > Pv$
Zone 2	Medium	$Pa > PA > Pv$
Zone 3	Highest	$Pa > Pv > PA$

Solution to Question 11:

$VA/Q = \infty$ means that there is no exchange of O_2 & CO_2 , which means the volume of the lung equals to dead space and the physiological effort of breathing is wasted.

VA represents the ventilation in alveoli and Q represents the blood flow through the alveolus. The ratio of VA and Q explains the respiratory exchange when there is an imbalance between alveolar ventilation and alveolar blood flow. The ventilation-perfusion ratio is considered normal if there is an equal amount of blood flow & ventilation through the alveolus.

VA/Q ratio is infinity when:

- There is adequate ventilation but no perfusion.
- No exchange of gases (O_2 and CO_2) through the respiratory membranes of corresponding alveoli.
- The alveolar air equilibrates the humidified inspired air.
- Partial pressures of the O_2 and CO_2 are 149 mmHg & 0 mm Hg respectively.

Solution to Question 12:

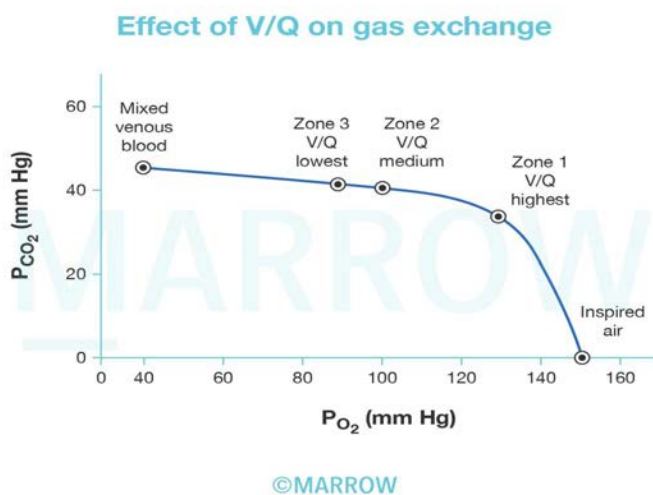
Decreased ventilation-perfusion ratio implies decreased oxygenation due to decreased ventilation with respect to perfusion. Hence, PAO_2 decreases and P_{CO_2} increases.

A decrease in the ventilation-perfusion ratio (V_A/Q) is depicted by moving to the left along the normal ventilation-perfusion line as shown in the figure (Zones here refer to the zones of variation in pulmonary blood flow and alveolar ventilation under influence of gravity)

- Zone 1 - Zone 1 has the lowest ventilation compared to Zones 2 and 3 but, in Zone 1 Ventilation (V) $\gg$ Perfusion (Q) hence V/Q is high
- PAO_2 is highest
- $PACO_2$ is lowest
- Zone 2 - Ventilation (V) almost equal to Perfusion (Q) but slightly lower so V/Q is average (~ 0.8)
- Zone 3 - Zone 3 has the highest ventilation compared to Zones 2 and 1 but, in Zone 3 Perfusion (Q) $\gg$ Ventilation (V) hence V/Q is low
- PAO_2 is lowest
- $PACO_2$ is highest.

* PAO_2 - Alveolar partial pressure of oxygen

* $PACO_2$ - Alveolar partial pressure of CO_2



Solution to Question 13:

Alveolar gas equation is used to calculate partial pressure of O_2 in alveolar air.

It is given by the formula:

Alveolar gas equation

$$PA_{O_2} = PI_{O_2} - \frac{PA_{CO_2}}{R} + \text{Correction factor}$$

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Where

- PA_{O_2} = Alveolar PO_2 (mm Hg)
- PI_{O_2} = PO_2 in inspired air (mm Hg)
- PA_{CO_2} = Alveolar PCO_2 (mm Hg), which is normally 40mmHg
- R = Respiratory exchange ratio or respiratory quotient (CO_2 production/ O_2 consumption)
- Correction factor is usually negligible.

Step 1: In the question PO_2 in inspired air is not given directly. It has to be calculated using the formula:

$$PI_{O_2} = FIO_2 (PB - PH_2O)$$

Where

- PI_{O_2} = PO_2 in inspired air (mm Hg)
- FIO_2 = Fraction of inspired O_2 (50% or 0.5)
- PB = Barometric pressure (760 mmHg)
- PH_2O = Water vapour pressure (47 mmHg)

Substituting the values:

$$PI_{O_2} = 0.5 (760 - 47) = 0.5 \times 713 = 356.5 \text{ mmHg}$$

Step 2:

$$PA_{O_2} = PI_{O_2} - (P_{CO_2}/R) = 356.5 - (40/0.8) = 356.5 - 50 = 306.5 \text{ mmHg.}$$

Solution to Question 14:

In emphysema, the destruction of the alveolar walls brings down the diffusion area.

Changes in Emphysema

- Due to the destruction of the alveolar walls, there is a decrease in surface/diffusion area.
- Cardiac output will be normal.
- Pulmonary artery pressure may be elevated because the destruction of alveoli leads to loss of surrounding capillaries which forces blood into remaining capillaries, which tends to increase pulmonary artery pressure.
- Alveolar P_{CO_2} will tend to increase due to difficulty in expiration.

Diffusion capacity of Lung for CO (DLCO):

- In case of emphysema – Decreased diffusion (DLCO) because of decreased surface area (trypsin destroys respiratory membrane)
- In case of fibrosis or pulmonary edema – Decreased diffusion (DLCO) because of increased thickness of membrane
- In case of anemia – Decreased DLCO
- Polycythemia, pulmonary hemorrhage – Increased DLCO.

Solution to Question 15:

During moderate exercise, there is no change in arterial pO_2 , arterial pCO_2 or arterial pH.

Ventilation increases abruptly at the onset of exercise due to psychic stimuli and afferent impulses from proprioceptors in muscles, tendons, and joints. This is followed by a brief pause and then a gradual increase in ventilation. This gradual increase in ventilation is believed to be due to multiple factors such as an increase in body temperature, increase in plasma K^+ level stimulating peripheral chemoreceptors, or the sensitivity of the neurons controlling the response to CO_2 .

In strenuous exercise, at a certain point, the O_2 uptake reaches its maximum levels, beyond which it cannot increase further. This leads to a rise in blood lactate levels. Buffering of lactic acid liberates CO_2 which further increases ventilation. The increased production of lactic acid, increase in ventilation and CO_2 production are proportional, and alveolar and arterial CO_2 changes little to nil. This is called isocapnic buffering.

Gas Transport in Blood

Question 1:

Which of the following is correct?

- a) $PT = nRV$
- b) $V = PnRT$
- c) $P = VnRT$
- d) $PV = nRT$

Question 2:

Transport of CO is diffusion limited because _____

- a) Alveolar membrane is less permeable to CO
- b) High affinity of CO to hemoglobin
- c) CO crosses epithelial barrier slowly
- d) None

Question 3:

What is the percentage of O₂ carried, chemically bound to Hb?

- a) 97%
- b) 3%
- c) 37%
- d) 66%

Question 4:

The presence of hemoglobin in normal arterial blood increases its O₂ concentration approximately by how many times?

- a) 70
- b) 10

- c) 30
- d) 40

Question 5:

O₂ delivery to tissues depends on all except_____

- a) Type of fluid administered
- b) Hemoglobin
- c) Ventilation
- d) Cardiac output

Question 6:

Each molecule of hemoglobin carries _____ molecules of O₂.

- a) 2
- b) 4
- c) 8
- d) 16

Question 7:

Most of the CO₂ transported in the arterial blood is in the form of_____

- a) HCO₃⁻
- b) Attached to haemoglobin
- c) Carbamino compounds
- d) Carbonic acid (H₂CO₃)

Question 8:

Which of the following statements about hemoglobin is false?

- a) The T configuration of Hb does not favor O₂ binding
- b) The R configuration of Hb does not favor O₂ binding
- c) The R configuration of Hb favors O₂ binding

- d) Each Hb molecules carries 4 oxygen molecules

Question 9:

The oxygen-haemoglobin dissociation curve is sigmoid shape because_____

- a) Binding of one O₂ molecule increases the affinity of binding of subsequent O₂ molecules
- b) Binding of one O₂ molecule decreases the affinity of binding of subsequent O₂ molecules
- c) Bohr effect
- d) Haldane effect

Question 10:

Hb would be expected to become 100% saturated with oxygen at P_{o2} of (in mmHg)_____

- a) 75
- b) 90
- c) 100
- d) 760

Question 11:

The normal value of P₅₀ in an adult is(mm Hg)_____

- a) 27
- b) 45
- c) 10
- d) 100

Question 12:

Decrease in pH shifts the oxygen dissociation curve to_____

- a) Right
- b) Left
- c) Doesn't affect
- d) On either side depending on other factors

Question 13:

Four molecules of myoglobin will bind how many molecules of oxygen?

- a) 4
- b) 16
- c) 1
- d) 8

Question 14:

Bohr effect

- a) Facilitates O₂ transport
- b) Facilitates CO₂ transport
- c) Facilitates chloride transport
- d) None

Question 15:

The level of 2,3 -DPG is decreased in _____

- a) Anemia
- b) Acidosis
- c) High altitude
- d) Exercise

Question 16:

High affinity of HbF to O₂ is due to _____

- a) Decreased binding with 2,3-DPG
- b) Increased binding with 2,3-DPG
- c) Increase in pH
- d) Decrease in concentration of Hb

Question 17:

Haldane effect is _____

- a) Binding of O₂ to Hb favours release of CO₂
- b) Binding of CO₂ to Hb favours release of O₂
- c) HCO₃⁻ leaving RBC in exchange for Cl⁻
- d) None of the above

Question 18:

You are comparing the packed cell volume (PCV) obtained from an arterial and venous blood sample of the same patient. Why is PCV higher in venous blood?

- a) Haldane effect
- b) Bohr effect
- c) Chloride shift
- d) Affinity of Hb to 2,3-DPG

Question 19:

What makes CPDA superior than ACD for blood storage:

- a) The fall in 2, 3 DPG is less
- b) Decreased release of O₂
- c) It has less P₅₀
- d) It is less acidic

Question 20:

The oxygen carrying capacity of a 18 year old boy with haemoglobin of 14g/dl is

- a) 22
- b) 16
- c) 14
- d) 18

Answer Key

Question No.	Correct Option
1	d
2	b
3	a
4	a
5	a
6	b
7	a
8	b
9	a
10	c
11	a
12	a
13	a
14	a
15	b
16	a
17	a
18	c
19	a
20	d

Detailed Explanations

Solution to Question 1:

Boyle's law states that, at a given temperature, the product of pressure time's volume for a gas is constant.

$$PV = nRT$$

Boyle's law: At a given temperature, the product of pressure times volume for a gas is constant $PV = nRT$. Charles law: At constant pressure, the volume occupied by a fixed mass of gas is proportional to temperature $V/T = \text{constant}$. Dalton's Law: It states that the partial pressure of a gas in a mixture of gases is the pressure that gas would exert if it occupied the total volume of the mixture. Partial pressure is the total pressure multiplied by the fractional concentration of dry gas. Henry's Law: The amount of gas dissolved is directly proportional to the partial pressure of the gas with which it is in equilibrium. Graham's Law: The rate of diffusion of a gas is inversely proportional to molecular weight.

Solution to Question 2:

Transport of CO is diffusion limited because of the high affinity of CO to hemoglobin.

Diffusion across the alveolocapillary membrane:

- Blood takes 0.75s to traverse pulmonary capillaries.
- CO diffuses into blood and combines with Hb with very high affinity. Thus the partial pressure of CO doesn't rise (Only dissolved form of gas can increase the partial pressure, Gas which combines with Hb or plasma proteins cannot). Partial pressure gradient necessary for diffusion of CO is maintained for the whole 0.75s and hence it is Diffusion limited.
- In the case of N₂O it does not react and reaches equilibrium in about 0.1s. The amount of N₂O taken up is not limited by diffusion but by the amount of blood flowing through the pulmonary capillaries; that is, it is Flow-limited.
- O₂ is intermediate between N₂O and CO; it is taken up by haemoglobin, but much less avidly than CO, and it reaches equilibrium with capillary blood in about 0.3 s. Thus, its uptake is Perfusion-limited.

In the case of perfusion limited gases, an increase in pulmonary blood flow increases the amount of gas transferred. Hence O₂ transport increases with an increase in pulmonary blood flow.

Solution to Question 3:

Percentage of O₂ carried chemically (bound to Hb) is 97%.

Oxygen is transported in the blood in two forms:

- physically 3% - dissolved in plasma
- chemically 97% - bound to hemoglobin.

Remember:

Gases are transported in 3 forms:

- Dissolved gas - This is the form which contributes to partial pressure. Nitrogen is carried only in dissolved form in blood.
- Bound gas - O₂, CO are bound to Hb in RBC; CO₂ is bound to Hb in RBC as well as plasma protein.
- Chemically modified gas - CO₂ in the form of HCO₃⁻.

Total gas concentration in solution is the sum of dissolved gas plus bound gas plus chemically modified gas.

Solution to Question 4:

The presence of hemoglobin in normal arterial blood increases its O₂ concentration approximately by 70 times

- Normal arterial blood has P_{O2} of about 100 mm Hg.

Amount of O₂ transported in dissolved form (Calculated by using Henry's law):

- Concentration of dissolved O₂ = Partial pressure of O₂ × solubility of O₂ = 100 mmHg × 0.003 mL O₂/100 mL blood per mm Hg = 0.3 mL O₂/100 mL
- Hence 1L of plasma has 3mL of O₂ in dissolved form.
- 5L of plasma can carry maximum 15mL of O₂.
- At rest, O₂ demand is around 250mL/min.
- Hence dissolved O₂ alone is not sufficient to meet body demands.

Amount of O₂ transported bound to Hb:

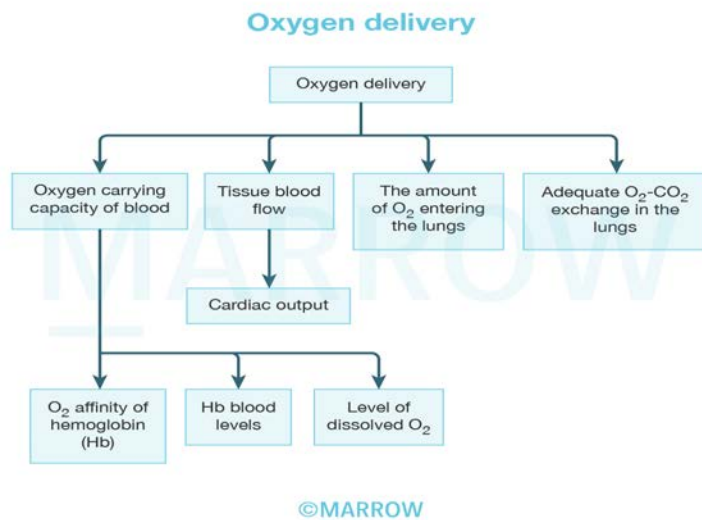
- Concentration of Hb is 15g/100 mL
- Each gram of Hb binds to 1.34 mL of O₂
- Hence, 15 g binds to 20.1mL of O₂/100 mL
- On average, the 15 g of hemoglobin in 100 mL of blood can combine with a total of about 20 mL of O₂.

Thus, this value (20.1) in the presence of Hb is about 70 times the value (0.3) in the absence of Hb.

Solution to Question 5:

Oxygen delivery is the volume of oxygen delivered to the systemic vascular bed per minute. It is the product of the cardiac output and the arterial oxygen concentration.

Oxygen delivery to tissue depends on (refer flow chart)



Solution to Question 6:

Hemoglobin has 4 subunits each of which has heme molecule and polypeptide chain. Ferrous iron binds to a molecule of O₂. Hence, hemoglobin binds to 4 molecules of O₂.

- Hemoglobin is a protein made up of four subunits.
- Each subunit contains a heme moiety attached to a polypeptide chain.
- Heme is a porphyrin ring complex that includes one atom of ferrous iron.
- In normal adults, most of the hemoglobin molecules contain two α and two β chains.
- Each of the four iron atoms in hemoglobin can reversibly bind one O₂. The iron stays in the ferrous state so that the reaction is oxygenation (not oxidation).

Solution to Question 7:

CO₂ is carried in the blood in three forms:

- Bicarbonate (HCO₃⁻) - Quantitatively the most important (70%)
- Carbaminohemoglobin (23%)
- Dissolved CO₂ (7%)

Solution to Question 8:

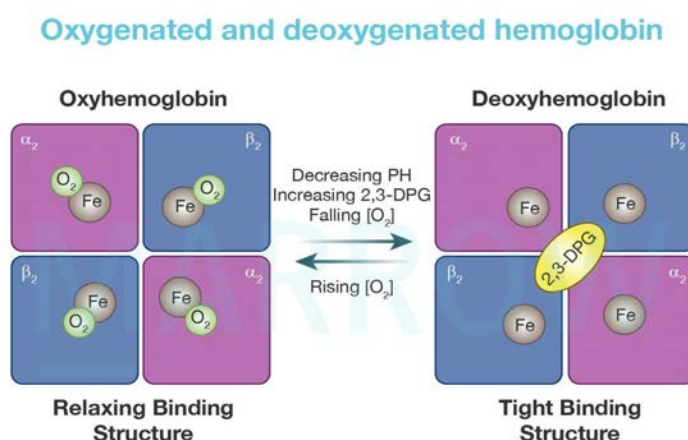
A relaxed (R) configuration, exposes more O₂ binding sites.

The quaternary structure of hemoglobin determines its affinity for O₂.

In deoxyhemoglobin, the globin units are tightly bound in a tense (T) configuration, which reduces the affinity of the molecule for O₂.

When O₂ is first bound, the bonds holding the globin units are released, producing a relaxed (R) configuration, which exposes more O₂ binding sites. The net result is a 500-fold increase in O₂

In tissues, these reactions are reversed, resulting in O₂ dissociation.



Solution to Question 9:

The oxygen-haemoglobin dissociation curve is sigmoid shape because binding of one O₂ molecule increases the affinity of binding of subsequent O₂ molecules.

The oxygen-hemoglobin dissociation curve relates percentage saturation of the O₂ carrying power of hemoglobin (abbreviated as SaO₂) to the Po₂. It has a characteristic sigmoid shape due to the T–R interconversion. Combination of the first heme in the Hb molecule with O₂ increases the affinity of the second heme for O₂ and oxygenation of the second increases the affinity of the third, and so on, such that the affinity of Hb for the fourth O₂ molecule is many times that for the first (Positive Co-operative Effect).

Option C: The effect of PCO₂ and pH on the O₂-hemoglobin dissociation curve is called the Bohr effect.

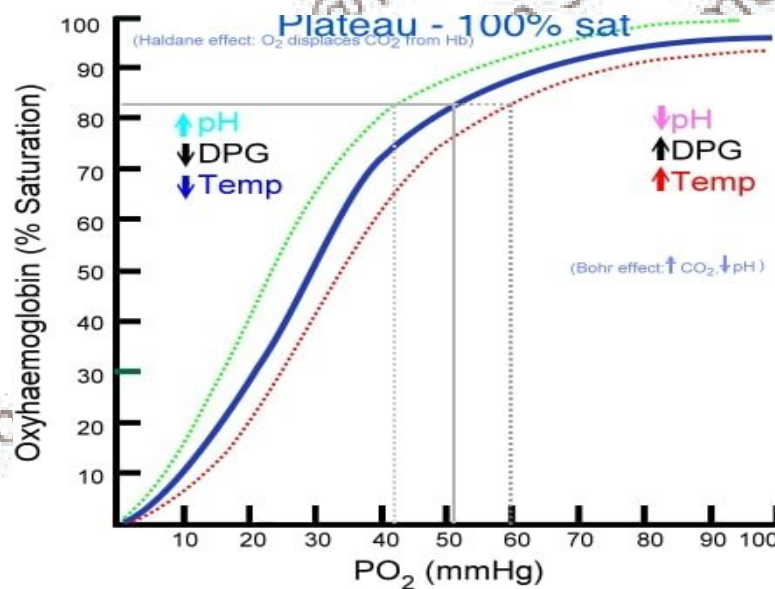
Option D: O₂ bound to hemoglobin changes its affinity for CO₂, such that when less O₂ is bound, the affinity of hemoglobin for CO₂ increases. This is referred to as the Haldane effect.

Solution to Question 10:

When blood is equilibrated with 100% O₂, the normal hemoglobin becomes 100% saturated.

- In vivo, the hemoglobin in the blood at the end of the pulmonary capillaries is about 97.5% saturated with O₂ (P_{O2} = 100 mm Hg).
- Because of a slight admixture with venous blood that bypasses the pulmonary capillaries (i.e, physiologic shunt), the hemoglobin in systemic arterial blood is only 97% saturated.
- Graph relates the percentage of saturated hemoglobin (SaO₂) to P_{O2} and dissolved O₂.

Partial pressure of O ₂	Corresponding S pO ₂
100	99 - 100%
60	89%
40	75%
27	50%



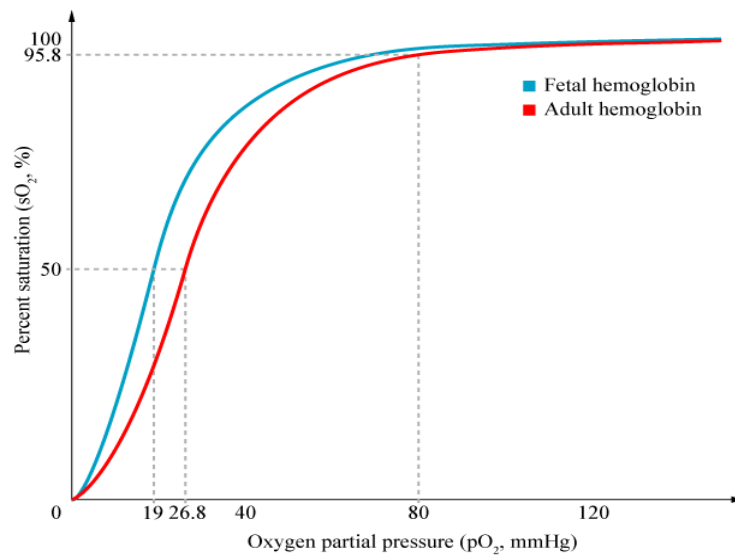
Solution to Question 11:

Normal P₅₀ is about 27 mm of Hg for adult Hb.

P₅₀ is the partial pressure of O₂ at which Hb is 50 % saturated. P₅₀ is an index of the affinity of Hb with O₂.

Higher the P₅₀ value, lower is the affinity: Oxygen-Hemoglobin Dissociation Curve shifts to right.

Lower the P₅₀ value, the higher is the affinity: Oxygen-Hemoglobin Dissociation Curve shifts to left.



Solution to Question 12:

The pH decreasing from the normal value shifts the O₂-hemoglobin dissociation curve, to the right.

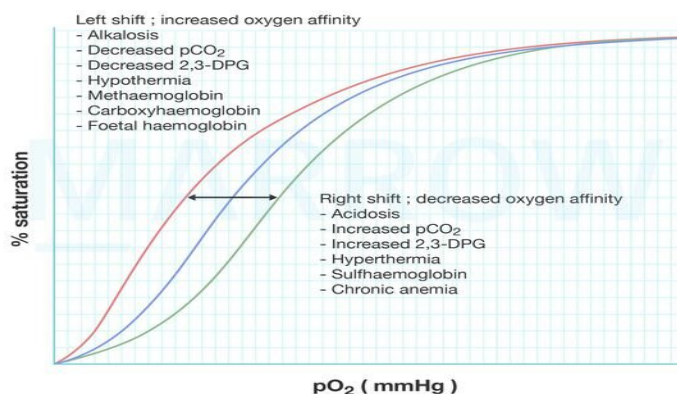
Oxygen-Hb dissociation curve (OHDC)

- This is a plot of the partial pressure of O₂ and the % saturation of Hb with O₂.
- It is normally sigmoid-shape.
- If OHDC is shifted to the right, it means that affinity of Hb for O₂ has become less (which favors O₂ delivery) i.e. Higher the P₅₀. (R - Right shift Releases oxygen)

Shift to the right is caused by: (Conditions with increased requirement of O₂ - metabolically active tissues)

- Hypoxia and Hypercarbia - Metabolically active tissues utilise O₂ and tissues produce CO₂
- ↓ pH - Secondary to increased CO₂ production
- ↑ in temperature, heat
- ↑ in 2,3-DPG - increased anaerobic glycolysis
- Growth hormone, thyroid hormone, androgen - by increasing 2,3 DPG
- Sickle cell anaemia - reduced affinity for O₂

Oxygen-hemoglobin dissociation curve

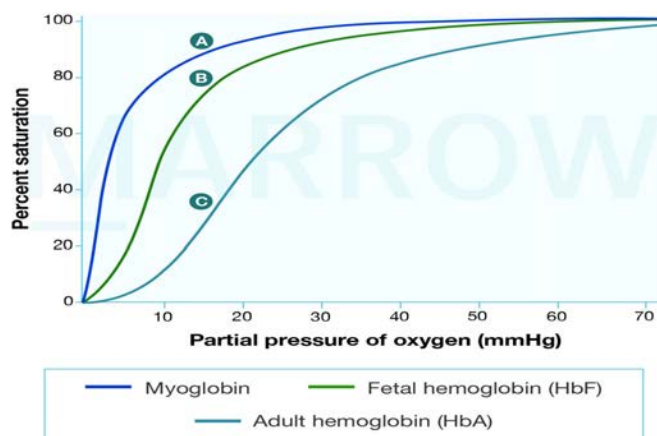


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Solution to Question 13:

4 molecules of myoglobin will bind 4 molecules of oxygen.

- Myoglobin: This is present in skeletal muscles.
- 1 molecule of myoglobin binds 1 molecule of O_2 . Thus, 4 molecules of myoglobin will bind 4 molecules of oxygen
- The shape of O_2 -myoglobin dissociation curve is the rectangular hyperbola. It lies to the left of Oxygen-Hemoglobin Dissociation Curve (OHDC)(refer figure).



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Solution to Question 14:

Facilitation of oxygen transport depends on the oxygen-hemoglobin dissociation curve, explained by the Bohr effect.

When the metabolic activity of tissues increases, the production of CO₂ increases. This CO₂ diffuses into the blood and in turn, raises the blood H₂CO₃ and causes an increase in the H⁺ concentration and a decrease in pH. This decreases the affinity of Hb for O₂, shifting the O₂-hemoglobin dissociation curve to the right. This facilitates the unloading of O₂ in the tissues. The effect of Pco₂ and pH on O₂ hemoglobin dissociation curve is called the Bohr Effect.

Conversely, in the lungs, CO₂ diffuses from the blood to the alveoli and the H⁺ concentration decreases and the O₂ hemoglobin dissociation curve shifts to the left increasing the affinity of Hb to O₂.

Note:

Double Bohr effect occurring in the placenta: Oxygen–hemoglobin dissociation curve of fetal blood and maternal blood move in opposite directions.

Mechanism:

- The fetal blood entering the placenta carries large amounts of carbon dioxide, but most of it diffuses from the fetal blood into the maternal blood.
- Fetal blood - loss of CO₂ - reduced H⁺ - shift to left - loading of O₂ i.e. oxygenation of fetal Hb.
- Maternal blood - increase in CO₂ - increase in H⁺ - shift to the right - Release of oxygen.

Solution to Question 15:

The level of 2,3-DPG is decreased in acidosis.

2,3-DPG is produced from 3-phosphoglycerate which is a product of glycolysis. Acidosis inhibits red cell glycolysis which decreases the levels of 2,3-DPG.

Solution to Question 16:

Increased affinity of HbF to O₂ is due to decreased affinity of γ chains of HbF to 2,3DPG.

HbF is the fetal variant of hemoglobin.

The β chains of adult hemoglobin (hemoglobin A) are replaced by γ chains in HbF. This modification results in increased affinity of hemoglobin for O₂, a left shift of the O₂-hemoglobin dissociation curve, and decreased P₅₀.

The mechanism of the left shift is based on the binding of 2,3-DPG. 2,3-DPG does not bind as avidly to the γ chains of HbF as it binds to the β chains of hemoglobin A. When less 2,3-DPG is bound, the affinity for O₂ increases and facilitates the movement of O₂ from the mother to the fetus.

This increased affinity is beneficial to the fetus, whose PaO₂ is low (approximately 40 mm Hg).

Solution to Question 17:

Binding of O₂ with hemoglobin tends to displace CO₂ from the blood. This effect is called the Haldane effect. (Remember - Haldane effect occurs in lungs and causes left shift of O₂-Hb dissociation curve)

Haldane Effect

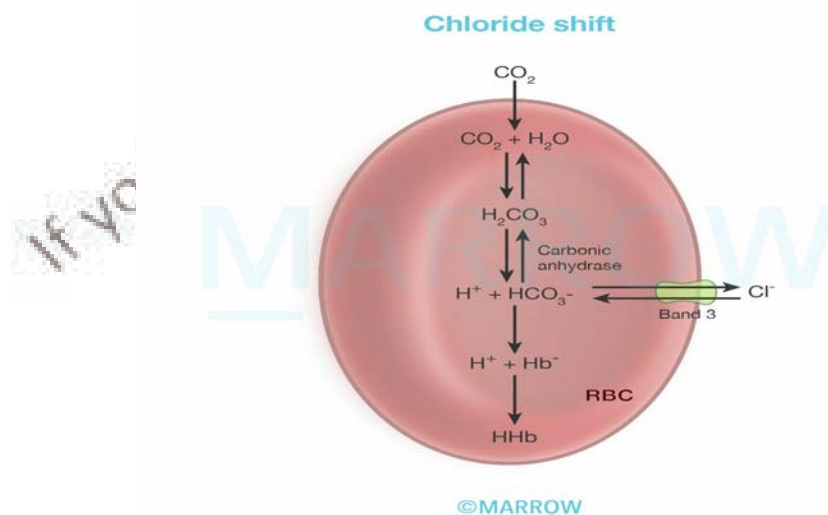
- Binding of O₂ with hemoglobin tends to displace CO₂ from the blood. This effect, called the Haldane effect, is important in promoting CO₂ transport.
- The Haldane effect results from the simple fact that the combination of O₂ with hemoglobin in the lungs causes the hemoglobin to become a stronger acid. This displaces CO₂ from the blood and into the alveoli.

Solution to Question 18:

Hematocrit (PCV) of venous blood is normally 3% greater than that of the arterial blood because of 2 reasons:

- Chloride shift
- A small amount of fluid in the arterial blood returns via the lymphatics rather than the veins.

Chloride shift:



- Upon entering the red blood cell, CO₂ is rapidly hydrated to H₂CO₃ by carbonic anhydrase.
- H₂CO₃ is in equilibrium with H⁺ and its conjugate base, HCO₃⁻.
- H⁺ can interact with deoxyhemoglobin, whereas HCO₃⁻ can be transported outside of the cell via anion exchanger 1 (AE1 or Band 3).
- In effect, for each CO₂ molecule that enters the red cell, there is an additional HCO₃⁻ or Cl⁻ in the cell.

- Because of this chloride shift, the Cl⁻ content of the red cells in venous blood is significantly greater than that in arterial blood.
- Note that for each CO₂ molecule added to a red cell, there is an increase of one osmotically active particles in the cell - either a HCO₃⁻ or a Cl⁻ ion. Consequently, the red cells take up water and increase in size.

Solution to Question 19:

The fall in 2, 3 DPG is less when the blood is stored in CPDA compared with ACD.

2, 3 DPG (2,3-diphosphoglycerate) binds to deoxygenated haemoglobin. Therefore, an increase in 2,3 DPG causes increased release of O₂ to the tissues. The equilibrium between Hb and O₂ can be understood from the following equation:



In stored blood, the 2,3 DPG levels usually fall. As a result, stored blood releases less oxygen to the tissues than the normal circulating blood, thus limiting its use in hypoxic patients (Eg: anaemia).

However, such a fall in 2,3 DPG is less when it is stored in a phosphate-containing solution like CPDA (Citrate, phosphate, dextrose, adenine), rather than ACD (Acid citrate dextrose). Therefore, the blood stored in CPDA, in comparison with blood stored in ACD has:

- Higher 2,3 DPG concentration
- More amount of deoxygenated blood bound to 2,3 DPG
- Decreased affinity of Hb to the oxygen.
- Increased release of oxygen to the tissues.
- Raised P₅₀ (PO₂ at which Hb is half saturated)
- Shift in O₂-Hb dissociation curve to the right

Solution to Question 20:

The oxygen carrying capacity of a boy with haemoglobin 14g/dl is 18.

When blood is saturated with 100% O₂, the haemoglobin is also 100% saturated and each gram of normal haemoglobin contains 1.39 mL of O₂.

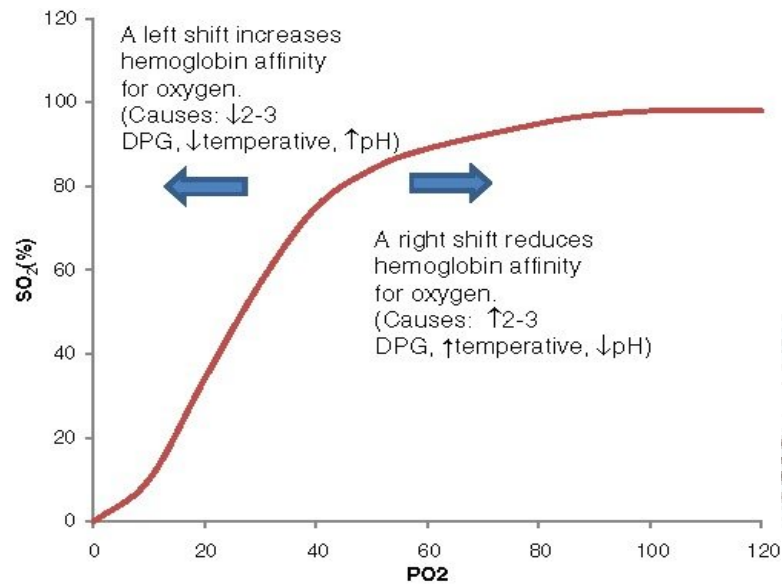
However, blood normally contains some inactive haemoglobin derivatives, and the measured value in vivo is thus slightly lower hence, each gram of haemoglobin in vivo contains 1.31 mL of O₂ (Hufner's constant).

According to the above question, when Hb=14g/dl, O₂= 1.31*14= 18.34

The oxygen-hemoglobin dissociation curve

- Demonstrates a progressive increase in the percentage of haemoglobin bound with oxygen as blood Po₂ increases, which is called the percent saturation of haemoglobin.

- Blood leaving the lungs and entering the systemic arteries usually has a P_{O_2} of about 95 mm Hg, the usual oxygen saturation of systemic arterial blood averages 97 per cent.
- Conversely, in normal venous blood returning from the peripheral tissues, the P_{O_2} is about 40 mm Hg, and the saturation of haemoglobin averages 75 per cent.



Lung Volumes and Lung Function Tests

Question 1:

Tidal volume is _____

- a) Inspiratory capacity minus inspiratory reserve volume
- b) Total lung capacity minus residual volume
- c) Functional residual capacity minus residual volume
- d) Vital capacity minus expiratory reserve volume

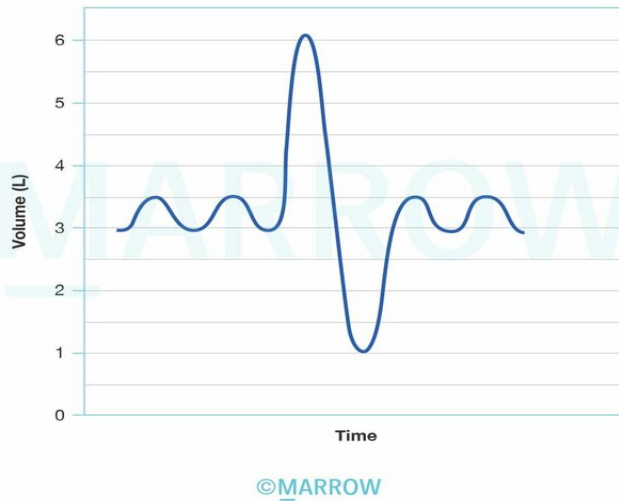
Question 2:

Functional residual capacity is:

- a) Volume expired after normal inspiration
- b) Volume that remains after forced expiration
- c) Expiratory reserve volume + residual volume
- d) Tidal volume + inspiratory reserve volume

Question 3:

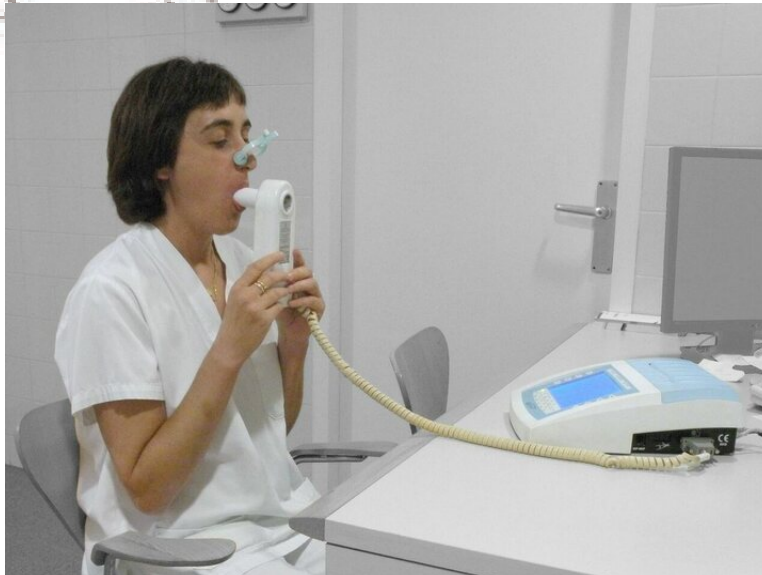
A 27-year-old man is breathing quietly. He then inhales as much air as possible and exhales as much air as he can, producing the spirogram shown in the figure. What is his expiratory reserve volume (in liters)?



- a) 2.0
- b) 2.5
- c) 3.0
- d) 3.5

Question 4:

A nurse in the pulmonology clinic administered a test using the following instrument. Which of the following parameters cannot be measured?



- a) Tidal volume

- b) Residual volume
- c) Vital capacity
- d) Inspiratory reserve capacity

Question 5:

Total lung capacity depends on_____

- a) Size of airways
- b) Closing volume
- c) Lung compliance
- d) Residual volume

Question 6:

Single breath nitrogen method is used for measuring _____

- a) Dead space volume
- b) FRC
- c) Tidal volume
- d) Diffusion capacity

Question 7:

Calculate the alveolar ventilation in a young man if the respiratory rate is 10/min and tidal volume is 600ml.

- a) 1.5 L/min
- b) 3.5 L/min
- c) 4.5 L/min
- d) 5.0 L/min

Question 8:

Normal $VD(Dead\ space\ volume) / VT (Tidal\ volume)$ is _____

- a) 20

- b) 0.3
- c) 4
- d) 50

Question 9:

A patient has a dead space of 150 mL, a functional residual capacity of 3L, tidal volume of 650 mL, expiratory reserve volume of 1.5 L, a total lung capacity of 8L, respiratory rate of 15 breaths/min. What is the rate of alveolar ventilation?

- a) 5 L/min
- b) 7.5 L/min
- c) 6.0 L/min
- d) 9.0 L/min

Question 10:

Closing volume of the lung is _____

- a) Above residual volume in upper portions of the lungs
- b) Above the residual volume in dependent portions of the lungs
- c) Above tidal volume in dependent portions of the lungs
- d) Above tidal volume in upper portions of the lungs

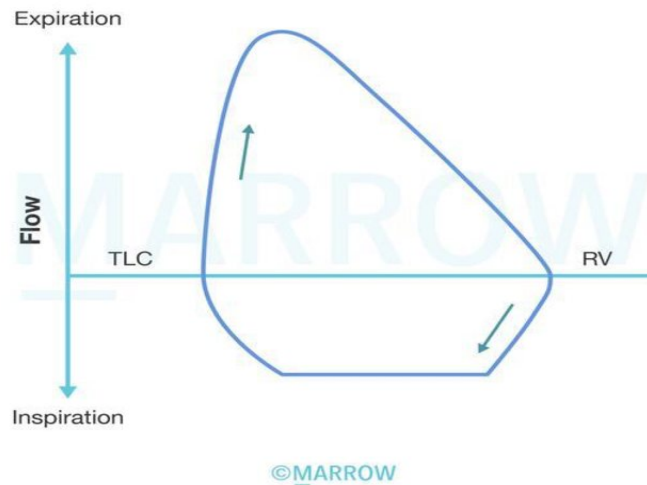
Question 11:

A 66-year-old man presented with gradually progressive dyspnea and a non-productive cough. On examination, he has bi-basilar inspiratory crackles and clubbing. Which of the following pulmonary function tests will be true if he is to be diagnosed with interstitial pulmonary fibrosis?

- a) $\uparrow$ FRC, $\uparrow$ Compliance
- b) $\uparrow$ FEV₁/FVC, $\downarrow$ Compliance
- c) $\downarrow$ FEV₁/FVC, $\uparrow$ Compliance
- d) $\uparrow$ Compliance

Question 12:

Flow volume given below indicates_____



- a) Normal
- b) Intrathoracic obstruction
- c) Extrathoracic obstruction
- d) Fixed

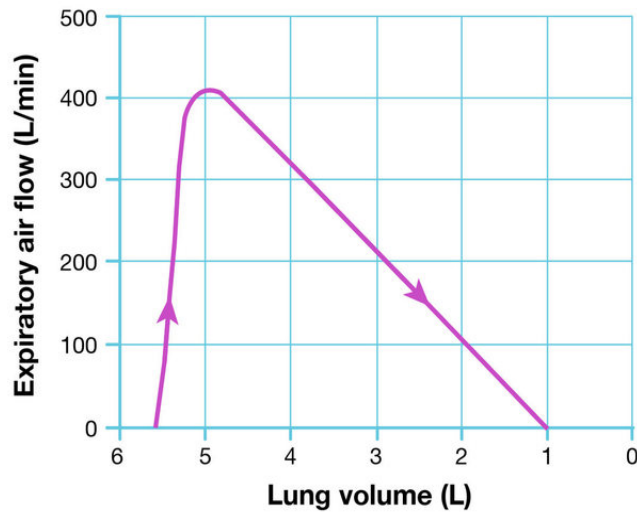
Question 13:

Which of the following best describes the changes in lung volume and thoracic volume in a case of pneumothorax, compared to normal?

- a) ↓lung volume, ↓thoracic volume
- b) ↓lung volume, ↑thoracic volume
- c) ↓lung volume, thoracic volume no change
- d) ↑lung volume, ↓thoracic volume

Question 14:

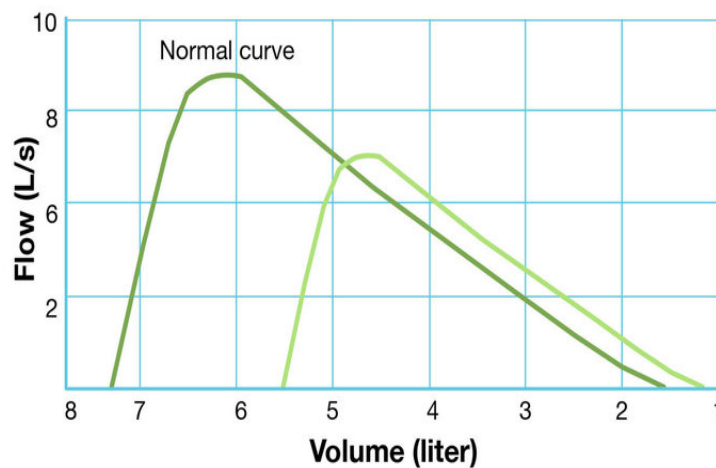
A 45-year-old man inhaled as much air as possible and then expired with a maximum effort until no more air could be expired. This produced the maximum expiratory flow-volume curve shown in the following diagram. What is the forced vital capacity of this man (in liters)?



- a) 1.5
- b) 2.5
- c) 3.5
- d) 4.5

Question 15:

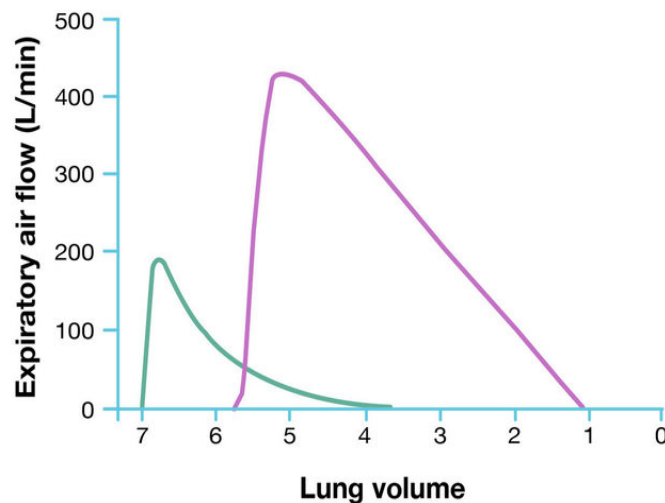
A 71-year-old woman presents with cough and shortness of breath. The physician conducts a pulmonary function screening test in his office, and the patient generates the maximum expiratory flow-volume curve shown to the right of the normal curve in the diagram below. These findings are consistent with which of the following conditions?



- a) Asbestosis
- b) Asthma
- c) Bronchospasm
- d) Emphysema

Question 16:

The maximum expiratory flow-volume (MEFV) curves shown in the figure below were obtained from a healthy man (purple curve) and a 57-year-old man with shortness of breath (green curve). What is the most probable diagnosis in this patient?



- a) Asbestosis
- b) Emphysema
- c) Kyphosis
- d) Tuberculosis

Question 17:

Which of the following increases in COPD?

- a) 1 only
- b) 1 and 4
- c) 1 and 3
- d) 1 and 2

Question 18:

Which of the following factors is most likely to stimulate ventilation after strenuous exercise?

- a) Collateral impulses from higher than brain centres
- b) Increased mean arterial H⁺
- c) Decreased mean arterial Po₂
- d) Decreased mean arterial Pco₂

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	b
5	c
6	a
7	c
8	b
9	b
10	b
11	b
12	c
13	b
14	d
15	a
16	b
17	d
18	b

Detailed Explanations

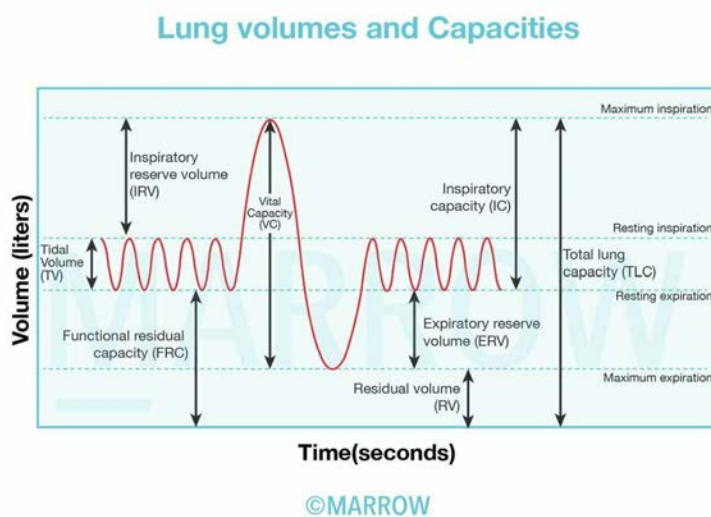
Solution to Question 1:

The tidal volume = inspiratory capacity - inspiratory reserve volume.

Solution to Question 2:

Functional Residual Capacity (FRC) is expiratory reserve volume + residual volume.

From the picture given below, (FRC = ERV + RV).



Lung Capacities:

The total lung capacity can be broken down into alternative capacities that help to define functioning lungs.

- The vital lung capacity refers to the maximum amount of air expired from the fully inflated lung or maximum inspiratory level.
- The inspiratory capacity is the maximum amount of air inspired from the end-expiratory level.
- The functional residual capacity represents the volume of the air remaining in the lungs after expiration of a normal breath.

Solution to Question 3:

Expiratory Reserve Volume = Functional Residual Capacity - Residual Volume

In the given spirogram,

- Functional Residual Capacity = 3L
- Residual Volume = 1L

Therefore,

Expiratory Reserve Volume = Functional Residual Capacity – Residual Volume = 3-1 = 2

Note:

The residual volume is the volume of air remaining in the lungs after the most forceful expiration.

The functional residual capacity equals the expiratory reserve volume plus the residual volume.

This capacity is the amount of air that remains in the lungs at the end of normal expiration

The expiratory reserve volume is the maximum extra volume of air that can be expired by forceful expiration after the end of a normal tidal expiration.

Solution to Question 4:

The given image shows spirometry. All of the given parameters can be measured by a spirometer except residual volume.

Residual volume is measured by:

- N₂ washout method
- Helium-dilution method
- Body plethysmography

Other parameters which cannot be measured by spirometry are:

- Functional residual capacity
- Total lung capacity

Solution to Question 5:

Total lung capacity depends on the distensibility of lungs (compliance).

Determinants of Lung Compliance

- Stretchability of the lung tissues, particularly their elastic connective tissue.
- Surface tension at the air-water interfaces within the alveoli, which is markedly reduced in the alveoli by pulmonary surfactant preventing their collapse.

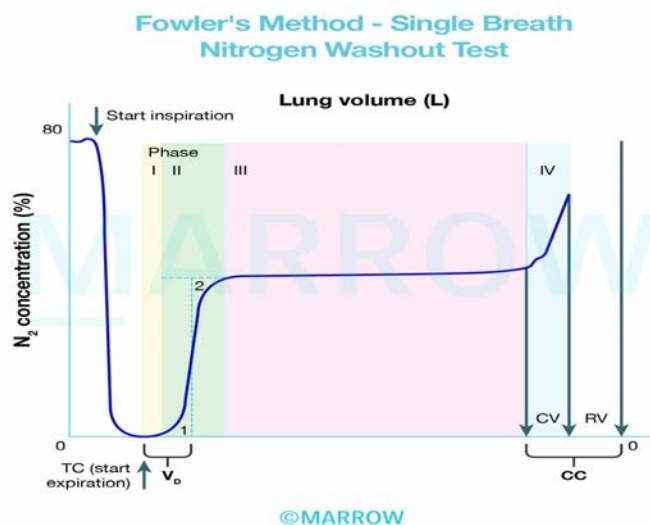
Option b: The Closing Volume is the lung volume above Residual Volume at which airways in the lower, dependent parts of the lungs begin to close off because of the lesser transmural pressure in these areas.

Solution to Question 6:

Anatomical dead space can be measured by the single-breath N₂ curve (Fowler's method).

- Anatomical dead space is the respiratory system volume exclusive of alveoli.

- Alveolar dead space is the volume of gas not equilibrating with blood i.e. wasted ventilation.
- Physiological/total dead space is the sum of anatomical and alveolar dead space.
- The equation used for the calculation of total/physiological dead space is the Bohr's equation.



The patient takes a deep breath of O₂ From mid-inspiration, then exhales steadily. The changes in the N₂ concentration of expired gas during expiration are plotted.

Region I - Dead Space (DS)

From I–II is a mixture of DS and alveolar gas.

The transition from III–IV is the closing volume (CV).

End of IV - Residual Volume (RV).

Factors affecting dead space:

Option B: Functional Residual Capacity (FRC) is measured using body plethysmography.

Option C: Tidal volume (TV) is measured using spirometry.

Option D: Diffusion Capacity is measured using carbon monoxide diffusion capacity (DLCO).

Factor	Effect
Posture: Upright/Supine	Increase/Decrease
Position of airway: Neck Extension/Neck Flexion	Increase/Decrease
Age	Increase
Artificial airway	Decrease
Positive pressure ventilation	Increase
Drug anti-cholinergic	Increase
Pulmonary perfusion: Pulmonary embolism/Hypotension	Increase/Increase

Factor	Effect
Pulmonary disease:Emphysema	Increase

Solution to Question 7:

$$\text{Alveolar ventilation} = \text{RR} \times (\text{VT} - \text{V D})$$

In the given case,

- Respiratory rate (RR)= 10/min
- Tidal volume (VT) = 600 mL
- Dead space volume (VD) = 150 mL in a young man

$$\text{Thus, Alveolar ventilation} = 10 \times (600-150) = 4500 \text{ mL/min} = 4.5 \text{ L/min}$$

Solution to Question 8:

- Dead space volume = VD = 150ml
- Tidal volume = VT = 500ml
- Normal VD/VT = 150/500 = 0.3

Solution to Question 9:

The rate of alveolar ventilation in this patient is 7.5 L/min.

Alveolar ventilation per minute is the total volume of new air entering the alveoli and adjacent gas exchange areas each minute. It is equal to the respiratory rate times the amount of new air that enters these areas with each breath.

$$\begin{aligned} \text{Alveolar ventilation} &= \text{Respiratory rate} \times (\text{Tidal volume} - \text{Physiologic dead space}) \\ &= 15 \times (650-150) \\ &= 7500 \text{ mL/min or } 7.5 \text{ L/min.} \end{aligned}$$

Solution to Question 10:

The closing volume (CV) is the lung volume above the residual volume (RV) at which the airways in the lower dependent parts of the lung begin to close off because of the lesser transmural pressure in these areas.

In young adults, the closing volume is about 10% of the vital capacity (VC). It increases steadily with age and is equal to about 40% of the VC. It equals FRC, at about the age of 65 years.

Diseases of the small airways apparently increase the closing volume

- Normal young adult: $CV \ll FRC$
- At 65 years of age: $CV = FRC$
- Small airway disease: $CV \gg FRC$
- In case of hyaline membrane disease, FRC is smaller than closing volume.

Solution to Question 11:

Interstitial pulmonary fibrosis is a restrictive lung disease that shows $\uparrow FEV_1/FVC$, $\downarrow$ Compliance when performing pulmonary function tests.

In a healthy normal adult male, FVC is approximately 5.0 L, FEV₁ is approximately 4.0 L, and thus, the calculated FEV₁/FVC is 80%.

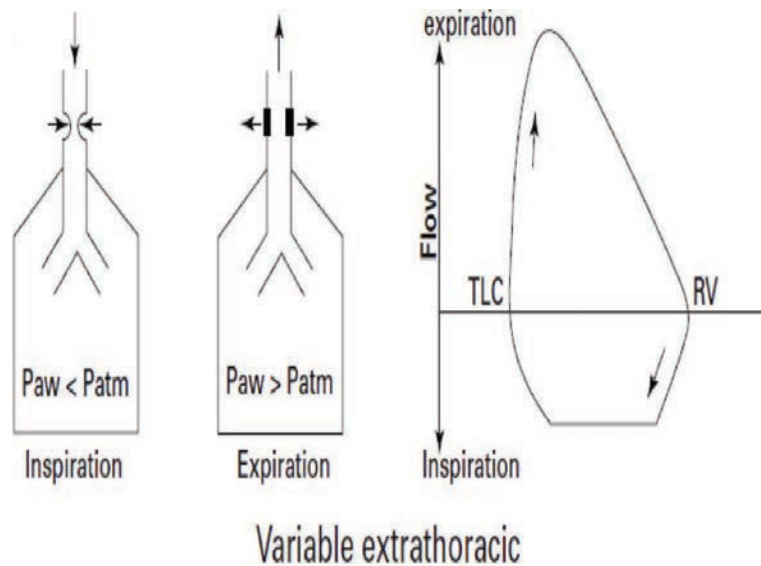
Obstructive disorders result in a marked decrease in both FVC and FEV₁/FVC, whereas restrictive disorders result in a loss of FVC without loss in FEV₁/FVC.

Compliance is reduced in cases of restrictive lung diseases. For example - interstitial lung diseases, fibrosis, sarcoidosis, asbestosis, and radiation exposure.

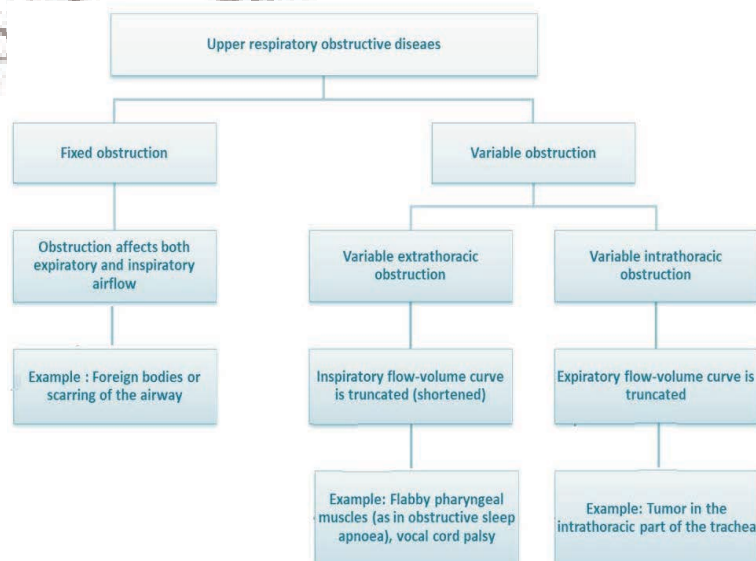
Parameter	Obstructive	Restrictive Parenchymal
Pathology	Airflow obstruction	Reduced compliance
FEF ₂₅₋₇₅	↓	N
FEV ₁	↓	N
FVC	↓	↓
FEV ₁ /FVC	↓	N to ↑
TLC & RV	↑	↓

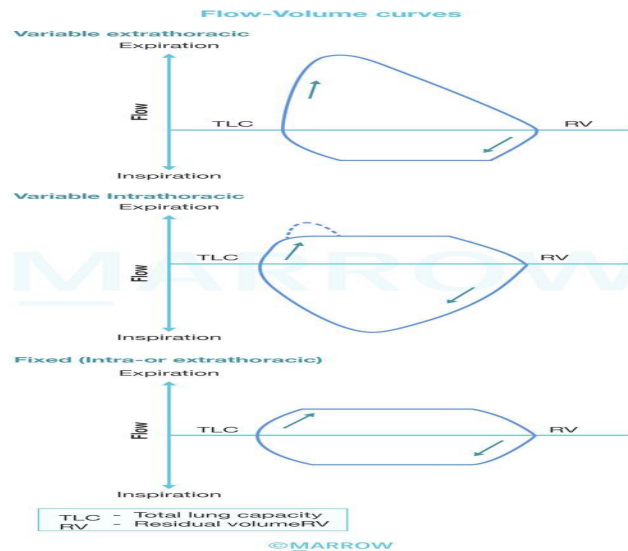
Solution to Question 12:

The inspiratory flow-volume curve is truncated in patients with variable extrathoracic obstructions.



- During a forced expiration, the cross-sectional area of a variable extrathoracic obstruction increases as the pressure inside the airway increases.
- The expiratory flow-volume curve is therefore nearly normal or not affected.
- However, during a forced inspiration, the pressure inside the upper airway decreases below atmospheric pressure, and unless the stability of the upper airway is maintained by reflex contraction of the pharyngeal muscles or by other structures, the cross-sectional area of the upper airway will decrease. Therefore, the inspiratory flow-volume curve is truncated in patients with variable extrathoracic obstructions.
- Flow-volume curves are very useful in assessing obstructions of upper airways and the trachea (refer figure and flowchart).





Solution to Question 13:

In pneumothorax, lung volume is decreased and thoracic volume is increased.

Both lungs and the chest wall (thoracic cage) are elastic structures. Under normal conditions, the elastic tendency of the lungs to collapse is exactly balanced by the elastic tendency of the chest wall to expand.

In pneumothorax, air is introduced into the pleural space, so the pleural pressure, which is normally subatmospheric, becomes equal to atmospheric pressure. The chest wall springs outward and the lungs collapse. This increases the thoracic volume and decreases the lung volume.

Solution to Question 14:

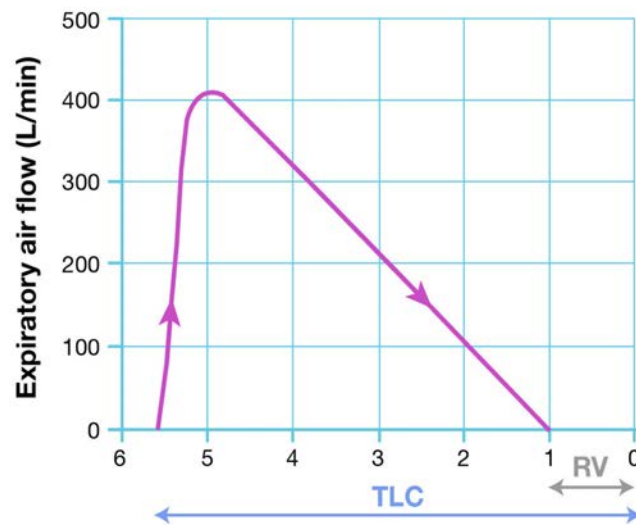
The forced vital capacity of this man is 4.5 liters.

The forced vital capacity (FVC), the largest amount of air that can be expired after a maximal inspiratory effort, is frequently measured clinically as an index of pulmonary function.

Forced vital capacity(FVC) = Total lung capacity (TLC) – Residual volume(RV)

- In the flow-volume curve, the point of intersection between abscissa and flow-volume curve gives the TLC and RV.
- From the given flow-volume curve we can note that:
- Total Lung Capacity (TLC) = 5.5 L
- Residual volume (RV)= 1.0 L
- Thus,

- $FVC = TLC - RV = 4.5 \text{ L}$



Solution to Question 15:

This is a case of asbestosis.

Asbestosis is a restrictive lung disease characterized by diffuse interstitial fibrosis. In restrictive lung disease, the maximum expiratory flow volume (MEFV) curve begins and ends at abnormally low lung volumes.

Note: In obstructive lung diseases, the airways are narrowed or radial traction of the airways is reduced, allowing them to close more easily. The lung volumes in these cases are higher than normal.

Solution to Question 16:

The green curve in the above image and the case scenario denotes that the diagnosis is likely to be emphysema.

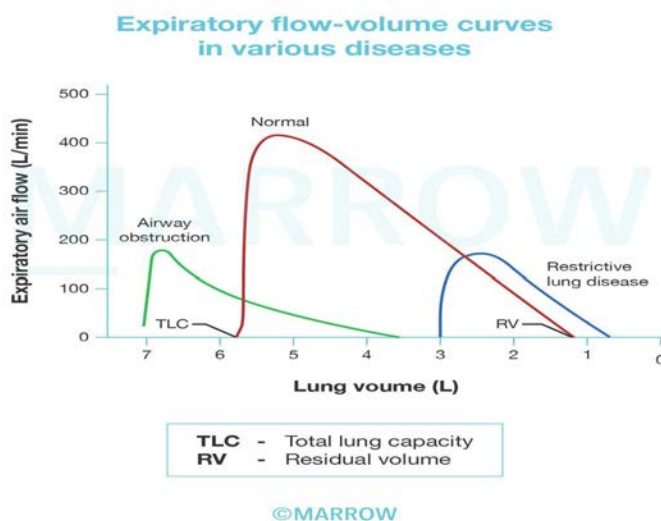
Maximum expiratory flow volume (MEFV): When a person expires with great force, the expiratory airflow reaches a maximum flow beyond which the flow cannot be increased anymore, even with greatly increased additional force.

In obstructive airway disease like Emphysema, it is usually much more difficult to expire than to inspire because the closing tendency of the airways is greatly increased by the extra positive pressure during expiration. Due to the obstruction of the airways and because they collapse more easily than normal airways, the maximum expiratory flow rate is greatly reduced (But, total lung capacity and residual volume is increased) which is shown in the green curve below.

Other options given above will cause constricted lungs where the lungs cannot expand to a normal maximum volume. Hence, even with the greatest possible expiratory effort, the maximum

expiratory flow rate will be reduced. (Note: here both the total lung capacity and the residual volume are reduced), which is shown as the blue curve in the graph below.

Following graph shows the MEFV of restrictive and obstructive lung diseases.



Solution to Question 17:

In COPD, the functional residual capacity increases. Total lung capacity may be increased or normal. Hence, both 1 and 2 are correct.

- In obstructive lung diseases, both FEV₁ and FVC reduces, where the decrease in FEV₁ is more than the reduction in FVC. As a result, the ratio of FEV₁/FVC also decreases.
- The total lung capacity is usually normal in chronic bronchitis, while increased in emphysema.
- Functional Residual Capacity (FRC) is the volume of air present in the lungs at the end of passive expiration. In COPD, FRC is increased.

Lung Capacities:

The total lung capacity can be broken down into alternative capacities that help to define functioning lungs.

- The vital lung capacity refers to the maximum amount of air expired from the fully inflated lung or maximum inspiratory level.
- The inspiratory capacity is the maximum amount of air inspired from the end-expiratory level.
- The functional residual capacity represents the volume of the air remaining in the lungs after expiration of a normal breath. Functional Residual Capacity (FRC) is expiratory reserve volume + residual volume.

Solution to Question 18:

The stimulus to ventilation after strenuous exercise is increased mean arterial H^+ .

The respiratory rate after exercise does not reach a basal level until the O_2 debt is repaid. The stimulus for ventilation is increased mean arterial H^+ due to an increase in blood lactic acid levels. Most of the lactic acid is converted to glycogen and some amount is converted to CO_2 and H_2O .

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you may have been scammed.

Respiratory Adaptations in Hypoxia, Anemia and with Pressure Changes

Question 1:

Which is the best parameter to analyze hypoxic hypoxia?

- a) Partial pressure of CO₂ in arterial blood (Paco₂)
- b) Partial pressure of O₂ in arterial blood (Pao₂)
- c) A-V difference
- d) Po₂ venous

Question 2:

Which type of hypoxia does not stimulate peripheral chemoreceptors?

- a) Anemic
- b) Histotoxic
- c) Stagnant
- d) Hypoxic

Question 3:

Which of the following is most likely in a patient with cyanosis?

- a) Anemic hypoxia
- b) Histotoxic hypoxia
- c) Carbon monoxide poisoning
- d) Stagnant hypoxia

Question 4:

Administration of O₂ has maximum benefit in:

- a) Anemic hypoxia
- b) Histotoxic hypoxia

- c) Hypoxic hypoxia
- d) Stagnant hypoxia

Question 5:

In which of the following situations is oxygen therapy most beneficial provided the lung function is normal?

- a) Anemia
- b) COPD
- c) Cyanide poisoning
- d) High altitude

Question 6:

In which of the following conditions is oxygen therapy not useful?

- a) Asthma
- b) Acute myocardial infarction
- c) Pulmonary edema
- d) Central respiratory depression

Question 7:

Which of the points on the following figures represent arterial blood in a severely anemic person?

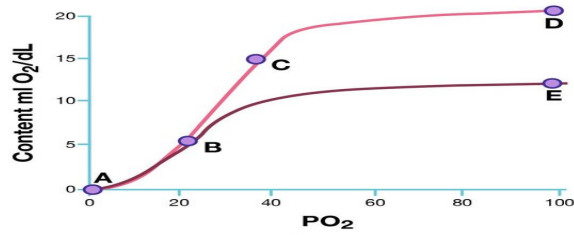


Figure 1

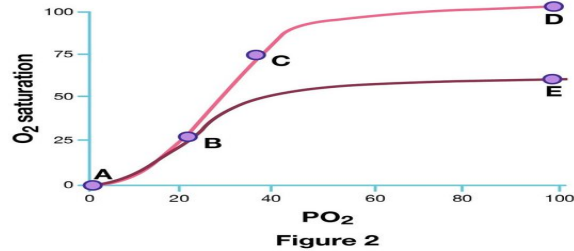


Figure 2

- a) Figure 1-D, Figure 2- D
- b) Figure 1-E, Figure 2- E
- c) Figure 1-D, Figure 2- E
- d) Figure 1-E, Figure 2- D

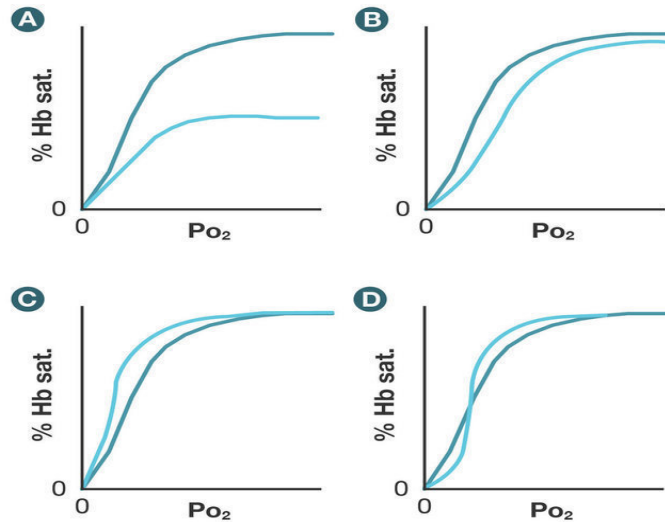
Question 8:

A 24-year-old woman is anemic with hemoglobin concentration of 6 g/dL. Which of the following set of changes have occurred in this case, compared to normal?

- a) ↓Arterial P_{O_2} , ↓Mixed venous P_{O_2} , ↑2,3-DPG
- b) ↓Arterial P_{O_2} , ↓Mixed venous P_{O_2} , Normal 2,3-DPG
- c) ↓Arterial P_{O_2} , Normal Mixed venous P_{O_2} , ↑2,3-DPG
- d) Normal Arterial P_{O_2} , ↓Mixed venous P_{O_2} , ↑2,3-DPG

Question 9:

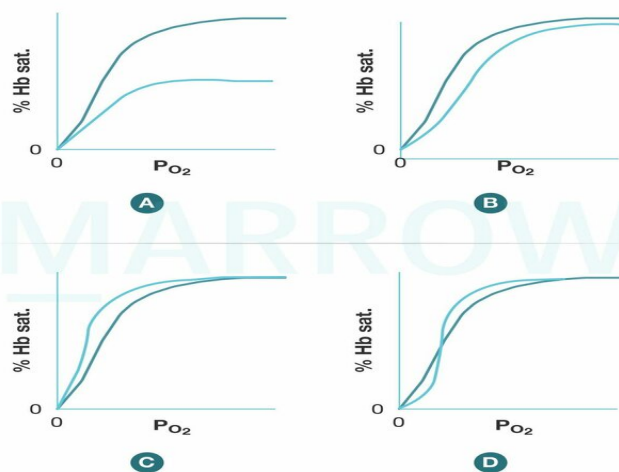
Which of the following oxygen-hemoglobin dissociation curves corresponds to blood from an adult (dark blue line) and blood from a fetus (light blue line)?



- a) A
- b) B
- c) C
- d) D

Question 10:

Which of the following oxygen-hemoglobin dissociation curves corresponds to blood during resting conditions (dark blue line) and blood during exercise (light blue line)?



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- a) A

- b) B
- c) C
- d) D

Question 11:

Decompression sickness is seen in_____

- a) All of the below
- b) Pilot
- c) Mountaineer
- d) Divers

Question 12:

Nitrogen narcosis is due to_____

- a) Nitrogen inhibits dismutase enzyme
- b) Increased production of nitrous oxide
- c) Increased solubility of nitrogen in nerve cell
- d) Decrease in free radicals

Question 13:

During acclimatization at higher altitude, all of the following changes take place except _____.

- a) ↑ Minute ventilation
- b) ↑ Sensitivity of central chemoreceptors
- c) ↑ Sensitivity of peripheral chemoreceptors
- d) Shift of Oxygen Haemoglobin Dissociation Curve to left

Question 14:

A person develops headache and breathlessness on a trekking expedition following rapid ascent to over 3000 metres above the sea level. Which of the following is not used in the treatment of this condition?

- a) IV digoxin
- b) Descent to lower altitude
- c) Administration of oxygen
- d) Tablet acetazolamide

Answer Key

Question No.	Correct Option
1	b
2	a
3	d
4	c
5	d
6	d
7	d
8	d
9	c
10	b
11	a
12	c
13	d
14	a

Detailed Explanations

Solution to Question 1:

Hypoxic hypoxia is characterized by low partial pressure of O₂ in arterial blood (Pao₂), while other parameters like partial pressure of CO₂ in arterial blood (Paco₂) may remain normal.

Ventilation–Perfusion Imbalance:

- The patchy ventilation–perfusion imbalance is the most common cause of hypoxic hypoxia or hypoxemia.
- If alveoli are not ventilated but perfusion is present - blood passing through that alveoli remains unoxygenated.

- When this unoxygenated blood mixes with blood from normal or overventilated alveoli, there is a right to left shunt created.
- This results in reduced partial pressure of O₂ in arterial blood (PaO₂) i.e., hypoxemia and consequently hypoxia.

Note:

- Hypoxemia can be defined as a condition where arterial oxygen tension or partial pressure of oxygen (PaO₂) is below normal.
- Hypoxia is defined as the reduction of oxygen supply at the tissue level, which is not measured directly by a laboratory value.

Scenario 1:

- Generally, the presence of hypoxemia suggests hypoxia i.e. if there is less oxygen in blood, O₂ supply to tissues will also be reduced.
- But, in case of cyanide poisoning, there is hypoxia in the absence of hypoxemia, i.e., arterial O₂ content is normal but tissues are not able to utilize it because cyanide interferes with aerobic metabolism in the tissues.

Scenario 2:

- Hypoxia may not be present in patients with hypoxemia if the patient compensates for a low PaO₂ by increasing oxygen delivery. This is typically achieved by increasing cardiac output (by raising the heart rate) or decreasing tissue oxygen consumption.
- Nevertheless, hypoxemia is definitely the most common cause of tissue hypoxia, and the correct diagnosis should be made because it is readily treatable.

Solution to Question 2:

In the case of anemic hypoxia, O₂ content is low but dissolved O₂ is within normal limits, so it does not stimulate peripheral chemoreceptors.

Peripheral chemoreceptors located in the carotid and aortic arteries sense changes in P_{O₂} (<60mm Hg), pH, and P_{CO₂} and they respond to changes in dissolved O₂ (PaO₂).

Histotoxic hypoxia will lead to metabolic acidosis that will lead to an increase in H⁺ concentration, which in turn will stimulate the peripheral chemoreceptors.

Hemorrhage that produces hypotension leads to chemoreceptor stimulation due to decreased blood flow to the chemoreceptors and consequent stagnant anoxia of these organs.

Note: In case of stagnant hypoxia, chemoreceptors can be stimulated if the resultant decreased oxygen delivery causes a significant drop in local tissue oxygen levels, which eventually can lower the PaO₂ detectable by chemoreceptors. Therefore, they may not be significantly stimulated unless the reduction in oxygen levels is severe. Therefore, anaemic hypoxia is a better answer.

Solution to Question 3:

In stagnant hypoxia, blood flow is very low; so, more oxygen is extracted causing an increase in reduced hemoglobin → cyanosis.

Cyanosis - Cyanosis is the dusky bluish discoloration of the tissues occurring when the reduced hemoglobin concentration is more than 5g/dL.

- Its occurrence depends upon
- The total amount of hemoglobin in the blood
- Degree of hemoglobin unsaturation
- The state of capillary circulation

Option a: In the case of anemia, the total hemoglobin content is low, so enough reduced hemoglobin cannot be formed.

Option b: In histotoxic hypoxia, the amount of reduced hemoglobin formed is very less because the defect is in the tissue O₂.

Option c: In CO poisoning, there is the formation of carboxyhemoglobin, which is red in color obscure the color of reduced hemoglobin.

Solution to Question 4:

Administration of oxygen has maximum benefit in hypoxic hypoxia.

Hypoxic hypoxia refers to a condition in which there is low atmospheric Po₂, leading to low arterial Po₂. Hence, supplemental O₂ can be very beneficial when the cause of hypoxia is due to limited oxygen availability.

Administration of oxygen-rich gas has limited value in anemic, histotoxic, and stagnant hypoxia because:

Anemic hypoxia- Oxygen delivery is not beneficial because despite increasing the exogenous oxygen delivery, an anemic person will not have the capacity to carry more oxygen due to the hemoglobin/ red blood cell deficit in anemia.

Histotoxic hypoxia- The most common example is cyanide poisoning. Cyanide inhibits important enzymes in the oxidative phosphorylation process, which is essential to the survival of our cells. Additional oxygen supplementation will not be able to overcome this inhibition, and so it will have limited benefit. Hyperbaric oxygen may be helpful to some extent.

Stagnant hypoxia- Stagnant hypoxia refers to ischemic hypoxia that results from inadequate circulation due to conditions such as heart failure and hypoxia in the setting of shock. In such conditions, administering intravenous fluids can improve the patient's condition because it is the circulation that is affected. If the shock is prolonged it can result in acute respiratory distress syndrome (ARDS), which can be treated with positive pressure ventilation and not simply by increasing oxygen administration.

Solution to Question 5:

With O₂ therapy, increasing the inspired Po₂ will result in an increased amount of dissolved O₂. At high altitudes, there is low barometric pressure and so less Po₂, hence, supplemental O₂ will help.

Option A: With anemia, the hemoglobin is fully saturated and O₂ will have minimal effect. Treatment of anemia is more beneficial than providing oxygen to an anemic patient.

Option B: With COPD there is CO₂ retention due to difficulty in the expiration, and there will be a decreased Po₂. So, there is a change in the regulation of respiration (drive for ventilation) from being under CO₂ control to being driven by the decrease in O₂. Giving supplemental O₂ to such a person will increase arterial Po₂, thus decrease the stimulus to breathe, resulting in patient stopping breathing.

Option C: Cyanide poisoning results in decreased O₂ utilization by the tissues and so additional O₂ would be of no benefit.

Solution to Question 6:

Oxygen therapy is not beneficial in central respiratory depression.

High dose oxygen reduces the output of peripheral chemoreceptors. So, if given to patients who already have central respiratory depression, it will indirectly reduce the hypoxic drive to breathe even further, leading to a reduction in respiratory frequency and minute ventilation.

Note: In acute severe asthma, a high concentration of oxygen should be given by face mask to achieve an oxygen saturation of >90%.

Solution to Question 7:

When a person is anemic, there is decrease in O₂ content of the blood (Fig 1-E) because of decreased number of RBCs and hence hemoglobin content in the blood. The oxygen saturation (Fig 2-D) of the available hemoglobin in the arterial blood is however normal and, the pO₂ is also not affected by the hemoglobin concentration.

Solution to Question 8:

Normal Arterial Po₂, ↓Mixed venous Po₂, ↑2,3-DPG are the set of changes that will be seen in the above patient.

The oxygen-carrying capacity of the blood is reduced in an anemic person, but the arterial Po₂ and oxygen saturation are both normal.

The decrease in arterial O₂ content is compensated for by an increase in the extraction of oxygen from hemoglobin, which reduces the Po₂ of the venous blood.

The unloading of oxygen at tissue level is enhanced by an increased level of 2,3-DPG in an anemic patient because 2,3-DPG causes a right shift of the oxygen-hemoglobin dissociation curve.

Solution to Question 9:

Curve C corresponds to blood from an adult (dark blue line) and blood from a fetus (light blue line). The fetal oxygen-hemoglobin dissociation curve is shifted to the left relative to the adult curve.

The fetal hemoglobin (Hb F) has a greater affinity for O₂ than adult hemoglobin (Hb A). This facilitates the movement of O₂ from the mother to fetus. The cause of this greater affinity is the poor binding of 2,3-DPG by α polypeptide chains that replace β chains in Hb F. The fetal oxygen-hemoglobin dissociation curve is thus shifted to the left relative to the adult curve. Typically, fetal arterial Po₂ is low, hence leftward shift enhances placental uptake of O₂.

Solution to Question 10:

The right shift of the oxygen-hemoglobin curve allows more oxygen to be released to the muscle at a given Po₂ in the blood. The oxygen dissociation curve shifts to the right when factors like:

- Increased temperature resulting from heat generated by the exercising muscle,
- Higher carbon dioxide levels released from the muscle tissue,
- Elevated 2,3-Bisphosphoglycerate (2,3-BPG), and
- Increased hydrogen ions concentration in the muscle capillary blood

Above factors decrease hemoglobin's oxygen affinity. This shift promotes easier oxygen release to muscles and other tissues.

Note: Despite the rightward shift, the maximum hemoglobin oxygen saturation at high oxygen levels remains unchanged. The plateau on the curve represents the maximal saturation achievable under optimal conditions, ensuring hemoglobin can still be fully saturated with oxygen.

Solution to Question 11:

Decompression sickness is seen in pilots, mountaineers, and divers.

Decompression sickness (DCS) or diver's disease, the bends, or Caisson disease is the name given to a variety of symptoms experienced by a person exposed to a reduction in the pressure surrounding their body.

It can be seen in

- Divers ascending rapidly from deep-sea (high pressure) to the surface of the earth (low pressure)
- Pilots traveling in unpressurized aircraft or mountaineers, rapidly ascending from the surface of the earth (high-pressure zone) to a high altitude (low-pressure zone).

Under high-pressure, gases get compressed and their solubility increases. When a person is in a high-pressure zone like the deep sea for a long time, an increased amount of nitrogen gets

dissolved in his or her body. If the person suddenly comes back to an area of atmospheric pressure or a low-pressure zone, the gases escape from a dissolved state and form bubbles. These bubbles are almost entirely made of nitrogen. The bubbles can plug many small blood vessels in different tissues.

Solution to Question 12:

Nitrogen narcosis is due to increased solubility of nitrogen in the nerve cell.

Concept:

- Below the sea level, atmospheric pressure increases at the rate of 1 atm for every 33 feet depth.
- According to Boyle's law, as pressure increases, volume decreases. Hence, lungs tend to collapse. To prevent this and to keep lung expanded, gas is delivered at high pressure to the lungs.
- According to Dalton's law, nitrogen contributes the maximum to barometric pressure and hence the amount of nitrogen entering alveoli increases.
- Henry's law: As the partial pressure of nitrogen in alveoli increases, its concentration in blood also increases (dissolved form). Nitrogen is 5 times more soluble in fat compared to water.
- As a result, nitrogen dissolves in the fatty substances in neuronal membranes, and because of its physical effect on altering ionic conductance through the membranes, it reduces neuronal excitability and produces the narcotic effect.

Solution to Question 13:

A shift of Oxygen Haemoglobin Dissociation Curve (OHDC) to the right is seen in acclimatization.

On ascension to higher altitudes over a period of days rather than a period of hours, the ability to breathe much more deeply and withstand lower atmospheric O₂ concentrations is improved. This phenomenon is called acclimatization.

Variation of pH in Acclimatization:

Immediate exposure to low pO₂ stimulates the arterial chemoreceptors. This increases alveolar ventilation. This immediate increase in pulmonary ventilation upon rising to a high altitude blows off large quantities of CO₂, reducing the pCO₂ and increases the pH of the body fluids.

These changes inhibit the brain stem respiratory center and thereby oppose the effect of low PO₂ to stimulate respiration through the peripheral arterial chemoreceptors in the carotid and aortic bodies.

There is a reduction of bicarbonate ion concentration in the cerebrospinal fluid, as well as in the brain tissues. This reduction, in turn, decreases the pH in the fluids surrounding the chemosensitive neurons of the respiratory center, thus increasing the respiratory stimulatory activity of the center. An important mechanism for the gradual decrease in bicarbonate concentration is compensation by the kidneys for the respiratory alkalosis. Gradually, the metabolic compensation of the kidney will shift the pH towards normal.

Solution to Question 14:

The symptoms of headache and breathlessness following rapid ascent indicate acute mountain/altitude sickness and IV digoxin is not used to treat this condition.

On ascent to high altitudes, the low partial pressure of oxygen leads to hypoxia and systemic arteriolar dilatation. Consequently, the capillary pressure rises with increased transudation of fluid into the brain tissue and the following symptoms are seen:

- Headache
- Irritation
- Insomnia
- Breathlessness
- Nausea and vomiting

Treatment involves administration of oxygen, descent to a lower altitude and administration of acetazolamide. Acetazolamide counteracts hypoxia by stimulating respiration and also reduces the formation of cerebrospinal fluid, thereby reducing intracranial tension.

Portable hyperbaric chambers are increasingly being used in the treatment of high-altitude illnesses when rapid descent is not possible. These chambers are available in armed forces hospitals in high altitude areas

Regulation of Respiration

Question 1:

Where is the pneumotaxic centre located?

- a) Pons
- b) Midbrain
- c) Medulla
- d) Cerebellum

Question 2:

Pacemaker generating the rhythm for breathing is _____

- a) Pneumotaxic centre
- b) Dorsal group of neurons in the medulla
- c) Pre-Bötzinger complex
- d) Apneustic centre

Question 3:

Which of the following is true about the main respiratory control neurons?

- a) Send out regular bursts of impulses to expiratory muscles during quiet respiration
- b) Are unaffected by pain
- c) Are unaffected by impulses from the cerebral cortex
- d) Send out regular bursts of impulses to inspiratory muscles during quiet respiration

Question 4:

The rate of inspiration is controlled by _____

- a) Cerebral cortex
- b) Ventral group of neurons in medulla
- c) Pneumotaxic centre

d) Apneustic centre

Question 5:

Depth of inspiration is increased by_____

- a) Pneumotaxic centre
- b) Ventral group of neurons in medulla
- c) Dorsal group of respiratory neurons
- d) Apneustic centre

Question 6:

A young couple was spending their afternoon on a picnic. Suddenly, the woman got down on her knee, pulled out a ring, and proposed. The man was taken aback and began to hyperventilate. The overdrive mechanism observed is due to _____

- a) Pneumotaxic centre
- b) Dorsal group of neurons (DRG) in the medulla
- c) Apneustic centre
- d) Ventral group of neurons (VRG) in the medulla

Question 7:

Stimulation of the apneustic centre causes_____

- a) Decrease in the inspiratory ramp signal
- b) Marked increase in inspiratory activity
- c) Stimulation of pneumotaxic centre
- d) Cessation of spontaneous respiration

Question 8:

A few nights before the respiratory physiology exam, a student was suffering from a recurring nightmare. In this nightmare, a knife-wielding horse was chasing him through the streets and threatening to cut off his brainstem at the mid-pontine level. Assuming the horse was proficient in its threat, and performed a bilateral vagotomy as well, which of the following

outcomes would occur?

- a) Ceasation of spontaneous respiration
- b) Continuation of regular breathing
- c) Irregular but rhythmic respiration
- d) Apneusis

Question 9:

Peripheral chemoreceptors _____

- a) Respond minimally to changes in PO_2 under normoxic conditions
- b) Respond to changes in arterial PO_2 but not pH
- c) Respond to changes on PCO_2 slower than the central chemoreceptors
- d) Are most important receptors causing an increased ventilation in response to a rise in PCO_2

Question 10:

Peripheral chemoreceptors are primarily sensitive to _____

- a) $PaCO_2$
- b) H^+
- c) PaO_2
- d) HCO_3^-

Question 11:

Peripheral chemoreceptors are stimulated maximally by _____

- a) Acidosis
- b) Hypercapnia
- c) Carbon monoxide
- d) Cyanide

Question 12:

Carotid body chemoreceptors are stimulated by all of the following types of hypoxia except_____

- a) Anemic
- b) Hypoxic
- c) Stagnant
- d) Histotoxic

Question 13:

Central chemoreceptors_____

- a) Respond to both $p\text{CO}_2$ and $p\text{O}_2$ of blood
- b) Are activated by changes in pH of the CSF
- c) For a given rise in $p\text{CO}_2$, the pH of the CSF falls less than that of blood
- d) HCO_3^- concentration in CSF doesn't affect output of central chemoreceptors

Question 14:

Central chemoreceptors are most sensitive to the following change in the blood:

- a) Decreased P_{O_2}
- b) Increased P_{CO_2}
- c) Increased H^+
- d) Decreased P_{CO_2}

Question 15:

The primary direct stimulus for excitation of central chemoreceptors which regulate ventilation is_____

- a) Increased H^+ in CSF
- b) Increased arterial PCO_2
- c) Decreased arterial PO_2
- d) Increased CO_2 in CSF

Question 16:

Peripheral and central chemoreceptors may both contribute to increased ventilation that occurs as a result of decreased_____

- a) Arterial PCO₂
- b) Arterial PO₂
- c) Arterial blood pressure
- d) H⁺

Question 17:

Which of the following do not stimulate either central or peripheral chemoreceptors?

- a) Hypocapnia
- b) Hypoxia
- c) Acidosis
- d) Low perfusion pressure

Question 18:

The most important stimulus controlling the level of resting ventilation is_____

- a) pH of CSF on central chemoreceptors
- b) pH on peripheral chemoreceptors
- c) Pco₂ on peripheral chemoreceptors
- d) Po₂ on peripheral chemoreceptors

Question 19:

Which of the following occurs after hyperventilation with 6% CO₂ in inspired air?

- a) Continued hyperventilation
- b) Apnoea
- c) Cheyne- Stokes breathing
- d) Kussmaul's breathing

Question 20:

Which of the following is true about the ventilatory response?

- a) CO₂ major factor controlling the normal level of ventilation
- b) Is not related to respiratory minute volume
- c) Depends only on central chemoreceptors
- d) Is increased if alveolar PO₂ is raised

Question 21:

Hering-Breuer reflexes are mediated by _____ .

- a) Slowly adapting myelinated vagal fibers
- b) Rapidly adapting myelinated vagal fibres
- c) Unmyelinated pulmonary C fibers
- d) Unmyelinated bronchial C fibers

Question 22:

All of the following statements about Hering-Breuer inflation reflex are true except_____

- a) The impulses travel to the brain via vagus nerve
- b) It results in further inspiratory efforts if the lung is maintained inflated
- c) It is seen in adults at high tidal volumes
- d) Abolishing the reflex in many animals causes slow deep breathing

Question 23:

Which of the following is true about Head's paradoxical reflex?

- a) Inhibits respiration
- b) Stimulated by hyperinflation of lung
- c) Plays an important role in normal respiration
- d) Mediated by tracheobronchial stretch receptors

Question 24:

The immediate response of J receptor stimulation is

- a) Tachycardia
- b) Hypertension
- c) Tachypnoea
- d) Apnoea

Question 25:

Moderate exercise being one of the most powerful stimulators of ventilation works by the way of _____

- a) Low arterial PO₂
- b) Low PO₂ in mixed venous blood
- c) Low arterial pH
- d) None of the above

Question 26:

The exercise-induced hyperventilation during vigorous exercise is produced due to the receptors present in _____

- a) Medulla oblongata
- b) Lung parenchyma
- c) Carotid bodies
- d) Trachea and large bronchi

Question 27:

In moderate exercise, the respiratory rate is increased at the beginning due to the response of _____

- a) J receptor stimulation
- b) Decrease in PO₂ in arterial blood
- c) Increased PCO₂ in arterial blood
- d) Joint proprioceptor stimulation

Question 28:

Which among the following is not a function of pneumotaxic centre ?

- a) Limitation of inspiration
- b) Increase in the rate of breathing
- c) Switching off the inspiratory ramp
- d) Generating respiratory rhythm

Question 29:

Which of the following ions is involved in peripheral oxygen sensing chemoreceptors?

- a) Potassium
- b) Calcium
- c) Sodium
- d) Chlorine

Question 30:

Which of the following neurons predominantly fire during forceful expiration?

- a) Dorsal group of neurons
- b) Ventral group of neurons
- c) Pneumotaxic center
- d) Chemoreceptors

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	c
5	d
6	d
7	b

8	d
9	a
10	c
11	d
12	a
13	b
14	b
15	a
16	c
17	a
18	a
19	a
20	a
21	a
22	b
23	b
24	d
25	d
26	c
27	d
28	d
29	a
30	b

Detailed Explanations

Solution to Question 1:

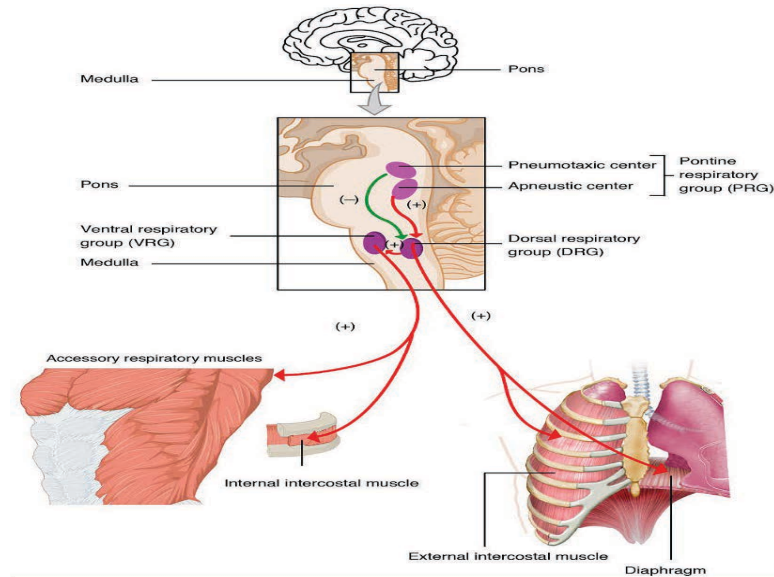
Pneumotaxic center is located in upper pons.

Neural regulation of respiration:

It involves set of neurons located in pons and medulla:

- Pons:
- Pneumotaxic centre
- Apneustic centre
- Medulla: Pre-Botzinger complex
- Dorsal respiratory group

- Ventral respiratory group



Solution to Question 2:

Pre-Botzinger complex generates the rhythm for automated breathing.

- These are also known as pacemaker cells located in the medulla.
- These are the first set of neurons that initiate inspiration.
- These neurons discharge rhythmically, and they produce rhythmic discharges in the phrenic motor neurons.

Breathing is under the control of both the following:

- Voluntary control: Located in the cerebral cortex (via corticospinal tract)
- Automatic control: Driven by pacemaker cells in the medulla (Pre-Botzinger complex). These activate motor neurons via cervical (Phrenic nerve innervating diaphragm) and thoracic spinal cord (external intercostal muscle).

There is a reciprocal innervation to expiratory and inspiratory muscles in that motor neurons supplying expiratory muscles are inactive when motor neurons supplying inspiratory muscles are active, and vice versa.

Inspiration	
Initiation	Pre-Botzinger complex (Pacemaker cells) in Medulla
Maintenance	Apneustic center in lower Pons stimulates Dorsal group of neurons in the medulla

Inspiration	
Termination	Apneustic center inhibited by Vagus and Pneumotaxic center

Regulation of expiration: During quiet breathing, expiration is a passive process; there is no firing of neurons. Ventral respiratory group of neurons are involved in forceful expiration like during exercise.

Mnemonic DIVE: Dorsal respiratory group for Inspiration and Ventral respiratory group for Expiration.

Solution to Question 3:

The main respiratory control neurons called the pre-Bötzinger complex (pre-BOTC), are located in the medulla.

Pre-Bötzinger complex (pre-BÖTC) is located on either side of the medulla, between the nucleus ambiguus and the lateral reticular nucleus

They send out regular bursts of impulses to inspiratory muscles via phrenic nerve during quiet respiration.

Option a: Expiration is passive during quiet breathing. There is no discharge of any neurons.

Option b: Pain stimulates respiration. There are NK1 receptors and μ -opioid receptors in Pre-Botzinger complex, and, in vivo, substance P stimulates and opioids inhibit respiration.

Option c: Impulses from cerebral cortex also influence the pre-Botzinger complex.

Solution to Question 4:

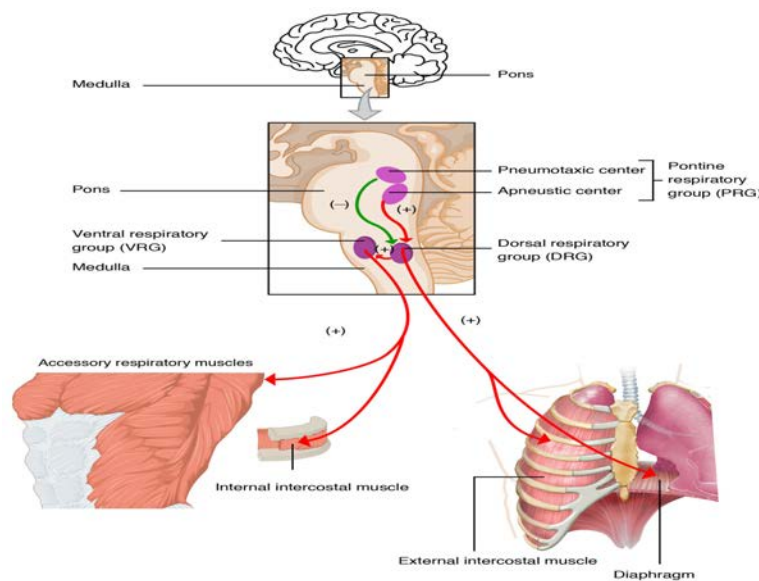
The rate of inspiration is controlled by the pneumotaxic center.

Pneumotaxic centre:

Location: Upper pons (dorsally in nucleus parabrachialis)

Function:

- Transmits signals to the inspiratory area, i.e., it inhibits the apneustic center and thus the dorsal respiratory group of neurons, to stop inspiration.
- It thus helps limit duration of inspiration and switching between inspiration and expiration.
- This has a secondary effect of increasing the rate of breathing because limiting the duration of inspiration also shortens the duration of expiration, and hence, the entire period of respiration.



Solution to Question 5:

Depth of inspiration is increased by the apneustic center.

The apneustic centre sends signals to prevent switching off of inspiration maintained by the dorsal respiratory group of neurons, thereby increasing depth of inspiration.

In contrast to this, the pneumotaxic center primarily limits inspiration by controlling the "switch off" point of the inspiratory ramp signal.

A note on inspiratory ramp signal:

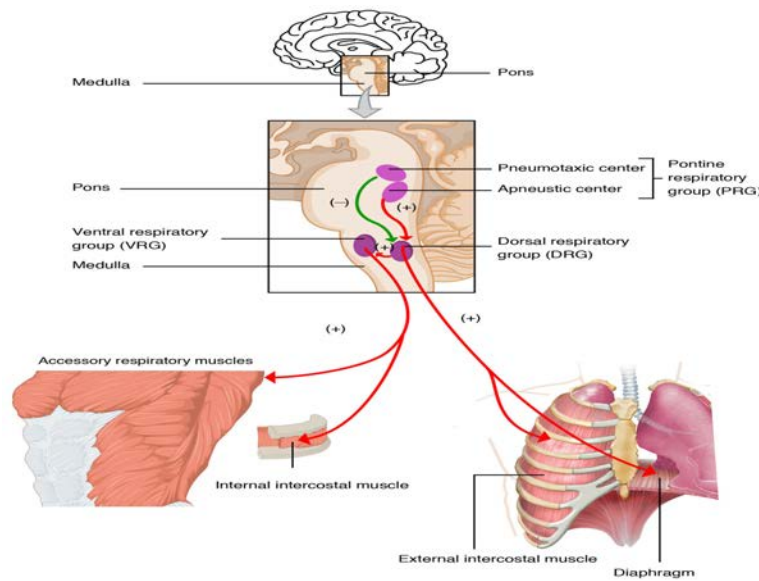
The neural signals transmitted to the inspiratory muscle, mainly the diaphragm, is not the instantaneous burst of action potentials. Instead, it begins weakly and increases steadily in a ramped manner. The advantage of the ramp is that it causes a steady increase in the volume of the lungs during inspiration, rather than inspiratory gasps.

Control of the limiting point at which the ramp suddenly ceases is the mechanism by which the respiratory centers control depth and rate of respiration.

Option A: Pneumotaxic center of the pons regulates the rate and depth of breathing.

Option B: Ventral group of neurons regulate expiration.

Option C: Dorsal group of neurons regulate inspiration.



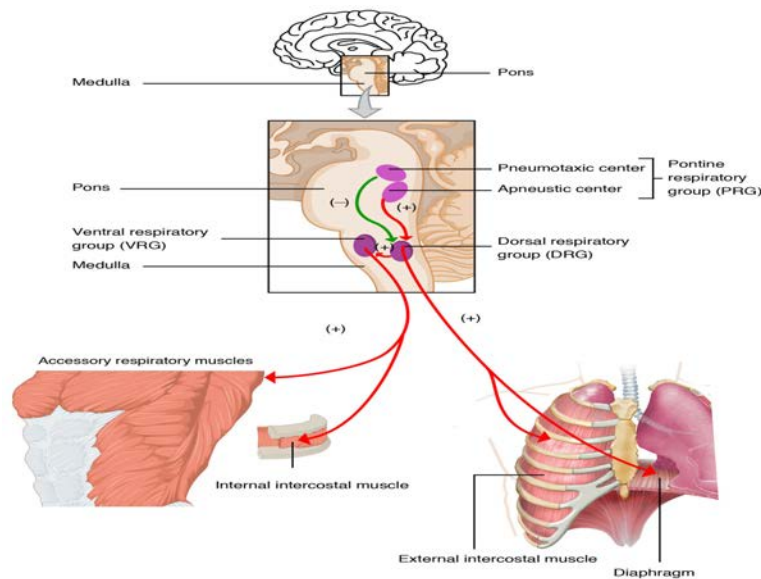
Solution to Question 6:

The ventral respiratory group contributes to the extra respiratory drive (overdrive mechanism)

- During quiet breathing: Expiration is a passive process and the neurons from the ventral respiratory group remain almost totally inactive during quiet respiration. Unlike the dorsal respiratory group (DRG; dorsal group of neurons), VRG (the ventral group of neurons) does not appear to participate in the basic rhythmic oscillations that control respiration.
- Extra-respiratory drive (overdrive mechanism): When the respiratory drive for increased pulmonary ventilation becomes greater than normal, for example, during exercise, respiratory signals spill over to VRG from DRG. So, VRG contributes to the extra respiratory drive (overdrive mechanism).
- The neurons of VRG contribute to both inspiration and expiration; especially, they are important in providing powerful expiratory signals to the abdominal muscles during very heavy expiration.

Dorsal and ventral respiratory group:

	Dorsal respiratory group	Ventral respiratory group
Location	Nucleus Tractus Solitarius in medulla	Nucleus Retro Ambiguus in medulla
Function	Inspiratory ramp signals - Stimulates inspiration	Inactive during quiet breathing; Active during forced expiration



Solution to Question 7:

The apneustic center is located in the lower pons and it stimulates the inspiratory center to increase inspiratory activity.

The apneustic center of pons send signals to the DRG in medulla to delay the 'switch off' signal of the inspiratory ramp provided by pneumotaxic center of pons. It gets feedback from vagal afferents and also from other respiratory centers.

The apneustic center is inhibited by vagal afferents from pulmonary stretch receptors aroused by inspiration so that stimulatory effect on the inspiratory neurons is switched off allowing expiration to supervene.

Solution to Question 8:

Transection of the brain stem at the mid-pontine level with bilateral vagotomy causes apneusis.

At mid pontine level lesion, the apneustic centre is intact while the pneumotaxic centre is separated. Concurrent removal of vagus inputs causes this kind of abnormal pattern of breathing characterized by deep, gasping inspiration with a pause at full inspiration followed by a brief insufficient release.

Solution to Question 9:

Peripheral chemoreceptors respond minimally to changes in PO_2 under normoxic conditions.

Because of the enormous blood flow per unit of tissue to the carotid bodies, the receptors are not stimulated in conditions such as anemia or carbon monoxide poisoning, in which the amount of

dissolved O₂ in the blood reaching the receptors is generally normal, even though the combined O₂ in the blood is markedly decreased.

Peripheral chemoreceptors are responsive to changes in arterial O₂, CO₂, and H⁺ ions.

They are excited by a decrease in arterial O₂. However, a graph plotted between the rate of firing of afferent neurons in the chemoreceptor vs. arterial pO₂ shows they are most sensitive to changes in pO₂ of less than 60mmHg. Thus, ventilation is driven by pCO₂ and H⁺ under normoxic conditions.

Other options :

- Option B: Peripheral chemoreceptors respond to changes in blood pH. They are excited by an increase in H⁺ ions (i.e decrease in pH)
- Option C: The response of peripheral chemoreceptors to changes in Pco₂ is faster than that of central chemoreceptors. Increase in arterial pCO₂ stimulates respiration. However, the stimulation by way of peripheral chemoreceptors occurs 5 times as rapidly as that from central chemoreceptors. This helps increase the rapidity of response to CO₂ at the onset of exercise.
- Option D: Central chemoreceptors are more important than the peripheral chemoreceptors in the ventilatory response to increased CO₂.

Note:

- Most potent stimulus for central chemoreceptor in blood: CO₂
- Most potent stimulus for central chemoreceptor in CSF: H⁺ ions
- Most potent stimulus for respiration: CO₂

Solution to Question 10:

The peripheral chemoreceptors, located in the carotid and aortic bodies, are primarily sensitive to decrease in arterial PO₂, although high PaCO₂ and low pH also stimulates them and enhance their sensitivity to hypoxia.

Carotid body is more sensitive than aortic body.

Solution to Question 11:

Cyanide causes histotoxic hypoxia, and thus, by preventing the O₂ utilization at the tissue level; it acts as the powerful stimulus for peripheral chemoreceptors. Even though Po₂ level of blood is normal in case of cyanide poisoning the sensors in carotid and aortic bodies are not able to sense oxygen in blood because of blockade of cellular respiration and hence, they go in for hypoxic state and gets stimulated.

- Option a: Acidosis (Increased H⁺ ions) stimulates peripheral chemoreceptors.
- Option b: Increased pCO₂ mainly stimulates central chemoreceptors.
- Option c: The peripheral chemoreceptors have an enormous blood flow per unit tissue (2000mL/100g of tissue/min) because of which the O₂ needs of the cells can be met by dissolved

O₂ Therefore, Carbon monoxide poisoning does not stimulate peripheral chemoreceptors because the amount of dissolved O₂ is generally normal, even though the combined O₂ in the blood is markedly decreased.

Solution to Question 12:

Anemic hypoxia does not stimulate carotid body chemoreceptors.

In anemia,

- Amount of dissolved O₂ reaching the receptors is generally normal
- Combined O₂ in the blood is markedly decreased.
- The blood flow per unit tissue of carotid body chemoreceptors is enormous (2000mL/100g of tissue/min). Hence, O₂ needs of the tissue can be met largely by dissolved O₂ alone. Therefore, the receptors are not stimulated by conditions such as anemia or carbon monoxide poisoning, in which the amount of dissolved O₂ reaching the receptors is generally normal, even though combined O₂ in the blood is markedly decreased. These peripheral chemoreceptors are stimulated when arterial pO₂ is low.

Solution to Question 13:

Central chemoreceptors are stimulated by changes in pH of the CSF. They are located on the ventral surface of medulla

Option A: Central chemoreceptors do not respond to pO₂ of the blood.

Option C: For a given rise in pCO₂, the CSF pH falls more than that of the blood because the CSF has less buffering. CO₂ combines with H₂O to form H₂CO₃ which dissociates into H⁺ and HCO₃⁻. H⁺ is buffered by combining with Hb. This buffering is less in CSF and hence pH falls.

Option D: The HCO₃⁻ concentration of the CSF can affect the output of the central chemoreceptors by buffering the changes in pH. HCO₃⁻ can combine with H⁺ and thus neutralise it and alter pH and thus alter output of central chemoreceptors.

Solution to Question 14:

Central chemoreceptors are most sensitive to increased Pco₂ in the blood.

Central chemoreceptors are most sensitive to increased Pco₂ in the blood because CO₂ moves into the CSF, where it is promptly hydrated to form H₂CO₃ that dissociates to increase the CSF H⁺ concentration. This stimulates the central chemoreceptors, which monitor CSF H⁺ concentration (not blood H⁺ concentration).

Other options:

- H^+ in arterial blood penetrates the blood-brain barrier very slowly and hence has little effect on central chemoreceptors.
- Arterial P_{O_2} has no effect on central chemoreceptors.

Solution to Question 15:

The central chemoreceptors are primarily stimulated by an increase in the H^+ in the CSF.

CO_2 readily penetrates the blood-brain barrier and is promptly hydrated in the CSF to form H_2CO_3 that dissociates so that local H^+ concentration rises. The H^+ concentration in the CSF parallels the arterial P_{CO_2} . So the effect of arterial P_{CO_2} and CO_2 in CSF is indirect or secondary.

Arterial P_{O_2} has no effect on central chemoreceptors.

Solution to Question 16:

Both peripheral and central chemoreceptors contribute to increased ventilation that occurs as a result of decreased arterial blood pressure.

- Decrease in arterial blood pressure $\rightarrow$ hypoperfusion $\rightarrow$ hypoxia $\rightarrow$ stimulation of peripheral chemoreceptors.
- Decrease in arterial blood pressure $\rightarrow$ CO_2 retention $\rightarrow$ increased arterial $PCO_2 \rightarrow \uparrow H^+$ concentration in the CSF $\rightarrow$ stimulation of central chemoreceptors

Option a : Decreased arterial P_{CO_2} doesn't stimulate chemoreceptors. It is increased P_{CO_2} that stimulates central chemoreceptors.

Option b : Decreased arterial P_{O_2} stimulates peripheral chemoreceptors but has no influence on central chemoreceptors.

Option d : Increased (not decreased) H^+ ions stimulate both peripheral and central chemoreceptors.

Solution to Question 17:

Hypercapnia is a stimulus for central and peripheral chemoreceptors and not hypocapnia.

Option b: The peripheral chemoreceptors are stimulated by hypoxia (hypoxic, stagnant and histotoxic but NOT anemic hypoxia) and exercise (increased K^+ levels).

Option c: The central chemoreceptors are stimulated by hypercapnia and resultant acidosis.

Option d: Low perfusion pressure produces hypoxia because the amount of O_2 delivered to the receptors per unit time is decreased. This stimulates the peripheral chemoreceptors.

Solution to Question 18:

The most important stimulus comes from the pH of the CSF on the central chemoreceptors. The CO₂ that enters the brain and CSF is hydrated to H₂CO₃ and dissociates, leading to rising H⁺ levels. The CSF H⁺ concentration hence parallels the arterial CO₂ levels.

The effect of pO₂ on peripheral chemoreceptors under normoxic conditions is very small. Changes in pCO₂ do affect the peripheral chemoreceptors, but the magnitude is less than that for the central chemoreceptors.

The effect of changes in the pH on peripheral chemoreceptors under normal conditions is small: and the changes in pO₂ do not affect the central chemoreceptors.

Solution to Question 19:

Hyperventilation with 6% CO₂ is followed by continued hyperventilation till alveolar Pco₂ drops towards normal.

Ventilatory responses to CO₂

There is a linear relationship between increase in % of CO₂ in inspired air and respiratory minute volume.

- The upper limit for this linearity is 6% of alveolar PCO₂.
- When the CO₂ content of the inspired gas is >7%, the alveolar and the arterial PCO₂ begin to rise abruptly in spite of hyperventilation leading to hypercapnia.
- Hypercapnia depresses the central nervous system, including respiratory centers leading to CO₂ narcosis (a headache, confusion and eventually coma).

Mechanism:

Increased CO₂ in inspired air → Increased Alveolar PCO₂ → Increased arterial PCO₂ → Stimulation of central chemoreceptors → Stimulation of ventilation → Alveolar PCO₂ drops toward normal

Solution to Question 20:

The normal level of ventilation is controlled by the ventilatory response to CO₂.

Option b: There is linear relationship between respiratory minute volume and the alveolar Pco₂, i.e. as alveolar Pco₂ increases respiratory minute volume increases.

Option c: The ventilatory response depends on peripheral chemoreceptors in addition to central chemoreceptors.

Option d: The ventilatory response to CO₂ is increased if the alveolar PO₂ is reduced

The ventilatory response is reduced also during sleep.

Solution to Question 21:

Hering-Breuer reflexes are mediated by slowly adapting pulmonary stretch receptors and the impulses travel in the vagus nerve via large myelinated fibers.

Hering-Breuer inflation reflex: "Prolonged inspiration inhibits further inspiration"

- Hyperinflation of lungs stimulates slowly adapting pulmonary stretch receptors and the impulses travel in the vagus nerve via large myelinated fibers.
- This signal inhibits the inspiratory ramp signals of the dorsal respiratory group of neurons (DRG), thus, inspiration is quickly inhibited and expiration follows.
- Advantages of Hering-Breuer inflation reflex: It cuts down the inspiration time, and therefore,
- It prevents overinflation of the lungs.
- Lungs operate at lower lung volumes, thereby preventing wastage of energy.
- The Hering-Breuer reflex is activated when the lung volume increases to more than three times the normal tidal volume (tidal volume = 0.5 liters/breath; So, lung volume $>$ 1.5 liters per breath activates the reflex).
- It signifies that the reflex plays a protective role preventing the overinflation of the lung rather than regulating the normal ventilation.

Solution to Question 22:

Hering-Breuer inflation reflex inhibits further inspiratory efforts if the lungs are maintained inflated.

Mechanism- Stretch receptors Vagus Inhibit Dorsal Respiratory Group Phrenic & Intercostal Nerves Inspiration shortened, Expiration prolonged

Option a: The impulses travel to the brain via the vagus nerve.

Option c: HB Reflex activated when TV $>$ 1.5L. The reflex is seen in adults at high tidal volumes.

Option d: Abolishing the reflex by cutting the vagal nerves in experimental animals causes slow deep breathing.

Solution to Question 23:

Head's paradoxical reflex is stimulated by hyperinflation of lungs. It refers to prolonged inspiration stimulating further inspiration. Hyperinflation of the lungs causes strong and long-lasting inspiratory effort (an effect opposite to the Hering-Breuer inflation reflex).

Physiological role:

- Occasional deep breaths which are thought to counteract any tendency to atelectasis in quiet breathing and possibly in reinforcing the gasps or inspiratory efforts.

- First cry after birth.
- The possible explanation for this could be the inhibition of vagus nerve at cold temperatures (due to the presence of pulmonary fluid in the unaerated newborn lungs).
- Lack of inhibitory impulses from the pulmonary stretch receptors and vagus has an effect opposite to Hering-Breuer's inflation reflex.

Solution to Question 24:

The immediate reflex response of J receptor stimulation is apnea. It is followed by rapid breathing, bradycardia, and hypotension (pulmonary chemoreflex).

Juxtacapillary receptors are the C fiber endings located close to the pulmonary vessels in the alveolar interstitium. They are not involved in quiet or normal respiration. They are stimulated by

- Hyperinflation of the lung
- Intravenous or intracardiac administration of chemicals like capsaicin P

Solution to Question 25:

Moderate exercise does not reduce the arterial P_{O_2} , increase the arterial P_{CO_2} , or reduce the arterial pH.

Option A: The brain, upon transmitting motor impulses to the exercising muscles, is believed to transmit collateral impulses into the brainstem at the same time to excite the respiratory center. This action is analogous to the stimulation of the vasomotor center of the brain stem during exercise that causes a simultaneous increase in arterial pressure.

Option B: The P_{O_2} of mixed venous blood does fall, but there are no known chemoreceptors that are stimulated as a result.

Option C:

Two points are important regarding pH during moderate exercise:

- $pH = pK + \log \left(\frac{[HCO_3^-]}{[CO_2]} \right)$
- The pH of the blood is dependent only on the ratio of the amount of CO_2 to the amount of HCO_3^- . The concentrations of both buffer components (HCO_3^- and CO_2) are very large compared to the amount of H^+ added to the blood during moderate exercise. Thus, any change in HCO_3^- due to the addition of H^+ during moderate exercise is very tiny compared to the amount of HCO_3^- already present in the blood. Thus, the pH of blood during moderate exercise remains constant.
- When sufficient oxygen is not available during exercise, there is increased accumulation of lactic acid, which initially leads to an increase in $PaCO_2$. However, this is counteracted by the stimulation of ventilation and as a result, $PaCO_2$ is decreased. This provides some respiratory compensation for further lactic acid production and prevents a decline in blood pH. Thus, blood pH remains constant in moderate exercise.

Solution to Question 26:

The exercise-induced hyperventilation during vigorous exercise is produced due to the receptors present in carotid bodies.

When exercise becomes more vigorous, buffering of the increased amounts of lactic acid that are produced liberates more CO_2 , and this further increases ventilation. The additional increase in ventilation produced by the acidosis depends on the carotid bodies and does not occur if they are removed.

Note: The abrupt increase at the start of exercise is presumably due to psychic stimuli and afferent impulses from proprioceptors in muscles, tendons, and joints.

Solution to Question 27:

In moderate exercise, the respiratory rate is increased at the beginning due to the response of joint proprioceptor stimulation.

Respiratory responses to exercise

- During exercise, there is an increase in ventilatory rate that matches the increase in O_2 consumption and CO_2 production by the body. The stimulus for increased ventilation rate is the activation of joint proprioceptors at the beginning of the exercise.
- The mean values for arterial PO_2 and PCO_2 do not change during exercise. Arterial pH does not change during moderate exercise, although it may decrease during strenuous exercise because of lactic acidosis.
- On the other hand, venous PCO_2 increases during exercise because the excess CO_2 produced by exercising muscle is carried to the lungs in venous blood.
- Pulmonary blood flow increases because cardiac output increases during exercise. As a result, more pulmonary capillaries are perfused, and more gas exchange occurs. The distribution of V/Q ratios throughout the lung is more even during exercise than when at rest, and there is a resulting decrease in the physiologic dead space.

The abrupt increase at the start of exercise is presumably due to

- Psychic stimuli
- Afferent impulses from proprioceptors in muscles, tendons, and joints.

The more gradual increase in ventilation is presumably due to

- Increase in O_2 consumption (Proportional increase)
- Increase in body temperature
- Increases in the plasma K^+ level (Stimulate the peripheral chemoreceptors).
- Sensitivity of the neurons controlling the response to CO_2 is increased or that the respiratory fluctuations in arterial PCO_2 increase so that, even though the mean arterial PCO_2 does not rise, it is CO_2 that is responsible for the increase in ventilation.

Solution to Question 28:

Respiratory rhythm generation is a function of dorsal respiratory group of neurons (and not pneumotaxic center).

Pneumotaxic center:

- It is located dorsally in the nucleus parabrachialis of the upper pons and it transmits signals to the inspiratory area.
- Primary effect of this center is to control the switchoff point of the inspiratory ramp, thus controlling the duration of the firing phase of the lung cycle.
- Function of the pneumotaxic center is primarily to limit inspiration, which has a secondary effect of increasing the rate of breathing, because limitation of inspiration also shortens expiration and the entire period of each respiration.

Solution to Question 29:

Potassium is involved in peripheral oxygen sensing chemoreceptors.

Type I glomus cells (carotid and aortic body) have O₂-sensitive K⁺ channels, whose conductance is reduced in proportion to the degree of hypoxia.

This reduces the K⁺ efflux, depolarizing the cell and causing Ca²⁺ influx.

The Ca²⁺ influx triggers action potentials and consequent excitation of the afferent nerve endings.

Solution to Question 30:

Ventral groups of neurons fire, when forceful expiration is needed like during heavy exercise.

Dorsal group of neurons control the normal inspiration and expiration.

Pneumotaxic center limits the duration of inspiration.

Vascular System and Regional Circulation I

Question 1:

The windkessel effect is seen in_____.

- a) Large elastic vessels
- b) Capacitance vessels
- c) Shunts
- d) Capillaries

Question 2:

The greatest total cross-sectional area in the circulatory system is of_____.

- a) Aorta
- b) Capillaries
- c) Venules
- d) Vene cavea

Question 3:

Largest percentage of blood volume in the circulatory system is in _____.

- a) Arteries
- b) Capillaries
- c) Veins
- d) Heart

Question 4:

Which of the following is not correct regarding capillaries?

- a) Have a single layer of cells bounding the lumen
- b) Greatest total cross sectional area
- c) Contains less blood than veins

- d) Contain 25% of blood

Question 5:

In the circulatory system, the highest velocity of blood flow is in the _____.

- a) Aorta
- b) Capillaries
- c) Venules
- d) Vene cava

Question 6:

The largest compliance in the circulatory system is of _____.

- a) Arteries
- b) Capillaries
- c) Veins
- d) Aorta

Question 7:

Bernoulli's principle states that _____.

- a) Sum of kinetic energy of flow and pressure energy is constant
- b) Low tones produces maximum stimulation at the apex of cochlea
- c) Magnitude of sensation felt is proportional to the intensity of stimulus
- d) Force of contraction is proportional to stretch of the cardiac muscle

Question 8:

The blood flow through a vessel varies directly with _____

- a) Resistance
- b) Viscosity
- c) Pressure difference
- d) Length of the vessel

Question 9:

Which of the following increases turbulence in blood flow?

- a) Increase in diameter of blood vessel
- b) Reynolds number less than 2000
- c) Decrease in density of blood
- d) Decrease in velocity of blood

Question 10:

The tendency for turbulent flow is greatest in _____.

- a) Arteries
- b) Capillaries
- c) Veins
- d) Aorta

Question 11:

The main site of peripheral vascular resistance is _____.

- a) Precapillary arterioles
- b) Precapillary sphincters
- c) Capillaries
- d) Windkessel vessels

Question 12:

The factor favoring filtration at the arterial end of the capillary bed is _____.

- a) Increase in interstitial hydrostatic pressure
- b) Increase in hydrostatic pressure of the capillary
- c) Increase in oncotic pressure of the capillary
- d) Decrease in oncotic pressure of the interstitium

Question 13:

Which of the following factors tends to increase the lymph flow?

- a) Increased capillary hydrostatic pressure
- b) Increased plasma oncotic pressure
- c) Decreased capillary permeability
- d) Precapillary constriction

Question 14:

The coronary blood flow is regulated by ____.

- a) Adenosine
- b) Bradykinin
- c) Prostaglandin
- d) Increased arterial pCO₂.

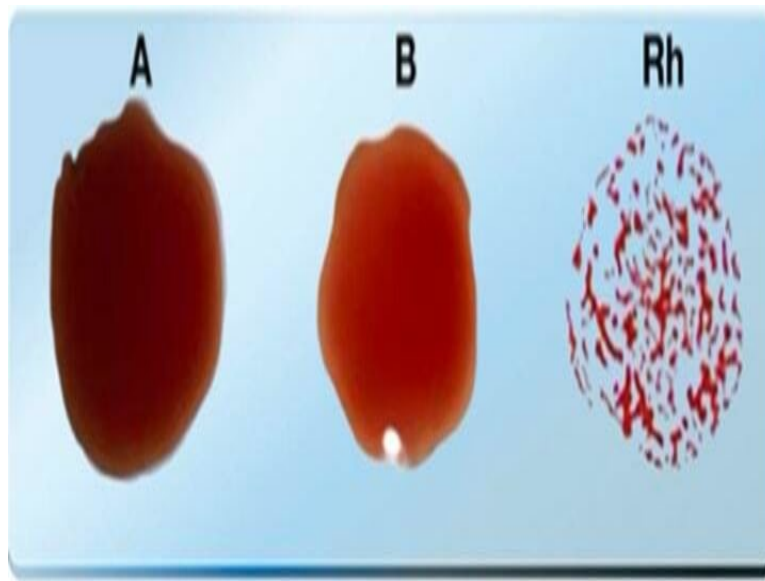
Question 15:

True about coronary flow during systole is:

- a) No change in flow with systole.
- b) Increased in flow with systole.
- c) Greater decrease in subendocardium than epicardium.
- d) Greater decrease in epicardium than in subendocardium.

Question 16:

Based on the agglutination test shown below, what is the blood group?



- a) B positive
- b) A positive
- c) O positive
- d) AB positive

Question 17:

Which of the following clotting factors would have decreased gamma carboxyglutamate residue in a patient on warfarin therapy?

- a) Factor 2
- b) Factor 11
- c) Tissue factor
- d) Factor 5

Question 18:

Which of the following has the highest oxygen concentration in fetal circulation?

- a) Inferior vena cava
- b) Right atrium
- c) Superior vena cava
- d) Ascending aorta

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	d
5	a
6	c
7	a
8	c
9	a
10	d
11	a
12	b
13	a
14	a
15	c
16	c
17	a
18	a

Detailed Explanations

Solution to Question 1:

The windkessel effect is seen in large elastic vessels.

The aorta, the pulmonary artery, and their major branches have a large amount of elastin in their walls. Hence, they have a high elastic recoil effect. This property of elastic recoil is called windkessel effect (windkessel is a German word for the elastic reservoir).

This recoil of the vessels that have been stretched during systole helps to propel the blood forward during diastole thus maintaining a continuous flow throughout the cardiac cycle.

Solution to Question 2:

The greatest total cross-sectional area in the circulatory system is of the capillaries.
The total cross-sectional area is minimum for the aorta.

Solution to Question 3:

Largest percentage of blood volume in the circulatory system is in veins.

Vessels and Percentage of Total Blood Volume

VESSELS	% OF TOTAL BLOOD VOLUME
Heart	12
Pulmonary circulation	18
Aorta	2
Artery	8
Arteriole	1
Capillary	5
Venule, vein, vena cava	54

Solution to Question 4:

Capillaries contain 5% of the total blood volume.

Solution to Question 5:

In circulatory system, the highest velocity of blood flow is in the aorta.

- The velocity of blood flow is inversely proportional to vascular cross-sectional area.

$$v = \frac{F}{A}$$

v= velocity

F= blood flow through the vessel

A=vascular cross-sectional area

- Aorta has the least cross sectional area (4.5 cm²) and capillaries have the maximum cross sectional area (4500 cm²). Thus, under resting conditions, the velocity is highest in aorta and least in capillaries.

Solution to Question 6:

The largest compliance in the circulatory system is of veins.

Vascular compliance/capacitance is the total quantity of blood that can be stored in a given portion of the circulation for each mmHg pressure rise.

Vascular Compliance = distensibility X volume.

The compliance of a systemic vein is about 24 times that of its corresponding artery because it is about 8 times as distensible and has a volume about 3 times as artery ($8 \times 3 = 24$).

Solution to Question 7:

The Bernoulli's theorem states that in a tube or blood vessel, the sum of the kinetic energy of flow and pressure energy is constant.

Other options:

- Option B: Travelling wave theory (Theories of hearing): Low tones produce maximum stimulation at the apex of the cochlea.

- Option C: Weber—Fechner law: The magnitude of sensation felt is proportional to the log of the intensity of the stimulus.
- Option D: Frank—Starling law of the heart: Force of contraction is proportional to stretch of the cardiac muscle.

Solution to Question 8:

The blood flow through a vessel varies directly with pressure difference.

The blood flow follows Ohm's law:

$$F = \frac{P}{R}$$

Where,

- F = Blood flow
- P = Pressure difference between two ends of the vessel
- R = Resistance

Solution to Question 9:

Turbulence in blood flow increases with increase in diameter of blood vessel.

The turbulence of flow depends upon Reynold's number:

$$Re = \frac{\rho v D}{\mu}$$

Where,

v – velocity of flow – cm/sec

ρ – density of blood

d – diameter of the vessel

μ – viscosity of blood in poise

Therefore, increase in diameter increases turbulence.

Reynold's number	Type of blood flow
<2000	Laminar
2000-3000	Transitional
>3000	Turbulent

Solution to Question 10:

The tendency for turbulent flow is greatest in the aorta.

The turbulence of flow depends upon Reynold's number:

$$Re = \frac{\rho v D}{\mu}$$

v – velocity of flow – cm/sec

ρ – density of blood

d – diameter of the vessel

μ – viscosity of blood in poise

In the proximal portions of the aorta and pulmonary artery, Reynold's number can rise to several thousand during the rapid phase of the ejection, which causes considerable turbulence due to the following:

- High velocity of blood flow
- Pulsatile nature of the flow
- Sudden change in vessel diameter
- Large vessel diameter

Reynold's number	Type of blood flow
<2000	Laminar
2000-3000	Transitional
>3000	Turbulent

Solution to Question 11:

The main site of peripheral vascular resistance is precapillary arterioles.

Poiseuille—Hagen formula explains that small changes in the calibre of vessels cause large changes in total peripheral resistance.

$$R = \frac{8\eta L}{\pi r^4}$$

Where,

R=resistance

η =viscosity

L=length of the tube

r=radius of the tube

The resistance varies inversely with the fourth power of radius.

Solution to Question 12:

The factor favoring filtration at the arterial end of the capillary bed is the increase in hydrostatic pressure of the capillary.

The rate of filtration at any point along the capillary bed depends on the balance of the Starling forces, which are responsible for fluid movement.

Net filtration pressure = $(P_c - P_i) - (\pi_c - \pi_i)$

where,

P_c =capillary hydrostatic pressure

P_i =interstitial hydrostatic pressure

π_c =capillary colloidal osmotic pressure

π_i =interstitial colloidal osmotic pressure

Glomerular filtration rate = $k \times \text{NFP}$

where,

k= capillary filtration coefficient

NFP= Net filtration pressure

Solution to Question 13:

The factor tend to increase lymph flow is increase in capillary hydrostatic pressure.

Since the lymph is derived from interstitial fluid, any factor that increases interstitial fluid pressure also increases lymph flow provided, the lymph vessels are functioning normally. The factors which increase interstitial fluid pressure include:

- Increased capillary hydrostatic pressure
- Increased interstitial fluid colloid osmotic pressure
- Increased permeability of the capillaries
- Decreased plasma colloid osmotic pressure

All of the above factors increase the filtration at the capillary-interstitium interface and therefore increase the interstitial fluid volume and pressure.

Solution to Question 14:

Coronary blood flow is regulated by adenosine.

Blood flow through the coronary system is regulated mostly by local arteriolar vasodilation in response to the nutritional needs of cardiac muscle.

$\uparrow$ local metabolism $\rightarrow$ $\downarrow$ in $O_2 \rightarrow$ breakdown of ATP to form adenosine $\rightarrow$ vasodilation $\rightarrow$ $\uparrow$ local coronary blood flow.

Solution to Question 15:

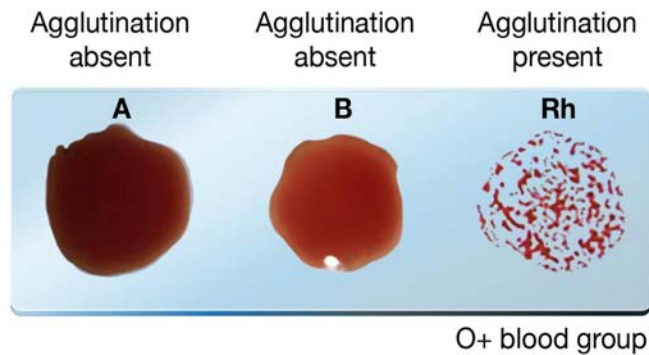
The coronary blood flow in systole has a greater decrease in subendocardium than epicardium.

The myocardial compression in systole compresses the subendocardial vessels to a greater extent, and the force is sufficiently dissipated in the epicardium to permit some flow in this region throughout the cardiac cycle.

Solution to Question 16:

Based on the agglutination test shown in the image, the blood group is O positive.

As agglutination is present only in 'Rh' and not with 'A' and 'B' hence the blood group is O Rh positive.



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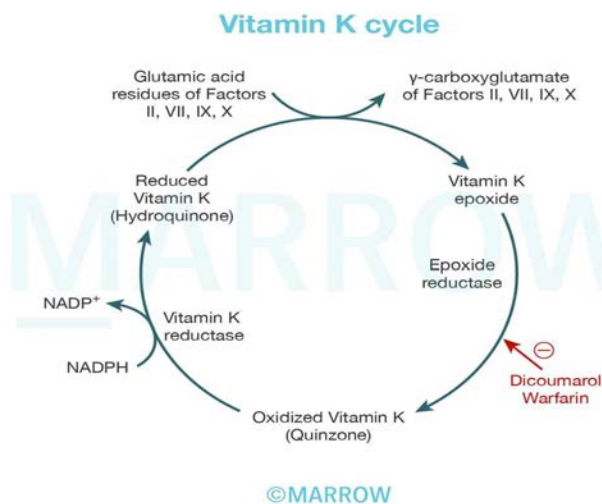
Solution to Question 17:

Factor 2 would have decreased gamma carboxyglutamate residue in a patient on warfarin therapy.

Activation of coagulation factors II, VII, IX, X is done by carboxylation of their glutamate residues. This is done by the enzyme γ -glutamyl carboxylase with the reduced form of vitamin K as a co-factor.

This reduced form of vitamin K needs to be constantly recycled back from its oxidized form for the reaction to continue. This recycling is done by the enzyme vitamin K epoxide reductase. This enzyme is inhibited by warfarin.

Hence, warfarin indirectly inhibits the γ -carboxylation of vitamin K dependent clotting factors - Factors II, VII, IX, X by directly inhibiting vitamin K epoxide reductase. Thus leading to decreased gamma carboxyglutamate residue in these factors.

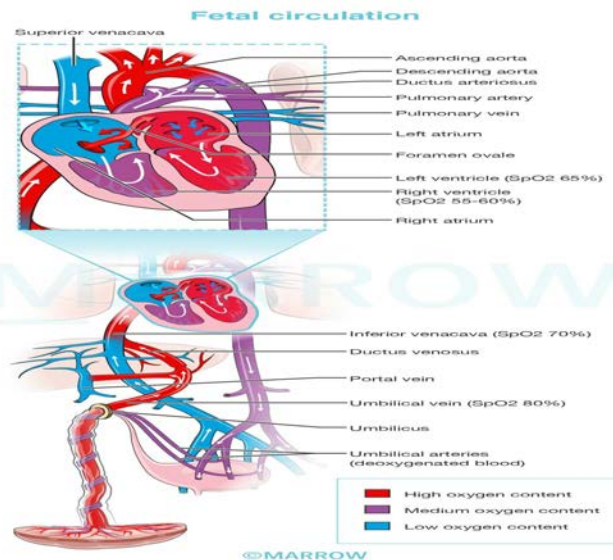


Solution to Question 18:

Inferior vena cava has the highest oxygen concentration in fetal circulation.

The blood gets oxygenated in the placenta and this passes to the umbilical vein. From the umbilical vein, 50% of the blood enters the hepatic circulation and 50% bypasses the liver to enter the inferior vena cava. From here, the blood enters the heart via the right atrium. The right atrium also receives deoxygenated blood from the superior vena cava. This mixed blood is distributed to the body via aorta.

Therefore, inferior vena cava has the highest concentration of oxygen in the fetal circulation.



Vascular System and Regional Circulation II

Question 1:

What is the most powerful activator of sympathetic vasomotor system?

- a) Posture of the body against gravity
- b) Chemoreceptors
- c) CNS ischemic response
- d) Baroreceptors

Question 2:

During exercise, the organ experiencing the smallest decrease in blood flow is ____.

- a) Brain
- b) Intestine
- c) Kidney
- d) Non exercising skeletal muscle

Question 3:

An elderly man, with congestive cardiac failure, complains of shortness of breath when lying down. What is the mechanism behind this symptom?

- a) Reservoir function of pulmonary veins.
- b) Reservoir function of veins in lower limbs.
- c) Pulmonary arterial hypertension.
- d) Systemic hypertension.

Question 4:

The interstitial fluid pressure is negative in ____.

- a) Skeletal muscle
- b) Lungs

- c) Kidney
- d) Brain

Question 5:

Blood flow per minute in pulmonary circulation is

- a) Less than flow through systemic circulation
- b) More than flow through systemic circulation
- c) Equal to flow through systemic circulation
- d) Independent of systemic circulation

Question 6:

A patient was receiving a red cell transfusion following hip arthroplasty. Midway through the transfusion, she developed signs and symptoms of shock. What is the type of shock seen in her?

- a) Obstructive shock
- b) Cardiogenic shock
- c) Distributive shock
- d) Hypovolemic shock

Question 7:

During foetal life, blood vessel with maximum concentration of oxygen is ____.

- a) IVC.
- b) Aorta.
- c) Umbilical vein.
- d) Umbilical artery.

Question 8:

Progressive hemorrhagic shock is characterized by _____.

- a) Blood clots in the small blood vessels

- b) Decreased lysosomal activity
- c) Increased active transport of sodium
- d) Tissue alkalosis

Question 9:

During rest, blood flow through skeletal muscle averages _____.

- a) 3 - 4 mL/ min /g of muscle
- b) 3 - 4 mL/min/100 g of muscle
- c) 4 - 6 mL/ min/g of muscle
- d) 4 - 6 mL/min/100 g of muscle

Question 10:

The type of shock that has high cardiac output is _____.

- a) Hemorrhagic shock
- b) Cardiogenic shock
- c) Septic shock
- d) Vasovagal shock

Question 11:

Thin-walled and delicate vessels like capillaries are less prone to rupture. This can be explained by _____.

- a) Poissulles law
- b) Laplace law
- c) Frank starlings law
- d) Ficks law

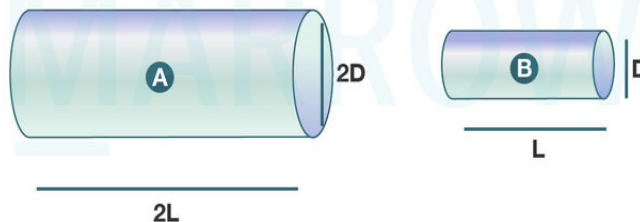
Question 12:

The relationship between the flow in a long narrow tube, the viscosity of the fluid, and the radius of the tube is expressed mathematically by_____.

- a) Laplace law
- b) Poiseuille's law
- c) Fick law
- d) Frank starling law

Question 13:

In the 2 vessels shown below, assume the pressure along both the vessels is the same and both follow linear flow pattern. What will be the amount of blood flow in A compared to B



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- a) 4 times
- b) 32 times
- c) 8 times
- d) 16 times

Question 14:

Which of the following statements is false about Cuthbertson's metabolic ebb and flow responses to trauma?

- a) The "ebb" phase is associated with a decline in body temperature and oxygen consumption.
- b) The "flow" phase occurs after resuscitation from a state of shock.
- c) The "ebb" phase is aimed at reducing posttraumatic energy depletion.

d) The "ebb" phase leads to activation of the innate immune system and induction of the hepatic acute-phase response.

Question 15:

Hering's nerve finally relays in _____.

- a)** Cortex
- b)** Midbrain
- c)** Pons
- d)** Medulla

Question 16:

Microcirculation consists of _____.

- a)** Capillaries
- b)** Arterioles
- c)** Venules
- d)** All of above

Question 17:

Mayer waves are due to _____.

- a)** Variation of arterial pressure during each systole.
- b)** Variation of arterial pressure due to reflex changes in nervous control of blood pressure
- c)** Variation of arterial pressure with changes in respiration.
- d)** Variation of arterial pressure with postural change.

Question 18:

Vasomotion refers to _____.

- a)** Venous drainage to RA by contraction of IVC.
- b)** Intermittent contraction of meta arterioles and precapillary sphincter.
- c)** Pulsatile action of arteries which help in lymphatic drainage.

d) Systolic pumping of blood by heart in to aorta.

Question 19:

Ejection fraction in resting healthy man is approximately____.

- a) 0.4
- b) 0.5
- c) 0.6
- d) 0.9

Question 20:

The relation between ventricular stroke volume and end-diastolic volume is explained by _____

- a) Frank starling law
- b) Laplace law
- c) Cushings law
- d) Poissulle's law

Question 21:

The best index of preload is_____.

- a) Atrial compliance
- b) Ventricular compliance
- c) left ventricular end diastolic volume
- d) right ventricular end diastolic volume

Answer Key

Question No.	Correct Option
1	c
2	a

3	a
4	b
5	c
6	c
7	c
8	a
9	b
10	c
11	b
12	b
13	c
14	d
15	d
16	d
17	b
18	b
19	c
20	a
21	c

Detailed Explanations

Solution to Question 1:

The most powerful activator of the sympathetic vasomotor system is CNS ischemic response.

CNS ischemic response:

When blood flow to the vasomotor center in lower brainstem (medulla) is severely decreased, the local concentration of CO₂ increases greatly and has a potent effect in stimulating the sympathetic vasomotor nervous control areas in the brain's medulla. Blood pressure may increase till 250 mmHg. The strong activation may cause some peripheral vessels to occlude.

It doesn't activate till arterial pressure drops to 60mmHg or less, hence it is principally a emergency mechanism.

Note: Baroreceptors are the most common activators, but not the most powerful.

Solution to Question 2:

During exercise, the organ experiencing the smallest decrease in blood flow is brain.

During exercise, the coronary and cerebral systems, are spared of vasoconstrictor effect because both these circulatory areas have poor vasoconstrictor innervation .

Solution to Question 3:

The given scenario describes orthopnea in congestive heart failure. It is due to reservoir function of pulmonary veins.

Orthopnea is dyspnea occurring in the recumbent position.

It results from redistribution of fluid from the splanchnic circulation and lower extremities (increased venous return) into the heart during recumbency. This increased venous return is not put out by the left ventricle and hence pools in the pulmonary veins that acts as capacitance vessels (reservoir function) adding to the already congested pulmonary vasculature resulting in breathing difficulty.

Solution to Question 4:

The interstitial fluid pressure is negative in the lungs.

The interstitial fluid pressure is slightly negative in the body except in the encapsulated organs.

It is positive in

- Rigid closed compartments- bone marrow, brain
- Encapsulated organs - kidney, skeletal muscle

This is due to the fact that in the kidneys, high-pressure vessels push the interstitial fluid against an unyielding fibrous capsule raising the pressure. The same principle applies to skeletal muscle surrounded by layers of fascia.

Solution to Question 5:

Blood flow per minute in the pulmonary circulation is equal to flow through the systemic circulation so as to provide continuous blood supply to the body.

If the flow rate is slower in the pulmonary circulation than systemic, then the left ventricle would soon empty completely. If the flow rate is faster in the pulmonary circulation then the left ventricle would be soon overloaded.

Please note that because some blood flows into the bronchial arteries, the actual flow through the pulmonary circulation is slightly lesser than the flow through the systemic circulation. However, this question aims to state that the blood flow through both pulmonary and systemic circulation is largely equal.

Solution to Question 6:

The shock due to a transfusion reaction is distributive shock.

During transfusion reaction, there is decreased systemic vascular resistance due to system-wide vasodilation mediated by release of histamine by mast cells in response to antigen bound IgE, leading to distributive shock.

Types of circulatory shock:

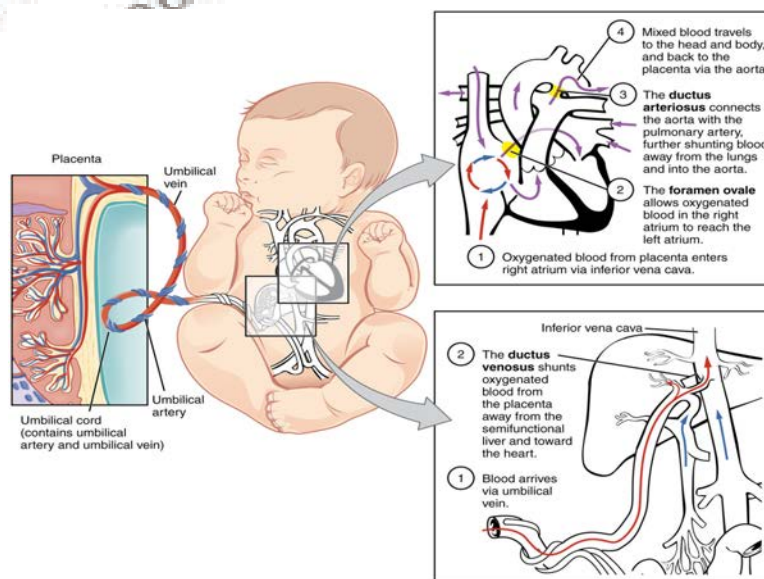
- Hypovolemic shock: due to decreased blood in vascular compartment. E.g., haemorrhage, dehydration.
- Cardiogenic shock: due to decreased capacity of heart to pump. E.g., any pathology of heart decreasing its ability to maintain cardiac output.
- Obstructive shock: due to conditions causing obstruction to blood flow in lungs and heart. E.g.: pulmonary embolism.
- Distributive/vasculogenic shock: excessive vasodilation leading to impaired organ perfusion. E.g.: septic shock, SIRS, transfusion reaction, Addisonian crisis, neurogenic shock.

Three stages of shock:

- Non-progressive: also called compensated shock, where normal compensatory mechanism eventually causes full recovery without help from outside therapy.
- Progressive: also called non-compensated, without therapy it gets steadily worse.
- Irreversible: even therapy is inadequate to save life.

Solution to Question 7:

During fetal life, blood vessel with maximum concentration of oxygen is umbilical vein.



Solution to Question 8:

Progressive hemorrhagic shock is characterized by the presence of blood clots in the small blood vessels.

The initiating cause of this blockage is sluggish blood flow in the microvessels. Tissue metabolites plus other deterioration products from the ischemic tissues causes minute clots in these vessels. Thus the microvasculature gets blocked by the sludged blood.

In progressive hemorrhagic shock, there is increased lysosomal activity, decreased active transport of sodium and tissue acidosis.

Solution to Question 9:

During rest, blood flow through skeletal muscle averages 3–4 mL/min/100 g of muscle.

The resting blood flow through various organs in a 63 kg adult man with a mean arterial pressure 90 mmHg and O₂ consumption 250 mL/min is given below.

Region	Blood flow (ml/min)
Liver	1500
Kidneys	1260
Brain	750
Skeletal muscle	840
Heart muscles	250

Solution to Question 10:

Septic shock is characterized by high cardiac output.

High cardiac output in septic shock is caused by

- arteriolar dilation in the infected tissues and
- high metabolic rate and vasodilation (resulting from bacterial toxin stimulating cellular metabolism and from high body temperature).

Solution to Question 11:

Thin-walled and delicate vessels like capillaries are less prone to rupture and this can be explained by Laplace law.

The principal reason for these vessels' relative invulnerability is their small diameter. The protective effect of small size, in this case, is an example of the law of Laplace.

Laplace law states that tension in the wall of a cylinder (T) is equal to the product of the transmural pressure (P) and the radius (r) divided by the wall thickness (w):

$$T = Pr/w$$

As the tissue pressure is low, P is equated to the pressure inside the blood vessel.

W can be ignored as well as capillaries are thin walled. Hence:

$$P = T/r$$

In a thin-walled viscus, $P = T$ divided by the two principal radii of curvature of the viscus: $P = T (1/r_1 + 1/r_2)$

In a cylinder such as a blood vessel, one radius is infinite, so $P = T / r$

Hence, smaller the radius, lower is the tension necessary in the wall to balance the distending pressure.

Solution to Question 12:

The relationship between the flow in a long narrow tube, the viscosity of the fluid, and the radius of the tube is expressed mathematically by Poiseuille's law.

Poiseuille-Hagen formula

$$F = (P_A - P_B) \times \left(\frac{\pi}{8}\right) \times \left(\frac{1}{\eta}\right) \times \left(\frac{r^4}{L}\right)$$

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where F = flow

$P_A - P_B$ = pressure difference between two ends of the tube

η = viscosity

r = radius of tube

L = length of tube

Solution to Question 13:

The relationship between the flow in a long narrow tube, the viscosity of the fluid, and the radius of the tube is expressed mathematically in the formula called as Poiseuille's law :

Poiseuille's law

$$Q = \frac{\Delta P \pi r^4}{8 \eta l}$$

Q - Volume flux	r - Pipe or vessel radius
ΔP - Change in pressure	η - Viscosity
	l - Pipe or vessel length

Blood flow (F) inside the vessel depends on:

- Pressure difference: directly proportional
- Viscosity: inversely proportional
- Radius: directly proportional to the fourth power
- Length of the vessel: inversely proportional

Here, the radius is doubled so the flow increases by 2 to the power of 4 = 16.

the length of the vessel is also doubled, so the flow decreases by half ($\div 2$) = 8

Solution to Question 14:

The "flow" phase (not "ebb") leads to activation of the innate immune system and induction of the hepatic acute-phase response.

Cuthbertson's ebb and flow response to trauma:

The "ebb" phase lasts for a brief duration and is associated with a decline in body temperature and oxygen consumption, aimed at reducing post traumatic energy depletion.

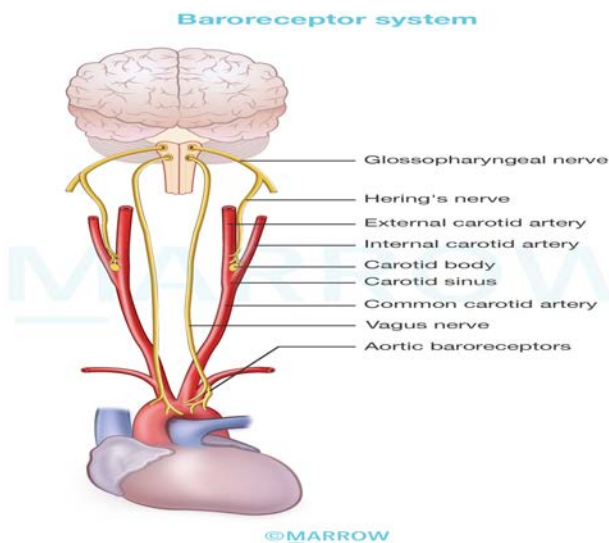
The "flow" phase occurs after resuscitation from a state of shock, which leads to an increased metabolic turnover, activation of the innate immune system and induction of the hepatic acute-phase response. This hypercatabolic condition leads to a significantly increased oxygen consumption and energy expenditure.

Solution to Question 15:

The Hering's nerve terminates in the medulla (Tractus solitarius).

Baroreceptor reflexes

- Carotid baroreceptors → Hering's nerve → the glossopharyngeal nerve → Tractus solitarius in medulla.
- Aortic baroreceptors → Vagus nerve → tractus solitarius in medulla



Solution to Question 16:

Microcirculation consists of capillaries, arterioles, venules and terminal lymphatics.

Solution to Question 17:

Mayer waves are due to the variation of arterial pressure due to reflex changes in nervous control of blood pressure.

The most important mechanisms are as follows:

I: Baroreceptor and chemoreceptor reflex:

High pressure excites the baroreceptors --&inhibits sympathetic nervous system and lowers the pressure --&The decreased pressure, in turn, reduces the baroreceptor stimulation --&elevating the pressure to a high value (the response is not instantaneous, it is delayed until a few seconds later). This high pressure then initiates another cycle, and the oscillation continues.

II: Oscillation of the CNS Ischemic Response:

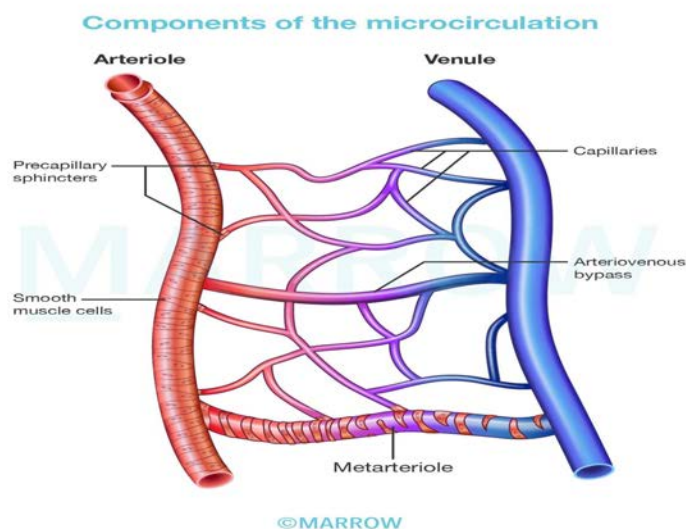
The cerebrospinal fluid pressure is raised to 160 mm Hg --> causes compression of cerebral blood vessels --> cerebral ischemia--> CNS ischemic pressure response raises the arterial pressure up to 200mm Hg -->the brain ischemia is relieved and the sympathetic nervous system becomes inactive -->the arterial pressure falls rapidly back to a much lower value, causing brain ischemia once again -->initiates another rise in pressure. This chain of events repeats itself cyclically as long as the cerebrospinal fluid pressure remains elevated.

Solution to Question 18:

Vasomotion means intermittent contraction of meta arterioles and precapillary sphincter.

Blood usually does not flow continuously through the capillaries. Instead, it flows intermittently, turning on and off every few seconds or minutes. This is vasomotion, which is responsible for flow of blood in capillaries .

The main regulatory factor for vasomotion is tissue oxygen concentration.

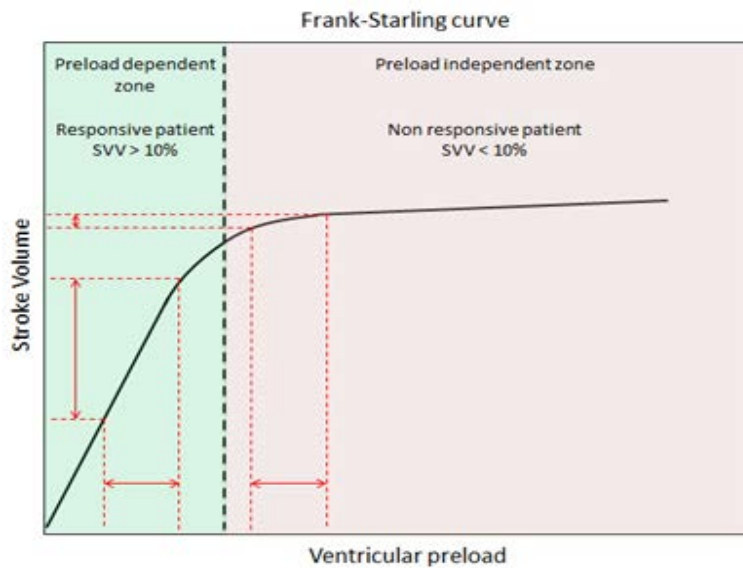


Solution to Question 19:

Ejection fraction in resting healthy man is approximately 0.6.

The fraction of the end-diastolic volume that is ejected is called the ejection fraction, usually equal to about 60 %.

Solution to Question 20:



The relation between ventricular stroke volume and end-diastolic volume is explained by Frank Starling's law.

Length–tension relationship in cardiac muscle is similar to that in skeletal muscle. when the muscle is stretched, the developed tension increases to a maximum and then declines as stretch becomes more extreme. Starling stated that “energy of contraction is proportional to the initial length of the cardiac muscle fibre”.

When cardiac output is regulated by changes in cardiac fibre length, this is referred to as heterometric regulation.

Conversely, regulation due to changes in contractility independent of length is called homometric regulation.

Solution to Question 21:

The best index of preload is left ventricular end diastolic volume.

Preload - load that the left ventricle carries before the systole.

Left ventricular end diastolic volume - blood remaining at the end of diastole, which determines stretch of the ventricle and thus force of contraction as explained by frank starlings law.

Cardiac Cycle and Cardiac Output

Question 1:

The basal cardiac output in adults is around _____

- a) 10 litres
- b) 5.5 litres
- c) 7.5 litres
- d) 6.5 litres

Question 2:

The cardiac output can be determined by all except:

- a) Fick's principle
- b) Thermodilution
- c) Echocardiography
- d) V/Q ratio

Question 3:

Cardiac output in litre per minute divided by heart rate gives _____

- a) Mean arterial pressure
- b) Cardiac index
- c) Cardiac efficiency
- d) Stroke volume

Question 4:

Ejection fraction is

- a) SV/ESV
- b) ESV/EDV
- c) EDV/SV

d) SV/EDV

Question 5:

A patient with cardiac output 5L/min and body surface area 1.7m². What will be the cardiac index (L/min/m²)?

- a) 3
- b) 4
- c) 5
- d) 6

Question 6:

Volume determining preload is

- a) End systolic volume
- b) Volume of blood in aorta
- c) Ventricular ejection volume
- d) End diastolic volume

Question 7:

Starling's law implies:

- a) Increased heart rate leads to increased cardiac output
- b) Increased discharge leads to increased cardiac output
- c) Increased venous return leads to increased cardiac output
- d) Increased blood pressure leads to increased cardiac output

Question 8:

Venous return to the heart during quiet standing is facilitated by all of the following factors except _____.

- a) Calf muscle contraction during standing
- b) Valves in perforators.

- c) Deep fascia.
- d) Increased arterial pressure due to gravitation.

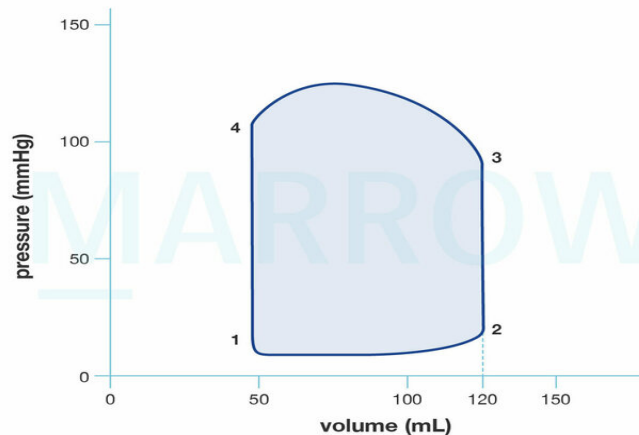
Question 9:

When a person changes position from standing to supine, _____

- a) Heart rate increases
- b) Venous return to heart rises immediately
- c) Cerebral blood flow increases
- d) Decrease in blood flow to the lung apices

Question 10:

A 30-year-old woman has a resting heart rate of 70-beats/min. Her pressure-volume diagram of the left ventricle is given below. What is her cardiac output (in mL/min)? (Point 2 corresponds to 120 ml)



- a) 2000
- b) 3000
- c) 4900
- d) 7000

Question 11:

A sister donated one of her kidneys to her brother. What would be the effect on her cardiac output?

- a) Decrease
- b) Increase
- c) No effect
- d) Variable effect

Question 12:

The duration of atrial diastole is _____

- a) 0.1 sec
- b) 0.3 sec
- c) 0.7 sec
- d) 0.5 sec

Question 13:

The indication of the end of the isovolumetric relaxation phase is when _____

- a) Ventricular pressure raise above atrial pressure
- b) Ventricular pressure falls below atrial pressure
- c) Ventricular pressure falls below aortic pressure
- d) ventricular pressure rises above aortic pressure

Question 14:

During the cardiac cycle, the opening of the aortic valve takes place at the _____

- a) End of isovolumterc relaxation.
- b) Begining of isovolumetric relaxation.
- c) Begining of isovolumetric contraction.
- d) End of isovolumetric contraction.

Question 15:

The effect of increased heart rate on cardiac cycle is that it_____.

- a) Decrease the duration of systole and diastole.
- b) Increase the duration of systole and diastole.
- c) Decrease the duration of systole more than diastole.
- d) Decrease the duration of diastole more than systole.

Question 16:

Electromechanical systole is the time interval between _____

- a) Q wave and T wave
- b) S wave and T wave
- c) R wave and first heart sound
- d) Q wave and second heart sound

Question 17:

The pressure-volume curve is shifted to the left in

- a) Aortic regurgitation
- b) Mitral regurgitation
- c) Aortic stenosis
- d) Mitral stenosis

Question 18:

Dicrotic notch is caused by _____

- a) Closing of mitral valve
- b) Closing of aortic valve
- c) Opening of mitral valve
- d) Opening of aortic valve

Question 19:

First heart sound and second heart sound are respectively due to _____

- a) Opening of semilunar and atrioventricular valves
- b) Closure of semilunar and atrioventricular valves
- c) Closure of atrioventricular and semilunar valves
- d) Opening of atrioventricular and semilunar valves

Question 20:

The fourth heart sound is caused by:

- a) Closure of aortic and pulmonary valves
- b) Vibrations in the ventricular wall during systole
- c) Ventricular filling
- d) Closure of mitral and tricuspid valves

Question 21:

“a” wave on JVP is for

- a) Atrial contraction
- b) Atrial filling
- c) Ventricular relaxation
- d) Ventricular filling

Question 22:

'c' wave in JVP is seen in

- a) Iso-volumetric contraction
- b) Slow filling at end of diastole
- c) End of systole
- d) Start of diastole

Question 23:

The x descent in jugular venous pressure corresponds to _____ phase of the cardiac cycle.

- a) Atrial systole

- b) Ventricular systole
- c) Ventricular diastole
- d) Joint diastole

Question 24:

C wave of JVP represents:

- a) Tricuspid bulge into right atrium during isovolumetric ventricular contraction
- b) Contraction of right atrium
- c) Passive filling of right atrium
- d) Passive emptying of right atrium

Question 25:

Reflex responsible for tachycardia during right atrial distension is:

- a) J reflex
- b) Cushing reflex
- c) Bezold jarisch reflex
- d) Bainbridge reflex

Question 26:

Arrange the following events in the sequence in which they occur, after 'p wave' in ECG:

- a) 1-2-3-4
- b) 1-2-4-3
- c) 3-1-2-4
- d) 1-3-2-4

Question 27:

Cardiac output is increased in:

- a) Pregnancy
- b) Sleep

- c) Sitting
- d) Standing

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	d
5	a
6	d
7	c
8	d
9	b
10	c
11	a
12	c
13	b
14	d
15	d
16	d
17	d
18	b
19	c
20	c
21	a
22	a
23	b
24	a
25	d
26	b
27	a

Detailed Explanations

Solution to Question 1:

The basal cardiac output in adults is around 5.5 litres.

Cardiac output is given by, $\text{Cardiac Output} = \text{Heart Rate} \times \text{Stroke Volume}$.

In a resting, supine man cardiac output averages about 5.0 L/min ($70 \text{ mL} \times 72 \text{ beats/min}$).

Solution to Question 2:

V/Q ratio not used to determine cardiac output.

Measurement of cardiac output:

- Fick principle
- Dye dilution/thermodilution
- Doppler plus echocardiograph
- MRI

Solution to Question 3:

Cardiac output in litre per minute divided by heart rate gives stroke volume.

Cardiac output is given by, $\text{Cardiac Output} = \text{Heart Rate} \times \text{Stroke Volume}$.

The stroke volume is the amount of blood pumped out of each ventricle per beat, is about 70 mL from each ventricle in a resting man in the supine position.

Solution to Question 4:

Ejection fraction is SV/EDV .

The amount of blood ejected by each ventricle per stroke at rest is 70–90 mL. The end-diastolic ventricular volume is about 130 mL. The ejection fraction is the percentage of the end-diastolic ventricular volume that is ejected with each stroke, is about 65%.

Solution to Question 5:

Cardiac index is the cardiac output per minute per square meter of body surface.

$\text{Cardiac Index} = \text{Cardiac output} / \text{body surface area}$.

Therefore Cardiac index = $5\text{L/min} / 1.7\text{m}^2 = 2.94\text{ L}$

Solution to Question 6:

Volume determining preload is End diastolic volume.

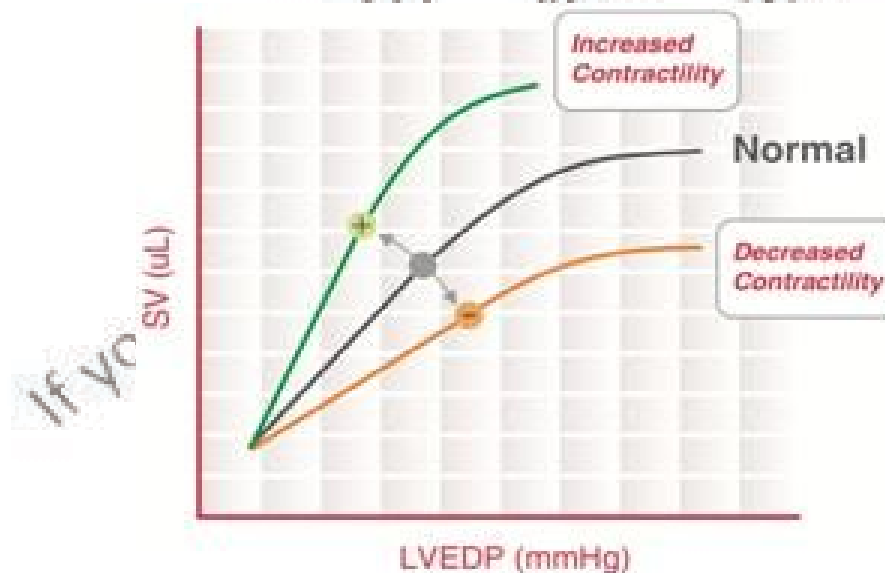
Preload is the initial stretch on muscle at the end of diastole. Preload is equal to End diastolic volume (EDV) and is determined by venous return to heart.

Solution to Question 7:

Starling's law implies increased venous return leads to increased cardiac output.

Frank-Starling law: The energy of contraction is proportional to the initial length of the cardiac muscle fiber.

The length of the muscle fibers is proportional to the end-diastolic volume, which is determined by venous return.



Solution to Question 8:

Venous return to the heart during quiet standing is not facilitated by increased arterial pressure.

Factors that facilitate venous return to the heart:

- Calf muscles contract and pump blood through veins.
- Valves don't allow the retrograde flow of blood through veins.
- Deep fascia by forming a supporting structure to veins.

Note: The pulsations of artery helps in increasing lymph flow .

Solution to Question 9:

When a person changes position from standing to supine, venous return to heart rises immediately. On standing, Gravitational pull causes venous pooling of blood in the legs. On supine position, this gravitational pull equalizes over body, causing the pooled blood to return back to heart.

Other options:

Option A: Heart rate decreases because of increased cardiac output and increased BP via baroreceptor reflex.

Option C: Cerebral blood flow will remain constant due to autoregulation.

Option D: Blood flow to the apex of lung will be high due to high cardiac output.

Solution to Question 10:

In this case, the patient has a heart rate of 70-beats/min and the stroke volume can be determined from the given pressure-volume diagram.

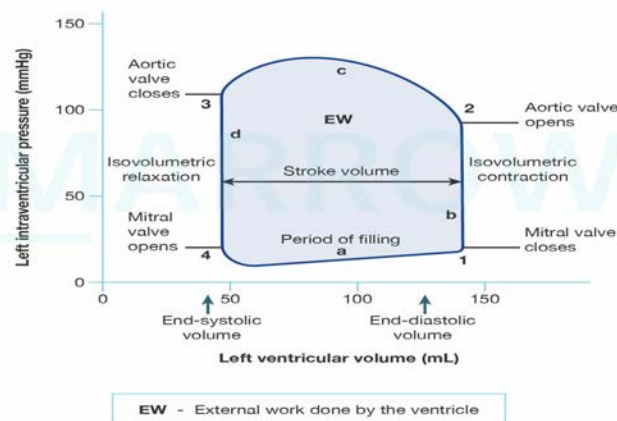
Point 2 corresponds to the EDV(End diastolic volume) = 120 mL

Point 1 corresponds to the ESV(End systolic volume) = 50 mL

Thus, stroke volume (SV) = EDV – ESV = 120 – 50 = 70 mL

Therefore, Cardiac output = HR X SV = 70 X 70 = 4900 mL/min.

Pressure-volume changes in cardiac cycle



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Solution to Question 11:

Surgical removal of kidney leads to decrease in cardiac output.

Amputation of a limb or surgical removal of organs like kidney removes a parallel circuit thus increasing total peripheral vascular resistance. As the resistance increases, afterload increases hence decreasing stroke volume.

$CO = HR \times SV$, therefore decrease in stroke volume leads to decrease in cardiac output.

Solution to Question 12:

The duration of atrial diastole is 0.7 sec.

Cardiac Cycle Events	Systole(s)	Diastole(s)	Total (s)
Ventricles	0.3	0.5	0.8
Atria	0.1	0.7	0.8

Solution to Question 13:

The indication of the end of the isovolumetric relaxation phase is when ventricular pressure falls below atrial pressure.

Events of the Cardiac Cycle

Diastole

- Protodiastole

Before the semilunar valve closes, even though ventricular muscle is relaxed, blood flow into great vessels is due to momentum of blood.

- Iso-volumetric Relaxation

The ventricular muscle continues to relax, even though the ventricular volume does not change, with all valves closed.

- Ventricular Filling

Isovolumetric relaxation ends when the ventricular pressure falls below the atrial pressure and the AV valves open, permitting the ventricles to fill.

- Atrial Systole

During the last third of diastole, the atria contract and give additional thrust to the inflow of blood into the ventricles. It correlates with 'a' wave in JVP.

Systole

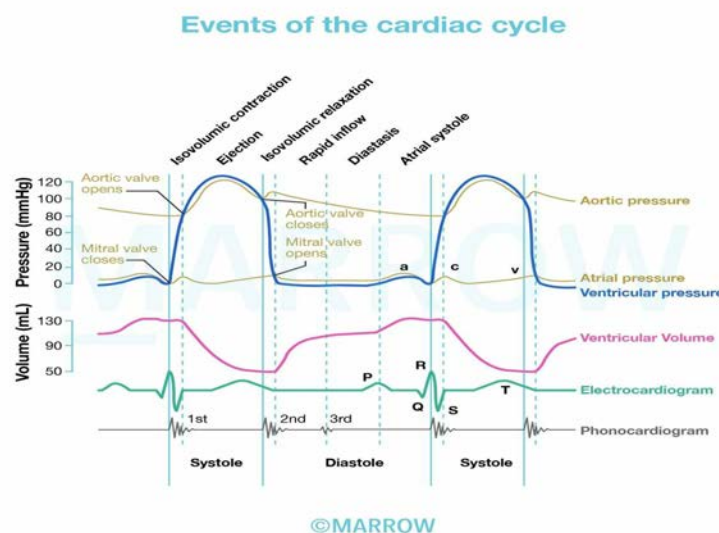
- Isovolumetric Contraction

The ventricular muscle continues to contract, even though the ventricular volume does not change, with all valves closed.

It correlates with the 'c' wave of JVP.

- Ventricular Ejection

When the ventricular pressures rise above aortic and pulmonary pressures the valves open, and ventricular ejection begins. Peak pressures in the left and right ventricles are about 120 and 25 mmHg, respectively.



Solution to Question 14:

Opening of the aortic valve takes place at the end of the isovolumetric contraction.

When the ventricular pressures rise above aortic and pulmonary pressures the semilunar valves (aortic and pulmonary valves) open, and ventricular ejection begins.

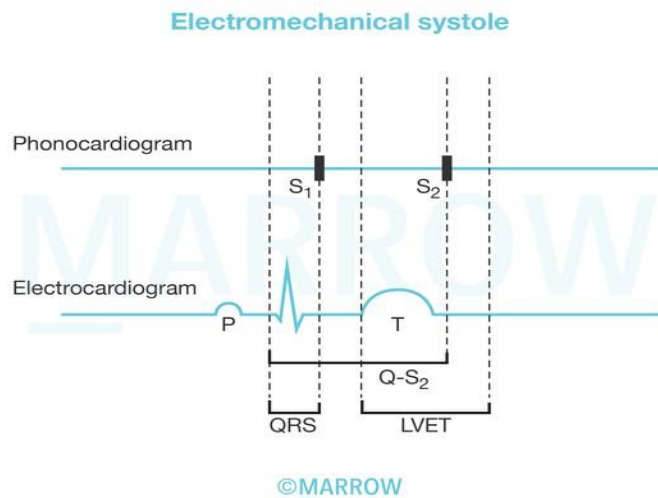
Solution to Question 15:

The effect of increased heart rate on cardiac cycle is that it decreases the duration of diastole more than systole.

The duration of systole is much more fixed than that of diastole, so when the heart rate is increased, diastole is shortened to a much greater degree. Coronary blood flow and ventricular filling occur in diastole.

Solution to Question 16:

Electromechanical systole is the time interval between Q wave and second heart sound.
To record this both ECG and phonocardiography are required.



Solution to Question 17:

The pressure-volume curve is shifted to the left in Mitral stenosis.

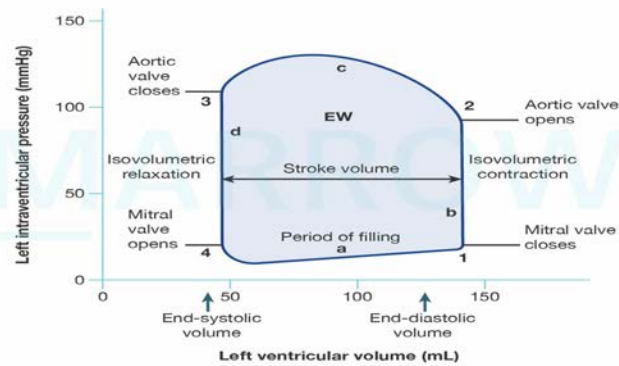
The PV loop is shifted to the right when there is volume overload (increased EDV).

The PV loop is shifted to the left in case of increased contractility or decreased compliance (decreased EDV).

Effect of Valvular Diseases on PV loop:

- Mitral stenosis: Left and down.
- Mitral regurgitation: Width increased and height decreased.
- Aortic stenosis: Tall with steep slope.
- Aortic regurgitation: Right and tall.

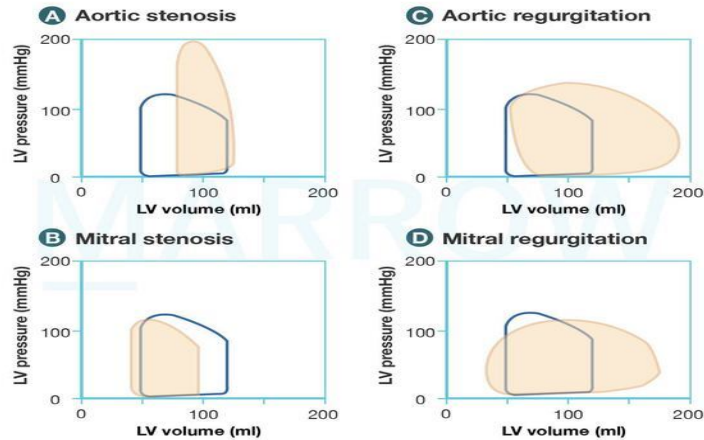
Pressure-volume changes in cardiac cycle



EW - External work done by the ventricle

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Pressure-Volume loops

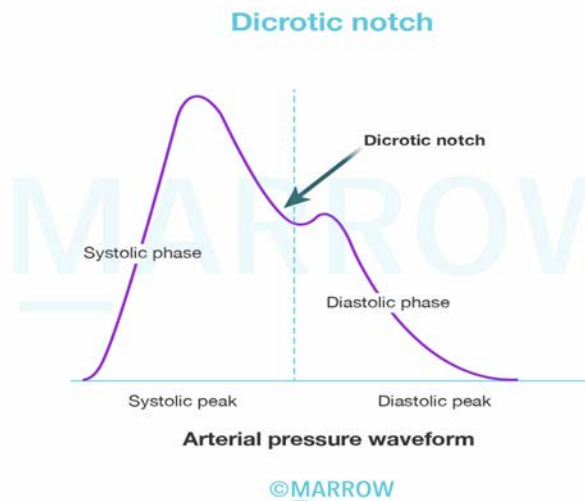


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Solution to Question 18:

Dicrotic notch is caused by closing of aortic valve.

The dicrotic notch, a small oscillation in the falling phase of the pulse wave caused by vibrations set up when the aortic valve shuts.



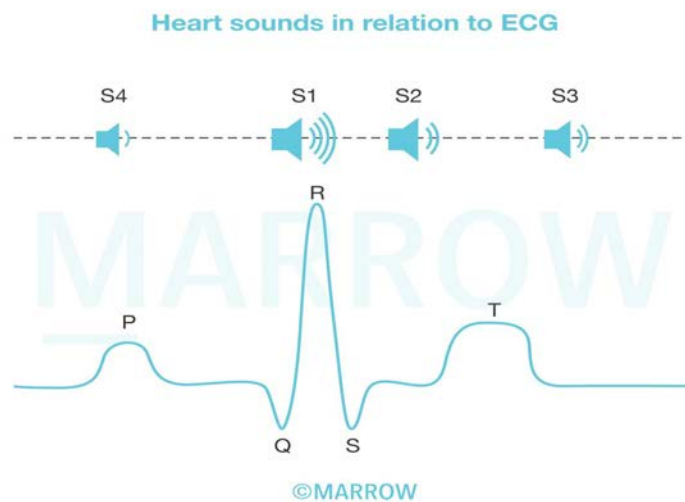
Solution to Question 19:

First heart sound and second heart sound are respectively due to closure of atrioventricular (AV) and semilunar valves.

Solution to Question 20:

The fourth heart sound is caused by ventricular filling.

A fourth sound is heard immediately before the first sound when atrial pressure is high or the ventricle is stiff in conditions such as ventricular hypertrophy. It is due to ventricular filling and is rarely heard in normal adults.

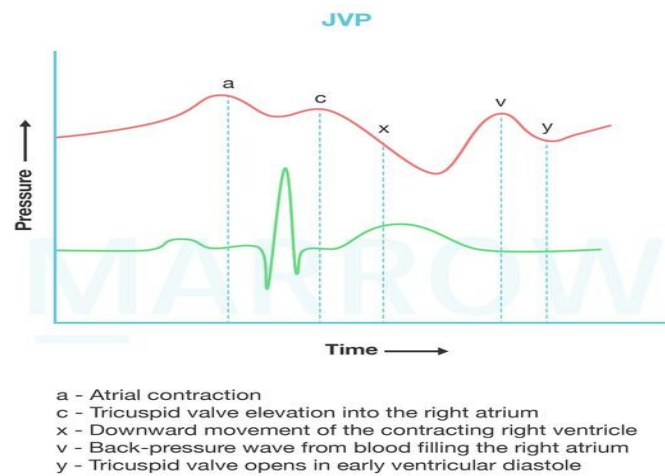


Solution to Question 21:

"a" wave in JVP represents atrial contraction.

Waves of JVP :

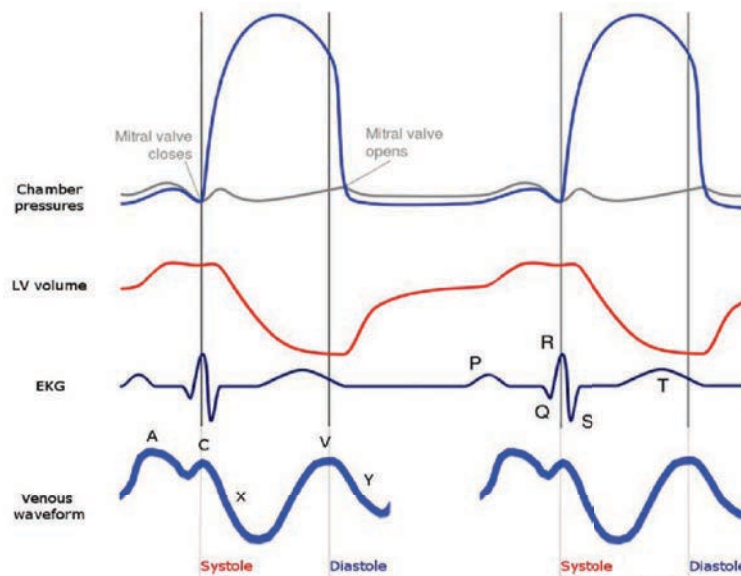
- a- atrial systole pushing the blood to the ventricles.
- c- isovolumetric contraction of ventricles causing the tricuspid valve to bulge.
- x'- fall in atrial pressure during ventricular systole due to pull of valve leaflets.
- v- venous filling which creates an increase in atrial pressure.
- y- passive filling of ventricles by a passive flow of the blood from the right atrium.



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Solution to Question 22:

'c' waves are produced by bulging of the tricuspid valve into the right atrium during the Iso-volumetric contraction of the right ventricle.

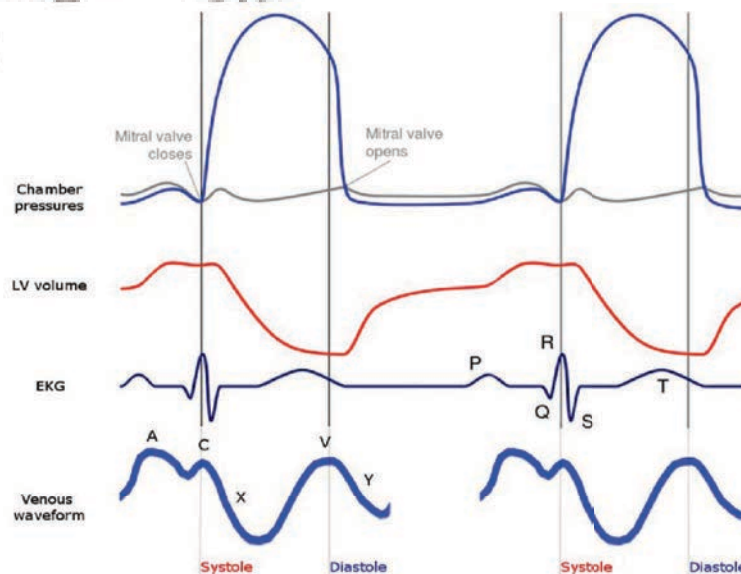


Solution to Question 23:

The x descent in jugular venous pressure corresponds to the ventricular systole of the cardiac cycle.

The jugular venous pulse

- The waves in JVP reflect the pressure changes in the right atrium.
- The internal jugular vein is preferred because the external jugular vein is valved and not directly in line with the superior vena cava and right atrium.



- a wave: Corresponds to right atrial contraction.
- c wave: Corresponds to the bulging of the tricuspid valve into the right atrium during isovolumetric ventricular contraction.

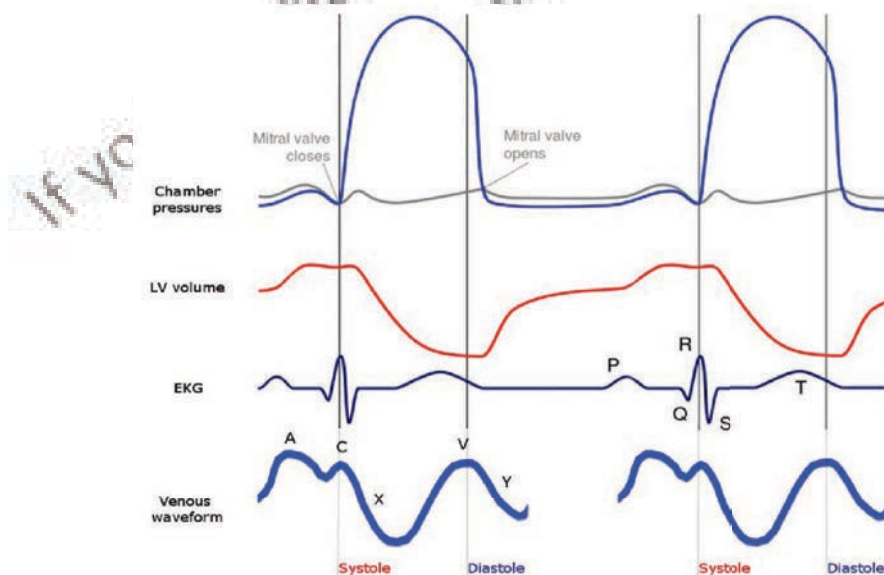
- x descent: Corresponds to the atrial relaxation as the ventricular systole pulls the tricuspid valve during ejection.
- v wave: Corresponds to the venous filling of the right atrium against a closed tricuspid valve.
- y descent: Corresponds to fall in right atrial pressure during passive emptying of the right atrium into the right ventricle.

Solution to Question 24:

'C' wave in JVP represents tricuspid bulge into right atrium during isovolumetric ventricular contraction.

Characteristics of JVP :

- a wave corresponds to atrial systole which causes regurgitation of some blood into the great veins.
- c wave occurs due to the increase in atrial pressure caused by bulging of the tricuspid valve into the right atrium during isovolumetric ventricular contraction.
- x descent occurs due to the fall in atrial pressure during ventricular systole due to pulling of valve leaflets.
- v wave is seen at the end of ventricular contraction and occurs due to the slow venous filling which causes an increase in atrial pressure.
- y descent occurs due to the passive emptying of the right atrium into the right ventricle.



Solution to Question 25:

Reflex responsible for tachycardia during right atrial distension is Bainbridge reflex.

Atrial Reflex Control of Heart Rate: The Bainbridge Reflex

- The stretch receptors of the atria that elicit the Bainbridge reflex transmit their afferent signals through the vagus nerves to the medulla of the brain.
- Then efferent signals are transmitted back through vagal and sympathetic nerves to increase heart rate and strength of heart contraction.

Bezold-Jarisch Reflex:

- Activation of chemosensitive vagal C fibres in the cardiopulmonary region (e.g. Juxta capillary region of alveoli, ventricles, atria, great veins, and pulmonary artery) causes profound bradycardia, hypotension and a brief period of apnea followed by rapid shallow breathing.

Cushing Reflex:

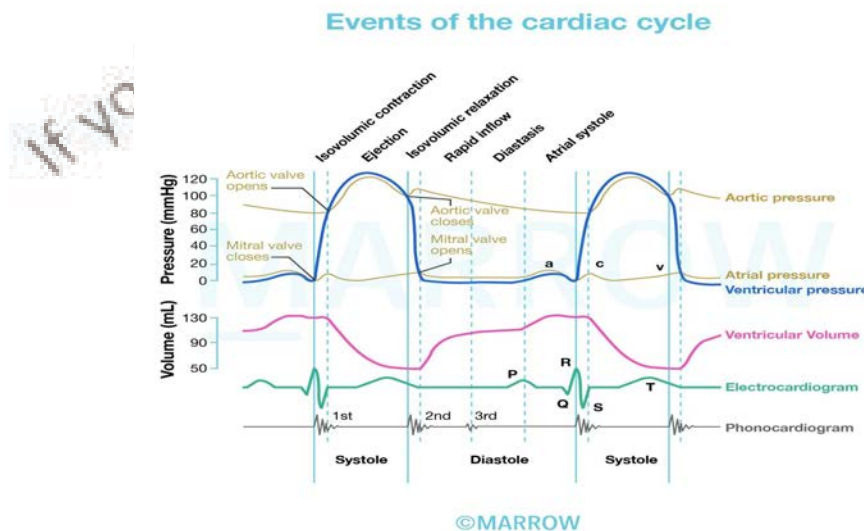
- When ICP is increased, the blood supply to RVLM neurons is compromised.
- The local hypoxia and hypercapnia causes an increase in discharge from the neurons, producing hypertension, bradycardia and irregular respiration.
- The resultant rise in systemic arterial pressure tends to restore the blood flow to the medulla.

Solution to Question 26:

The correct sequence of events that occur after the p-wave is 1-2-4-3, ie:-

a wave in JVP → First heart sound → T wave in ECG → Rapid filling of ventricles.

The sequence of the events in this question can be better understood with the following diagram:



- The p wave in the ECG represents atrial depolarization, which causes contraction of the atria, represented by the 'a' wave in the JVP. Atrial contraction causes final filling of the ventricles (fourth phase of ventricular diastole)

- Atrial contraction is followed by the closure of the mitral valve, which causes the first-heart sound.
- This is followed by the isovolumetric contraction of the ventricle, which causes c wave in JVP, due to bulging of the tricuspid valve. This is followed by the rapid ejection phase of the ventricle.
- The T wave of ECG occurs during ventricular repolarization, which is followed by the opening of the mitral valve, followed by rapid filling phase of the ventricle (second phase of ventricular diastole).

Note: Passive rapid filling occurs as the mitral valve opens just before atrial contraction occurs.

Solution to Question 27:

Cardiac output is increased in pregnancy.

Effect of various conditions on cardiac output:

No change	SleepModerate changes in environmental temperature
Increase	Anxiety and excitementEatingExerciseHigh environmental temperaturePregnancyEpinephrine
Decrease	Sitting or standing from lying positionRapid arrhythmiasHeart disease

Electrophysiology of the Heart

Question 1:

The SA Node acts as the pacemaker of the heart because, it ____.

- a) Generates impulse spontaneously
- b) Has rich sympathetic innervation
- c) Generates impulses at highest rate
- d) Has poor cholinergic innervation

Question 2:

In which of the following the conduction velocity is least?

- a) AV node
- b) Ventricles
- c) Bundle of His
- d) Purkinje fibers

Question 3:

The velocity of conduction of impulse is maximum in ____.

- a) Atrial fibers
- b) SA node
- c) AV node
- d) Purkinje fibers

Question 4:

The AV nodal delay is ____ (seconds).

- a) 0.2
- b) 0.1
- c) 0.5

d) 0.01

Question 5:

If the SA node discharges at 0.00 sec, when will the action potential normally arrive at the epicardial surface at the base of the left ventricle?

- a) 0.16 sec
- b) 0.18 sec
- c) 0.22 sec
- d) 0.12 sec

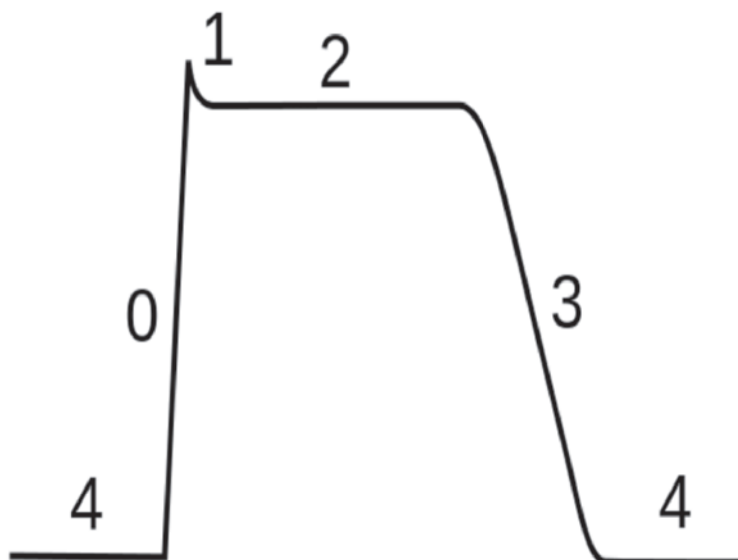
Question 6:

The order of depolarization after stimulation of Purkinje fibers is:

- a) Septum→ Endocardium→ Epicardium
- b) Endocardium→ Epicardium→ Septum
- c) Epicardium→ Endocardium→ Septum
- d) Septum→ Epicardium→ Endocardium

Question 7:

Phases of the ventricular muscle action potential are shown below. At which phase of the below image, the action potential is most dependant on sodium permeability?



- a) Phase 3
- b) Phase 2
- c) Phase 0
- d) Phase 1

Question 8:

Plateau phase of ventricular muscle is due to opening of :

- a) Na channel
- b) Ca channel
- c) K channel
- d) Cl channel

Question 9:

Repolarization of cardiac muscle proceeds from:

- a) Left to right
- b) Endocardium to epicardium
- c) Epicardium to endocardium
- d) Base to the apex

Question 10:

Phase 0 of pacemaker potential is initiated by:

- a) Opening of voltage-gated Na⁺ channel
- b) Opening of voltage gated Ca²⁺ channel
- c) Inactivation of voltage gated Ca²⁺ channel
- d) Inactivation of voltage gated Na⁺ channel

Question 11:

You are researching the benefits of vagus nerve stimulation as a potential treatment for epilepsy. Which of the following parameters can be increased due to vagal stimulation?

- a) Heart rate
- b) R-R interval in ECG
- c) Cardiac output
- d) Force of contraction

Question 12:

In recording Lead I, the right arm is negative electrode and the positive electrode is

- a) Left arm
- b) Left leg
- c) Right leg
- d) Left arm + left leg

Question 13:

Einthoven law states that

- a) $II = I + III$
- b) $I = II + III$
- c) $III = I + II$
- d) Any of the above

Question 14:

In Einthoven's triangle, what is the value of lead III when lead I = 1mV and lead II = 2mV ?

- a) 1
- b) 2
- c) 3
- d) 4

Question 15:

In a standard ECG, an augmented limb lead measures the electrical potential difference between:

- a) 2 limbs
- b) 1 limb and 2 other limbs
- c) One limb and neutral
- d) 2 limbs and 2 other limbs

Question 16:

Which part of the ECG corresponds to ventricular repolarization?

- a) The P wave
- b) The QRS duration
- c) The T wave
- d) The U wave

Question 17:

The PR interval indicates ____.

- a) Atrial depolarisation
- b) Ventricular repolarization
- c) AV node conduction time
- d) Both A and C

Question 18:

The P wave is absent in ____.

- a) Atrial fibrillation
- b) Atrial flutter
- c) CCF
- d) WPW

Question 19:

The normal mean QRS axis lies between _____

- a) $+30^{\circ}$ and $+100^{\circ}$.

- b) -30° and -100° .
- c) -30° and $+110^\circ$
- d) -30° and -110°

Question 20:

The sympathetic innervation to AV node is by_____.

- a) Right Thoracic ganglion
- b) Left thoracic ganglion
- c) Right stellate ganglion
- d) Left stellate ganglion

Question 21:

You are following up with a patient on a sympathomimetic drug. Which of the following effects is unlikely to be due to sympathetic stimulation?

- a) Increased conduction velocity
- b) Increased refractory period
- c) Increased heart rate
- d) Increased contractility of heart

Question 22:

The AH interval on His Bundle Electrocardiogram (HBE) corresponds to the conduction of _____ on ECG.

- a) Through His –Purkinje system
- b) From AV node to bundle of His
- c) Purkinje to ventricular fibers
- d) Through ventricular fibers

Answer Key

Question No.	Correct Option
1	c
2	a
3	d
4	b
5	c
6	a
7	c
8	b
9	c
10	b
11	b
12	a
13	a
14	a
15	b
16	c
17	d
18	a
19	c
20	d
21	b
22	b

Detailed Explanations

Solution to Question 1:

The SA node acts as the pacemaker of the heart as it generates impulses at the highest rate (60-100/min), compared to AV node (40-60/min) and Purkinje fibres (15-40). The impulses generated by SA node are conducted through AV node and Purkinje fibres repeatedly that it will prevent AV node or Purkinje fibres reach their own thresholds for self-excitation.

Solution to Question 2:

The conduction velocity is least in AV node.

The slow conduction in the AV node is caused mainly by diminished numbers of gap junctions between successive cells in the conducting pathways.

Solution to Question 3:

The velocity of conduction of impulse is maximum in Purkinje fibers.

Purkinje fibers have the maximum number of gap junctions hence the maximum conduction velocity (4m/s).

Solution to Question 4:

The AV nodal delay is 0.1 second.

The slow conduction is caused mainly by diminished numbers of gap junctions between successive cells in the conducting pathways. It ensures that the atria have ejected their blood into the ventricles first before the ventricles contract.

Solution to Question 5:

The duration of action potential to arrive at the epicardial surface at the base of the left ventricle is 0.22sec.

Impulse initiated from SA node at 0.00s

↓

Impulse reaches AV node at 0.03s

↓

Delay of impulse at AV node by 0.13s resulting in a total time of 0.16s (0.03+0.13) to reach the beginning of Purkinje fibres

↓

Impulse reaches base of Purkinje fibres by 0.03s resulting in a total of time of 0.19s (0.16+0.03) to reach the endocardial surface of the base of the ventricle

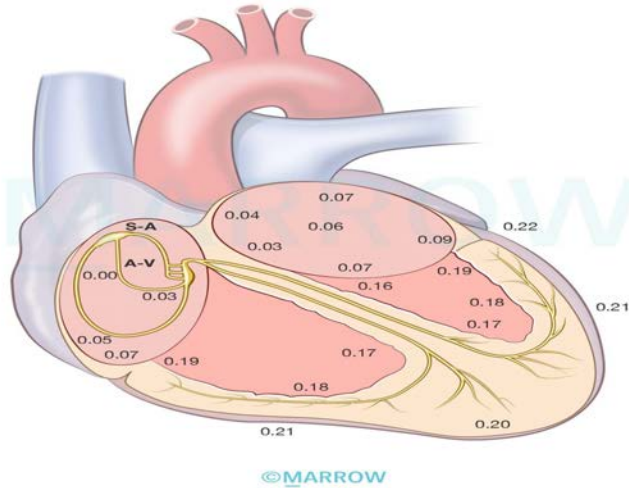
↓

Impulse travels from endocardial to epicardial surface in 0.03s (due to arrangement of myocardial fibres)

↓

Hence, a total time of 0.22s (0.19+0.03) is taken for the impulse to travel from SA node to epicardial surface of the base of ventricles

Transmission of cardiac impulse



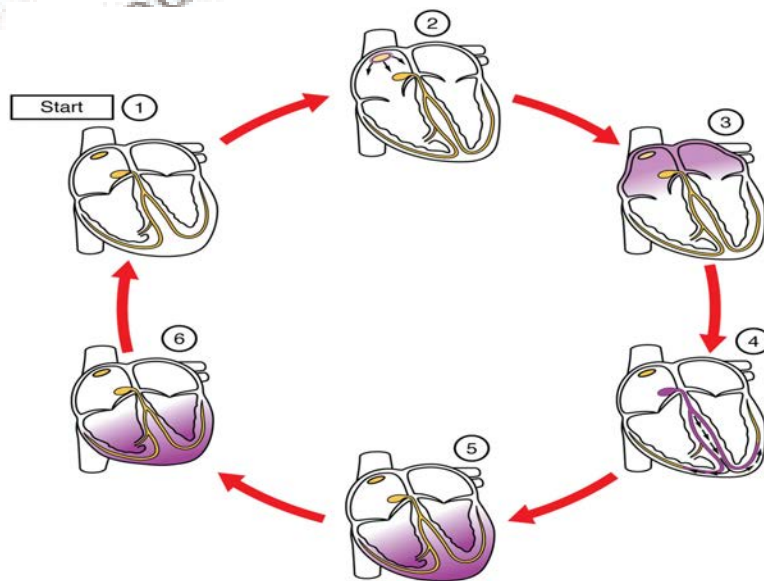
Solution to Question 6:

The order of depolarization is from septum → endocardium → epicardium.

The spread of Cardiac Excitation:

Atrial depolarization is initiated from SA node spreads radially through atria, then converges on the AV node. A delay of about 0.1s (AV nodal delay) occurs before excitation spreads to the ventricles.

Ventricular depolarization starts at the left side of the interventricular septum and moves first to the right across the midportion of the septum. It moves along the ventricular walls to the AV groove, proceeding from the endocardial to the epicardial surface.

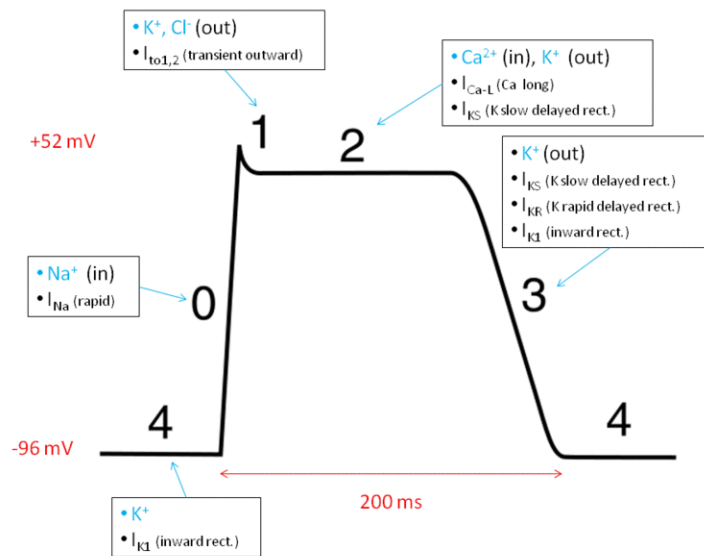


Solution to Question 7:

The action potential is most dependant on sodium permeability in phase 0.

Action potential of myocardium:

- Phase 0: Depolarisation: Influx of Na^+ into myocyte.
- Phase 1: Early repolarization: Efflux of K^+ from myocyte.
- Phase 2: Plateau: Ca^{2+} enters myocardial cells via voltage regulated L-type Ca^{2+} channels.
- Phase 3: Repolarisation: Efflux of K^+ .
- Phase 4: Restoration of RMP: Restoration of Na^+ and K^+ by Na-K ATPase.

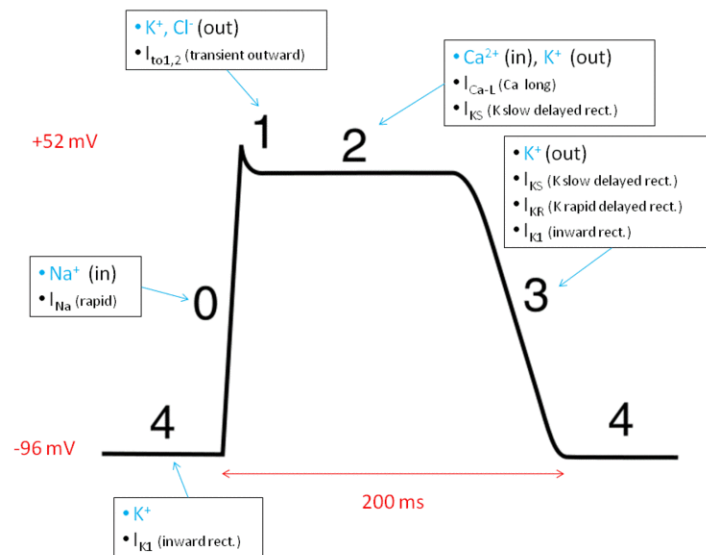


Solution to Question 8:

Plateau phase of ventricular muscle is due to opening of Ca channel.

Action potential of myocardium:

- Phase 0: Depolarisation: Influx of Na^+ into myocyte.
- Phase 1: Early repolarization: Efflux of K^+ from myocyte.
- Phase 2: Plateau: Ca^{2+} enters myocardial cells via voltage regulated L-type Ca^{2+} channels.
- Phase 3: Repolarisation: Efflux of K^+ .
- Phase 4: Restoration of RMP: Restoration of Na^+ and K^+ by Na-K ATPase.



Solution to Question 9:

Repolarization of cardiac muscle proceeds from epicardium to endocardium.

Depolarisation proceeds from endocardium to epicardium in an intact cardiac muscle.

Solution to Question 10:

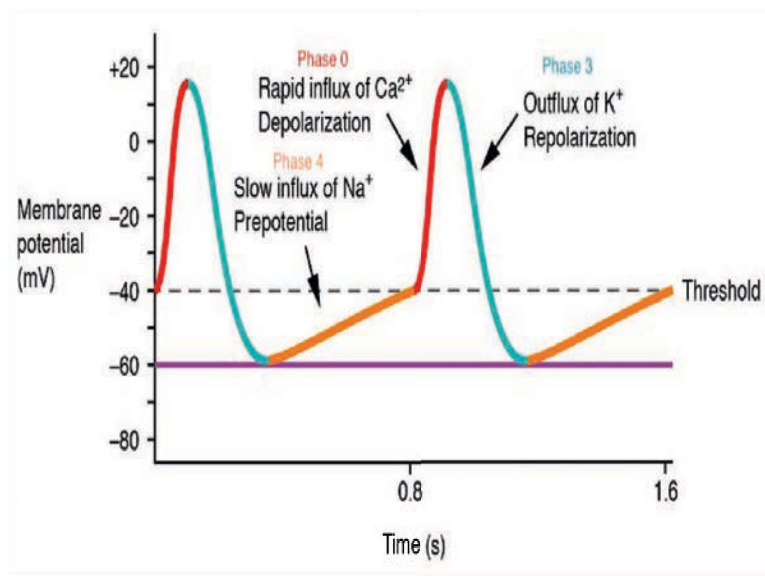
Phase 0 of pacemaker potential is initiated by opening of voltage-gated Ca^{2+} channels.

The pacemaker potential consists of:

- Phase 0 - this is upstroke seen due to the opening of voltage-gated Ca^{2+} channels. This causes an influx of calcium leading to the upstroke.
- Phase 3 - This is followed by inactivation of the calcium channels and simultaneous activation of K^{+} channels. This leads to efflux of K^{+} bringing about repolarization.
- Phase 4- As the K^{+} efflux declines, a channel permeable to both Na^{+} and K^{+} opens leading to an influx of both Na^{+} and K^{+} . This is the I_f channel. This is responsible for slow spontaneous diastolic depolarization. This accounts for the automaticity of SA and AV node.

The slope of Phase 4 in the SA node determines the heart rate.

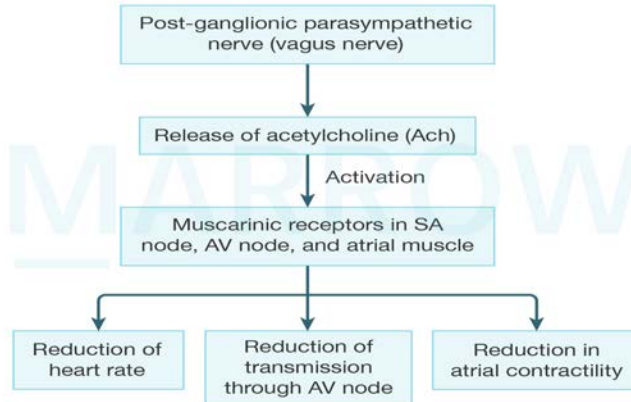
Note: Fast voltage-gated Na^{+} channels seen in myocyte potential are permanently inactivated because of the less negative resting potential of the pacemaker cells.



Solution to Question 11:

Vagal stimulation can cause an increase in the R-R interval in the ECG as it decreases heart rate.

Vagus nerve effects on the heart



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Solution to Question 12:

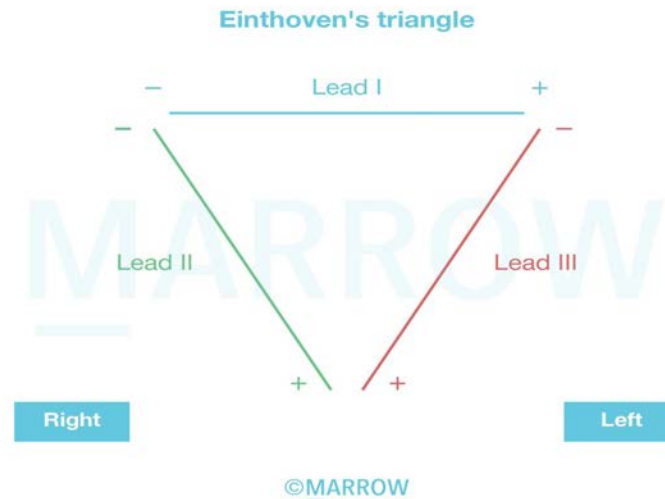
In recording Lead I, the right arm is negative electrode and the positive electrode is left arm.

Bipolar Limb leads for ECG recording

- LA → left arm
- RA → right arm

- LL→ left foot

Lead type	Positive input (+)	Negative input (-)
Lead I	LA	RA
Lead II	LL	RA
Lead III	LL	LA



Solution to Question 13:

Einthoven's law states $\text{Lead I} + \text{Lead III} = \text{Lead II}$.

Einthoven's triangle is an imaginary formation of three limb leads in a triangle used in electrocardiography.

Solution to Question 14:

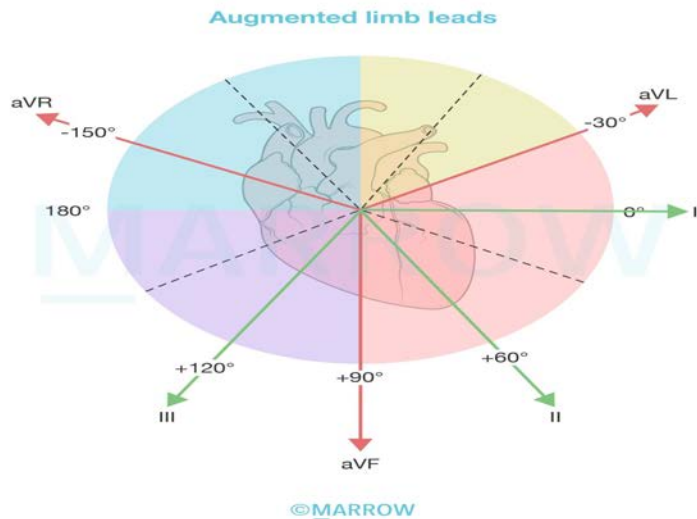
According to Einthoven's law,

$\text{Lead I} + \text{Lead III} = \text{Lead II}$.

$\text{Lead III} = \text{Lead II} - \text{Lead I} = (2-1) = 1 \text{ mV}$.

Solution to Question 15:

In a standard ECG, an augmented lead measures the electrical potential difference between one limb and the other two limbs. aVR, aVL, aVF are the augmented limb leads.



Solution to Question 16:

The part of the ECG that corresponds to ventricular repolarization is T wave.

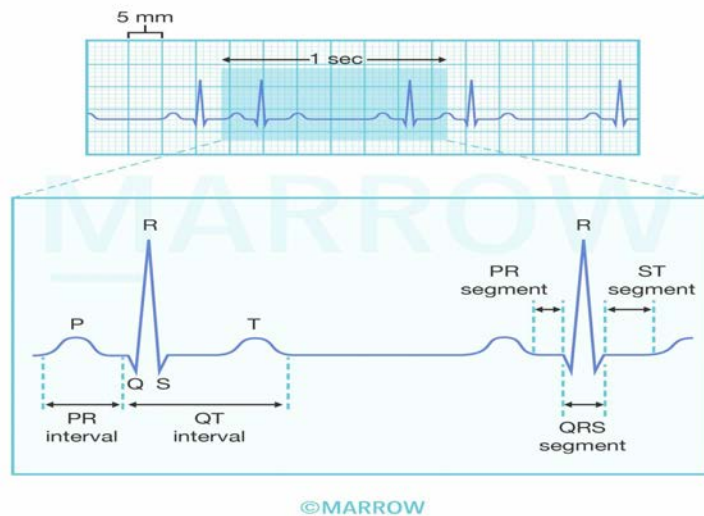
ECG waves	Events in the heart
P wave	Atrial depolarization
QRS complex	Ventricular depolarization
T wave	Ventricular repolarization
U wave	Repolarization of ventricular myocytes with long action potential



Solution to Question 17:

The PR interval indicates atrial depolarisation(P wave) and AV node conduction time(PR segment).

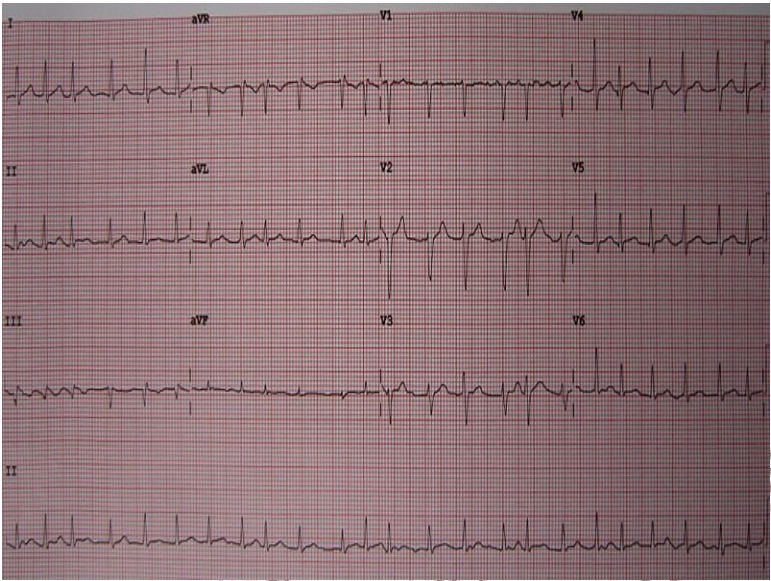
PR interval is the time between the beginning of the P wave and the beginning of the QRS complex. It is the interval between the beginning of electrical excitation of the atria and the beginning of excitation of the ventricles. Normal duration is 0.12-0.20s.



Solution to Question 18:

The P wave is absent in atrial fibrillation.

In atrial fibrillation, numerous small depolarization waves spread in all directions through the atria. The waves are weak and many of them are of opposite polarity at any given time, they almost electrically neutralize one another. Therefore, P waves are absent from ECG in atrial fibrillation.



Solution to Question 19:

The normal mean QRS axis lies between -30° and $+110^{\circ}$.

The QRS axis represents the major vector of ventricular activation, which is the overall direction of electrical activity.

Solution to Question 20:

The sympathetic innervation to AV node is by Left stellate ganglion.

Innervation of SA Node and AV Node:

The SA node develops from structures on the right side of the embryo and the AV node from structures on the left. Therefore -

ANS	SA node	AV node
Parasympathetic	Right vagus	Left vagus
Sympathetic	Right stellate ganglion	Left stellate ganglion

Solution to Question 21:

Sympathetic stimulation causes faster conduction, therefore decreased refractory period.

Effects of sympathetic stimulation:

- Chronotropic- It increases the rate of sinus nodal discharge.
- Dromotropic- It increases the rate of conduction.
- Bathmotropic- It increases the level of excitability in all portions of the heart.
- Inotropic- It increases the force of contraction of all the cardiac musculature.

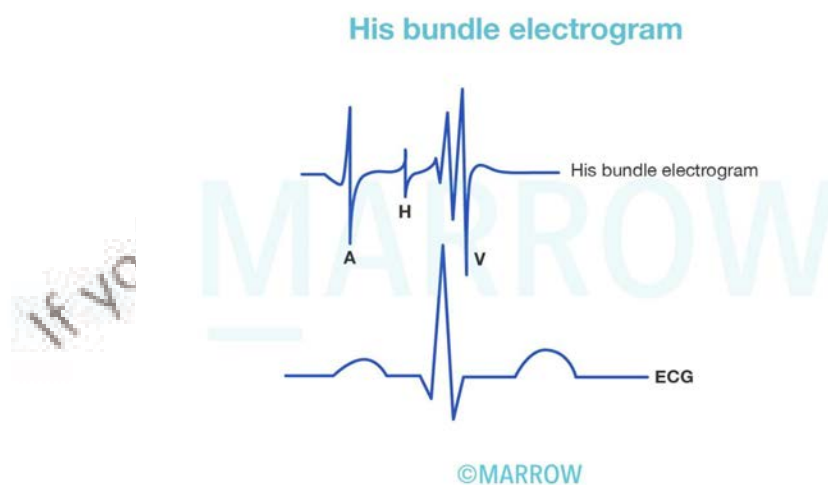
Solution to Question 22:

The AH interval corresponds to conduction from AV node to bundle of His on an ECG.

His bundle electrogram (HBE)

The record of the electrical activity obtained with the catheter passed through a vein to the right side of the heart and placed close to the tricuspid valve is HBE.

It shows 3 deflections as shown in the figure:



With HBE and standard ECG leads, 3 intervals can be timed accurately

His bundle waves	Description
A	Denotes AV node activation
H	Denotes His bundle activation
V	Denotes ventricular depolarization

Interval	Recording	Corresponding event
PA interval	Time from the first appearance of atrial depolarization to the A wave in HBE	Conduction time from SA node to AV node
AH interval	From A wave to start of H spike	AV nodal conduction time
HV interval	Time from start of H spike to start of QRS deflection in ECG	Conduction in Bundle of His and bundle branches

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If you purchased this from someone else,
you may have been scammed.

Blood Pressure and Regulation

Question 1:

What is blood pressure?

- a) SBP X Pulse rate
- b) DBP X pulse rate
- c) Pulse pressure X pulse rate
- d) Cardiac output X peripheral resistance

Question 2:

What is the SI unit of blood pressure?

- a) pascal
- b) mmHg
- c) torr
- d) psi

Question 3:

What is mean arterial pressure?

- a) Systolic + (Diastolic BP)/2
- b) Diastolic + 1/3rd Pulse pressure
- c) Systolic + 1/3rd Pulse pressure
- d) Diastolic + (Systolic BP)/2

Question 4:

Korotkoffs sounds are due to _____.

- a) Increased laminar flow on application of pressure
- b) Closure of aortic valve in diastole
- c) Arterial turbulence on occlusion of vessel

- d) Normal sound that exist even without application of cuff

Question 5:

When measuring blood pressure, which of the following is unlikely to show a spuriously high BP?

- a) Auscultatory gap
- b) Small cuff
- c) Thick calcified vessel
- d) Obesity

Question 6:

The systolic pressure in the right ventricle is _____

- a) 25 mmHg
- b) 85 mmHg
- c) 95 mmHg
- d) 100 mmHg

Question 7:

During diastole, the arterial pressure is maintained by:

- a) Musculature of arteries
- b) Elastic recoil of aorta
- c) Constriction of capillaries
- d) Contraction of left ventricle

Question 8:

The shape of arterial pulse is influenced by _____

- a) Viscosity of blood
- b) Velocity of blood
- c) Cross-sectional area of artery

- d) Arterial wall expansion

Question 9:

Carotid sinus baroreceptor is most sensitive to _____

- a) Mean BP
- b) Diastolic BP
- c) Systolic BP
- d) Pulse pressure

Question 10:

Baroreceptor stimulation would result in _____

- a) Decreased Vagal activity
- b) Increased heart rate
- c) Decreased sympathetic discharge to heart
- d) Increased vasomotor tone

Question 11:

The maximum blood pressure to which carotid baroreceptors respond is

- a) 180 mmHg
- b) 160 mmHg
- c) 200 mmHg
- d) 220 mmHg

Question 12:

Discharge from Baroreceptors cause inhibition of:

- a) Rostral ventrolateral medulla
- b) Caudal ventrolateral medulla
- c) Nucleus of tractus solitarius
- d) Nucleus ambiguus

Question 13:

Baroreceptor provides _____

- a) Negative feedback
- b) Positive feedback
- c) Both negative and positive feedback
- d) Feedforward control

Question 14:

Afferent fibers from baroreceptors for blood pressure control goes to?

- a) Nucleus ambiguus
- b) Rostral-ventro lateral medulla
- c) Nucleus Tractus Solitarius
- d) Caudal ventrolateral medulla

Question 15:

The low pressure receptors are _____

- a) Affected by cardiac output
- b) Stimulated by atrial systole and diastole
- c) Stimulated by left ventricular contraction
- d) Stimulated by aortic pressure

Question 16:

Occlusion of common carotid arteries on both sides leads to _____

- a) Increase in heart rate and BP
- b) Increase in BP and decrease in heart rate
- c) Decrease in heart rate and BP
- d) No effect on heart rate and BP

Question 17:

The hormone involved in the regulation of blood pressure is _____

- a) Serotonin
- b) Histamine
- c) Angiotensin II
- d) Prostaglandin

Question 18:

Marey's law states that _____

- a) Heart rate varies directly with blood pressure
- b) Heart rate varies inversely with blood pressure
- c) Heart rate do not have any relation with blood pressure
- d) Heart rate is directly proportional to cardiac output

Question 19:

Which of the following is NOT caused by sympathetic stimulation?

- a) Increased BP
- b) Increased HR
- c) Increased venous capacitance
- d) Increased total peripheral resistance

Question 20:

Which of the following is false regarding nitric oxide?

- a) Derived from endothelium
- b) Acts by increasing cAMP levels
- c) Vasodilator
- d) Derived from arginine

Question 21:

All of the following are true regarding the actions of endothelin-1 except _____

- a) Bronchodilation
- b) Vasoconstriction
- c) Decreased GFR
- d) Has inotropic effect

Question 22:

A 55-year-old man comes to his primary-care clinician complaining of erectile dysfunction. He is given a prescription for sildenafil, and on follow-up, reports that his ability to sustain an erection has been improved markedly by this treatment. The action of which of the following vasoactive mediators would primarily be prolonged in this patient?

- a) Histamine
- b) Endothelin-1
- c) Prostacyclin
- d) Nitric oxide

Question 23:

Reflex responsible for tachycardia during right atrial distension is:

- a) Bezold-Jarisch reflex
- b) Bainbridge reflex
- c) Cushing reflex
- d) J reflex

Question 24:

A 45-year-old male met with a road traffic accident and suffered a head injury. On examination, he was drowsy and his blood pressure is high. Which reflex is responsible for elevated blood pressure in this condition?

- a) J reflex
- b) Bainbridge reflex
- c) Bezold-Jarisch reflex
- d) Cushing reflex

Question 25:

All of the following are rapidly acting blood pressure control mechanisms except:

- a) Chemoreceptor reflex
- b) Central nervous system ischemic mechanism
- c) Renin angiotensin vasoconstrictor mechanism
- d) Baroreceptor reflex

Question 26:

Which of the following neurotransmitters is correctly matched with an autonomic pathway?

- a) Glutamate is released by Caudal Ventrolateral Medulla (CVLM) neurons projecting to the intermediolateral nucleus (IML)
- b) GABA is released by Nucleus Tractus Solitarius (NTS) neurons projecting to the nucleus ambiguus
- c) GABA is released by CVLM neurons projecting to the Rostral Ventrolateral Medulla (RVLM)
- d) Glutamate is released by CVLM neurons projecting to the NTS

Question 27:

Stimulation of peripheral chemoreceptors causes:

- a) Bradycardia
- b) Tachycardia
- c) Either bradycardia or tachycardia
- d) Neither bradycardia nor tachycardia

Question 28:

Which of the following can be responsible for Bezold-Jarisch reflex?

- a) Serotonin
- b) Histamine
- c) Prostaglandin
- d) Angiotensin

Question 29:

Which of the following events occur immediately when a person changes his/her posture from supine to standing?

- a) Decrease in firing from baroreceptor
- b) Vagal output to cardiac fibres increase
- c) Mesenteric Vasoconstriction
- d) Increase cardiac contractility

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	c
5	a
6	a
7	b
8	d
9	d
10	c
11	a
12	a
13	a
14	c
15	b
16	a
17	c
18	b
19	c
20	b
21	a

22	d
23	b
24	d
25	c
26	c
27	c
28	a
29	a

Detailed Explanations

Solution to Question 1:

Blood pressure is defined as cardiac output X peripheral resistance.

Solution to Question 2:

The SI unit of blood pressure is Pascal (Pa). It is conventionally expressed in mm of Hg.

Note: 1 mmHg = 133 Pa.

Solution to Question 3:

Mean arterial pressure is diastolic + 1/3rd pulse pressure.

The mean pressure is the average pressure throughout the cardiac cycle. It is not equal to the average of systolic and diastolic pressure because, at normal heart rates, a greater fraction of the cardiac cycle is spent in diastole than in systole. The mean arterial pressure is therefore determined by about 60 percent by the diastolic pressure and 40 percent by the systolic pressure.

Mean BP= DBP + 1/3 Pulse pressure

$$= \text{DBP} + \frac{1}{3}(\text{SBP} - \text{DBP})$$

$$= \frac{(3\text{DBP} + \text{SBP} - \text{DBP})}{3}$$

$$= \frac{2}{3} \text{DBP} + \frac{1}{3} \text{SBP}$$

Solution to Question 4:

Korotkoffs sounds are due to arterial turbulence on occlusion of the vessel.

The streamline flow in the unstricted artery is silent. But when the artery is narrowed, the velocity of flow through the constriction exceeds the critical velocity and turbulent flow results. Turbulent flow in the partly occluded brachial artery sets up the vibrations (oscillations) of the vessel wall which are heard through the stethoscope and are known as Korotkoffs sounds.

Solution to Question 5:

The auscultatory gap may give a falsely low recording of systolic BP and overestimation of diastolic blood pressure not spuriously high BP.

In older patients with wide pulse pressure and in some hypertensive patients, the Korotkoff sounds may become inaudible between SBP and DBP, and reappear as cuff deflation continues, this is known as auscultatory gap.

Small cuff size, obesity, and thick calcified arteries can give rise to falsely high BP as in these conditions, vessels are difficult to compress and higher cuff pressure is required to compress them.

Solution to Question 6:

The systolic pressure in the right ventricle is about 25 mmHg.

Solution to Question 7:

During diastole, the arterial pressure is maintained by the elastic recoil of aorta.

There is stretching of aorta and arteries during systole due to ejection of blood from ventricle. During diastole, the previously stretched aorta and arteries recoil. The volume of blood that is displaced by the recoil furnishes continuous capillary flow throughout diastole.

Solution to Question 8:

The shape of arterial pulse is influenced by arterial wall expansion.

Arterial Pulse

- The blood forced into the aorta during systole not only moves the blood in the vessels forward but also sets up a pressure wave that travels along the arteries. The pressure wave expands the arterial walls as it travels and the expansion is palpable as the pulse.
- The two major factors that influence arterial pulse are- stroke volume of the heart and distensibility (compliance) of the vessels.

Solution to Question 9:

The carotid sinus baroreceptor is most sensitive to pulse pressure. They fire more rapidly to a constantly changing pressure than a stationary one.

Baroreceptors

- Stretch receptors located in the heart and adventitia of the blood vessels
- Types based on location
- Carotid sinus and aortic arch receptors (high-pressure regions)
- Cardiopulmonary receptors (low-pressure regions)
- Afferents
- Carotid sinus – a branch of Glossopharyngeal nerve (IX CN) – the carotid sinus nerve (sinus nerve of Hering)
- Aortic arch receptor – a branch of the Vagus nerve (X CN) – aortic depressor nerve
- Center for Reflex
- Medulla (Nucleus Tractus solitarius)
- Efferents
- Sympathetic and parasympathetic neurons to heart and vasculature
- Most sensitive to pulsatile pressure: They work to maintain BP in the range of mean BP, but it is the changes in the pressure they are the most sensitive to.
- The threshold for excitation: 50 mmHg
- Maximal activity: 180 mmHg
- Responses when one carotid sinus is isolated and perfused and the other baroreceptors are denervated
- Perfusion pressure <30mm Hg: no response
- 70–110mm Hg: the linear relationship between perfusion pressure and the fall in systemic BP and HR
- >150mm Hg: no further increase in response to the degree of inhibition of the sympathetic nerve is maximal.

Solution to Question 10:

Baroreceptor stimulation would result in decreased sympathetic discharge to heart.

The baroreceptors are stretch receptors in the walls of the heart and blood vessels.

Baroreceptor stimulation, by distension of the structures they are located in, inhibits the tonic discharge of sympathetic nerves and excites the vagal innervation of the heart. Thus leading to:

- Vasodilation and venodilation
- Hypotension

- Bradycardia
- Decrease in cardiac output

Solution to Question 11:

The maximum blood pressure to which carotid baroreceptors respond is 180 mmHg (arterial pressure here and not MAP).

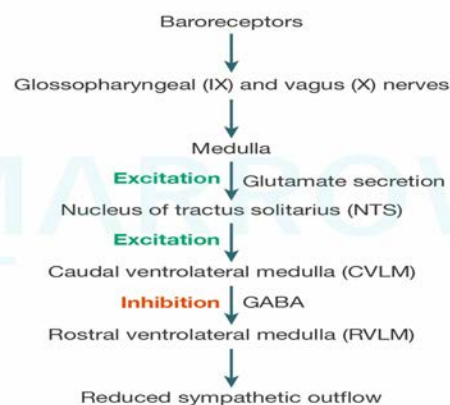
The blood pressure range to which carotid baroreceptors respond is 70- 150 mmHg of mean arterial pressure.

Solution to Question 12:

Discharge from Baroreceptors cause inhibition of Rostral ventrolateral medulla.

The baroreceptors are stimulated by distension of the structures in which they are located. Their afferent fibers pass via the glossopharyngeal and vagus nerves to the medulla. These fibers release glutamate to excite neurons in the nucleus of the tractus solitarius (NTS). These NTS neurons then excite neurons in the caudal ventrolateral medulla (CVLM). These CVLM neurons release the inhibitory neurotransmitter γ -aminobutyric acid (GABA) into the RVLM to reduce the firing rate of these. Thus, reducing the sympathetic outflow.

Neural pathway of baroreceptors



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Solution to Question 13:

Baroreceptors provide negative feedback in response to an increase in blood pressure.

Increased baroreceptor discharge inhibits the tonic discharge of sympathetic nerves and excites the vagal innervation of the heart.

Baroreceptor reflex

- Increased BP → stimulation of baroreceptors → stimulation of NTS → inhibition of RVLM → inhibition of sympathetic system
- Stimulation of NTS also stimulates CVC → parasympathetic stimulation
- Net effect = sympathetic inhibition and parasympathetic stimulation → decreased HR and BP

NTS - Nucleus tractus solitarius

CVC - Cardiovagal center

RVLM - Rostral ventrolateral medulla

HR- Heart rate

BP - Blood pressure

Solution to Question 14:

Afferent fiber for blood pressure control goes to the Nucleus Tractus Solitarius (NTS).

Increased BP stimulates the baroreceptors which provide negative feedback in response to the increase in blood pressure.

Baroreceptor reflex:

- Increased BP → stimulation of baroreceptors → stimulation of NTS → excitation of CVLM → inhibition of RVLM → inhibition of sympathetic system
- Stimulation of NTS also stimulates CVC → parasympathetic stimulation
- Net effect = sympathetic inhibition and parasympathetic stimulation → decreased HR and BP

NTS - Nucleus tractus solitarius

CVC - Cardiovagal centre

CVLM - Caudal ventrolateral medulla

RVLM - Rostral ventrolateral medulla

HR - Heart rate

BP - Blood pressure

Solution to Question 15:

The low-pressure receptors are stimulated by atrial systole and diastole

The low-pressure receptors are those in the great veins, right and left atria and pulmonary vessels.

Atrial stretch receptors are of two types:

- Type A: discharge primarily during atrial systole
- Type B: discharge primarily during late diastole

Distension of the atrial walls → stimulation of atrial receptors → Vasodilation, fall in BP and increased heart rate.

Solution to Question 16:

Occlusion of both common carotid arteries causes an increase in HR and BP.

Reduced sinus pressure → response carried to nucleus tractus solitarius via glossopharyngeal and vagus nerve → less inhibitory effect on the vasomotor center.

This leads to:

- Decreased parasympathetic activity to the heart - Increases heart rate
- Increased sympathetic activity to the heart and blood vessels - Increases heart rate, contractility, constricts arterioles - Increases TPR, constricts veins - increases venous return

Note: The carotid baroreceptors are located in the walls of the carotid sinus, which is located at or slightly above the bifurcation of the common carotid arteries. So, the occlusion of common carotid arteries is occlusion proximal to the carotid sinus.

If the occlusion is distal to the carotid sinus → increase in sinus pressure → activation of baroreceptor reflex → decrease in BP and HR.

Solution to Question 17:

The hormone involved in the regulation of blood pressure is Angiotensin II.

Angiotensin II is a powerful vasoconstrictor that increases the total peripheral resistance and decreases Na⁺ and H₂O excretion by kidney, thereby causing an increase in arterial BP.

Solution to Question 18:

Marey's law states that the heart rate varies inversely with blood pressure.

Solution to Question 19:

Increased venous capacitance is NOT caused by sympathetic stimulation.

Important effects of increased sympathetic stimulation:

- Arteriolar vasoconstriction
- Increased stroke volume
- Increased heart rate
- Venoconstriction and decrease in venous capacitance

- The increase in mean systemic filling pressure is caused mainly by venous contraction.
- All these effects together bring about an increased arterial pressure.

Solution to Question 20:

Nitric oxide acts by increasing cGMP not by increasing cAMP.

Nitric Oxide

- Also known as Endothelium-derived relaxing factor (EDRF).
- Synthesized in endothelial cells from arginine in a reaction catalyzed by nitric oxide synthase (NOS).
- Mechanism of action:

NO causes activation of guanylyl cyclase $\rightarrow \uparrow$ cGMP $\rightarrow$ relaxation of vascular smooth muscle $\rightarrow$ vasodilation.

- Effects of NO:
- Vasodilation.
- Neuromodulator in brain.
- Antimicrobial and cytotoxic activity of various inflammatory cells.
- In GIT, relaxation of smooth muscle.
- Penile erection.

Solution to Question 21:

Endothelin-1

- Potent vasoconstrictor agent.
- Produced by endothelial cells.
- They are secreted locally and act in a paracrine fashion.

Actions of Endothelin-1 (ET-1)

- Potent vasoconstrictor.
- Platelet activation.
- Inotropic effect.
- Closure of ductus arteriosus.
- Have a role in the pathogenesis of hypertension, myocardial infarction, and congestive heart failure.
- In the brain, regulate the transport of substances across blood brain barrier (BBB).
- In the kidneys, they are involved in tubuloglomerular feedback $\rightarrow \downarrow$ GFR.

- In the lungs, it causes bronchoconstriction (mediated via ET-B receptors).
- In GIT, they are involved in the migration of myenteric plexus to distal colon (failure leads to Hirschsprung's/megacolon).
- Deletion of both alleles of ET-1 gene→craniofacial abnormalities and respiratory failure at birth.

Solution to Question 22:

The action of nitric oxide would primarily be prolonged in this patient.

Sildenafil: It is used for the treatment of erectile dysfunction, which inhibits cGMP-specific phosphodiesterase-5 (PDE-5), an enzyme that degrades cGMP, thereby effectively prolonging the action of NO.

Penile erection is produced by release of nitric oxide (NO), with consequent vasodilation and engorgement of the corpora cavernosa.

Solution to Question 23:

Reflex responsible for tachycardia during right atrial distension is Bainbridge Reflex.

The stretch receptors of the atria that elicit the Bainbridge reflex transmit their afferent signals through the vagus nerves to the medulla of the brain. Then efferent signals are transmitted back through vagal and sympathetic nerves to increase heart rate and strength of heart contraction. Thus, this reflex helps prevent damming of blood in the veins, atria, and pulmonary circulation.

Other options:

Option A: Bezold–Jarisch Reflex: Activation of chemosensitive vagal C fibers in the cardiopulmonary region causes profound bradycardia, hypotension and a brief period of apnoea followed by rapid shallow breathing.

Option C: Cushing Reflex: When intracranial pressure is increased, the blood supply to Rostral Vento-Lateral Medulla (RVLM) neurons is compromised. The local hypoxia and hypercapnia cause an increase in discharge from the neurons, producing hypertension, bradycardia, and irregular respiration.

Option D: J Reflex: The stimulation of the J-receptors (C receptors/juxta-pulmonary capillary receptors) causes apnea followed by rapid breathing, bradycardia, and hypotension.

Solution to Question 24:

The reflex responsible for elevated blood pressure(BP) in head injury patients is the Cushing reflex.

When intracranial pressure is increased, the blood supply to rostral ventrolateral neurons is compromised, and the local hypoxia and hypercapnia increase the discharge of the central chemoreceptors. The resultant rise in systemic arterial pressure (Cushing reflex) tends to restore the blood flow to the medulla.

Cushing reflex causes a triad of symptoms:

- Reflex bradycardia (via arterial baroreceptors)
- Hypertension (with widened pulse pressure)
- Irregular respiration (bradypnea)

Solution to Question 25:

Renin-angiotensin vasoconstrictor mechanism is not a rapidly acting pressure control mechanism, it is an intermediate-acting mechanism.

Regulation of Arterial Blood Pressure:

- Rapidly acting pressure control mechanism (acting within sec or min)
- The chemoreceptor reflex.
- The central nervous system ischemic mechanism.
- The baroreceptor reflex.
- Intermediate-acting mechanisms (act after minutes)
- The capillary fluid shift mechanism.
- Stress-relaxation of the vasculature.
- The renin-angiotensin vasoconstrictor mechanism.
- Long-term mechanisms (after days)
- Renin-angiotensin-aldosterone system (RAAS)

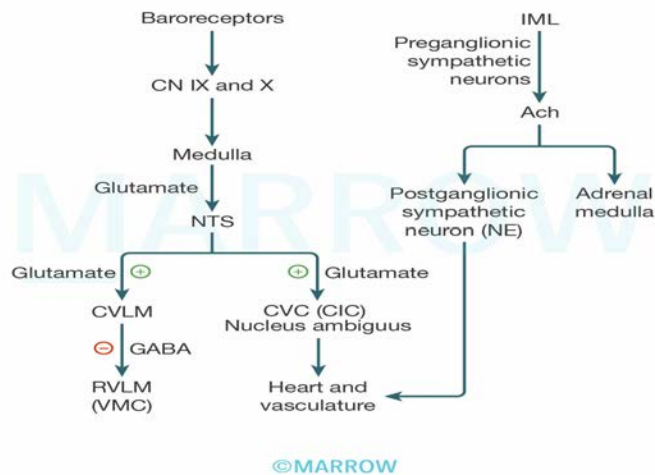
Note:

- Renin-angiotensin vasoconstrictor mechanism is an intermediate-acting mechanism, but water conservation by RAAS is a long-term mechanism in the regulation of BP.
- Most rapidly acting: Baroreceptor reflex > Chemoreceptor reflex

Solution to Question 26:

Basic Neural Pathway involved in Regulation of BP

Neural Pathway involved in regulation of BP



- NTS: Nucleus Tractus Solitarius
- CVLM: Caudal Ventrolateral Medulla
- RVLM (VMC): Rostral Ventrolateral Medulla (Vasomotor center)
- CVC (CIC): Cardiovagal center (Cardioinhibitory center)
- intermediolateral nucleus (IML)

Solution to Question 27:

Peripheral chemoreceptors in the carotid and aortic bodies can produce variable effects on heart rate-

- Direct Effect
- vagal stimulation → bradycardia
- Indirect effect
- Hypoxia → peripheral chemoreceptor stimulation → hyperventilation and increased catecholamine secretion → tachycardia and increased cardiac output

Chemoreceptor reflex

- Receptors
- Peripheral chemoreceptors in carotid and aortic bodies
- Central chemoreceptors in the medulla
- Stimulus
- decrease in- PaO_2
- increase in- Pa CO_2 , H^+ ion conc
- Afferent

- IX (sinus nerve of Hering) and X (vagus N)
- Center for reflex
- Medulla
- Efferent
- Sympathetic and parasympathetic neurons to heart vasculature and adrenal medulla
- Response
- Direct effect- bradycardia
- Indirect effect- hyperventilation, tachycardia, vasoconstriction, increased cardiac output and increased BP

Solution to Question 28:

Serotonin can be responsible for Bezold–Jarisch reflex.

Activation of chemosensitive vagal C fibers located in the cardiopulmonary region (ex: ventricles, atria, great veins, pulmonary artery, juxta-capillary region of alveoli) causes profound bradycardia, hypotension, and apnea followed by rapid shallow breathing. This response is called Bezold-Jarisch reflex.

Other substances that can cause Bezold–Jarisch reflex are capsaicin, phenyl biguanide, and veratridine.

Solution to Question 29:

There is a decreased firing from the baroreceptors.

- When a person stands up from a lying down position, it causes pooling of the blood in the legs, due to gravity. This reduces the venous return and causes a decrease in cardiac output.
- Reduced cardiac output, leads to a fall in blood pressure, which in turn causes reduced firing from baroreceptors.
- As a result, the vagal tone is suppressed leading to increased heart rate.

Gastrointestinal Secretion and Gastrointestinal Hormones

Question 1:

Ptyalin is secreted by which of the following glands?

- a) Gastric gland
- b) Duodenal gland
- c) Salivary gland
- d) Pancreatic gland

Question 2:

Which of the following statements is false?

- a) Meissner's plexus is present in submucosa
- b) Myenteric plexus is present between inner circular and outer longitudinal muscle layer
- c) Muscularis mucosa is associated with peristalsis
- d) Villi are absent in large intestine

Question 3:

Salivary amylase is inactivated by which of the following?

- a) Enteropeptidase
- b) Low pH of stomach
- c) High pH of intestine
- d) Chloride ions

Question 4:

Which of the following gland secretes lingual lipase?

- a) Parotid gland
- b) Ebner's gland

- c) Sublingual gland
- d) Submandibular gland

Question 5:

Which of the following fluids have the highest daily secretion of potassium?

- a) Bile
- b) Pancreatic secretion
- c) Jejunal secretion
- d) Saliva

Question 6:

The secretomotor fibers to the parotid gland causing a patient to salivate, pass through which of the following ganglions?

- a) Genuiculate ganglion
- b) Sphenopalatine ganglion
- c) Otic ganglion
- d) Lesser ganglion

Question 7:

Rennin is present in _____

- a) Gastric juice
- b) Pancreatic secretions
- c) Kidney
- d) Saliva

Question 8:

A patient presents with anemia and achlorhydria. After evaluation, pernicious anemia is diagnosed. Damage to which of the following cells is responsible for this condition?

- a) Chief cells

- b) Parietal cells
- c) Enterochromaffin cells
- d) Goblet cells

Question 9:

Which of the following is secreted by the Goblet cells?

- a) HCl
- b) Mucus
- c) Pepsin
- d) Serotonin

Question 10:

In the stomach, H^+ ions are secreted in exchange for which of the following ions?

- a) Na^+
- b) K^+
- c) Ca^{2+}
- d) Cl^-

Question 11:

A patient diagnosed with gastroesophageal reflux disease is found to have excess gastric secretion. All of the following are likely to increase the secretion, except:

- a) Histamine
- b) Acetyl choline
- c) Gastrin
- d) HCl

Question 12:

A patient with recurrent abdominal pain is found to have multiple duodenal ulcers. After a detailed evaluation, he is diagnosed with Zollinger-Ellison syndrome. Stimulation of which of the following can inhibit gastric secretion in this patient?

- a) Somatostatin
- b) Histamine
- c) Vagal stimulation
- d) Gastrin

Question 13:

Maximum gastric secretion occurs in which of the following phases?

- a) Cephalic phase
- b) Gastric phase
- c) Intestinal phase
- d) Secretion is equal in all phases

Question 14:

Which of the following is responsible for increasing the HCl secretion in the gastric phase?

- a) Pepsinogen
- b) CCK
- c) Gastrin
- d) Secretin

Question 15:

Which of the following is true about the secretion from the exocrine pancreas?

- a) It is hypertonic
- b) Secretion is inhibited by Ach
- c) It is hypotonic
- d) Secretion is stimulated by cholecystikinin

Question 16:

Bicarbonate-rich pancreatic juice is secreted under the influence of which of the following stimulants?

- a) Secretin
- b) GIP
- c) Pancreatozymin
- d) Gastrin

Question 17:

Which among the following is the most alkaline secretion?

- a) Saliva
- b) Gastric secretion
- c) Brunner's gland secretion
- d) Small intestine secretion

Question 18:

Which of the following is the most potent choleretic?

- a) CCK
- b) Fatty acids
- c) Bile salts
- d) Pancreatic juices

Question 19:

Primary bile acids are synthesized from_____

- a) Cholesterol
- b) Amino acids
- c) Bilirubin
- d) Fatty acids

Question 20:

Enterohepatic circulation is necessary for the secretion of_____

- a) Bile salt

- b) Intestinal juice
- c) Pancreatic juice
- d) Bilirubin

Question 21:

Which of the following does not belong to the secretin family?

- a) GIP
- b) Glucagon
- c) CCK
- d) VIP

Question 22:

A patient underwent resection of the gastric antrum. A few weeks later, endoscopy revealed atrophy of the gastric mucosa. Serum levels of gastrin were found to be decreased. Which of the following is true regarding this GI hormone?

- a) Hormone secreted by pepsinogen cells
- b) A polypeptide
- c) primarily released in cephalic phase
- d) Stimulated by secretin

Question 23:

What is the effect of cholecystokinin on GIT?

- a) Contraction of sphincter of Oddi
- b) Inhibits gastric emptying
- c) Inhibits pancreatic enzyme secretion
- d) Relaxes gall bladder

Question 24:

What stimulates secretin secretion?

- a) Epinephrine
- b) Acid
- c) Fat
- d) Acetylcholine

Question 25:

Liver synthesizes all of the following except:

- a) Albumin
- b) Coagulation factors
- c) Acute phase proteins
- d) Immunoglobulins

Question 26:

How much is the daily secretion of gastric juice?

- a) 500-1000 ml
- b) 1500-2000 ml
- c) 3000-3500 ml
- d) 3500-5000 ml

Question 27:

Which of the following statement is correct regarding secretin?

- a) ↑ bicarbonate secretion
- b) Inhibit bicarbonate secretion
- c) ↑ gastric acid secretion
- d) Stimulates gastrin release

Question 28:

Which of the following is secreted by the I cells of duodenum?

- a) Secretin

- b) Gastrin
- c) CCK
- d) Motilin

Question 29:

A 40-year-old man presented with abdominal pain and diarrhea. On investigations, multiple ulcers were seen in the proximal and distal duodenum and in the jejunum. The gastric pH was found to be 1.2 with a serum gastrin concentration of 2000 pg/ml. Which of the following is the primary site for the production of GI hormone raised in this patient?

- a) Pylorus
- b) Antrum
- c) Pancreas
- d) Small intestine

Question 30:

Which of the following hormones do not act on the pancreas?

- a) Secretin
- b) Gastrin
- c) Cholecystokinin
- d) GIP

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	b
5	d
6	c
7	a

8	b
9	b
10	b
11	d
12	a
13	b
14	c
15	d
16	a
17	c
18	c
19	a
20	a
21	c
22	b
23	b
24	b
25	d
26	b
27	a
28	c
29	b
30	b

Detailed Explanations

Solution to Question 1:

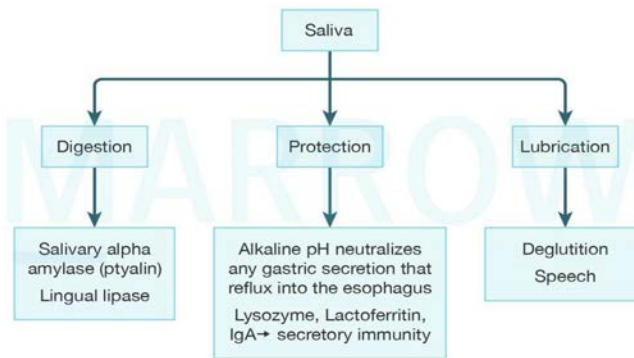
Ptyalin (salivary α -amylase) is secreted by the salivary gland.

There are 3 pairs of salivary glands

- Parotid
- Submandibular
- Sublingual glands

Functions of saliva:

Functions of saliva



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Solution to Question 2:

Muscularis propria is associated with peristalsis and not muscularis mucosa.

Contraction of muscularis mucosa alters absorptive and secretive surface area.

Meissner's plexus (submucosal plexus) is present in the submucosa.

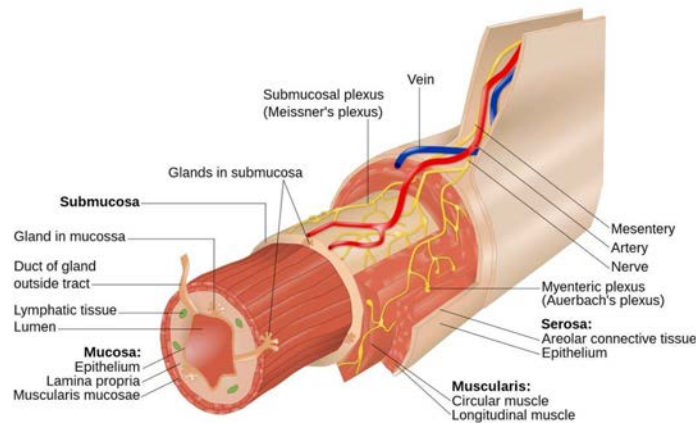
The myenteric plexus (Auerbach plexus) is present between the inner circular and outer longitudinal muscle layer.

Villi are absent in the large intestine but present in the small intestine.

Layers of intestinal wall from outside to inside are as follows:

- Serosa
- The longitudinal smooth muscle layer
- The circular smooth muscle layer
- Submucosa
- Mucosa

Layers of the intestine



Solution to Question 3:

Salivary amylase is inactivated by the low pH of the stomach. It is rapidly denatured at pH 4 (gastric pH is 1 to 3.5).

Salivary amylase is activated by chloride.

Salivary α amylase (Ptyalin):

- Digests starch: cleaves internal α -1,4-glycosidic bonds in starch
- Optimum pH: 7
- Denatured at pH 4

Since a large portion of a meal often remains unmixed for a considerable length of time in the stomach, the salivary enzyme may account for the digestion of as much as 75% of the starch present before it is denatured by gastric acid.

Note: Even in the absence of salivary amylase, there is no defect in carbohydrate digestion because pancreatic amylase is sufficient to digest all of the ingested starch.

Solution to Question 4:

Lingual lipase is secreted by von Ebner's gland.

von Ebner's gland is a serous salivary gland situated around circumvallate and foliate papillae in front of the terminal sulcus.

Lingual lipase hydrolyse dietary lipid. The optimal pH is 3-6. Unlike pancreatic lipase, it is not inhibited by bile salts, medium-chain FA, and lecithin. Hence, it remains active through the stomach and into the intestine.

Solution to Question 5:

The highest daily secretion of K^+ is into the saliva.

In the process of formation of saliva, Na^+ and Cl^- are reabsorbed from the saliva, and K^+ and HCO_3^- are secreted into the saliva.

Note: The highest concentration of K^+ is in the rectal fluid.

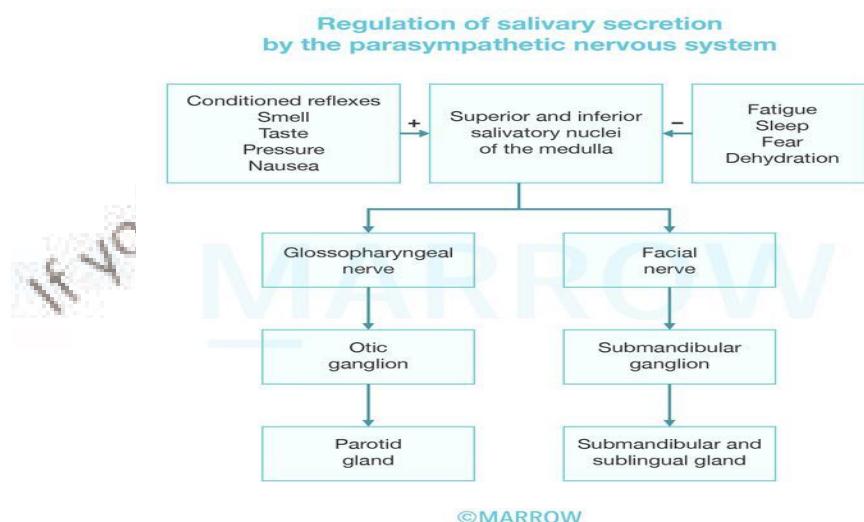
Solution to Question 6:

The secretomotor fibers to the parotid gland pass through the Otic ganglion.

Salivary secretion is almost entirely controlled by the parasympathetic nervous system. Volume is mainly mediated by parasympathetic stimulation while the composition of saliva (increased proteinaceous content) is slightly modified by sympathetic input.

The parasympathetic preganglionic fibers are delivered by the glossopharyngeal and facial nerves to otic and submandibular ganglion respectively.

Regulation of salivary secretion by the parasympathetic nervous system:



Solution to Question 7:

Rennin is present in the gastric juice. It is also called chymosin.

It is a protein-digesting enzyme (endopeptidase) found in the stomach of newborn cows. It transforms (curdles) caseinogen into insoluble casein.

In humans and in some animals that lack rennin, milk is coagulated by the action of pepsin.

Note: Renin is produced by juxtaglomerular cells of the kidney, which hydrolyzes angiotensinogen to angiotensin I.

Solution to Question 8:

Pernicious anemia develops due to autoimmune damage to the parietal cells in gastric glands.

The parietal cells secrete both HCl and intrinsic factor. Therefore, when the parietal cells are destroyed, both achlorhydria, as well as pernicious anemia, develop.

Pernicious anemia occurs due to the deficiency of the intrinsic factor of Castle, which is important for the absorption of vitamin B₁₂.

Solution to Question 9:

Goblet cells secrete mucus. These cells are located on the surface of the crypts of Lieberkuhn and the villi of the small intestine.

Cells present in the crypts of Lieberkuhn:

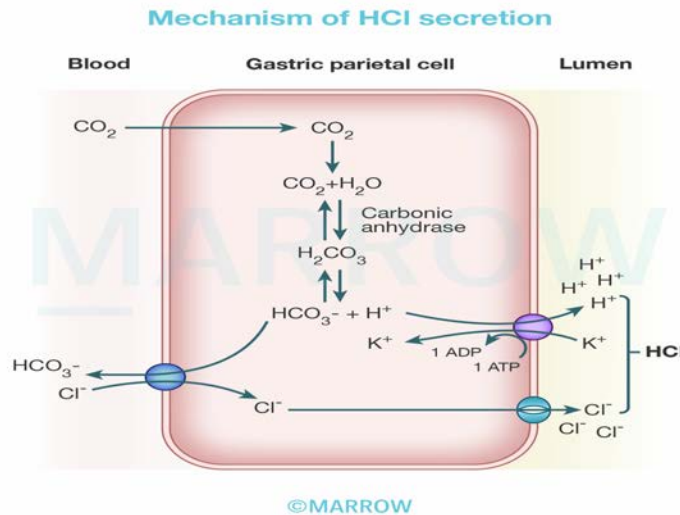
- Goblet cells (similar to surface cells of the stomach) - secrete mucus that acts as a lubricant and protects the intestinal surface
- Epithelial cells - secrete succus entericus (intestinal secretion)
- Paneth cells - secrete lysozymes and defensins (antibacterial substances)

Solution to Question 10:

H⁺ ions are secreted into the lumen of the stomach in exchange for K⁺ by the H⁺-K⁺ pump (H⁺, K⁺-ATPase).

Mechanism of HCl formation:

- In the parietal cells, CO₂ and H₂O are converted to H⁺ and HCO₃⁻, catalyzed by carbonic anhydrase.
- H⁺ is secreted into the lumen of the stomach by the H⁺-K⁺ pump in exchange for K⁺.
- HCO₃⁻ moves out of the cell into the bloodstream in exchange for Cl⁻ (Cl⁻-HCO₃⁻ exchange). This Cl⁻ moves into the lumen of the stomach.
- H⁺ and Cl⁻ combine to form HCl.



As HCO_3^- is added to the venous blood, the pH of the blood increases. This is called alkaline tide. Eventually, this HCO_3^- will be secreted in the pancreatic secretions to neutralize H^+ in the small intestine.

Proton pump inhibitors inhibit H^+-K^+ pump.

Solution to Question 11:

Gastric secretion is not increased by HCl. HCl decreases gastric secretion via a negative feedback mechanism.

Low pH due to HCl in the stomach inhibits gastrin secretion, which in turn, inhibits H^+ secretion. Hence, the presence of HCl inhibits gastric secretion.

Stimuli that increase acid and gastrin secretion:

- Distension of stomach
- Peptides
- Calcium
- Alcohol and caffeine
- Increased vagal discharge

Stimuli that decrease acid and gastrin secretion:

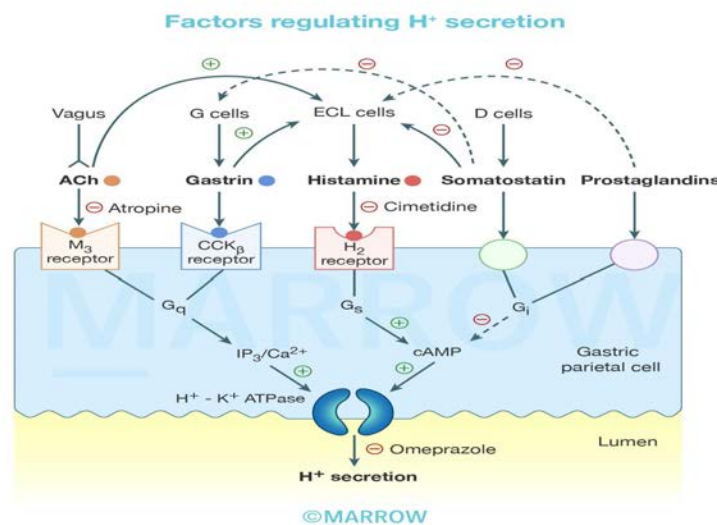
- Gastric acid
- Secretin family - secretin, glucagon, VIP, GIP
- Somatostatin
- CCK, Calcitonin
- PGE_2

Solution to Question 12:

Gastric secretion is inhibited by somatostatin.

It inhibits gastric secretion via both direct and indirect pathways:

- Direct pathway - somatostatin acts directly on the acid-producing parietal cells by binding to receptors on parietal cells that are coupled to adenylyl cyclase via a G_i protein. This inhibits adenylyl cyclase, and cAMP levels are reduced. In this way, somatostatin antagonizes the stimulatory effect of histamine on H^+ secretion.
- Indirect pathway - somatostatin inhibits the release of histamine and gastrin and thus decreasing H^+ secretion indirectly.



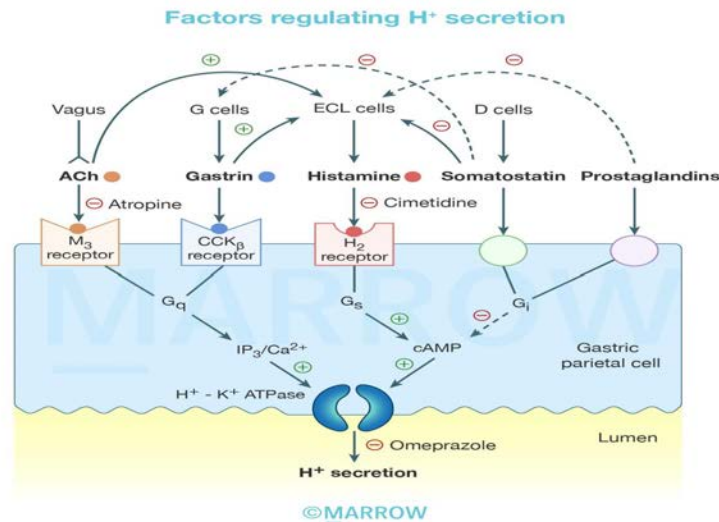
Solution to Question 13:

Maximum gastric secretion occurs in the gastric phase.

Solution to Question 14:

HCl secretion in the gastric phase is increased by gastrin.

Gastrin stimulates H^+ secretion by interaction with cholecystokinin B (CCKB) receptor on the parietal cell which secretes HCl.



Solution to Question 15:

Pancreatic secretion is stimulated by cholecystokinin (CCK) and ACh. It is always isotonic.

It has virtually the same Na⁺ and K⁺ concentrations as in plasma. It is rich in several digestive enzymes like amylase, trypsin, chymotrypsin, lipase, elastase, etc.

Solution to Question 16:

Bicarbonate-rich pancreatic juice is secreted under the influence of secretin.

Secretin is secreted by the S cells of the duodenum. Its primary function is to promote the secretion of bicarbonate-rich pancreatic juice by the ductal cells of the pancreas and duodenum. This secretion is poor in enzymes.

Other functions:

- Augments the effects of CCK
- Inhibits gastrin release
- Decreases gastric acid secretion
- Contraction of the pyloric sphincter
- Exerts trophic effect on the pancreas

Solution to Question 17:

The most alkaline secretion is Brunner's gland secretion with a pH of 8.0 – 8.9.

Brunner's gland is located in the first few centimeters of the duodenal wall. It secretes large amounts of alkaline mucus in response to tactile stimuli on the duodenal mucosa, vagal stimulation, and GI hormones, especially secretin. It is inhibited by sympathetic stimulation.

The function of the mucus secreted by Brunner's glands is to protect the duodenal wall from digestion by the highly acidic gastric juice emptying from the stomach.

Note: The second most alkaline secretion in the GI tract is the pancreatic juice.

Solution to Question 18:

Bile salts (chemical choleretic) are the most potent choleretics.

Any substance that increases bile secretion is called a choleretic. Bile secretion may be increased by chemical (bile salts), hormonal (Secretin), and neural (vagal stimulation) mechanisms.

Note: Cholagogues cause contraction of the gallbladder. Eg. CCK

Solution to Question 19:

Primary bile acids are synthesized from cholesterol by hepatocytes.

Bile contains bile acids, bile salts, bile pigments, phospholipids, water, and electrolytes. It is secreted by the liver and concentrated in the gall bladder.

Two primary bile acids (cholic acid and chenodeoxycholic acid) are synthesized from cholesterol by hepatocytes. When these primary bile acids are secreted into the lumen of the intestine, a portion of each is converted to secondary bile acids (deoxycholic acid and lithocholic acid) by bacteria.

The liver conjugates primary and secondary bile acids with glycine or taurine to their respective bile salts. Ex: Glycocholic acid, Taurodeoxycholic acid.

Rate of bile secretion is 30 - 60 mL/hour (500 - 1000 mL/day).

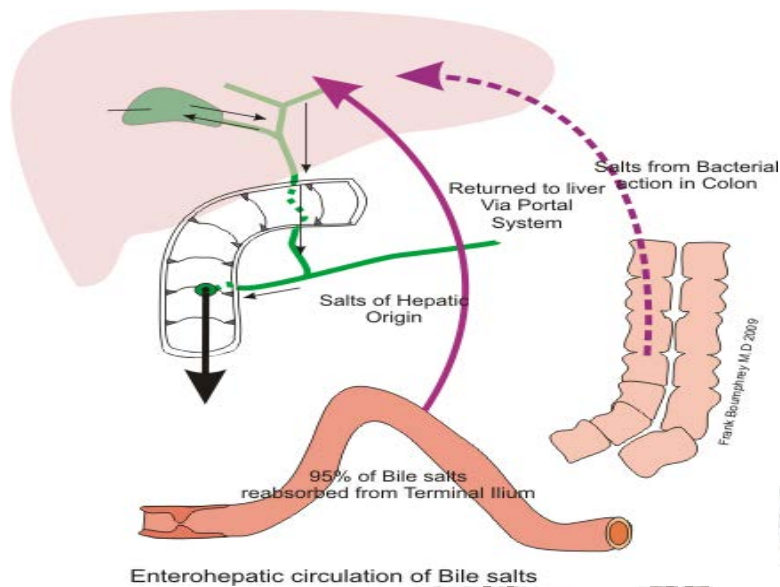
Note: Bile has no digestive enzymes.

Solution to Question 20:

Enterohepatic circulation is necessary for the secretion of bile salts.

The enterohepatic circulation of bile salts is the recycling of bile salts between the small intestine (terminal ileum) and the liver by means of the portal vein. It occurs 6-8 times a day.

The recirculation of bile salts to the liver reduces the demand to synthesize new bile salts.



Solution to Question 21:

CCK (Cholecystokinin) belongs to the gastrin family, not the secretin family.

Classification of GI hormones based on structure and function:

- Gastrin family - Gastrin, Cholecystokinin
- Secretin family - Secretin, Glucagon, VIP, GIP
- Others - Motilin, Ghrelin, Somatostatin, etc

Solution to Question 22:

Gastrin is a peptide hormone.

It is secreted by the G cells of the gastric mucosa present in the antrum of the stomach, primarily during the gastric phase of gastric acid secretion.

It causes gastric secretion at 1000-1500 mL/day. Its primary stimulus is gastric distension.

Primary functions of Gastrin:

- Gastric acid (HCl) secretion
- Pepsin secretion
- Trophic action on the gastric mucosa

Other functions:

- Stimulates gastric motility
- Prevents gastroesophageal reflux by contraction of LES

Conditions associated with increased gastrin secretions (Ex: Zollinger-Ellison syndrome), will cause an increase in H^+ secretion and hypertrophy of the gastric mucosa.

Conditions associated with reduced gastrin secretion (Ex: Resection of the gastric antrum), will cause a decrease in H^+ secretion and atrophy of the gastric mucosa.

Solution to Question 23:

Cholecystokinin (CCK) inhibits gastric emptying.

It is secreted by the I cells of the duodenal and jejunal mucosa in response to monoglycerides, fatty acids, and amino acids.

Primary functions:

- Stimulation of pancreatic enzyme secretion
- Contraction of the gallbladder
- Relaxation of the sphincter of Oddi

Other functions:

- Augmentation of the action of secretin in producing alkaline (bicarbonate rich) pancreatic juice
- Inhibition of gastric emptying
- Trophic (growth) effect on the exocrine pancreas
- Increases the synthesis of enterokinase
- It enhances the motility of the small intestine and colon
- Along with secretin, it augments the contraction of the pyloric sphincter and thus preventing the reflux of duodenal contents into the stomach
- It promotes satiety

Solution to Question 24:

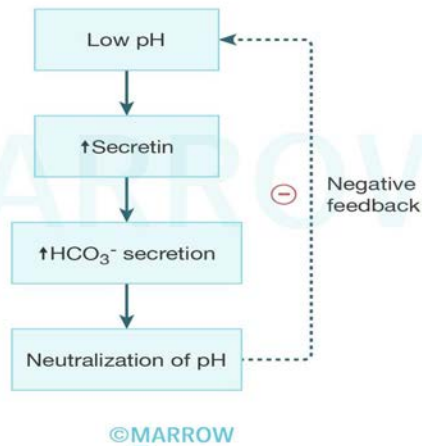
Secretin secretion is stimulated by acid.

Stimuli that increase secretin secretion:

- Low duodenal pH (2-4.5) due to acid in the chyme
- Products of protein digestion in the upper small intestine

Secretin secretion is regulated by negative feedback:

Secretin secretion



Solution to Question 25:

Immunoglobulins are not synthesized by the liver. They are produced by plasma cells.

Plasma proteins synthesized by the liver are:

- Albumin
- Coagulation/clotting factors
- Acute-phase proteins
- Steroid binding and other hormone-binding proteins

Solution to Question 26:

Daily secretion of gastric juice is between 1500-2000 mL.

Note: This is a controversial question- Daily secretion of gastric juice is

- 1500 mL according to Guyton and Hall Textbook of Medical Physiology, 13th edition, Page no: 819
- 2500 mL according to Ganong's Review Of Medical Physiology - 25th Edition, Page no: 464

Choose 2500 mL if given in the options. If not, choose within the range of 1500-2500 mL.

Solution to Question 27:

Secretin increases HCO₃⁻ secretion by ductal cells of the pancreas and duodenum.

It is secreted by the S cells in the upper small intestine.

Functions of secretion are:

- Primary function
- $\uparrow$ HCO_3^- secretion by ductal cells of pancreas and duodenum
- Other functions
- Augments the effects of CCK
- Inhibits gastrin release
- $\downarrow$ gastric acid secretion
- Contraction of pyloric sphincter
- Exerts trophic effect on the pancreas

Solution to Question 28:

I cells of the duodenum secrete CCK.

Solution to Question 29:

The given clinical scenario is suggestive of Zollinger-Ellison syndrome. In this condition, serum gastrin concentration is raised to around >1000 pg/ml. Gastrin is primarily secreted by G cells present in the antrum of the stomach.

Solution to Question 30:

Gastrin does not act upon the pancreas. Gastrin stimulates gastric acid secretion and gastric mucosal growth.

GI hormones acting on the pancreas include:

GIP is also known as Glucose-dependent insulinitropic peptide.

	CCK(Cholecystokinin)	Secretin	GIP(Gastric inhibitory peptide)
Stimulates	Pancreatic enzyme secretion Pancreatic bicarbonate secretion Gallbladder contraction Growth of exocrine pancreas	Pepsin secretion Pancreatic bicarbonate secretion Biliary bicarbonate secretion Growth of exocrine pancreas	Insulin release
Inhibits	Gastric emptying	Gastric acid secretion	Gastric acid secretion

Digestion & Absorption

Question 1:

Salivary amylase is activated by which of the following ions?

- a) Na^+
- b) K^+
- c) HCO_3^-
- d) Cl^-

Question 2:

Where does the digestion of disaccharides occur?

- a) Mouth
- b) Stomach
- c) Small intestine
- d) Large intestine

Question 3:

The mechanism by which glucose is absorbed in the intestine is:

- a) Primary active transport
- b) Secondary active transport
- c) Facilitated diffusion
- d) Any of the above

Question 4:

Which of the following is a true statement?

- a) Fructose is absorbed in the intestine by secondary active transport
- b) SGLT2 is involved in absorption of monosaccharide in intestine
- c) GLUT2 is involved in transport of glucose from enterocyte into interstitium

d) GLUT4 is responsible for basal glucose uptake in many organs

Question 5:

Which of the following is the main enzyme involved in the hydrolysis of 1,6° linkage of starch?

- a) Maltase
- b) Isomaltase
- c) Sucrase
- d) Lactase

Question 6:

What is chymotrypsinogen?

- a) Zymogen
- b) Carboxypeptidase
- c) Transaminase
- d) Elastase

Question 7:

Which of the following processes converts trypsinogen to trypsin?

- a) Combination of 2 molecules of trypsin
- b) Phosphorylation
- c) Removal of a few amino acids
- d) Addition of alkali group

Question 8:

Which of the following protects the normal pancreas from autodigestion?

- a) Secretion of bicarbonate
- b) Protease inhibitors
- c) Proenzymes

d) Resistance of pancreatic cells

Question 9:

Which of the following is not an endopeptidase?

- a) Trypsin
- b) Chymotrypsin
- c) Elastase
- d) Carboxypeptidase

Question 10:

Peptide transporter 1 is involved in the absorption of which of the following?

- a) Polypeptides
- b) Dipeptide
- c) Amino acids
- d) Immunoglobulin A

Question 11:

Which of the following is passively absorbed in the gut?

- a) Glucose
- b) galactose
- c) Lipids
- d) Amino acids

Question 12:

Short-chain fatty acids produced by bacteria are maximally absorbed in which part of the intestine?

- a) Duodenum
- b) Jejunum
- c) Ileum

d) Colon

Question 13:

Which of the following is not a function of short-chain fatty acids in the colon?

- a) Trophic effect on the colonic epithelial cells
- b) Combat inflammation
- c) Maintain acid–base equilibrium
- d) Secretion of Na^+

Question 14:

Which of the following patients is most likely to have a major electrolyte imbalance?

- a) A patient with duodenal resection due to a primary tumor
- b) A patient who underwent jejunal resection due to an obstruction
- c) A patient with ileal resection due to recurrent Crohn's disease
- d) A patient with sigmoid resection due to a recurrent fistula

Question 15:

What is the amount of fluid absorbed in the intestine in a day?

- a) 5L
- b) 11L
- c) 2L
- d) 8L

Question 16:

Which of the following is not absorbed in the colon?

- a) Acetate
- b) Cl^-
- c) Na^+
- d) K^+

Question 17:

Which of the following is false about vitamin absorption?

- a) Fat-soluble vitamin digestion needs cholesterol esterase
- b) Vitamin B12 is absorbed in ileum
- c) Folate is absorbed by carriers that are Na⁺ cotransporters
- d) Most vitamins are absorbed in upper small intestine

Question 18:

A patient with anemia is prescribed iron tablets. The absorption of the supplement is likely to take place in which part of the intestine?

- a) Duodenum
- b) Jejunum
- c) Ileum
- d) Stomach

Question 19:

Which of the following stimulates ferritin synthesis?

- a) Ceruloplasmin
- b) Hepcidin
- c) Iron
- d) Transferrin

Question 20:

Which of the following is inhibited by hepcidin?

- a) Cobalamin absorption
- b) Iron absorption
- c) Folic acid synthesis
- d) Copper absorption

Question 21:

While prescribing iron to a patient with iron deficiency anemia, which of the following would you recommend to increase its absorption?

- a) Milk
- b) Ascorbic acid
- c) Antacid
- d) Tannins

Question 22:

All of the following regulates intestinal absorption of iron except:

- a) Dietary intake of iron
- b) Daily loss of iron
- c) State of iron stores in the body
- d) State of erythropoiesis in bone marrow

Question 23:

What is the end product after the action of colonic bacteria on dietary fiber?

- a) Free radicals
- b) Glycerol
- c) Butyrate
- d) Sucrose

Question 24:

Where are the cell bodies of the orexinergic neurons are present in a patient?

- a) Locus ceruleus
- b) Dorsal raphe
- c) Lateral hypothalamic area
- d) Hippocampus

Question 25:

In a patient who is fasting, the stomach releases a hormone that increases food intake. Identify the hormone secreted.

- a) Orexin
- b) Leptin
- c) CCK
- d) Ghrelin

Question 26:

Which of the following is false regarding ghrelin?

- a) Increases appetite
- b) Inhibits leptin release
- c) Decreases gastric acid secretion
- d) Stimulates growth hormone release

Question 27:

What will be the trend of leptin levels in a patient with a BMI of 34 kg/m².

- a) High
- b) Normal
- c) Low
- d) May be low or normal

Question 28:

Which of the following is not a dietary fiber?

- a) Cellulose
- b) Inulin
- c) Pectin
- d) Gums

Question 29:

All of the following have a significant role in lipid digestion except:

- a) Pancreatic lipase
- b) Colipase
- c) Lingual lipase
- d) Bile acids

Question 30:

Following complete ileal and partial jejunal resection, the patient is most likely to have

- a) Folic acid deficiency
- b) Vitamin B12 deficiency
- c) Gastric ulcer
- d) Constipation

Question 31:

Transport molecule for Fe^{2+} into the enterocytes is

- a) Ferroportin 1
- b) Hephaestin
- c) Divalent Metal Transporter 1
- d) Divalent Metal Transporter 2

Question 32:

Which part of the liver stores vitamin A?

- a) Ito cells
- b) Hepatocytes
- c) Endothelial cells
- d) Kupffer cells

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	c
5	b
6	a
7	c
8	c
9	d
10	b
11	c
12	d
13	d
14	b
15	d
16	d
17	c
18	a
19	c
20	b
21	b
22	b
23	c
24	c
25	d
26	c
27	a
28	d
29	c
30	b
31	c
32	a

Detailed Explanations

Solution to Question 1:

Salivary amylase is activated by Cl^- .

Human salivary and pancreatic amylases are activated by chloride (Cl^-). They have optimum activities near-neutral pH (7.4). They hydrolyze internal 1:4° linkages but spare 1:6° linkages and terminal 1:4° linkages in starch.

Solution to Question 2:

Digestion of disaccharides occurs in the small intestine.

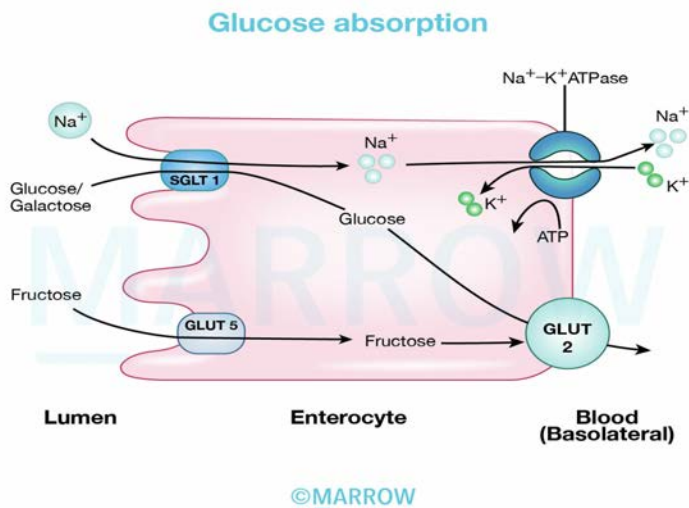
The enzymes maltase, sucrase, isomaltase, lactase, and trehalase are located in the enterocytes covering the intestinal microvilli brush border. The disaccharides are digested into their constituent monosaccharides as they come in contact with these enterocytes.

Solution to Question 3:

Glucose is absorbed in the intestine by secondary active transport.

On the basolateral side, there is active transport of sodium ions out of the cell into the interstitial fluid. This depletes sodium inside the cell. Because of this Na^+ concentration difference, more Na^+ binds to the SGLT on the luminal side. This increases the affinity of SGLT for glucose and glucose binds to it.

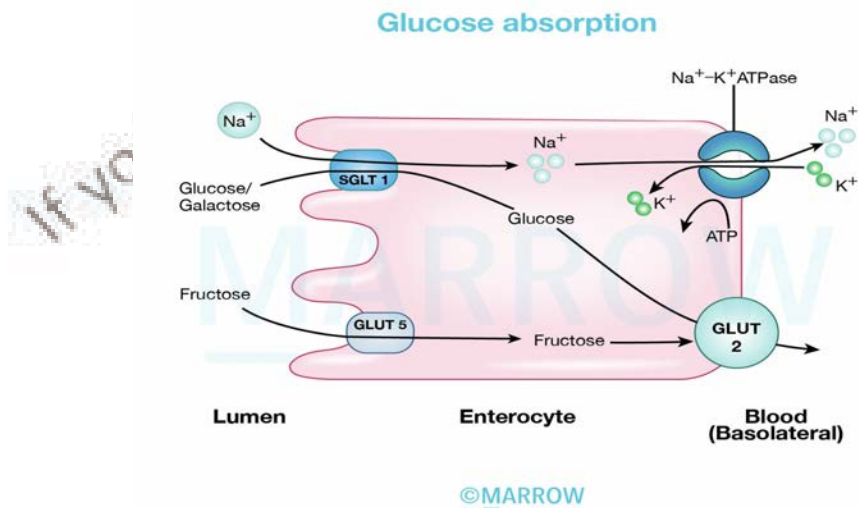
Then, both the Na^+ and glucose are transported together to the interior of the cell. Thus, glucose is absorbed in the intestine by secondary active transport. The energy for glucose transport is provided indirectly, by the active transport of Na^+ out of the cell. The glucose moves out of the cell into the blood across the basolateral membrane by facilitated diffusion mediated via GLUT.



Solution to Question 4:

GLUT2 is involved in the transport of glucose from the enterocyte into the interstitium (or out of the enterocytes).

The image below shows the function of GLUT 2 on enterocytes.



Option A: Fructose absorption is through facilitated diffusion from the intestinal lumen into the enterocyte by GLUT5 and out of the enterocytes into the interstitium by GLUT2.

Option B: SGLT1 is involved in the absorption of glucose into the enterocyte. SGLT2 is responsible for glucose transport out of the renal tubules.

Option D: GLUT1 is responsible for basal glucose uptake in many organs.

Solution to Question 5:

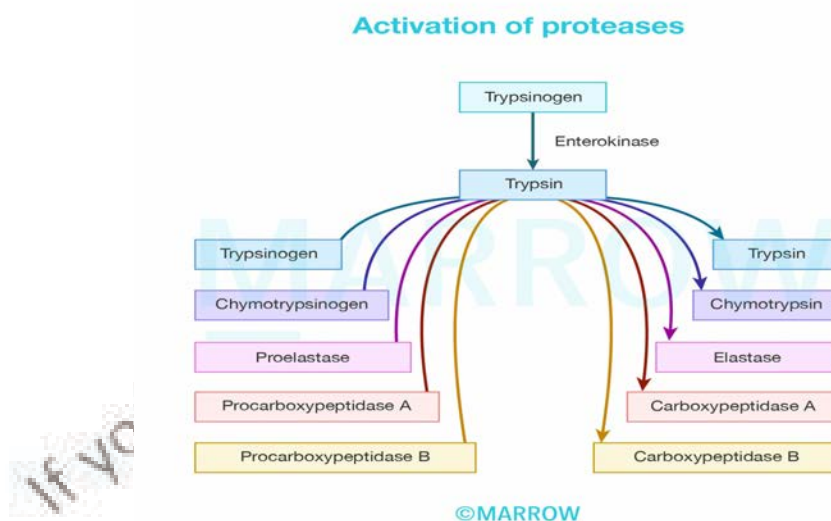
The main enzyme involved in the hydrolysis of 1,6° linkage of starch is isomaltase.

Both salivary and the pancreatic α -amylases hydrolyze internal 1:4° linkages but spare 1:6° linkages and terminal 1:4° linkages of starch.

Solution to Question 6:

Chymotrypsinogen is a zymogen.

Zymogens (proenzymes) are inactive precursors of the enzymes. Most of the peptidases (proteases) are present in the form of proenzymes. Trypsinogen, chymotrypsinogen, proelastase, procarboxypeptidase A, procarboxypeptidase B are zymogens.

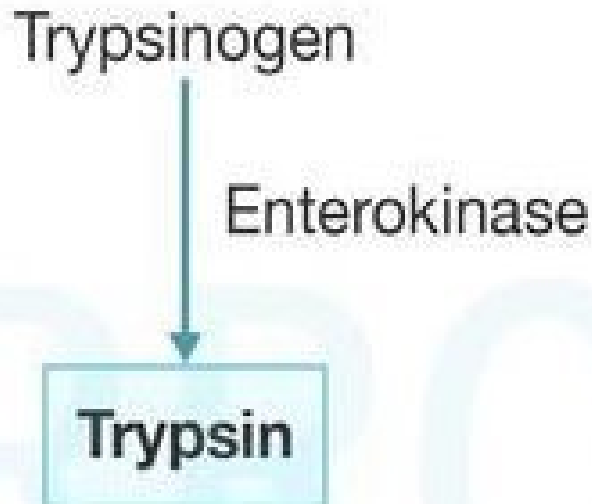


Solution to Question 7:

Trypsinogen is converted to trypsin by the removal of a few amino acids.

All proenzymes are converted to their active form by selective proteolysis (removal of a few amino acids).

Enterokinase cleaves trypsinogen to yield active trypsin. It is located on the brush border of duodenal enterocytes.



Solution to Question 8:

The synthesis and secretion of proteases as catalytically inactive proenzymes protect the normal pancreas from autodigestion.

All the protease proenzymes present in the pancreatic juice get converted to their active form only in the small intestinal lumen (site of action).

If proenzymes get activated in the pancreas (site of origin), they can digest the entire pancreas within a few hours, giving rise to the condition called acute pancreatitis.

Solution to Question 9:

Carboxypeptidase is an exopeptidase, not endopeptidase.

Peptidases (proteases) are of two types:

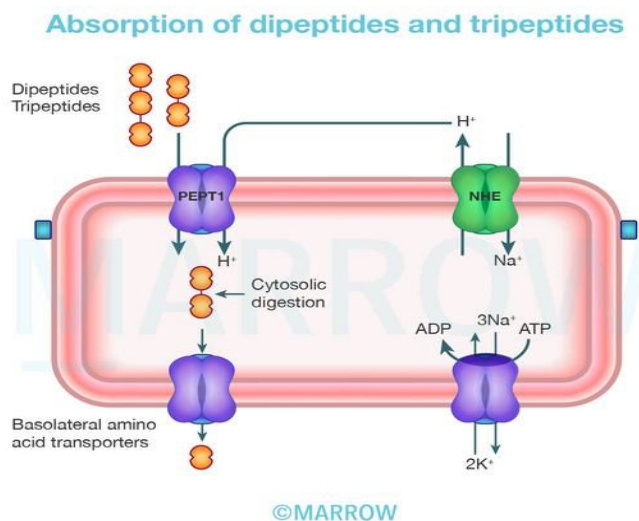
- Endopeptidases - they act at interior peptide bonds in the peptide molecules. Ex- pepsin, trypsin, chymotrypsin, elastase.
- Exopeptidases - they hydrolyze the amino acids at the carboxyl ends of the polypeptides, one at a time. Ex- carboxypeptidase, aminopeptidase.

Solution to Question 10:

Peptide transporter 1 (PepT1) is involved in the absorption of the dipeptide.

The dipeptides and tripeptides are transported along with H^+ into enterocytes by a system known as peptide transporter 1 (PepT1).

These peptides undergo intracellular hydrolysis to release amino acids. These amino acids are absorbed into the circulation from the basolateral membrane via basolateral amino acid transporter. H^+ moves out of the cell via Na^+-H^+ exchanger on the luminal side.



Solution to Question 11:

Lipids are passively absorbed in the gut.

The products of lipid digestion (monoglycerides and free fatty acids) are incorporated into micelles that are formed by bile salts and diffuse across the intestinal epithelium. Short and medium-chain fatty acids can directly diffuse across the intestinal epithelium.

Note: Recent evidence suggests that specific carrier proteins (membrane transporters) may be involved in the transport of long-chain fatty acids.

Solution to Question 12:

Short-chain fatty acids produced by bacteria are absorbed maximally in the colon by secondary active transport.

Almost all the iron absorption occurs in the duodenum.

Vitamin B12 is absorbed in the ileum.

Folate is absorbed in the upper small intestine (duodenum and jejunum).

Maximum water absorption takes place in the jejunum.

Solution to Question 13:

Short-chain fatty acids (2–5-carbon) in the colon promote the absorption of Na^+ , not secretion.

Short-chain fatty acids (SCFA) in the colon are formed by the action of colonic bacteria on complex carbohydrates, resistant starches, and other components of the dietary fiber i.e the material that escapes digestion in the upper gastrointestinal tract and enters the colon.

Functions:

- Exert a trophic effect on the colonic epithelial cells
- Combat inflammation
- Helps in maintaining acid-base equilibrium

Solution to Question 14:

The resection of jejunum leads to major electrolyte imbalance because most of the fluid and electrolyte reabsorption occurs in this segment.

The maximum water absorption takes place in the jejunum.

The amount of water absorbed in the intestines follows this order: Jejunum > Ileum > Colon.

Almost all the iron absorption occurs in the duodenum.

Vitamin B12 is absorbed in the ileum.

Folate is absorbed in the upper small intestine (duodenum and jejunum).

Solution to Question 15:

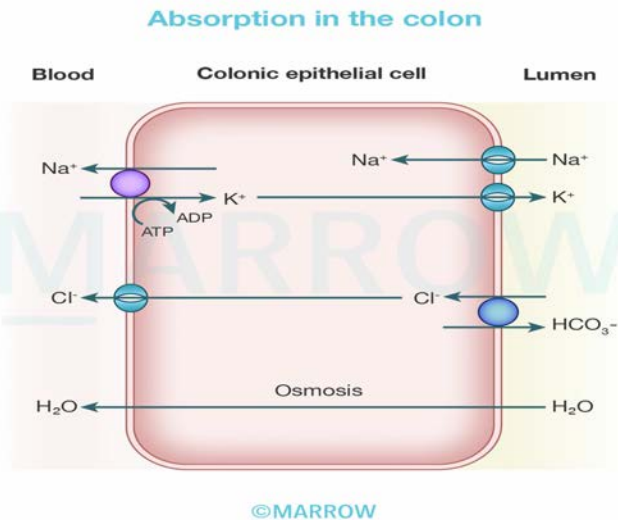
The amount of fluid absorbed in the intestine in a day is around 8L.

About 2L of ingested water and 7L of various secretions enter the gastrointestinal tract. Out of this, the majority (8800 mL) is absorbed in the small intestine and colon. The small remaining volume that is not absorbed (100–200 mL) is excreted in feces.

Solution to Question 16:

K^+ is not absorbed in the colon. It is secreted in the rectal fluid.

Electrolytes (Na^+ , Cl^-), short-chain fatty acids (acetate, propionate, butyrate), and water are absorbed in the colon.



Solution to Question 17:

Folate and vitamin B12 absorption is Na^+ -independent.

The remaining water-soluble vitamins—thiamin, riboflavin, niacin, pyridoxine, pantothenate, biotin, and ascorbic acid are absorbed by carriers that are Na^+ cotransporters.

Option A: Fat-soluble vitamins are ingested in the form of esters. Digestion needs hydrolysis by cholesterol esterase and absorbed after incorporating into micelles.

Options B and D: Most vitamins are absorbed in the upper small intestine, but vitamin B12 is absorbed in the ileum. Vitamin B12 binds to intrinsic factor, a protein secreted by the parietal cells of the stomach. This complex is absorbed across the ileal mucosa.

Solution to Question 18:

Almost all the iron absorption occurs in the duodenum.

Most of the iron in the diet is in the ferric (Fe^{3+}) form, whereas it is the ferrous (Fe^{2+}) form that is absorbed.

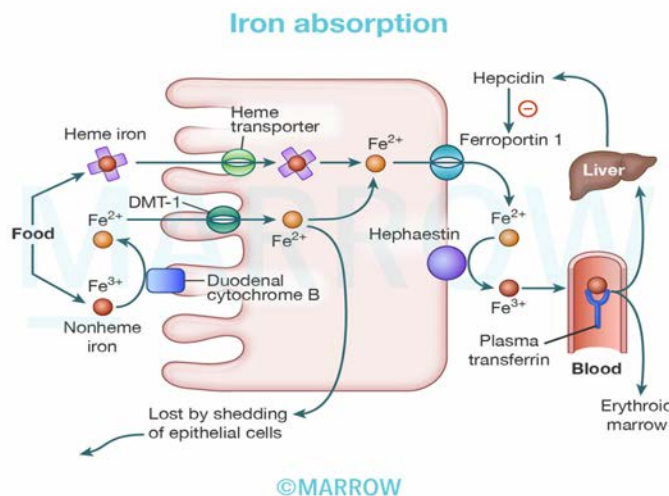
Note: Carbohydrates, proteins, lipids are absorbed throughout the length of the small intestine. Bile acids are present throughout the small intestine, but active absorption occurs only in the ileum, and active absorption of folate occurs in the upper small intestine (duodenum and jejunum).

Solution to Question 19:

Elemental iron stimulates ferritin synthesis.

It synthesizes ferritin by causing the release of a cytoplasmic protein that binds to a specific region in the ferritin mRNA. This disruption activates ferritin mRNA and results in its translation.

Hepcidin binds to ferroportin and triggers its internalization and degradation. This results in decreased iron absorption and decreased circulating iron levels which reduces ferritin synthesis.



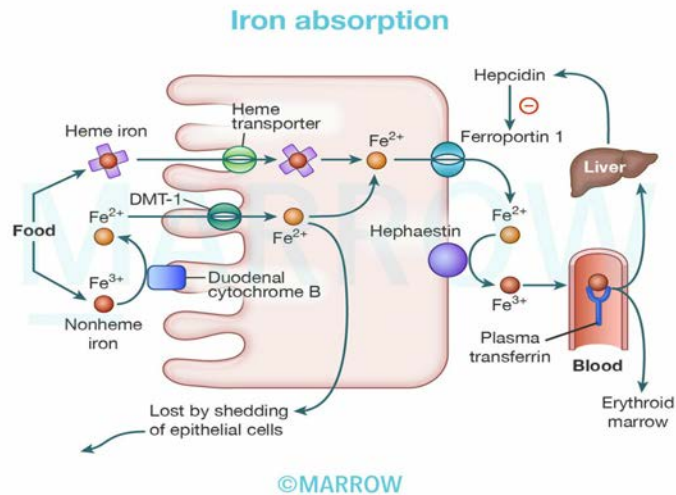
Note: Ferritin sequesters Fe^{2+} , a potentially toxic molecule.

Solution to Question 20:

Hepcidin inhibits iron absorption.

It is named for its originally described property as a hepatocellular protein with bactericidal activities.

Iron metabolism is regulated by hepcidin, a protein and acute phase reactant produced by the liver. It binds to ferroportin and degrades it. This inhibits the transfer of iron from the enterocyte to the plasma via ferroportin. As a result, iron becomes trapped within the duodenal cells (enterocytes) in the form of mucosal ferritin and is lost in feces when these cells are shed off.



Solution to Question 21:

Ascorbic acid is given with iron to increase its absorption.

Ascorbic acid (vitamin C) forms soluble complexes with iron and reduces iron from the ferric to the ferrous state, thereby enhancing iron absorption.

Option A: Milk decreases iron absorption.

Option C: Gastric secretions increase iron absorption by dissolving the iron and allowing it to form soluble complexes with ascorbic and other substances. This helps in its reduction to the Fe^{2+} form. Antacid decreases the absorption of iron.

Option D: Tannins (present in tea), phytates, phosphates, and oxalates form insoluble complexes with iron and lower its absorption.

Solution to Question 22:

Intestinal absorption of iron is not regulated by the daily loss of iron.

The intestinal absorption of iron is regulated by three factors:

- Recent dietary intake of iron
- State of the iron stores in the body
- State of erythropoiesis in the bone marrow

Solution to Question 23:

The action of colonic bacteria on dietary fiber would give butyrate.

Colonic bacteria act on complex carbohydrates, resistant starches, and dietary fiber to form short-chain fatty acids such as acetate, propionate, and butyrate.

SCFAs are 2–5-carbon weak acids that have an average normal concentration of about 80 mmol/L in the lumen.

Solution to Question 24:

Cell bodies of orexigenic neurons are present in the lateral hypothalamic area. Orexin stimulates food intake.

Role of hypothalamus in gastrointestinal and feeding regulation:

- Hunger center - lateral hypothalamic area
- Satiety center - ventromedial nuclei
- Feeding reflexes such as licking the lips and swallowing - mammillary bodies.

Note: The arcuate nucleus contains at least two different types of neurons that when stimulated lead to either increased or decreased appetite.

Solution to Question 25:

The hormone secreted by the stomach that increases food intake is ghrelin.

Ghrelin, the hunger hormone is produced by the endocrine cells in the oxyntic glands of the stomach. It increases food intake. Its levels increase during fasting, peaks just before the beginning of food intake, or when the person is expecting a meal. The levels fall within 1 hour of the start of feeding.

Option A: Orexin is synthesized from lateral hypothalamic neurons. It increases food intake.

Option B: Leptin is produced by adipocytes. It decreases food intake.

Option C: CCK is released from the small intestine. It decreases food intake.

Solution to Question 26:

Ghrelin increases gastric acid secretion.

Ghrelin is mainly produced by the endocrine cells in the oxyntic gland area of the stomach. It is also secreted by the pancreas and adrenal glands.

Functions:

- Increases appetite/food intake (orexigenic)
- Inhibits leptin release

- Stimulates growth hormone secretion by directly acting on anterior pituitary (ghrelin-growth hormone-releasing peptide)
- Increases gastric acid secretion and gastrointestinal motility to prepare the body for food intake
- Regulates the distribution and rate of use of energy

Solution to Question 27:

A patient with a BMI of 34 kg/m² is categorized as obese. Leptin levels are high in obesity.

Leptin is an appetite-suppressing hormone synthesized from adipocytes. The larger the fat stores, the more leptin is released into the blood.

Leptin resistance in obese persons may be due to its failure to cross the blood-brain barrier and reach the hypothalamus to serve its appetite-suppressing function.

Note: A recent study has indicated that triglycerides which are elevated in obesity, inhibit the transport of leptin across the blood-brain barrier.

Solution to Question 28:

Gum is the best answer among the given options. Even though all of the given options are dietary fibers, the consumption of gums through diet is very low and hence, it is the best answer.

Dietary fibers consist of both soluble and insoluble forms and include lignin, pectin, inulin, cellulose, hemicellulose, etc.

Advantages of fibers:

- Forms bulk of the stool and reduces stool transit time
- Reduces incidence of carcinoma colon, DM, CAD, diverticulitis

Solution to Question 29:

Lingual lipases are of little quantitative significance for lipid digestion other than in the setting of pancreatic insufficiency.

Lingual lipase is secreted by Ebner glands on the dorsal surface of the tongue in some species, and the stomach also secretes lipase.

Digestion of Lipids

- Pancreatic lipase is the most important enzyme involved in lipid digestion. The main products of its action are FFA and 2-monoglycerides (2-monoacylglycerols)
- Colipase is an accessory factor secreted in the pancreatic juice that helps to stabilize pancreatic lipase in an active conformation

- Bile acids emulsify the insoluble lipids and make them reach the surface of the mucosal cells. Bile acids also activate another pancreatic lipase called cholesterol esterase
- Cholesterol esterase hydrolyzes cholesterol esters, esters of fat-soluble vitamins, phospholipids, and triglycerides

Solution to Question 30:

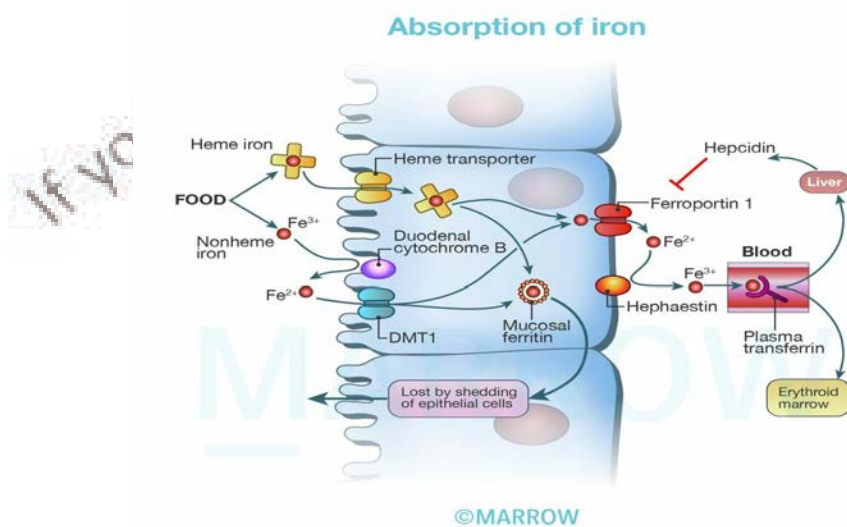
After complete ileal and partial jejunal resection, the patient is likely to have vitamin B12 deficiency.

Vitamin B12 is absorbed in the ileum. Vitamin B12 binds to intrinsic factor, a protein secreted by the parietal cells of the stomach, and the complex is absorbed across the ileal mucosa. Hence after ileal resection, there is impaired absorption of vitamin B12.

Solution to Question 31:

Transport of Fe^{2+} into the enterocytes occurs via divalent metal transporter 1 (DMT1), through an active transport process.

Protein DMT-1 (Divalent Metal Transporter-1) is expressed on the apical surface of enterocytes in the initial part of the duodenum. DMT-1 is not specific to iron & transports other substances like zinc, copper, cobalt & manganese.



Solution to Question 32:

Vitamin A is stored in the liver perisinusoidal stellate (Ito) cells as retinyl ester (retinol palmitate).

Kupffer cells, also known as stellate macrophages and Kupffer-Borowicz cells; are specialized macrophages located in the liver that form part of the mononuclear phagocyte system. These are present, lining the walls of the sinusoids, not the perisinusoidal space (Space of Disse).

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you may have been scammed.

GI Peristalsis and Motility

Question 1:

Slow waves are generated by:

- a) Myenteric neurons
- b) Smooth muscle cells
- c) Interstitial cells of Cajal
- d) Parasympathetic neurons

Question 2:

Least frequency of slow waves is seen in:

- a) Duodenum
- b) Ileum
- c) Jejunum
- d) Stomach

Question 3:

Slow waves are maximum in which part of the intestine?

- a) Stomach
- b) Ileum
- c) Colon
- d) Rectum

Question 4:

As a part of cranial nerves examination, swallowing reflex is performed on a patient. Which of the following is not involved in afferent limb of this reflex?

- a) Trigeminal nerve
- b) Facial nerve

- c) Glossopharyngeal nerve
- d) Vagus nerve

Question 5:

Chyme is propelled forward in small intestine by_____

- a) Segmentation
- b) Haustration
- c) MMC
- d) Peristalsis

Question 6:

Which of the following is the major initiating response for peristalsis?

- a) Hormonal
- b) Local stretching of gut
- c) Neural
- d) Gastric acid

Question 7:

Which of the following is seen in a patient on parasympathetic stimulation?

- a) Sphincter closure of gall bladder
- b) Increased peristalsis
- c) Decreased GI motility
- d) Decreased pancreatic secretion

Question 8:

Which neurotransmitter is responsible for anterograde dilatation of the small intestine?

- a) VIP
- b) Acetylcholine
- c) Substance P

d) Serotonin

Question 9:

Which of the following neurotransmitters acts on the muscle, behind a bolus of food, in the process of peristalsis?

- a) NO
- b) VIP
- c) Serotonin
- d) Substance P

Question 10:

All of the following stimulates gastric motility except:

- a) Enkephalin
- b) Angiotensin
- c) Substance P
- d) Substance K

Question 11:

Gastric emptying is mainly regulated by which of the following mechanisms?

- a) Neural reflexes
- b) Enteric reflexes
- c) Local hormones in stomach
- d) Local hormones in duodenum

Question 12:

Which among the following increases gastric emptying?

- a) Secretin
- b) CCK
- c) GIP

- d) Distension of stomach

Question 13:

Which of the following does not delay gastric emptying?

- a) Fat in duodenum
- b) Acid in duodenum
- c) Gastrin
- d) Secretin

Question 14:

In which of the following reflexes, gastric emptying is delayed when acidic content enters the duodenum?

- a) Gastrocolic
- b) Enterocolic
- c) Gastroileal
- d) Enterogastric

Question 15:

Enterogastric inhibitory reflexes are stimulated by all of the following except:

- a) Alkaline content of small intestine
- b) Hyperosmolarity of chyme
- c) Distension of duodenum
- d) Duodenal mucosal irritation

Question 16:

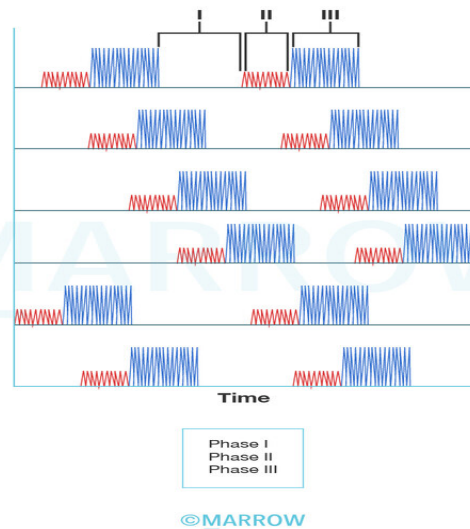
A concerned mother brings her infant to the OPD for advice regarding nutrition. She says, soon after every meal, her kid defecates. What is the cause of colonic contraction in this situation?

- a) Gastroileal reflex
- b) Enterogastric reflex

- c) Increased circulation of CCK
- d) Gastrocolic reflex

Question 17:

The following image shows which type of GI motility?



- a) Peristalsis
- b) Segmentation and mixing
- c) Migrating motor complex
- d) BER

Question 18:

Migrating motor complexes occurs at the rate of _____ and at an interval of _____

- a) 90 cm/min, 5 min
- b) 5 cm/min, 100 min
- c) 90 cm/sec, 5 min
- d) 5 cm/sec, 100 min

Question 19:

Which of the following is true about gastroduodenal motor activity?

- a) Migrating motor complex/MMC in the fasting state lasts for 50 minutes
- b) Duodenal slow waves are limited to the duodenum
- c) From the duodenum, the MMC moves distally at 5-10 cm/min
- d) The amplitude of contraction decreases to a minimum in phase III of MMC

Question 20:

Which of the following does not increase small intestinal motility?

- a) Secretin
- b) Gastroenteric reflex
- c) CCK
- d) Serotonin

Question 21:

Which part of the GIT has the longest transit time?

- a) Stomach
- b) Jejunum
- c) Colon
- d) Ileum

Question 22:

The maximum motility of the colon is seen in which of the following part?

- a) Ascending colon
- b) Transverse colon
- c) Descending colon
- d) Sigmoid colon

Question 23:

A patient is undergoing sigmoidoscopy as a part of his routine colon cancer screening. While doing the procedure, the rectum is inflated with gas. Which part of the intestine would you likely see increased peristalsis in this patient?

- a) Whole colon
- b) Proximal colon
- c) Distal colon
- d) Whole intestine

Question 24:

Which of the following controls the secretions of GIT?

- a) Myenteric plexuses
- b) Auerbach plexuses
- c) Meissner plexuses
- d) Extrinsic nerve

Question 25:

You will feel the first urge to defecate when rectal pressure is increased to:

- a) 12 mmHg
- b) 14 mmHg
- c) 16 mmHg
- d) 18 mmHg

Answer Key

Question No.	Correct Option
1	c
2	d
3	b
4	b
5	d
6	b
7	b
8	a
9	d

10	a
11	b
12	d
13	c
14	d
15	a
16	d
17	c
18	b
19	c
20	a
21	c
22	d
23	c
24	c
25	d

Detailed Explanations

Solution to Question 1:

The interstitial cells of Cajal undergo cyclic changes in membrane potential due to unique ion channels that periodically open and produce inward (pacemaker) currents that generate slow-wave activity.

Interstitial cells of Cajal are the pacemaker cells of the GIT that initiate the basic electrical rhythm (BER). These are stellate mesenchymal pacemaker cells with smooth muscle-like features.

In the stomach and the small intestine, these cells are located in the circular smooth muscle layer near the myenteric plexus. In the colon, they are at the submucosal border of the circular muscle layer.

The slow waves usually do not by themselves cause muscle contraction in most parts of the gastrointestinal tract, except perhaps in the stomach. Instead, they mainly excite the appearance of intermittent spike potentials, and the spike potentials in turn actually excite the muscle contraction.

Note: Gastrointestinal stromal tumors arise from the interstitial cells of Cajal.

Solution to Question 2:

Among the given options, the least frequency of slow waves is seen in the stomach (4/min).

The Basal electrical rhythm (BER) also called slow-wave, is a rhythmic fluctuation of membrane potential in the gastrointestinal smooth muscle. It is initiated by the interstitial cells of Cajal. The membrane potential fluctuates between -65mV and -45mV.

The BER is seen in all sites of gastrointestinal smooth muscle except in the esophagus and proximal stomach.

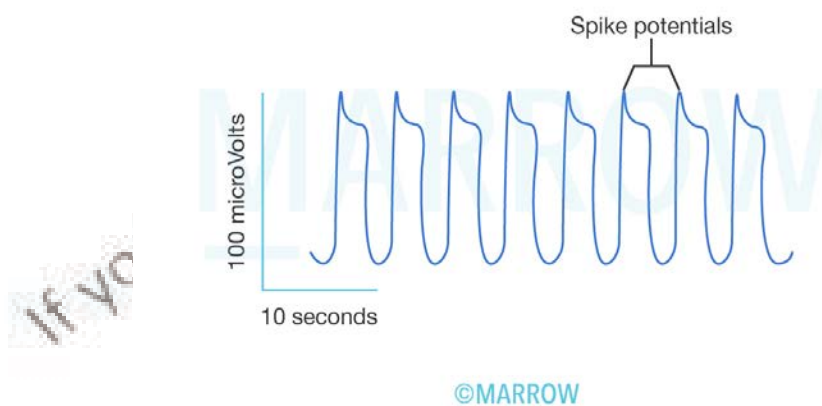
The rate of BER in various sites is as follows:

- Stomach: 4/min
- Duodenum: 12/min
- Distal ileum: 8/min
- Cecum: 2/min
- Sigmoid colon: 6/min

The BER does not cause contraction of the smooth muscle. However, when the BER is superimposed by the spike potentials, it increases muscle tension.

Acetylcholine causes an increase in the number of spike potentials and muscle tension. Epinephrine causes a decrease in the number of spike potentials and muscle tension.

Basal electrical rhythm of the GIT



Muscle contraction can occur only when the motor stimulus is provided during the depolarisation part of the BER.

Thus, BER does not cause muscle contraction by itself, but controls the rate and organization sites of the muscle contractions, as in coordination of the peristaltic activity and setting the rhythm of segmentation.

Solution to Question 3:

Among the given options, the frequency of 'slow waves' is maximum in the ileum (8/min).

Normal intestinal motility is determined by two patterns of electrical activity - slow waves and spike potentials.

Slow waves, also known as basic electrical rhythm are partial depolarizations of smooth muscle cells, spreading along the GIT. They originate from the interstitial cells of Cajal. The smooth muscles are in a hyperpolarized state, having a resting membrane potential of -50 to -60mV.

These waves are not action potentials themselves and they do not cause any contractions. Slow waves raise the membrane potential by 5-15mV, bringing it closer to the threshold enabling spike potential (true action potential) to occur. Therefore, they set the basic rhythm of GIT.

Solution to Question 4:

The facial nerve is a part of an efferent limb (and not afferent) of the swallowing reflex.

Swallowing/ deglutition reflex	
Afferents	Trigeminal nerve Glossopharyngeal nerve Vagus nerve
Centre	Nucleus of the tractus solitarius Nucleus ambiguus
Efferents	Trigeminal nerve Facial nerve Hypoglossal nerve

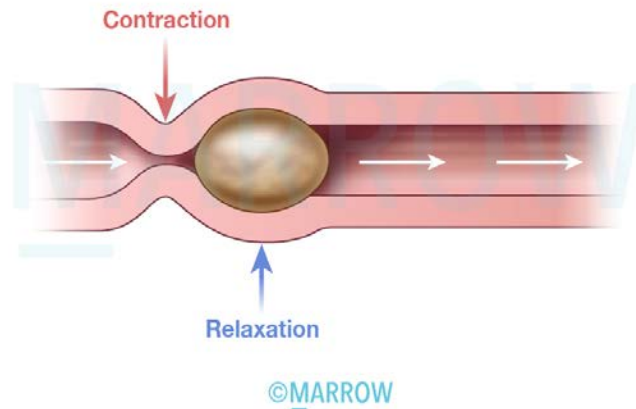
Solution to Question 5:

Chyme is propelled forward in the intestine by the peristaltic movement.

Chyme is a thick semifluid mass of partially digested food and digestive secretions that is expelled by the stomach into the duodenum.

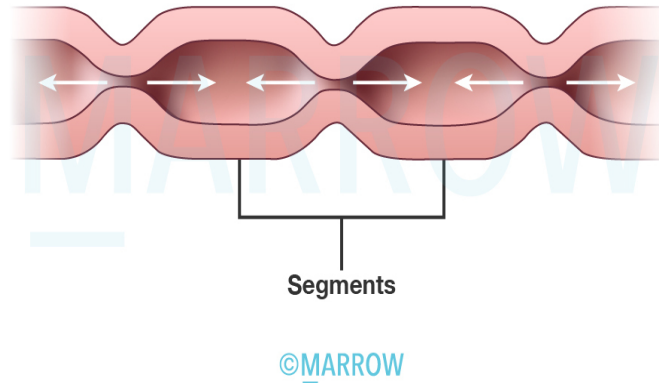
Peristalsis is a propulsive movement, which consists of a segment of contraction behind and relaxation in front.

Peristalsis



Segmentation consists of a segment of bowel contracting at both ends, and then a second contraction occurs in the center of the segment to force the chyme both backward and forward. It is the small intestine's primary method of motility during digestion of a meal.

Segmentation



Migrating motor complexes (MMC) are the peristaltic motor patterns that occur during fasting. They migrate from the stomach to the distal ileum.

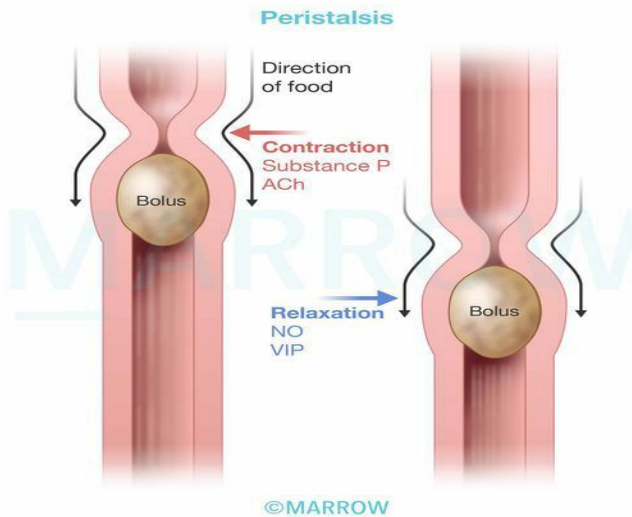
Haustrations are the characteristic saclike segments seen in the large intestine.

Solution to Question 6:

The major initiating response for peristalsis is local stretching of the gut.

Peristalsis is a reflex response that is initiated when the gut wall is stretched by the contents of the lumen. It occurs in all parts of the gastrointestinal tract from the esophagus to the rectum.

The stretch initiates a circular contraction behind the stimulus and an area of relaxation in front of it. The wave of contraction then moves in an oral-to-caudal direction. This propels the contents of the lumen forward at rates that vary from 2 to 25 cm/s.



Solution to Question 7:

Parasympathetic stimulation results in increased peristalsis.

Generally, the stimulation of parasympathetic nerves (PNS) causes an increase in the activity of the entire enteric nervous system. This, in turn, enhances the activity of most gastrointestinal functions.

PNS stimulation causes:

- Contraction of smooth muscle in the wall
- Relaxation of sphincters
- ↑ Salivary secretion
- ↑ Gastric secretion
- ↑ Pancreatic secretion

SNS stimulation causes:

- Relaxation of smooth muscle in the wall
- Contraction of sphincters

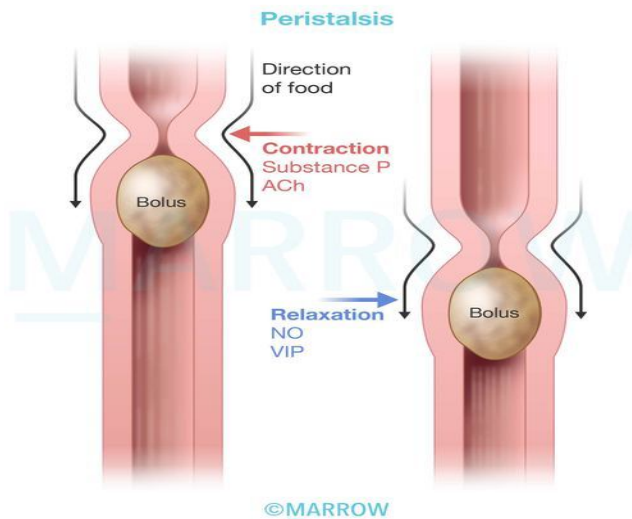
Solution to Question 8:

VIP is responsible for anterograde dilatation of the small intestine.

Peristalsis involves the contraction of a segment of intestine behind the bolus of food and relaxation of a segment of the intestine in front of the bolus of food.

Cholinergic neurons passing in anterograde direction produce NO and VIP which cause relaxation of a segment of the intestine in front of the bolus.

Cholinergic neurons passing in retrograde direction produce substance P and acetylcholine which cause contraction of a segment of the intestine behind the bolus.



Note: In achalasia cardia, there is decreased release of NO and VIP in the distal part of the esophagus.

Solution to Question 9:

Substance P and acetylcholine are released from neurons behind the bolus of food, causing smooth muscle contraction, which propels the food downwards.

NO and VIP are released by neurons in the anterograde direction from the bolus which produces relaxation in front of the stimulus.

Solution to Question 10:

Enkephalin inhibits the motility of the stomach.

ECL: enterochromaffin-like

Function	Source
Stimulate secretion	

Function	Source
Gastrin	G-cells
Histamine	ECL cells
Acetylcholine	Neurones
Gastrin-releasing peptide	Neurones and mucosa
Cholecystokinin (CCK)	Duodenal endocrine cells
Inhibit secretion	
Somatostatin	D-cells and neurones
Secretin	Duodenal endocrine cells
Enteroglucagon	Small intestinal endocrine cells
Prostaglandins	Mucosa
Neurotensin	Neurones
GIP	Duodenal and jejunal endocrine cells
PYY	Small intestinal endocrine cells
Stimulate motility	
Acetylcholine	Neurones
5-HT	Neurones
Histamine	ECL cells
Substance P	Neurones
Substance K	Neurones
Motilin	Neurones
Gastrin	G-cells
Angiotensin	
Inhibit motility	
Somatostatin	D-cells and neurones
VIP	Neurones
Nitric oxide	Neurones and smooth muscle
Noradrenaline	Neurones
Enkephalin	Neurones
Dopamine	Neurones

Solution to Question 11:

Gastric emptying is mainly regulated by enteric reflexes.

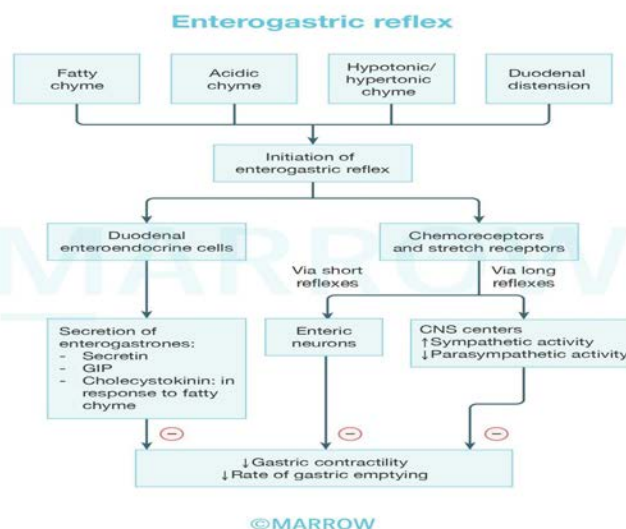
Regulation of gastric emptying is mainly by the inhibitory feedback signals from the duodenum that includes enterogastric reflex and hormonal feedback by cholecystokinin.

Enterogastric reflex: When food enters the duodenum, multiple nervous reflexes are initiated from the duodenal wall. They strongly inhibit the pyloric pump propulsive contractions and increase the tone of the pyloric sphincter.

These reflexes are mediated by three routes:

- Directly from the duodenum to the stomach through the enteric nervous system in the gut wall (major role)
- Through inhibitory sympathetic nerve fibers to the stomach (minor role)
- Vagus nerves to the brain stem where they inhibit the normal excitatory signals transmitted to the stomach (minor role)

Hormonal feedback: Cholecystokinin is released from the mucosa of the jejunum in response to fatty substances in the chyme. This hormone acts as an inhibitor to block increased stomach motility caused by gastrin.



Solution to Question 12:

The distension of the stomach increases gastric emptying.

Distension activates the mechanoreceptors that trigger long and short reflexes in the stomach. This increases the force of antral contraction and gastric emptying.

Solution to Question 13:

Gastrin increases gastric emptying. It does not delay it.

Gastrin has mild to moderate stimulatory effects on motor functions in the body of the stomach. Thus, gastrin likely promotes stomach emptying.

Solution to Question 14:

In the enterogastric reflex, gastric emptying is delayed when acidic content enters the duodenum. When the food enters the duodenum, multiple reflexes are initiated from the duodenal wall which passes back to the stomach. These reflexes slow or stop the stomach emptying if the volume of the chyme in the duodenum becomes too much. These reflexes are called enterogastric reflexes.

This reflex can be triggered due to the following reasons:

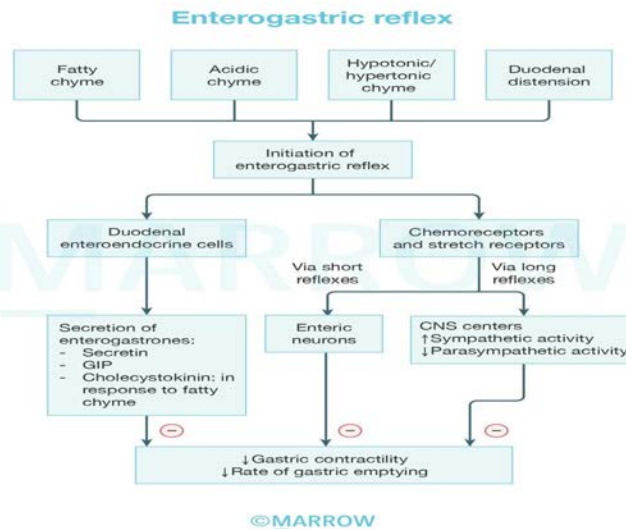
- Presence of irritants and acids in duodenal chyme (most sensitive factor)
- Distension of duodenum due to the presence of an excessive volume of chyme
- Rapid entry of hypertonic or hypotonic fluids into the duodenum
- Presence of breakdown products of protein digestion and sometimes fat

Solution to Question 15:

Enterogastric inhibitory reflexes are stimulated by acidic contents, not alkaline.

The duodenal factors that can initiate enterogastric inhibitory reflexes (enterogastric reflexes) are:

- Distention of the duodenum
- Presence of any irritation of the duodenal mucosa
- Acidity of the duodenal chyme
- Osmolality of the chyme - either hypotonic or hypertonic fluid initiate enterogastric inhibitory reflexes. Thus, the flow of non-isotonic fluids into the small intestine is regulated. This prevents rapid changes in electrolyte concentrations in the extracellular fluid during the absorption of the intestinal contents.
- Presence of certain breakdown products in the chyme, especially proteins and to a lesser extent, of fats



Solution to Question 16:

The cause of colonic contraction in the given situation is due to gastrocolic reflex.

Distension of the stomach by food initiates contractions of the rectum and frequently, a desire to defecate. This response is called the gastrocolic reflex. Because of this response, defecation after meals is common in children. This response may be amplified by the action of gastrin on the colon.

Solution to Question 17:

The given image shows the migrating motor complex.

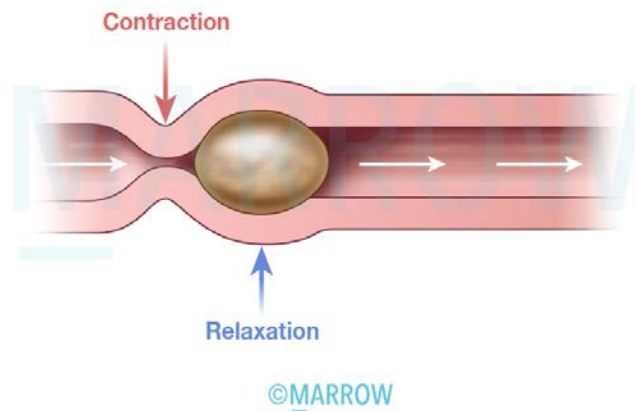
The image shows 3 phases:

- Phase I - quiescent period
- Phase II - period of irregular electrical & mechanical activity
- Phase III - ends with bursts of regular activity

This occurs during the inter-digestive period at intervals of about 100 min and passes in the aboral direction at the rate of 5 cm/min. Its main function is to clear intestinal contents and prepare for the next meal.

Option A: Peristalsis is a propulsive movement, which consists of a segment of contraction behind and relaxation in front.

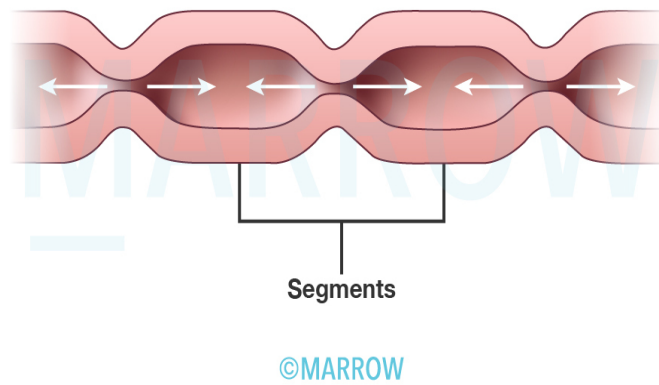
Peristalsis



Option B: Segmentation and mixing help in mixing intestinal contents with digestive juices and provides time for digestion and absorption.

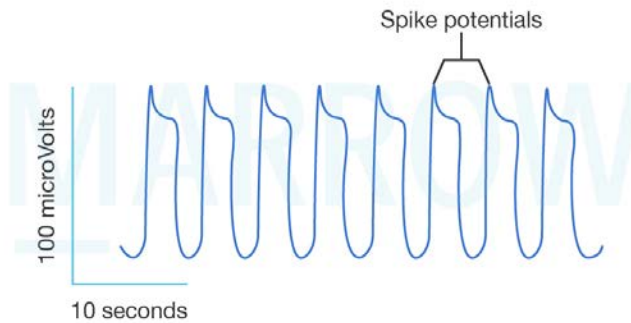
It consists of a segment of bowel contracting at both ends, and then a second contraction occurs in the center of the segment to force the chyme both backward and forward.

Segmentation



Option D: Basal Electrical Rhythm is fluctuation in the resting membrane potential of smooth muscle cells of the intestine as shown below.

Basal electrical rhythm of the GIT



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Solution to Question 18:

Migrating motor complex (MMC) migrates from the stomach to the distal ileum at a rate of about 5 cm/min and also occurs at intervals of approximately 100 minutes.

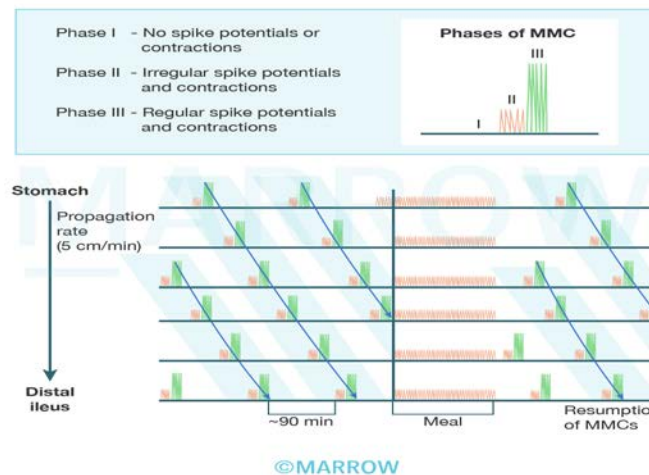
MMC is a type of gastrointestinal motility that occurs during fasting between periods of digestion. It is initiated by motilin. It helps to clear the stomach and small intestine of luminal contents in preparation for the next meal. Gastric secretion, bile flow, and pancreatic secretion increase during each MMC.

After ingestion of a meal, secretion of motilin is suppressed and the MMC is abolished.

Each cycle of MMC has 3 phases

- Phase I - quiescent period
- Phase II - period of irregular electrical & mechanical activity
- Phase III - ends with bursts of regular activity

Phases of Migrating Motor Complex (MMC)



Note: Disorders of MMC have been associated with gastroparesis, intestinal pseudo-obstruction, and small intestinal bacterial overgrowth.

Solution to Question 19:

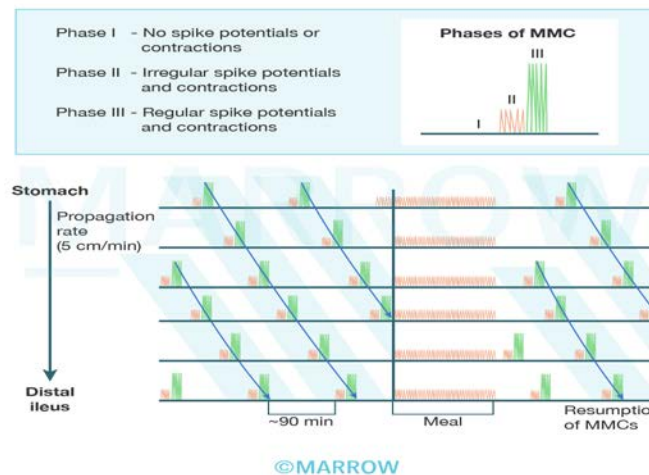
From the duodenum, the migrating motor complex (MMC) moves distally at 5–10 cm/min, reaching the terminal ileum after 1.5 hours.

In the fasting state, there is a period of quiescence in the small bowel, lasting for approximately 40 minutes (phase I). This is followed by a series of waves, also lasting for about 40 minutes, propagating from the fundus of the stomach in a caudal direction at a rate of about 3 per minute (phase II).

Duodenal slow waves are generated in the duodenum at a rate of about 10 per minute, which carry down the small bowel.

The amplitude of these contractions increases to a maximum in phase III, which lasts for about 10 minutes. This 90-minute cycle of activity is then repeated.

Phases of Migrating Motor Complex (MMC)



Solution to Question 20:

Secretin inhibits small intestinal motility.

Factors that increase the peristaltic activity of the small intestine:

- Hormonal - gastrin, CCK, insulin, motilin, and serotonin.
- Neural - duodenal distension by chyme, gastroenteric reflexes, gastroileal reflexes, parasympathetic (Ach) stimulation.

Factors that decrease the peristaltic activity of the small intestine:

- Hormonal - secretin and glucagon.
- Neural - sympathetic stimulation.

Gastroileal reflex - On reaching the ileocecal valve, the chyme is sometimes blocked for several hours until the person eats another meal. At that time, a gastroileal reflex intensifies peristalsis in the ileum and forces the remaining chyme through the ileocecal valve into the cecum of the large intestine.

Solution to Question 21:

The longest transit time is seen in the colon.

Average transit time:

Segment	Transit time
Esophagus	2-3 seconds

Segment	Transit time
Stomach	2-5 hours
Small intestine	3-6 hours
Caecum	4 hours
First third of colon	6 hours
Second third of colon	9 hours
Sigmoid colon	12 hours

Solution to Question 22:

The maximum motility of the colon is seen in the sigmoid colon.

The rate of BER at sigmoid colon is 6/min.

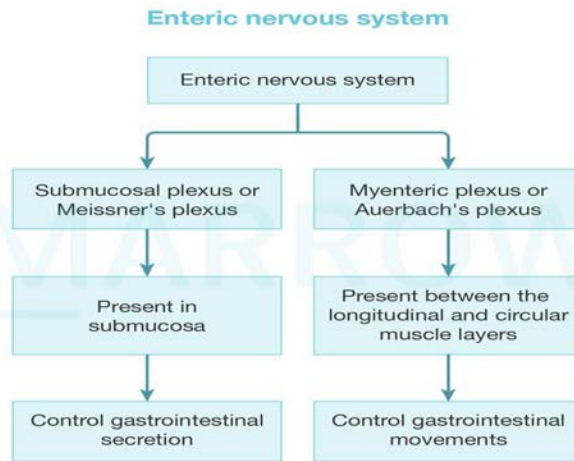
Solution to Question 23:

While doing sigmoidoscopy if the rectum is inflated with gas, increased peristalsis is seen in the distal colon.

The distention of the rectal wall initiates afferent signals. These signals spread through the myenteric plexus to initiate peristaltic waves in the descending colon, sigmoid (distal colon), and rectum.

Solution to Question 24:

Secretions of GIT are controlled by the Meissner plexus.



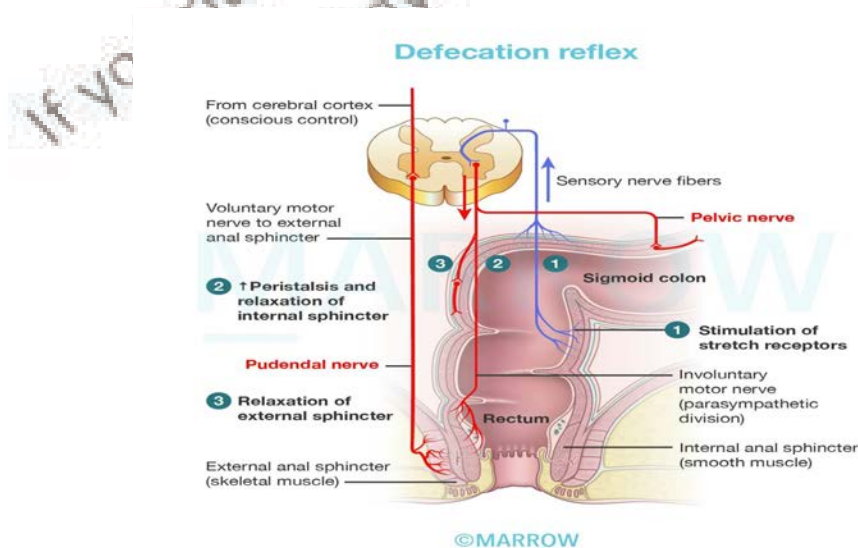
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Solution to Question 25:

The first urge to defecate occurs when rectal pressure is increased to 18 mm Hg.

When this pressure reaches 55 mm Hg, the external and the internal sphincter relax, and there is reflex expulsion of the contents of the rectum.

Defecation reflex: Distension of the rectum and colon stimulates stretch receptors in the rectal wall. Then, the defecation reflexes travel from the colon and rectum to the spinal cord and back again to produce the powerful colonic, rectal, and abdominal contractions required for defecation.



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Glomerular Filtration Rate, Renal Blood Flow, and Renal Clearance

Question 1:

Principal cells and intercalated cells are present in which part of the nephron?

- a) Distal collecting tubule and collecting duct
- b) Glomerulus and proximal collecting tubule
- c) Proximal collecting tubule and loop of Henle
- d) Loop of Henle and distal collecting tubule

Question 2:

How does DCT differ from PCT?

- a) Brush border in luminal membrane
- b) Carbonic anhydrase in its luminal membrane
- c) 'Tight' tight junctions
- d) No $\text{Na}^+ - \text{K}^+$ ATPase in its basolateral membrane

Question 3:

In comparison to cortical nephron, juxtamedullary nephron_____

- a) Are more in number
- b) Has long loop of Henle
- c) Has low filtration rate
- d) Have no vasa recta

Question 4:

About what percentage of the cardiac output do the kidneys receive?

- a) 5%
- b) 10%

- c) 15%
- d) 20%

Question 5:

What is the normal resting GFR?

- a) 125 ml/min
- b) 90 ml/min
- c) 60 ml/min
- d) 150 ml/min

Question 6:

What is the renal blood flow in a resting adult?

- a) 1-1.5 L/min
- b) 1.5-2 L/min
- c) 2-2.5 L/min
- d) 2.5-3 L/min

Question 7:

What is the normal renal filtration fraction?

- a) 0.3
- b) 0.4
- c) 0.2
- d) 1.5

Question 8:

Which of the following does not form filtration barrier?

- a) Podocytes
- b) Mesangium
- c) Endothelial cell

d) Basement membrane

Question 9:

Two particles share the same diameter. The one which will pass through the glomerular membrane has _____

- a) Positive charge
- b) Negative charge
- c) No charge
- d) Filtration across GBM does not depend on charge

Question 10:

What size of neutral substances are permitted by the filtration barriers?

- a) 8-10 nm
- b) 4-8 nm
- c) 8-10 μm
- d) 4-8 μm

Question 11:

Which of the following is not a part of the juxtaglomerular apparatus?

- a) Lacis cells
- b) Macula densa
- c) JG cells
- d) Glomerular mesangial cells

Question 12:

In patients with renal diseases, albumin is the first to appear in the urine. What is the reason behind this?

- a) Its high concentration in plasma
- b) Its effective molecular diameter

- c) Its high albumin: globulin ratio
- d) Epithelial cells are sensitive

Question 13:

Calculate the net filtration pressure across the GBM for a patient with Glomerular ultrafiltration coefficient = 6 Glomerular hydrostatic pressure = 45 Glomerular capillary oncotic pressure = 20 Bowman's capsule hydrostatic pressure = 10 Bowman's capsule oncotic pressure = 0

- a) 90 mmHg
- b) 60 mmHg
- c) 30 mmHg
- d) 15 mmHg

Question 14:

How much is the inulin clearance?

- a) 55 ml/ min
- b) 625 ml/ min
- c) 125 ml/ min
- d) 40 ml/min

Question 15:

In a patient, urine flow rate is 10 mL/min; plasma inulin is 2 mg/mL; and urine inulin is 25 mg/mL. Which of the following is true assuming GFR to be normal?

- a) Inulin clearance = GFR
- b) Inulin clearance $>$ GFR
- c) Inulin clearance $<$ GFR
- d) GFR cannot be calculated

Question 16:

A patient's urine flow rate is 2 ml/min. What is the clearance of a substance x whose plasma concentration is 10mg/mL and in urine is 100mg/mL?

- a) 0.2 ml/min
- b) 2 ml/min
- c) 20 ml/min
- d) 200 ml/min

Question 17:

What is the effect of sympathetic stimulation on GFR?

- a) ↑ GFR with afferent arteriolar dilation
- b) ↑ GFR with efferent arteriolar dilation
- c) ↓ GFR with afferent arteriolar constriction
- d) ↓ GFR with Efferent arteriolar constriction

Question 18:

If a patient is on a high protein diet, what will be the effect on the GFR?

- a) Increases GFR
- b) Decreases GFR
- c) Has no effect
- d) Is unpredictable

Question 19:

Which of the following would cause an increase in both GFR and RPF?

- a) Hyperproteinemia
- b) Constriction of efferent arteriole
- c) Dilation of efferent arteriole
- d) Dilation of the afferent arteriole

Question 20:

Which of the following causes relaxation of the mesangial cells?

- a) cAMP
- b) Endothelin
- c) PGF₂
- d) Vasopressin

Question 21:

What is the effect of prostaglandins on renal blood flow?

- a) Increase blood flow to cortex and medulla
- b) Increase blood flow to cortex and decrease blood flow to medulla
- c) Decrease blood flow to cortex and increase blood flow to medulla
- d) Decrease blood flow to cortex as well as medulla

Question 22:

Which part of the proximal tubule is most susceptible to ischemic injury?

- a) S₁
- b) S₂
- c) S₃
- d) S₄

Question 23:

What is the effective renal plasma flow?

- a) 425 mL/min
- b) 625 mL/min
- c) 525 mL/min
- d) 725 mL/min

Question 24:

Which of the following formulae is correct?

- a) $ERPF = RPF / \text{Extraction ratio}$
- b) $RPF = ERPF / \text{Extraction ratio}$
- c) $ERPF = \text{Extraction ratio} / RPF$
- d) $RPF = \text{Extraction ratio} / ERPF$

Question 25:

In a patient with renal disease, you want to measure the filtration fraction. Which of the following substances will you use?

- a) Inulin and mannitol
- b) Urea and diodrast
- c) PAH and phenol red
- d) Inulin and PAHA

Question 26:

According to myogenic hypothesis of renal autoregulation the afferent arterioles contract in response to stretch-induced by_____

- a) NO
- b) Noradrenaline
- c) Opening of calcium channels
- d) Adenosine

Question 27:

Tubuloglomerular feedback is mediated by the sensing of_____

- a) Increased Na + concentration in thick ascending loop of Henle
- b) Increased Cl - concentration in thick ascending loop of Henle
- c) Increased Na + concentration in macula densa
- d) Increased Cl- concentration in macula densa

Question 28:

Renal autoregulation is absent in which of the following cases?

- a) Denervated kidney
- b) Isolated, perfused kidney
- c) Paralysis of vascular smooth muscle
- d) All of the above

Question 29:

Kidney responds to shock by _____.

- a) Increasing renal blood flow
- b) Increasing afferent arteriole resistance
- c) Increasing GFR
- d) Both B and C

Question 30:

What is the function of tubuloglomerular feedback?

- a) Regulate GFR
- b) Decrease plasma sodium
- c) Increase plasma volume
- d) Decrease tubular secretion

Question 31:

The concentration of a substance is 2 mg % in afferent arteriole and 0 mg% in the efferent arteriole of a patient. This substance is

- a) Freely filtered
- b) Secreted
- c) Absorbed in PCT
- d) Impermeable to loop of henle

Question 32:

The protein in the glomerular basement membrane responsible for charge-dependent filtration is?

- a) Fibronectin
- b) Heparan sulfate
- c) Collagen Type IV
- d) Entactin

Question 33:

Efferent arteriolar constriction causes all except:

- a) Decrease in GFR
- b) Decreases renal blood flow
- c) Decreases oncotic pressure in glomerular capillaries
- d) Increases hydrostatic pressure in glomerular capillaries

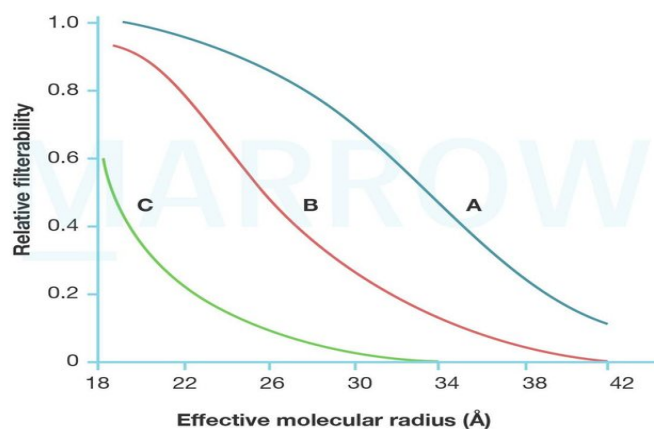
Question 34:

GFR is increased by all except

- a) Increased renal blood flow
- b) Efferent arteriolar constriction
- c) Ureteric stone
- d) Decreased oncotic pressure

Question 35:

This is the graph depicting the filterability of dextran through the kidney, 1 representing complete filtration and 0 representing no filtration. Choose the substance A, B, C respectively.



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- a) Polycationic, Neutral, Polyanionic
b) Polyanionic, Neutral, Polycationic
c) Neutral, Polycationic, Polyanionic
d) Polyanionic, Polycationic, Neutral

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	d
5	a
6	a
7	c
8	b
9	a
10	b
11	d
12	b
13	d
14	c

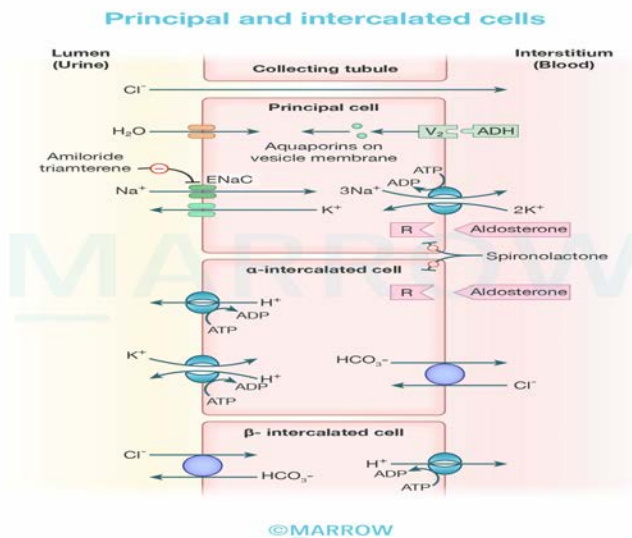
15	a
16	c
17	c
18	a
19	d
20	a
21	b
22	c
23	b
24	b
25	d
26	c
27	c
28	c
29	b
30	a
31	a
32	b
33	c
34	c
35	a

Detailed Explanations

Solution to Question 1:

Principal cells and intercalated cells are present in the distal collecting tubule and collecting duct.

	P cell	I cell
Functions	Na ⁺ reabsorption, Water reabsorption, K ⁺ secretion	Type A-H ⁺ secretion, HCO ₃ ⁻ reabsorption in acidosis. Type B-HCO ₃ ⁻ secretion, acid reabsorption in alkalosis.



Note: Principal cells are the primary sites of action of potassium-sparing diuretics.

Solution to Question 2:

DCT has 'tight' tight junctions, while PCT has 'leaky' tight junctions.

The 'tight' tight junctions can maintain the high electrochemical gradients produced by active transcellular transport. This configuration is used to produce either highly concentrated or diluted secretions, as in the distal nephron. Hence, it can produce urine with osmolarity severalfold higher or lower than plasma.

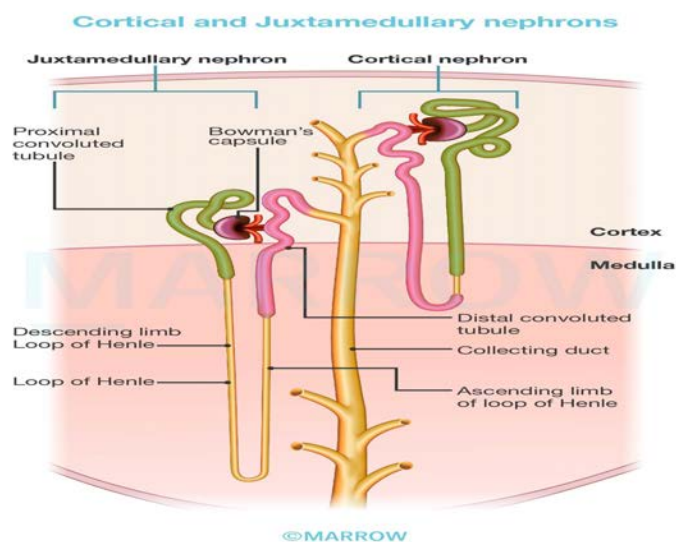
In contrast, epithelia with 'leaky' tight junctions move large amounts of isosmotic fluids. A good example is a transport in PCT.

Na⁺- K⁺- ATPase is present throughout most parts of the nephron.

PCT	DCT
Brush border present in the apical membrane	No brush border
Cytoplasmic carbonic anhydrase (type 2 & type 4) present	Cytoplasmic carbonic anhydrase (type 2 only)
Has 'leaky' tight junctions	Has 'tight' tight junctions
Paracellular transport is possible via leaky tight junctions	No paracellular transport

Solution to Question 3:

In comparison to cortical nephrons, juxtamedullary nephrons are characterized by a long loop of Henle that descends deep into the inner medulla. It is essential for the concentration of urine.



Solution to Question 4:

Kidneys receive approximately 20 to 25% of the cardiac output, which is approximately 1000 mL per minute.

Region	Blood flow (ml/min)
Liver	1500
Kidneys	1260
Brain	750
Skeletal muscle	840
Heart muscles	250

Solution to Question 5:

The GFR in a healthy adult of average size is approximately 125 mL/min.

Glomerular filtration rate (GFR) is the volume of fluid filtered from the renal glomerular capillaries into the Bowman's capsule per unit time.

The GFR is determined by

- Sum of the hydrostatic and colloid osmotic forces across the glomerular membrane, which gives the net filtration pressure
- Glomerular Kf (glomerular ultrafiltration coefficient)

Mathematically, it can be expressed as follows:

$$\text{GFR} = K_f [\text{forces favoring filtration} - \text{forces opposing filtration}]$$

$$\text{GFR} = K_f [\text{PG} + \pi_B - \text{PB} - \pi_G]$$

Forces Favoring Filtration

PG: Glomerular hydrostatic pressure

π_B : Bowman's capsule oncotic pressure

Forces Opposing Filtration

PB: Bowman's capsule hydrostatic pressure

π_G : Glomerular capillary oncotic pressure

Solution to Question 6:

In a resting adult, the kidneys receive 1.2–1.3 L of blood per minute or just under 25% of the cardiac output. Among the given options, 1–1.5 L/min is the closest range.

Renal blood flow is determined by the pressure gradient across the renal vasculature (the difference between the renal artery and renal vein hydrostatic pressures), divided by the total renal vascular resistance. It is expressed as follows:

$$\text{Renal blood flow} = \frac{\text{Renal artery pressure} - \text{Renal vein pressure}}{\text{Total renal vascular resistance}}$$

Note: Renal blood flow = Renal plasma flow / 1 - Hematocrit

Solution to Question 7:

Normal renal filtration fraction (i.e. the fraction of renal plasma flow that is filtered) averages about 0.2.

It means that 20% of the plasma flowing through the kidney is filtered through the glomerular capillaries. The remaining 80% leaves the glomerular capillaries by the efferent arterioles and becomes the peritubular capillary circulation.

It is calculated by the given equation:

$$\text{Filtration fraction} = \frac{\text{GFR}}{\text{RPF}}$$

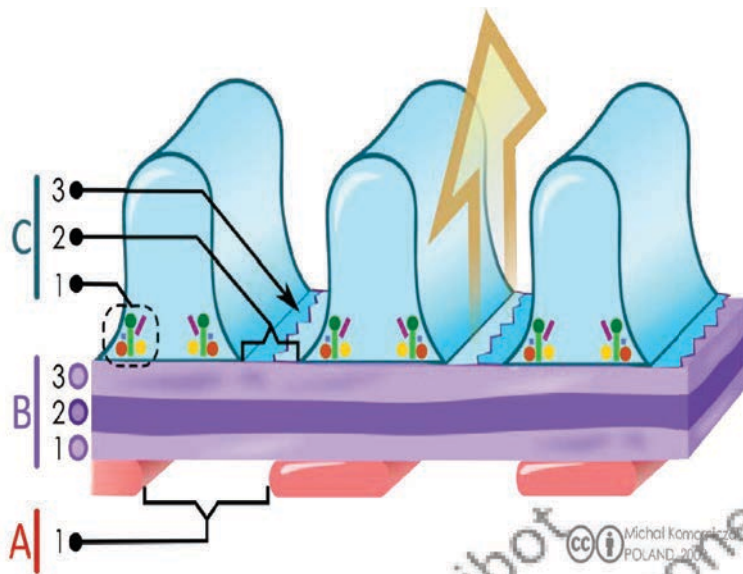
When there is a fall in systemic blood pressure, the GFR falls less than the RPF because of efferent arteriolar constriction, and consequently, the filtration fraction rises.

Solution to Question 8:

Mesangium does not form a filtration barrier. It is made up of mesangial cells, supporting glomerular capillaries.

The filtration barrier consists of:

- Endothelium of the capillary
- Basement membrane:
- Lamina densa - electron-dense layer. It contains type IV collagen
- Lamina rara interna - contains glycosaminoglycan heparan sulfate which has a negative charge
- Lamina rara externa - contains glycosaminoglycan heparan sulfate which has a negative charge
- Layer of epithelial cells (podocytes) surrounding the outer surface of the capillary basement membrane



A. The endothelial cells of the glomerulus; 1. Pore (fenestra)

B. Glomerular basement membrane: 1. Lamina rara interna 2. Lamina densa 3. Lamina rara externa

C. Podocytes: 1. Enzymatic and structural protein 2. Filtration slit 3. Diaphragm.

Solution to Question 9:

For any given molecular radius, positively charged molecules are filtered much more readily than are negatively charged molecules.

This is because the sialoproteins in the glomerular capillary wall are negatively charged, and this negative charge repels the negatively charged substances in the blood.

Studies have showed that for tracer molecules presumed to be of similar size, charges which are filtered are positive charge > neutral charge > negative charge.

Solution to Question 10:

Filtration barriers permit neutral substances that have the size of 4-8 nm.

Neutral substances with diameters:

- <4 nm - freely filtered
- >8 nm - not filtered
- Between 4-8 nm - the permeability is inversely proportional to the diameter

Clinical correlation:

- Genetically abnormal basement membranes: Alport's syndrome (° 3 and ° 4 chains on the long arm of chromosome 2 involved in autosomal dominant and recessive pattern; ° 5 chain on X

chromosome involved in X-linked inheritance)

- Autoimmune response to basement membrane collagen IV (° 3 NC1 domain): Goodpasture's syndrome

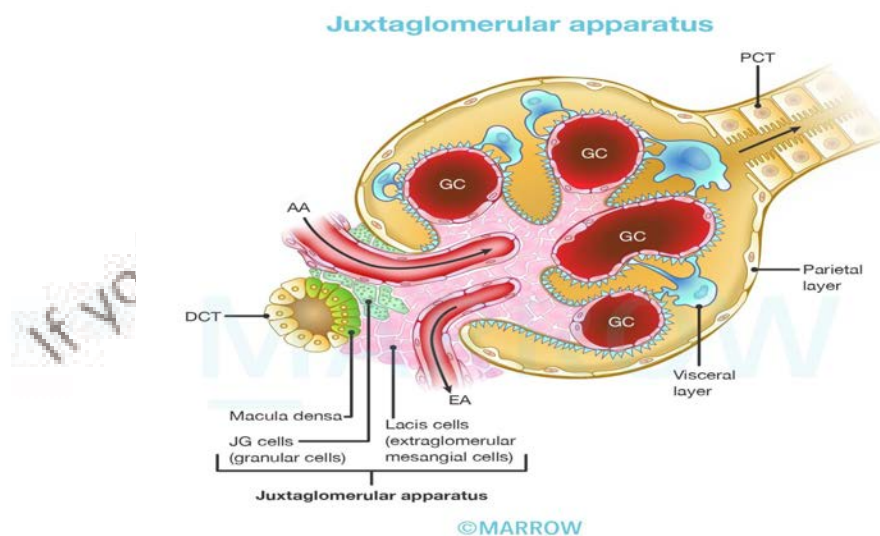
Solution to Question 11:

Glomerular mesangial cells are not a part of the juxtaglomerular apparatus.

Juxtaglomerular apparatus contains:

- Macula densa: Specialized cells at the end of the thick ascending limb of Henle forms the macula densa. They detect sodium concentration of the fluid in the tubule (Chemoreceptor). In response to elevated sodium, the macula densa cells trigger contraction of the afferent arteriole, reducing flow of blood to the glomerulus and the glomerular filtration rate.
- Lacis cells: Extraglomerular mesangial cells situated between afferent and efferent arteriole.
- Granular cells/JG cells: Modified smooth muscle cells of the afferent arteriole, that secrete renin. Fall in pressure in afferent arteriole stimulates renin release (mechanoreceptor), which, in turn, increases blood pressure.

Note: The distal convoluted tubule starts at the macula densa.



Solution to Question 12:

In renal diseases, albumin is the first to appear because of its effective molecular diameter.

In minimal change nephropathy, the negative charges on the basement membrane are lost even before there are noticeable changes in kidney histology. As a result of this loss of negative charges on the basement membranes, some of the lower molecular weight proteins, especially albumin (effective molecular diameter of 7 nm), are filtered and appear in the urine, a condition known as

proteinuria or albuminuria.

Solution to Question 13:

Net filtration pressure = Hydrostatic pressure in the glomerular capillaries - Pressure in Bowman's capsule - Oncotic pressure in the glomerular capillaries.

$$\text{Net filtration pressure} = [P_G - P_B - \pi_G + \pi_B]$$

Therefore,

$$\begin{aligned}\text{Net filtration pressure} &= (45 - 10 - 20 + 0) \\ &= 25 - 10 \\ &= 15 \text{ mmHg}\end{aligned}$$

Note:

Here, just the net filtration pressure is asked and not the GFR.

$$\text{GFR} = K_f [P_G - P_B - \pi_G + \pi_B]$$

Solution to Question 14:

Inulin clearance is equal to 125 ml/min. Since it is used for the measurement of GFR, inulin clearance will be equal to GFR, i.e. 125 ml/min.

Renal clearance is the volume of plasma completely cleared of a substance by the kidneys per unit time.

A higher renal clearance suggests the substance may be cleared almost completely in one pass through the kidneys while a low value describes a substance that may not be eliminated by the kidneys at all.

Clearance refers to the volume of plasma that would be necessary to supply the amount of substance excreted in the urine per unit of time. Stated mathematically,

$$C_x \cdot P_x = U_x \cdot V$$

where C_x is the clearance rate of a substance x

P_x is the plasma concentration of x

U_x is the urine concentration of that x

V is the urine flow rate

Rearranging this equation, clearance can be expressed as

$$\text{Clearance} = \frac{[U]_x \cdot V}{[P]_x}$$

Solution to Question 15:

In the given scenario, inulin clearance = GFR

Inulin clearance is calculated by the formula:

$$C_{\text{inulin}} = \frac{U_{\text{inulin}} V}{P_{\text{inulin}}}$$

Substituting the values given,

- $U_{\text{inulin}} = 25 \text{ mg/mL}$
- $V_{\text{inulin}} = 10 \text{ mL/min}$
- $P_{\text{inulin}} = 2 \text{ mg/mL}$

$$C = \frac{25 \times 10}{2} = 125 \text{ mL/min}$$

Thus, Inulin clearance=GFR (normal GFR= 125 ml/min)

Inulin is a polymer of fructose, clearance of inulin can be used for measurement of GFR, because of the following reasons:

- Freely filtered
- Neither reabsorbed nor secreted in the tubules
- Nontoxic
- Not metabolized by the body

Solution to Question 16:

The clearance in the given scenario is 20 ml/ min.

$$\text{Clearance} = \frac{[U] \times V}{[P] \times}$$

C = Clearance (mL/min)

[U]x = Urine concentration of substance x (mg/mL)=100mg/mL

V= Urine flow rate per minute (mL/min)=2mL/min

[P]x = Plasma concentration of substance x (mg/mL)=10mg/mL

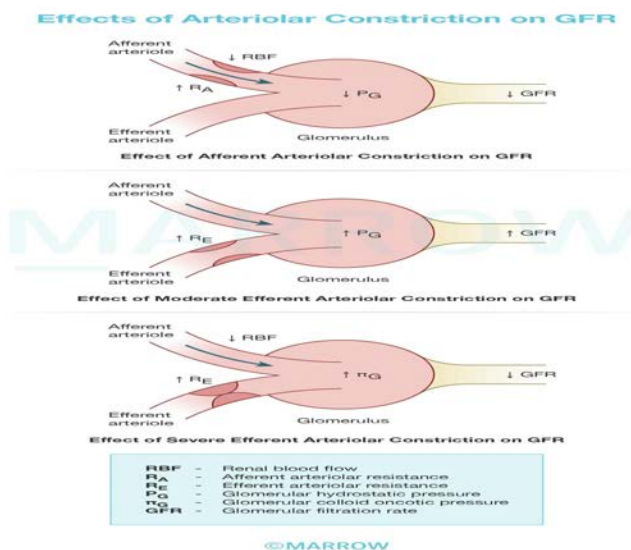
Substituting the values,

$$\text{Clearance} = \frac{100 \times 2}{10} = 20 \text{ mL/min}$$

Solution to Question 17:

GFR decreases with sympathetic stimulation as there is afferent arteriolar constriction.

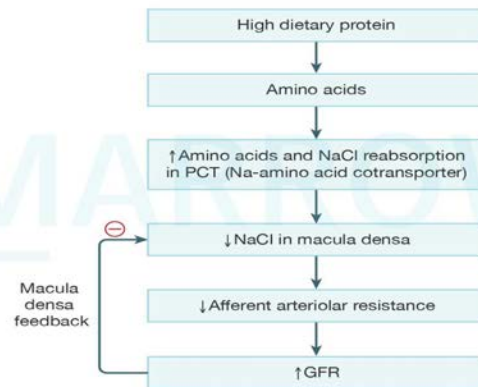
Strong activation of the renal sympathetic nerves constrict the afferent and efferent renal arterioles through the α_1 receptors. As the afferent arterioles have far more α_1 receptors than the efferent, it results in decreased renal blood flow and GFR.



Solution to Question 18:

High protein intake increases GFR.

Effect of protein intake on GFR



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Note: Increase in the plasma protein concentration is not the same as high protein intake. An increase in the plasma protein concentration decreases GFR due to a raised oncotic pressure. A high protein intake raises the amino acid concentration in the blood, not the plasma protein concentration.

Solution to Question 19:

Dilation of the afferent arteriole will increase both GFR and RPF.

- Increase renal plasma flow (RPF) - because renal vascular resistance is decreased
- Increase glomerular filtration rate (GFR) - because glomerular capillary hydrostatic pressure is increased.
- Filtration fraction (FF) remains normal

Dilation of the efferent arteriole will increase RPF but decrease GFR.

Solution to Question 20:

cAMP causes relaxation of the mesangial cells. The other agents mentioned in the options cause contraction of the mesangial cells.

Agents causing relaxation of mesangial cells (mnemonic - DANCE)

- Dopamine
- ANP
- cAMP
- PGE₂

Angiotensin II is an important regulator of mesangial contraction, and there are angiotensin II receptors in the glomeruli.

Solution to Question 21:

Prostaglandins increase blood flow in the renal cortex and decrease blood flow in the renal medulla.

PGE₂ is partially responsible for the vasodilatation of the afferent arteriole, which increases blood flow to the glomerulus. This phenomenon describes why NSAIDs (which inhibit prostaglandin synthesis) result in afferent arteriole vasoconstriction. It can cause a reduced GFR in patients who rely heavily on prostaglandin synthesis to maintain renal blood flow.

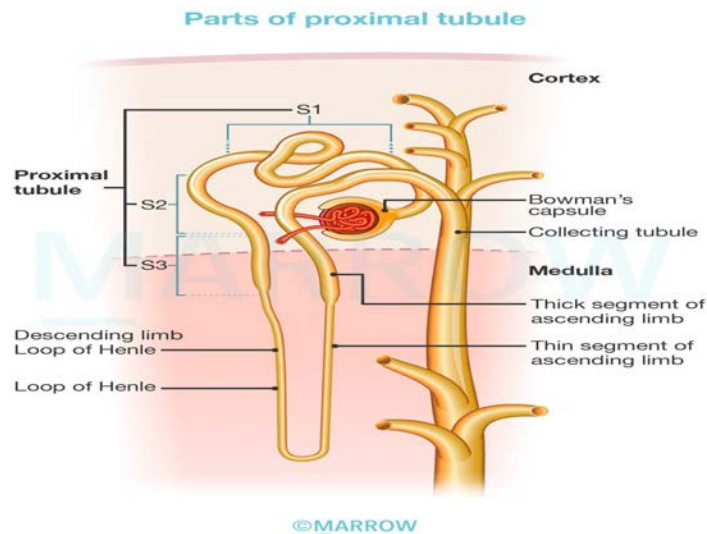
Solution to Question 22:

Regions of the kidney most prone to ischemic injury are the S₃ segment of the proximal tubule and the medullary thick ascending limb of the loop of Henle (mTAL). These tubular areas exist physiologically in relatively lower oxygen conditions.

The S₃ segment, in particular, has a much lower capability of producing energy through glycolysis under anaerobic conditions than the mTAL.

Based on the ultrastructure proximal tubule is divided into three segments:

- S₁ segment: 1st half of PCT
- S₂ segment: 2nd half of PCT and 1st half of proximal straight tubule
- S₃ segment: 2nd half of proximal straight tubule



Solution to Question 23:

In humans, the effective renal plasma flow (eRPF) averages about 625 mL/min.

Measurement of renal plasma flow (RPF) is done by measuring the clearance of para-aminohippuric acid (PAH). PAH is filtered and secreted by the renal tubules, hence it can be used for RPF measurement.

$$RPF = C_{PAH} = \frac{U_{PAH}V}{P_{PAH}}$$

where:

- RPF = renal plasma flow (mL/min or mL/24 hour)
- CPAH = clearance of PAH (mL/min or mL/24 hour)
- UPAH = urine concentration of PAH (mg/mL)

- V = urine flow rate (mL/min or mL/24 hour)
- PPAH = plasma concentration of PAH (mg/mL)

Note: Renal blood flow = Renal plasma flow / (1 - Hematocrit)

Solution to Question 24:

True renal plasma flow (RPF) is obtained by dividing effective renal plasma flow (ERPF) by extraction ratio.

$$RPF = \frac{ERPF}{\text{Extraction Ratio}}$$

where,

$$\begin{aligned} \text{a. ERPF} &= \text{effective Renal Plasma flow} \\ &= \text{Clearance of PAH} \\ &= \frac{U_{(PAH)} V}{P_{(PAH)}} \end{aligned}$$

$$\begin{aligned} \text{b. Extraction Ratio} &= \text{The percentage of PAH removed from the blood} \\ &= 90\% = 0.9 \end{aligned}$$

When we inject para-aminohippuric acid, 90% of it is filtered through the glomerulus at the rate of 625mL per minute - which is the effective renal plasma flow (ERPF). Using that, to we calculate the true RPF, we divide it by the extraction ratio to account for the 10% that is not filtered by dividing the ERPF by 0.9, hence getting the final value of ~ 700mL per minute

When peripheral venous plasma concentration of PAH is used for calculation of RPF, RPF obtained is called effective renal plasma flow (ERPF). This is because PAH concentration is essentially identical to that in the arterial plasma reaching the kidney.

Solution to Question 25:

Inulin and PAHA are used to determine the filtration fraction. The filtration fraction is the ratio of GFR and RPF

Inulin, a polymer of fructose can be used to measure GFR. Para-aminohippuric acid (PAHA) is used to measure RPF.

Solution to Question 26:

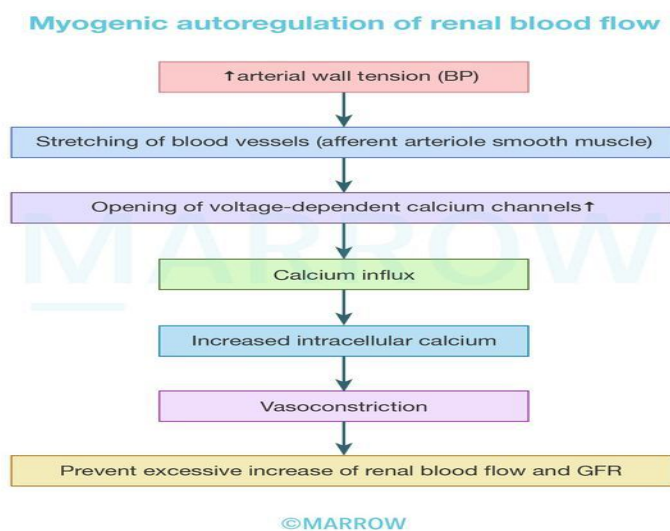
The afferent arterioles contract in response to stretch induced by the opening of calcium channels. The mechanism is called myogenic autoregulation of renal blood flow and GFR.

The stretch of the vascular wall allows increased movement of calcium ions from the extracellular fluid into the cells, causing them to contract. This contraction prevents excessive stretch of the vessel. This prevents excessive increase of renal blood flow and GFR by raising vascular resistance.

This mechanism may be more important in protecting the kidney from hypertension-induced injury.

Autoregulation has two main mechanisms:

- Metabolic mechanism: Tissues produce vasodilatory metabolites that regulate flow e.g. prostaglandins, NO, CO₂, H⁺ and K⁺
- Myogenic mechanism



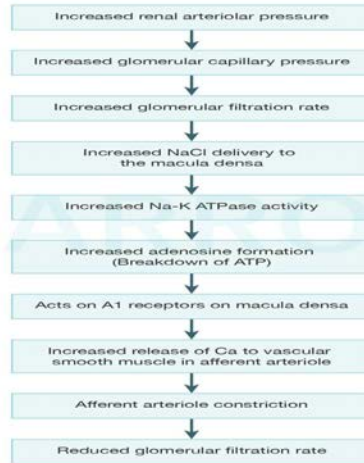
Solution to Question 27:

Tubuloglomerular feedback is mediated by the sensing of increased Na⁺ concentration in macula densa.

Macula densa are the specialized cells present at the distal end of the thick ascending loop of Henle where it reaches the glomerulus. Distal convoluted tubule starts from macula densa. Macula forms part of the juxtaglomerular apparatus.

While both Na⁺ and Cl⁻ are delivered to the macula densa cells, it is the increased concentration of Na⁺ in the cells that increases the Na⁺-K ATPase activity leading to the feedback.

Tubuloglomerular feedback



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Solution to Question 28:

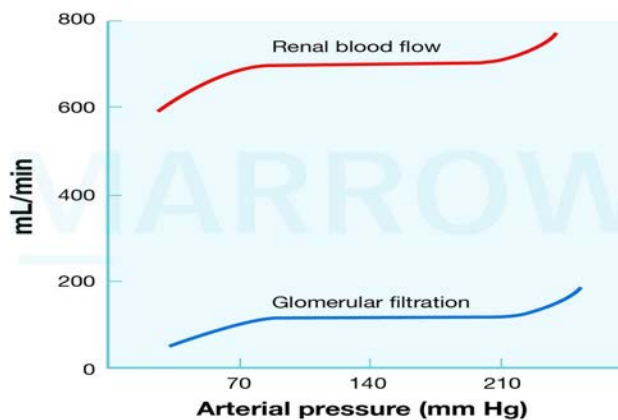
Renal autoregulation is absent in case of paralysis of vascular smooth muscle.

Paralysis of vascular smooth muscles abolishes its action i.e., autoregulation of renal blood flow and GFR. Vascular smooth muscles act through myogenic mechanism of renal autoregulation. Paralysis of vascular smooth muscles can be brought about by administration of drugs.

Renal autoregulation is present in denervated and in isolated, perfused kidneys.

Feedback mechanisms intrinsic to the kidneys keep the renal blood flow and GFR relatively constant, despite marked changes in arterial blood pressure. This relative constancy of GFR and renal blood flow is referred to as autoregulation.

Renal autoregulation



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Solution to Question 29:

Kidney responds to shock by increasing afferent arteriolar resistance.

During shock, both afferent and efferent arterioles in kidney are constricted, however, the efferent vessels are constricted much more.

The glomerular filtration rate and the plasma renal flow is decreased during shock. As plasma renal flow is decreased to a greater extent, the filtration fraction increases. (Option 1 & 3)

Solution to Question 30:

Tubuloglomerular feedback regulates glomerular filtration rate (GFR).

As the rate of flow through the ascending limb of the loop of Henle and first part of the distal tubule increases, glomerular filtration in the same nephron decreases, and, conversely, a decrease in flow increases the GFR.

Solution to Question 31:

A concentration of 0 mg% in efferent arteriole indicates the substance is freely filtered.

Afferent arterioles → glomerulus (filtration) → efferent arterioles → peritubular capillaries (reabsorption+secretion).

Note: The question compares the concentration in afferent and efferent arterioles. Reabsorption and secretion happen between the tubular lumen of the nephron and the peritubular capillaries and not the efferent arteriole.

Solution to Question 32:

The protein in the glomerular basement membrane responsible for charge-dependent filtration is heparan sulfate. It prevents filtration by repelling negatively charged proteins.

Collagen type IV, fibronectin, entactin are structural components of basement membrane.

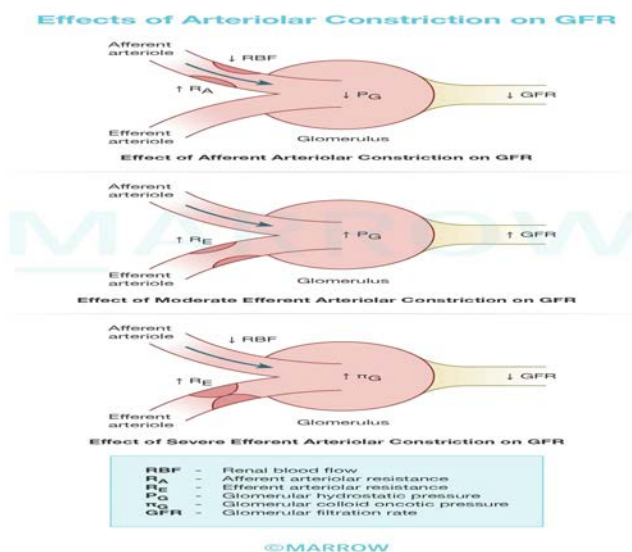
Solution to Question 33:

Efferent arteriolar constriction increases oncotic pressure in glomerular capillaries.

Efferent arteriolar constriction has a biphasic response on GFR depending on the severity of resistance.

Moderate efferent arteriolar constriction causes increased glomerular hydrostatic pressure and slightly increased GFR.

However, in cases of severe efferent arteriolar constriction (more than threefold increase in efferent arteriolar resistance), the renal blood flow is significantly reduced. This results in increased plasma protein concentration, thereby increasing the glomerular colloid oncotic pressure. As the increase in colloid oncotic pressure exceeds the increase in hydrostatic pressure, the net force for filtration decreases, resulting in decreased GFR.



Solution to Question 34:

Ureteric stone causes a reduction in glomerular filtration rate (GFR).

Following ureteral obstruction, there is a progressive fall in glomerular filtration rate (GFR) due to a reduction in a single-nephron glomerular filtration rate (SNGFR) and a reduced number of filtering nephrons.

All the other factors/options increase GFR.

Solution to Question 35:

In the given graph, the order of filterability is A > B > C, which likely indicates that the substances are Polycationic, Neutral, Polyanionic.

The glomerular filtration barrier is negatively charged, which will repel most of the negative charged (anionic substances). Whereas, the cationic substances are easily filtered.

Hence, the order of filterability would be: Polycationic > Neutral > Polyanionic, represented by A, B, and C in the graph shown.

The glomerular filtration barrier consists of:

- Endothelium of the capillary
- Basement membrane:
- Lamina densa - electron-dense layer. It contains type IV collagen
- Lamina rara interna - contains glycosaminoglycan heparan sulfate which has a negative charge
- Lamina rara externa - contains glycosaminoglycan heparan sulfate which has a negative charge
- Layer of epithelial cells (podocytes) surrounding the outer surface of the capillary basement membrane

The capillary endothelium is perforated by thousands of small holes called fenestrae, which measure from 4 nm up to 8nm. Therefore, neutral substances with molecular diameters of less than 4 nm are freely filtered, and filtration of those with diameters of more than 8 nm approaches zero.

However, in addition to the size of the particles, the filtration also depends mainly on the charge of the particle being filtered. The glomerular capillary wall consists of highly negatively charged proteins called sialoproteins, which repel the negatively charged substances (anions). The property can be studied by using charged dextran.

When the filtration of the negatively and positively charged dextrans was studied, the results indicated that the negatively charged dextrans in the blood are repelled, with the result that filtration of anionic substances 4 nm in diameter is less than half that of neutral substances of the same size.

This explains, why albumin despite having an effective molecular diameter of approximately 7 nm, has a very low filtration rate.

Renal Tubular Functions, Urine Concentration and Dilution

Question 1:

What is implied if a drug has more renal clearance than GFR?

- a) Drug is reabsorbed by the renal tubules
- b) Drug is secreted in the renal tubules
- c) Drug is excreted in renal tubules
- d) Drug is neither excreted nor reabsorbed in renal tubules

Question 2:

Choose the correct formula regarding renal physiology.

- a) Filtration fraction = $(RPF / GFR) \times 100$
- b) Filtration rate = $GFR \times \text{urine concentration}$
- c) Excretion rate = $U \times V$
- d) Clearance = UP / V

Question 3:

Choose the correct option.

- a) Inulin is completely secreted and never reabsorbed.
- b) Glucose is completely reabsorbed in DCT.
- c) Creatinine is partially secreted.
- d) PAH at any plasma concentration is completely filtered and secreted.

Question 4:

Renal plasma flow is being calculated using PAH in a patient. The technician inadvertently uses PAH thrice the recommended dose. What will be the estimated effective renal plasma flow?

- a) False high

- b) False low
- c) Both
- d) Correct and is not affected

Question 5:

Which of the following transporter is present in the ascending loop of Henle?

- a) Na^+ - Cl^- co transporter
- b) ENaC channel
- c) Na^+ - 2Cl^- - K^+ co transporter
- d) Na^+ - aminoacid co transporter

Question 6:

The principal site of absorption of Na^+ is _____

- a) Proximal convoluted tubule
- b) Distal convoluted tubule
- c) Thick ascending limb of Loop of Henle
- d) Collecting duct

Question 7:

Potassium is maximally absorbed in _____

- a) PCT
- b) DCT
- c) Collecting duct
- d) Loop of henle

Question 8:

Bicarbonate is reabsorbed across the luminal membrane with the help of _____

- a) Na^+ - HCO_3^- antiport
- b) Na^+ - HCO_3^- Co transport

- c) Na^+ - H^+ exchanger
- d) Na^+ - K^+ exchanger

Question 9:

True about renal tubular transport_____

- a) Maximum reabsorption in distal tubule
- b) Urea is absorbed 100% in PCT
- c) H^+ is absorbed in proximal tubule
- d) Amino acid is reabsorbed in proximal tubule

Question 10:

Which of the following occurs along with glucose transport into the cell?

- a) Sodium symport
- b) Calcium antiport
- c) Potassium transport
- d) Amino acid antiport

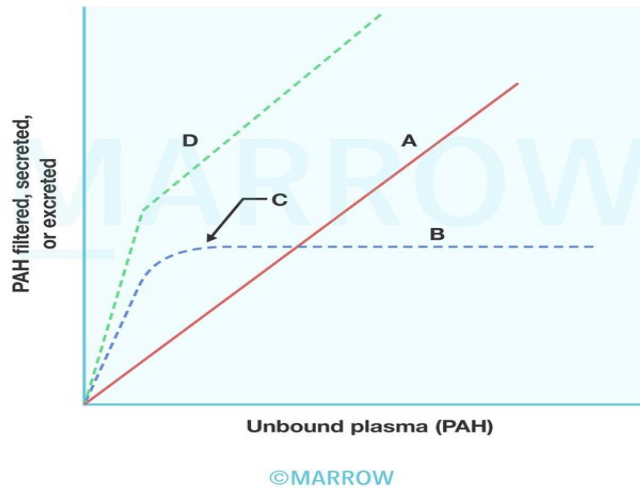
Question 11:

Normal renal threshold (g/L) and transport maxima (mg/min) for glucose respectively are _____, _____

- a) 200, 375
- b) 375, 200
- c) 2, 375
- d) 375, 2

Question 12:

The image of PAH excretion is depicted below. Match the labellings.



- a) a. Filtered, b. Excreted, c. T_m , d. Secreted
- b) a. Filtered, b. Secreted, c. T_m , d. Excreted
- c) a. Secreted, b. Filtered, c. T_m , d. Excreted
- d) a. Excreted, b. Filtered, c. T_m , d. Secreted

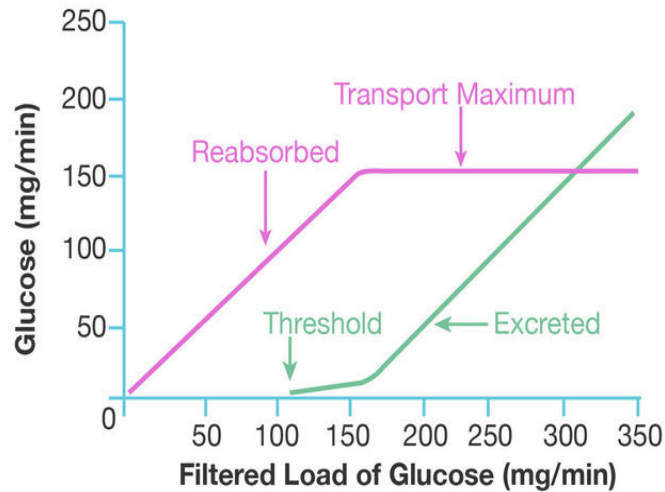
Question 13:

In a patient with uncontrolled diabetes, what is the value of plasma clearance of glucose?

- a) Zero
- b) Equal to inulin clearance
- c) Greater than PAH clearance
- d) Greater than zero

Question 14:

A patient has a blood glucose level of 200 mg/dL and GFR of 90 ml/min. What is the amount of glucose excreted if the transport maximum of the patient is as shown below?



- a) 80 mg/min
- b) 50 mg/min
- c) 40 mg/min
- d) 30 mg/min

Question 15:

Choose the correct statement regarding the concentration of solutes at the end of proximal convoluted tubule

- a) Concentration of glucose in tubular fluid is more than that of plasma
- b) Concentration of urea in tubular fluid is more than that of plasma
- c) Concentration of bicarbonate in tubular fluid is more than that of plasma
- d) Concentration of sodium in tubular fluid is less than that of plasma

Question 16:

The mechanism by which majority of water is reabsorbed in PCT is

- a) Active
- b) Secondary active
- c) Facilitated
- d) Osmosis

Question 17:

A dehydrated patient is excreting hypertonic urine. Absorption of water in which part of the nephron causes this?

- a) Collecting duct
- b) PCT
- c) Ascending loop
- d) Descending loop

Question 18:

A patient with diabetes insipidus is prescribed vasopressin. This drug has its maximum effect on which part of the renal tubule?

- a) Proximal tubule
- b) Distal tubule
- c) Loop of henle
- d) Collecting duct

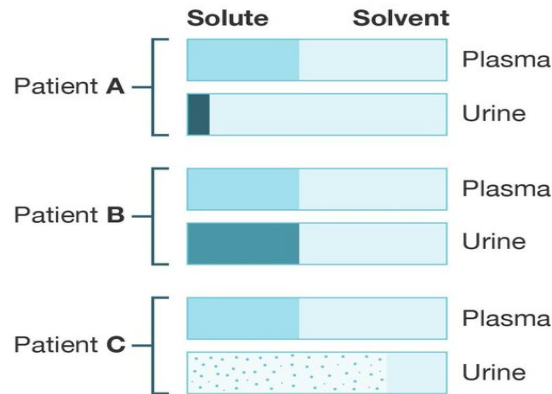
Question 19:

Which of the following is true about free water clearance?

- a) Regulated by ADH
- b) Regulated by aldosterone
- c) Unaffected by furosemide
- d) None of the above

Question 20:

The following picture shows the composition of solute in the urine and plasma sample of 3 patients. Identify the true statement.



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- a) A shows negative free water clearance, B is isotonic, C is positive free water clearance
- b) A shows positive free water clearance B is isotonic C is positive free water clearance
- c) A shows positive free water clearance B is isotonic C is negative free water clearance
- d) A shows negative free water clearance B is isotonic C is negative free water clearance

Question 21:

All are true except that _____.

- a) PCT is permeable to water
- b) Descending limb of loop of henle is permeable to water
- c) Thick ascending limb is permeable to water
- d) Distal tubule is permeable to water

Question 22:

The average urine osmolality in mosm/kg is _____.

- a) 100-800
- b) 500-1400
- c) 200-400
- d) 50-1400

Question 23:

What are the requirements for a patient to excrete concentrated urine?

- a) High ADH levels and hypoosmolar renal medulla
- b) Low ADH levels and hypoosmolar renal medulla
- c) High ADH levels and hyperosmolar renal medulla
- d) Low ADH levels and hyperosmolar renal medulla

Question 24:

Maximum osmotic gradient is found in _____

- a) Outer medulla
- b) Inner medulla
- c) Outer cortex
- d) Inner cortex

Question 25:

Hyperosmolarity of renal medulla is due to _____

- a) K^+
- b) Na^+
- c) Glucose
- d) H^+

Question 26:

Which of the following is not a component of the countercurrent mechanism?

- a) Thick ascending loop of Henle
- b) Proximal convoluted tubule
- c) Vasa recta
- d) Descending limb of loop of Henle

Question 27:

Maximum osmolality that can be reached at the tip of medulla is proportional to

- a) Distribution of Na-K-2Cl channel in the thick ascending loop of Henle
- b) Permeability of descending loop to water
- c) Length of loop of Henle
- d) All of the above

Question 28:

Counter current blood flow is seen in all except _____

- a) Kidney
- b) Testes
- c) Eye
- d) Small intestine

Question 29:

All of the following are components responsible for the countercurrent mechanism in the kidney except _____.

- a) Sodium outflow in thick ascending limb
- b) Water outflow in thin descending limb
- c) Sodium outflow in thin ascending limb
- d) Flow of tubular fluid from PCT to DCT

Question 30:

Which of the following is the diluting segment of the kidney?

- a) PCT
- b) Collecting duct
- c) Ascending thick loop of Henle
- d) Descending loop of Henle

Question 31:

As filtrate flows through PCT, the concentration of all of the following decreases except:

- a) Aminoacid
- b) Glucose
- c) Chloride
- d) Bicarbonate

Question 32:

Macula densa is formed by which part of the nephron?

- a) PCT
- b) DCT
- c) Afferent arteriole
- d) Bowman's capsule

Answer Key

Question No.	Correct Option
1	b
2	c
3	c
4	b
5	c
6	a
7	a
8	c
9	d
10	a
11	c
12	b
13	d
14	d
15	b
16	d

17	a
18	d
19	a
20	c
21	c
22	a
23	c
24	b
25	b
26	b
27	c
28	c
29	c
30	c
31	c
32	b

Detailed Explanations

Solution to Question 1:

If drug has more renal clearance than GFR, then drug is secreted in renal tubules.

Excretion rate = (filtration rate - reabsorption rate) + secretion rate.

The clearance of the substance:

- Equals the GFR if there is no net tubular secretion or reabsorption.
- Exceeds the GFR if there is net tubular secretion.
- Less than the GFR if there is net tubular reabsorption.

Solution to Question 2:

The correct formula is Excretion rate = UV.

- Filtration fraction = $(\text{GFR} / \text{RPF}) \times 100 = 20\%$ (or) 0.2 (approx)
- Filtration rate = GFR * Plasma concentration
- Clearance = $U \times V / P$

Where,

GFR = Glomerular filtration rate

RPF = Renal plasma flow

U = Concentration of the substance in urine

V = Urine flow rate

P = Concentration of the substance in plasma

Solution to Question 3:

Creatinine is partially secreted and never reabsorbed. Clearance of creatinine (CCr) can also be used to determine GFR. Since creatinine is secreted by the tubules, the clearance of creatinine will be slightly higher than inulin. Hence, it overestimates GFR.

Inulin is completely filtered and never reabsorbed. Hence, it can be used for GFR estimation.

Glucose is completely reabsorbed below the renal threshold (i.e. $\sim 180\text{mg/dl}$) in PCT.

PAH (para amino hippuric acid) at low plasma concentrations is completely filtered and secreted. Hence, it can be used for estimation of renal blood flow.

Solution to Question 4:

Effective renal plasma flow in this patient will be falsely low.

It is equal to the clearance of para-aminohippurate (PAH) as shown by the following equation.

$$ERPF = \frac{U_{PAH} V}{P_{PAH}} = C_{PAH}$$

Where,

- ERPF = Effective Renal Plasma Flow
- U_{PAH} = [PAH] in urine

- V = Urine flow rate
- $PPAH$ = $[PAH]$ in plasma
- $CPAH$ = Clearance of PAH

At high concentrations of PAH, as given in this question, tubular secretion attains saturation after which the clearance of PAH falls. Hence, a low $ERPF$ will be seen.

Solution to Question 5:

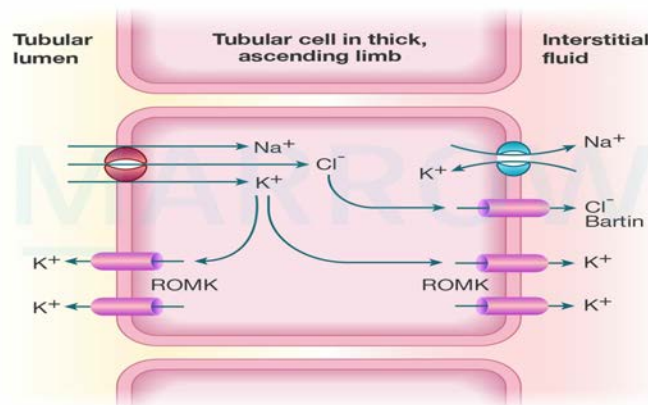
Thick ascending limb of loop of Henle has $Na^+ - 2Cl^- - K^+$ co transporter that cotransports one Na^+ , one K^+ , and $2Cl^-$ from the tubular lumen into the tubular cells. This is an example of secondary active transport.

Na^+ is transported out of the cell into the interstitium by Na, K ATPase in the basolateral membrane of the cell.

Cl^- exits in basolateral $ClC-Kb$ Cl^- channels. Barttin, a protein in the cell membrane, is essential for normal $ClC-Kb$ function.

K^+ moves from the cell to the interstitium and the tubular lumen by ROMK (renal outer medullary potassium channel) and other K^+ channels.

NaCl transport in loop of Henle



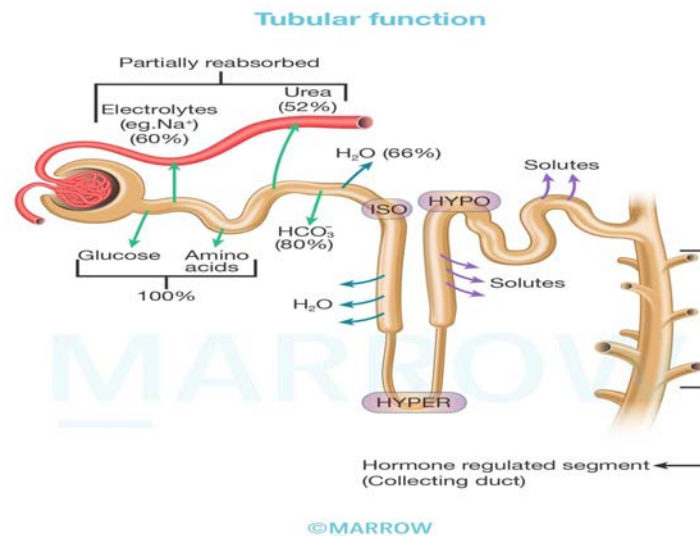
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Solution to Question 6:

The principal site of absorption of Na^+ is PCT. About 60-65% of filtered sodium is reabsorbed in the proximal convoluted tubule (PCT), coupled with the movement of H^+ , glucose, amino acids, organic acids, phosphate, and other electrolytes and substances.

All segments of nephron are involved in sodium reabsorption except thin descending loop of Henle.

% of filtered Na reabsorbed	Part of nephron	Main transporter involved
60	PCT	$\text{Na}^+ - \text{H}^+$ exchanger
30	Thick ascending limb of the loop of Henle	$\text{Na}^+ - 2\text{Cl}^- - \text{K}^+$ cotransporter
7	DCT	$\text{Na} - \text{Cl}$ cotransporter
3	Collecting ducts	ENaC channels (regulated by aldosterone to permit homeostatic adjustments)



Solution to Question 7:

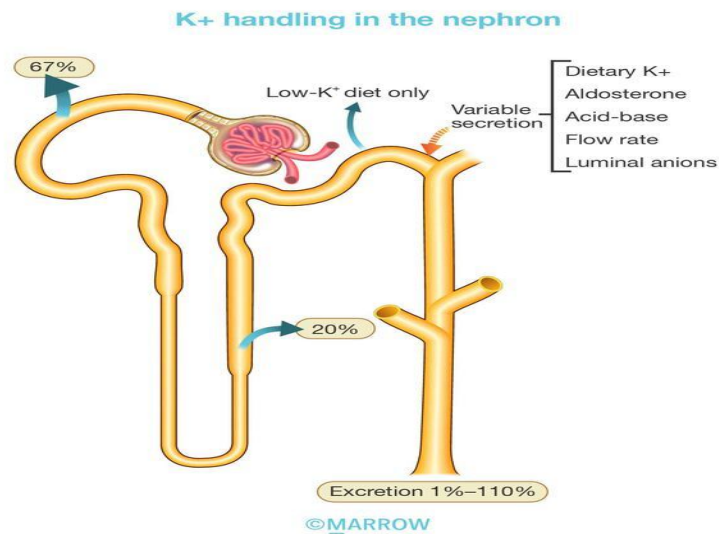
Potassium is maximally absorbed in PCT.

Potassium is freely filtered by the glomerulus. The bulk of filtered K^+ is reabsorbed in the proximal tubule. In the proximal tubule, K^+ absorption is primarily passive and proportional to Na^+ and water through paracellular transport.

K^+ reabsorption in the thick ascending limb of Henle occurs through both transcellular ($\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ cotransport) and paracellular pathways.

K^+ secretion begins in the early distal convoluted tubule and progressively increases along the distal nephron into the cortical collecting duct. Most urinary K^+ can be accounted for by K^+ secretion mediated by principal cells. Under conditions of K^+ depletion, reabsorption of K^+ occurs in the collecting duct.

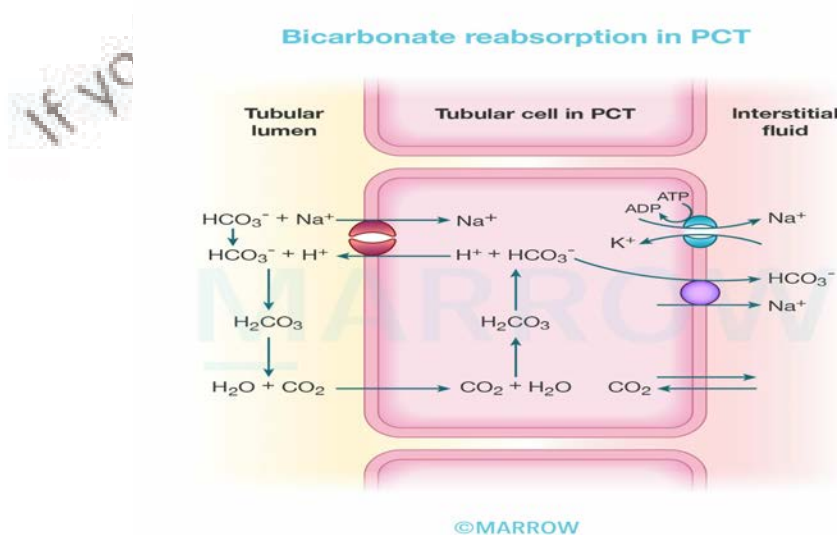
It is the only electrolyte to be reabsorbed as well as secreted.



Solution to Question 8:

Bicarbonate is reabsorbed at the luminal membrane in the form of CO₂ and H₂O with the help of Na⁺ - H⁺ exchanger.

In the tubular cell of the PCT, CO₂ and H₂O recombine with the help of carbonic anhydrase II, and then dissociate into H⁺ and HCO₃⁻. From here, the H⁺ moves back into the tubular lumen via the Na⁺ - H⁺ exchanger while the HCO₃⁻ is reabsorbed at the basolateral membrane via the Na⁺ - HCO₃⁻ co-transporter.



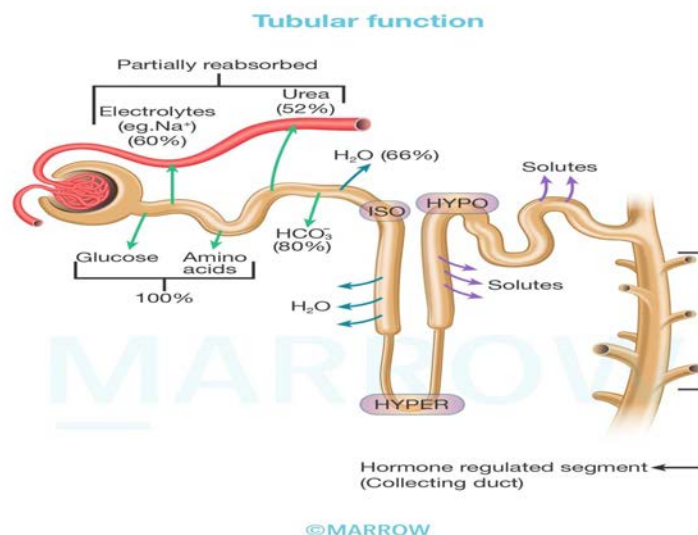
Solution to Question 9:

Glucose and amino acids are completely reabsorbed in the proximal tubule.

Option A: Maximum reabsorption occurs in proximal tubule

Option B: Urea is reabsorbed and secreted. It is reabsorbed in PCT and in collecting duct. It is secreted in thin ascending loop of henle.

Option C: H^+ is secreted in PCT as well as in collecting ducts.



Solution to Question 10:

Sodium symport occurs along with glucose transport into the cell.

Sodium and glucose are co-transported across the luminal membrane of proximal tubular cells. This occurs via sodium glucose co-transporters (SGLT2 and SGLT1). The energy derived from the movement of Na^+ is used to move glucose against its concentration gradient into the tubular cell. This is referred to as secondary active transport.

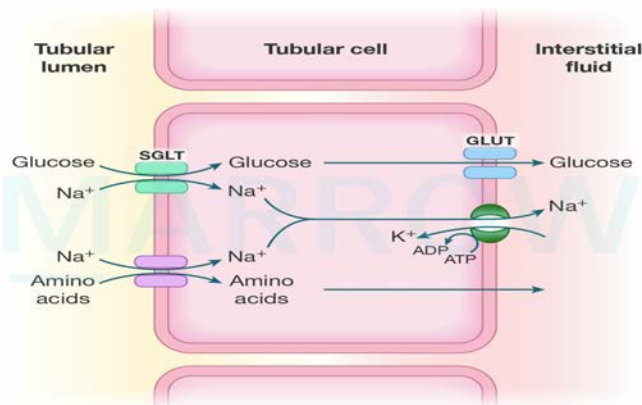
The Na^+ is then pumped out of the cell into the interstitium, and the glucose exits by facilitated diffusion via glucose transporter (GLUT)-2 along the basolateral membrane into the interstitial fluid.

Clinical correlation:

Fanconi-Bickel syndrome: Defective GLUT2

Isolated renal glycosuria: Defective SGLT2

Sodium-glucose co-transport mechanism



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Solution to Question 11:

The actual renal threshold is about 200 mg/dL (i.e., 2 g/L) (1g/L=100mg/dl) and the transport maximum for glucose is 375 mg/min.

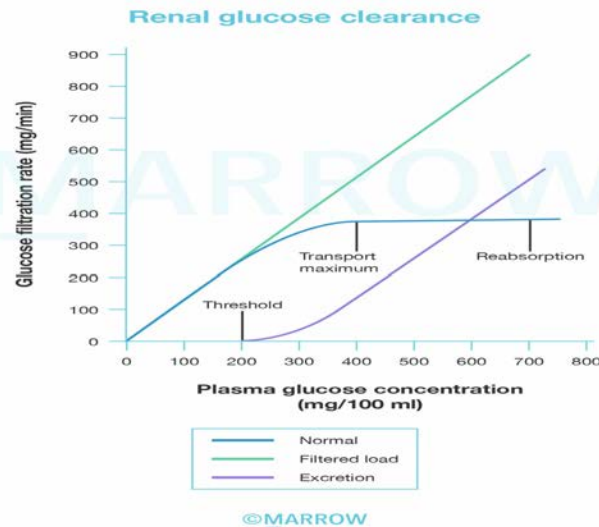
For substances that are actively reabsorbed or secreted, there is a limit to the rate at which the solute can be transported, which is referred to as the transport maximum. This limit is due to saturation of the specific transport systems involved.

Normally, measurable glucose does not appear in the urine because essentially all the filtered glucose is reabsorbed in the proximal tubule.

The transport maximum for glucose is about 375 mg/min, whereas the filtered load of glucose is only about 125 mg/min. With large increases in GFR and/or plasma glucose concentration that increase the filtered load of glucose above 375 mg/min, the excess glucose filtered is not reabsorbed and passes into the urine.

When the plasma glucose concentration is 100 mg/100 ml and the filtered load is at its normal level (125 mg/min) there is no loss of glucose in the urine. However, when the plasma concentration of glucose rises above about 200 mg/100 ml, increasing the filtered load to about 250 mg/min, a small amount of glucose begins to appear in the urine. This point is termed as the threshold for glucose.

Splay is due to separation between filtered load and excretion. It occurs due to different transport maximums of different nephrons.



Solution to Question 12:

The labellings depicted are a. Filtered, b. Secreted, c. T_m , d. Excreted

Filtered Load: The filtered load of PAH increases linearly as the unbound concentration of PAH increases (filtered load = $GFR \cdot [P]PAH$).

Secretion: At low concentrations of PAH, many carriers are available and secretion increases linearly as the plasma concentration increases. When the PAH concentration increases to a level where the carriers are saturated, T_m is reached. After this point, no matter how much the PAH concentration increases, there can be no further increase in the secretion rate.

Excretion. For a secreted substance such as PAH, excretion is the sum of filtration and secretion. At low PAH concentrations (below T_m), excretion increases steeply with increases in plasma PAH concentration because both filtration and secretion are increasing. At PAH concentrations above T_m , excretion increases less steeply (and parallels the curve for filtration) because only the filtration component increases as concentration increases; secretion is already saturated.

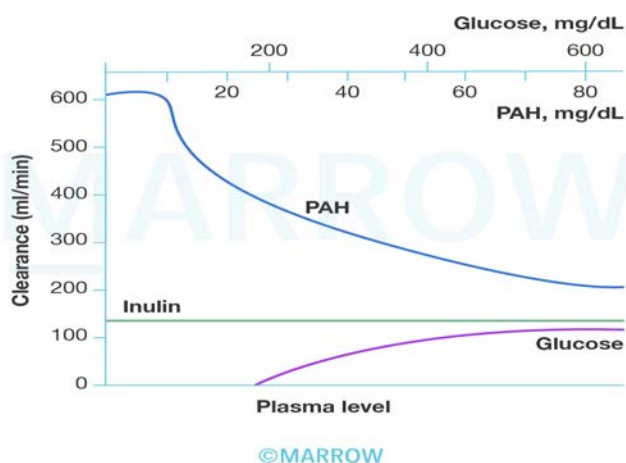
Solution to Question 13:

In uncontrolled diabetes, the plasma clearance of glucose is greater than zero.

Normally, all the glucose that is filtered is reabsorbed and no glucose is excreted in the urine. Hence, the clearance is zero. In uncontrolled diabetes, since the plasma glucose concentration is greater than the renal threshold for reabsorption, excess glucose is excreted in the urine and the clearance is greater than zero.

Clearance of different substances can be shown in a graph as follows:

Renal clearance of various substances



The above graph depicts renal handling of glucose, inulin, and para-aminohippurate (PAH) and can be explained as follows:

Glucose clearance: Normally, all the glucose that is filtered is reabsorbed and no glucose is excreted in the urine. Hence, the clearance is zero. In uncontrolled diabetes, since the plasma glucose concentration is greater than the renal threshold for reabsorption, excess glucose is excreted in the urine and the clearance is greater than zero.

Inulin clearance: Inulin is freely filtered and is neither reabsorbed nor secreted. Hence, it is constant and equal to the glomerular filtration rate.

PAH clearance: PAH is both filtered and secreted. When plasma PAH is low, clearance is high. However, as plasma PAH rises above the threshold level for PAH secretion, clearance falls progressively.

Solution to Question 14:

The amount of glucose excreted in the above scenario is 30 mg/min.

Calculation:

Step 1: Calculate total filtered glucose:

In this question,

- GFR = 90 ml/min
- Plasma glucose concentration = 200 mg/dL = 2 mg/ml (1 dL=100 ml).
- Total filtered glucose = GFR X Plasma glucose concentration.

$$= 90 \text{ ml/min} \times 2 \text{ mg/ml}$$

$$= 180 \text{ mg/min.}$$

Step 2: Check transport maximum:

In the above question transport maximum is 150 mg/min.

Step 3: Calculate the excretion:

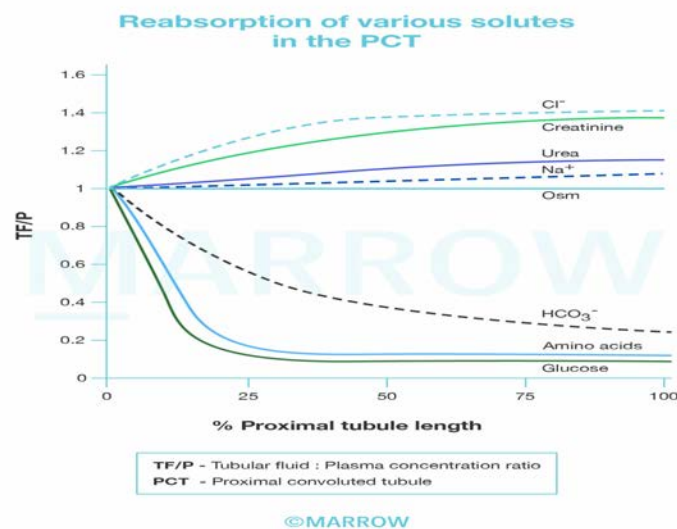
Excretion = Total filtered glucose - Total reabsorbed glucose (Transport Maximum)

= 180-150 mg/min

= 30 mg/min

Solution to Question 15:

The concentration of urea in tubular fluid is more than that of plasma.



[TF/P]>1, when solute is reabsorbed less quickly than water or when solute is secreted. Ex- PAH, Creatinine, Inulin, Urea.

[TF/P]=1, when solute and water are reabsorbed at the same rate Ex - Na.

[TF/P]<1, when solute is reabsorbed more quickly than water. Ex- glucose, amino acid, bicarbonate.

Solution to Question 16:

Water reabsorption in PCT mainly occurs by osmosis.

The proximal tubule reabsorbs 67% of the filtered water. The driving force for water reabsorption is a transtubular osmotic gradient established by reabsorption of solutes.

Solution to Question 17:

Hypertonic urine is excreted due to absorption of water in collecting duct.

The final concentration of urine is determined by the amount of water reabsorption that occurs in the collecting ducts. The changes in osmolarity of the urine (hypo/hypertonic) depends on the amount of ADH (vasopressin) acting on the collecting ducts.

The basic requirements for forming a concentrated urine (hypertonic urine) are

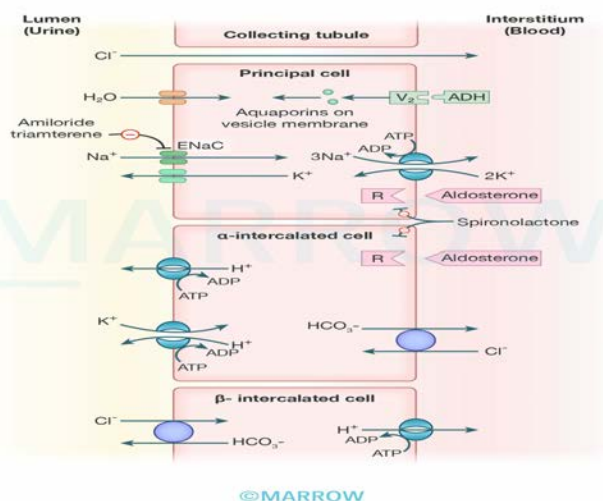
- A high level of ADH, which increases the permeability of the distal tubules and collecting ducts to water, thereby allowing these tubular segments to avidly reabsorb water.
- A high osmolarity of the renal medullary interstitial fluid, which provides the osmotic gradient necessary for water reabsorption to occur in the presence of high levels of ADH.

Solution to Question 18:

Vasopressin has the maximum effect on the collecting ducts.

Vasopressin acts by inserting aquaporin 2 present in cytoplasmic vesicles of principal cells on to the apical membrane. The effect is mediated via the vasopressin V₂ receptor, cyclic adenosine 5-monophosphate (cAMP), and protein kinase A. When vasopressin is absent, the collecting duct epithelium is relatively impermeable to water.

Principal and intercalated cells



Solution to Question 19:

Free water clearance is regulated by ADH.

Free water clearance is defined as the amount of solute-free water generated by the kidneys, either when dilute urine or when concentrated urine is formed.

Free-water clearance is positive, when excess water is being excreted (dilute urine).

Free-water clearance is negative, when excess solutes are being removed and water is being conserved (concentrated urine).

Solution to Question 20:

Patient A shows positive free water clearance B is isotonic C shows negative free water clearance.

Free water clearance is defined as the amount of solute-free water generated by the kidneys, either when dilute urine or when concentrated urine is formed.

Patient A has positive free-water clearance, since excess water is being excreted (dilute urine).

Patient C has negative free-water clearance, since excess solutes are being removed and water is being conserved (concentrated urine).

Solution to Question 21:

The ascending limb is always impermeable to H₂O.

Solution to Question 22:

The average urine osmolality is 100-800 mosm/kg.

Solution to Question 23:

The requirements for excreting concentrated urine are high ADH levels and hyperosmolar renal medulla.

A high level of ADH, which increases the permeability of the distal tubules and collecting ducts to water, thereby allowing these tubular segments to avidly reabsorb water, and a high osmolarity of the renal medullary interstitial fluid, which provides the osmotic gradient necessary for water reabsorption to occur in the presence of high levels of ADH.

Solution to Question 24:

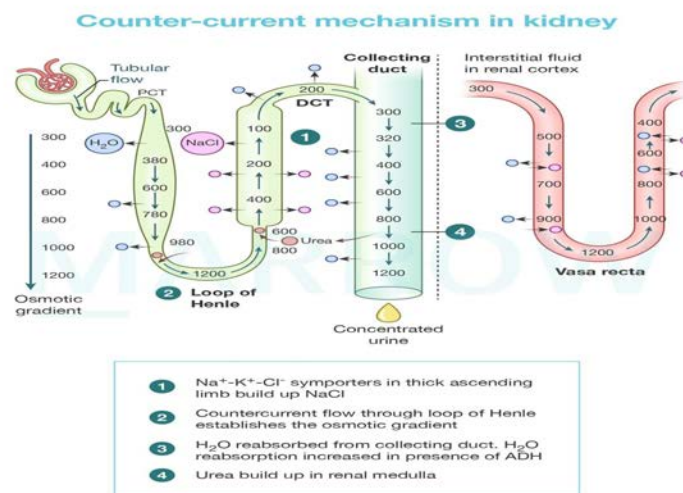
Maximum osmotic gradient is found in the inner medulla. The gradient of osmolarity from the cortex (300 mosm/L) to the inner medulla (1200 mosm/L) is produced by the loops of Henle as countercurrent multipliers and maintained by operation of the vasa recta as countercurrent exchangers.

Solution to Question 25:

Hyperosmolarity of the medullary interstitium is mainly due to NaCl and urea.

The major factors that contribute to the buildup of solute concentration into the renal medulla are

- Active transport of Na^+ and co-transport of K^+ , Cl^- , and other ions out of the thick portion of the ascending limb of the loop of Henle into the medullary interstitium.
- Active transport of ions from the collecting ducts into the medullary interstitium.
- Facilitated diffusion of urea from the inner medullary collecting ducts into the medullary interstitium.
- Diffusion of only small amounts of water from the medullary tubules into the medullary interstitium

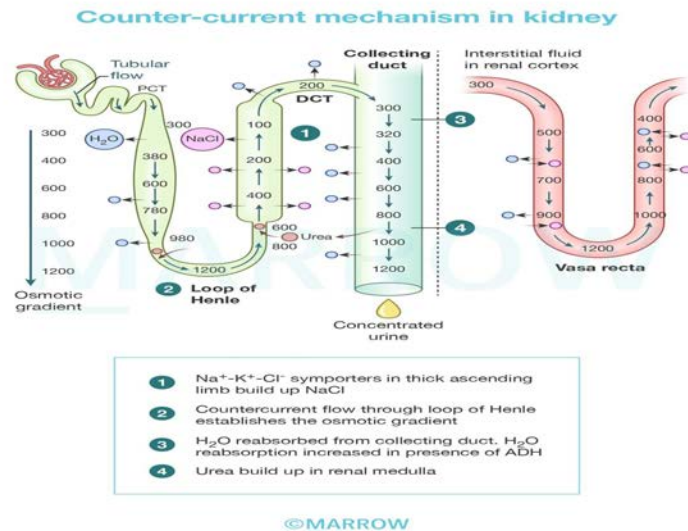


Solution to Question 26:

Proximal convoluted tubule (PCT) does not play a role in countercurrent mechanism.

The countercurrent mechanism consists of 2 components:

- Countercurrent multiplier:
 - a. Absorption of water via Aquaporin 1 in thin descending limb
 - b. Active transport of Na^+ - K^+ - 2Cl^- in thick ascending limb
 - c. Flow of fluid from PCT to DCT
- Countercurrent exchanger: Vasa recta are involved in this process. They help in maintaining the osmotic gradient in the renal medulla. This is a passive process.



Solution to Question 27:

The maximum osmolality that can be reached at the tip of the medulla is proportional to the length of the loop of Henle, because the Loop of Henle acts as Countercurrent multiplier to maintain the gradient in medulla. Greater is the length of loop of Henle, greater will be the osmolality that can be reached at the tip of the medulla.

In case of juxtamedullary nephrons with longer loops and thin ascending limbs, the osmotic gradient is spread over a greater distance and the osmolality at the tip of the loop is greater.

Solution to Question 28:

The eye does not have a countercurrent blood flow system.

Heat and substances, including gases, steroids and peptide hormones, can pass from venous blood, interstitial fluid and lymph to the arterial blood; the process is called local counter-current transfer.

Examples of countercurrent blood flow

- Heat exchange between the arteries and venae comitantes of the limbs.
- Vasa recta in the kidneys
- Testes
- Small intestine

Solution to Question 29:

Thin ascending limb is not a component of the countercurrent mechanism.

The loop of Henle function as countercurrent multipliers and vasa recta as countercurrent exchangers.

The functioning of the loop of Henle depends upon:

- High permeability of the thin descending limb to water (via aquaporin-1)
- Active transport of Na^+ and Cl^- from the thick ascending limb
- Inflow of tubular fluid from proximal tubule and outflow to the distal tubule.

Note: The thin ascending limbs of juxtamedullary nephrons are permeable to Na^+ and Cl^- and hence do contribute passively to countercurrent multiplication. However, among the options provided, Option C is the best answer as the other options are directly responsible for the countercurrent mechanism.

Solution to Question 30:

The ascending thick limb of the loop of Henle is called the diluting segment.

In the ascending limb, there is the movement of Na^+ and Cl^- out of the tubular lumen, and the tubular fluid becomes more dilute.

When the fluid reaches the top of the ascending limb, it becomes hypotonic to plasma.

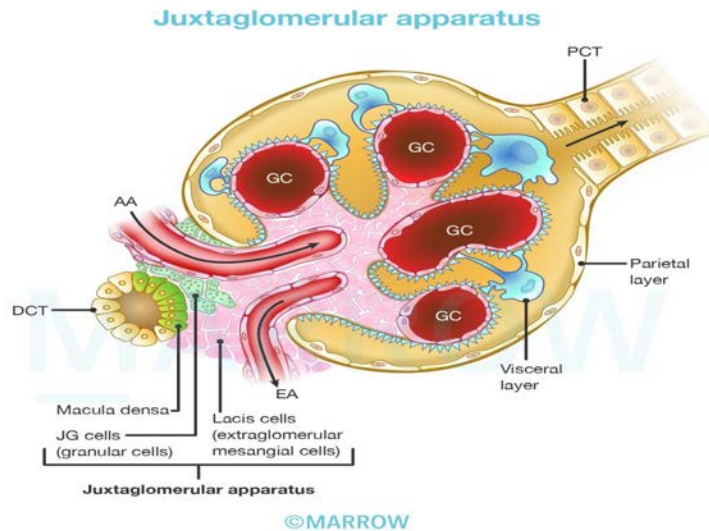
Solution to Question 31:

As the filtrate flows through the proximal convoluted tubule, the concentration of the chloride increases.

- The concentration of chloride in the early part of the pCT is $\approx 105 \text{ mEq/L}$.
- As the sodium is reabsorbed, it preferentially carries glucose, amino acids, and bicarbonate.
- This leaves behind the solute, that has a higher concentration of the chloride, which as it reaches the second half of the PCT is $\approx 140 \text{ mEq/L}$.

Solution to Question 32:

Macula densa is formed by specialized tubular epithelial cells of the distal convoluted tubule.



The juxtaglomerular apparatus consists of the following parts:

- **Macula densa:** At the end of the thick ascending limb of Henle, and at the beginning of the distal tubule, the tubular epithelial cells are specialized to form the macula densa. They detect sodium concentration of the fluid in the tubule (chemoreceptor). In response to elevated sodium, the macula densa cells trigger the contraction of the afferent arteriole, reducing the flow of blood to the glomerulus and the glomerular filtration rate. This happens as follows: When there is increased sodium, there is increased sodium-potassium ATPase activity which leads to increased ATP hydrolysis causing more adenosine formation. The adenosine receptors on the macula densa cells stimulate it to release more calcium, so that there is vasoconstriction of the afferent arteriole, thereby decreasing the GFR.
- **Lacis cells:** Extraglomerular mesangial cells situated between afferent and efferent arteriole.
- **Granular cells/JG cells:** Modified smooth muscle cells of the afferent arteriole, they secrete renin. Fall in pressure in afferent arteriole stimulates renin release (mechanoreceptor), which, in turn, increases blood pressure.

Acid-Base Balance, Renal Hormones and Micturition Reflex

Question 1:

Principal site of acidification of urine _____

- a) Distal convoluted tubule
- b) Proximal convoluted tubule
- c) Loop of henle
- d) Collecting duct

Question 2:

Most important extracellular buffer _____

- a) Bicarbonates
- b) Plasma protein
- c) Phosphates
- d) Nitrates

Question 3:

Normal anion gap is mostly due to _____

- a) Unmeasured proteins
- b) Unmeasured Na^+
- c) Unmeasured Cl^-
- d) None of the above

Question 4:

Normal anion gap is _____ mEq / L.

- a) 4-8
- b) 8-16

- c) 0-4
- d) 16-20

Question 5:

Endocrine functions associated with kidney include all of the following except _____

- a) Erythropoietin secretion
- b) ANP secretion
- c) 1,25-DHCC formation (Vitamin D₃)
- d) Renin secretion

Question 6:

The major prostanoid synthesized in kidney is _____

- a) PGE₁
- b) PGE₂
- c) PGI₂
- d) PGF₂^o

Question 7:

All of the following results in increased renin secretion except _____.

- a) Renal ischemia
- b) Increased amount of Na⁺ in DCT
- c) Decreased amount of Na⁺ in PCT
- d) Decreased afferent arteriolar pressure

Question 8:

Angiotensin II causes all of the following except _____

- a) Stimulation of thirst
- b) Aldosterone secretion
- c) Increased ADH secretion

d) Vasodilation

Question 9:

Vasopressin secretion is increased by_____

- a) Alcohol**
- b) Decreased ECF volume**
- c) Decreased oncotic pressure**
- d) Procainamide**

Question 10:

A patient who got into an accident is brought to the casualty and is bleeding from his right leg. His blood pressure is 90/60 mm Hg. Which of the following plays the most important role for sodium and water retention in this patient?

- a) Renin-angiotensin system**
- b) ADH**
- c) Aldosterone**
- d) ANP**

Question 11:

Atriopeptin (ANP) acts at _____

- a) Collecting duct**
- b) Loop of Henle**
- c) Early distal tubule**
- d) Proximal tubule**

Question 12:

ANP secretion is raised by all except_____

- a) Hypovolemia**
- b) Administration of isotonic saline**

- c) Raised central venous pressure
- d) ↑blood pressure

Question 13:

Several hormones regulate the tubular reabsorption of water and electrolytes at different sites in the nephron physiologically in a patient. Which of the following combination is correct?

- a) Angiotensin in distal tubule
- b) Aldosterone in collecting ducts
- c) ADH in proximal tubule
- d) ANP in loop of Henle

Question 14:

The areas associated with micturition reflex is/are _____

- a) Pons
- b) Paracentral lobule
- c) Both a and b
- d) none

Question 15:

Hypoxia-induced factors stimulate _____ cells of the kidney to initiate erythropoiesis.

- a) Intercalated cells
- b) Lacis cells
- c) Interstitial cells
- d) Macula densa

Question 16:

Choose the correct option mentioning the correct site of product conversion

- a) Cholecalciferol → 25 hydroxy vitamin D3 @ skin
- b) Cholecalciferol → 25 hydroxy vitamin D3 @ kidney

- c) Angiotensin 1 → angiotensin 2 @ liver
- d) Angiotensin 1 → angiotensin 2 @ lungs

Question 17:

ADH acts through which aquaporin channel?

- a) 1
- b) 2
- c) 3
- d) 4

Question 18:

Ureteric peristalsis is mainly due to _____

- a) Sympathetic innervation
- b) Parasympathetic innervation
- c) Both a and b
- d) Pacemaker activity of smooth muscles of renal pelvis

Question 19:

In micturition reflex, the first change to occur is _____

- a) Trigone relaxation
- b) Detrusor contraction
- c) Relaxation of perineal muscles
- d) Decreased urethral pressure

Question 20:

How much urine in the bladder will cause the first urge to void in a patient?

- a) 150ml
- b) 250ml
- c) 350ml

d) 100ml

Question 21:

In a patient, pO_2 is 85mmHg, pCO_2 is 50mmHg, pH is 7.2, and HCO_3^- is 36mmol/L. Identify the condition.

- a) Respiratory acidosis with compensatory metabolic alkalosis
- b) Uncompensated respiratory acidosis
- c) Respiratory alkalosis with compensatory metabolic acidosis
- d) Respiratory acidosis with compensatory metabolic acidosis

Question 22:

A patient with salicylic acid poisoning has the following arterial blood gas analysis report: pH = 7.1, pCO_2 = 18mmHg, HCO_3^- = 12mmol/L. Identify the acid-base disorder here.

- a) Metabolic acidosis with respiratory alkalosis
- b) Respiratory acidosis with compensatory metabolic alkalosis
- c) Metabolic acidosis with uncompensated respiratory alkalosis
- d) Respiratory acidosis with metabolic alkalosis

Question 23:

Which of the following patients will have metabolic alkalosis?

- a) A patient suffering from diarrhoea
- b) A patient with chronic renal failure who is recommended dialysis
- c) A patient who was diagnosed with hyperaldosteronism
- d) A patient who took a whole bottle of aspirin to get rid of the headache

Question 24:

In a patient with metabolic acidosis, which of the following changes are seen?

- a) Increased K^+ excretion
- b) Decreased K^+ excretion

- c) Increased Na⁺ excretion
- d) Increased Na⁺ reabsorption

Question 25:

A normal anion gap metabolic acidosis occurs in patients with_____

- a) Diarrhoea
- b) Diabetic ketoacidosis
- c) Methyl alcohol poisoning
- d) Acute renal failure

Answer Key

Question No.	Correct Option
1	d
2	a
3	a
4	b
5	b
6	b
7	b
8	d
9	b
10	a
11	a
12	a
13	b
14	c
15	c
16	d
17	b
18	d
19	c
20	a

21	a
22	a
23	c
24	b
25	a

Detailed Explanations

Solution to Question 1:

Principal site of acidification of urine is collecting duct. Acidification occurs in both distal tubule and collecting duct, but collecting duct is more important.

The urinary pH of 4.5 (limiting pH) is normally reached in the collecting ducts.

In the late distal tubules and collecting ducts, the tubular epithelium secretes H^+ by primary active transport. Primary active secretion of H^+ occurs in type A intercalated cells of the late distal tubule and in the collecting tubules. H^+ concentration can be increased as much as 900-fold in the collecting tubules.

Solution to Question 2:

The most important extracellular buffer is bicarbonate. The source of bicarbonate is plasma by filtration. It is a non inducible buffer.

Proteins: Are the most abundant intracellular buffer.

Phosphate: minor buffer in kidney. The source is plasma by filtration. It is a noninducible buffer.

Ammonia is a minor buffer, locally produced by tubular cells. Its production increases in metabolic acidosis. It is the only inducible buffer.

Solution to Question 3:

Normal anion gap is mostly due to unmeasured proteins.

Normally in the plasma, the concentrations of anions and cations are equal to maintain electrical neutrality. Thus no real "anion gap" exists in the plasma and this is purely a diagnostic concept.

Diagnostically, when the electrolytes are measured in the blood, routinely only certain cations and anions are measured. These are Na^+ , Cl^- and HCO_3^- . The anion gap is the difference between the unmeasured anions and unmeasured cations. This is estimated as :

Plasma anion gap= $[Na^+] - [Cl^-] - [HCO_3^-]$

The unmeasured cations include calcium, magnesium and potassium.

The unmeasured anions include albumin, phosphate, sulphate and organic anions.

This is primarily used in measurement of different causes of metabolic acidosis.

Solution to Question 4:

The normal anion gap ranges between 8 and 16 mEq/L.

The concentrations of anions and cations in plasma must be equal to maintain electrical neutrality. Therefore, there is no real “anion gap” in the plasma.

However, only certain cations and anions are routinely measured in the clinical laboratory. The cation normally measured is Na^+ , and the anions are usually Cl^- and HCO_3^- .

The “anion gap” (which is only a diagnostic concept) is the difference between unmeasured anions and unmeasured cations and is estimated as

$$\text{Plasma anion gap} = [\text{Na}^+] - [\text{HCO}_3^-] - [\text{Cl}^-] = 144 - 24 - 108 = 12 \text{ mEq/L}$$

The anion gap will increase if unmeasured anions rise or if unmeasured cations fall.

The most important unmeasured cations include calcium, magnesium and potassium and the major unmeasured anions are albumin, phosphate, sulfate and other organic anions.

Usually the unmeasured anions exceed the unmeasured cations and the anion gap ranges between 8 and 16 mEq/L.

Solution to Question 5:

ANP is secreted by the heart.

Endocrine/Paracrine Functions of the Kidney:

Substance	Site of secretion
Renin	Granular cells or juxtaglomerular cells (JG cells) of the juxtaglomerular apparatus (JGA)
Erythropoietin	Interstitial cells in the peritubular capillary bed
Prostaglandin E ₂ (PGE ₂)	Renal medullary interstitial cells (RMICs), macula densa, and by cells of collecting duct
PGI ₂ and other prostaglandins	Arterioles and glomeruli
1,25-Dihydroxycholecalciferol (calcitriol or vitamin D ₃)	Cells of proximal tubules of the kidneys.

Solution to Question 6:

PGE₂ is the major prostanoid synthesized in the kidney.

It is secreted by:

- RMICs (Renal Medullary Interstitial Cells)
- Macula densa
- Cells in the collecting ducts.

Prostacyclin (PGI₂) and other prostaglandins are secreted by the arterioles and glomeruli.

Renal medullary interstitial cells (RMICs) are specialized fibroblast-like cells which are major site of cyclooxygenase 2 (COX-2) and prostaglandin synthase (PGES) expression.

Solution to Question 7:

The increased amount of Na⁺ and Cl⁻ reabsorption across macula densa (DCT) is inhibitory to renin secretion. All other factors increase renin secretion.

Renin leads to a series of events through the Renin-Angiotensin-Aldosterone pathway that finally, in simple terms, leads to salt and water retention and increases blood pressure. Therefore, it is easy to remember that all factors that cause decreased effective volume, pressure, or sodium delivery will increase the renin levels and vice versa.

Note: Renin is secreted by the juxtaglomerular cells (granular cells) which are epitheloid cells located in the media of afferent arterioles of the glomerulus.

Solution to Question 8:

Angiotensin II is one of the most potent vasoconstrictors in the body.

It constricts efferent arteriole, thereby increasing filtration fraction and glomerular filtration rate, when there is decreased renal bloodflow

Solution to Question 9:

Vasopressin (ADH) secretion increases in response to decreased ECF volume to bring about water retention to restore the ECF volume.

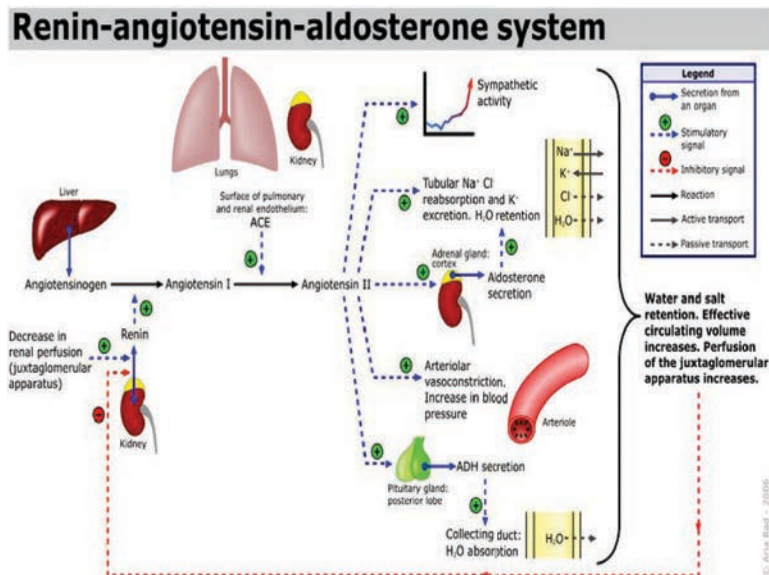
ADH exerts its antidiuretic effect by acting on the V₂ receptors and involves the insertion of aquaporin 2 (AQP2) into the apical (luminal) membranes of the principal cells of the collecting ducts.

Solution to Question 10:

The renin-angiotensin system plays the most important role for sodium and water retention in this patient as the patient has hypovolemia due to blood loss. (refer to figure)

The plasma concentration of renin is the rate-limiting step in the renin-angiotensin system; it is the major determinant of plasma angiotensin II concentration.

Angiotensin II stimulates the aldosterone and ADH secretion; it also constricts the blood vessels and causes thirst, which helps in maintaining blood pressure.



Solution to Question 11:

ANP acts on the collecting duct.

ANP is secreted by the atria in response to an increase in ECF volume. It decreases Na⁺ reabsorption in the late distal tubule and collecting ducts. It causes natriuresis.

ANP causes dilation of afferent arterioles and constriction of efferent arterioles resulting in an increased RBF (Because the dilatory effect of ANP on afferent arterioles is more than the constrictor effect on efferent arterioles, there is an overall decrease in renal vascular resistance and resulting

increase in RBF) and increased GFR.

Solution to Question 12:

It is ↑ in ECF volume and not hypovolemia that causes increased secretion of ANP.

Stimuli that increase the secretion of ANP

- ↑atrial stretch or ventricular stretch
- ↑isotonic saline administration
- ↑ECF volume

Stimuli that decrease ANP secretion

- Rising from supine to standing position (as this decreases the central venous pressure).

Solution to Question 13:

The combination which is correct is Aldosterone in collecting ducts.

Aldosterone acts on principal cells and intercalated cells, which are present in collecting ducts. It regulates ECF volume and Na⁺ content. Major stimulus for aldosterone secretion is Hyperkalemia. The functions of aldosterone are Na⁺ and H₂O reabsorption, K⁺ and H⁺ excretion.

Option A: Angiotensin II acts on efferent arteriole. It causes vasoconstriction increasing BP. It causes constriction of efferent arteriole, increases filtration fraction, preserving GFR when RBF ↓.

Option B: ADH acts on principal cells in collecting ducts. It primarily regulates serum osmolarity. Stimulates reabsorption of water in collecting ducts via Aquaporin-2.

Option D: ANP is released from atria and BNP is released from ventricles in response to ↑ volume, it inhibits RAAS, relaxes mesangial cells and promotes natriuresis.

Solution to Question 14:

The micturition center is located in the pons. The paracentral lobule is also associated with micturition.

The micturition reflex is an autonomic spinal cord reflex, but it can be inhibited or facilitated by centers in the brain.

These centers include-

- A strong facilitative and inhibitory centers in the brain stem, located mainly in the pons.
- Centers located in the cerebral cortex that are mainly inhibitory but can become excitatory.

Solution to Question 15:

Hypoxia induced factors stimulate interstitial cells to initiate erythropoiesis

Erythropoietin is produced by interstitial cells in the peritubular capillary bed of the kidney.

Erythropoietin in turn initiates erythropoiesis

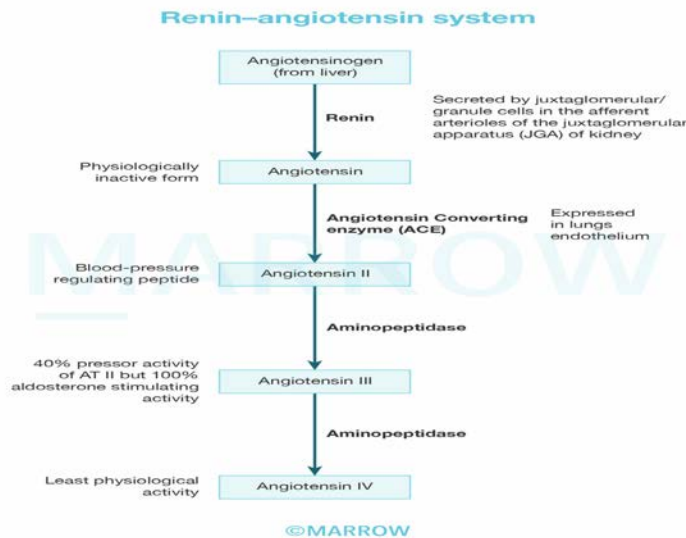
Solution to Question 16:

Angiotensin 1 → angiotensin 2 @ lungs

Cholecalciferol → 25 hydroxy vitamin D3 @ liver

25 hydroxy vitamin D3 → 1,25 dihydroxy vitamin D3 @ Kidney

Angiotensin I is converted to Angiotensin II by Angiotensin Converting Enzyme (ACE) which is expressed in the lungs.



Solution to Question 17:

The antidiuretic hormone acts through aquaporin channel 2.

ADH binds to V2 receptors in the late distal tubules, collecting tubules, and collecting ducts, and stimulates the movement of an intracellular protein, called aquaporin-2 (AQP-2), to the luminal side of the cell membranes.

Solution to Question 18:

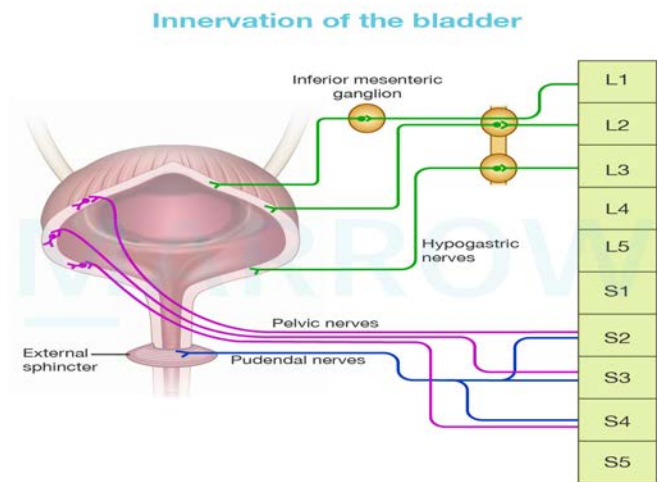
Urine flowing from the collecting ducts into the renal calyces stretches the calyces and increases their inherent pacemaker activity, which in turn initiates peristaltic contractions occurring 1 to 5 times per minute that spread to the renal pelvis and then downward along the length of the ureter, thereby forcing urine from the renal pelvis toward the bladder.

- The walls of the ureters contain smooth muscle and are innervated by both sympathetic and parasympathetic nerves, as well as by an intramural plexus of neurons and nerve fibers that extends along the entire length of the ureters.
- As with other visceral smooth muscle, peristaltic contractions in the ureter are
- Enhanced by parasympathetic stimulation
- Inhibited by sympathetic stimulation

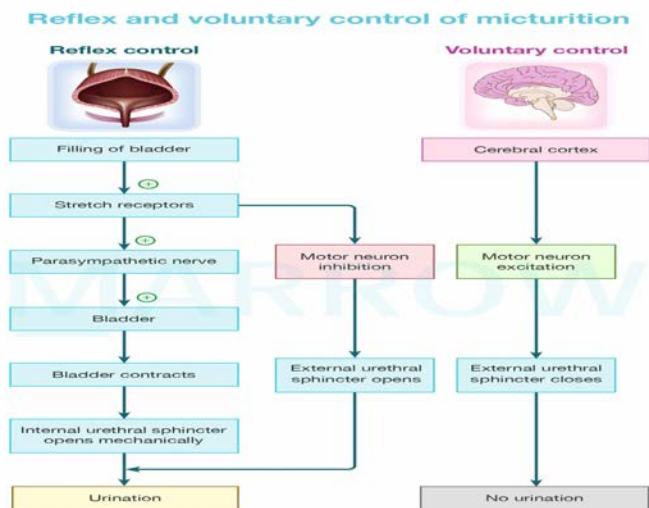
Solution to Question 19:

- One of the initial events is the relaxation of the muscles of the pelvic floor, and this may cause a sufficient downward tug on the detrusor muscle to initiate its contraction.
- Micturition Reflex
- Micturition is fundamentally a spinal reflex facilitated and inhibited by higher brain centers and is subject to voluntary facilitation and inhibition.
- These centers include
- strong facilitative and inhibitory centers in the brain stem, located mainly in the pons
- several centers located in the cerebral cortex that is mainly inhibitory but can become excitatory.
- The micturition reflex is the basic cause of micturition, but the higher centers normally exert final control of micturition.
- Micturition Reflex is the result of a stretch reflex initiated by sensory stretch receptors in the bladder wall, especially by the receptors in the posterior urethra when this area begins to fill with urine at the higher bladder pressures.
- Sensory signals from the bladder stretch receptors are conducted to the sacral segments of the cord through the pelvic nerves and then reflexively back again to the bladder through the parasympathetic nerve fibers by way of these same nerves. (refer figure)
- As the bladder continues to fill, the micturition reflexes become more frequent and cause greater contractions of the detrusor muscle.
- Once a micturition reflex begins, it is “self-regenerative.”
- It is a single complete cycle of
- The progressive and rapid increase of pressure,
- A period of sustained pressure, and
- Return of the pressure to the basal tone of the bladder.
- As the bladder becomes more and more filled, micturition reflexes occur more and more often and more and more powerfully.
- Once the micturition reflex becomes powerful enough, it causes another reflex, which passes through the pudendal nerves to the external sphincter to inhibit it. If this inhibition is more potent in the brain than the voluntary constrictor signals to the external sphincter, urination will occur. If not, urination will not occur until the bladder fills still further and the micturition reflex

becomes more powerful.



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Solution to Question 20:

The first urge to void is felt at a bladder volume of about 150 mL, and a marked sense of fullness at about 400 mL.

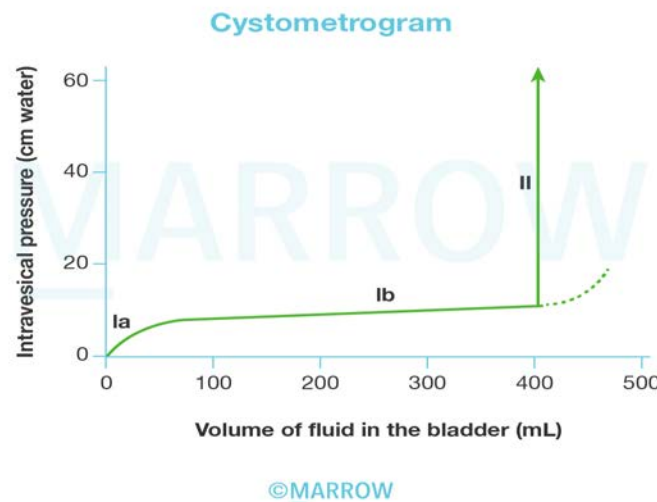
Cystometrogram (refer figure)

A plot of intravesical pressure against the volume of fluid in the bladder is called a cystometrogram.

The curve shows

- An initial slight rise in pressure when the first increments in the volume are produced(Ia);
- A long, nearly flat segment as further increments are produced(Ib);

- A sudden, sharp rise in pressure as the micturition reflex is triggered(II).
- The flatness of segment Ib is a manifestation of the law of Laplace.
- This law states that the pressure in a spherical viscus is equal to twice the wall tension divided by the radius.
- $P = \frac{2T}{R}$
- In the case of the bladder, the tension increases as the organ fills but so does the radius. Therefore, the pressure increase is slight until the organ is relatively full.



Solution to Question 21:

This patient has respiratory acidosis with compensatory metabolic alkalosis.

pH is decreased (<7.35) so it is acidosis.

The CO_2 levels are increased ($>45\text{mmHg}$), which is in line with the change in pH (in respiratory acidosis CO_2 is expected to raise), thus the primary problem is respiratory.

The HCO_3^- levels are increased ($>30\text{mmol/L}$), which is opposite to change in pH (i.e. there is acidosis and HCO_3^- levels are high), thus, there is compensatory metabolic alkalosis.

Thus, this is a case of respiratory acidosis with compensatory metabolic alkalosis.

Solution to Question 22:

The acid-base disorder here is a mixed disorder of metabolic acidosis and respiratory alkalosis.

It should not be confused with metabolic acidosis with uncompensated respiratory alkalosis. This can be understood by the following calculation:

Step 1: The pH in this patient is 7.1 (pH <7.35) which indicates acidosis

Step 2: HCO_3^- is 12 mmol/L (Normal HCO_3^- - 22–30 mmol/L). In metabolic acidosis, HCO_3^- levels decrease.

Hence, this is a case of metabolic acidosis.

Step 3: As per Winter's formula, the expected respiratory compensation in metabolic acidosis can be calculated using,

$$p\text{CO}_2 = (1.5 \times [\text{HCO}_3^-]) + 8 \pm 2$$

In this case, Expected $p\text{CO}_2 = (1.5 \times [12]) + 8 \pm 2$

$$= 26 \pm 2$$

$$= 24 - 28 \text{ mmHg}$$

The $p\text{CO}_2$ in this patient is 18 mmHg, which implies there is an overcompensation of $p\text{CO}_2$. This overcompensation can happen only if there is an additional component of respiratory alkalosis contributing to reduced $p\text{CO}_2$.

Hence, this patient has a mixed disorder - metabolic acidosis and respiratory alkalosis.

Solution to Question 23:

Metabolic alkalosis is seen in patients with hyperaldosteronism.

In hyperaldosteronism: increased aldosterone levels cause increased H^+ secretion by the distal tubule and increased reabsorption of $\text{HCO}_3^- \rightarrow$ Metabolic alkalosis.

Solution to Question 24:

K^+ secretion into the renal tubules is decreased when the H^+ secretion is increased (Intracellular Buffering of H^+). Thus in metabolic acidosis, there is decreased K^+ excretion.

Solution to Question 25:

Diarrhoea is a cause of normal anion gap metabolic acidosis produced as a result of increased HCO_3^- loss in feces.

This is because a decrease in HCO_3^- is replaced by an increase in the concentration of Cl^- . This is a measurable anion and hence the anion gap is normal.

Extra - edge : Siggard-Anderson curve normogram is used to plot the acid-base composition of the blood

Pituitary and Thyroid

Question 1:

Which of the following is secreted from acidophilic cells of pituitary?

- a) GH
- b) TSH
- c) ACTH
- d) FSH

Question 2:

A 40-year-old man is suffering from hypertension, buffalo hump and hyperpigmentation of the skin, including oral mucosa. He is diagnosed with Cushing syndrome due to pituitary cause. Which of the following is responsible for the hyperpigmentation seen in this patient?

- a) TSH
- b) FSH
- c) ACTH
- d) MSH

Question 3:

Inhibitory control by the hypothalamus is exerted on:

- a) Prolactin only
- b) Prolactin and growth hormone
- c) Growth hormone only
- d) Neither

Question 4:

A patient with a known brain tumour learns that his pituitary stalk has been affected. Secretion of which of the following hormones is increased after the sectioning of the pituitary stalk?

- a) Prolactin
- b) TSH
- c) ACTH
- d) FSH

Question 5:

Which of the following is not a cause of elevated prolactin level?

- a) Sleep
- b) Pregnancy
- c) Stress
- d) L-dopa

Question 6:

Which of the following is the effect of continuous GnRH administration?

- a) Stimulates hypothalamic pituitary axis
- b) Suppresses hypothalamic pituitary axis
- c) May suppress or stimulate
- d) Has no effect

Question 7:

Which of the following actions of GH is mediated by IGF-1?

- a) Lipolysis
- b) Decreased insulin sensitivity
- c) Antilipolysis
- d) Na⁺ retention

Question 8:

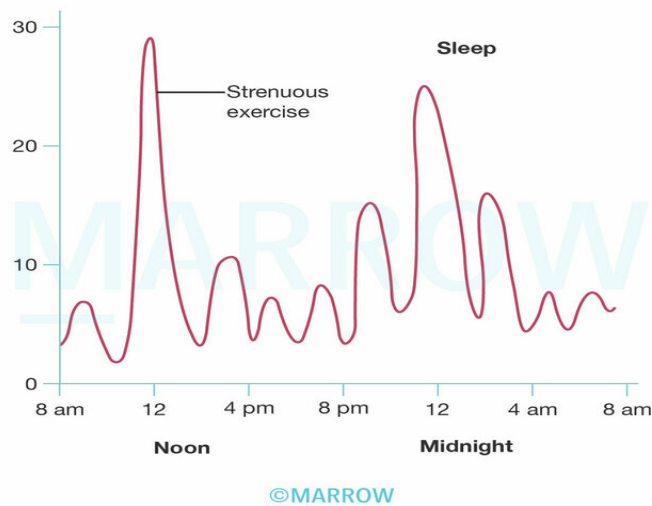
GH has its effect on bone growth through_____

- a) Directly

- b) IGF-1
- c) Thyroxine
- d) Both directly and via IGF-1

Question 9:

Which of the following hormones demonstrates the fluctuations depicted below?



- a) ACTH
- b) GH
- c) TSH
- d) Prolactin

Question 10:

Growth hormone level is decreased in which of the following conditions?

- a) Hypoglycemia
- b) Fasting
- c) REM Sleep
- d) Exercise

Question 11:

A 35 yr old man presents with hypertension, coarse facial features, frontal bossing, large hands and feet, mandibular enlargement and a large fleshy nose. His age-matched serum IGF-I levels are elevated. What is the most likely cause of his condition?

- a) Excess of GH
- b) Deficiency of GH
- c) Excess of somatostatin
- d) Deficiency of somatostatin

Question 12:

What does the neurohypophysis release?

- a) Oxytocin
- b) Prolactin
- c) MSH
- d) ACTH

Question 13:

Which pair of vasopressin receptors is incorrect?

- a) V1a-smooth muscles
- b) V2-collecting ducts
- c) V1b-anterior pituitary
- d) V4-CNS

Question 14:

"Vasopressin escape" in SIADH results due to:

- a) Downregulation of aquaporin 2 production
- b) Increase in urine osmolarity
- c) Decrease in ADH secretion
- d) Upregulation of aquaporin 2 production

Question 15:

Sodium-iodide symport is absent in which of the following sites?

- a) Pituitary
- b) Placenta
- c) Parotid
- d) Thyroid

Question 16:

A 35-year-old lady is diagnosed with Hashimoto thyroiditis and a positive antibody titer against thyroid peroxidase enzyme. Which of the following is not a function of the affected enzyme?

- a) Coupling
- b) Oxidation
- c) Organification
- d) Iodide trapping

Question 17:

A patient with hyperthyroidism is planned for thyroidectomy. In order to achieve a euthyroid status preoperatively, he is started on Lugol's iodine solution. Which of the following best describes its mechanism of action?

- a) Suppression of TSH secretion
- b) Decreased iodination of MIT
- c) Decreased T₃ to T₄ conversion
- d) Inhibition of organification and synthesis of thyroid hormones

Question 18:

How long is thyroxine stored in thyroid follicles?

- a) 2-3 weeks
- b) 2-3 days
- c) 2-3 months
- d) 2-3 years

Question 19:

The major thyroxine-binding protein is _____.

- a) Albumin
- b) Transthyretin
- c) Thyroxine-binding globulin
- d) Transferrin

Question 20:

Which of the following is false regarding thyroid hormone?

- a) T₃ is more avidly bound to nuclear receptor than T₄
- b) T₄ has more plasma concentration
- c) T₃ is more active than T₄
- d) T₄ has shorter half- life than T₃

Question 21:

Half-life of T₄_____

- a) 10 hours
- b) 6-7 days
- c) 6 weeks
- d) 2-3 months

Question 22:

A 35 yr old woman presents with heat intolerance and menstrual abnormalities. On examination, a diffusely enlarged thyroid gland is present. Which of the following features will not be present in her?

- a) Increased heart rate
- b) Increased stroke volume
- c) Decreased peripheral vascular resistance
- d) Decreased protein breakdown

Question 23:

Prolactin level is highest during :

- a) 24 hours after ovulation
- b) 24 hours after delivery
- c) REM Sleep
- d) After running for 1 hour

Question 24:

Which of the following conditions would have the highest levels of ACTH and cortisol?

- a) Early morning
- b) Normal individuals with dexamethasone administration
- c) Addison's disease
- d) Cushing's disease

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	a
5	d
6	b
7	c
8	d
9	b
10	c
11	a
12	a
13	d
14	a
15	a

16	d
17	d
18	c
19	c
20	d
21	b
22	d
23	b
24	d

Detailed Explanations

Solution to Question 1:

Growth hormone is secreted from acidophilic cells of anterior pituitary.

Acidophilic cells secrete growth hormone and prolactin.

Basophilic cells secrete TSH, ACTH, FSH and LH.

Solution to Question 2:

ACTH is more responsible than MSH for the hyperpigmentation seen in a patient with Cushing syndrome due to pituitary cause (ACTH hypersecretion).

ACTH is a derivative of POMC (Pro-opiomelanocortin). POMC is the precursor of ACTH, melanocyte stimulating hormone (MSH), β lipotropin and β endorphin. Since the melanocyte-stimulating hormone is the byproduct of ACTH synthesis from POMC, any disease process which leads to ACTH overproduction will also lead to an increased level of MSH. ACTH has mild MSH activity, hence elevated levels of ACTH acts on melanotropin-1 receptors causing generalized hyperpigmentation.

A similar manifestation will also be seen in Addison's disease where the ACTH level is high.

Solution to Question 3:

Inhibitory control by the hypothalamus is exerted mainly on prolactin and growth hormone.

The major hypothalamic releasing hormones are:

- Thyrotropin releasing hormone (TRH)
- Corticotropin releasing hormone (CRH)

- Growth hormone releasing hormone (GHRH)
- Gonadotropin releasing hormone (GnRH)

The major hypothalamic inhibitory hormones are:

- Prolactin inhibitory hormone or dopamine
- Growth hormone inhibitory hormone (GHIH), also called Somatostatin.

Solution to Question 4:

The secretion of prolactin is increased after the sectioning of the pituitary stalk.

Prolactin is unique among the pituitary hormones because the predominant central control mechanism is inhibitory, i.e., dopamine inhibits prolactin secretion. Hence, sectioning of the pituitary stalk (Ex: In head trauma or compressive skull lesions) prevents this inhibition and leads to increased secretion of prolactin.

Note:

If the pituitary stalk is cut above the pituitary gland but the entire hypothalamus is left intact, the posterior pituitary hormones (ADH & Oxytocin) show a transient decrease because they are initially synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and are then transported down to the nerve endings to the posterior pituitary gland, requiring several days to reach the gland.

Solution to Question 5:

L- Dopa is not a cause of increased prolactin level, it decreases prolactin secretion by increasing the formation of dopamine (prolactin inhibitory hormone).

Solution to Question 6:

Continuous GnRH administration suppresses the hypothalamic-pituitary axis.

Pulsatile GnRH stimulates the pulsatile release of LH and FSH.

Note: Long-acting GnRH agonists are used to suppress gonadotropin levels in children with precocious puberty and in men with prostate cancer.

Solution to Question 7:

Antilipolysis is mediated by IGF-1.

The actions of growth hormone-mediated via IGF-1 (somatomedin-C) are the indirect effects of growth hormone and include:

- Antilipolytic activity
- Insulin-like activity
- Protein synthesis
- Epiphyseal growth

The direct effects of GH include:

- Sodium retention
- Decreased insulin sensitivity
- Lipolysis
- Protein synthesis
- Epiphyseal growth

Solution to Question 8:

Growth hormone (GH) has both direct and indirect effects on bone growth.

GH stimulates prechondrocytes to differentiate into chondrocytes. GH strongly stimulates osteoblasts and chondrocytes causing increased deposition of proteins → bone growth. GH also increases insulin-like growth factor (IGF) levels, which stimulate bone growth.

Note: The pygmies of Africa have a congenital inability to synthesize significant amounts of somatomedin C (IGF-I). Therefore, even though their plasma concentration of growth hormone is either normal or high, they have diminished amounts of IGF-I in the plasma, which accounts for the short stature. Some other dwarfs (e.g., the Levi-Lorain dwarf) also have this problem.

Solution to Question 9:

Growth hormone demonstrates the depicted fluctuations. Growth hormone is secreted in a pulsatile fashion, with increase during strenuous exercise and during the early period of sleep, (Non-REM Sleep) as shown in the graph. It is reduced during the later stages of sleep. (REM sleep)

Other options:

Option A: ACTH levels are high in the early morning and low in the late evening.

Option C: TSH is secreted in pulses, but the fluctuations are less in comparison to other pituitary hormones because of the low amplitude of the pulses and the relatively long half-life of TSH.

Option D: Prolactin is secreted in a pulsatile fashion, with the highest secretory peak between 4:00 and 6:00 a.m. (REM Sleep)

Solution to Question 10:

Growth hormone(GH) level is decreased during deep sleep. Specifically, it is reduced during REM sleep as the levels of GHRH reduces and levels of cortisol are higher compared to slow wave sleep.

Hypoglycemia, fasting, and exercise cause an increase in GH secretion. GH levels are increased during non-REM sleep.

GH levels also decline with age and obesity.

Solution to Question 11:

The given clinical scenario is suggestive of acromegaly and is due to excess growth hormone (GH).

Acromegaly occurs when excessive GH is secreted after puberty (after fusion of long bone epiphysis). Features include

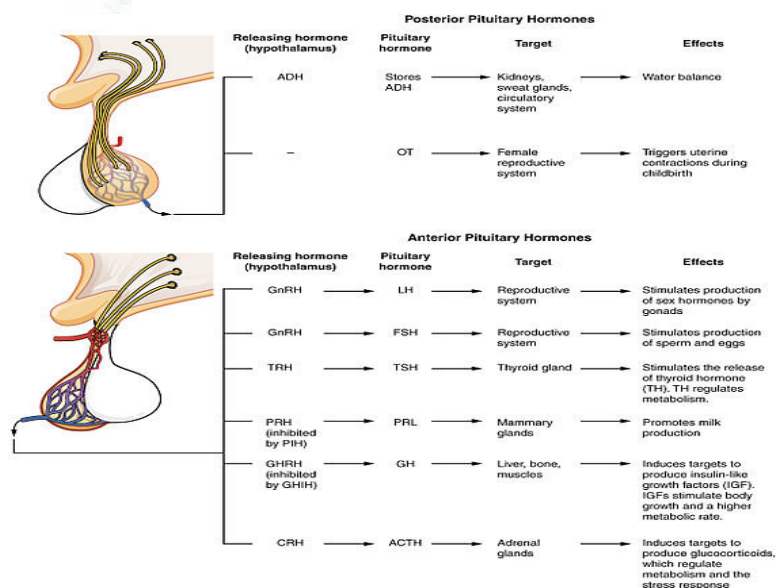
- Prognathism (protrusion of jaw), hypertrophy of frontal bones
- Acral enlargement in hands and feet (due to thickening of soft tissues)
- Visceromegaly: Enlargement of most organs especially the tongue, thyroid, salivary gland, liver, spleen, kidney, and prostate

Serious complications include cardiomyopathy with arrhythmias, diabetes mellitus, obstructive sleep apnoea, and increased risk of colonic polyps and cancer. Age-matched IGF-I levels are elevated.

Solution to Question 12:

The posterior pituitary or neurohypophysis releases oxytocin and anti-diuretic hormone (ADH or vasopressin)

Note: ADH and oxytocin are synthesised by the supraoptic and paraventricular nuclei of the hypothalamus. Thereafter, they are stored and released from the posterior pituitary.



Solution to Question 13:

The pair V4-CNS is incorrect.

Vasopressin mediates its effect via 3 kinds of receptors- V1a, V1b and V2. There is no V4.

Solution to Question 14:

Vasopressin escape in SIADH (Syndrome of inappropriate ADH secretion) results due to downregulation of aquaporin 2 production.

Prolonged exposure to elevated levels of vasopressin (ADH) leads eventually to downregulation of aquaporin-2 receptors. This prevents excessive reabsorption of water, despite high serum ADH levels and hence maintains euolemia. This is known as vasopressin escape. This helps prevent excessive dilutional hyponatremia in SIADH.

Solution to Question 15:

Sodium-iodide symport is absent in the pituitary.

Na⁺/I⁻ symporter is present in

- Basolateral membranes of thyrocytes facing the capillaries.
- Salivary glands
- Gastric mucosa
- Placenta
- Ciliary body of the eye
- Choroid plexus
- Mammary glands

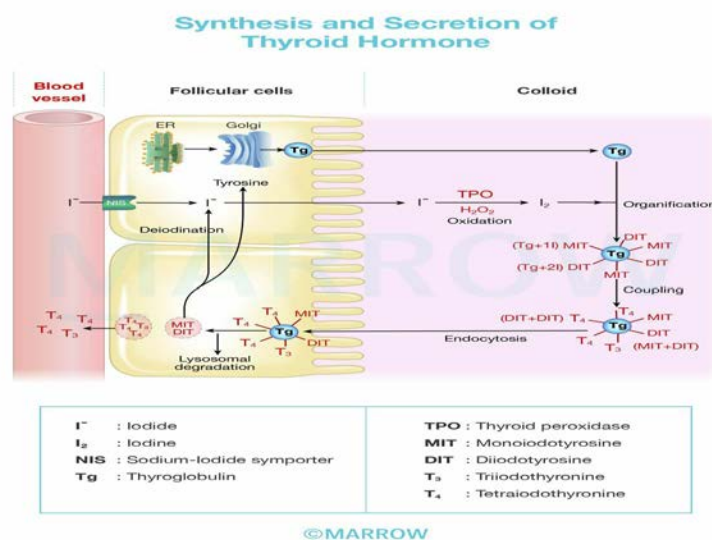
Solution to Question 16:

Iodide trapping (iodide uptake) is not catalyzed by thyroid peroxidase. Oxidation, organification, and coupling are catalyzed by it.

Steps in synthesis and secretion of thyroid hormone:

- Iodide uptake: The basolateral sodium-iodide symporter (NIS) actively takes up I⁻(iodide) from the circulation into the follicular cells. It is then transported into the colloid through a pendrin (chloride-iodide) transporter

- **Thyroglobulin synthesis:** Thyroglobulin (Tg), is a large glycoprotein rich in tyrosine residues. It is synthesized by follicular cells.
- **Iodination of thyroglobulin:** This is regulated by the thyroid peroxidase (TPO) enzyme. TPO has three functions:
 - **Oxidation:** TPO oxidizes I⁻ to I₂ (iodine) in the presence of H₂O₂
 - **Organification:** TPO links tyrosine residues of thyroglobulin with iodine (I₂) to form moniodotyrosine (MIT), diiodotyrosine (DIT)
 - **Coupling reactions:** TPO combines iodinated tyrosine residues to make MIT + DIT = T₃ (triiodothyronine), DIT + DIT = T₄ (tetraiodothyronine)
- **Storage:** Thyroid hormones are bound to thyroglobulin and stored in the follicular lumen as colloids. The colloid has hormone reserves for 2-3 months
- **Release:** Iodinated thyroglobulin is taken up by the thyroid follicles via endocytosis and fuses with lysosomes. It is cleaved by lysosomal proteolytic enzymes into MIT, DIT, T₃, and T₄. T₄ (80%) and T₃ (20%) are released into the blood
- **Peripheral T₃ conversion:** T₄ undergoes deiodination in extrathyroidal tissue to form T₃, which is the active form of the hormone



Solution to Question 17:

Introduction of exogenous iodine in the form of lugol's iodine solution leads to transient inhibition of organification and synthesis of thyroid hormones. This is known as the wolff-chaikoff effect.

In individuals with a normal thyroid, the gland escapes from this inhibitory effect and iodide organification resumes; but the suppressive action of high iodide may persist in patients with hyperthyroidism due to underlying autoimmune thyroid disease.

Note: Jod-Basedow effect is iodine-induced hyperthyroidism, usually occurring after the use of an I-based contrast agent. It is often seen in the presence of pre-existing thyroid conditions like multinodular goitre.

Solution to Question 18:

Thyroxine is stored in the thyroid follicles for 2-3 months, bound to thyroglobulin.

Serves as a reserve of T₃ and T₄, thus delaying the occurrence of hypothyroidism, if thyroid secretion ceases.

Solution to Question 19:

Thyroxine-binding globulin (TBG) is the major thyroxine-binding protein.

Other thyroxine-binding proteins are:

- Albumin
- Transthyretin (TTR)/ thyroxine-binding pre-albumin (TBPA)
- Lipoproteins

Solution to Question 20:

T₄ doesn't have a shorter half-life than T₃, rather T₃ has a shorter half-life and more rapid action than T₄.

The table compares T₄ and T₃:

T ₃	T ₄
More potent (4x)	Less potent
More active	Less active
Rapid onset of action	Slow onset of action
Shorter half-life (1 day)	Longer half-life (7 days)
Less plasma concentration	More plasma concentration
Less in colloid	More in colloid
Responsible for physiological actions	Used for pharmacotherapy

Solution to Question 21:

The half-life of T₄ is 6 - 7 days.

T₃ has the shorter half-life of 1 day.

Solution to Question 22:

The given clinical scenario is suggestive of hyperthyroidism, which does not cause decreased protein breakdown. Rather it causes increased protein breakdown.

Solution to Question 23:

Among the given options, the prolactin level is highest 24 hours after delivery.

The maximum level of prolactin is seen during pregnancy and in the immediate postpartum period. During pregnancy, the secretion is increased reaching a peak at the time of delivery. After delivery, the plasma concentration falls to nonpregnant levels in a few weeks. Suckling produces surges of prolactin release.

Solution to Question 24:

Among the given conditions, the levels of ACTH and cortisol will be highest in Cushing's disease.

Cushing's disease is caused by an ACTH secreting pituitary adenoma. The increased ACTH causes increased cortisol production.

Note: In normal physiologic state, ACTH levels are higher in the early morning but not as high as in Cushing's disease.



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The Pancreas

Question 1:

Which scientists were awarded the Nobel Prize for the discovery of insulin?

- a) Banting and Best
- b) Sanger
- c) Banting and McLeod
- d) Collip and McLeod

Question 2:

For cells of islets of Langerhans, which of the following is correctly matched?

- a) δ -somatostatin
- b) D- glucagon
- c) F- pancreatic polypeptide
- d) A – insulin

Question 3:

A 21-year-old female is found in a semiconscious state. On initial evaluation, her CBG is 23mg/dL. Her parents tell that the patient is an athlete and was trying to gain weight recently. An evaluation of which of the following will help in establishing a diagnosis?

- a) Pancreatic polypeptide
- b) Glucagon
- c) Somatostatin
- d) C peptide

Question 4:

The half-life of insulin is _____ minutes?

- a) 1-2
- b) 4-6

- c) 10-12
- d) 12-16

Question 5:

A 40-year-old diabetic man is initiated on a new hypoglycemic drug whose mechanism of action is extremely similar to the glucose-mediated insulin release mechanism. Which of the following best represents its mode of action?

- a) Closes ATP sensitive K⁺ channels
- b) Increases cAMP level in cell
- c) Participates in carrier mediated transport
- d) Causes receptor phosphorylation

Question 6:

Insulin mainly facilitates glucose uptake in which of the following?

- a) Brain
- b) Skeletal muscle
- c) RBC
- d) Kidney

Question 7:

Insulin acts on glucose metabolism by_____

- a) ↑ facilitated diffusion of glucose
- b) ↓ permeability of glucose across cell membrane
- c) ↑ reabsorption of glucose from renal tubules
- d) ↑ glucose reabsorption from intestine

Question 8:

A 45-year-old diabetic man develops a gradually increasing demand for exogenous subcutaneous insulin therapy for glycemic control. His current insulin requirement is 130 IU/day. Downregulation of which of the following is associated with his condition?

- a) GLUT 1
- b) GLUT 2
- c) GLUT 3
- d) GLUT 4

Question 9:

A 40-year-old man presented with diabetic ketoacidosis. After the initial evaluation, he is started on intravenous insulin therapy via an infusion pump. He is at increased risk of developing which of the following electrolyte abnormalities?

- a) Hyperkalemia
- b) Hypokalemia
- c) Hypernatremia
- d) Hyponatremia

Question 10:

Which of the following increases insulin secretion?

- a) Arginine
- b) Propranolol
- c) Epinephrine
- d) K⁺ depletion

Question 11:

Sympathetic stimulation has the following effect on insulin secretion_____

- a) Inhibition
- b) Stimulation
- c) Inhibition followed by stimulation
- d) No effect

Question 12:

Which of the following decreases the secretion of both insulin and glucagon?

- a) Epinephrine
- b) Somatostatin
- c) ↑ blood glucose
- d) None

Question 13:

Insulin secretion is normally stimulated by _____

- a) GLP 1
- b) GLP 2
- c) VIP
- d) Adrenergic receptor

Question 14:

Which of the following compounds antagonizes the action of insulin?

- a) Neuropeptide Y
- b) Growth Hormone
- c) Substance P
- d) Vasoactive Intestinal Peptide (VIP)

Question 15:

In fetus, the insulin secretion begins by _____(months)

- a) 5
- b) 3
- c) 7
- d) 9

Question 16:

How many parts are there in insulin receptor?

- a) 1

- b) 2
- c) 3
- d) 4

Question 17:

Regarding the action of insulin, which of the following is true?

- a) Decreased K^+ uptake in adipose tissue
- b) Increased protein catabolism in Muscle
- c) Activation of hormone sensitive lipase
- d) Decreased ketogenesis in liver

Question 18:

On giving a patient insulin, which of the following is expected to occur first?

- a) Increased transport of K^+ into adipocytes
- b) Stimulation of protein synthesis
- c) Activation of glycolysis
- d) Increase in mRNA for lipolysis

Question 19:

Which of the following is incorrectly paired?

- a) Epinephrine: increased glycolysis
- b) Insulin: increased protein synthesis
- c) Glucagon: increased gluconeogenesis
- d) Thyroxine: decrease blood glucose

Question 20:

A meal rich in insulin-stimulating amino acids but low in carbohydrates does not cause hypoglycemia because _____

- a) Cortisol in the circulation prevents glucose from entering muscle.

- b) Glucagon secretion is also stimulated by the meal.
- c) The amino acids in the meal are promptly converted to glucose.
- d) Insulin does not bind to insulin receptors if the plasma concentration of amino acids is elevated.

Question 21:

Which of the following would be least likely seen 14 days after a rat is injected with a drug that kills all of its pancreatic β cells?

- a) A rise in the plasma H^+ concentration
- b) A rise in the plasma amino acid concentration
- c) A fall in the plasma amino acid concentration
- d) A rise in plasma osmolality

Question 22:

Which of the following is an action of insulin?

- a) Activation of liver phosphorylase
- b) Decreased activity of glucokinase
- c) Decreased activity of glycogen synthase
- d) Inhibition of glucose 6-phosphatase

Question 23:

A 30 yr old male presents with recurrent attacks of sweating and dizziness. Further workup reveals hypoglycemia, elevated insulin and C-peptide levels. He is diagnosed with insulinoma and given diazoxide. Which of the following is the mechanism of action of this drug?

- a) Opening of the ATP Sensitive K^+ channels
- b) Closing of the ATP Sensitive K^+ channels
- c) Increase in number of GLUT-4 receptors
- d) Opening of the voltage sensitive Ca^{2+} channels

Question 24:

Coma occurs at a glucose level below _____.

- a) 30 mg/dl
- b) 45 mg/dl
- c) 15 mg/dl
- d) 5 mg/dl

Question 25:

Which of the following hormones use receptor tyrosine kinase signalling?

- a) Insulin
- b) TSH
- c) TRH
- d) MSH

Answer Key

Question No.	Correct Option
1	c
2	c
3	d
4	b
5	a
6	b
7	a
8	d
9	b
10	a
11	a
12	b
13	a
14	b
15	a
16	d

17	d
18	a
19	d
20	b
21	c
22	d
23	a
24	a
25	a

Detailed Explanations

Solution to Question 1:

Banting and McLeod were awarded the Nobel Prize for the discovery of insulin.

In 1921, Banting took his idea to Professor John Macleod at the University of Toronto, who was a leading figure in the study of diabetes in Canada. Macleod gave Banting a laboratory with a minimum equipments and ten dogs. Banting also got an assistant, a medical student by the name of Charles Best.

In 1923, the Nobel Committee decided to award Banting and Macleod the Nobel Prize in Physiology or Medicine. The decision of the Nobel Committee made Banting furious. He felt that the prize should have been shared between him and Best.

Solution to Question 2:

The correctly matched pair is F-Pancreatic polypeptide.

Cells of the endocrine pancreas are divided into:

The close interrelations among these cells allow direct control of secretion of some of the hormones by the other hormones. For example:

- Insulin inhibits glucagon secretion,
- Amylin inhibits glucagon secretion, and
- Somatostatin inhibits secretion of insulin and glucagon.

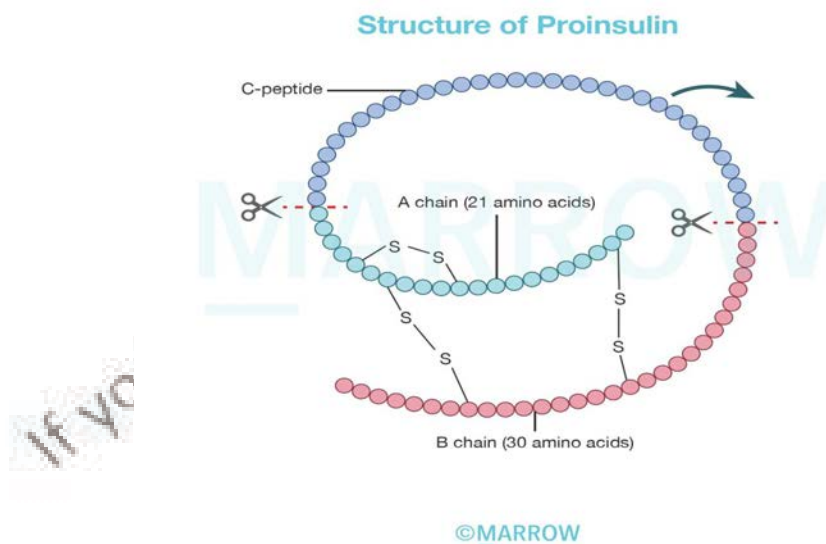
Cell type	What they secrete
A (alpha)	Glucagon
B (beta)	Insulin and amylin

Cell type	What they secrete
D (delta)	Somatostatin
F or PP	Pancreatic polypeptide

Solution to Question 3:

The given clinical vignette strongly suggests hypoglycemia due to surreptitious use of exogenous insulin, characterized by a low C-peptide level.

Insulin abuse is a method for increasing muscle bulk and performance enhancement in athletes, although less common than anabolic steroid abuse. Exogenous insulin does not contain C-peptide. Therefore blood level of C-peptide will be low in such patients. On the other hand, endogenous production of insulin will be characterized by high C-peptide level in blood.



Insulin is composed of two chains A and B, and a connecting peptide called C peptide. The insulin and C peptide are packaged in the secretory granules and secreted in equimolar amounts.

Importance of C-peptide:

- C peptide is measured by radioimmunoassay, and its level in the blood provides an index of beta-cell function in patients receiving exogenous insulin. If they are unable to produce insulin, they have greatly decreased levels of C peptide.
- C peptide levels can be used to differentiate between an insulinoma (high C-peptide) and exogenous insulin overdose (low C-peptide) in a patient with hypoglycemia.

Solution to Question 4:

The half-life of insulin is 4-6 minutes.

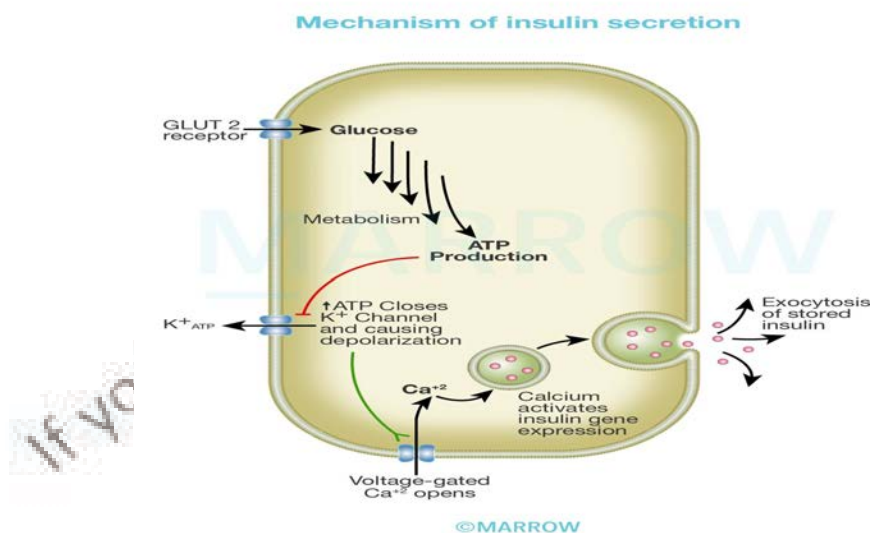
Insulin is degraded by the insulinase enzyme in the liver, and to a lesser extent, in the kidneys and muscles.

Note: Half-life of C-peptide is 20-30 minutes.

Solution to Question 5:

Glucose-mediated insulin release occurs through the closure of ATP sensitive K^+ channels. Therefore the new drug being initiated is most probably a sulfonylurea which acts via a similar mechanism.

Increased blood glucose concentration is the primary controller of insulin secretion. Inside the beta cells, glucose is converted to glucose-6-phosphate and later ATP is generated. This ATP closes ATP sensitive potassium channels, leading to a rise in K^+ and depolarization. This in turn causes the opening of voltage-gated Ca^{2+} channels and influx of Ca^{2+} . This ultimately leads to the secretion of insulin via exocytosis.



Diazoxide keeps the ATP sensitive K^+ channels open and prevents depolarization, resulting in a decreased insulin release. It is used in patients with insulinoma.

Sulfonylureas cause closure of the ATP sensitive K^+ channels, leading to increased insulin secretion. It is used in patients with diabetes mellitus.

Solution to Question 6:

Insulin mainly stimulates glucose uptake by skeletal muscles.

Glucose uptake by skeletal muscle, cardiac muscle and adipose tissues is via GLUT-4 which is insulin-dependent. In the presence of insulin, GLUT-4 receptors translocate to the cell membrane

and hence, glucose uptake by the cell occurs.

Most of the other GLUT transporters are not insulin sensitive and are constitutively expressed in the cell membrane. Ex: GLUT-1 in RBCs, brain and kidneys. (GLUT-3 is also present in the brain and kidneys)

Note: Insulin promotes glucose uptake in the liver. However, this is done by increasing the activity of glucokinase and inactivating glycogen phosphorylase. The net effect is the synthesis of glycogen and decreased breakdown of glycogen.

Solution to Question 7:

Insulin stimulates the glucose entry into the cells via facilitated diffusion by increasing the number of glucose transporters (GLUTs) in the cell membrane.

This type of insulin-dependent glucose transport via GLUT 4 is observed in skeletal muscle, cardiac muscle, and adipose tissues.

Solution to Question 8:

The given clinical vignette is suggestive of insulin resistance, which is characterized by the downregulation of GLUT-4.

GLUT-4 is an insulin-dependent GLUT. Other GLUTs are not insulin sensitive and are constitutively expressed in the cell membrane.

An arbitrary but clinically useful benchmark considers patients insulin resistant if they require greater than 1 unit/kilogram/day of exogenous insulin to maintain glycemic control. Patients requiring greater than 200 units of exogenous insulin per day are considered severely insulin resistant.

Solution to Question 9:

In patients who require intravenous insulin therapy, a rapid insulin infusion results in hypokalemia.

Therefore such patients should undergo serum electrolyte level assessment at frequent intervals while on insulin infusion for prompt diagnosis and early correction of hypokalemia.

Insulin increases the activity of Na-K ATPase in cell membranes so that more K⁺ is pumped into cells. Hence, serum K⁺ falls.

Note: Since insulin decreases serum potassium level, insulin-glucose infusions are used to treat hyperkalemia.

Solution to Question 10:

Arginine increases insulin secretion.

Option B: β -adrenergic blockers like propranolol inhibits insulin secretion.

Option C: Epinephrine both stimulates (by beta-adrenergic agonistic action) and inhibits (by alpha-adrenergic agonistic action) the secretion of insulin. However, the inhibitory effect of epinephrine is more dominant than the stimulatory effect.

Option D: Insulin promotes glucose uptake by cells, causing reduced serum K^+ . Low serum K^+ does not cause an increase in insulin secretion.

Solution to Question 11:

Sympathetic stimulation inhibits insulin secretion.

The regulation of insulin secretion can be controlled by alpha-adrenergic mediated response and beta-adrenergic mediated response. The stimulation of alpha-adrenergic receptor (α_2) plays an inhibitory role in insulin secretion. On the other hand, the stimulation of beta-adrenergic receptor (β_2) plays a stimulatory role in insulin secretion.

Stimulation of the sympathetic nerves to the pancreas releases norepinephrine which predominantly acts on α_2 -adrenergic receptors, causing a decrease in insulin secretion.

Note: If α -adrenergic receptors are blocked, stimulation of the sympathetic nerves causes increased insulin secretion mediated by β_2 -adrenergic receptors.

Solution to Question 12:

Somatostatin inhibits insulin and glucagon secretion.

Note: Other actions of somatostatin:

- Inhibits growth hormone secretion. Somatostatin is also known as growth hormone inhibitory hormone (GHIH).
- Decreases motility of the stomach, duodenum, and gallbladder.
- Decreases both secretion and absorption in the gastrointestinal tract.

Solution to Question 13:

Insulin secretion is normally stimulated by GLP-1.

Glucagon-like peptide-1 (GLP-1) and glucose-dependent insulintropic peptide (GIP) are called incretins and are potent stimulators of insulin secretion. They also inhibit glucagon secretion. They are released in the gastrointestinal tract after a person eats a meal. They then cause an "anticipatory" increase in blood insulin in preparation for the glucose and amino acids to be

absorbed from the meal.

Note: Drugs that mimic the action of incretins are used in the management of diabetes.

Solution to Question 14:

Growth hormone is a hyperglycemic hormone and antagonizes the effect of insulin.

Hyperglycemic Hormones

- Glucagon
- Epinephrine
- Cortisol
- Growth hormone
- Thyroid Hormone

Solution to Question 15:

In a fetus, the pancreatic islets are formed in the third month of the fetal life, while insulin secretion begins at about 5th month of fetal life.

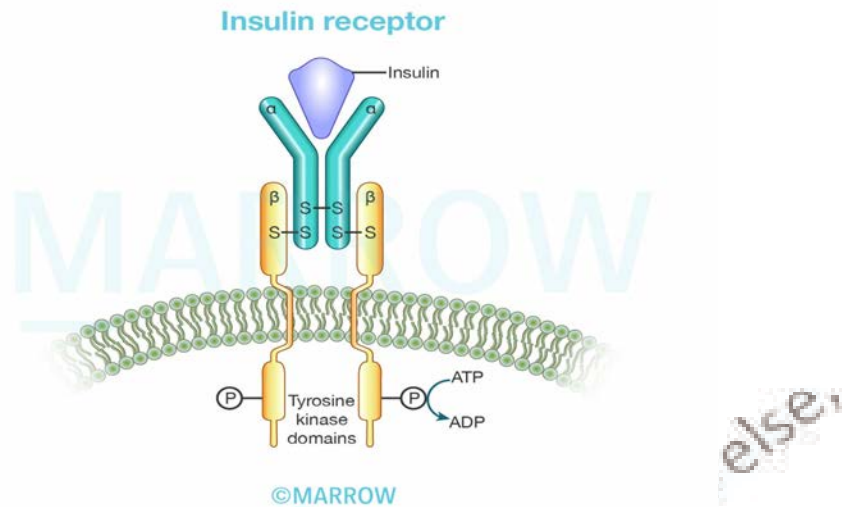
The exocrine pancreas starts functioning after birth.

Solution to Question 16:

An insulin receptor has 4 parts. It is a tetramer made up of two α and two β subunits.

The α subunits are extracellular, whereas the β subunits penetrate through the cell membrane into the cytoplasm. The intracellular portions of the β subunits have tyrosine kinase activity.

Once insulin binds to the alpha subunits, the portions of the beta subunits protruding into the cell become autophosphorylated. Autophosphorylation of the beta subunits, in turn, activates a local tyrosine kinase, which causes phosphorylation of multiple intracellular enzymes. The net effect is to activate some of these enzymes while inactivating others.



Solution to Question 17:

Insulin decreases ketogenesis in the liver. In the absence of insulin (Type-1 DM), diabetic ketosis may occur.

Effects of insulin on various tissues:

Tissue	Increases	Decreases
Adipose tissue	Increased glucose entry Increased fatty acid synthesis Increased glycerol phosphate synthesis Increased triglycerides deposition Activation of lipoprotein lipase Increased K^+ uptake	Inhibition of hormone-sensitive lipase
Muscle	Increased glucose entry Increased glycogen synthesis Increased amino acid uptake Increased protein synthesis in ribosome Increase ketone uptake Increase K^+ uptake	Decrease protein catabolism Decrease release of glucogenic amino acids
Liver	Increased protein synthesis Increased lipid synthesis Increased glycogen synthesis & glycolysis	Decrease ketogenesis Decreased glucose output due to decreased gluconeogenesis
General	Increased cell growth	

Solution to Question 18:

Increased transport of K^+ into adipocytes will occur first on giving an injection of insulin.

Based on time on onset, actions of insulin are classified as:

Rapid (seconds): Increased permeability, leading to:

- Increased transport of glucose, amino acids, and K^+ into insulin sensitive cells

Intermediate (minutes): Changes in activity levels of intracellular enzymes, leading to:

- Stimulation of protein synthesis
- Inhibition of protein degradation
- Activation of glycolytic enzymes and glycogen synthase
- Inhibition of phosphorylase and gluconeogenic enzymes

Delayed (hours to days): Changed rates of transcription of DNA, leading to:

- Increase in mRNAs for lipogenic and other enzymes

Solution to Question 19:

The incorrect pair is thyroxine - decreased blood glucose. Thyroxine is a diabetogenic hormone leading to an increase in blood glucose levels.

Effects Of Hormones On Carbohydrate Metabolism:

- Catecholamines: Activation of glycogen phosphorylase and therefore glycogenolysis occurs. Hepatic glucose output is increased, producing hyperglycemia.
- Glucocorticoids: Glucocorticoids from the adrenal cortex elevate blood glucose by stimulating gluconeogenesis and inhibiting glucose utilization by the cells.
- Growth Hormone: The effects of growth hormone are partly direct and partly mediated via IGF-I (Insulin-like Growth Factor-I). The growth hormone mobilizes FFA from adipose tissue, thus favoring ketogenesis. It decreases glucose uptake into some tissues ("anti-insulin action"), increases hepatic glucose output, and may decrease tissue binding of insulin.

Note: Infusing catecholamine (Epinephrine) mainly promotes glycogenolysis and it is also associated with enhanced rates of aerobic glycolysis in skeletal muscles during exercise (resulting in adenosine triphosphate production).

Solution to Question 20:

A meal rich in arginine and alanine stimulates both insulin and glucagon secretion. Hence, hypoglycemia does not occur.

Increased blood amino acids stimulate secretion of glucagon, which in turn causes the amino acids to get converted to glucose. So, the mechanism by which hypoglycemia is prevented after a protein rich meal is secretion of glucagon.

Solution to Question 21:

A fall in plasma amino acid concentration is least likely to be seen in this case.

Injury to the beta cells of the pancreas impairs insulin production rendering the rat insulin deficient. Due to insulin deficiency, catabolism of proteins increases and amino acids in plasma increase.

Insulin deficiency increases blood glucose levels, which lead to glycosuria and osmotic diuresis with extracellular dehydration. There is increase in the plasma osmolality.

Insulin deficiency causes increased utilization of fats, excess production of ketoacids and metabolic acidosis. Metabolic acidosis - a rise in the plasma H^+ develops.

Solution to Question 22:

Insulin inhibits glucose-6-phosphatase.

Insulin promotes liver uptake, storage, and the use of glucose. After a meal, insulin causes glucose to be rapidly stored in the liver in the form of glycogen. This occurs by:

- Inactivation of liver phosphorylase, which is the principal enzyme that causes liver glycogen to split into glucose.
- Enhanced uptake of glucose by the liver cells by increasing the activity of the enzyme glucokinase which causes phosphorylation of glucose after it enters the liver cells. Once phosphorylated, the glucose is temporarily trapped inside the liver cells because phosphorylated glucose cannot diffuse back through the cell membrane.
- Increased activity of enzymes that promote glycogen synthesis, like glycogen synthase.

When insulin levels drop (Ex: Fasting), glucose phosphatase gets activated. This causes phosphate to get detached from glucose phosphate, allowing free glucose from the liver cells to re-enter circulation.

Solution to Question 23:

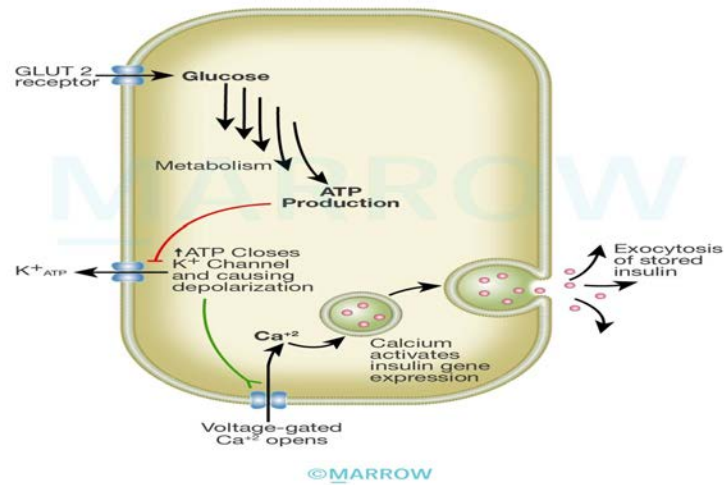
Diazoxide acts by opening ATP sensitive K^+ channels and causing a decrease in insulin secretion.

Glucose enters beta cells of the pancreas via GLUT-2 receptors. Once inside, it is phosphorylated by glucokinase (rate-limiting step) to glucose-6-phosphate which is oxidized to form adenosine triphosphate (ATP). This ATP causes closure of ATP-sensitive potassium channels and hence depolarization. This leads to opening voltage-gated calcium channels and Ca^{2+} influx. This stimulates fusion of the insulin-containing vesicles with the cell membrane and secretion of insulin into the extracellular fluid by exocytosis.

Diazoxide keeps the ATP sensitive K^+ channels open and prevents depolarization and subsequent insulin release.

Sulfonylureas cause closure of the ATP sensitive K^+ channels, leading to increased insulin secretion.

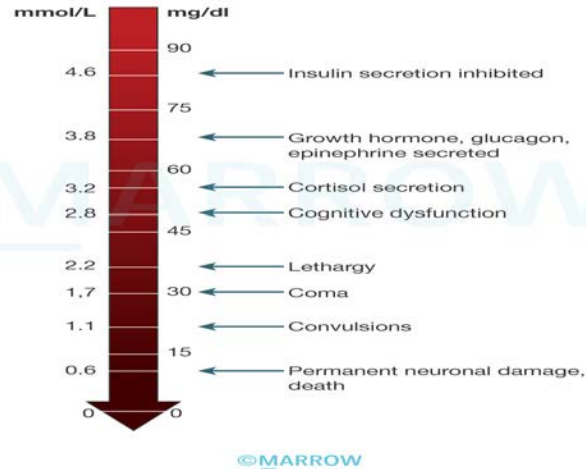
Mechanism of insulin secretion



Solution to Question 24:

Coma occurs when glucose levels fall below 30 mg/dL.

Effects of hypoglycemia and plasma glucose levels



Solution to Question 25:

Insulin uses the receptor tyrosine kinase signalling to exert its functions.

Other hormones acting in a similar fashion includes

- Growth hormone
- Growth factors like VEGF, PDGF, FGF, NGF

- Prolactin
- Leptin

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The Adrenals

Question 1:

Which of the following hormones is secreted by the zona glomerulosa?

- a) Aldosterone
- b) Testosterone
- c) Cortisol
- d) Epinephrine

Question 2:

Which of the following is not required for the formation of estradiol?

- a) Lyase
- b) 11β hydroxylase
- c) Aromatase
- d) Hydroxysteroid dehydrogenase

Question 3:

Which of the following statements is false?

- a) 90% of the cortisol is bound to plasma proteins
- b) Cortisol has a half-life of 60-90 minutes.
- c) Aldosterone has a half-life of about 20 minutes.
- d) 80% of the aldosterone is bound to plasma proteins

Question 4:

Which of the following is not an action of cortisol?

- a) Stimulates gluconeogenesis
- b) $\uparrow$ protein catabolism
- c) Stimulates lipolysis

- d) ↑ number of T-lymphocytes

Question 5:

All of the following conditions increase cortisol except _____

- a) Surgery
- b) Infection
- c) Heat stroke
- d) Sleep

Question 6:

A 60-year-old man is being treated for severe COVID with dexamethasone. Which of the following mechanisms best explain the rationale for starting steroids in this patient?

- a) Inhibition of phospholipase A₂
- b) Inhibition of cyclo-oxygenase
- c) ↑ activity of lipoprotein lipase
- d) Inhibition of lipoxygenase

Question 7:

Which of the following effects is produced only at high levels of glucocorticoids?

- a) Maintenance of normal vascular reactivity
- b) Increased excretion of a water load
- c) Inhibition of the inflammatory response
- d) Increased differentiation of adipose tissue

Question 8:

ACTH level is higher during _____

- a) Morning
- b) Evening
- c) Afternoon

d) Night

Question 9:

ACTH secretion is inhibited by_____

- a) Aldosterone
- b) Epinephrine
- c) Glucocorticoid
- d) CRH

Question 10:

Aldosterone secretion is maximally stimulated by_____

- a) Hyperkalemia
- b) ACTH
- c) Hypernatremia
- d) Exogenous steroid

Question 11:

Mineralocorticoid receptors are present in all except_____

- a) Hippocampus
- b) Kidney
- c) Colon
- d) Liver

Question 12:

In which of the following ways does aldosterone react with renal tubular cells?

- a) Increases transport of ENaCs from the cytoplasm to the cell membrane
- b) Binds to steroid receptor inside the nucleus of renal tubular cells
- c) Binds to ENaC's on the cell membrane and increase Na^+ transport
- d) Binds to G-protein coupled receptors on the cell surface

Question 13:

Which of the following has the greatest effect on plasma osmolality?

- a) Progesterone
- b) Cortisol
- c) Vasopressin
- d) Aldosterone

Question 14:

A 30-year-old man is admitted with nausea, vomiting, and dizziness. On evaluation, he is hypotensive and has mucosal hyperpigmentation. Laboratory workup reveals the diagnosis of hypoadosteronism. Which of the following is least likely to be found in this patient?

- a) Hyponatremia
- b) Hyperkalemia
- c) Hypotension
- d) Alkalosis

Question 15:

A 35-year-old female presents to the OPD with headaches and high blood pressure. On investigation, increased aldosterone levels with low renin levels are seen and a diagnosis of Conn's syndrome is made. Which of the following is least likely to be found in this patient?

- a) Slight increase in ECF volume
- b) Rise of $> 7\text{meq/L}$ in plasma Na^+ level
- c) Hypertension
- d) Hypokalemia

Question 16:

Which of the following decreases aldosterone secretion?

- a) Renin
- b) Hyperkalemia

- c) ACTH
- d) Hypernatremia

Question 17:

Hormones that are increased in plasma during periods of stress include all of the following except _____

- a) Glucocorticoids
- b) Endorphins
- c) Vasopressin
- d) Insulin

Question 18:

Which of the following is the major adrenal androgen ?

- a) Testosterone
- b) Androstenedione
- c) Dehydroepiandrosterone
- d) Cortisol

Question 19:

A 16-year-old female patient presents to the OPD with hirsutism and masculinization. Which of the following hormones of the adrenal cortex is the likely culprit?

- a) Dehydroepiandrosterone
- b) Aldosterone
- c) Cortisol
- d) Epinephrine

Question 20:

Which of the following is not directly controlled by ACTH?

- a) Glucocorticoids

- b) Androstenedione
- c) Dehydroepiandrosterone
- d) Epinephrine

Question 21:

Which of the following hormones is exclusively secreted by adrenal gland but not by the ovaries of females?

- a) Progesterone
- b) DHEA
- c) Estrogen
- d) DHEA-S

Answer Key

Question No.	Correct Option
1	a
2	b
3	d
4	d
5	d
6	a
7	c
8	a
9	c
10	a
11	d
12	a
13	c
14	d
15	b
16	d
17	d
18	c

19	a
20	d
21	d

Detailed Explanations

Solution to Question 1:

The zona glomerulosa secretes aldosterone.

The adrenal cortex has three zones. From outer to inner, they are:

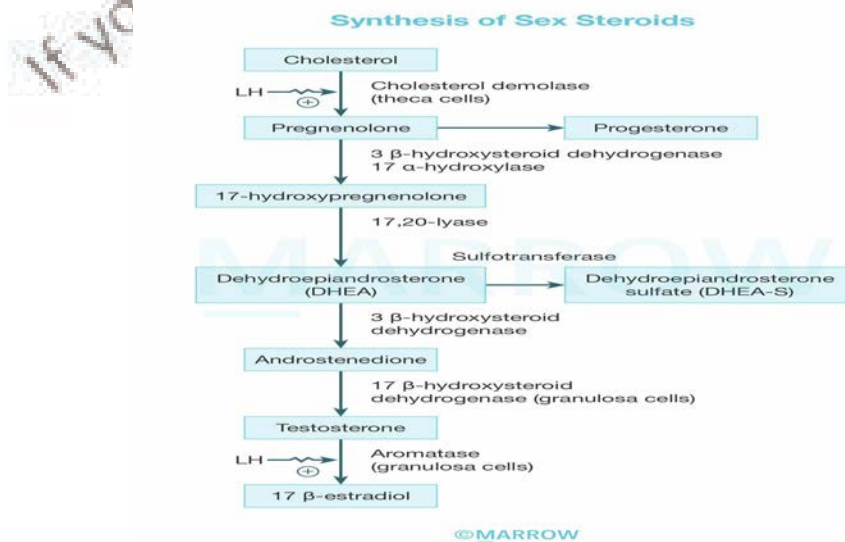
- Zona glomerulosa: Secretes mineralocorticoids (Aldosterone)
- Zona fasciculata: Secretes glucocorticoids (Cortisol and corticosterone)
- Zona reticularis: Secretes sex steroids (dehydroepiandrosterone, androstenedione, estrogens, and glucocorticoids)

The adrenal medulla secretes catecholamines.

Solution to Question 2:

11 β hydroxylase is not required for estradiol synthesis.

11 β hydroxylase is required for the conversion of 11-deoxycorticosterone to corticosterone, a compound formed in the mineralocorticoid synthesis pathway.



Solution to Question 3:

60% of aldosterone is bound to plasma proteins, not 80%.

90% of cortisol is bound to plasma proteins. Most of it is bound to transcortin, a corticosteroid-specific γ globulin, and the rest to albumin. This high degree of binding to plasma proteins slows the elimination of cortisol. Therefore, cortisol has a longer half-life of 60 to 90 minutes.

Since only about 60% of aldosterone is in bound form, it has a relatively short half-life of about 20 minutes.

Solution to Question 4:

Cortisol suppresses the proliferation of T-lymphocytes and decreases the number of T-Lymphocytes.

Actions of Cortisol:

Metabolic effects:

- Effect on carbohydrates: Exert an anti-insulin effect, leading to hyperglycemia
- Increases gluconeogenesis
- Increases glycogen synthesis
- Decreases utilization of glucose in peripheral tissues
- Effect on fat and appetite: promotes mobilization of fatty acids from adipose tissue in acute states. This mobilization increases the concentration of free fatty acids in the plasma, which also increases their utilization for energy, causing lipolysis. However, in chronic states cortisol increases appetite and food intake by inducing the synthesis of neuropeptide-Y which leads to an apparent lipogenic effect seen as buffalo humping and central obesity.
- Effect on protein: Exerts a catabolic effect and causes proteolysis
- Effect on water and electrolytes: Sodium water retention and diuresis

Anti-inflammatory and immunosuppressive effects:

- Induce the synthesis of lipocortin, an inhibitor of phospholipase A₂
- Inhibit the release of histamine and serotonin
- Inhibit the production of IL-2 and T lymphocytes, and depress cellular immunity

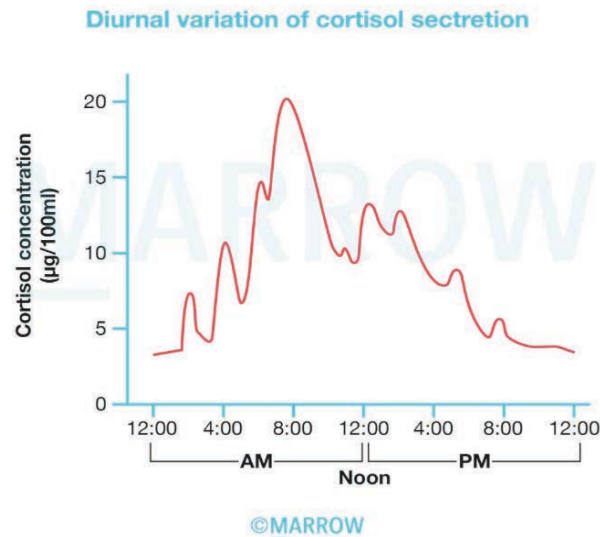
Maintenance of vascular responsiveness to catecholamines:

- Cortisol up-regulates α_1 receptors on arterioles, increasing their sensitivity to the vasoconstrictor effect of norepinephrine

Solution to Question 5:

Cortisol secretion is low during sleep.

Any type of stress, physical or mental, can increase cortisol release.



Solution to Question 6:

Inhibition of phospholipase A2 is an anti-inflammatory effect of steroids, thus being the rationale for starting steroids in this patient.

Patients with severe COVID are at high risk of developing a systemic inflammatory response. Steroids help in mitigating these deleterious effects.

Phospholipase A2 is a precursor of arachidonic acid. Arachidonic acid is the precursor for prostaglandins and leukotrienes, which are involved in inflammation. Steroids induce the synthesis of lipocortin, which inhibits phospholipase A2.

Other anti-inflammatory and immunosuppressive effects of steroids are:

- Inhibit the production of interleukin-2 (IL-2) and proliferation of T lymphocytes
- Inhibit the release of histamine and serotonin from mast cells and platelets

Solution to Question 7:

Inhibition of inflammatory response requires high levels of glucocorticoids.

Anti-inflammatory and anti-allergic responses (inhibition of histamine from mast cells and basophils) are produced only at high levels of glucocorticoids. These effects cannot be produced by normal circulating glucocorticoids in the body and thus require exogenous supplementation of steroids to produce them.

Solution to Question 8:

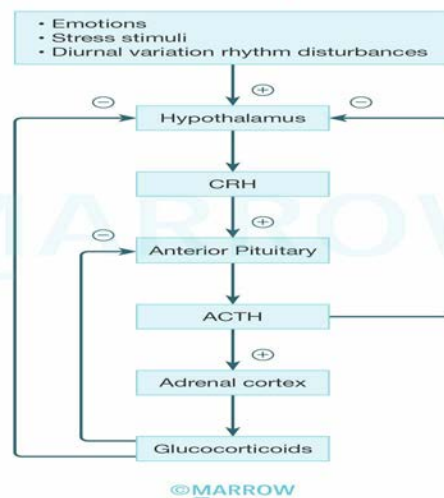
ACTH level is higher during the early morning and low during late evening. 75% of the ACTH production occurs between 4:00 AM to 10:00 AM.

Note: Cortisol levels are also high in the early morning and low in the late evening.

Solution to Question 9:

Glucocorticoids inhibit ACTH secretion by a direct negative feedback mechanism. They also inhibit CRH secretion.

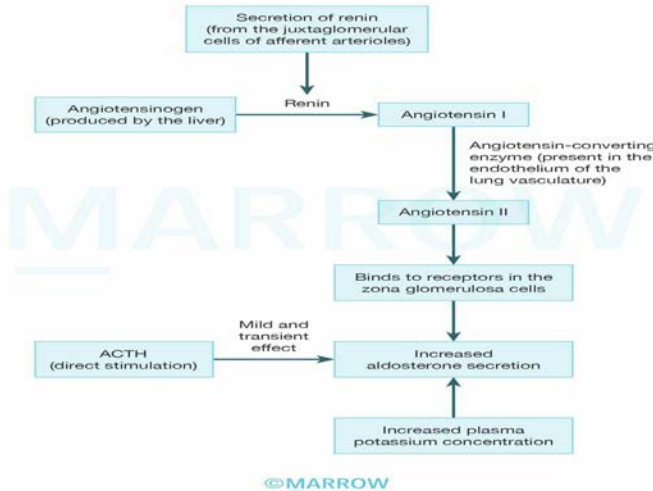
Hypothalamo-Pituitary-Adrenal Axis



Solution to Question 10:

Hyperkalemia and activation of renin-angiotensin system are the most potent stimulators of aldosterone secretion.

Regulation of Aldosterone



Solution to Question 11:

Mineralocorticoid receptors are not present in the liver. They are present in kidneys, colon, hippocampus, salivary glands, and sweat glands.

Mineralocorticoid activity increases the reabsorption of Na^+ and potassium excretion from urine, sweat, saliva, and contents of the colon. Along with the Na^+ , there is water reabsorption and inturn ECF volume expansion.

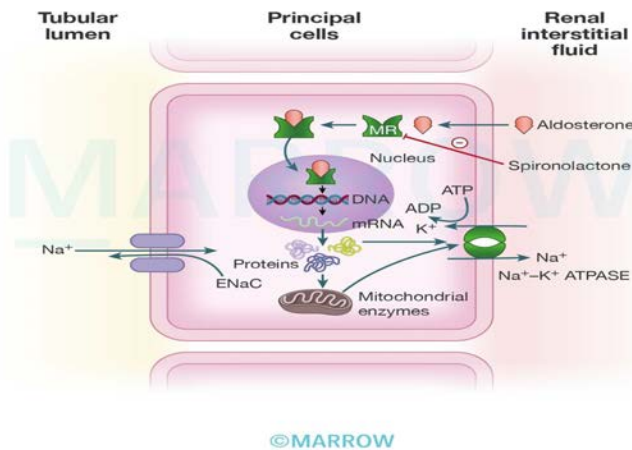
Solution to Question 12:

Aldosterone increases transport of ENaCs (epithelial sodium channels) from the cytoplasm to the cell membrane. This is a rapid effect of aldosterone.

Mechanism of action of aldosterone:

- Aldosterone diffuses into the renal tubular epithelial cells and combines with a mineralocorticoid receptor (MR) in the cytoplasm.
- The aldosterone-receptor complex diffuses into the nucleus and induces transcription.
- The mRNA diffuses back into the cytoplasm and undergoes translation to form proteins. These have two effects - Rapid and Slow.
- The slower effect is increase in synthesis of ENaCs.
- The important products formed are:
 - $\text{Na}^+\text{K}^+\text{ATPase}$
 - membrane transport proteins like epithelial sodium channel

Aldosterone mechanism of action



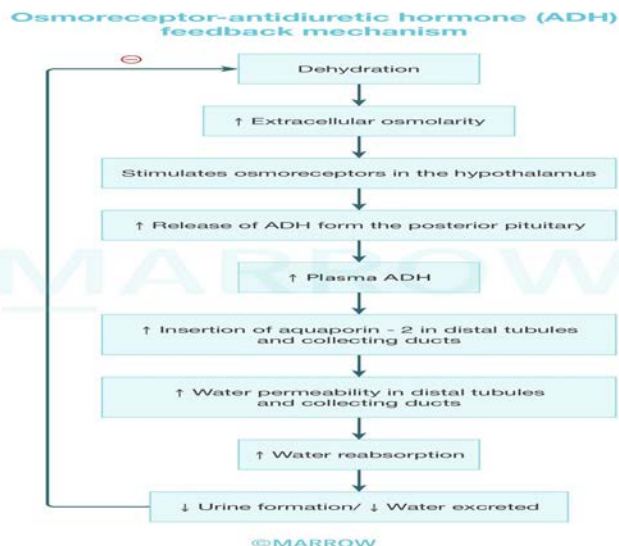
Solution to Question 13:

Vasopressin (also called ADH or anti-diuretic hormone) has the greatest effect on plasma osmolality.

Two primary systems are especially involved in regulating the concentration of sodium and osmolality of extracellular fluid:

- The osmoreceptor-ADH system and
- The thirst mechanism.

Although aldosterone decreases the rate of sodium excretion by the kidneys, the concentration of sodium in the extracellular fluid rises only by a few mEq. The reason for this is that when sodium is reabsorbed by the tubules, simultaneous absorption of equivalent amounts of water occurs. So, aldosterone does not cause a significant change in the plasma osmolality.



Solution to Question 14:

In a patient with hypoaldosteronism, alkalosis is least likely to be found. Hypoaldosteronism is rather responsible for causing metabolic acidosis.

Patients with hypoaldosteronism have:

- Hyponatremia- due to increased Na^+ loss in urine
- Hyperkalemia- due to K^+ retention
- Hypotension- due to reduced ECF volume
- Mild metabolic acidosis because hydrogen ions are not secreted in the urine

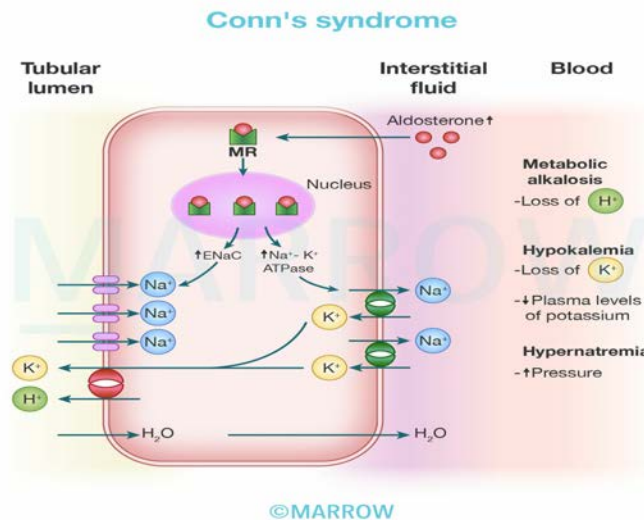
Solution to Question 15:

Rise in plasma Na^+ level of $> 7\text{mEq/L}$ than the normal value is least likely to be found in this patient.

Excess secretion of aldosterone causes only a modest increase in plasma sodium, usually a rise of less than $4\text{-}6\text{ mEq/L}$. When sodium gets reabsorbed, water also accompanies it, resulting in plasma sodium remaining fairly normal.

Conn's syndrome or primary hyperaldosteronism is due to an aldosterone-producing tumour of the zona glomerulosa, causing:

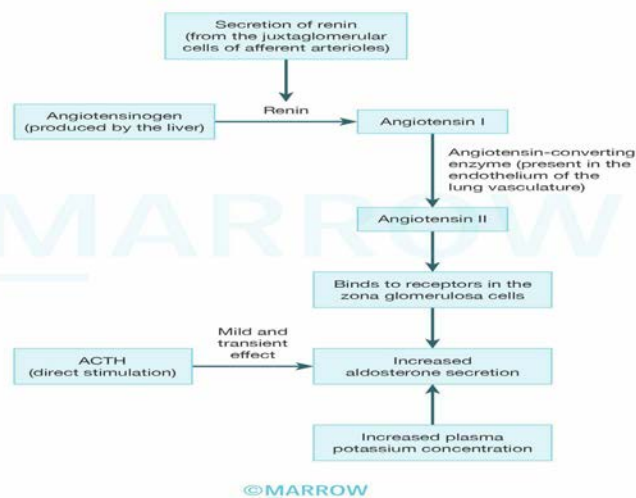
- Loss of potassium by the kidneys leads to hypokalemia.
- Mild metabolic alkalosis.
- A slight increase in extracellular fluid volume and blood volume.
- A modest increase in plasma sodium concentration.
- Hypertension.



Solution to Question 16:

Hypertension decreases aldosterone secretion.

Regulation of Aldosterone



Solution to Question 17:

Insulin does not increase during stress.

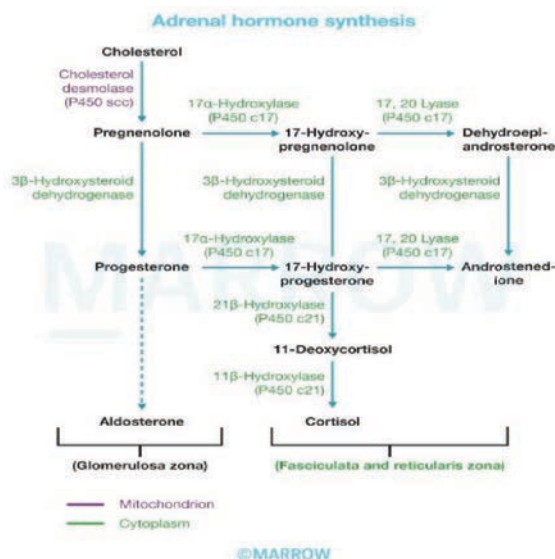
In response to stress, the usual physiological functions like gluconeogenesis, lipolysis, and inhibition of insulin actions occurs, resulting in increased blood levels of fuel sources (glucose, fatty acids) needed to cope.

Hormones that are increased in the blood by stress are CRH, ACTH, cortisol, β -endorphin(to reduce pain), vasopressin, aldosterone, growth hormone, and glucagon.

Solution to Question 18:

Dehydroepiandrosterone (DHEA) is the major adrenal androgen.

The androgens secreted from the adrenal gland are dehydroepiandrosterone and androstenedione.



Solution to Question 19:

Hirsutism and masculinization in a female suggest excessive androgens like dehydroepiandrosterone (DHEA).

Adrenogenital syndrome is due to an adrenocortical tumor secreting excessive quantities of androgens that cause intense masculinizing effects. In boys, it presents as precocious puberty. In women, virile characteristics develop, including:

- Growth of a beard
- Deeper voice
- Masculine distribution of hair on the body and the pubis and
- Growth of the clitoris

The excretion of 17-ketosteroids (which are derived from androgens) in the urine maybe 10 to 15 times elevated. This finding can be used in diagnosing the disease.

Solution to Question 20:

The secretion of the adrenal medullary hormones (epinephrine, norepinephrine, and dopamine) is not controlled directly by ACTH.

	Hormones secreted	Regulated by
Zona glomerulosa	Mineralocorticoids (aldosterone)	Angiotensin II, K ⁺ , ACTH (mild transient effect)
Zona fasciculata	Glucocorticoids (cortisol, corticosterone)	ACTH
Zona reticularis	Sex steroids (DHEA, DHEA-S)	ACTH
Adrenal medulla	Catecholamines (epinephrine, norepinephrine)	Diffuse sympathetic discharge (stress)

Solution to Question 21:

DHEAS is exclusively secreted from the adrenals but not by ovaries because ovaries lack the enzyme dehydroepiandrosterone-sulphotransferase needed to form DHEAS.

90% of DHEA is synthesised by the adrenals and 10% comes from the ovaries.

Calcium Homeostasis

Question 1:

Which of the following vitamins have a hormonal action?

- a) Niacin
- b) Pyridoxine
- c) Vitamin D
- d) Riboflavin

Question 2:

The active form of vitamin D is _____

- a) Cholecalciferol
- b) 24,25(OH) Vitamin D
- c) 1,25 (OH)₂ Vitamin D
- d) 25 (OH) Vitamin D

Question 3:

Which of the following is false regarding actions of vitamin D?

- a) Vit D receptor-retinoid X receptor complex binds to DNA and activates transcription
- b) Vit D increases absorption of calcium from the intestine by increasing calbindin
- c) Phosphate flux through the gastrointestinal epithelium is enhanced by vitamin D
- d) Vit D decreases phosphate reabsorption by the renal tubules

Question 4:

A 6-year-old boy was brought to the OPD with craniotables, rachitic rosary and bowed legs. Diagnosis of rickets was made. To investigate the cause of rickets and to find the status of vitamin D in the body, which of the following should be measured?

- a) Cholecalciferol
- b) 25 hydroxyvitamin D

- c) 1, 25 dihydroxyvitamin D
- d) 24, 25 dihydroxyvitamin D

Question 5:

Which of the following decreases the conversion of 25-hydroxycholecalciferol to 1,25 dihydroxycholecalciferol?

- a) Diet low in Ca^{2+}
- b) Hypocalcemia
- c) Hyperparathyroidism
- d) Chronic renal failure

Question 6:

Which organ is not involved in calcium homeostasis?

- a) Kidneys
- b) Skin
- c) Intestine
- d) Lung

Question 7:

Increased calcium levels lead to _____

- a) $\uparrow$ 1,25 dihydroxycholecalciferol
- b) $\uparrow$ 24,25 dihydroxycholecalciferol
- c) $\downarrow$ calcitonin
- d) $\uparrow$ parathormone

Question 8:

What would be seen in a patient whose diet is deficient in calcium?

- a) $\uparrow$ phosphate
- b) $\uparrow$ calcium

- c) ↑ parathormone
- d) Activation of 24,25 dihydrocholecalciferol

Question 9:

In an experiment, a rat is infused with a small volume of a calcium chloride solution. Compared to the control condition, which of the following would result from the calcium load?

- a) Bone demineralization
- b) Decreased secretion of calcitonin
- c) Decreased blood coagulability
- d) Increased formation of 24,25-dihydroxycholecalciferol

Question 10:

Calcium absorption takes place mainly from which of the following sites?

- a) Proximal small intestine
- b) Middle small intestine
- c) Distal ileum
- d) Ascending colon

Question 11:

Which of the following is an enhancer of calcium absorption from the intestine?

- a) Protein
- b) Oxalates
- c) Phytic acid
- d) Phosphates

Question 12:

Which of the following is the active form of calcium?

- a) Ionized

- b) Albumin bound
- c) Phosphate
- d) Protein bound

Question 13:

Which of the following cells play a key role in bone resorption?

- a) Osteoblasts
- b) Osteoclasts
- c) Stem cells
- d) Cytotoxic cells

Question 14:

On infusing parathyroid hormone in experimental mice, there is increased resorption of bone due to stimulation of the osteoclastic activity. Which of the following is true regarding the cellular mechanism of osteoclastic activation?

- a) The bone-resorbing osteoclast cells do not have PTH receptors
- b) Osteoclasts produce osteoprotegerin (OPG) a cytokine that increases bone resorption
- c) RANK ligand and M-CSF are necessary for formation of mature osteoblasts
- d) Glucocorticoids promote osteoblast activity and inhibit bone resorption

Question 15:

A mouse is engineered to lack a transcription factor necessary for the normal development of osteoclasts. Compared to normal littermate mice, which of the following would be reduced in the knocked-out mouse?

- a) Phosphate deposition in trabecular bone
- b) Hydroxyapatite levels in bone
- c) Osteoblast proliferation
- d) Secretion of acid proteases

Question 16:

Which of the following is true regarding calcium homeostasis?

- a) 90% of calcium is filtered from the glomerulus
- b) Calcium is reabsorbed mainly from the late distal convoluted tubule (DCT)
- c) Parathyroid hormone (PTH) promotes calcium reabsorption from the renal tubules
- d) Calcitonin promotes action of parathyroid hormone

Question 17:

Which of the following is not an effect of parathormone?

- a) ↑ bone resorption
- b) ↑ calcium reabsorption in kidney
- c) ↑ phosphate reabsorption in kidney
- d) ↑ calcitriol (vitamin D₃) synthesis

Question 18:

Which of the following will be most likely seen in a patient with parathyroid deficiency after inadvertent damage to the parathyroid glands during thyroid surgery?

- a) High plasma phosphate, low Ca²⁺ levels and tetanus
- b) Low plasma Ca²⁺ level, increased muscular excitability, and Trousseau sign
- c) High plasma phosphate and Ca²⁺ levels and bone demineralization
- d) Increased muscular excitability, a high plasma Ca²⁺ level, and bone demineralization

Question 19:

Which of these cause hypocalcemia?

- a) Thyroxine
- b) Calcitonin
- c) Parathyroid hormone
- d) Cholecalciferol

Question 20:

A 57-year-old lady with metastatic breast cancer develops nausea, vomiting and confusion. Further evaluation reveals severe hypercalcemia. IV hydration is initiated and calcitonin is added to her treatment plan. Which of the following best represents the role of calcitonin in lowering serum calcium level?

- a) ↓ Excretion of calcium by kidney
- b) ↓ Intestinal absorption of calcium
- c) ↓ Bone resorption
- d) ↑ Renal absorption of calcium

Question 21:

A 30-year-old lady undergoes thyroidectomy due to suspected malignancy. A week after the surgery she develops a spasm of facial muscles on tapping over the facial nerve in front of the tragus. Which of the following is the reason behind her muscle hyperexcitability?

- a) ↑Ca²⁺
- b) Prevent K⁺ release
- c) Prevent Na⁺ and K⁺ release
- d) ↑permeability of Na⁺ ions

Question 22:

Which of the following is false regarding confounders affecting serum calcium level?

- a) Total calcium concentration is altered by variations in the serum albumin concentration.
- b) Forearm exercise can lead to falsely elevated ionized calcium levels
- c) Hemoconcentration due to tourniquet results in a rise of serum calcium
- d) Acidosis causes a decrease in the ionized calcium levels

Question 23:

Which of the following cells is involved in the initiation and control of osteoclastic activity?

- a) Osteocytes
- b) Osteoblasts
- c) Mononuclear cells
- d) Phagocytes

Question 24:

Which of these is not a marker of bone resorption?

- a) Urine hydroxyproline
- b) Serum cross-linked c telopeptide
- c) Urine total free deoxypyridinoline
- d) Serum bone specific alkaline phosphatase

Question 25:

Which of the following is not a cause of secondary hyperparathyroidism?

- a) Parathyroid adenoma
- b) Lithium intake
- c) Vitamin D deficiency
- d) Chronic kidney disease

Answer Key

Question No.	Correct Option
1	c
2	c
3	d
4	b
5	d
6	d
7	b
8	c
9	d
10	a
11	a
12	a
13	b

14	a
15	d
16	c
17	c
18	b
19	b
20	c
21	d
22	d
23	b
24	d
25	a

Detailed Explanations

Solution to Question 1:

Vitamin D has hormonal action. In conjunction with parathyroid hormone, it is a major regulatory hormone for Ca^{2+} and phosphate metabolism.

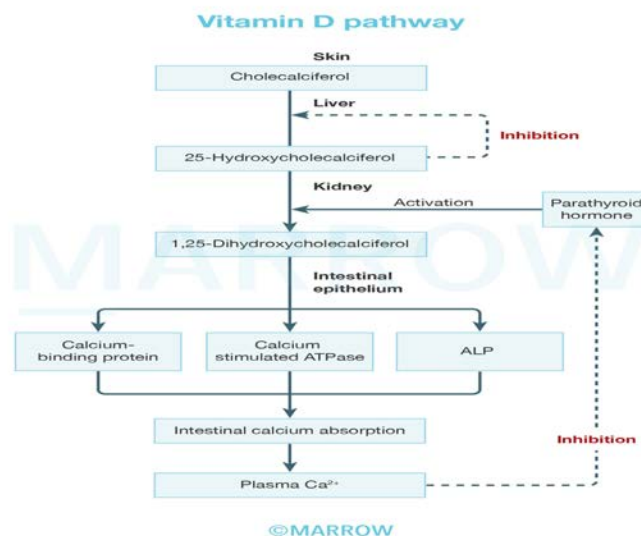
Solution to Question 2:

The active form of Vit D is 1,25-dihydroxycholecalciferol.

Cholecalciferol is formed in the skin and converted to 25 hydroxycholecalciferol in the liver. This is hydroxylated in the kidney to form 1,25-dihydroxycholecalciferol.

Solution to Question 3:

Vit D does not decrease phosphate reabsorption by the renal tubules. It increases calcium and phosphate reabsorption by the renal tubules, leading to a decrease in excretion of these substances in the urine.



Solution to Question 4:

Measurement of 25 hydroxyvitamin D (calcidiol) provides more accurate information about vitamin D levels.

PTH is increased in vitamin D deficiency. PTH is a major stimulus for the 1° -hydroxylase, so there is increased synthesis of the active hormone, 1,25-dihydroxyvitamin D. Paradoxically, levels of this hormone are often normal in severe vitamin D deficiency. Therefore, measurements of 1,25-dihydroxyvitamin D are not accurate reflections of vitamin D stores and should not be used to diagnose vitamin D deficiency in patients with normal renal function.

Solution to Question 5:

The conversion of 25-hydroxycholecalciferol to 1,25 dihydroxycholecalciferol is done by enzyme 1° -hydroxylase which is present in the kidney. In chronic renal failure (CRF), there is a deficiency of this enzyme and formation of the active form of vitamin D does not take place.

Factors which increase activity of 1° -hydroxylase are:

- Ca²⁺ deficiency
- Increased parathyroid hormone (PTH)

Solution to Question 6:

The lung is not involved in calcium homeostasis.

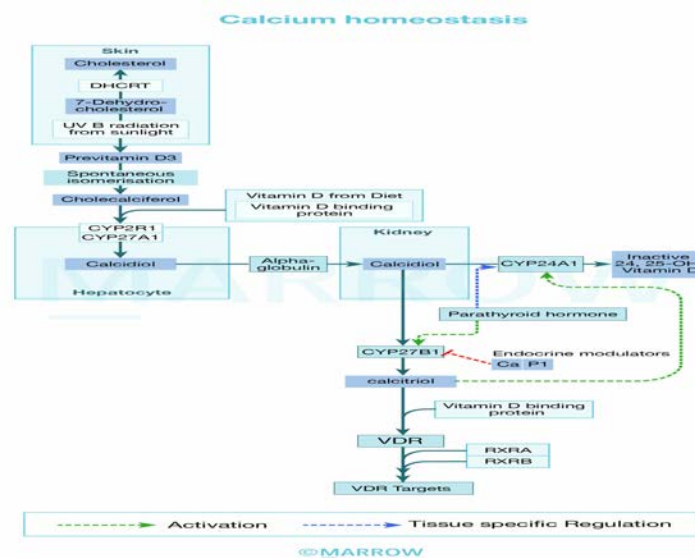
There are two sources of cholecalciferol in the body:

- Diet
- Synthesized in the skin from 7-dehydrocholesterol in the presence of ultraviolet light.

Cholecalciferol is hydroxylated in the liver to form 25-hydroxycholecalciferol, which is further hydroxylated in the kidney to form 1,25-dihydroxycholecalciferol. At higher calcium concentrations, when PTH is suppressed, the 25-hydroxycholecalciferol is converted to a different compound—24,25-dihydroxycholecalciferol—that has almost no vitamin D effect.

Note: This is a controversial question because new evidence suggests an association between calcium homeostasis and pulmonary fibroblasts. Therefore:

- If Lung is given in the options and spleen is provided as one of the choices instead of Kidneys, Skin, Liver or Intestine, then the best answer would be the spleen.
- If other options are Kidneys, Skin, Liver, or Intestine, then the best answer would be Lung as the role of these organs in calcium metabolism is beyond doubt.

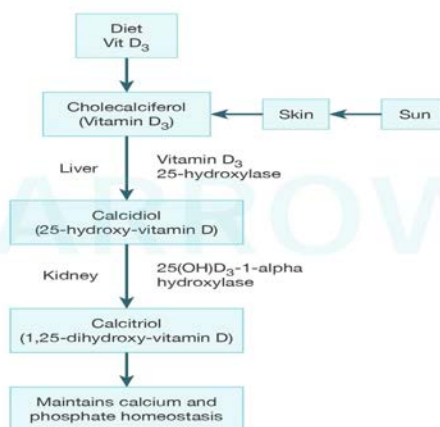


Solution to Question 7:

Increased calcium levels lead to an increase in 24,25 dihydroxycholecalciferol.

The levels of calcium control formation of 1,25-dihydroxycholecalciferol. When calcium is low, PTH (parathyroid hormone) promotes the conversion of 25hydroxycholecalciferol to 1,25-dihydroxycholecalciferol in the kidneys. At higher calcium concentrations, when PTH is suppressed, the 25hydroxycholecalciferol is converted to a different compound—24,25-dihydroxycholecalciferol—that has almost no vitamin D effect.

Formation of 1,25-dihydroxycholecalciferol



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Solution to Question 8:

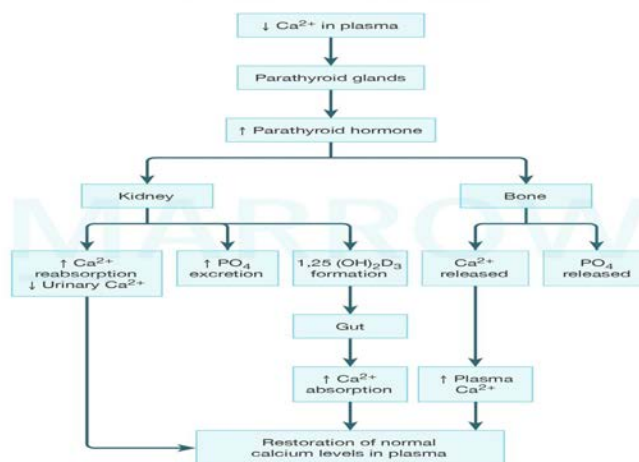
Low Ca^{2+} diet leads to hypocalcemia, which stimulates PTH release. Hence, this patient will have high PTH levels.

Increased PTH causes

- Hypophosphatemia due to increased excretion of phosphate via the kidneys.
- PTH acts on the kidneys to increase the absorption of calcium.
- PTH also increases the level of the enzyme 1° hydroxylase which leads to more production of the active form of Vitamin D- 1,25 dihydrocholecalciferol.

In the presence of hypocalcemia, serum calcitonin levels decrease and active form of vitamin D increases.

Calcium homeostasis



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Solution to Question 9:

In the given scenario, increased formation of 24,25-dihydroxycholecalciferol will take place.

Infusion of calcium chloride solution will lead to raised calcium concentration in blood. At higher calcium concentrations, PTH is also suppressed. So the 25-hydroxycholecalciferol is converted to a different compound : 24,25dihydroxycholecalciferol— that has almost no vitamin D effect.

When plasma calcium concentration is high, formation of 1,25 dihydroxycholecalciferol is also greatly depressed. Lack of 1,25 dihydroxycholecalciferol, in turn, decreases the absorption of calcium from the intestines, bones, and renal tubules, thus causing the calcium ion concentration to fall back toward its normal level.

Option A: Since PTH levels are suppressed bone demineralisation does not take place.

Option B: Calcitonin is a calcium regulatory hormone whose level increases when the calcium concentration in plasma is high.

Option C: Calcium ions are a part of the coagulation cascade and the blood coagulability either increases or remains the same with high plasma Ca^{+2} levels.

Solution to Question 10:

Calcium absorption is mainly from the proximal small intestine (duodenum).

Vit D and PTH are important factors that control calcium absorption.

Solution to Question 11:

Ca^{2+} absorption is enhanced by proteins.

It is inhibited by phosphates, phytic acid, and oxalates because these anions form insoluble salts with Ca^{2+} in the intestine.

Solution to Question 12:

Ionic calcium is the active form that is important for most functions of calcium in the body, including the effect of calcium on the heart, the nervous system, and bone formation.

Most of the calcium (99%) is stored in the bones. Only about 0.1 percent of the total body calcium is in the extracellular fluid and 1 percent is in the cells and its organelles.

From the 0.1% Ca^{+2} in the plasma, the distribution of calcium is as follows:

- About 41 % is combined with the plasma proteins (non-diffusible through the capillary membrane)

- About 9 % is combined with anionic substances of the plasma and interstitial fluids like citrate and phosphate in such a manner that it is not ionized (diffusible through the capillary membrane); and
- The remaining 50 % of the calcium in the plasma is both diffusible through the capillary membrane and ionized.

Solution to Question 13:

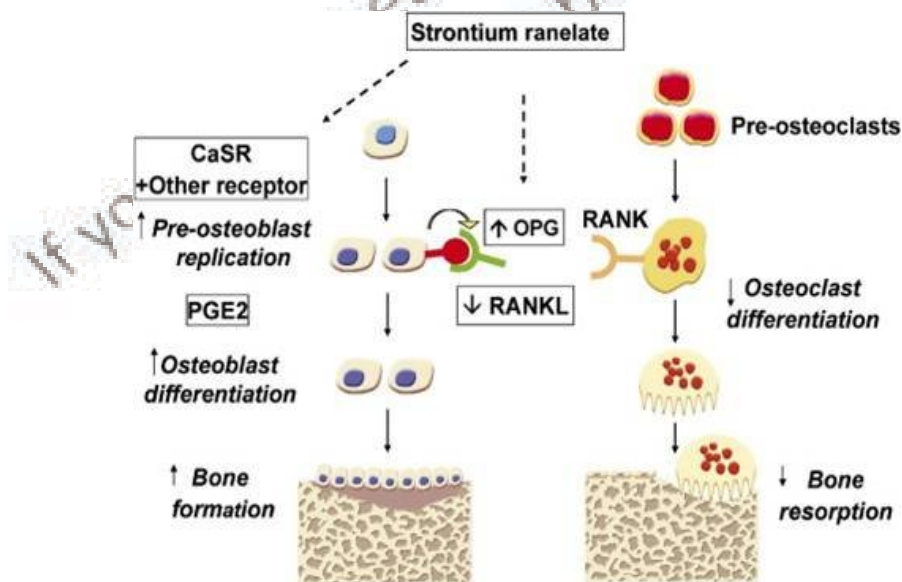
Osteoclasts play a key role in bone resorption.

The cells responsible for bone formation are osteoblasts and the cells responsible for bone resorption are osteoclasts.

Osteoblasts are modified fibroblasts. Osteoclasts are the members of the monocyte family

Osteoclasts erode and absorb the previously formed bone

- They become attached to the bone via integrins forming a sealing zone where the bone resorption takes place
- Proton pumps (H^+ -ATPases) then move from the endosomes of the osteoclasts into the cell membrane as opposed to the sealing zone and acidify the area to pH 4.0.
- The acidic pH dissolves hydroxyapatite, and the acid proteases secreted by the cell break down collagen.



Solution to Question 14:

Osteoclast cells do not have PTH receptors. PTH stimulates osteoclast activity and bone resorption by an indirect mechanism.

PTH has receptors on the osteoblasts in the bone. When PTH binds to these receptors, osteoblasts release cytokines like RANK-ligand and M-CSF. These cytokines allow the differentiation of macrophages to osteoclasts.

Osteoclasts, in turn, cut the collagen in the bone, which is bound to calcium. This causes the release of calcium, phosphorus, and collagen breakdown products in the blood.

Osteoblasts produce osteoprotegerin (OPG), or osteoclastogenesis inhibitory factor, a cytokine that inhibits bone resorption by preventing interaction of RANKL with its receptor. PTH decreases the production of osteoprotegerin (OPG).

Glucocorticoids promote osteoclast activity and increase bone resorption by increasing RANKL production and decreasing the formation of OPG. On the other hand, the hormone estrogen stimulates OPG production.

Solution to Question 15:

Secretion of acid proteases would be decreased in the knocked-out mouse. Osteoclasts produce proteolytic enzymes and several acids necessary for bone resorption. The enzymes digest or dissolve the organic matrix of the bone, and the acids cause dissolution of the bone salts.

A knocked-out mouse is a genetically modified mouse in which an existing gene is inactivated or 'knocked out' by replacing it with an artificial one. The control mice used in the above case are littermate mice i.e mice with genetically modified embryonic stem (ES) cell strains used to test whether specific genes control a disease or immunological phenomena (eg: normal osteoclastic development)

Option B: The acidic pH caused by the osteoclasts dissolves hydroxyapatite. In the knocked-out mouse, hydroxyapatite levels would therefore increase.

Option A and C: Since the lineage of osteoclasts and osteoblasts is different, there is no decrease in the proliferation of osteoblasts. As the osteoblast levels are normal, phosphate deposition in bone also remains normal.

Solution to Question 16:

PTH promotes the reabsorption of calcium from the renal tubules.

Option A: About 41 percent of the plasma calcium is bound to plasma proteins and is therefore not filtered by the glomerular capillaries. The rest of the calcium is filtered through the glomeruli into the renal tubules.

Option B: Of the filtered calcium, approximately 90 percent is reabsorbed in the proximal tubules, loops of Henle, and early distal tubules. However, when calcium levels are low and PTH levels rise, PTH promotes calcium reabsorption from the later DCT.

Option D: Calcitonin tends to decrease plasma calcium concentration and, in general, has effects opposite to those of PTH.

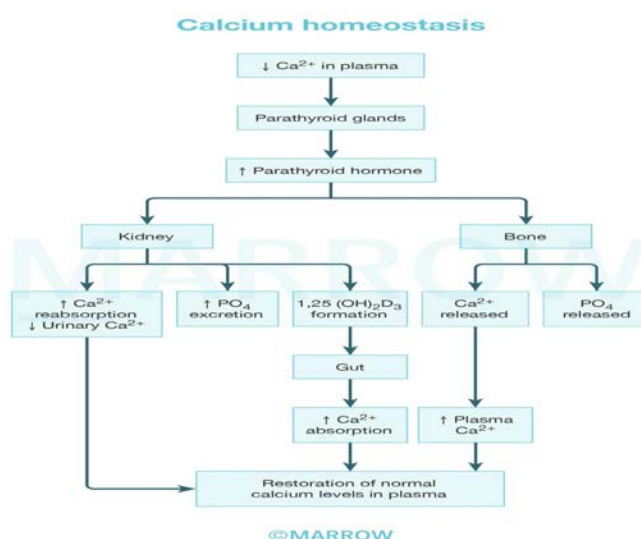
Solution to Question 17:

PTH does not increase phosphate reabsorption in kidney, rather it causes decreased phosphate reabsorption in the kidney.

Solution to Question 18:

In a patient with parathyroid deficiency due to inadvertent damage to the parathyroid glands, the clinical evaluation will show low Ca^{2+} level, tetany, trousseau sign, and high phosphate levels.

PTH is released in response to low calcium levels. It increases the calcium levels by the following actions:



Following signs can be elicited in hypoparathyroidism due to hypocalcemia (tetany):

- Chvostek sign: Contraction facial muscles on tapping over facial nerve in front of the tragus of the ear
- Trousseau sign: accoucheur's hand position and painful carpopedal spasm by applying a sphygmomanometer cuff to the arm and keeping the pressure above systolic blood pressure for 5 minutes

Note - In option A, tetanus is different from tetany. Tetanus is caused by infection with the bacteria *Clostridium tetani*

The image below is showing Trousseau sign.



Solution to Question 19:

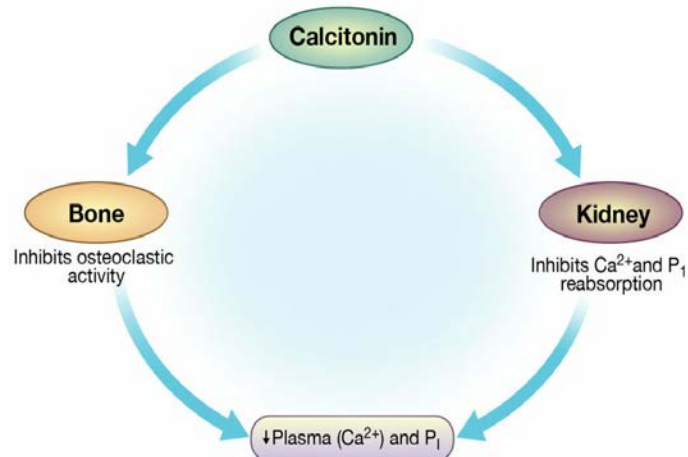
Calcitonin has calcium-lowering effects and may cause hypocalcemia. It does so by inhibiting bone resorption (direct effect)

Calcitonin is synthesized and secreted by parafollicular (clear) cells of the thyroid. Actions of calcitonin are as follows:

- Inhibits osteoclasts, thereby decreasing bone resorption (direct effect)
- Increases renal calcium excretion
- Decreases calcium absorption by the kidneys and intestines

Under normal physiological conditions, calcitonin is more important in young individuals (compared to adults), and it may play a role in skeletal development.

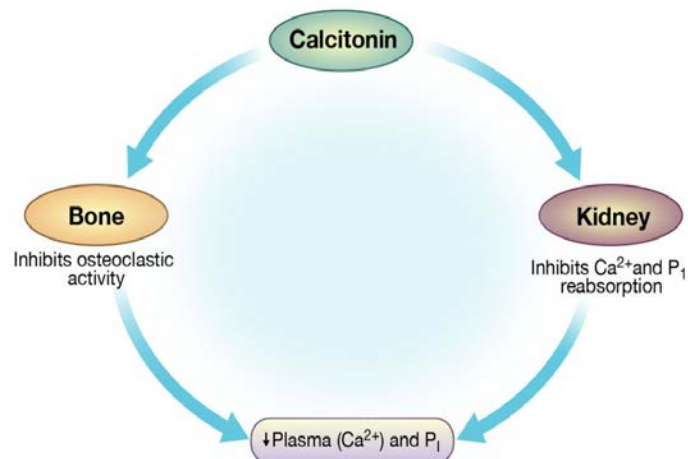
In addition, it may protect the bones of the mother from excess calcium loss during pregnancy. Bone formation in the infant and lactation are major drains on Ca^{2+} stores. They would cause bone loss in the mother if bone resorption were not simultaneously inhibited by an increase in the plasma calcitonin level.



Solution to Question 20:

The given clinical vignette is suggestive of hypercalcemia of malignancy, most probably due to metastatic breast cancer. Calcitonin exerts its calcium-lowering effect mainly by inhibiting bone resorption and helps in the management of hypercalcemia.

This calcium lowering action of calcitonin is direct, by inhibiting the activity of osteoclasts. However, calcitonin also increases renal calcium excretion and decreases calcium absorption by the kidneys and intestines.



Solution to Question 21:

The given clinical vignette is suggestive of Chvostek sign, seen in hypocalcemia due to accidental parathyroid injury during thyroid surgery. The muscle hyperexcitability seen in hypocalcemia is due to increased permeability of cells to Na^+ .

During thyroid surgery, parathyroid gland is prone to accidental injury due to proximity with the thyroid gland. Damage to the parathyroid gland will lead to a low PTH level and low Ca^{2+} level. When the ECF concentration of Ca^{2+} falls below normal, the nervous system becomes progressively more excitable. Because there is increased membrane permeability to Na^+ . This leads to easy initiation of the action potentials.

The peripheral nerve fibres become so excitable that they begin to discharge spontaneously, initiating impulses that pass to the skeletal muscles to elicit tetanic muscle contraction. Tetany occurs usually when calcium concentration falls to 6 mg/dL.

Solution to Question 22:

Acidosis does not cause a decrease in the ionized calcium levels. Rather, alkalosis causes a decrease in the ionized calcium levels.

Plasma proteins are more ionized when the pH is high (alkalosis), providing more protein anions to bind with and sequester free Ca^{2+} . This causes a decrease in ionized calcium and may trigger symptoms of hypocalcemia.

Option A: Total calcium concentration is altered by variations in the serum albumin concentration as 41% of calcium is bound to plasma proteins like albumin, hence changes in albumin concentration affects the serum calcium levels. The corrected calcium levels can be calculated as:

Corrected calcium (mg/dL) = measured total Ca (mg/dL) + 0.8 (4.0 - serum albumin [g/dL]), where 4.0 represents the average albumin level.

Option B: Fist-clenching or forearm exercise can lead to falsely elevated ionized (free) calcium levels.

Option C: A tourniquet is usually avoided while drawing blood for calcium measurements since hemoconcentration also leads to an increase in serum calcium levels.

Solution to Question 23:

Osteoblasts are involved in the initiation and control of osteoclastic activity.

Osteoblasts produce receptor activator for nuclear factor κB ligand (RANKL) and macrophage colony stimulating factor (M-CSF), which are necessary for formation of mature osteoclasts and hence osteoclastic activity.

Specific receptors for parathormone are present only on the osteoblasts and not on osteoclasts.

Solution to Question 24:

Serum bone specific alkaline phosphatase is a marker for bone formation.

Other markers of bone formation are:

-

Serum osteocalcin

- Serum type 1 procollagen (C-terminal/N-terminal)

Markers for bone resorption are:

-

Urinary hydroxyproline

-

Urinary total pyridinoline (PYD)

-

Urinary free deoxypyridinoline (DPD)

-

Urinary collagen type 1 cross-linked N-Telopeptide (NTX)

-

Urinary or serum collagen type 1 cross-linked C-Telopeptide (CTX)

-

Tartrate-resistant acid phosphatase 5b

International Osteoporosis Foundation has proposed serum CTX and serum PINP (Procollagen 1 N-terminal peptide) to be used as reference markers of bone resorption and formation, respectively, for the assessment of fracture risk and monitoring therapy in clinical settings.

Solution to Question 25:

Parathyroid adenoma is a benign tumor of the parathyroid gland, which causes primary hyperparathyroidism (not secondary).

Secondary hyperparathyroidism is characterized by high levels of parathyroid hormone, to compensate for the low levels of serum calcium, due to some other underlying condition.

Causes of secondary hyperparathyroidism, include:

- Chronic kidney disease: The damaged kidneys are unable to produce sufficient amounts of the active form of vitamin D, 1,25-dihydroxycholecalciferol. This further causes hypocalcemia and triggers PTH secretion.
- Vitamin D deficiency: Causes hypocalcemia, which triggers the increased secretion of PTH.
- Lithium: May cause an increased set-point of calcium for parathyroid hormone suppression, which leads to parathyroid hyperplasia.

Male Reproductive Physiology

Question 1:

A 32 year old lady is undergoing evaluation for polycystic ovary syndrome. Her sex hormone-binding globulin (SHBG) is low. The decrease in sex hormone-binding globulin will have the maximum effect on which of the following hormones?

- a) Estrogen
- b) Testosterone
- c) DHEA
- d) Progesterone

Question 2:

Sertoli cells in testes have receptors for _____

- a) FSH
- b) LH
- c) Inhibin
- d) GnRH

Question 3:

The negative feedback mechanism involving the Sertoli cells causes _____

- a) ↓ LH
- b) ↓FSH
- c) ↓TRH
- d) ↓CRH

Question 4:

In men, LH controls secretion of _____

- a) Inhibin
- b) AMH

- c) Testosterone
- d) ABP

Question 5:

Testosterone production is mainly contributed by _____

- a) Leydig cells
- b) Sertoli cells
- c) Seminiferous tubules
- d) Epididymis

Question 6:

All of the following are androgens except _____

- a) Testosterone
- b) Dihydrotestosterone
- c) Androstenedione
- d) 17° hydroxy-progesterone

Question 7:

Which of the following cells secrete anti mullerian hormone?

- a) Theca cells
- b) Leydig cells
- c) Sertoli cells and Granulosa cells
- d) Muller cells

Question 8:

The blood-testis barrier is formed by _____

- a) Tight junctions between adjacent leydig cells
- b) Tight junctions between adjacent sertoli cells
- c) Basal lamina and spermatogonia

- d) Basal lamina and leydig cells

Question 9:

Spermiogenesis refers to _____

- a) Formation of spermatozoa from spermatogonia
- b) Formation of spermatozoa from spermatid
- c) Formation of spermatids from sperm
- d) Formation of secondary spermatocyte from primary spermatocyte

Question 10:

Development of spermatozoa from spermatogonium takes how much time?

- a) 30-35 days
- b) 40-45 days
- c) 50-55 days
- d) 70-75 days

Question 11:

Spermatogenesis is maintained by which hormone?

- a) Testosterone
- b) FSH
- c) LH
- d) All of the above

Question 12:

Role of growth hormone in spermatogenesis is _____

- a) Stimulation of sertoli cells
- b) Early division of spermatogonia
- c) Formation of acrosome
- d) Stimulation of leydig cells

Question 13:

Which of the following provides a negative feedback in spermatogenesis?

- a) ABP(androgen- binding protein)
- b) Inhibin
- c) Progesterone
- d) GnRH

Question 14:

A 35-year-old man is being evaluated for suspected infertility. His semen analysis shows a decrease in sperm count and motility. On physical examination, a 'bag of worms'-like structure is palpated in his scrotum, suggestive of varicocele. Which of the following is the most likely reason behind his decreased sperm count?

- a) Lowered scrotal temperature, compared to core temperature
- b) Rise in scrotal temperature, compared to core temperature
- c) Blockage in epididymal ducts
- d) Blockage in vas deferens

Question 15:

Sperm acquires motility in _____

- a) Seminal vesicle
- b) Testes
- c) Epididymis
- d) Ejaculatory duct

Question 16:

At which location does the capacitation of sperm take place?

- a) Seminiferous tubules
- b) Uterine tubes
- c) Epididymis

d) Ejaculatory duct

Question 17:

Which of the following factors increases sperm motility?

- a) ↑ Temperature
- b) Acidic environment
- c) Vaginal secretions
- d) While passing through seminiferous tubule

Question 18:

A 70-year-old man suffering from prostate cancer is undergoing long-term androgen deprivation therapy. He is being treated with long acting GnRH agonist. After prolonged injection of GnRH, which of the following will most likely be seen in this patient?

- a) ↑ FSH
- b) ↓ LH
- c) ↑ spermiogenesis
- d) ↑ Testosterone

Question 19:

The major contribution to semen is from which of the following organs?

- a) Seminal vesicle
- b) Cowpers gland
- c) Prostate
- d) Urethral gland

Question 20:

Normal pH of human semen is _____

- a) 7
- b) 6.5

- c) 7.4
- d) 8

Question 21:

Prostaglandins in semen are secreted by which of the following organs?

- a) Seminal vesicle
- b) Cowpers gland
- c) Prostate
- d) Urethral gland

Question 22:

A couple is undergoing evaluation for their inability to conceive. In order to rule out male factor infertility, the consultant orders a semen analysis for the husband. Which of the following best estimates the normal sperm count in an adult?

- a) 20-40 million/mL
- b) 40-60 million/mL
- c) 60-80 million/mL
- d) 90-120 million/mL

Question 23:

In which part of the fallopian tube does fertilization take place?

- a) Interstitial part
- b) Isthmus
- c) Fimbria
- d) Ampulla

Question 24:

For how many hours does human sperm remain fertile in the female genital tract?

- a) 6-8 h

- b) 12-24 h
- c) 24-48 h
- d) 72-96 h

Question 25:

For how long does the ovum remain viable following ovulation?

- a) 12 hours
- b) 24 hours
- c) 48 hours
- d) 72 hours

Question 26:

Which of the following hormones has no role for the attainment of puberty?

- a) GH
- b) Leptin
- c) Testosterone
- d) Insulin

Question 27:

Which of the following hormones is responsible for pubarche?

- a) GH
- b) Progesterone
- c) Testosterone
- d) Estrogen

Question 28:

Which of the following hormones play a permissive role in the attainment of puberty?

- a) GnRH
- b) Leptin

- c) Insulin
- d) GH

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	c
5	a
6	d
7	c
8	b
9	b
10	d
11	d
12	b
13	b
14	b
15	c
16	b
17	c
18	b
19	a
20	c
21	a
22	d
23	d
24	c
25	b
26	d
27	c
28	b

Detailed Explanations

Solution to Question 1:

The decrease in sex hormone-binding globulin will have the maximum effect on testosterone.

Around 98% of the circulating testosterone is bound to plasma proteins such as sex hormone-binding globulin (65%) and albumin. So, the decrease in sex-hormone-binding globulin (SHBG) or gonadal steroid-binding globulin (GBG) will have the maximum effect on testosterone levels by increasing the levels of its free form.

Therefore patients with polycystic ovary syndrome will show features of hyperandrogenism (due to elevated free testosterone).

Solution to Question 2:

Sertoli cells express FSH receptor, whereas Leydig cell express LH receptor.

Sertoli cells are the true epithelial cells of the seminiferous epithelium and extend from basal lamina to the lumen.

Functions of Sertoli cells:

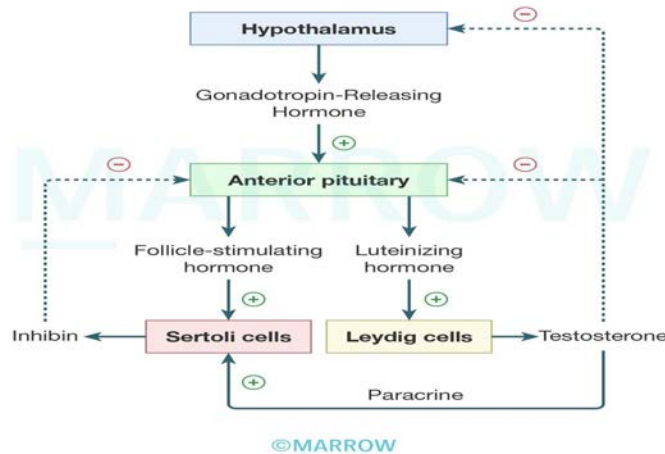
- Express FSH receptor and androgen receptor.
- Forms the blood-testis barrier.
- Surround the sperm cells and provide structural support within the epithelium.
- Guide sperm cells towards the lumen as they advance to later stages of spermatogenesis.
- Produce inhibin and Mullerian-inhibiting substance (MIS).
- Produce androgen-binding proteins that maintain high androgen levels within the lumen of the seminiferous tubules and the proximal part of the male reproductive tract.
- Perform phagocytic function by engulfing residual bodies (cytoplasm shed by spermatozoa during spermiogenesis) and dead sperm cells.
- Express aromatase, which converts testosterone to potent estrogen.

Solution to Question 3:

The negative feedback mechanism involving the Sertoli cells causes ↓FSH.

FSH stimulates inhibin production by the Sertoli cells. This inhibin inhibits FSH production by directly acting on the anterior pituitary. Thus, inhibin keeps FSH levels within a set point.

Negative feedback mechanism of Hypothalamic-pituitary axis in males



Solution to Question 4:

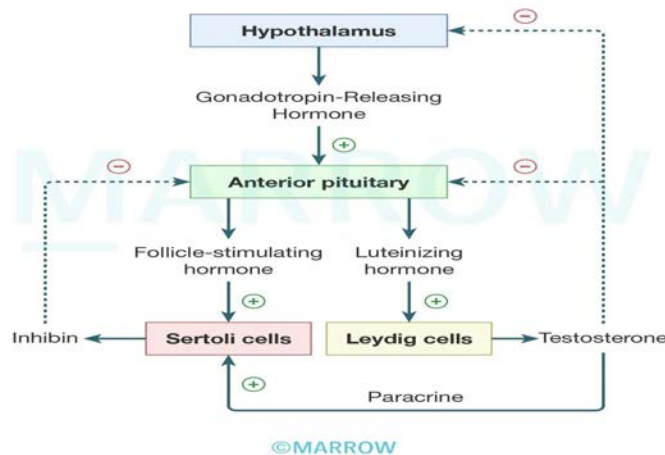
In men, LH controls the secretion of testosterone.

LH stimulates Leydig cells to secrete testosterone, which in turn inhibits LH by a negative feedback mechanism. FSH stimulates Sertoli cells to secrete inhibin, which in turn inhibits FSH by a negative feedback mechanism.

Testosterone inhibits LH secretion by acting directly on the anterior pituitary and by inhibiting the secretion of GnRH from the hypothalamus. Inhibin acts directly on the anterior pituitary to inhibit FSH secretion.

FSH also stimulates the secretion of ABP (androgen-binding protein) and inhibin. FSH and androgens maintain the gametogenic function of the testes. Hypothalamic lesions in animals and hypothalamic disease in humans lead to atrophy of the testes and loss of their function.

Negative feedback mechanism of Hypothalamic-pituitary axis in males



Solution to Question 5:

Testosterone production is mainly contributed by Leydig cells.

Testosterone is synthesized from cholesterol in the Leydig cells and also from androstenedione secreted by the adrenal cortex.

The diagnosis of testosterone deficiency should be made only on the basis of clinical signs and symptoms consistent with androgen deficiency, and consistently subnormal serum testosterone concentrations at 8 to 10 AM on three occasions.

Testosterone has been used to treat primary or secondary hypogonadism in adult men.

Solution to Question 6:

17° hydroxy-progesterone is a progestin, not androgen.

The term androgen means any steroid hormone that has masculinizing effects. Testosterone, Dihydrotestosterone, androstenedione, and dehydroepiandrosterone are androgens.

The sites of production:

- Testes.
- Adrenal gland.
- Ovary (not significant).
- Tumors of the adrenal androgen-producing cells etc.

Both in the testes and in the adrenals, the androgens can be synthesized either from cholesterol or directly from acetyl coenzyme A.

Solution to Question 7:

(Anti Mullerian Hormone) is secreted by both sertoli cells and granulosa cells.

Sertoli cells produce Anti Mullerian Hormone (Mullerian inhibitory substance) during development. It induces the regression of the mullerian duct that is programmed to give rise to a female reproductive tract.

Granulosa cells of the ovary produce AMH during the reproductive years. It limits the formation of primary follicles by inhibiting excessive follicular recruitment by FSH.

Solution to Question 8:

The blood-testis barrier is formed by the tight junctions between adjacent sertoli cells.

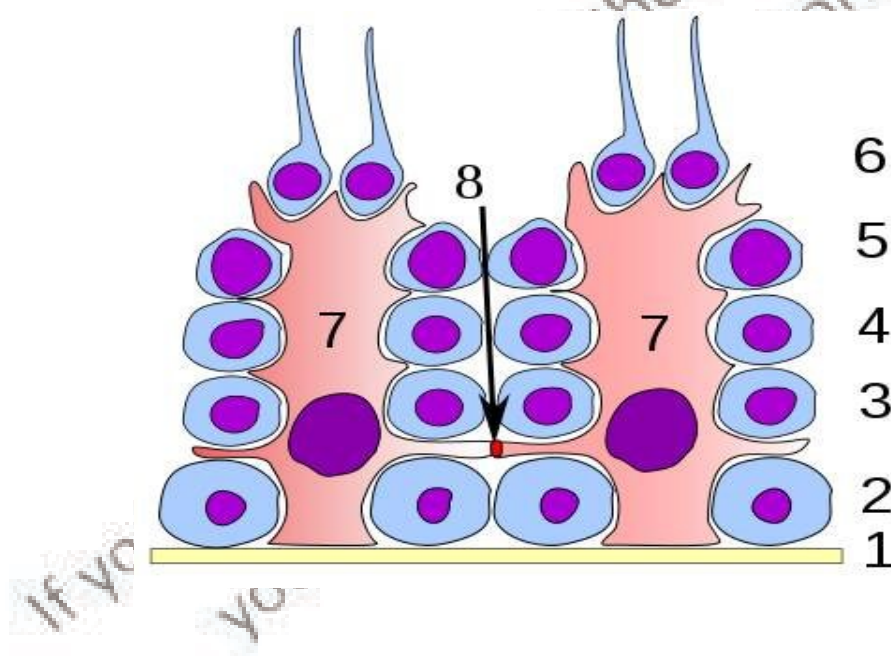
This Sertoli-Sertoli cell tight junctions divide the seminiferous epithelium into 2 compartments:

- Basal compartment - contains the spermatogonia and early-stage primary spermatocytes.
- Adluminal compartment - contains later-stage primary spermatocytes and all subsequent stages of sperm cells.

It imparts a selective permeability, allowing few substances such as testosterone to cross but preventing noxious substances that might damage the developing sperm. It also protects sperms from antibodies by creating a specialized, immunologically safe microenvironment for developing sperm.

BELOW DIAGRAM -

1 basal lamina, 2 spermatogonia, 3 spermatocyte 1st order, 4 spermatocyte 2nd order, 5 spermatid, 6 mature spermatid, 7 Sertoli cell, 8 tight junctions(Blood- Testis Barrier)



Solution to Question 9:

Spermiogenesis refers to the formation of spermatozoa from spermatids.

Spermiogenesis is a process by which there is a formation of extremely specialized mobile spermatozoa from spermatids. This process requires extensive remodeling or packaging of cell elements. It occurs in the recesses formed by invaginations of the sertoli cell plasma membrane.

Maturation from spermatids to spermatozoa depends on androgen acting on sertoli cells. When sperm maturation is complete, the cytoplasm of the sertoli cell around sperm retracts and sperms are released into the lumen to be bathed by the luminal fluid.

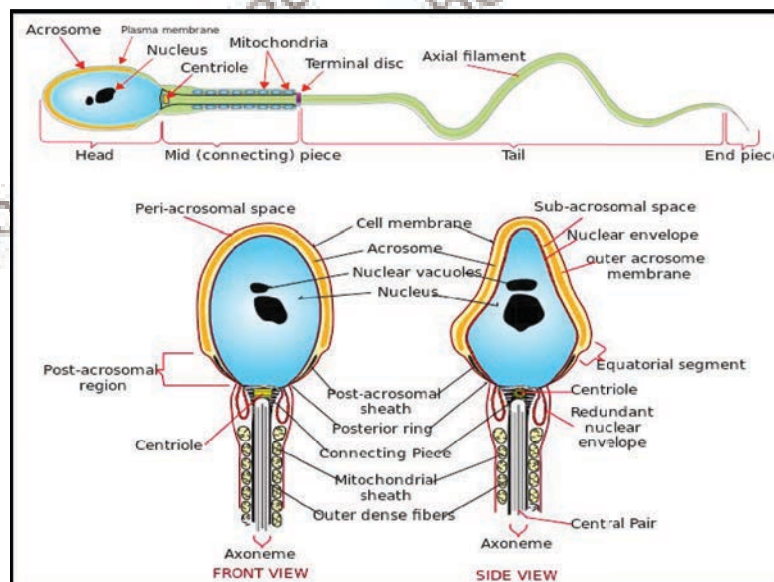
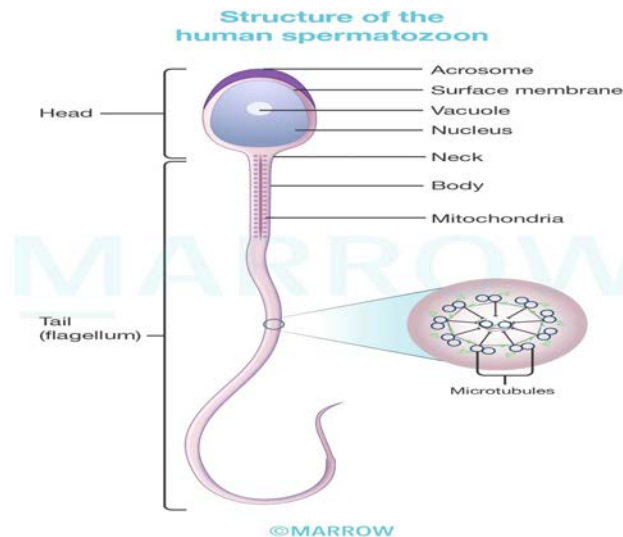
The following changes occur during the formation of spermatozoa from spermatids :

- The nucleus becomes eccentric and decreases in size, and the chromatin becomes condensed.
- The cytoplasmic content is redistributed and discarded.

- The acrosome buds from the golgi apparatus.
- The centrioles migrate to the caudal pole and form a long axial filament made of microtubules. This becomes the axoneme or major portion of the tail.

Spermiation is the release of the sperm from the sertoli cells. During spermiation, most of the remaining cytoplasm is shed in the form of residual bodies.

Spermatogenesis refers to the formation of spermatozoa from spermatogonia.



Solution to Question 10:

The time required to produce mature spermatozoa from the earliest stage of spermatogonia is 65 to 70 days.

Since several developmental stages of spermatogenic cells occur during this time frame, the stages are collectively known as the spermatogenic cycle.

Solution to Question 11:

All three of them (testosterone, FSH, LH) are required for spermatogenesis.

Pulses of hypothalamic GnRH stimulate the anterior pituitary to secrete FSH and LH, which then act on the testes

FSH acts on the Sertoli cells to stimulate spermatogenesis and inhibin secretion. LH acts on the Leydig cells to stimulate testosterone secretion. Testosterone, acting locally on the Sertoli cells, is essential for maintaining spermatogenesis.

Solution to Question 12:

The growth hormone specifically promotes the early division of spermatogonia.

Hormonal factors that stimulate spermatogenesis:

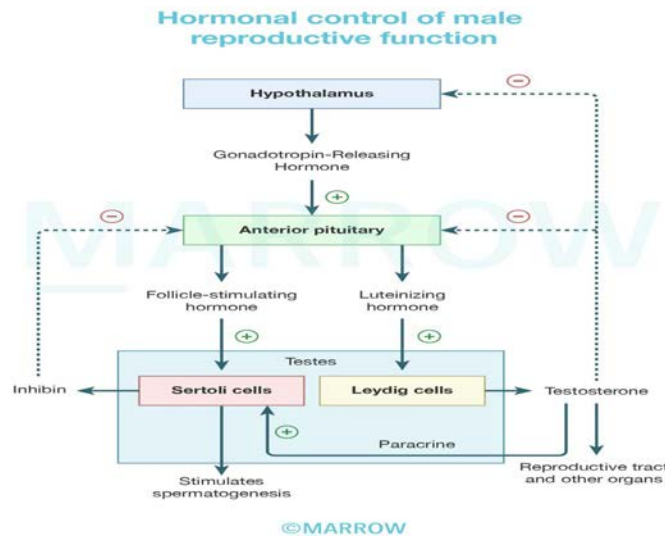
- Testosterone, secreted by the Leydig cells of the testis, is essential for growth and division of the testicular germinal cells, which is the first stage in forming sperm.
- Luteinizing hormone, secreted by the anterior pituitary gland, stimulates the Leydig cells to secrete testosterone.
- Follicle-stimulating hormone, secreted by the anterior pituitary gland, stimulates the Sertoli cells. Without this stimulation, the spermiogenesis will not occur.
- Estrogens, formed from testosterone by the Sertoli cells when stimulated by FSH, are also essential for spermiogenesis.
- Growth hormone is necessary for controlling the background metabolic functions of the testes. It specifically promotes early division of the spermatogonia themselves. In its absence, as in pituitary dwarfs, spermatogenesis is severely deficient or absent, thus causing infertility.

Solution to Question 13:

Negative feedback in spermatogenesis is by inhibin.

FSH stimulates inhibin production by the Sertoli cells. This inhibin inhibits FSH production by directly acting on the anterior pituitary.

LH stimulates Leydig cells to secrete testosterone, which in turn inhibits LH. Testosterone inhibits LH secretion by acting directly on the anterior pituitary and by inhibiting the secretion of gonadotropin-releasing hormone (GnRH) from the hypothalamus.



Solution to Question 14:

The most likely reason behind decreased sperm count in this patient is rise in scrotal temperature, secondary to varicocele.

Normally, the testes occupy the scrotum, which lies outside the body cavity and is maintained at 35° to 36°C, or 1° to 2°C below body temperature. This lower temperature is essential for normal spermatogenesis. It is maintained by a countercurrent arrangement of testicular arteries and veins, which facilitates heat exchange.

Enlarged veins around the testes, called varicoceles, cause infertility by raising the temperature in the scrotum and decreasing sperm production.

Solution to Question 15:

Sperm is stored in the epididymis, where they undergo a process of maturation before they are capable of progressive motility and fertilization.

Changes during the sperm maturation in the epididymis:

- Progressive increase in forward motility
- Increase ability to fertilize
- Maturation of acrosome
- Molecular reorganization of plasma membrane: Lipid (stabilization of plasma membrane) protein (shedding as well as the acquisition of new proteins)
- Ability to bind to zona pellucida
- Acquisition of receptors for proteins of the zona pellucida

- Increase disulfide bond between cysteine residues in sperm nucleoproteins
- Topographic regionalization of glycosidic residues
- Accumulation of mannosylated residues on periacrosomal plasma membrane
- Decreased cytoplasm and cell volume

Solution to Question 16:

Capacitation of sperm takes place in uterine tubes.

Once ejaculated into the female genital tract, the spermatozoa move up the uterus to the isthmus of the uterine tubes, where they slow down and undergo final conditioning called capacitation.

This process involves two components:

- Increasing the motility of the spermatozoa
- Facilitating their preparation for the acrosomal reaction.

Solution to Question 17:

Sperm motility is increased by vaginal secretions.

On coming in contact with the fluids of the female genital tract, multiple changes occur that activate the sperm for the final processes of fertilization. These collective changes are called capacitation of the spermatozoa, which normally requires from 1 to 10 hours.

The following changes are believed to occur:

- The uterine and fallopian tube fluids wash away the various inhibitory factors that suppress sperm activity in the male genital ducts.
- While the spermatozoa remain in the fluid of the male genital ducts, they are continually exposed to many floating vesicles from the seminiferous tubules containing large amounts of cholesterol. This cholesterol is continually added to the cellular membrane covering the sperm acrosome, toughening this membrane and preventing the release of its enzymes.
- After ejaculation, the sperm deposited in the vagina swim away from the cholesterol vesicles upward into the uterine cavity, and they gradually lose much of their other excess cholesterol during the next few hours. In doing so, the membrane at the head of the sperm (the acrosome) becomes much weaker.
- The membrane of the sperm also becomes much more permeable to calcium ions, so calcium now enters the sperm in abundance and changes the activity of the flagellum, giving it a powerful whiplash motion in contrast to its previously weak undulating motion. In addition, the calcium ions cause changes in the cellular membrane that cover the leading edge of the acrosome, making it possible for the acrosome to release its enzymes rapidly and easily as the sperm penetrates the granulosa cell mass surrounding the ovum, and even more so as it attempts to penetrate the zona pellucida of the ovum.

Solution to Question 18:

Prolonged injection of GnRH will lead to decreased LH levels.

A constant administration of GnRH produces an initial agonistic action known as "flare", followed by a downregulation of GnRH receptors (decrease in GnRH receptor concentration). This desensitizes pituitary to the action of GnRH and a hypogonadotropic-hypogonadal state is induced.

This is characterized by low LH, low FSH levels and thereby reduced testosterone production (i.e., chemical castration).

Solution to Question 19:

The major contribution to semen is from seminal vesicles.

The seminal vesicles contribute ~60-70% of the volume of semen. Apart from sperm, the remaining 20-40% represents epididymal fluids, as well as secretions of the prostate gland and bulbourethral glands.

The seminal plasma originates primarily from the accessory glands (the seminal vesicles, the prostate gland, and the bulbourethral glands).

Solution to Question 20:

The normal pH of human semen is 7.4 (isotonic).

The pH in the lumen of the epididymis is relatively acidic (6.5 to 6.8) as the result of H⁺ secretion by clear cells that are analogous to intercalated cells in the nephron. Addition of the relatively alkaline secretions of the seminal vesicles raises the final pH of seminal plasma to between 7.2 and 7.8.

Spermatozoa generally tolerate alkalinity better than acidity.

A pH near neutrality or slightly higher is optimal for the motility and survival of sperm cells in humans and in other species as well.

Solution to Question 21:

Prostaglandins in semen are secreted by seminal vesicle.

The seminal vesicles secrete fluid that is rich in fructose, citrate, prostaglandins, and fibrinogen.

The prostaglandins present in the seminal fluid may assist in fertilization in two ways:

- Prostaglandins react with cervical mucus to make it more penetrable by sperm.

- Prostaglandins induce peristaltic contractions in the female reproductive tract (i.e., the uterus and fallopian tubes) to propel the sperm up the tract.

Solution to Question 22:

Normal sperm count on an average is 100 million/ml (90-120 million/mL).

Note: The WHO 2010 minimal criteria for semen analysis requires a minimum sperm count ≥ 15 million/mL or ≥ 40 million/per ejaculate.

Solution to Question 23:

In humans, fertilization of the ovum by the sperm usually occurs in the ampulla of the fallopian tube.

Solution to Question 24:

After ejaculation, sperms have a maximal lifespan of 24-48 hours in the female genital tract.

The activity and rate of metabolism of sperm increase markedly with increasing temperature causing its life to be considerably shortened.

Sperm can be cryopreserved for years if agents such as glycerol are used to prevent ice crystal formation during freezing.

Note: The ovum remains viable and capable of being fertilized, probably no longer than 24 hours after it is expelled from the ovary.

Solution to Question 25:

The ovum remains viable up to 24 hours following ovulation.

If not fertilized, the ovum degenerates in the fallopian tube without leaving behind any trace after 24 hours.

Solution to Question 26:

Insulin has no role in the attainment of puberty. GH, leptin and testosterone are all required for the onset of puberty.

The onset of puberty is marked by enhanced androgenic activity. The growth spurt of adolescent males is influenced by a complex interplay between androgens, growth hormone (GH), nutrition, and genetic factors.

Leptin, the satiety-producing hormone secreted by fat cells, may be the link between body weight and puberty.

Solution to Question 27:

Pubarche or the development of axillary and pubic hair is due to testosterone.

The growth of pubic and axillary hair in both sexes is primarily due to androgens.

Solution to Question 28:

The hormone that plays a permissive role in the attainment of puberty is leptin.

It now appears that leptin may be the link between body weight and puberty. Leptin is a satiety-producing hormone secreted by adipocytes. High leptin levels reduce appetite and increase energy expenditure.

Evidence for which is given by:

- Obese ob/ob mice that cannot make leptin are infertile, and their fertility is restored by injections of leptin.
- Leptin treatment also induces precocious puberty in immature female mice.
- However, the way that leptin fits into the overall control of puberty remains to be determined.

Leptin also accelerates GnRH pulsatility in neurons, fastening puberty.

Female Reproductive Physiology

Question 1:

Which of the following hormones is not secreted by the ovary?

- a) Gonadotrophins
- b) Progesterone
- c) Estrogen
- d) Inhibin B

Question 2:

A 30-year-old lady has regular, infrequent menstrual cycles occurring at an interval of 35-36 days. What is the usual timing of ovulation in this lady?

- a) 14 days before menstruation
- b) Just before LH surge
- c) Just after corpus luteal maturation
- d) 22 days before menstruation

Question 3:

The corpus luteum starts regressing after how many days post ovulation?

- a) 5
- b) 10
- c) 24
- d) 17

Question 4:

Which of the following is not seen in humans?

- a) Estrous cycle
- b) Menstrual cycle

- c) Endometrial cycle
- d) Ovarian cycle

Question 5:

Which of the following is the dominant hormone in the luteal phase?

- a) Estrogen
- b) Progesterone
- c) Prolactin
- d) Oxytocin

Question 6:

Which of the following hormones is responsible for the maintenance of endometrial thickness in the post-ovulatory phase?

- a) Estrogen
- b) Progesterone
- c) LH
- d) FSH

Question 7:

A couple is planning to conceive by artificial intrauterine insemination (IUI). In order to ensure optimal timing for intrauterine transfer of husband sperm, the wife is advised to use a home-based ovulation predictor kit. Detecting which of the following hormones in urine is the purpose of this kit?

- a) Estrogen
- b) Prolactin
- c) hCG
- d) LH

Question 8:

Positive feedback action of estrogen for inducing luteinizing hormone surge is associated with which of the following steroid hormone ratio in peripheral circulation?

- a) High estrogen: low progesterone
- b) Low estrogen: high progesterone
- c) High estrogen: high progesterone
- d) Low estrogen: low progesterone

Question 9:

Which of the following best explains the mechanism behind an increase in LH secretion just before ovulation ?

- a) Positive feedback by progesterone
- b) Positive feedback by estrogen
- c) Positive feedback by FSH
- d) Positive feedback by relaxin

Question 10:

Corpus luteum of pregnancy is maintained by_____

- a) Estrogen
- b) Progesterone
- c) Oxytocin
- d) HCG

Question 11:

FSH receptor is present in_____

- a) Theca cells
- b) Granulosa cells
- c) Leydig cells
- d) Basement membrane of follicle

Question 12:

A 30-year-old lady is using the basal body temperature method for contraception. She takes note of a rise in her body temperature for abstinence during the fertile period. The increase in basal body temperature in the mid-luteal phase is due to which of the following hormones?

- a) FSH peak
- b) LH peak
- c) Estradiol
- d) Progesterone

Question 13:

A blood sample is drawn on day 20 of the menstrual cycle of a young woman undergoing evaluation for irregular menstruation. Serum level of which of the following hormones help in assessing whether her cycle was ovulatory or not?

- a) FSH
- b) LH
- c) Estradiol
- d) Progesterone

Question 14:

Secretion of estrogen is maximum _____

- a) Just before menopause
- b) At puberty
- c) At menstruation
- d) Before ovulation

Question 15:

Which of the following hormones inhibit FSH and LH secretion in the luteal phase?

- a) Growth hormone
- b) GnRH
- c) Estrogen
- d) Insulin like growth factor

Question 16:

Which of the following statement can be regarded as primary action of inhibin?

- a) It inhibits secretion of prolactin
- b) It stimulates synthesis of estradiol
- c) It stimulates secretion of TSH
- d) It inhibits secretion of FSH

Question 17:

Inhibin B is secreted by_____

- a) Ovarian follicle
- b) Stroma
- c) Surface epithelium of ovary
- d) Corpus luteum

Question 18:

Source of progesterone during normal menstrual cycle_____

- a) Corpus luteum
- b) Stroma
- c) Surface epithelium of ovary
- d) Anterior pituitary

Question 19:

Which of the following is the source of estrogen?

- a) Theca cells
- b) Granulosa cells
- c) Leydig cells
- d) Theca interna

Question 20:

Regarding the two-cell—two-gonadotropin hypothesis which of the following is true?

- a) FSH and LH inhibits the release of estrogen and progesterone
- b) FSH acts on granulosa cells and LH on theca cells
- c) FSH stimulates estrogen while LH inhibits estrogen
- d) FSH feedback inhibits hypothalamus while LH feedback inhibits pituitary

Question 21:

Regarding the action of progesterone which of the following is true?

- a) Increases sensitivity of uterus to oxytocin
- b) Inhibits LH secretion
- c) Decrease in body temperature
- d) Causes marked proliferative changes in uterus

Question 22:

Which of the following is the most potent type of estrogen found in women of reproductive age group?

- a) Estrone
- b) Estriol
- c) Estradiol
- d) None

Question 23:

Which is the predominant estrogen in the plasma of premenopausal non-pregnant women?

- a) Estradiol
- b) Estriol
- c) Estrone
- d) Androstenedione

Question 24:

Which of the following is false about estrogen action?

- a) Development of secondary sexual characters in female
- b) Stimulates osteoclast
- c) ↓ LDL
- d) ↑ NO

Question 25:

A 12-year-old girl is undergoing pubertal changes. Her tanner stage is stage 4 for breast development and pubic hair scale. Which of the following pubertal changes is not due to estrogen?

- a) Vaginal cornification
- b) Pubic and axillary hair growth
- c) Cervical mucus production
- d) Menstruation

Question 26:

β HCG is secreted by which of the following organs?

- a) Ovary
- b) Pituitary
- c) Corpus luteum
- d) Placenta

Question 27:

Progesterone has how many carbon atoms?

- a) 18
- b) 19
- c) 20
- d) 21

Question 28:

Which of the following is false about milk ejection?

- a) The neuroendocrine part of anterior pituitary is involved.
- b) Its secretion occurs by contraction of lactiferous sinus.
- c) Oxytocin hormone is involved.
- d) It is affected by emotion.

Question 29:

Which of the following hormones is responsible for milk ejection?

- a) Prolactin
- b) Oxytocin
- c) Progesterone
- d) FSH

Question 30:

Which of the following is not a function of oxytocin?

- a) Lactogenesis
- b) Milk ejection
- c) Uterus contraction
- d) Myoepithelial cell contraction

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	a
5	b
6	b
7	d

8	a
9	b
10	d
11	b
12	d
13	d
14	d
15	c
16	d
17	a
18	a
19	b
20	b
21	b
22	c
23	a
24	b
25	b
26	d
27	d
28	a
29	b
30	a

Detailed Explanations

Solution to Question 1:

Gonadotropins (FSH, LH) are not secreted by the ovary. They are secreted by the anterior pituitary.

Ovaries secrete estrogen, progesterone, and inhibin B.

Solution to Question 2:

Ovulation occurs 14 days before menses, regardless of cycle length.

If we calculate the day of menstruation from the beginning of the cycle, in a 28-day cycle, ovulation occurs on day 14 and in a 35-day cycle, ovulation will occur on day 22.

Option B: LH surge leads to ovulation. Ovulation does not occur before LH surge.

Option C: The ruptured ovarian follicle (after ovulation) is converted into corpus luteum. Ovulation does not occur after corpus luteal maturation.

Option D: Ovulation will occur 14 days before menstruation, i.e. on day 22 of a 35-day cycle. It does not occur 22 days before menstruation.

Solution to Question 3:

The human corpus luteum starts regressing ~10 days after ovulation.

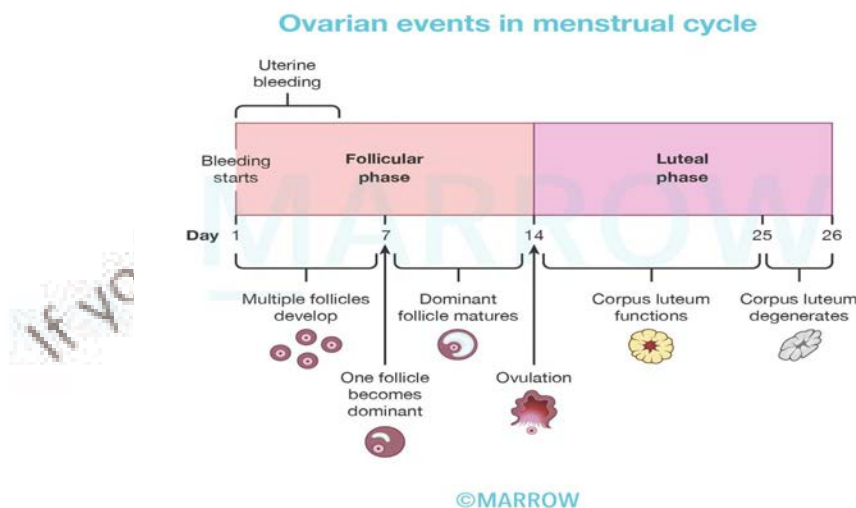
Ovulation occurs on day 14, and the day corpus luteum begins to regress on day 25 of the menstrual cycle.

Therefore,

Days after ovulation when corpus luteum starts regressing = $25 - 14 = 11$ days

The human corpus luteum is programmed to live for 14 days, plus or minus 2 days (corpus luteum of menstruation).

Figure Events of the menstrual cycle:



Solution to Question 4:

Estrous cycle is not seen in humans.

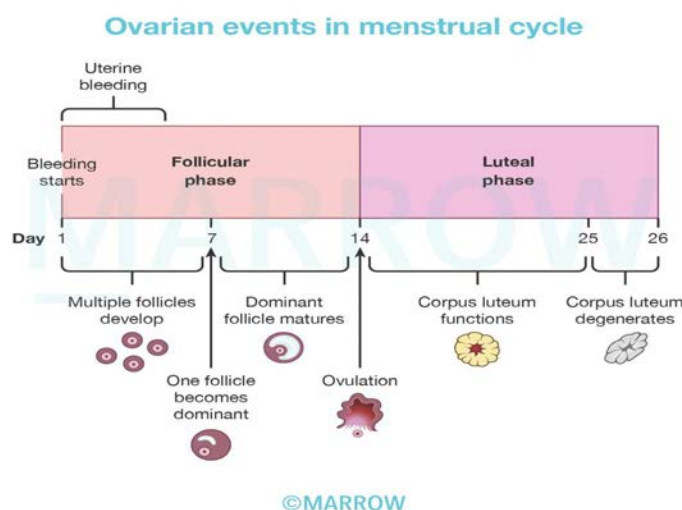
Mammals other than primates do not menstruate, and their sexual cycle is called an estrous cycle. It is named for the conspicuous period of "heat" (estrus) at the time of ovulation, normally the only time during which the sexual interest of the female is aroused.

In spontaneously ovulating species with estrous cycles, such as the rat, no episodic vaginal bleeding occurs, but the underlying endocrine events are essentially the same as those in the menstrual cycle.

In other species, ovulation is induced by copulation (reflex ovulation).

Solution to Question 5:

- Progesterone is the dominant hormone during the luteal phase.
- The corpus luteum in the non-pregnant woman secretes large quantities of progesterone and a small amount of estrogen, as well as inhibin.
- In terms of ovarian function, the menstrual cycle may be divided into two phases approximately equal in length and separated by ovulation (Figure):
- The follicular phase, during which a mature follicle and secondary oocyte develop; and
- The luteal phase begins after ovulation and lasts until the death of the corpus luteum.



Follicular phase (days 0–14)

- A primordial follicle develops to the Graafian stage, with atresia of neighboring follicles.
- LH and FSH receptors are up-regulated in theca and granulosa cells.
- Estradiol levels increase and cause proliferation of the uterus.
- FSH and LH levels are suppressed by the negative feedback effect of estradiol on the anterior pituitary.
- Progesterone levels are low.

Ovulation (day 14)

- Occurs 14 days before menses, regardless of cycle length. Thus, in a 28-day cycle, ovulation occurs on day 14; in a 35-day cycle, ovulation occurs on day 22.

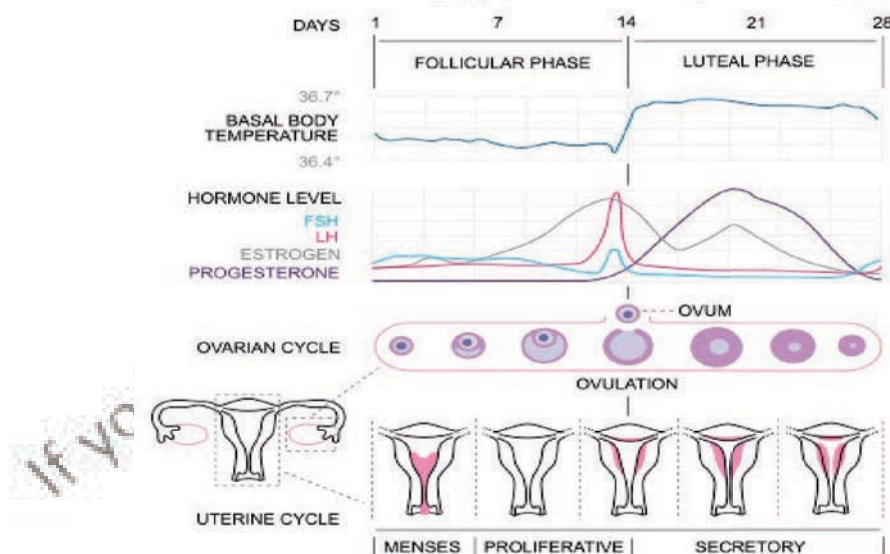
- A burst of estradiol synthesis at the end of the follicular phase has a positive feedback effect on the secretion of FSH and LH (LH surge).
- Ovulation occurs as a result of the estrogen-induced LH surge.
- Estrogen levels decrease just after ovulation (but rise again during the luteal phase).
- Cervical mucus increases in quantity; it becomes less viscous and more penetrable by sperm.

Luteal phase (days 14–28)

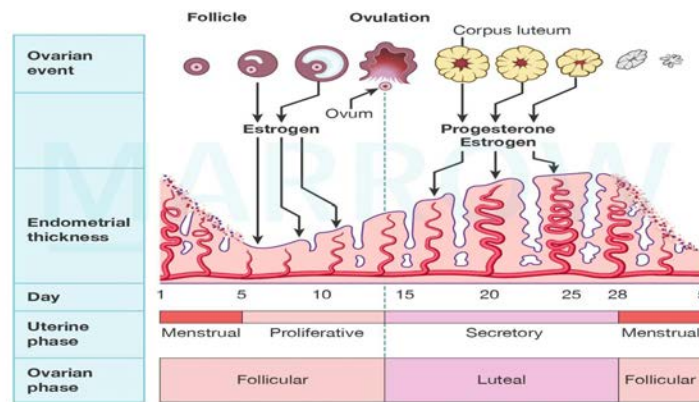
- The corpus luteum begins to develop, and it synthesizes estrogen and progesterone. Vascularity and secretory activity of the endometrium increase to prepare for receipt of a fertilized egg.
- Basal body temperature increases because of the effect of progesterone on the hypothalamic thermoregulatory center.
- If fertilization does not occur, the corpus luteum regresses at the end of the luteal phase. As a result, estradiol and progesterone levels decrease abruptly.

Menses (days 0–4)

- The endometrium is sloughed because of the abrupt withdrawal of estradiol and progesterone



Relationship between ovarian and uterine changes during the menstrual cycle



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Solution to Question 6:

Progesterone is mainly responsible for the maintenance of endometrial thickness in the post-ovulatory phase.

The menstrual cycle consists of three phases: the menstrual phase, the proliferative phase, and the secretory, or progesterational, phase.

Menstrual phase: This phase is characterized by the discharge of blood and endometrial debris from the vagina.

- It coincides with the end of the ovarian luteal phase and onset of a new follicular phase.
- Withdrawal of progesterone on the degeneration of the corpus luteum (CL) leads simultaneously to sloughing of the endometrium (menstruation) and development of new follicles in the ovary under the influence of rising gonadotropic hormone levels.

Proliferative Phase: Thus, menstrual flow ceases, and the proliferative phase of the uterine cycle begins concurrently with the last portion of the ovarian follicular phase.

- When the menstrual flow ceases, a thin endometrial layer less than 1-mm thick remains. Estrogen stimulates proliferation of epithelial cells, glands, and blood vessels in the endometrium, increasing this lining to a thickness of 3 to 5 mm.
- The estrogen-dominant proliferative phase lasts from the end of menstruation to ovulation. Peak estrogen levels trigger the luteinizing hormone (LH) surge responsible for ovulation.

Secretory or Progesterational Phase: After ovulation, when a new corpus luteum is formed, the uterus enters the secretory, or progesterational, phase, which coincides in time with the ovarian luteal phase. The corpus luteum secretes large amounts of progesterone and estrogen. Progesterone converts the thickened, estrogen-primed endometrium to a richly vascularized, glycogen-filled tissue.

- This period is called either the secretory phase, because the endometrial glands are actively secreting glycogen into the uterine lumen for early nourishment of a developing embryo before it

implants, or the progestational (“before pregnancy”) phase, referring to the development of a lush endometrial lining capable of supporting an early embryo after implantation.

- If fertilization and implantation do not occur, the corpus luteum degenerates and a new follicular phase and menstrual phase begin again.

Solution to Question 7:

Detection of an increased LH level in urine will be performed by this ovulation predictor kit.

The most important physiological marker of imminent ovulation is luteinizing hormone (LH) surge, which is characterized by an increased level of LH in serum and urine. A peak level of 75 ng/mL of LH in serum is required for ovulation. LH surge occurs 24-36 hours before ovulation. LH peak lasts 24 hours.

LH surge causes the following:

- Completion of meiosis of ovum
- Ovulation
- Development of corpus luteum

Solution to Question 8:

Positive feedback action of estrogen for inducing luteinizing hormone (LH) surge is associated with high estrogen: low progesterone.

Phases	Hormone	Type of feedback	Site	FSH and LH secretion
Follicular	Estrogen	Negative	Anterior pituitary	Both suppressed
Midcycle	Estrogen	Positive	Anterior pituitary	Increased LH surge
Luteal	Estrogen Progesterone	Negative Negative	Anterior pituitary Anterior pituitary	Both are decreased

Solution to Question 9:

Increased luteinizing hormone (LH) secretion just before ovulation is due to positive feedback by estrogen.

Phases	Hormone	Type of feedback	Site	FSH and LH secretion
Follicular	Estrogen	Negative	Anterior pituitary	Both suppressed
Midcycle	Estrogen	Positive	Anterior pituitary	Increased LH surge
Luteal	Estrogen Progesterone	Negative Negative	Anterior pituitary Anterior pituitary	Both are decreased

Solution to Question 10:

Human chorionic gonadotrophin (HCG) causes persistence of the corpus luteum and prevents menstruation in pregnancy.

The fate of the corpus luteum

- If the ovum is not fertilized, about 2 days before the end of the monthly cycle, the corpus luteum in the ovary involutes
- If fertilization occurs, under the influence of human chorionic gonadotropin, the corpus luteum in the mother's ovary grows to about twice its initial size by a month or so after pregnancy begins
- The corpus luteum secretes estrogens and progesterone and maintains the decidual nature of the uterine endometrium, which is necessary for the early development of the fetus
- The corpus luteum of pregnancy involutes slowly after the 13th to 17th week of gestation. After that time, the placenta secretes sufficient quantities of progesterone and estrogens to maintain pregnancy for the remainder of the gestation period

Solution to Question 11:

- Granulosa cells express the FSH receptor.
- Theca interna cells express LH receptor.

Theca interna cells have many LH receptors. LH acts via cAMP to increase conversion of cholesterol to androstenedione. The theca interna cells supply androstenedione to the granulosa cells. The granulosa cells make estradiol when provided with androgens.

Granulosa cells have many FSH receptors. FSH facilitates their secretion of estradiol by acting via cAMP to increase their aromatase activity. Mature granulosa cells also acquire LH receptors, and LH also stimulates estradiol production.

Solution to Question 12:

An increase in basal body temperature in the mid-luteal phase is due to the effect of progesterone on the hypothalamic thermoregulatory centre.

Immediately after ovulation, corpus luteum is formed, which secretes progesterone in preparation of the body for potential pregnancy during that cycle. This progesterone acts on the hypothalamus and increases basal body temperature. Charting the daily basal body temperature will show an increase of approximately 0.5 to 1 degrees fahrenheit shortly after ovulation.

Since the rise in basal body temperature coincides with ovulation, abstaining from sexual intercourse until at least 3 days after the woman's temperature has risen above her regular temperature significantly lowers the chance of pregnancy. Alternatively, using barrier methods (e.g. condom, diaphragm) during this time will prevent pregnancy.

However several factors (e.g fever, sleep deprivation etc.) that change the basal body temperature, can affect the efficacy and reliability of this method.

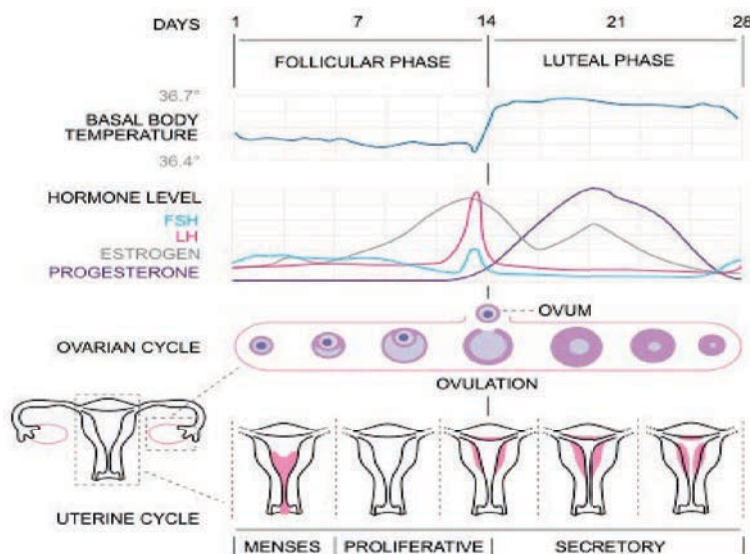
Solution to Question 13:

Whether her cycle was ovulatory or not can be validly assessed by the reported serum level of progesterone.

Day 20 usually corresponds to the luteal phase of the menstrual cycle. In case of anovulatory cycles, no corpus luteum will be formed and so progesterone levels will be low. Hence progesterone levels are valuable to predict if the cycles were ovulatory or not.

Solution to Question 14:

- Estrogen comes from the ovary, and two peaks of secretion occur one just before ovulation and one during the mid-luteal phase.
- After menopause, estrogen secretion declines to low levels.



Solution to Question 15:

Estrogen inhibits FSH and LH secretion in the luteal phase.

Feedback regulation of the hypothalamic-pituitary-ovarian axis

Luteal phase: the corpus luteum secretes large quantities of progesterone and estrogen. They exert a negative feedback effect on the anterior pituitary gland causing the suppression of both FSH and LH secretion.

Preovulatory period: The high level of estrogens at this time (or the beginning of progesterone secretion by the follicles) causes a positive feedback stimulatory effect on the anterior pituitary leading to a surge of LH and to a lesser extent, FSH.

Follicular phase: secretion of estrogen, progesterone from the corpus luteum decreases which leads to loss of negative feedback effect of these hormones on the anterior pituitary. Therefore, pituitary secretion of FSH begins to increase again, as much as twofold. LH secretion increases slightly as well.

GnRH stimulates FSH and LH secretion. Growth hormone and IGF may cause an increase in FSH and LH levels.

Solution to Question 16:

The inhibins inhibit FSH secretion by the gonadotrophs of the anterior pituitary (hence the name inhibin) in a classic negative feedback arrangement.

Solution to Question 17:

The inhibins are produced by the granulosa cells of the ovarian follicle, as well as other tissues, including the pituitary, the brain, the adrenal gland, the kidney, the bone marrow, the corpus luteum, and the placenta. Inhibin A is produced by the corpus luteum whereas inhibin B is mainly produced by granulosa cells of the follicle

FSH specifically stimulates the granulosa cells to produce inhibins.

Just before ovulation, after the granulosa cells acquire LH receptors, LH also stimulates the production of inhibin by granulosa cells.

Solution to Question 18:

The source of progesterone during the normal menstrual cycle is corpus luteum.

Progesterone is also secreted by the placenta, and the ovarian follicle (in small amounts).

Solution to Question 19:

Estrogen are secreted primarily by the granulosa cells of the ovarian follicles, the corpus luteum, and the placenta .

Solution to Question 20:

The two-cell—two-gonadotropin hypothesis:

- Estradiol synthesis requires the contribution of two distinct cell types:
- The theca and granulosa cells within the follicle and
- The theca-lutein and granulosa lutein cells within the corpus luteum
- The follicular theca cells, under the control of LH, produce androgens that diffuse to the follicular granulosa cells, where they are converted to estrogens via an FSH-supported aromatization reaction. The granulosa cells cannot convert progesterone to androstenedione because of the lack of the enzyme 17^o-hydroxylase.

Solution to Question 21:

Progesterone inhibits LH secretion.

Actions of progesterone

- Has negative feedback effects on FSH and LH secretion during the luteal phase.
- Maintains secretory activity of the uterus during the luteal phase.
- Maintains pregnancy.
- Raises the uterine threshold to contractile stimuli during pregnancy, i.e decreases sensitivity of the uterus to oxytocin.
- Participates in the development of the breasts.
- Increases body temperature
- It stimulates respiration, and the alveolar PCO₂ in women during the luteal phase of the menstrual cycle is lower than that in men.
- Large doses of progesterone produce natriuresis, probably by blocking the action of aldosterone on the kidney.

Solution to Question 22:

- Estradiol is the most abundant in the reproductive age group and is 12 and 80 times more potent than estrone and estriol respectively.
- The principal sex steroids in females are estrogen, progestin, and androgen. Three estrogens are present in significant quantities—estradiol, estrone, and estriol.

Solution to Question 23:

Estradiol is the predominant estrogen in the plasma of premenopausal non-pregnant women. It is primarily secreted from ovaries.

Estrone is synthesized from dehydroepiandrosterone in adipose tissue stroma. Estrone is the abundant circulating form of estrogen in postmenopausal women.

In pregnancy, the placenta uses dehydroepiandrosterone and its 16 alpha hydroxyl derivative to form large amounts of estrone and estriol.

Solution to Question 24:

Estrogen does not stimulate osteoclast. Rather it prevents osteoporosis by inhibiting osteoclastic activity.

Actions of Estrogen

- Causes maturation and maintenance of the fallopian tubes, uterus, cervix, and vagina.
- Causes proliferation and development of ovarian granulosa cells.
- Proliferation of endometrium
- Causes the development of female secondary sex characteristics at puberty.
- Maintains pregnancy.
- Lowers the uterine threshold to contractile stimuli during pregnancy.
- Causes the ductal development of the breasts.
- Has both negative and positive feedback effects on FSH and LH secretion.
- Up-regulates estrogen, LH, and progesterone receptors.
- Increases secretion of angiotensinogen and thyroid binding globulin
- Epiphyseal closure of long bones
- Increase libido by acting on the hypothalamus.
- Salt and water retention before menstruation
- ↓LDL cholesterol
- ↑ the local production of NO→vasodilation
- Inhibit osteoclast, prevent osteoporosis

Solution to Question 25:

Pubarche or development of axillary and pubic hair is not due to estrogen. It is due to testosterone (in both the sexes).

Puberty in females :

- Involves the beginning of menstrual cycles (menarche), breast development (thelarche), and an increase in adrenal androgen secretion (adrenarche).
- Estradiol induces the development of secondary sex characteristics, including the breasts and reproductive tract, and increased fat in the hips.
- Estrogens also regulate the growth spurt at puberty, vaginal cornification and cervical mucus production, induce closure of the epiphyses, have a positive effect in maintaining bone formation, and can antagonize the degrading actions of parathyroid hormone on bone.

Solution to Question 26:

β HCG (human chorionic gonadotropin) is secreted by the syncytiotrophoblast of the placenta.

Some important features of β HCG:

- hCG is a glycoprotein that contains galactose and hexosamine.
- It is produced by the syncytiotrophoblast of the placenta.
- It is made up of α and β subunits.
- hCG- α is identical to the α subunit of LH, FSH, and TSH.
- hCG is primarily luteinizing and luteotropic and has little FSH activity.
- Its presence in early pregnancy in urine is the basis of the various laboratory tests for pregnancy, and it can sometimes be detected in the urine as early as 14 days after conception.

Solution to Question 27:

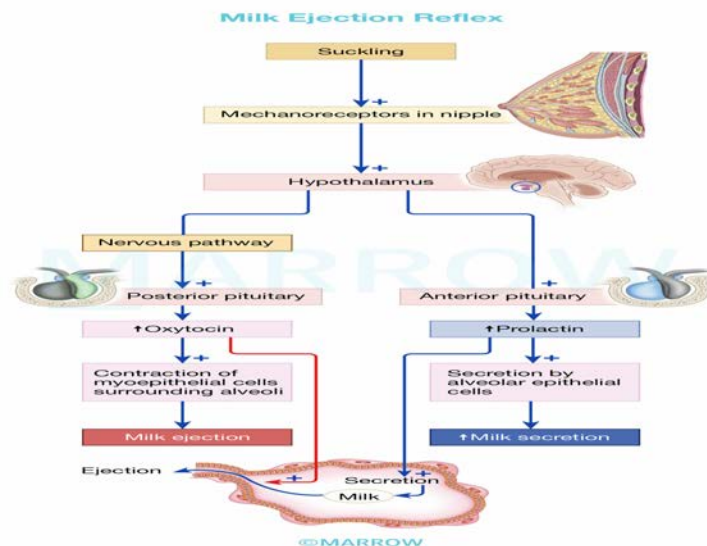
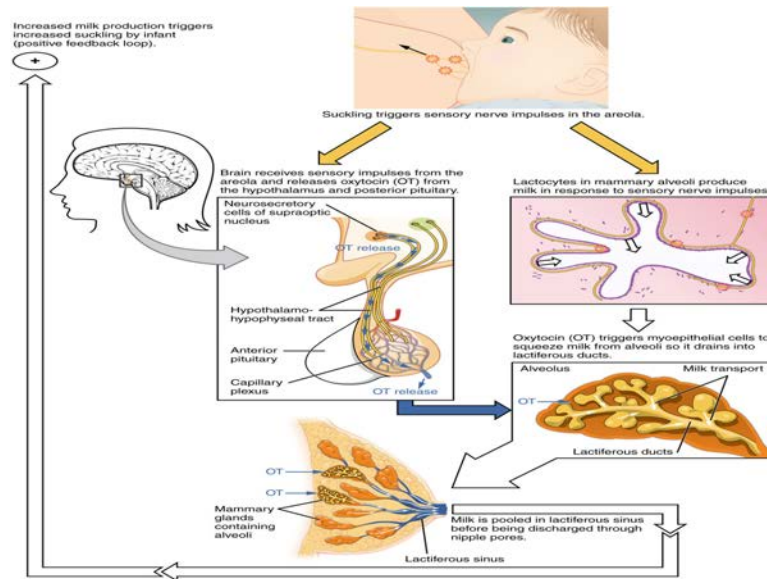
Progesterone has 21 carbon atoms.

Solution to Question 28:

The neuroendocrine part of the anterior pituitary is not involved in milk ejection. The posterior pituitary is involved in milk ejection reflex.

The posterior pituitary secretes oxytocin, which is responsible for milk ejection. The anterior pituitary produces prolactin, which is responsible for lactogenesis.

Figure: Milk Ejection Reflex



Solution to Question 29:

- Oxytocin causes contraction of the myoepithelial cells lining the mammary ducts and ejects milk during suckling. But, it has no galactogenic effect.
- Prolactin is responsible for milk secretion.
- Estrogens are primarily responsible for the proliferation of the mammary ducts and the progesterone for the development of lobules.

Solution to Question 30:

Lactogenesis is not a function of oxytocin. It is caused by prolactin.

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Exercise Physiology

Question 1:

The maximum contractile force of any muscle per square cm cross-sectional area is _____

- a) 3-4 kg/cm²
- b) 5-6 kg/cm²
- c) 10-12 kg/cm²
- d) 15-17 kg/cm²

Question 2:

A budding athlete who is keen to participate in long races visits his dietician for a discussion on his exercise capacity. When an athlete runs at a speed typical for the marathon race, the time until complete exhaustion is maximal with which of the following diets?

- a) High-fat diet
- b) High-carbohydrate diet
- c) Mixed diet
- d) High sodium diet

Question 3:

A professional athlete uses phosphocreatine to improve his sprinting capacity and maximize muscle power during a 100 m dash race. Which of the following energy systems is utilized by the muscles of the athlete's body during a dash race?

- a) Phosphagen system
- b) Glycogen Lactic acid system
- c) Aerobic system
- d) Pentose phosphate pathway

Question 4:

Which of the following is false regarding muscles adapted for endurance running?

- a) Have relatively more mitochondria
- b) Have relatively more myoglobin
- c) Have relatively more capillary content
- d) Twice as large in diameter compared to fast twitch fibers

Question 5:

Regarding respiratory adjustment seen during exercise, which of the following statements is true?

- a) Activity of joint proprioceptors is important factor in increasing ventilation
- b) Arterial pCO₂ rises significantly
- c) Arterio-venous difference in oxygen concentration remains constant
- d) Arterial pH falls

Question 6:

Exercise at high altitudes is not hindered by which of the following factors?

- a) Fall in blood bicarbonate levels
- b) Fall in the total oxygen supply available
- c) Respiratory alkalosis
- d) High blood lactate levels

Question 7:

A 30-year-old man who is extremely motivated to participate in high-intensity exercises is advised to quit smoking by this physician. Which of the following is not a reason behind smoking-related decrease in exercise capacity?

- a) Constriction of the terminal bronchioles
- b) Increased fluid secretion into the bronchial tree
- c) Paralysis of cilia on the surfaces of the respiratory epithelial cells
- d) Desensitization of the chemoreceptors

Question 8:

Which of the following is seen in response to exercise?

- a) Increased cerebral blood flow due to increase in systolic blood pressure
- b) Fall in arterial pH
- c) Increased blood flow in muscles after at least 15 minutes of activity
- d) Increase in body temperature

Question 9:

Which of the following circulatory systems demonstrates an increase in blood supply during exercise?

- a) Cutaneous circulation
- b) Hepato-splanchnic circulation
- c) Coronary circulation
- d) Cerebral Circulation

Question 10:

Which of the following is not true regarding exercise-induced physiological changes?

- a) Left shift of Hb-O₂ dissociation curve
- b) Increase in blood supply to muscle
- c) Increased O₂ extraction
- d) Unaltered arterial pH

Question 11:

Which of the following is the main cause of increased blood flow to exercising muscles ?

- a) Raised blood pressure
- b) Vasodilatation due to local metabolites
- c) Increased sympathetic discharge to peripheral vessels
- d) Increased heart rate

Question 12:

Isometric (static) exercise differs from isotonic (dynamic) exercise in that it causes a greater increase in _____.

- a) Venous return
- b) Muscle blood flow
- c) Diastolic blood pressure
- d) Stroke volume

Question 13:

Regarding circulatory adjustments made during exercise, _____.

- a) The cardiac output increases more than ten-fold.
- b) Increase in cardiac output is mainly due to an increase in stroke volume.
- c) The mean systemic filling pressure rises at the onset of heavy exercise.
- d) In severe exercise, the systolic blood pressure falls.

Question 14:

Which of the following is not true regarding the increase in the blood supply of muscles during exercise?

- a) Decreased beta adrenergic stimulation
- b) Local metabolites
- c) Increased blood pressure
- d) Raised Heart Rate

Question 15:

Which of the following is not seen in strenuous exercise?

- a) Hypokalemia
- b) Increased release of endogenous opioids
- c) Lactic acidosis
- d) Right shift of O₂-Hb dissociation curve

Question 16:

Calculate the respiratory quotient (RQ) in a patient with 50kg body weight, if the CO₂ exhaled is 200ml/min and O₂ consumed is 250 ml/min.

- a) 0.8
- b) 1
- c) 1.25
- d) 1.5

Answer Key

Question No.	Correct Option
1	a
2	b
3	a
4	d
5	a
6	a
7	d
8	d
9	c
10	a
11	b
12	c
13	c
14	a
15	a
16	a

Detailed Explanations

Solution to Question 1:

The strength of a muscle is determined mainly by its size, with a maximal contractile force between 3 and 4 kg/cm² of muscle cross-sectional area.

- The female muscle can achieve almost exactly the same maximal force of contraction as that of the male muscle per square centimeter of cross-sectional area.
- Most of the difference in total muscle performance between males and females lies in the extra percentage of the male body that is muscle, which is caused partly by endocrine differences:
- Testosterone secreted by the male testes has a powerful anabolic effect in causing greatly increased deposition of protein in the muscles.
- Estrogen increases the deposition of fat in the female, especially in the breasts, hips, and subcutaneous tissue.

Solution to Question 2:

When athletes run at speeds typical for the marathon race, the time that they can sustain the race until complete exhaustion (also known as endurance) is maximal with a high-carbohydrate diet.

A person, who consumes a high-carbohydrate diet, stores far more glycogen in muscles than does a person who consumes either a mixed diet or a high-fat diet. Therefore, endurance is enhanced by a high-carbohydrate diet.

When athletes run at speeds typical for the marathon race, their endurance is approximately the following:

Diet	Minutes
High-carbohydrate diet	240
Mixed diet	120
High-fat diet	85

Solution to Question 3:

The phosphagen system is utilized by the muscles of the athlete's body during a dash race.

The combined amounts of cell ATP and cell phosphocreatine are called the phosphagen energy system. These substances together can provide maximal muscle power for 8 to 10 seconds, almost enough for the 100-meter run. Thus, the energy from the phosphagen system is used for maximal short bursts of muscle power.

Energy systems:

- The phosphocreatine-creatine system: The combined amounts of cell ATP and cell phosphocreatine are called the phosphagen energy system. These substances together can provide maximal muscle power for 8 to 10 seconds.
- The glycogen–lactic acid system: can provide 1.3 to 1.6 minutes of maximal muscle activity.

- The aerobic system: The aerobic system is the oxidation of foodstuffs in the mitochondria to provide energy. It can sustain muscle activity for an unlimited time (as long as nutrients last).

Solution to Question 4:

Slow-twitch fibres, which are adapted for endurance running, are not larger in diameter than fast-twitch fibres. Fast-twitch fibres are about twice as large in diameter compared with slow-twitch fibres.

Fast-Twitch and Slow-Twitch Muscle Fibers

- Fast-twitch fibres can deliver extreme amounts of power from a few seconds to a minute.
- The enzymes that promote rapid release of energy from the phosphagen and glycogen–lactic acid energy systems are more active in fast-twitch fibres.
- Fast-twitch fibres are about twice as large in diameter compared with slow-twitch fibres.
- Gastrocnemius muscle has a higher preponderance of fast-twitch fibres, which gives it the capability of forceful and rapid contraction of the type used in jumping.
- Slow-twitch fibres are mainly organized for endurance, especially for the generation of aerobic energy.
- They have far more mitochondria than do the fast-twitch fibres.
- They contain considerably more myoglobin. The extra myoglobin increases the rate of diffusion of oxygen throughout the fibre.
- The enzymes of the aerobic metabolic system are considerably more active in slow-twitch fibres.
- The number of capillaries is greater in the vicinity of slow-twitch fibres.
- Soleus muscle has a higher preponderance of slow-twitch muscle fibres and therefore is used to a greater extent for prolonged lower leg muscle activity.

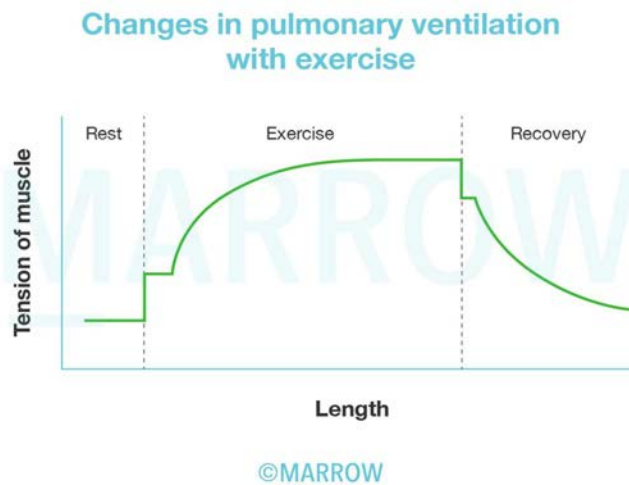
Solution to Question 5:

Signals from joint proprioceptors play a role in stimulating the respiratory drive during exercise.

Regulation of respiration in exercise

- During strenuous exercise, O₂ consumption and CO₂ formation can increase as much as 20-fold. Yet in the healthy athlete, the arterial PO₂, PCO₂, and pH remain almost exactly normal. This is because the alveolar ventilation increases along with the increased level of oxygen metabolism.
- Part of this stimulation of alveolar ventilation results from direct stimulation of the respiratory center by nervous signals. An additional part is believed to result from sensory signals transmitted into the respiratory center from the contracting muscles and joint proprioceptors.
- During strenuous exercise, the utilization of O₂ in the entire body can increase to 75% to 85%. So, essentially all the oxygen is given to the exercising muscles leading to an increased

arterio-venous difference in oxygen concentration.



Solution to Question 6:

The fall in blood bicarbonate levels is compensatory, and it helps increase the exercise effort, not hinder it.

Adaptation at High Altitudes

The work capacity is reduced at high altitudes and this reduction is in direct proportion to the decrease in O₂ uptake. (Option B)

Immediate exposure to low PO₂ at higher altitudes stimulates the arterial chemoreceptors, and this stimulation increases alveolar ventilation.

This immediate increase in pulmonary ventilation upon rising to a high altitude blows off large quantities of CO₂, reducing the PCO₂ and increasing the pH of the body fluids (Respiratory alkalosis). This raised pH inhibits the brain stem respiratory center and opposes the increased ventilation. So, respiratory alkalosis will hinder the maximal exercise effort. (Option C).

To compensate for this respiratory alkalosis, kidneys reduce hydrogen ion secretion and increase bicarbonate excretion. This reduces plasma and cerebrospinal fluid bicarbonate concentration and brings the pH towards normal. This removes the inhibitory effect of alkalosis on respiration. Hence the fall in blood bicarbonate facilitates the exercise effort.

NOTE

Lactic acid causes extreme fatigue. (Option D)

Ordinarily, the pyruvic acid from glycolysis enters the mitochondria of muscle cells and reacts with oxygen to form many ATP molecules.

However, when there is insufficient oxygen (at high altitudes), most of the pyruvic acid then is converted into lactic acid, which diffuses out of the muscle cells into the interstitial fluid and blood.

Solution to Question 7:

Smoking causes no change to the sensitivity of the chemoreceptors.

Effect of Smoking on Pulmonary Ventilation in Exercise

- Nicotine causes constriction of the terminal bronchioles of the lungs, which increases the resistance of air flow into and out of the lungs.
- The irritating effects of the smoke causes increased fluid secretion into the bronchial tree and some degree of swelling of the epithelial linings, which again increases airflow resistance.
- Nicotine paralyzes the cilia on the surfaces of the respiratory epithelial cells. As a result, much debris accumulates in the passageways, and this adds further to the difficulty in breathing.
- All these factors together make even a light smoker experience respiratory strain during maximal exercise, and their level of performance may be reduced.
- In persons with severe emphysema, as much as four-fifths of the respiratory membrane can be destroyed; then, even the slightest exercise can cause respiratory distress. Chronic smoking is an important cause of emphysema.

Solution to Question 8:

Exercise causes an increase in body temperature.

Almost all the energy released by the body's metabolism of nutrients is eventually converted into body heat. Since metabolism rises significantly during exercise there is an increase in the body temperature. This increase in body temperature is significant and even lethal conditions such as heatstroke can easily develop in an athlete.

Other options:

Option A: Cerebral blood flow is "autoregulated" extremely well between arterial pressure limits of 60 and 140 mm Hg. That is, the mean arterial pressure can be increased to as high as 140 mm Hg without significant change in cerebral blood flow.

Option B: The arterial P_{O_2} , P_{CO_2} , and pH remain almost exactly normal even though during strenuous exercise, O_2 consumption and CO_2 formation can increase as much as 20 fold.

Option C: The increase in blood flow to exercising muscles is immediate and not after 15 minutes.

Solution to Question 9:

Coronary circulation increases during exercise.

During exercise, blood flow

- Decreases in inactive skeletal muscles, kidney, liver, and GIT

- Remains unaltered in brain (due to autoregulation)
- Increases in exercising muscles, coronary circulation
- Initially decreases due to redistribution, but later increases due to an increase in temperature in the skin

Solution to Question 10:

Left shift of Hb-O₂ dissociation curve is not seen during exercise.

During exercise, several factors shift the Hb-O₂ dissociation curve considerably to the right, thus delivering extra amounts of O₂ to the active, exercising muscle fibers.

The right shift of Hb-O₂ dissociation curve is due to the following:

- Release of large quantities of CO₂ by the exercising muscles
- Increase in the hydrogen ion concentration in the muscle capillary blood
- Increase in the temperature of the muscle by 2° to 3°C

Solution to Question 11:

The tremendous increase in muscle blood flow that occurs during skeletal muscle activity is caused mainly by chemicals acting directly on the muscle arterioles to cause dilation.

These chemicals include:

- Potassium ions,
- Adenosine triphosphate (ATP),
- Lactic acid, and
- Carbon dioxide.

Other factors leading to increased muscle flow include

- sympathetic nervous system activation
- increase in arterial pressure, and
- increase in cardiac output.

However, the effect of local metabolites is greater than these factors in increasing the muscle blood flow. Hence, the best answer is local metabolites.

Solution to Question 12:

There is a sharp rise in the diastolic blood pressure in isometric exercise. This is due to the sustained compression of the blood vessels by the contracting muscles. In isotonic exercise, there

is much vasodilatation in exercising muscles. So, even though the muscles are contracting, there is no rise in the diastolic blood pressure.

Option (a) Since the muscle pump is more active during isotonic exercise, the venous return is higher in isotonic exercise.

Option (b) The increase in the muscle blood flow is less compared to isotonic exercise since inflow of blood is obstructed by sustained compression exerted by the contracting muscle.

Option (d) Isometric exercise increases the after load of the heart leading to less increase in the stroke volume of the heart.

Solution to Question 13:

The mean systemic filling pressure rises tremendously at the onset of heavy exercise.

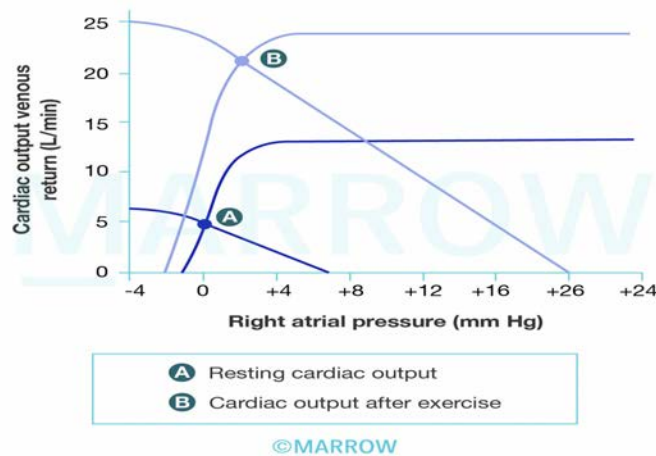
This occurs as a result of

- The sympathetic stimulation contracts the veins.
- Tensing of the abdominal and other skeletal muscles of the body compresses many of the internal vessels, thus providing more compression of the entire vascular system and causing a still greater increase in mean systemic filling pressure.

Cardiovascular Changes in Exercise

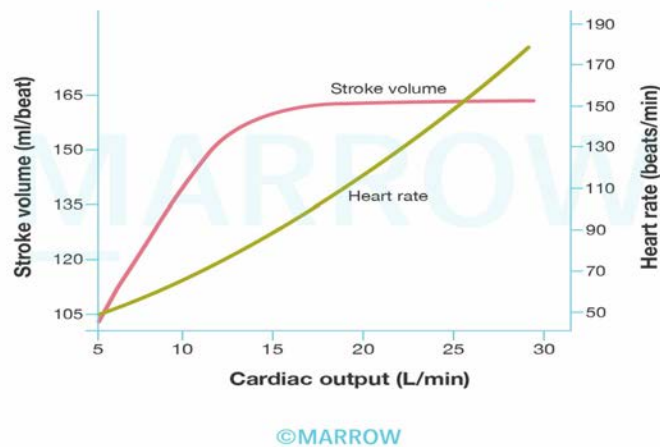
- The maximum increase of cardiac output is 4-fold that can commonly be achieved by the untrained runner and 7-fold that can be achieved in some marathon runners.
- The increase in heart rate accounts for a greater proportion of the increase in cardiac output than the increase in stroke volume during sustained strenuous exercise.
- Exercise also causes an increased venous return due to
 - Increased activity of the respiratory pump
 - Increased activity of the muscular pump
 - Increased venous tone due to sympathetic stimulation
 - Increased blood flow from arteries to veins
- In severe exercise, there is increased systolic and mean blood pressure due to
 - vasoconstriction of the arterioles and small arteries in most tissues of the body
 - increased pumping activity by the heart
 - a great increase in mean systemic filling pressure caused mainly by venous contraction.

Cardiovascular changes in exercise



- As seen in the figure, point A is the resting cardiac output. On exercise, there is a rise in the mean systemic filling pressure shown on the X-axis.
- The venous return also rises which causes the cardiac output to reach point B, which is 4 times the normal resting cardiac output.

Relationship of stroke volume and heart rate to cardiac output



Solution to Question 14:

During exercise, there is increased beta-adrenergic stimulation.

Medullae of the two adrenal glands secrete large amounts of norepinephrine plus epinephrine into the circulating blood during strenuous exercise. The epinephrine often has a slight vasodilator effect because it excites more of the beta-adrenergic receptors of the vessels, which are vasodilator receptors, in contrast to the alpha vasoconstrictor receptors excited especially by norepinephrine.

Solution to Question 15:

Strenuous exercise can cause hyperkalemia by releasing potassium from skeletal muscle.

NOTE: Exercise leads to release of endogenous opioids and is associated with mood enhancement.

Solution to Question 16:

The respiratory quotient is 0.8.

The respiratory quotient is the ratio of carbon dioxide production to oxygen utilization.

In the given question, the CO₂ production is 200mL/min, while the O₂ consumption is 250mL/min.

Therefore,

$$\text{Respiratory quotient} = 200/250 = 0.8$$

The respiratory quotient (RQ) is used for determining the metabolic utilization of various foods.

- When carbohydrates are metabolized with one molecule of oxygen, one carbon dioxide molecule is released. Therefore, the RQ of carbohydrates = 1.
- When fats are oxidized, for every 100 molecules of oxygen consumed, 70 molecules of CO₂ are released. Therefore, the RQ of fats is 0.70.
- Similarly, the respiratory quotient for proteins is 0.80.